

Essentials of Medicine for Dental Students

Anil K Tripathi
Kamal K Sawlani

SECOND EDITION

JAYPEE

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ESSENTIALS OF MEDICINE FOR DENTAL STUDENTS



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Second Edition

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Essentials of Medicine for Dental Students

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► PREFACE TO SECOND EDITION

It is a matter of immense pleasure and gratification that we are bringing the second edition of the book *Essentials of Medicine for Dental Students*. The book has received tremendous response all over the country. We have received compliments as well as valuable suggestions. We are thankful to readers and all those who have sent their views.

We have tried to adhere to the latest curriculum prescribed by Dental Council of India. For ready reference, recent curriculum of General Medicine as prescribed by Dental Council of India has been added in the book. Accordingly, we have included some new topics or added more details such as halitosis, dysphagia, infectious mononucleosis, herpes, acute respiratory distress syndrome, pneumothorax, lung cancer, interstitial lung disease, shock, cardiac arrest, cardiorespiratory resuscitation, arrhythmias, Addison's disease, etc. In particular, a chapter on "Medical Emergencies in Dental Practice" has been specially added keeping in view of ever-growing need for dental surgeons to learn and practice emergency medicine.

There was feedback from the students that the text in some of the chapters was too concised to comprehend and remember. Hence, in this edition, we have reframed the language of the text, which has now become comfortable and easy to retain. The running text has also been written in point-wise fashion. More figures and tables have been added for better illustration of the subject.

The medical science is ever evolving. Newer understanding of the diseases, newer diagnostics and drugs need to be included in the book to enable us to provide state-of-the art treatment. Hence, we have modified definitions, classifications and treatment protocols in order to include recent advancements in the subject.

We sincerely hope that the present edition of the book will be even more useful for the students as well as practicing dental surgeons. We look forward to receiving valuable views, comments and suggestions from our esteemed readers.

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► PREFACE TO THE FIRST EDITION

Over the years, a customized and concise textbook of General Medicine for dental students and dental practitioners has been much awaited. Students need a textbook that is easily readable, concise and in accordance with the prescribed curriculum.

The primary objective of this book is to provide basic and practical information on the pertinent topics that will enable the reader to understand and improve his/her diagnostic and therapeutic skills. Special attempt has been made to make the book succinct without compromising on the required details.

The chapters are chosen according to the Dental Council of India guidelines. However, additional chapters on other important subjects such as HIV/AIDS, malaria, stroke, pleural diseases have also been included. Emphasis has been given on the clinical methods including history taking and physical examination which are described in the beginning of each system. This will make a wholesome reading without much need to go for other books on clinical methods.

Each chapter contains relevant “Multiple choice questions” and “Fill in the blanks” to help the reader self-assess their knowledge. In addition to them, a “Model test paper” is also provided at the end of the book which will enable the readers to prepare for the competitive entrance tests.

The essence of learning General Medicine by dental students is in applying the concept and knowledge while they deal with patients suffering from various medical disorders. In such endeavor, each chapter is followed by a section “Implications on dental practice” which describes how the presence of medical disorders affects the management decisions.

While the book is intended primarily for dental undergraduate and postgraduate students, this should also be useful for medical students and practitioners.

Although every attempt has been made to avoid any error or controversy, shortcomings are inevitable. Readers are requested to offer their valuable comments and suggestions that will be of great help in improving the next edition.

AK Tripathi
KK Sawlani

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We owe to our family members for their constant support and encouragement without which the task could never have been possible.

We are also indebted to our teachers and students who have been constant source of inspiration and learning. We acknowledge the contributions of various experts and readers who provided us with their valuable comments and suggestions.

Finally, we thank and highly appreciate the efforts of M/s Jaypee Brothers Medical Publishers (P) Ltd., New Delhi for excellent and outstanding quality of the book.

CONTENTS

Clinical Methods 1

CHAPTER 1

- History 1
- Physical Examination 2
- Scheme of the General Examination 12
- Definitions 13
- Self Assessment 13

Gastrointestinal and Hepatobiliary System 15

CHAPTER 2

- Symptoms and Signs of Gastrointestinal Diseases 15
- Stomatitis and Oral Ulcers 18
- Gingival Hyperplasia 20
- Dysphagia 20
- Gastritis 22
- Peptic Ulcer 23
- Diarrhea, Dysentery and Food Poisoning 25
- Malabsorption 28
- Liver: Structure and Function 30
- Jaundice 32
- Acute Hepatitis 35
- Acute Viral Hepatitis 35
- Chronic Hepatitis 39
- Cirrhosis of Liver 40
- Portal Hypertension 42
- Upper Gastrointestinal Bleeding 43
- Hepatic Encephalopathy 43
- Ascites 44
- Hepatomegaly 46
- Implications on Dental Practice 47
- Self Assessment 47

Hematological System 50

CHAPTER 3

- Anemia 50
- Iron Deficiency Anemia (IDA) 51
- Megaloblastic Anemia 53
- Aplastic Anemia 55

- Hemolytic Anemia 56
- Anemia of Acute Blood Loss 58
- Leukemia 58
- Lymphomas 61
- Multiple Myeloma 64
- Normal Hemostasis 64
- Platelet Disorders 67
- Von Willebrand's Disease 69
- Coagulation Disorders 69
- Splenomegaly 73
- Lymphadenopathy 74
- Implications on Dental Practice 75
- Self Assessment 76

Cardiovascular System 78

CHAPTER 4

- Symptoms and Signs of Cardiovascular Diseases 78
- Cardiac Examination 81
- Investigations 82
- Acute Rheumatic Fever 83
- Rheumatic Valvular Heart Disease 84
- Aortic Stenosis (AS) 87
- Aortic Regurgitation (AR) 88
- Infective Endocarditis 89
- Hypertension 92
- Ischemic Heart Disease/Coronary Artery Disease 96
- Stable Angina 97
- Unstable Angina 99
- Acute Myocardial Infarction or STEMI 100
- Heart Failure 103
- Acute Pulmonary Edema (cardiogenic) 107
- Congenital Heart Disease 108
- Syncope 110
- Arrhythmia 112
- Bradyarrhythmias 113
- Tachyarrhythmias 114
- Implications on Dental Practice 118
- Self Assessment 118

Respiratory Diseases 120

CHAPTER 5	○ Symptoms	120
	○ Examination	122
	○ Investigations	125
	○ Pneumonia	126
	○ Lung Abscess	128
	○ Bronchiectasis	129
	○ Bronchial Asthma	130
	○ Chronic Obstructive Pulmonary Disease (COPD)	134
	○ Pulmonary Eosinophilia	136
	○ Pulmonary Embolism	137
	○ Tuberculosis	139
	○ Respiratory Failure	144
	○ Pleural Diseases	145
	○ Pneumothorax	146
	○ Interstitial Lung Disease	147
	○ Lung Cancers (Bronchogenic Carcinoma)	148
○ Implications on Dental Practice	150	
○ Self Assessment	151	

Renal Diseases 153

CHAPTER 6	○ Renal Syndromes	153
	○ Investigations in Renal Disorders	154
	○ Nephrotic Syndrome	154
	○ Nephritic Syndrome	156
	○ Acute Renal Failure (ARF)	158
	○ Chronic Renal Failure (CRF)	159
	○ Implications on Dental Practice	160
	○ Self Assessment	160

Nervous System 162

CHAPTER 7	○ Examination of Cranial Nerves	162
	○ Palatal Paralysis	166
	○ Examination of a Comatose Patient	167
	○ Epilepsy	169
	○ Meningitis	174
	○ Headache	178
	○ Facial Pain	180
	○ Facial Nerve Palsy	181
	○ Cerebrovascular Diseases	184
	○ Implications on Dental Practice	185
	○ Self Assessment	186

Endocrine and Metabolic Disorders 188

CHAPTER 8	○ Thyroid Disorders	188
	○ Calcium Metabolism	193
	○ Parathyroid Disorders	195
	○ Pituitary Gland	196
	○ Implications on Dental Practice	208
	○ Self Assessment	209

Infections 211

CHAPTER 9	○ Measles (Rubeola)	211
	○ Mumps	212
	○ Rubella (German Measles)	213
	○ Chickenpox (Varicella) and Herpes Zoster (Shingles)	214
	○ Herpes Simplex	216
	○ Infectious Mononucleosis	218
	○ Diphtheria	220
	○ Enteric Fever (Typhoid Fever)	221
	○ Gonorrhea	223
	○ Syphilis	224
	○ Amoebiasis	226
	○ Malaria	228
	○ Human Immunodeficiency Syndrome/ Acquired Immunodeficiency Syndrome (HIV/AIDS)	231
	○ Implications on Dental Practice	238
	○ Self Assessment	238

Medical Emergencies in Dental Practice 241

CHAPTER 10	○ Postural Hypotension (Orthostatic Hypotension)	242
	○ Hypertensive Crisis	242
	○ Acute Pulmonary Edema	243
	○ Chest Pain	243
	○ Asthmatic Attack	243
	○ Airways Obstruction	243
	○ Hyperventilation	244
	○ Seizures	244
	○ Stroke (Cerebrovascular Accident)	245

○	Anaphylaxis	245
○	Hypoglycemia	245
○	Adrenal Crisis	245
○	Excessive Bleeding	246
○	Dental Procedures in Pregnant Women	246
○	Excessive Bleeding	245
○	Dental Procedures in Pregnant Women	245

Critical Care 247

CHAPTER 11	○	Shock	247
	○	Acute Respiratory Distress Syndrome (ARDS)	252
	○	Cardiovascular Collapse	253
	○	Cardiac Arrest	253
	○	Cardiopulmonary Resuscitation	253

Anaphylaxis and Drug Allergy 256

CHAPTER 12	○	Anaphylaxis	256
	○	Drug Allergy	259
	○	Implications on Dental Practice	260
	○	Self Assessment	260

Nutrition 262

CHAPTER 13	○	Diet and Nutrition	262
	○	Protein Energy Malnutrition (PEM)	263
	○	Vitamins	264
	○	Implications on Dental Practice	269
	○	Self Assessment	270

Preoperative Evaluation 271

CHAPTER 14	○	Cardiovascular Evaluation	271
	○	Pulmonary Evaluation	271
	○	Hematological Evaluation	271
	○	Endocrinal Evaluation	271
	○	Nutritional Evaluation	272
	○	Evaluation of Other Organs	272
	○	Preoperative Laboratory Evaluation	272

Test Paper	273
Reference Laboratory Values	281
Answers	285
Index	289

THEORY SYLLABUS

1. Aims of Medicine

Definitions of Signs, Symptoms, Diagnosis, Differential Diagnosis, Treatment and Prognosis.

2. GIT and Hepatobiliary System

CORE TOPICS (Must Know)

- Stomatitis and Oral Ulcers
- Gingival Hyperplasia
- Dysphagia
- Gastritis
- Peptic Ulcer
- Jaundice
- Acute Viral Hepatitis
- Cirrhosis of Liver
- Ascites
- Hepatomegaly

COLLATERAL TOPICS (Desirable to Know)

- Diarrhea
- Dysentery
- Amoebiasis
- Malabsorption

3. Cardiovascular System

CORE TOPICS (Must Know)

- Acute Rheumatic Fever
- Rheumatic Valvular Heart Disease
- Infective Endocarditis
- Hypertension
- Ischemic Heart Disease
- Congestive Heart Failure
- Acute Pulmonary Edema
- Congenital Heart Disease
- Common Arrhythmias

4. Respiratory System

CORE TOPICS (Must Know)

- Pneumonia
- COPD
- Pulmonary Tuberculosis
- Bronchial Asthma

COLLATERAL TOPICS (Desirable to Know)

- Lung Abscess
- Bronchiectasis
- Pleural Diseases (Pleural Effusion, Pneumothorax)
- Lung Cancer

5. Hematology

CORE TOPICS (Must Know)

- Anemias
- Bleeding and Clotting Disorders
- Leukemias
- Lymphomas
- Agranulocytosis
- Splenomegaly
- Oral Manifestations of Hematological Disorders
- Lymphadenopathy

6. Renal System

CORE TOPICS (Must Know)

- Acute Nephritis
- Nephrotic Syndrome

COLLATERAL TOPICS (Desirable to Know)

- Renal Failure

7. Central Nervous System

CORE TOPICS (Must Know)

- Facial Nerve Palsy
- Facial Pain including Trigeminal Neuralgia
- Epilepsy
- Headache including Migraine

COLLATERAL TOPICS (Desirable to Know)

- Meningitis
- Examination of Comatose Patient
- Examination of Cranial Nerves

8. Infections

CORE TOPICS (Must Know)

- Enteric Fever (Typhoid fever)
- HIV/AIDS

- Herpes Zoster
- Herpes Simplex
- Syphilis
- Diphtheria

COLLATERAL TOPICS (Desirable to Know)

- Infectious Mononucleosis
- Measles (Rubeola)
- Mumps
- Rubella (German Measles)
- Malaria

9. Endocrinal System

CORE TOPICS (Must Know)

- Diabetes Mellitus
- Thyroid Disorders (Hypothyroidism, Thyrotoxicosis)
- Calcium Metabolism
- Parathyroid Disorders
- Pituitary Gland (Acromegaly)

COLLATERAL TOPICS (Desirable to Know)

- Addison's Disease
- Cushing's Syndrome

10. Nutrition

CORE TOPICS (Must Know)

- Avitaminosis

COLLATERAL TOPICS (Desirable to Know)

- Balanced Diet and Nutrition
- Protein Energy Malnutrition

11. Critical Care

CORE TOPICS (Must know)

- Syncope
- Cardiac arrest
- Cardiopulmonary resuscitation (CPR)
- Shock

COLLATERAL TOPICS (Desirable to know)

- Acute left ventricular failure (LVF)
- Adult respiratory distress syndrome ARDS)

12. Emergencies in Dental Practice

- Myocardial Infarction
- Status Epilepticus
- Status Asthmaticus
- Syncope
- Anaphylaxis
- Bleeding
- Arrhythmia

13. Anaphylaxis and Drug Allergy

- Anaphylaxis
- Drug Allergy

▶ LIST OF ABBREVIATIONS

DRUG ADMINISTRATION

b.d.	twice daily
IM	intramuscular
IV	intravenous
o.d.	daily, once a day
q.i.d.	four times a day
q4h	every four hours
SC	subcutaneous
SL	sublingual
stat	immediately
t.i.d.	three times a day, alternative t.d.s.

UNITS

cm	centimeter(s)
d	per day
dL	deciliter(s)

fL	femtoliter(s)
g	gram(s)
h	hour (s)
hrly	hourly
Hg	mercury
IU	international unit (s), alternative U
Kg	kilogram (s)
L	liter(s)
µg	microgram(s)
µL	microliter(s)
mEq	milliequivalent(s)
mg	milligram(s)
ml	milliliter(s)
mm	millimeter(s)
mmol	millimole(s)
min	minute(s)
pg	picogram(s)

Chapter

1

Clinical Methods

Clinical methods form the basis of the approach to a patient by which a proper diagnosis is achieved. The skill of a clinician depends on his knowledge of theoretical as well as practical aspects of the clinical methods. This skill is acquired and refined with experience.

HISTORY

History is the physician's abstraction of certain facts developed in the course of the patient's interview and arranged in a manner that facilitates diagnosis.

Proper history is important for making a correct diagnosis. A careful evaluation by competent clinicians reveals that 82% of diagnoses are made by history, 9% by physical examination and 9% by the laboratory tests. A history is not simply a collection of facts. It must also contain information. Facts are the true statements made by the patient while information consists of facts arranged in useful manner. A general format is followed while taking a history of the patient. The contents of history are recorded in the patient's version and no part in the history should be distorted or omitted. However, the focus and contents may vary from patient to patient and also with the experience of the clinician.

The standard format used for history taking is as follows;

1. Patient's details (name, age, sex, marital status, occupation, address)
2. Presenting complaints with duration
3. History of present illness
4. History of past illnesses
5. Treatment history
6. Personal history
7. Family history
8. Menstrual history (in females)

Presenting Complaints

This is also known as chief complaints. The patient is asked about the main problems for which he has come to the doctor. These main symptoms/problems are listed in a chronological order (noted in the order of their appearance). Generally the patient is allowed to tell by himself. Leading questions are avoided. The list should not be too long.

History of Present Illness

The patient is then asked to narrate individual symptoms in details. Once this is over with, leading questions are asked to clarify certain points or associations related to different symptoms. For example, if the patient is complaining of pain, details should be recorded about the site, severity, character, radiation, duration and timing, relieving and aggravating factors.

- a. *Site:* The exact site of pain is noted. Whether it is localized or diffuse pain.
- b. *Severity:* Does the pain interfere with routine daily activities or keep the patient awake at night? Is the patient in severe agony or is he shouting?
- c. *Character:* Description of character of pain such as burning, stabbing, pricking, colicky, and dull ache are helpful. Colicky pain is the waxing and waning type of pain and may cause the patient to roll about. Colicky pain suggests obstruction of hollow structure like intestine, ureter or common bile duct.
- d. *Timing and duration:* When does it start and when does it stop?
- e. *Relieving factors and aggravating factors:* Cardiac pain occurs on exertion and is relieved by rest and nitrates. Pain of duodenal ulcer is relieved by eating. Musculoskeletal pain may be relieved by change in the

posture and by simple analgesics. Anginal pain is relieved by sublingual nitrates.

The typical anginal (ischemic) pain is described in Table 1.1.

Symptoms pertaining to different systems are asked. Important symptoms regarding disorders of various systems are given in Table 1.2.

History of Past Illness

A detailed account is noted about any illness which occurred in the past. A disease or symptom which has occurred in the past could be a part of the present disease process or related to the present problem.

For example, a patient with liver cirrhosis may give a history of jaundice or blood transfusion.

TABLE 1.1: Characteristics of chest pain in stable angina

Site of the pain	Retrosternal or precordial
Character	Squeezing, constricting, piercing, feeling of heaviness or pressure
Precipitating factors	Physical exertion, cold exposure, heavy meals, emotional stress, anemia, thyroid disease, vivid dreams (nocturnal angina)
Associated features	Feeling of impending death, breathlessness, apprehension, nausea, vomiting
Relieving factors	Rest, sublingual nitroglycerin
Radiation	Left shoulder, both arms, jaw, neck
Duration	Typically 2-10 minutes (>30 minutes suggests infarction)

TABLE 1.2: Common symptoms in various systemic disorders

General
Fever, weight loss, weakness, bodyache, headache
Respiratory system
Cough, sputum, hemoptysis, dyspnea, chest pain and wheezing
Cardiovascular system
Chest pain, dyspnea, orthopnea, palpitation, edema, cough
Gastrointestinal system
Anorexia, nausea, vomiting, heart burn, dysphagia, diarrhea, constipation, jaundice, pain in abdomen
Hematological system
Pallor, weakness, fever, dyspnea, bleeding, lymph gland enlargement
Urinary system
Dysuria, hematuria, polyuria, oliguria, anuria, retention
Nervous system
Headache, seizures, stroke

Treatment History

The details of medications taken are noted. History of any adverse effects of drugs is also asked. It is imperative to know what drugs the patient is taking currently so that drug interactions may be avoided. The history of intake of certain drugs may help in knowing the cause of disease. For example, steroids and NSAIDs can cause gastric erosion and hematemesis.

It is noted whether the patient has been compliant or not. If not, the reason for the drug non-compliance is discussed.

Personal History

The patient is asked about the consumption of alcohol, tobacco, or smoking. His occupation should also be noted. Certain occupations are associated with a higher incidence of a particular disease, e.g. persons working in a silica factory are prone to develop silicosis. A history of stress at home and office should also be recorded. Financial status of the patient is also an important fact to be noted in the history.

Family History

Any history of genetic disorders in the family is enquired about. Any history of similar illness in other family members and cause of death of immediate relative should be recorded. History of hypertension, diabetes mellitus, tuberculosis, cardiovascular diseases, and bleeding diatheses in other family members should be noted.

Menstrual History

Women should be asked about menstruation. Regularity of the cycle, duration of cycle, and amount of bleeding are noted. Obstetric history is also important. Many drugs are contraindicated or avoided during pregnancy. Migraine can be triggered by menstruation and heart failure may become worse during pregnancy. Excessive bleeding during menstruation (menorrhagia) may be due to bleeding disorder and amenorrhea can occur in certain diseases.

PHYSICAL EXAMINATION

- Proper physical examination needs cooperation of the patient.
- The patient should be comfortable and relaxed.

- The nature and need of such an examination should be explained to him.
- Examination is performed in a quiet and well-lit room. Day light is always better than artificial light as changes in skin color may be masked in the latter.
- Examination is carried out as gently as possible.

The examination is carried out in a routine manner. However, the information from the history may suggest which part or system should be particularly examined in greater detail.

The examination is customarily divided into general and systemic examination. **Systemic examination is described in specific chapters.**

General Physical Examination

General examination of the patient starts even as the history is being taken. A standard scheme should be followed to avoid any omissions. Points that should be noted are given in Table 1.3.

Mental and Emotional Status

History taking and simple observations can assess the mental, emotional status and intelligence of the patient. State of consciousness is noted.

- In a *confusional state*, the patient is subdued, drowsy and physically inactive. He is also disoriented about time, place and person.
- *Delerium* is a confusional state accompanied by agitation, hallucination and illusion. These always indicate disease of the nervous system.

TABLE 1.3: Points for general physical examination

- Mental and emotional state
- Built of the body
- Temperature
- Pulse
- Blood pressure
- Respiration
- Anemia
- Jaundice
- Cyanosis
- Oral cavity and throat
- Neck veins
- Thyroid
- Lymphadenopathy
- Clubbing
- Peripheral edema
- Skin and mucous membrane

- *Stupor* state is lesser degree of altered consciousness from which patient can be awakened by vigorous stimuli.
- *Coma* is a deep sleep-like state from which the patient cannot be aroused. The patient does not respond to external stimulus or to inner needs.
- In *dementia*, there is a loss of previously acquired intellectual functions but in the absence of impairment of consciousness. Memory is the most common intellectual function lost in dementia.

Built of the Body

This can be assessed by general inspection. The physique may be short, tall, obese, muscular, thin or asthenic.

- Dwarfism is found in hypopituitarism, hypothyroidism, and achondroplasia.
- Height is increased in Marfan's syndrome and hyperpituitarism (Gigantism).
- Weight loss may occur in malnutrition, malabsorption, thyrotoxicosis, chronic infections (tuberculosis), diabetes mellitus, malignancies, depression, anxiety, and anorexia nervosa. Weight loss despite normal or increased food intake suggests diabetes mellitus, thyrotoxicosis or malabsorption.
- Weight gain may occur due to hypothyroidism or fluid retention.

The most widely used method to measure obesity is body mass index (BMI). BMI is calculated as weight in kg divided by the square of height in meters (kg/m^2). BMI upto 25 is normal, 25-29.9 is overweight and above 30 is obesity. Abdominal obesity (increased waist-hip ratio: >0.9 in females, >1.0 in males) is an important risk factor for coronary artery disease.

Temperature

Temperature is measured with a thermometer. Thermometer is placed in the mouth or in the axilla in adults while it is placed in the fold of the groin with thigh flexed or in the rectum in case of small children.

- Mouth temperature is 0.5°C higher than that of groin or axilla. Rectal temperature is about 0.4°C (0.7°F) higher than mouth temperature.
- The evening (pm) temperature may be up to 0.5°C or 0.9°F higher than the morning (am) temperature in normal persons.

- The maximum normal is 37.2°C (98.9°F) at 6 am and 37.7°C (99.9°F) at 4 pm.
- A fever of more than 41.5°C (106.7°F) is known as *hyperpyrexia*. A temperature less than 35°C (95°F) is called *hypothermia*.

The fever may be continued, remittent or intermittent. These classical patterns of fever are less commonly seen due to early initiation of treatment with antipyretics and antibiotics.

- Fever which at no time touches the normal and does not fluctuate more than 1°C during 24 hours is called *continued fever*.
- When the daily fluctuation in the temperature is more than 2°C, the fever is of the *remittent* type.
- Fever which occurs only for several hours during 24 hours is called *intermittent fever*. Intermittent fever can be quotidian (occurs daily), tertian (occurs on alternate days) or quartan (occurs every third day). Infection with *P. falciparum* causes intermittent quotidian fever, *P. vivax* and *P. ovale* cause tertian fever and *P. malariae* causes quartan fever.

Pulse

Arterial pulse should be examined mainly for following things:

- Rate
- Rhythm
- Volume
- Character
- Radio-femoral delay

The rate and rhythm are assessed by palpating the radial artery (Fig. 1.1). The character of the pulse is better assessed by palpating the carotid artery (Fig. 1.2). Other peripheral arteries like brachial, popliteal, posterior tibial and dorsalis pedis can also be palpated. The pulse may be *absent or weak* in obstruction in the proximal part of the artery due to thromboembolism and atherosclerosis.

Rate: The *pulse rate* is determined by counting it for at least 30 seconds. The normal pulse rate varies from 60-100 per minute.

- Bradycardia is defined as pulse rate <60/min. Important causes are raised intra-cranial pressure, heart blocks and sinus node disease, cholestatic jaundice, hypothy-



FIGURE 1.1: Palpation of the radial artery; the forearm of the patient is in semiprone position with wrist semiflexed



FIGURE 1.2: Palpation of the left carotid artery with the right thumb

roidism and drugs (beta blockers, verapamil, digoxin). Bradycardia may also be present in athletes.

- Tachycardia (pulse rate >100/min) occurs due to fever, exercise, anxiety, thyrotoxicosis, anemia, tachyarrhythmias, shock and drugs.
- Pulse is slower than would be expected from the height of fever in typhoid fever (*relative bradycardia*).

Rhythm: Normally the rhythm of the pulse is regular. An *irregular rhythm* is seen in atrial fibrillation (irregularly irregular) and frequent ectopic beats (regularly irregular).

Character:

- A low volume and slow rising pulse (*parvus et tardus*) is found in aortic stenosis (AS).
- A large bounding pulse (*hyperkinetic pulse*) is seen in hyperkinetic states (anemia, fever, anxiety, exercise), patent ductus arteriosus, ventricular septal defect, and aortic regurgitation (AR).
- *Bisferiens pulse* which has two systolic peaks is found in mixed lesion of AS and AR.
- Alternating strong and weak pulse (*pulsus alternans*) is present in severe left ventricular failure.
- Normally there is a fall in systolic arterial pressure of <10 mmHg during inspiration. An accentuation in this phenomenon can lead to weakening or disappearance of pulse during inspiration (*paradoxical pulse*). This is found in cardiac tamponade and obstructive airway disease.

Radio-femoral delay: The femoral pulse is weak and delayed as compared with the radial pulse (*radio-femoral delay*) in coarctation of aorta.

Blood Pressure (BP)

Blood pressure is measured with the help of a sphygmomanometer. A mercury sphygmomanometer is more accurate than aneroid one.

- BP should be measured in both arms and also in the lower limb. In coarctation of aorta, the arterial pressure in the upper limb is much higher than in the lower limbs.
- The patient should be sitting at ease.
- The cuff should be applied closely to the upper arm; it should not be loose or tight. The lower border of the cuff must be one inch (2.5 cm) above the cubital fossa.
- The instrument should be placed at the same level as the cuff on the patient's arm and the observer's eye.
- The standard cuff width for adults is 12.5 cm. The size of cuff is also important since a small cuff may record false high blood pressure.
- The blood pressure must be recorded when the patient is resting quietly as anxiety, exertion, excitement, smoking and intake of coffee and tea within last half an hour will give rise to false readings.
- In elderly and patients on drugs (for hypertension) BP should be recorded in standing and lying down position to detect the occurrence of postural hypotension.

Initially the assessment of systolic BP is made by palpatory method. The radial or brachial artery is palpated while the cuff is inflated to raise pressure about 30 mm Hg above the level at which radial/brachial pulse disappears. The stethoscope is placed over the brachial artery and cuff is deflated slowly (Figs 1.3A and B). The level at which Korotkoff sounds appear (phase 1) is the systolic pressure and the level at which they disappear completely (phase 5) is the diastolic pressure. When the *pulse pressure* (the difference between systolic and diastolic blood pressure) is increased as in cases with hyperdynamic circulation (aortic regurgitation, pregnancy, thyrotoxicosis, anemia, arteriovenous fistula) the sounds may not disappear



FIGURES 1.3A and B: (A) Localization of brachial artery (B) Measurement of blood pressure; note the position of cuff and the stethoscope

completely even at 0 level. In such cases the level at which sounds become suddenly muffled (phase 4) is taken as diastolic blood pressure.

The patient is said to be hypertensive if the systolic BP is >140 mmHg and/or diastolic BP is >90 mmHg (Table 1.4). Sometimes the blood pressure recorded by the clinician at clinic or hospital is high while normal readings are obtained at home or when BP is measured under casual circumstances. This is known as *white coat hypertension* and is the result of the anxiety upon visiting a physician or a hospital.

Jugular Venous Pulse

Pulsations and pressure in internal jugular vein in the neck are noted (Fig. 1.4).

- **Venous vs arterial pulsation:** Venous pulsations must be differentiated from carotid artery pulsations. Venous pulsations are better seen while arterial pulsations are better palpable. The upper level of venous pulsation varies with the change in posture and phases of respiration.
- Jugular veins are distended and pulsatile in congestive heart failure and pericardial effusion. Neck veins are also distended in cases of mediastinal tumors and retrosternal goiter but these are not pulsatile.
- Normally there is fall in the jugular venous pressure (JVP) during inspiration. There may be a paradoxical rise in the JVP during inspiration in constrictive pericarditis and cardiac tamponade (*Kussmaul's sign*).

The venous pulse has three positive waves, a, c, and v, and two negative waves or descents, x and y. The a wave is due to atrial contraction. This is followed by x descent (due to descent of tricuspid valve ring) which is interrupted by a small c wave. The v wave is due to passive filling of blood from veins into the right atrium during ventricular systole.

TABLE 1.4: Classification of blood pressure for adults (>18 yrs)

Category	Systolic blood pressure (mmHg)	Diastolic blood pressure (mmHg)
Normal	<120	<80
Prehypertension	120-139	80-89
Hypertension		
Stage 1	140-159	90-99
Stage 2	>160	>100



FIGURE 1.4: Prominent jugular vein

This is followed by y descent due to rapid flow of blood from the right atrium to the right ventricle when the tricuspid valve is open.

The “a” wave is absent in atrial fibrillation while it is prominent in tricuspid stenosis (TS). Prominent “Y” descent is seen in tricuspid regurgitation.

Respiration

Normal rate of respiration is 12-16 per minute in adults. The causes of fast breathing (tachypnea) are given in Table 1.5. Dyspnea is an abnormally uncomfortable awareness of breathing. This could be due to respiratory diseases, cardiac diseases, anemia, acidosis, and psychogenic. Dyspnea, orthopnea and paroxysmal nocturnal dyspnea are described in detail in Chapters 4 and 5.

- Noisy breathing may occur due to obstruction of the respiratory passages at various levels. Obstruction at the level of larynx and trachea causes inspiratory stridor and obstruction in bronchi and bronchioles produces wheezing.
- Rapid and deep respiration (Kussmaul's breathing) is present in metabolic acidosis while rapid shallow breathing is a feature of restrictive lung disease.
- Cheyne-Stokes respiration is characterized by cyclical waxing and waning of rate and depth of respiration

TABLE 1.5: Causes of tachypnea

- | |
|--|
| <ul style="list-style-type: none"> • Recent exertion • Anxiety • Fever • Metabolic acidosis • Hysterical over-breathing • Pulmonary and cardiac conditions causing hypoxia • Cerebral disturbance |
|--|

intervened with periods of apnea. It is observed in narcotic overdose and severe left heart failure.

Pallor

- The presence of pallor depends on the thickness and quality of the skin, amount of blood in the capillaries and quality of the blood in the capillaries.
- The evidence of pallor is looked at palpebral conjunctiva and mucous membrane of the mouth (Fig. 1.5). Other sites are nailbed and palmar creases (Fig. 1.6).
- Generalized pallor is present in anemia. Pallor can also be found in hypopituitarism, thick or opaque skin, and diminished capillary blood flow as in shock, syncope, left heart failure.

Jaundice

A yellowish discoloration of the skin and mucous membrane due to deposition of bilirubin is known as jaundice (icterus). The deposition of bilirubin in tissues occurs when the serum bilirubin level is raised (hyperbilirubinemia). Sclerae have a high affinity for bilirubin due to their rich elastin content.

- The normal total serum bilirubin level is 0.3-1.0 mg/dL. Jaundice is clinically apparent in sclera when the bilirubin level is raised above 3 mg/dL (Fig. 1.7).
- The clinical detection of jaundice is difficult in artificial light. Hence, it should be examined preferably in day light. Besides sclera, other sites to be looked for the evidence of jaundice are mucosa of oral cavity underneath the tongue and skin.
- Yellow discoloration of the skin can also occur in *carotenemia* (carotenoderma) and exposure to *quinacrine* or *phenols*. Sclera is typically not involved in carotenemia.

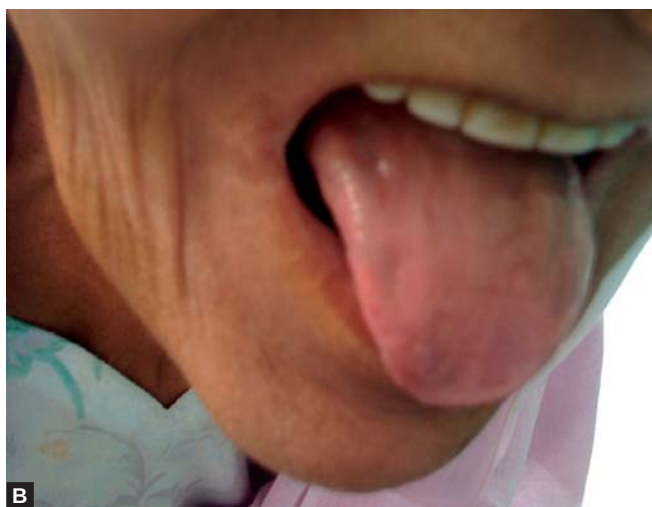
**FIGURE 1.5:** Palpebral conjunctivae showing pallor**FIGURE 1.6:** A. Pallor of the palm; compare with palm of normal person (left) B. pale tongue



FIGURE 1.7: Jaundice visible over sclera

Clubbing

The enlargement of the distal portion of the fingers and toes, due to proliferation of connective tissues, is known as clubbing (Fig. 1.8A). The clubbing is graded as follows;

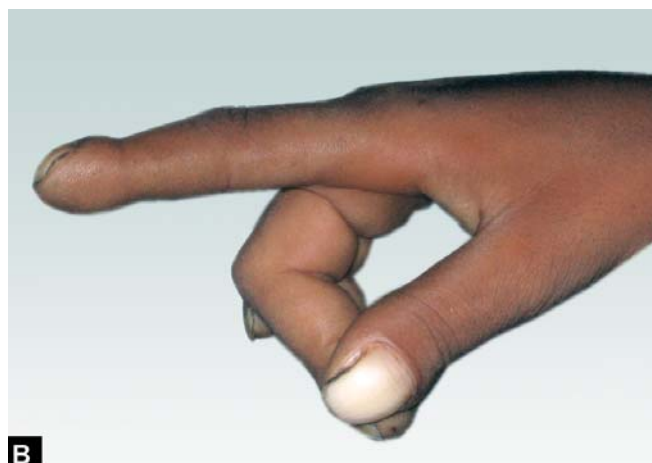
- Grade I: There is thickening of tissues at the nail base.
- Grade II: In addition to the features of grade I, the angle between nail base and the adjacent skin fold of the finger is obliterated. There is reduction in the space between thumb nails when placed in apposition (*Schamroth's window test*).
- Grade III: In addition to the features of grade I and grade II, the shape of the nail becomes convex in both horizontal and vertical directions. In severe cases there is bulbous enlargement of the distal segment of the fingers (drumstick appearance) (Fig. 1.8B).
- Grade IV: Along with the clubbing, there may be swelling above the wrist and ankles due to periosteitis of long bones (*hypertrophic osteoarthropathy*).

The exact mechanism of clubbing is clearly not known. However, it is thought to be due to some humoral substances leading to increased vascularity in the nailbed.

Clubbing may be present since birth (congenital), or acquired. Acquired causes of clubbing are given in Table 1.6.

Cyanosis

Cyanosis is bluish discoloration of the skin and mucous membrane caused by an increased quantity of reduced hemoglobin ($> 4g\%$) in superficial blood vessels. The bluish



FIGURES 1.8A and B: (A) Marked digital clubbing (B) Severe clubbing (drum stick appearance)

discoloration can also be seen in methemoglobinemia and sulfhemoglobinemia where the patient is cyanosed but not breathless. A cherry red discoloration is caused by carboxyhemoglobin in carbon monoxide poisoning (not true cyanosis).

Cyanosis is looked for at lips, nailbeds, malar area, ear lobes and mucous membrane of the oral cavity (Fig. 1.9).

Cyanosis is classified into central and peripheral types (Table 1.7).

- The imperfect oxygen saturation or abnormal hemoglobin derivatives lead to *central cyanosis* which is seen in both the mucous membrane (tongue) and skin and also nailbeds of the limbs. The extremities are warm.
- Peripheral cyanosis* is due to excessive extraction of oxygen from the capillaries when the flow of blood is

TABLE 1.6: Causes of clubbing**Respiratory diseases**

Chronic suppurative lung diseases

- Lung abscess
- Bronchiectasis
- Empyema

Bronchogenic carcinoma

Mesothelioma (pleural neoplasm)

Pulmonary tuberculosis

Fibrosing alveolitis

Cardiac diseases

Congenital cyanotic heart diseases

- Fallot's tetralogy
- Eisenmenger syndrome

Subacute bacterial endocarditis

Gastrointestinal diseases

Inflammatory bowel diseases

- Ulcerative colitis
- Crohn's disease

Hepatic cirrhosis

Idiopathic**FIGURE 1.9:** Central cyanosis

slow. The extremities are cyanosed and cold while mucous membrane of the oral cavity and tongue are spared. Warming of the cyanotic extremity may increase blood flow and abolish peripheral (but not central) cyanosis.

- c. Cyanosis due to heart failure is of *mixed type*, both central and peripheral.

TABLE 1.7: Causes of cyanosis**Central cyanosis**

- High altitude
- Respiratory diseases (COPD, extensive pneumonia, pulmonary edema, massive pulmonary embolism)
- Cardiac diseases (congenital cyanotic heart diseases, Eisenmenger syndrome, heart failure)
- Abnormal hemoglobin (methemoglobinemia, sulfhemoglobinemia)

Peripheral cyanosis

- Cold exposure
- Heart failure (reduced cardiac output)
- Arterial obstruction
- Venous obstruction

Edema

Edema is the presence of an excess of fluid in interstitial space causing swelling of the tissues.

1. Edema may be localized or generalized. Generalized edema is known as anasarca, in which the fluid may also accumulate in the pleural cavity (hydrothorax) and peritoneal cavity (ascites). Edema over feet is known as pedal edema. Causes of pedal edema are given in Table 1.8.

2. Edema may be of the pitting or non-pitting type. Pitting edema means formation of an indentation or pit following the application of firm pressure for a sustained period over the area of swelling (Figs 1.10A and B).

The mechanisms of edema can be described as follows:

- a. The hydrostatic pressure in vascular system and tissue colloid oncotic pressure tend to drive fluid from the vascular to the extravascular space. On the contrary, colloid oncotic pressure maintained by plasma proteins in the vascular system and hydrostatic pressure in the interstitial fluid promote the movement of fluid in the vascular compartment. The development of edema is a result of the imbalance between these “Starling forces”. For example, the edema in congestive heart failure is due to an increase in the vascular hydrostatic pressure. A decrease in the plasma colloid oncotic pressure is the cause of edema in hypoalbuminemic states like nephrotic syndrome, malnutrition, and liver disease. The edema is of the pitting type.

TABLE 1.8: Causes of pedal edema**Bilateral pedal edema***Pitting type*

- Congestive heart failure
- Nephrotic syndrome, acute nephritis
- Liver cirrhosis
- Malnutrition
- Epidemic dropsy
- Drugs (calcium channel blockers, NSAIDs, steroids)

Non-pitting type

- Myxedema

Unilateral edema

- Filariasis
- Thrombophlebitis
- Cellulitis
- Trauma
- Regional lymph node resection

**FIGURES 1.10A and B:** (A) Pressure applied over edematous limb (B) pitting edema

- Edema may result from damage to the capillary endothelium which causes exudation of fluid and protein due to increased permeability. Injury to capillary endothelium may occur due to drugs, infections, and trauma. Capillary permeability is also increased in hypersensitivity reactions. This type of edema is usually localized, non-pitting and may be accompanied by other signs of inflammation.
- In many forms of edema, the effective arterial blood volume is reduced. This in turn initiates physiological mechanisms to restore the volume by renal salt and water retention, which further adds up to the edema. Compensatory physiological responses are activation of rennin-angiotensin-aldosterone system, and increased secretion of vasopressin.

Edema generally appears first over the periorbital area and is more marked in the mornings, in nephrotic syndrome and acute nephritis (see Fig. 6.1). In heart failure, the edema is more marked during the evenings and present over the ankles and dorsum of the feet. In these conditions, edema may become generalized later on. In bed-ridden patients, the edema first appears in the presacral region.

Localized edema in a single limb is generally due to either venous or lymphatic obstruction. For example, edema of the leg may occur due to thrombosis of the popliteal or femoral vein. Compression of axillary vein due to malignant lymph nodes may cause edema of the arm.

Lymphatic obstruction due to resection of regional lymph nodes or in filariasis leads to non-pitting edema. Generalized non-pitting edema is found in myxedema.

Lymph Nodes

Palpation of lymph nodes is an important part of general examination (Fig. 1.11). Lymph nodes are examined for;

- size
- number
- texture
- tenderness
- mobility
- signs of inflammation over the nodes

Important groups of lymph nodes which must be examined are submental, submandibular, preauricular, postauricular, cervical (anterior and posterior chains), supraclavicular, axillary and inguinals. For other details, see Chapter 3.



FIGURE 1.11: Palpation of the submandibular lymph nodes

Halitosis

Halitosis is an unpleasant odor or smell emerging from the mouth or nostrils that is detected by the patient or others. It is also referred as bad breath, breath malodor, puppy breath, and dragon mouth. About 35% of world population is suffering from bad breath problem. Important causes of halitosis are given in Table 1.9.

Types of Halitosis

1. *Physiologic bad breath:* This affects all normal healthy persons and is caused by anaerobic bacterial overgrowth mainly on the tongue deep in the papillae.
2. *Pathologic bad breath:* Occurs due to oral infections including carious teeth.
3. *Halitophobia:* Some patients may complain of bad breath in spite of treatment being given to them. It may be due to psychiatric illness.
4. *Transitory bad breath:* This type occurs after consumption of certain foods like garlic, onions, and certain medications. It usually lasts for hours/days.

Pathophysiology of Halitosis

Anaerobic bacteria are responsible for bad breath. These bacteria are abundantly present in the oral cavity, tonsils and throat. Bacteria react with food, medications in the presence of acidic environment and produce volatile sulphur compounds (VSC) such as hydrogen sulphide (HS) which smells like rotten eggs, methyl mercaptans (smells like gym socks), putrescine and cadverin (smells like old garbage).

TABLE 1.9: Causes of halitosis

Dental causes
• Dental decay-carries, exposed teeth
• Gum diseases
• Oral infections-abscess
• Oral cancer
• Xerostomia (dry mouth)
• Mouth breathing habit
• Tongue coating
Medical causes
• Sinus infections, cough and cold
• Allergies, post nasal drip
• Lung abscess
• Diabetic ketoacidosis (sweet and fruity)
• Renal failure (ammonical, urinary)
• Hepatic failure (fishy, mousy)
• Hiatus hernia
• Menstruation
• Medications
Miscellaneous
• Certain foods- fish, dairy products, garlic, eggs
• Smoking
• Alcohol
• Stress
• Certain professions
• High protein diets (Atkins diet)

Foods Causing Bad Breath

1. *Sugars:* bacteria cause breakdown of sugar and produce acidic environment.
2. *Dense proteins/milk products:* Cheese, yoghurt, ice cream stay on the tongue and between teeth. Anaerobic bacteria break down the proteins and produce VSC.
3. *Acidic foods:* Anaerobic bacteria multiply very fast in acidic medium and produce high levels of VSC. Acidic foods include coffee, tomato juice, citrus fruit juices aerated drinks.
4. *Onion, garlic, cabbage:* When taken raw, smell comes from mouth because of sulphur compounds present in them.

Examination/Tests for Halitosis

These are following scientifically proven ways to check breath:

- a. *Using Halimeter:* It measures the concentration of sulphides in the breath. Reading above 75 ppb (parts per billion) indicates bad breath.
- b. *Using bad breath detective:* It measures the amount of VSC coming from the tongue by simply swabbing the

back of tongue and placing it into the test tube that comes with bad breath detective.

- c. Other methods used to detect halitosis are:
1. Lick the back surface of hand, let it dry and smell after 15-20 seconds.
 2. Use floss interdentally and smell it.
 3. If the back surface of tongue is whitish it indicates that person has bad breath.
 4. If friends, colleagues and relatives, move away or offer mint while person is talking to them.

Myth about Halitosis

Bad breath comes from stomach is myth. There is no open tube connecting the stomach or intestines to mouth as there are valves, sphincters and muscles etc, that keep digested food at its place.

Tongue and Bad Breath

Tongue is responsible for almost 85% of bad breath that comes from the mouth. Tongue has glossy surface so the food gets accumulated here and bacteria utilize the food to produce the volatile sulphur compounds (VSC). Tongue cleaning is more effective than brushing in stopping bad breath. Tongue cleaning reduces oral bacteria thereby decreasing chances of bad breath and plaque formation. It also improves taste sensitivity and quality.

Treatment of Bad Breath

Treatment of bad breath is possible if the cause is removed. It can be done in multiple visits to dental clinic (fresh breath clinics). It includes following:

- Thorough dental and oral checkup
- Medical history to rule out any medical cause
- Dietary analysis and counseling
- Bad breath testing using halitometer
- Oral hygiene instructions and techniques

Mouth wash: Sugar, saccharin and alcohol containing mouth washes should be avoided. Alcohol causes dryness of mouth. Special mouthwashes are used which release oxygen that kills the anaerobic bacteria.

Xylitol chewing gums: sugar free xylitol chewing gums are also useful.

SCHEME OF THE GENERAL EXAMINATION

General Physical Examination

General appearance

Mental state

Built

Height and weight

Decubitus

Vitals:

- Pulse – Rate and rhythm (radial)
– Character and volume (carotids)
– Symmetry
- Blood pressure
- Temperature
- Respiration

Eyes:

- Exophthalmos, ptosis, eye movements
- Conjunctiva—pallor, icterus
- Pupils—size, reaction to light and accommodation

Face: Symmetry, puffiness, cyanosis, parotid gland

Oral cavity: odour, lips, tongue, teeth, gum, buccal mucous membrane

- Lips: color, eruption
- Teeth: denture, other abnormalities
- Gums: swelling, bleeding, inflammation, ulcers
- Tongue: color, appearance, ulcers
- Buccal mucosa: color, ulcers

Pharynx: tonsils, oropharynx.

Neck:

- JVP
- Thyroid
- Lymph nodes (cervical)

Upper limbs:

- Nails-clubbing, koilonychia, pallor, cyanosis
- Pulse
- Blood pressure
- Lymph nodes (axillary)

Lower limbs:

- Edema of feet, ankles

DEFINITIONS

Certain terms are frequently used in clinical medicine. These have profound effects on presentations. Some important terms are defined below.

Diagnosis

Diagnosis is an act or process of identifying or determining the nature of a disease by way of examination and assessment of the symptoms and signs.

Diagnosis is an art wherein scientific methods are applied to the elucidation of problems presented by a patient. A concept is formed about the etiology, pathology, and organ dysfunctions which constitutes the patient's disease.

Diagnosis provides a firm basis for the treatment and prognosis of the individual patients.

Clinical Diagnosis

Diagnosis made by bedside methods without the help of laboratory tests.

Differential Diagnosis

The recognition of a particular condition from amongst others which closely resemble it in certain aspects.

Prognosis

Prognosis is a considered opinion of the probable development and outcome of the disease based upon all the relevant available facts of the case.

Treatment

Treatment is the course of action adopted to deal with illness and control of the patient.

Illness

Illness is defined by the totality of effects, predicaments, and repercussions of the disease, deformity, or circumstances produced in the patient.

SELF ASSESSMENT

Multiple Choice Questions

1. Rate and rhythm of the pulse is best appreciated by palpating:
 - A. Brachial artery
 - B. Radial artery
 - C. Popliteal artery
 - D. Femoral artery
2. Following is not true in pulsus paradoxus:
 - A. Found in cardiac tamponade
 - B. Disappears during expiration
 - C. Inspiratory fall in blood pressure
 - D. May be present in bronchial asthma
3. Following conditions are associated with bradycardia except:
 - A. Hypothyroidism
 - B. Athletes
 - C. Hypotensive shock
 - D. Raised intracranial tension
4. Edema in both lower limbs can occur in:
 - A. Filariasis
 - B. Cellulitis
 - C. Nephrotic syndrome
 - D. Popliteal vein thrombosis
5. Cyanosis in CHF is of following type:
 - A. Central
 - B. Peripheral
 - C. Both, central and peripheral
 - D. Not found
6. Which of the following is not matched correctly:
 - A. Pulsus Parvus et tardus — aortic stenosis
 - B. Pulsus bisferiens—severe mitral stenosis
 - C. Water hammer pulse—aortic regurgitation
 - D. Hyperkinetic pulse—thyrotoxicosis
7. Clubbing can be found in the following except:
 - A. Bronchiectasis
 - B. Lung abscess
 - C. Pneumonia
 - D. Bronchogenic carcinoma
8. Yellowish discoloration of sclera and skin occurs in the following:
 - A. Carotinemia
 - B. Hyperbilirubinemia
 - C. Quinacrine ingestion
 - D. Both A and B
 - E. All of the above
9. Cyanosis is seen in:
 - A. Fallot's tetralogy
 - B. Methemoglobinemia
 - C. Sulfhemoglobinemia
 - D. All of the above
10. All of the following is true in peripheral cyanosis except:
 - A. It improves on warming
 - B. Best seen in oral mucous membrane
 - C. Occurs in cases with low cardiac output
 - D. May occur following exposure to cold
11. Following is not matched properly:
 - A. CHF _____ pedal edema
 - B. Thyrotoxicosis _____ tachycardia
 - C. High arterial CO₂ _____ cyanosis
 - D. Orthopnea _____ mitral stenosis
12. Early morning periorbital edema suggests the disease of following system:
 - A. Cardiac
 - B. Renal
 - C. Hepatic
 - D. All of the above
13. "a" wave in JVP is absent in:
 - A. Pericardial tamponade
 - B. Complete heart block

C. Atrial fibrillation

D. Hypotension

14. Distended but nonpulsatile neck veins are found in:

A. Right heart failure

B. Tricuspid stenosis

C. Mediastinal tumor

D. Constrictive pericarditis

15. JVP is best examined in:

A. External jugular vein

B. Internal jugular vein

C. Subclavian vein

D. Any one of the above

16. Sweet fruity odour is found in the oral cavity in case of:

A. Renal failure

B. Hepatic failure

C. Diabetic ketoacidosis

D. All of the above

17. Blood pressure is generally measured by auscultating over following artery:

A. Radial artery

B. Brachial artery

C. Carotid artery

D. Any of the above

18. The diastolic BP corresponds best with:

A. First appearance of Korotkoff sound

B. Disappearance of Korotkoff sound

C. Muffling of Korotkoff sound

D. In between appearance and disappearance of korotkoff sound

19. The following can be measured by sphygmo-manometer and palpating the artery:

A. Systolic blood pressure

B. Diastolic blood pressure

C. Both

D. None

20. In coarctation of aorta, following is true:

A. BP in lower limbs is higher than in upper limbs

B. BP is equal in lower and upper limbs

C. BP in upper limb is higher than in lower limbs

D. BP is generally not recordable in upper limbs

21. Cyanosis is accompanied with clubbing in the following except:

A. Eisenmenger's syndrome

B. Fallot's tetralogy

C. Interstitial lung disease

D. Conditions with peripheral cyanosis

22. Clubbing may be present in the following except:

A. Lung cancer

B. Crohn's disease

C. Infective endocarditis

D. Left to right cardiac shunts

23. Following is not properly matched:

A. Clubbing_____Fallot's tetralogy

B. Cyanosis_____pulmonary edema

C. Eisenmenger's syndrome_____cardiac shunts with left to right flow

D. Pulmonary osteoarthropathy_____lung cancer.

Fill in the Blanks

1. Radio-femoral delay is found in _____.

2. Pulsus alternans is present in _____.

3. Cyanosis appears when amount of reduced Hb exceeds _____ g/dL.

4. Rise in JVP during inspiration in constrictive pericarditis is called _____ sign.

5. Prominent Y descent in JVP is seen in _____.

6. Bradycardia is defined as pulse rate less than _____ per minute.

7. Tachycardia is defined as pulse rate more than _____ per minute.

8. Normal respiratory rate in adults is _____ per minute.

9. Regularly irregular pulse is found in _____.

10. Fishy mousy odour in the oral cavity suggests _____.

11. Cherry red discoloration of skin is found in _____.

12. Waxing and waning respiration with intervening periods of apnea is called _____.

Chapter 2

Gastrointestinal and Hepatobiliary System

SYMPTOMS AND SIGNS OF GASTROINTESTINAL DISEASES

The gastrointestinal (GI) system extends from the mouth to the anus. The symptoms arising from GI tract diseases are complex and varied. Some important symptoms are:

- Abdominal pain
- Nausea and vomiting
- Heart burn
- Altered bowel habits (diarrhea, constipation)
- Abdominal distension
- Bleeding
- Jaundice

Symptoms

Abdominal Pain

- The points to be noted in relation to the abdominal pain are site, nature, severity, radiation, time of onset, aggravating and relieving factors.
- The pain may be localized or generalized. The nature could be colicky or a diffuse dull ache. Colicky pain is spasmodic in nature with episodes of pain lasting for a few seconds or even up to minutes intervening with pain free periods. This is typical of bowel obstruction.
- Important causes of abdominal pain are:
 - Appendicitis
 - Gallstones
 - Liver abscess
 - Pancreatitis
 - Peptic ulcer
 - Intestinal obstruction
 - Renal stones
 - Gynecological diseases

- The extra-abdominal diseases like myocardial infarction, pneumonia, herpes zoster and spinal diseases may present with abdominal pain.

Heart Burn

Heart burn is the burning pain in the epigastrium, chest and neck due to the reflux of acid into the esophagus. There may be sour eructation or a bitter taste in the mouth. The pain can be confused with angina. Heart burn occurs more frequently when the patient lies flat in bed or bends forward.

Nausea and Vomiting

Nausea and vomiting due to gastrointestinal causes are generally associated with abdominal pain. Vomiting and nausea may also be due to causes such as pregnancy, medications, toxins, infections, central nervous system disorders and motion sickness.

Altered Bowel Habits

Altered bowel habits include constipation and diarrhea.

- Constipation is referred by patients as incomplete evacuation of stool, passage of hard stools, defecation with straining, or infrequent defecation (less than 3 times a week). The common causes of constipation are intestinal obstruction, medications, motility disorders and hypothyroidism.
- Passage of unformed and liquid stools and/or increased frequency of stools is called diarrhea. Acute diarrhea generally occurs due to infections. Chronic diarrhea (of more than 4 weeks) raises the possibility of inflammatory bowel diseases and malabsorption.

Abdominal Distension

- Generalized distension of the abdomen may occur due to fat, fluid, flatus, feces or fetus. (Fig. 2.1). Flatulence (excessive wind) usually represents functional bowel disease where large amount of air is swallowed. Certain foods may also produce flatulence.
- Localized distension may result from organ enlargement (liver, spleen, kidneys and ovary) or small bowel obstruction.

GI Bleeding

Bleeding may occur in the upper or lower gastrointestinal tracts.

- Upper GI bleeding (bleeding from esophagus, stomach and duodenum) commonly presents with hematemesis or melena. *Hematemesis* is vomiting of red blood or “coffee grounds” material while *melena* is foul smelling, black tarry stools. Common causes of upper GI bleeding are peptic ulcer, variceal rupture and erosive gastritis.
- Passage of bright red blood in stool (*hematochezia*) indicates lower GI bleeding (small intestine and colon), arising below the ligament of Treitz. Common causes of lower GI bleeding are hemorrhoids, anal fissures, neoplasms, diverticula, inflammatory bowel diseases, infections and ischemia of the colon.
- Occult GI bleeding refers to bleeding that is not visible but can be identified by a positive fecal occult blood test. Chronic blood loss of < 100 ml/day may not cause any change in appearance of stools although it may present as iron deficiency anemia.

Jaundice

Jaundice may result from prehepatic, intrahepatic and posthepatic causes (see chapter on *Jaundice*).

Other Symptoms

- Difficulty in swallowing (*dysphagia*) and painful swallowing (*odynophagia*) are features of esophageal disease.
- *Hiccups* can arise due to distension or irritation of the upper GI tract. However, they may also occur because of non-GI causes.
- *Weight loss, anorexia and fatigue* are non-specific manifestations and can be seen in malignancy, malabsorption, inflammatory and psychiatric conditions.

- *Dyspepsia* generally refers to symptoms of upper abdominal discomfort, bloating, belching, burning, fullness, early satiety, nausea and vomiting. This is usually benign and occurs due to overeating, high fat diet, alcohol, coffee and medicines. Dyspepsia may be the symptom of peptic ulcer or gastric cancer. Functional or non-ulcer dyspepsia is the most common cause of chronic dyspepsia. In this condition no obvious organic cause is found.

General Examination

Certain features in general examination are important and should be recorded.

One should look for the presence of jaundice, signs of chronic liver disease (spider nevi, palmer erythema, gynecomastia, testicular atrophy, parotid swelling), clubbing (seen in inflammatory bowel disease, cirrhosis), edema, anasarca and anemia. Scratch marks (pruritus) may suggest cholestasis. Assessment of nutritional status (weight loss) is an important finding in chronic malabsorption, chronic liver disease, malignancies and severe protein loss.

Careful examination of lips, teeth, gums, tongue and buccal mucosa is performed for the presence of glossitis, bald tongue, ulcers, cheilitis and angular stomatitis.

Abdominal Examination

The examination of the abdomen should be performed in a systemic manner. This includes inspection, palpation, percussion and auscultation.

Inspection

The patient should be supine and the abdomen should be adequately exposed. The inspection should be performed under proper lighting conditions. Following points should particularly be noted;

- Shape of the abdomen:* The abdomen may be distended or sunken (scaphoid). The generalized distension of the abdomen may occur due to fat, fluid, flatus, feces or fetus. Localized distension could be due to organ enlargement or small bowel obstruction. Sunken abdomen is seen in starvation and malignancy.
- Umbilicus:* Umbilicus is everted and horizontal in ascites. Umbilical hernia may occur in massive ascites (Fig. 2.1).



FIGURE 2.1: Massive ascites with umbilical hernia



FIGURE 2.2: Prominent veins over abdomen



FIGURE 2.3: Abdominal stria in a patient with ascites

- c. *Movements of the abdominal wall:* Movements of abdominal wall are absent in peritonitis. Visible pulsations can be seen normally in thin persons or can also be due to aortic aneurysms. Visible peristalsis may be present in gastric outlet obstruction and small bowel obstruction.
- d. *Prominent veins:* Prominent superficial veins may be present in inferior vena cava obstruction (Fig. 2.2) or portal hypertension (caput medusae).
- e. *Skin:* Skin over the abdomen is shiny and smooth in marked distension. Purple striae are seen in Cushing's Syndrome. Striae atrophica or gravidorum are pink or white linear marks produced by gross stretching of abdomen as in ascites and pregnancy (Fig. 2.3).
- f. *Inspection of groin, genitalia and hernial sites* should also be done.

Palpation

- The patient is asked to bend the knees to relax abdominal muscles and to breathe deeply.
 - The palpation begins anti-clockwise from left iliac fossa.
 - The palpation should be gently performed with warm hands. Initially superficial palpation is done to find out tender areas, which are to be examined in the end.
- a. Organs such as liver, spleen, kidneys, ovaries, uterus and urinary bladder are palpated to detect any enlargement. Further details about the enlarged organs like size, surface, borders, consistency, and tenderness are noted.
 - *Enlarged liver* is palpable in the right hypochondrium. It moves with respiration and the finger cannot be insinuated between the costal margin and the lump (Fig. 2.4). The upper border of the liver should also be demarcated by percussion in order to assess the size. (see hepatomegaly described elsewhere in this chapter).
 - *Spleen* is palpated in the left hypochondrium. It moves with respiration and the finger cannot be insinuated between the costal margin and the lump (Fig. 2.5). A notch is felt at the medial border of the spleen. This is not palpable bimanually.
 - *Kidneys* are bimanually palpable and ballotable (it can be pushed from the one hand to the other).
 - b. Any area showing tenderness or rigidity should be noted. *Murphy's sign* may be present in acute cholecystitis.



FIGURE 2.4: Palpation of the liver



FIGURE 2.5: Palpation of the spleen



FIGURE 2.6: Method to elicit fluid thrill

c. *Fluid thrill* detects the presence of free fluid in the abdomen (ascites). Place one hand flat in the lumbar region of one side. Ask an assistant to put the side of hand in the midline of the abdomen. Tap the opposite lumbar region. A wave or thrill is felt by the hand held flat in the other lumbar region (Fig. 2.6). The assistant's hand does not allow the transmission of impulse through the abdominal wall. Fluid thrill is a sign of tense and massive ascites.

Percussion

The normal note of the abdominal percussion is tympanitic (resonant). Percussion is useful in confirming the enlargement of liver and spleen and detecting fluid in the peritoneal cavity (ascites).

Shifting dullness: Shifting dullness is a sign of moderate ascites and may be absent when the ascites is tense. Percussion is performed in supine patient from the midline of the abdomen toward the flanks till dullness is detected (Figs 2.7A and B). Keeping the hand on the abdomen in the same position (at the point of dullness), patient is rolled laterally to the opposite side. Percussion is repeated after a minute from flank toward the umbilicus. In case of fluid (as in ascites), the previously dull point at flanks becomes resonant because of shifting of fluid towards the umbilicus. To confirm, the test is repeated on the other side of the abdomen.

Auscultation

The bowel sound is exaggerated in small bowel obstruction. These may be absent in paralytic ileus. Vascular bruit may be heard in case of stenosis or aneurysm of the vessels.

STOMATITIS AND ORAL ULCERS

The inflammation of oral mucosa is called *stomatitis*. This can lead to disruption of mucosa leading to *oral ulcers*. Important causes of stomatitis/oral ulcers are given in Table 2.1.

Recurrent aphthous ulcers and herpes simplex ulcers are among the commonest causes of oral ulcers.

Angular Stomatitis (Cheilosis)

Cheilosis is characterized by the presence of cracks or fissures at the corners of the mouth. Important causes are:



FIGURES 2.7A and B: Method to elicit shifting dullness (A) supine position and (B) lateral position)

- Ill fitting dentures (in elderly)
- Severe iron deficiency
- Vitamin B complex deficiency
- Candidiasis

Ulcerative Stomatitis (Aphthous Ulcer)

- Ulcerative stomatitis is a common condition.
- Etiology is unknown (idiopathic), however, human herpes virus-6 has been associated with this condition.
- Aphthous ulcers are recurrent single or multiple, superficial painful lesions with central yellow grey slough surrounded by erythematous borders.
- These involve non-keratinized oral mucosa such as buccal mucosa, labial mucosa, floor of the mouth, soft palate, lateral and ventral tongue.

TABLE 2.1: Causes of oral ulcers

<i>Aphthous ulcer</i>
<i>Infections:</i>
• Viral (Herpes, CMV, EBV, HIV)
• Fungal (Candida)
• Bacterial (Vincent's infection, syphilis)
<i>Dermatological:</i>
• Pemphigus, pemphigoid, lichen planus
<i>Drugs:</i>
• Chemotherapy drugs
• Erythema multiforme, Stevens-Johnson syndrome
<i>Systemic diseases:</i>
• Behçet's syndrome, SLE
<i>Nutritional:</i>
• Vitamin deficiency (Vitamin B and C), iron deficiency
<i>Neoplasia:</i>
• Leukemia, squamous cell carcinoma, Kaposi's sarcoma
<i>Gastrointestinal:</i>
• Crohn's disease, celiac disease
<i>Traumatic:</i>
• Dentures
<i>Chemical or thermal burns:</i>
• Corrosives, hot liquids

- The painful stage lasts for 7-10 days followed by complete healing within a week.
- The size of the ulcers is generally less than 1-5 mm. In cases with large and persistent ulcers, biopsy is needed to differentiate them from other causes such as erythema multiforme, herpes simplex, pemphigus, pemphigoid, Behçet's disease and inflammatory bowel disease.
- Local anesthetic gel or mouth washes give symptomatic relief. Topical steroids (triamcinolone or fluocinonide) can affect healing. Severe cases may need a course of oral prednisolone.

Vincent's Infection (Necrotizing Ulcerative Stomatitis, Trench Mouth)

- There are painful, sloughing deep ulcers which primarily involve gums.
- There is severe inflammation and necrosis of gingiva with bleeding.
- It may be associated with halitosis, fever and cervical lymphadenopathy.
- The causative organisms are fusiform bacilli and spirochetes.
- Malnutrition, poor oral hygiene and immunosuppression (such as AIDS) predispose to this condition.

- Treatment includes debridement and hydrogen peroxide mouth wash. Antibiotic (penicillin) is helpful in acutely ill patients.

Herpetic Stomatitis

- Herpetic gingivostomatitis is caused by herpes simplex virus (HSV) type I and rarely by HSV type 2.
- This occurs mainly in children and young adults.
- Lesions involve lips and oral mucosa. Labial lesions are in the form of vesicles that rupture and crust. Intraoral vesicles are very painful and they rapidly ulcerate.
- There may also be fever, malaise, halitosis and cervical lymphadenopathy.
- Labial lesions are common in recurrent herpes simplex (herpes labialis).
- Lesions usually heal within 10-14 days.
- Topical or oral antiviral (acyclovir, valacyclovir, famcyclovir) drugs enhance healing in severe cases. Daily acyclovir may be needed to prevent recurrent herpetic lesions, particularly in immunocompromised cases.

Oral Candidiasis

Candida albicans is normal mouth commensal. It can produce thrush in babies, diabetics, patients on corticosteroids or broad spectrum antibiotics and immunosuppressed states (AIDS, cancer chemotherapy). This can involve any part of the mouth. Oral candidiasis can present in following forms:

- Pseudomembranous type (thrush)*: Creamy white curd-like patches are seen over erythematous mucosa. These can be easily scraped and reveal raw bleeding surface. Painful deglutition (odynophagia) suggests pharyngeal and esophageal involvement.
- Erythematous type*: These are flat, red, sore areas in the oral cavity.
- Candidal leukoplakia*: There is non-removable white thickening of mucosal epithelium due to *Candida*.
- Angular cheilitis*: Sore fissures at the corner of the mouth can be caused by *Candida*.

Diagnosis: The diagnosis can be made clinically and is confirmed by the demonstration of spores and mycelia on KOH wet preparation or biopsy of the lesion.

Treatment: Clotrimazole lozenges or nystatin mouth washes are effective in oral candidiasis. Ketoconazole

(200-400 mg oral daily) or fluconazole (100 mg oral daily) is given for 1-2 weeks in oro-esophageal candidiasis and in immunocompromised patients. Itraconazole is given in refractory cases.

Nutritional Deficiency

Swollen bleeding gums and ulcers are common in vitamin C deficiency. Deficiency of vitamin B complex (B_2 , B_{12} , folic acid) can cause glossitis, oral ulceration and cheilosis. Cheilosis, glossitis and dysphagia are also found in iron deficiency cases.

Hematological Diseases

All forms of leukemia, particularly acute myelo-monocytic leukemia can produce oral ulcers, gingival swelling and bleeding. Oral ulcers are also found in agranulocytosis. Mucositis and ulcers are complications of chemotherapy and radiation therapy.

GINGIVAL HYPERPLASIA

Gingival hyperplasia is swelling or enlargement of gingiva. Apart from oro-dental causes, gingival swelling can be manifestation of systemic disease. Hence, understanding various systemic causes of gingival hyperplasia is very important.

Gingival swelling can be;

- Generalized
- Localized

Important causes of generalized gingival swelling are scurvy (Vitamin C deficiency), acute myeloid leukemia and drug induced (phenytoin, cyclosporine and calcium channel blockers). Causes of gingival hyperplasia are given in Table 2.2.

Diagnosis can be made by proper history including drug intake and the underlying features of causative disease. Hematological investigations can rule out the presence of leukemia.

Treatment includes management of underlying cause and withdrawal of offending drug.

DYSPHAGIA

Dysphagia is defined as difficulty in swallowing. Odynophagia is painful swallowing while phagophobia is

TABLE 2.2: Causes of gingival hyperplasia**Generalized gingival hyperplasia***Acquired*

- Acute Myeloid Leukemia (M₄ & M₅ type)
- Vitamin C deficiency
- Drugs: Phenytoin, Cyclosporin, Calcium channel blockers (nifedipine, verapamil, diltiazem, amlodipine), Sodium valproate, Tranexamic acid
- Wegener's disease (strawberry gums)

Congenital

- Mucopolysaccharidosis
- Primary amyloidosis
- Hereditary gingival fibromatosis

Localized gingival hyperplasia*Acquired*

- Lymphomas
- Multiple myeloma
- Squamous cell carcinoma
- Kaposi sarcoma
- Pregnancy
- Sarcoidosis
- Wegener's disease
- Giant cell epulis (primary hyperparathyroidism)

Congenital

- Fabry's syndrome
- Tuberous sclerosis
- Sturge-Weber angiomatosis

fear of swallowing or refusal to swallow (in hysteria, rabies, tetanus, pharyngeal paralysis).

Classification

Dysphagia is divided into following types;

- Oropharyngeal dysphagia:** There is problem in transferring food from mouth to the esophagus. This is associated with nasal regurgitation and pulmonary aspiration during swallowing. The causes of oropharyngeal dysphagia can be subgrouped into (a) mechanical and (b) motor dysphagia. Important causes are given in Table 2.3.
- Esophageal dysphagia:** Difficulty in swallowing the food down the esophagus. Patients with mechanical obstruction complain of dysphagia mainly for solids whereas those with motility disorders have dysphagia for both solids and liquids. Causes are given in Table 2.4.

Odynophagia

Odynophagia is a painful swallowing that may limit oral intake. Important causes are infectious esophagitis due to

candida, herpes, or CMV. It can also be due to corrosive injury (due to caustic ingestion) or pill induced ulcers (pill esophagitis).

Diagnosis

Following investigations are helpful in making the diagnosis.

- Endoscopy (esophagogastroscope)
- Video esophagography
- Barium esophagography
- Esophageal manometry
- Esophageal pH recording
- Imaging studies (CT scan)
- Specific tests to rule out neuromuscular disorders

Treatment*General*

- Modification of diet
- Enteral feeding through a gastrostomy tube

TABLE 2.3: Causes of oropharyngeal dysphagia**Mechanical**

Oropharyngeal tumours
Post surgical/ radiation changes
Zenker's diverticulum
Retropharyngeal abscess/mass
Thyroid disorders
Plummer-Vinson syndrome
Inflammatory lesions

Motor

Brain injury, cerebral palsy, parkinsonism, stroke
Rabies, tetanus
Cranial nerve palsy, Guillain Barre syndrome
Myasthenia gravis, botulinum toxin
Myositis, myopathies

TABLE 2.4: Causes of esophageal dysphagia**Mechanical**

Esophageal cancer
Peptic stricture
Inflammatory esophagitis
Schatzki's ring (lower esophageal mucosal ring)
Posterior mediastinal mass
Aortic aneurysm

Motor

Achalasia
Scleroderma
Diffuse esophageal spasm

- c. Endoscopic removal of obstructing food bolus in acute dysphagia
- d. Nutrition counselling

Medical

- a. Proton pump inhibitors (for mucosal inflammation in reflux disease)
- b. Antimicrobial agents (infectious esophagitis)
- c. Viscous lidocaine solution for symptomatic relief
- d. Anticholinergic medications in cases with drooling of saliva and oropharyngeal dysphagia

Endoscopic Therapy

- a. Esophageal dilatation
- b. Esophageal stent placement
- c. Pneumatic dilatation of lower esophageal sphincter (LES) for achalasia.
- d. Botulinum toxin injection to LES in achalasia

Surgical

- a. Laproscopic myotomy in achalasia

GASTRITIS

Gastritis is a histological diagnosis characterized by inflammation of the gastric mucosa.

Acute Erosive and Hemorrhagic Gastritis

Acute gastritis is usually erosive and hemorrhagic. Important causes of erosive and hemorrhagic gastritis are:

- drugs (aspirin, NSAIDs)
- alcohol
- stress due to severe illness
- congestive gastropathy due to portal hypertension

Stress ulceration in patients with head trauma is called *Cushing's ulcer* and in severe burns, *Curling's ulcer*. Stress injury is not characterized by inflammation, thus the term 'gastritis' is a misnomer.

Erosive gastritis is usually asymptomatic. It may present with anorexia, nausea, vomiting and abdominal pain. Hematemesis and melena are most important manifestations of erosive gastritis.

Diagnosis is made by endoscopy which reveals superficial hemorrhages and erosions. Biopsy may be required to differentiate it from peptic ulcer and cancer.

Management

1. The treatment of acute gastritis consists of administration of sucralfate suspension (1 g 4-6 hrly) and H₂ receptor antagonist (ranitidine 150 mg twice daily, famotidine 20 mg twice daily) or proton pump inhibitors (omeprazole 20 mg once daily, rabeprazole 20 mg once daily, pantoprazole 40 mg once daily, lansoprazole 30 mg once daily).
2. The consumption of alcohol and drugs (NSAIDs) should be avoided.
3. Stress gastritis can be prevented by the administration of sucralfate, H₂ receptor antagonist or proton pump inhibitor in critically ill patients.

Acute Gastritis Due to Infections

A variety of infections can lead to acute inflammatory changes in the gastric mucosa. This presents as sudden onset of epigastric pain, nausea and vomiting.

H. pylori infection can lead to acute gastritis which generally progresses to chronic gastritis.

Acute bacterial infections (aerobic and anaerobic bacteria) can lead to progressive life-threatening necrotizing gastritis (phlegmonous gastritis). The treatment includes antibiotics and emergency gastrectomy.

Herpes simplex virus, CMV and Candida can cause gastritis in immunocompromized patients such as AIDS.

Chronic Gastritis

Histologically chronic gastritis is characterized by chronic inflammation with predominant infiltration by lymphocytes and plasma cells. The early stage of chronic gastritis is superficial gastritis. This is followed by the stage of atrophic gastritis. The final stage is gastric atrophy. The subsequent development of metaplasia may be precancerous. The two main types of chronic gastritis are autoimmune gastritis and *H. pylori* related chronic gastritis.

Autoimmune Gastritis (Type A Gastritis)

- This is characterized by the involvement of fundus and body of the stomach, *sparing antrum*.
- Circulating autoantibodies are found against parietal cells and intrinsic factor.

- This type of gastritis is generally asymptomatic. However, it may be associated with vitamin B₁₂ deficiency (*pernicious anemia*).
- Features of other *autoimmune involvement* (such as thyroid disease) may be present.
- There is four-fold increase in the incidence of gastric cancer.
- The treatment of pernicious anemia includes regular parenteral B₁₂ supplementation.

Helicobacter Pylori Gastritis (Type B Gastritis)

- *H. pylori* is the most common cause of chronic gastritis.
- The incidence of *H. pylori* gastritis increases with age.
- Initially, the antrum is predominantly involved, later pangastritis occurs.
- The majority of patients are asymptomatic with no sequelae.
- It is associated with peptic ulcer disease, with a 2-6 fold increase in the risk of *gastric adenocarcinoma* and low grade B-cell gastric lymphoma (MALT lymphoma).
- Eradication of *H. pylori* is routinely not recommended. Patients with peptic ulcer disease and MALT (mucosa associated lymphoid tissue) lymphoma are tested and treated for *H. pylori*. Antibiotics are given to treat *H. pylori* (see peptic ulcer).

Uncommon Types of Gastritis

Other types of gastritis are granulomatous gastritis (tuberculosis, sarcoidosis, candidiasis, syphilis, Crohn's disease), eosinophilic gastritis and lymphocytic gastritis.

PEPTIC ULCER

Peptic ulcer is defined as the presence of ulcer in the lower esophagus, stomach and duodenum. The ulcer is a breach in the mucosa due to an imbalance between mucosal defensive factors and luminal factors such as acid and pepsin. A break in the mucosal surface of more than 5 mm and depth to the submucosa are characteristics of an ulcer.

Duodenal ulcer is more common than gastric ulcer and occurs at younger age group (30-55 years) as compared to gastric ulcer (55-70 years).

Etiology

The common etiological factors of peptic ulcer are:

- *H. pylori* infection
- NSAID
- smoking
- acid hypersecretory states (Zollinger-Ellison syndrome)

Multiple factors may be responsible for the ulcer in a particular patient. Whatever the cause, the common abnormality is an imbalance between mucosal defensive factors and the aggressive factors (acid, pepsin).

H. Pylori Infection

H. pylori infection is a very common and important factor in the etiology of peptic ulcer. The prevalence of *H. pylori* infection is around 75% in duodenal ulcer and 30-60% in gastric ulcer. The infection leads to a state of hypergastrinemia, increased gastric acid secretion and decreased duodenal bicarbonate secretion. These factors result in patches of gastric metaplasia in the duodenal bulb. Colonization of these patches by *H. pylori* subsequently causes inflammation and formation of duodenal ulcer. The gastric ulcer in *H. pylori* infection occurs predominantly due to reduced gastric mucosal resistance.

NSAIDs Induced Ulcers

The chronic use of NSAIDs is more commonly associated with gastric ulcer than duodenal ulcer. Users of NSAIDs are also more likely to suffer from serious ulcer-related complications. These drugs reduce prostaglandins synthesis in the gastric mucosa by inhibiting enzyme cyclo-oxygenase 1 (COX-1). Prostaglandins play an important role in the gastric cytoprotection and repair. Drugs which selectively inhibit COX-2 at the site of inflammation (valdecoxib, celecoxib, etoricoxib) without affecting COX-1 activity are less likely to cause gastric ulcer.

Smoking and Diet

Smokers are more likely to develop peptic ulcers and related complications. The healing of the ulcer and response to therapy are diminished in smokers.

There is no association of any specific diet, alcohol or caffeine with the formation of ulcers, although certain foods can cause dyspepsia.

Clinical Features

- Recurrent abdominal pain is the most common symptom.
- Pain is localized to epigastrium and is burning or gnawing type. It may be dull aching, vague or described as hunger pain. Pain in duodenal ulcer is relieved by taking meals and antacids. Nocturnal pain (at 3-4 am) usually indicates duodenal ulcer. Pain in gastric ulcer may increase after meals.
- Nausea and weight loss are commonly present in gastric ulcer.
- Physical examination may be normal or may reveal epigastric tenderness in uncomplicated peptic ulcer.

Complications of peptic ulcer are given in Table 2.5. Constant pain not relieved by food or antacids suggests penetration. Severe pain (acute abdomen) may occur in case of perforation of peptic ulcer. Melena or hematemesis occurs in case of bleeding.

Investigations

1. Endoscopy is the investigation of choice for the diagnosis of peptic ulcer. Biopsy of the lesion to detect malignancy and *H. pylori* infection can also be done.
2. Barium studies of the upper GI tract can be performed alternatively. However, endoscopy is subsequently needed in most cases.
3. *H. pylori* infection can be diagnosed by noninvasive tests like serology, fecal antigen assay and urea breath test. Rapid urease test, histology and culture can be performed upon the biopsy sample.
4. There may be anemia. Stool may be positive for occult blood. Leukocytosis suggests complications such as penetration or perforation.

Treatment

General Measures

1. Dietary restriction does not help; hence, it is not required.
2. Meals should be taken at regular intervals.
3. Smoking should be stopped completely.

TABLE 2.5: Complications of peptic ulcer

- GI hemorrhages
- Perforation
- Ulcer penetration (into pancreas, liver)
- Gastric outlet obstruction

Medical Treatment

The medical treatment of peptic ulcer can be divided into three categories:

1. Acid neutralizing or inhibitory drugs
2. Mucosal protective agents
3. Eradication of *H. pylori*.

Acid Neutralizing or Inhibitory Drugs

- a. *Antacids*: These are commonly a mixture of aluminium hydroxide and magnesium hydroxide. Antacids are used in the initial phase of treatment because they provide rapid relief.
- b. *H₂ receptor antagonist*: Ranitidine (300 mg daily) or famotidine (40 mg daily) in single or divided dosage can cause healing of the ulcer in 85% cases in 6-8 weeks.
- c. *Proton pump inhibitor (PPI)*: These agents are preferred over H₂ receptor antagonists because of superior efficacy. The healing of the ulcer occurs in over 90% cases in case of gastric ulcer in 8 weeks and in case of duodenal ulcer in 4 weeks. Agents used are omeprazole or rabeprazole 20 mg, lansoprazole 30 mg, esomeprazole or pantoprazole 40 mg daily half an hour before breakfast.

Mucosal Protective Agents

Protective agents (sucralfate, bismuth, misoprostol) promote ulcer healing by enhancing the mucosal defensive mechanism. However, these are not used as the first line therapy in active ulcers because other agents (PPI and H₂ receptor blockers) are more efficacious and better tolerated.

- Sucralfate (1 g 4 times daily) is used in addition to other drugs in refractory ulcers.
- Bismuth containing compounds are given along with antibiotics to eradicate *H. pylori*.
- Misoprostol is used to prevent ulcers due to NSAIDs.

H. Pylori Eradication Therapy

Triple drug therapy is employed to eradicate *H. pylori* infection. The regimen includes two antibiotics and one PPI. Single antibiotic is not given to avoid the problem of resistance.

- Commonly employed regimen includes Amoxycillin 1 g twice daily plus Clarithromycin 500 mg twice daily

plus twice a day PPI (omeprazole or rabeprazole 20 mg, lansoprazole 30 mg, pantoprazole 40 mg) for 14 days. Metronidazole can be used in place of Amoxicillin in cases of penicillin allergy.

- Eradication of *H. pylori* is achieved in 85-90% cases. Successful eradication reduces the recurrence of ulcer.
- If infection persists after giving triple therapy, quadruple therapy (PPI, bismuth, tetracycline, metronidazole) is given.

Therapy in Specific Cases

- H. pylori* associated ulcers if uncomplicated are treated with triple drug combination for 14 days. Thereafter PPI or H₂ receptor blocker must be continued for additional 2-6 weeks.
- Treatment of active ulcer due to NSAIDs includes immediate withdrawal of the offending agent and administration of PPI or H₂ receptor antagonist. All such patients should also be tested for *H. pylori* infection. If positive, eradication therapy should also be given.
- Preventive treatment with PPI or misoprostol is required in high risk patients on NSAIDs such as age >60 years, history of ulcer disease, concurrent therapy with corticosteroids or anticoagulants and serious underlying medical illness. Use of COX-2 selective NSAIDs reduces injury to the gastric mucosa.

Surgical Treatment

- Emergency surgery is performed in case of perforation and persistent hemorrhage.
- Elective surgery is done in gastric outflow obstruction and recurrent ulcer despite medical treatment.

DIARRHEA, DYSENTERY AND FOOD POISONING

Diarrhea is defined as passage of stool weighing more than 200-250 g. Practical definition is passage of liquid or unformed stools at an increased frequency. Normal frequency varies from 1-3/day to once in 3 days.

Approximately 9-10 L of fluid enters the small intestine daily. The majority of this fluid is absorbed in the small intestine and only about 1.5 L enters the colon. Fluid absorption also occurs in the colon and normally only 100-200 ml fluid is excreted daily in the stool.

Diarrhea is classified into:

- Acute (< 2 weeks)
- Persistent (2-4 weeks)
- Chronic (> 4 weeks)

Acute Diarrhea

Acute diarrhea is caused mainly by infections (90%). It may also be caused by drugs, ischemia, toxins and other conditions. Causes of acute diarrhea are given in Table 2.6.

Incubation Period

The incubation period varies from few hours to days. It is few hours (1-6 hrs) in case of preformed toxin induced diarrhea while 12-24 hours in infective diarrhea.

Pathogenic Mechanisms

Pathogens can cause diarrhea via various mechanisms. These are described as below:

- Toxin production:** Bacterial toxins either preformed or produced in the gut can cause diarrhea. Such toxins are exotoxins and include enterotoxins, cytotoxins, and neurotoxins. Enterotoxins disturb normal secretory mechanisms and cause profuse watery diarrhea whereas cytotoxins lead to inflammatory diarrhea by causing destruction of mucosal cells. Some bacteria produce exotoxins with both enterotoxin and cytotoxin activities. Neurotoxins produced by *S. aureus* and *B. cereus* act on the nervous system to produce vomiting.
- Invasion:** Inflammatory diarrhea or dysentery results from the invasion and destruction of mucosal cells by *Shigella* or Enteroinvasive *E. coli*. Intraepithelial multiplication and spread to adjacent cells also occurs.
- Penetration:** *Salmonella typhi* and *Yersinia enterocolitica* penetrate intestinal mucosa and multiply in Peyer's patches or intestinal lymph nodes. They disseminate from these lymph nodes and cause fever.

Clinical Manifestations

- Fever, abdominal pain and bloody diarrhea (dysentery) suggest inflammatory type such as shigellosis, salmonellosis, amoebiasis, *C. difficile* and Enterohemorrhage *E. coli*.
- Watery non-bloody diarrhea with nausea, vomiting and abdominal bloating is indicative of noninflammatory diarrhea caused by toxin producing bacteria, giardia or viruses (Table 2.6).

TABLE 2.6: Causes of acute diarrhea**Infectious**

- Viral
 - Rotavirus
 - Norwalk agents
 - Cytomegalovirus*
- Bacterial
 1. Preformed toxin
 - *S. aureus*
 - *B. cereus*
 - *Clostridium perfringens*
 2. Enterotoxin induced
 - Enterotoxigenic *E. coli* (ETEC)
 - *Vibrio cholerae*
 3. Cytotoxin production*
 - Enterohemorrhagic *E. coli* (EHEC)
 - *Clostridium difficile*
 4. Mucosal invasion*
 - *Shigella*
 - *Campylobacter jejuni*
 - *Salmonella*
 - Enteroinvasive *E. coli* (EIEC)
 - *Yersinia enterocolitica*
- Protozoal
 - *Entameba histolytica**
 - *Giardia lamblia*
 - *Cryptosporidium*

Non-infectious

- Diverticulitis
- Inflammatory bowel disease (ulcerative colitis, Crohn's disease)
- Metabolic (DKA, carcinoid)
- Sepsis
- Drugs (NSAIDs, antibiotics)
- Ischemic colitis

* Causes associated with inflammatory diarrhea

The diarrhea may be profuse leading to dehydration.

- a. Thirst, dry mouth, decreased sweating, oliguria and mild weight loss suggest mild dehydration.
- b. Orthostatic hypotension, sunken eyes, sunken fontanelles in infants and loss of skin turgor indicate moderate dehydration.
- c. Severe dehydration may result in hypotension, tachycardia, altered sensorium and shock.

Investigations

Stool examination is the basic investigation. This may reveal ova or parasites, fecal leukocytes and increased fecal lactoferrin, blood and mucus. Presence of blood, mucus and increased fecal lactoferrin suggests inflammatory

diarrhea. Microbiological investigations include culture for viral and bacterial pathogens and immunoassays for toxins.

Lower GI endoscopy and biopsy of the intestine may be needed in cases with persistent diarrhea and non-infectious diarrhea. Upper GI endoscopy and duodenal aspirate examination may also be required.

Dysentery

Dysentery is defined as diarrhea due to acute inflammation of the large intestine characterized by the presence of blood and mucus in the stool. The two main types of dysentery are *bacillary dysentery* and *amoebic dysentery*. Important causes of bacillary dysentery are *shigella*, enteroinvasive *E. coli* (EIEC), and *Yersinia enterocolitica*. Amoebic dysentery is caused by *E. histolytica*.

Clinical features of dysentery include diarrhea, fever, abdominal pain and tenesmus. Stools are usually small and contain blood or purulent material. The colon is tender to palpate. Diagnosis depends on stool examination and culture.

Food Poisoning

- Food poisoning is gastroenteritis of infective or non-infective origin.
- The important infective causes are *S. aureus*, *salmonella*, *B. cereus* and *E. coli*.
- Non-infective causes are allergy to sea foods, fish or fungal toxins (Table 2.7).
- The presentation is in the form of vomiting, diarrhea or both which usually occurs within 1-48 hours of consumption of contaminated drinks or food.
- The incubation period is short (from minutes to hours) in case of noninfective causes or due to ingestion of food with preformed toxins.
- The concurrent occurrence of illness in more than one member of the family, group or institution suggests the possibility of food poisoning.

Investigations

The stool, vomitus or suspected food should be sent for culture. The tests for the presence of specific toxins should also be performed.

TABLE 2.7: Causes of food poisoning

Infective
<i>Toxin mediated</i>
• <i>S. aureus</i> (1-6 H)* • <i>C. perfringens</i> (8-16H) • <i>C. botulinum</i> • <i>E. coli</i> (EHEC, ETEC) (>16 H) • <i>Bacillus cereus</i> (1-6, 8-16H) • <i>Vibrio cholerae</i> (>16H)
<i>Non-toxin mediated</i>
• <i>Salmonella</i> (>16H) • <i>Shigella</i> (>16H) • <i>Campylobacter jejuni</i> (>16H) • <i>Bacillus anthracis</i> • <i>Listeria monocytogenes</i> • Viruses (rotavirus)
Non-infective causes
<i>Allergic</i>
• Shellfish, strawberries
<i>Non-allergic</i>
• Fish (Ciguatoxin, scombotoxin) • Fungi (<i>Amanita phalloides</i>) • Chemicals, metals

* Incubation period in hours

Management

Most cases are mild and do not require specific therapy. However evaluation and treatment are required in:

- Profuse diarrhea
- Bloody stools
- Presence of fever
- Severe abdominal pain
- Duration of more than 48 hours without improvement
- Elderly
- Immunocompromised patients

Fluid and Electrolyte Management

Fluid and electrolyte management is the cornerstone in the treatment of diarrhea. In most cases with non-inflammatory diarrhea, no treatment is required except adequate rehydration. The disease is mostly self limiting. The patient is advised to take fluids orally to maintain hydration and electrolytes. *Oral rehydration solutions* (ORS) are available for fluid and electrolyte replacement. This usually contains 3.5 g of sodium chloride, 2.5 g of sodium bicarbonate, 1.5 g of potassium chloride and 20 g of glucose to be dissolved in one liter of water. In severe dehydration and in infants and the elderly, intravenous fluids are required.

Diet

Liquids, semisolids, soft and easily digestible foods are permitted, while the intake of milk, high fiber, fat, caffeine and alcohol is to be avoided.

Antimotility/antisecretory Agents

Antimotility/antisecretory agents are used in noninflammatory diarrhea and avoided if diarrhea is bloody and there is fever.

- Antimotility/antisecretory agents such as codeine phosphate, loperamide and bismuth subsalicylates may be used to reduce the frequency and fluidity of stools.
- Racecadotril is a newer antisecretory agent useful in acute watery diarrhea.
- Diphenoxylate should be avoided in acute diarrhea.

Antimicrobial Agents

Antibiotics are not used routinely even in inflammatory diarrhea which is generally self-limiting. However, empirical antibiotics are given in patients with fever, bloody diarrhea, tenesmus and in elderly or immunocompromised patients.

- The antibiotics include fluoroquinolones (ciprofloxacin 500 mg, ofloxacin 400 mg or norfloxacin 400 mg twice daily) for 5-7 days. Alternatively, doxycycline 100 mg twice daily or trimethoprim-sulfamethoxazole 160/800 mg twice daily may be used.
- Metronidazole, tinidazole or ornidazole can empirically be given if giardiasis or amoebiasis is suspected.
- Specific antimicrobial treatment is needed in shigellosis, cholera, salmonellosis, *traveller's* diarrhea, *C. difficile* infection, giardiasis and amoebiasis.

Chronic Diarrhea

Diarrhea lasting for more than 4 weeks is known as chronic diarrhea. It is mostly caused by noninfectious causes. A number of pathologic mechanisms can lead to chronic diarrhea (Table 2.8).

Secretory Diarrhea

This is due to alteration in fluid and electrolyte transport across the mucosa. Watery, large volume diarrhea which persists on fasting is characteristic. Important causes include medications (laxatives), hormone-mediated (VIPoma, Carcinoid), villous adenoma and bile salt malabsorption.

TABLE 2.8: Causes of chronic diarrhea

Inflammatory causes
• Ulcerative colitis
• Crohn's disease
• Malignancies (lymphoma, adenocarcinoma)
Chronic infections
• Giardia, Entamoeba
• AIDS related
– CMV
– <i>Microsporidium</i> , <i>Cryptosporidium</i>
– <i>Isospora belli</i>
Motility disorders
• Diabetes
• Hyperthyroidism
• Irritable bowel syndrome
Osmotic diarrhea
• Medications (lactulose, sorbitol)
• Lactose intolerance
Secretory diarrhea
• Medications
• VIPoma
• Carcinoid
• Zollinger-Ellison syndrome
• Villous adenoma
Malabsorption syndromes

Osmotic Diarrhea

Osmotic diarrhea occurs when an ingested nutrient is not absorbed and drags fluid along with it. This type of diarrhea ceases on fasting. Osmotic laxatives (antacids, lactulose, sorbitol) and carbohydrate malabsorption, lactase deficiency) are important causes of osmotic diarrhea.

MALABSORPTION

Disorders of digestion and diminished absorption of dietary nutrients (one or more) are referred as malabsorption syndromes. Various diseases with varied etiologies can lead to malabsorption and may present with different clinical manifestations.

Normal digestion and absorption may be divided into three phases and malabsorption can result from abnormalities in one or more of these phases (Table 2.9).

Intraluminal Phase

There is inadequate hydrolysis and solubilization of dietary nutrients (protein, fat and carbohydrates) leading to malabsorption. This is mainly due to insufficient bile or pancreatic enzymes. The important causes are pancreatic

TABLE 2.9: Causes of malabsorption

Intraluminal maldigestion
<i>Pancreatic enzyme deficiency</i>
• Chronic pancreatitis
• Pancreatic carcinoma
• Cystic fibrosis
<i>Reduced bile acids</i>
• Liver diseases
• Intestinal bacterial overgrowth (fistula, blind loop)
• Decreased enterohepatic circulation (ileal resection, Crohn's disease)
Mucosal defects
• Intestinal resection
• Infiltration, inflammation or infection of mucosa
1. Crohn's disease
2. Celiac sprue
3. Tropical sprue
4. Whipple's disease
5. Giardiasis
6. Lymphoma
• Genetic diseases (lactase deficiency)
Impaired nutrient uptake
• Lymphatic obstruction (lymphoma, lymphangiectasia)
• CHF, pericarditis
Miscellaneous
• Diabetes mellitus
• Hyperthyroidism
• Hypoparathyroidism

diseases, biliary obstruction, cholestatic liver diseases and decreased enterohepatic circulation of bile salts.

Mucosal Phase

The damage to the intestinal epithelium or resection of a part of small intestine diminishes the surface area for absorption. The brush border enzyme defects may also lead to malabsorption.

Absorptive Phase

Lymphatic obstruction prevents proper uptake and transport of absorbed lipoproteins and chylomicrons. Increased pressure in lymphatics may cause leakage of absorbed nutrients back into the intestinal lumen leading to steatorrhea and protein losing enteropathy.

The deficiency of any specific nutrient and its manifestations depend on the site and extent of the intestinal involvement. Iron, folic acid and calcium are absorbed in the proximal intestine while vitamin B₁₂ and bile salts are absorbed in the ileum. For example, the disease of terminal

ileum may lead to vitamin B₁₂ deficiency and involvement of proximal small intestine may cause iron deficiency.

Clinical Manifestations

Diarrhea and weight loss despite normal dietary intake should prompt one to evaluate the patients for malabsorption. In most patients, there is steatorrhea that is an increase in excretion of more than 7 g per day fat in the stool. Bulky, pale and offensive stool which floats on water is characteristic of steatorrhea. Malabsorption syndromes should be considered in the differential diagnosis of chronic diarrhea.

Diarrhea may result due to following pathogenetic mechanisms:

- Osmotic diarrhea*: It occurs due to decreased absorption of dietary nutrients. It improves on prolonged fasting.
- Nonabsorbed fatty acids in cases of steatorrhea interfere with intestinal ion transport leading to diarrhea.
- Secretory diarrhea*: It occurs due to increased intestinal fluid and electrolytes secretion in response to exotoxins or increased gut hormones (VIP). The diarrhea does not improve on fasting.

Other features are abdominal distension, cramps in abdomen and presence of undigested food in stool. Increased flatus formation occurs due to bacterial fermentation of unabsorbed carbohydrates. Symptoms related to specific deficiency of minerals, vitamins or other nutrients can also be present. Weakness, lethargy and malaise may also be present (Table 2.10).

Investigations

Routine laboratory studies: Tests are performed to detect any nutrient deficiency (one or many). These tests reveal malabsorption of particular nutrient/nutrients but do not establish the cause. Some important tests are given in Table 2.11.

Specific Tests

- Fecal fat estimation*: This is a reliable test to confirm steatorrhea and fat malabsorption. Sudan III stain may show an increase in the stool fat. Quantitative estimation of fat in the stool is more reliable and sensitive. A 72

TABLE 2.10: Pathologic basis of symptoms of malabsorption

Symptoms/signs	Malabsorbed nutrients
Anemia	Iron, folic acid, Vit B ₁₂
Bleeding	Vitamin K, Vit C
Glossitis, stomatitis	Iron, folic acid, Vit B ₁₂ , Vit A
Night blindness, xerophthalmia	Vit A
Tetany, paresthesia	Calcium, magnesium, potassium
Bone pain	Calcium, Vit D, protein
Neuropathies	Vit B ₁₂ and Vit B ₁
Dermatitis	Vit A, zinc, essential fatty acids
Azotemia, hypotension	Fluid and electrolytes
Edema	Protein
Muscle wasting	Protein
Bleeding gums	Vitamin C

TABLE 2.11: Routine blood tests in malabsorption

Complete blood count	Anemia (iron, folate and cobalamin deficiency)
General blood picture	
• Microcytic	Iron deficiency
• Macrocytic	Folate and cobalamin deficiency
Prothombin time	Vitamin K deficiency
Total serum protein and albumin	Protein malabsorption
Alkaline phosphatase	Vitamin D deficiency
Serum carotene	Vitamin A deficiency
Serum cholesterol	Fat malabsorption
Serum iron, folate, cobalamin	Iron, folate, cobalamin malabsorption
Serum calcium	Calcium or Vitamin D malabsorption

hour stool collection, while the patient is on a defined diet, is used for fat estimation. Excretion of more than 10 g fat per day suggests fat malabsorption.

- Schilling test*: This is useful in the diagnosis of cobalamin malabsorption and its cause. Cobalamin metabolism and absorption depends on the normal gastric, pancreatic and ileal functions. Cobalamin malabsorption may occur in pernicious anemia, chronic pancreatitis, achlorhydria and bacterial overgrowth. Schilling test can be used to assess the functional status of these organs and to determine the pathological processes responsible for malabsorption. Radio-labeled cobalamin (1 mg ⁵⁸Co) is given orally and its excretion in urine is measured. One mg cobalamin is administered intramuscularly to saturate hepatic binding sites so that all radio-labeled cobalamin is excreted in the urine. The test is abnormal if less than

10% of the radio-labeled cobalamin is excreted in the urine in 24 hours. If the test is abnormal, it is repeated by giving radio-labeled cobalamin in combination with intrinsic factor or pancreatic enzymes. It may also be repeated after a 5 days course of antibiotics. This will help in differentiating the various defects responsible for malabsorption of cobalamin.

- c. *D-Xylose test* is performed to detect carbohydrate malabsorption. 25 g D-Xylose is given orally and its excretion is measured in urine. Excretion of less than 4.5 g in 5 h is indicative of malabsorption.
- d. *Upper GI endoscopy and biopsy of small intestinal mucosa*: This is essential for the diagnosis of conditions like tropical sprue, celiac sprue, Whipple's disease and Crohn's disease.
- e. *Radiological assessment* of the small intestine with barium contrast is helpful in evaluation of structural abnormalities.
- f. *Pancreatic exocrine functions* are assessed in patients with steatorrhea.
- g. *Serological studies*: Autoantibodies are detected in some conditions such as celiac sprue and pernicious anemia.

Treatment

1. Deficient nutrients are replaced.
2. Specific therapy depends on the cause:
 - a. Gluten diets (wheat, barley) should be avoided in celiac sprue.
 - b. Tropical sprue is treated with tetracycline and folic acid for 6 months.
 - c. Cotrimoxazole double strength tablet daily is given for 1 yr in Whipple's disease.
 - d. A one to two week course of antibiotics (fluoroquinolone, tetracycline, metronidazole) is given in malabsorption with bacterial overgrowth.
 - e. Milk products are avoided in lactase deficiency.

LIVER: STRUCTURE AND FUNCTION

The liver is the largest organ of the body. It is situated in the right hypochondrium under the diaphragm. Anatomically it is divided into right and left lobes. It has dual blood supply, 80% comes from the portal vein and the remainder from the hepatic artery. Two-third of the mass of the liver is constituted by hepatocytes. Other important cell types are

Kupfer cells (RE system), stellate cells (Ito or fat storing cells) and endothelial cells. Histologically the liver is composed of lobules with portal areas at the periphery and central vein in the center of the lobule.

The liver has numerous functions necessary for good health. The main functions can be classified into following groups:

1. *Synthetic function*: Liver is the site for the synthesis of albumin, coagulation factors, carrier proteins, hormones and growth factors. Almost all coagulation factors (except factor VIII) are synthesized in the liver. The synthesis of factors II, VII, IX, X requires Vitamin K.
2. *Metabolic functions*: Liver is an important site for the metabolism of carbohydrates, proteins and lipids. It also metabolizes drugs, alcohol and bilirubin. Ammonia is metabolized in the liver into urea.
3. *Excretory functions*: The bile and its contents (bile acid, bile salts) are synthesized and excreted by the liver.
4. *Storage functions*: Liver is the storage site for vitamins (A, D, B₁₂, folate and K) and minerals (iron and copper). Iron is stored in the form of ferritin and hemosiderin.

Liver Function Tests

The liver has numerous functions and no single test is enough to assess all its functions. However, certain tests are performed to assess some of the main functions of the liver (serum albumin, serum bilirubin and prothrombin time) while other tests indicate the severity of damage (aminotransferases) or obstruction to bile flow (alkaline phosphatase).

Tests to Assess Excretory Functions

- a. *Serum bilirubin*: The normal level of serum total bilirubin is 0.3-1.0 mg/dL. It is present in two forms, unconjugated (0.2-0.7 mg/dL) and conjugated bilirubin (0.1-0.3 mg/dL). A rise in the level of conjugated bilirubin suggests liver or biliary tract disease. Isolated elevation of unconjugated bilirubin is rare in liver disease and it indicates hemolysis.
- b. *Urine bilirubin*: Conjugated bilirubin (not the unconjugated bilirubin) is excreted in the urine when

the plasma level of conjugated bilirubin is raised. Hence, the presence of bilirubin in the urine suggests liver disease.

- c. *Blood ammonia level:* Blood ammonia level may be raised in severe hepatic dysfunction and hepatic encephalopathy. It may also be high due to portal shunting in portal hypertension.

Tests to Assess Synthetic Functions

- a. *Serum albumin:* Albumin is exclusively synthesized in the liver. The normal value of serum albumin is 3.5-5.5 g/dL. It has a long half life of about 20 days, therefore a low serum albumin suggests chronic liver disease.
- b. *Coagulation factors:* The half life of most coagulation factors is short; factor VII has the shortest half life (six hours). Hence, measurement of prothrombin time (PT) is helpful in the diagnosis and the prognosis of acute parenchymal liver disease. Prothrombin time may also be prolonged in obstructive jaundice due to vitamin K malabsorption. Correction of PT after administration of vitamin K suggests vitamin K deficiency.
- c. *Miscellaneous:* Other tests used to measure synthetic functions of the liver include serum levels of α -1 anti-trypsin, α -fetoprotein and ceruloplasmin.

Serum Enzymes

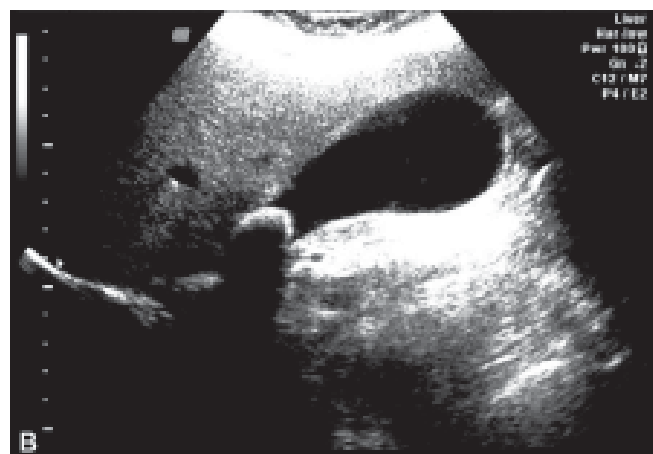
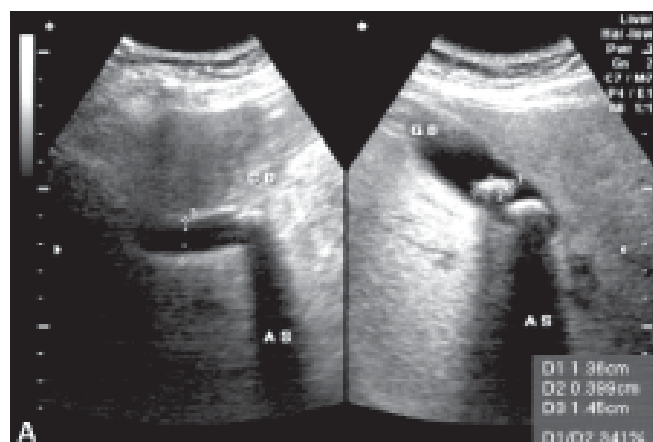
The liver contains thousands of enzymes. Many are present in the serum in very low quantity. The elevation of these enzymes in the serum indicates either damage to hepatocytes or cholestasis.

- a. *Aminotransferases:* Serum alanine aminotransferase or serum glutamate pyruvate transaminase (ALT or SGPT) and aspartate aminotransferase (AST or SGOT) are raised in acute liver cell injury. The normal serum levels of ALT and AST are 0-35 Unit/L. However, its elevation does not correlate with the severity of the disease. A rise in ALT is more specific to liver cell injury since AST can also be raised in other conditions such as diseases of myocardium, skeletal muscles, kidneys and brain. The ratio of serum AST to ALT is less than one in viral hepatitis whereas it is greater than two in alcoholic liver disease. They are usually not significantly elevated in obstructive jaundice.

- b. *Alkaline phosphatase:* A significant rise in serum alkaline phosphatase (ALP) suggests cholestasis (obstruction in the bile flow). It may be mildly raised in other liver diseases. The normal serum level is 30-120 units/L. Alkaline phosphatase is also present in other body tissues like bone, intestine, placenta and leukocytes.
- c. *Serum 5' nucleotidase and gamma glutamyl transpeptidase (GGT)* are also raised in cholestasis. Hence, a concomitant rise in serum 5' nucleotidase or gamma glutamyl transpeptidase (GGT) along with ALP suggests hepatic origin of alkaline phosphatase.

Imaging Techniques

- a. *Ultrasonography:* It is useful to screen for the evidence of intrahepatic or extrahepatic cholestasis and gallstones (Fig. 2.8). In addition, it can detect liver cysts, abscesses



FIGURES 2.8A and B: (A) Ultrasound showing stones in the gall- bladder (B) Ultrasound showing impacted stone in the neck of gallbladder

and masses. Biopsy of a lesion can be done under ultrasonographic monitoring. Ultrasonography may also detect portal hypertension (dilated portal vein, ascites, splenomegaly and collateral vessels).

- b. *CT scan and MRI*: These are useful in the evaluation of parenchymal liver disease. Contrast enhancement can be used to differentiate the nature of space occupying lesions. With MRI, vessels can be visualized without use of IV contrast.
- c. *Endoscopic retrograde cholangiopancreatography (ERCP) and Percutaneous transhepatic cholangiography (PTC)*: These help in the detection of the cause, location and the extent of the biliary obstruction. ERCP can be used for therapeutic interventions such as stone extraction from common bile duct and for placing a stent. Magnetic resonance cholangiopancreatography (MRCP) is a sensitive and noninvasive technique of visualizing the biliary tree.

Liver Biopsy

Biopsy is a definitive method to diagnose the cause and severity of hepatocellular diseases. This is usually done through percutaneous route. Transjugular route can be used in patients with ascites or a risk of bleeding.

JAUNDICE

A yellowish discoloration of the skin and mucous membrane due to deposition of bilirubin is known as jaundice (icterus). The deposition of bilirubin in tissues occurs when the serum bilirubin level is raised (hyperbilirubinemia).

- The normal total serum bilirubin level is 0.3-1.0 mg/dL. With the use of Van den Bergh's method, it has been shown that up to 30% of the total bilirubin (0.3 mg/dL) may be the conjugated or direct reacting type. However, newer techniques have shown that in normal individuals, almost all bilirubin in the serum is unconjugated.
- Sclerae have a high affinity for bilirubin due to their rich elastin content. Jaundice is clinically apparent in sclera when bilirubin level is raised above 3 mg/dL.
- The clinical detection of jaundice is difficult in artificial light. Hence, it should be examined preferably in day light.

- Other sites to be looked for the evidence of jaundice are mucosa of oral cavity underneath the tongue and skin.
- Urine is dark yellow in color due to excretion of conjugated bilirubin.
- Yellow discoloration of the skin can also occur in carotenemia (carotenoderma) and exposure to quinacrine or phenols. Sclera is typically not involved in carotenemia.

Bilirubin Metabolism

- Bilirubin is a product of heme metabolism. The breakdown of old red blood cells in the reticuloendothelial system (primarily spleen and liver) leads to the release of heme (Fig. 2.9). Heme is further metabolized into biliverdin and subsequently to bilirubin. About 70-80% of bilirubin is derived from this mechanism. The rest comes from the breakdown of premature erythroid cells in bone marrow (ineffective erythropoiesis) and the catabolism of myoglobin and cytochromes.
- This insoluble bilirubin binds reversibly and noncovalently to albumin (unconjugated or indirect bilirubin) and is transported to the liver where it is detached from albumin and is taken up by hepatocytes.
- In hepatocytes, the bilirubin is solubilized by conjugation to glucuronic acid that yields bilirubin monoglucuronide and diglucuronide (conjugated or direct bilirubin). This reaction is catalyzed by the enzyme UDP-glucuronyl transferase.
- The conjugated bilirubin is then transported to the duodenum via bile. Colonic bacteria metabolize the conjugated bilirubin into stercobilinogen which may be further oxidized to stercobilin. Stercobilinogen and stercobilin are excreted in stool. A small amount of stercobilinogen is absorbed to reach the liver through portal system and is re-excreted into the bile. A fraction escapes liver uptake and passes into urine as urobilinogen and its oxidized form, urobilin.
- The unconjugated bilirubin, because it is bound to albumin, is not filtered through kidneys and therefore does not appear in urine. Hence, in unconjugated hyperbilirubinemia (as in hemolysis), bilirubin is absent in the urine (acholuric jaundice). However, urobilinogen is increased in the urine. The presence of bilirubin in

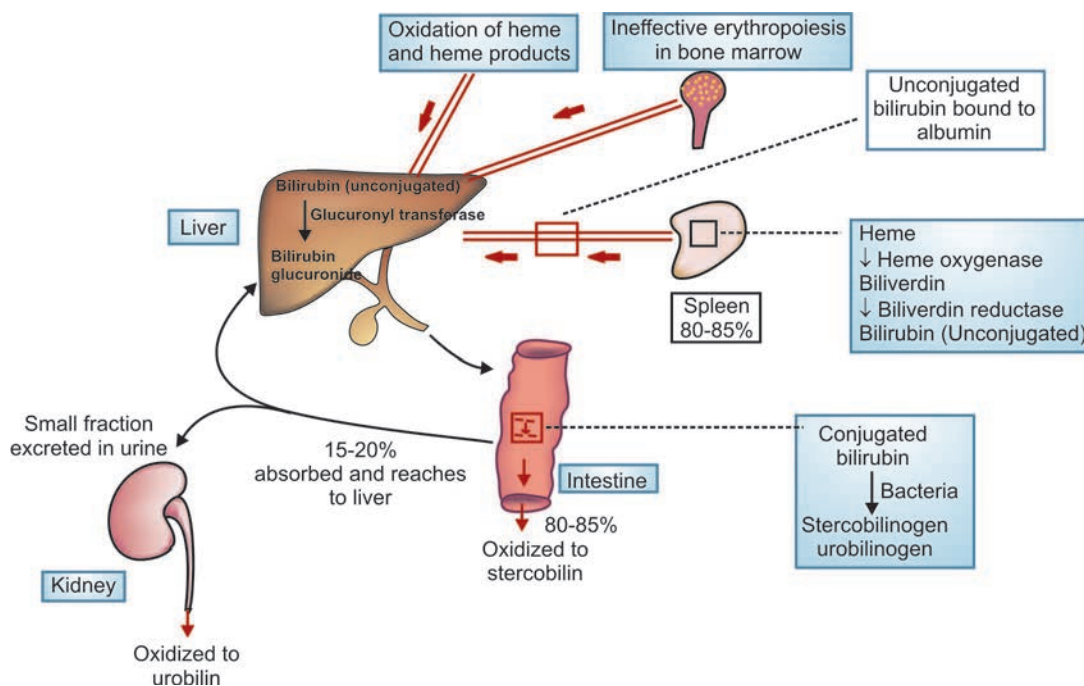


FIGURE 2.9: Bilirubin metabolism

TABLE 2.12: Urinary findings in different types of jaundice

Urine	Hemolytic findings	Hepato-cellular	Obstructive	Normal
Urobilinogen	+++	++	-	+
Bilirubin	-	++	++	-

urine suggests predominant conjugated hyperbilirubinemia (liver disease, obstructive jaundice). Urobilinogen is absent in the urine in cases of obstructive jaundice because bilirubin is not available in the intestine to be metabolized into stercobilinogen (Table 2.12).

Types of Jaundice

Jaundice is classically divided into two broad types, unconjugated hyperbilirubinemia and conjugated hyperbilirubinemia.

Unconjugated hyperbilirubinemia: This is characterized by predominantly high levels of serum unconjugated bilirubin.

- The color of urine is normal as there is no bilirubin.
- The jaundice is generally mild (serum bilirubin <6 mg/dL).
- Additionally patients may have anemia and splenomegaly.

- The raised unconjugated bilirubin (>20 mg/dL) can lead to central nervous system manifestations in neonates (kernicterus).
 - Liver transaminases and alkaline phosphatase are normal. Unconjugated hyperbilirubinemia may result either from overproduction of bilirubin or from impaired hepatic uptake or conjugation of bilirubin. Table 2.13 shows the clinical and laboratory features in different types of jaundice.
 - Overproduction of bilirubin (prehepatic):** This may occur in hemolytic disorders such as hereditary spherocytosis, sickle cell anemia, G6PD deficiency, paroxysmal nocturnal hemoglobinuria and autoimmune hemolytic anemia. Other causes include ineffective erythropoiesis, hemolytic reaction and resolution of hematoma.
 - Impaired hepatic metabolism:** The hepatic uptake of bilirubin is impaired because of some drugs (like rifampicin and probenecid) and in Gilbert's syndrome. The conjugation of bilirubin is impaired in Gilbert's syndrome and Crigler-Najjar syndromes because of decreased activity of the enzyme, glucuronyl transferase.
- Most neonates develop mild rise in unconjugated bilirubin because of incompletely developed hepatic

TABLE 2.13: Features of different types of jaundice

	Prehepatic (hemolytic)	Hepatic (Hepatocellular)	Posthepatic (obstructive)
History	Recurrent jaundice, positive family history, anemia	Prodrome: fever, anorexia, nausea	Pain in abdomen, may be recurrent
Skin color	Lemon yellow	Yellow	Dark yellow
Pruritus	Absent	Occasional (early phase)	Present
Spleen	Enlarged	Occasional	Not enlarged
Gallbladder	Not palpable	Not palpable	May be palpable
Feces	Normal	Pale (if cholestasis)	Pale
Urine color	Colorless, yellow on standing (urobilinogen)	Dark (bilirubin and urobilinogen)	Very dark (high bilirubin)
Tests for hemolysis	Positive	Negative	Negative
Serum bilirubin	Unconjugated (<6 mg/dL)	Mixed	Conjugated
ALT (SGPT)	Normal	Increased (+++)	Minimally increased (+)
Alkaline phosphatase	Normal	Minimally high (+)	High (+++)
Serum albumin	Normal	Low (in chronic disease)	Normal
Prothrombin time	Normal	Prolonged	Prolonged (correctible with vit K)
Important causes	Hemolysis, Gilbert's syndrome	Viral hepatitis, drugs, alcohol	Gallstones, carcinoma pancreas

functions (*physiologic neonatal jaundice*). Marked elevation of unconjugated bilirubin in premature infants can lead to kernicterus.

Gilbert's syndrome, an autosomal dominant disorder, is a common disorder with mild jaundice. Conjugation of bilirubin is also impaired due to decreased activity of glucuronyl transferase enzyme. Serum bilirubin is generally less than 6 mg/dL. No treatment is required.

Criggler-Najjar syndromes are characterized by complete absence of activity of glucuronyl transferase (Type I) or reduced activity of the enzyme (Type II). The jaundice is severe and death occurs early in type I Crigglar-Najjar syndrome. Most patients with type II Crigglar-Najjar syndrome survive up to adulthood.

Conjugated Hyperbilirubinemia: Serum conjugated bilirubin is raised in inherited conditions such as Dubin-Johnson syndrome and Rotor syndrome, hepatocellular diseases and cholestatic conditions.

- Dubin-Johnson and Rotor syndromes*: These are rare inherited disorders characterized by asymptomatic jaundice due to altered excretion of bilirubin in the bile ducts. The liver transaminases and alkaline phosphatase levels are normal.
- Hepatocellular diseases*: The bilirubin uptake, conjugation and its transport to bile canaliculi can all be affected due to parenchymal liver disease. Besides conjugated bilirubin, unconjugated bilirubin may also be raised. Levels of amino-

transferases (ALT, AST) are raised out of proportion to alkaline phosphatase. Clinical features of hepatocellular jaundice are given in the Table 2.13. Causes of hepatocellular jaundice are given in Table 2.14.

- Cholestatic jaundice (Obstructive jaundice)*: This type of jaundice results from the obstruction in the bile flow which may be intrahepatic or extrahepatic. Clinical features of cholestatic jaundice are given in the Table 2.13. Causes of obstructive jaundice are given in Table 2.15.

TABLE 2.14: Causes of hepatocellular jaundice

Viral hepatitis: A,B,C,D,E, EBV, CMV
 Alcohol
 Drugs: Isoniazid, paracetamol
 Toxins: Vinyl chloride, *Amanita phalloides*.
 Metabolic: Wilson's disease
 Immune: Autoimmune hepatitis

TABLE 2.15: Causes of obstructive jaundice

Intrahepatic

- Hepatitis
- Primary biliary cirrhosis
- Drugs: Chlorpromazine, anabolic steroids, contraceptives

Extrahepatic

- Stone in bile duct
- Sclerosing cholangitis
- Neoplasms of gallbladder and pancreas
- Compression by enlarged lymph nodes

- Conjugated bilirubin enters into blood and appears in urine.
- The stool becomes pale due to reduced formation of stercobilinogen in the intestine.
- Retention of bile salts leads to its deposition in the skin and severe pruritus (itching). Patients usually have scratch marks on skin.
- Reduced availability of bile in the intestine may lead to malabsorption of fat and fat soluble vitamins (vitamin K and vitamin D). This may result in bleeding diathesis and osteomalacia.
- Serum alkaline phosphatase is raised out of proportion as compared to aminotransferases.

ACUTE HEPATITIS

Acute hepatitis can be mainly caused by viral infections, alcohol and drugs. Important causes are given in Table 2.16. Acute viral hepatitis is described below in detail.

ACUTE VIRAL HEPATITIS

Acute viral hepatitis is a systemic viral infection characterized by the predominant involvement of liver. It is caused by one of the specific hepatitis viruses. Rarely, other viruses may also cause hepatitis particularly in immunocompromised host (Table 2.17).

Infection by any of the viruses mentioned in the Table 2.17 results in similar clinical features, which may vary from asymptomatic anicteric presentation to fulminant hepatic failure. The infection due to HBV, HCV, and HDV can lead to chronic hepatitis, cirrhosis and hepatocellular carcinoma.

Pathology

The pathological features are generally the same in all types of hepatitis.

TABLE 2.16: Important causes of acute hepatitis

- Viral hepatitis
- Alcohol
- Drugs (INH, rifampicin, phenytoin, methyldopa)
- Autoimmune hepatitis
- Weil's disease (leptospirosis)
- Toxins (CCl₄, yellow phosphorus, *Amanita phalloides*)
- Wilson's disease
- Hemochromatosis

TABLE 2.17: Causes of viral hepatitis

Specific hepatitis viruses

- Hepatitis A virus (HAV)
- Hepatitis B virus (HBV)
- Hepatitis C virus (HCV)
- Hepatitis D virus (HDV)
- Hepatitis E virus (HEV)

Other viruses

- Cytomegalovirus
- Epstein Barr virus
- Herpes simplex virus
- Yellow fever virus

- Acute hepatitis is characterized by generalized mononuclear cell infiltration, hepatic cell damage (swelling and necrosis), Kupffer cell hyperplasia and variable degree of cholestasis.
- The hepatocytes are swollen and granular. Hepatic cells undergo acidophilic degeneration. These are called Councilman bodies.
- More severe damage leads to the collapse of the reticulin framework bringing the central vein and portal tracts closer. This is known as *bridging or subacute hepatic necrosis*.
- Widespread massive hepatic necrosis occurs in fulminant hepatitis.

Hepatitis A

Hepatitis A virus (HAV) is a RNA virus of the picornavirus family.

- The transmission is by the feco-oral route. An epidemic can occur due to contamination of food and drinking water by HAV. Poor sanitation and overcrowding facilitate its spread.
- Infected persons excrete the virus in their feces about two weeks before the onset and during the first two weeks of illness. Rarely, the HAV can also be transmitted by blood and via the sexual route.
- The incubation period of HAV infection is about 4 weeks.
- Infection and subsequent immunity occurs usually in childhood.
- Mortality is low and fulminant hepatitis is uncommon.
- There is no carrier state and chronic hepatitis does not occur.

Diagnosis: Anti-HAV of IgM type is useful for the diagnosis of acute hepatitis A. Anti-HAV of IgG type appears later and indicates previous infection and immunity.

Hepatitis B

Hepatitis B virus (HBV) is a DNA virus.

- It is transmitted through infected blood and blood products or by sexual contact. Infected mother can transmit infection to the child at delivery (vertical transmission). Tattooing and acupuncture can also spread the disease.
- High-risk groups for HBV infection include staff and patients of hemodialysis center, physicians, surgeons, dentists, paramedical staff and persons working in laboratory and blood bank.
- Patients are infectious during the incubation period and the illness and as long as they are positive for HBsAg.
- Fulminant hepatitis can occur in less than 1% patients. Acute infection can progress to the chronic phase in about 1-10% adults and 90% children who have acquired the infection from the mother at birth. Cirrhosis may develop in 15-20% patients with chronic HBV infection after 5-20 years. Hepatocellular carcinoma may also occur. The risk is higher when infection is acquired in early life.

Diagnosis: HBV has a number of antigens. The three important antigens are hepatitis B surface antigen (HBsAg), core antigen (HBcAg) and hepatitis e antigen (HBeAg). The various HBV antigens and antibodies at various stages of illness are presented in Table 2.18.

- Appearance of Hepatitis B surface antigen (HBsAg) in serum is the first evidence of infection (Fig. 2.10). It normally persists for 3-4 weeks but can persist upto 6 months. After disappearance of HBsAg, antibody against HBsAg (Anti-HBs) appears and persists for years and confers immunity. Presence of Anti-HBs antibody means either previous infection or vaccination.
- The HBcAg is not found in the blood. However, antibody to it (anti-HBc) appears early during the illness. Presence of IgM anti-HBc indicates acute infection and IgG anti-HBc suggests chronic infection (when HBsAg is positive) or recovery (when anti-HBs is positive).
- The presence of HBeAg indicates active viral replication and high degree of infectivity. Anti-HBe appears as HBeAg disappears and its presence suggests low level of viral replication and decreased infectivity.

TABLE 2.18: HBV antigens and antibodies at various stages of infections

	HBsAg	Anti-HBc	Anti-HBs	Others
Acute hepatitis B	+	IgM type	-	HBeAg +
Chronic hepatitis	+	IgG type	-	HBeAg/ anti-HBe +
Recovery	-	IgG type	+	Anti-HBe +/-
Vaccination	-	-	+	-

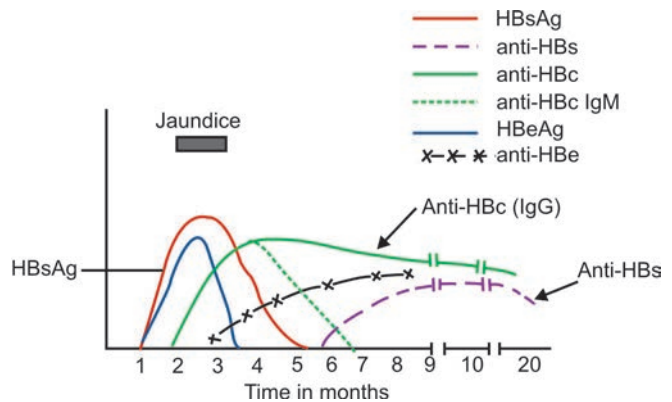


FIGURE 2.10: Serological findings in acute viral hepatitis B

- The presence of HBV DNA in the serum runs parallel with HBeAg. However, it is a more sensitive and precise marker of viral replication and infectivity.

Hepatitis C

Hepatitis C virus (HCV) is a RNA virus and the only member of genus Hepacivirus of the flavivirus family. Six genotypes of HCV have been identified.

- The route of transmission is parenteral (transfusion of blood and blood products, IV drug abuse). Sexual and vertical spread is less common than in hepatitis B infection. In the past it was responsible for 90% of post-transfusion hepatitis.
- Acute HCV infection is usually subclinical. Chronic infection occurs in 70-80% cases. Cirrhosis and hepatocellular carcinoma can also occur.
- Extrahepatic manifestations like vasculitis, arthritis, glomerulonephritis and cryoglobulinemia may occur.

Diagnosis: HCV contains several antigens leading to antibody formation which are used in its diagnosis. The antibody is not protective and does not confer immunity. The most commonly used screening tests detect anti-HCV.

These antibodies generally appear late and thus identify chronic infection. PCR test can detect HCV-RNA in the serum 1-2 weeks after infection and is used for the confirmation of diagnosis and for the monitoring of therapy.

Hepatitis D

Hepatitis D virus (HDV) is a RNA-defective virus .

- It requires HBV for replication and causes hepatitis only in the presence of hepatitis B infection.
- HDV infection can occur simultaneously with HBV (co-infection) or it can cause infection in patients suffering from chronic hepatitis B (superinfection).
- Chronic HBV infection superimposed with HDV can rapidly progress to cirrhosis.

Diagnosis: The diagnosis is made by detecting antibody to hepatitis D antigen (anti-HDV) or by finding HDV-RNA in serum.

Hepatitis E

Hepatitis E virus (HEV), a RNA virus, spreads by feco-oral route. It leads to water borne epidemics of hepatitis.

- The clinical features are similar to HAV infection.
- It does not cause chronic infection.
- Acute fulminant hepatic failure occurs at high frequency if the hepatitis occurs during pregnancy and is associated with high mortality.

Diagnosis: The diagnosis is made by detecting anti-HEV antibodies, IgM type during early phase and IgG type after recovery.

Clinical Features of Acute Hepatitis

The clinical features of acute hepatitis may be described under various stages:

- Prodromal phase:** Development of jaundice is usually preceded by a prodromal phase during which nonspecific systemic symptoms like anorexia, nausea, vomiting, headache, fatigue, malaise, myalgia and arthralgia may occur. There may be distaste for smoking. Low grade fever is common in hepatitis A and E. There may be mild pain in the right upper abdomen.
- Icteric phase:** Many patients with acute hepatitis may never become icteric (Anicteric hepatitis). Prodromal symptoms usually diminish with the onset of clinical jaundice. Patients may notice dark urine and yellowish

discoloration of eyes and skin. Clay-colored stool and pruritus suggest cholestasis.

- Recovery phase:** The icteric phase is followed by an improvement in general symptoms and a diminution of jaundice. Complete clinical and biochemical recovery occurs within 1-2 months in hepatitis A and E and 3-4 months in the majority of patients with hepatitis B. In about 5% cases recovery is delayed.
- Signs:** The sclera is yellow and the skin may show scratch marks due to pruritus. Tender hepatomegaly is present in more than 50% cases. Splenomegaly and lymphadenopathy may occur in 10-20% cases.

Complications

The most feared complication of acute hepatitis is fulminant hepatitis. It is primarily seen in hepatitis B and D, hepatitis E during pregnancy and rarely in hepatitis A. Relapsing jaundice and cholestasis may complicate hepatitis A. A serum sickness like syndrome (arthralgia, rash, angioedema) may occur in hepatitis B. Extrahepatic complications are seen in hepatitis B and C. Important complications are given in Table 2.19.

Investigations

- During the early phase of hepatitis, there is significant (>400 units/L) increase in the plasma ALT and AST.
- This is followed by the rise in serum bilirubin level. However, in anicteric hepatitis, the rise in ALT and AST is not associated with any rise in bilirubin.

TABLE 2.19: Complications of acute hepatitis

<i>Hepatic</i>
• Acute fulminant hepatitis • Relapsing hepatitis • Cholestatic hepatitis • Chronic hepatitis • Cirrhosis • Hepatocellular carcinoma
<i>Extrahepatic</i>
• Aplastic anemia • Henoch-Schönlein purpura • Glomerulonephritis • Papular acrodermatitis • Myelitis and neuropathy • Arthritis • Cryoglobulinemia

- High alkaline phosphatase level suggests cholestasis.
- Serum albumin concentration is normal.
- Prolongation in prothrombin time (PT) is a reliable indicator of severe liver damage and correlates with the prognosis.
- The total leukocyte count is normal or low. There may be relative lymphocytosis.
- Serological tests (as mentioned above) are performed to identify the cause of the hepatitis.

Differential Diagnosis

Viral hepatitis should be differentiated from drug-induced hepatitis and alcoholic hepatitis. Important causes of acute hepatitis are given in Table 2.16.

- History of alcoholism and intake of hepatotoxic drugs should be noted.
- Other viruses (CMV, EBV, Herpes simplex) can lead to acute hepatitis and can be diagnosed by appropriate serological tests.
- The presence of Kayser-Fleischer (KF) ring in the cornea and low serum ceruloplasmin are indicative of Wilson's disease.
- Serum ferritin is very high in hemochromatosis.

Treatment

- a. Physical activity is restricted. However, bed rest is recommended only in severe cases.
- b. A high calorie diet is recommended. A good protein intake should be encouraged. The contents of the diet should be palatable and according to the wish and acceptability of the patient. Hospitalization and intravenous fluid (10% glucose) are indicated if oral intake is not adequate or there is marked nausea and vomiting.
- c. Drugs which are hepatotoxic or those that are metabolized in the liver should be avoided.
- d. Alcohol intake should be avoided.
- e. Bile salt sequestering agent (cholestyramine) reduces pruritus in cases with cholestasis.
- f. Patients with features of severe hepatic failure such as alteration in mental status (hepatic encephalopathy) and prolonged PT/bleeding should be hospitalized.
- g. No specific therapy is recommended for acute viral hepatitis except in acute HCV infection. Subcutaneous

interferon alpha has been shown to reduce the rate of chronicity in acute HCV hepatitis.

- h. Liver transplantation may be required in cases with acute fulminant hepatic failure.

Prevention of Viral Hepatitis

- The prophylaxis for HAV and HBV are available. However, currently no prophylaxis is available against HCV and HEV. HDV is prevented by taking prophylactic measures against HBV.
- Improvement in sanitation and provision for safe drinking water are helpful in preventing waterborne infections like hepatitis A and E. A thorough hand washing is mandatory after bowel movements and before taking meals.
- Careful handling and disposal of used needles, use of safe blood and blood products and universal work precautions help in prevention of hepatitis B, D and C.

Hepatitis A

Active immunization: Hepatitis A vaccine (formalin inactivated vaccine) is given intramuscularly followed by a booster dose at 6-12 months. The dose is 1440 ELU for adults and half the dose for the children. Travellers to endemic area should receive it at least 4 weeks before the date of travel. This is also recommended for patients with chronic hepatitis B or C to prevent hepatitis A infection.

Passive immunization: Immune serum globulin is administered intramuscularly in the dosage of 0.02 ml/kg to the contacts soon after the exposure.

Hepatitis B

Active immunization (pre-exposure prophylaxis):

- Recombinant vaccines containing HBsAg are available.
- The standard regimen in adults is 20 µg IM in the deltoid region at 0, 1 and 6 months. For rapid immunity a schedule of 0, 1, 2 and 12 months is followed. Children need half the dose (10 µg) while immunocompromised patients need 40 µg.
- The vaccine is particularly indicated in those at high risk of getting HBV infection (Table 2.20).
- Many countries have included HBV vaccination in their infant immunization program.

TABLE 2.20: High risk individuals who require hepatitis B vaccination

• Medical, dental and nursing students • Persons on hemodialysis • Patients requiring multiple transfusions • IV drug abusers • Medical, nursing, other health workers and laboratory staff • Newborns of HBsAg positive mothers • Persons with multiple sexual partners
--

Passive immunization (post-exposure prophylaxis):

- This is given to nonvaccinated persons who have exposure to HBV through mucous membrane and breaks in the skin such as accidental needle stick injury.
- This is also indicated for newborn infants of HBsAg positive mothers and in individuals who had sexual contact with persons with HBV infection.
- A dose of 0.06 ml/kg hepatitis B immunoglobulin (HBIG) is given intramuscularly as soon as possible.
- Active immunization with vaccine is also initiated simultaneously if the person is unvaccinated. These are injected at different sites.

Prognosis

- Clinical recovery is generally complete in 3-6 weeks. However, laboratory recovery may be delayed. Overall mortality is less than 1%.
- Hepatitis A patients recover completely without progressing to chronic liver disease. Some patients may have relapses up to one year.
- The mortality rate in hepatitis B is low but rises higher if superimposed with hepatitis D infection.
- Acute hepatitis is less severe and more likely to be anicteric in hepatitis C.
- Hepatitis B and C can progress to chronic phase in 1-2% and 80% cases respectively. However, 90% infants and neonates with hepatitis B infection develop chronicity.
- Mortality in hepatitis E is particularly high (10-20%), if it occurs during pregnancy.

CHRONIC HEPATITIS

Chronic hepatitis is defined as persistent inflammation and necrosis of liver for at least 6 months. Progression is slow in mild forms but severe forms progress rapidly and lead

TABLE 2.21: Causes of chronic hepatitis

- | |
|--|
| <ul style="list-style-type: none"> • Viral hepatitis <ul style="list-style-type: none"> Hepatitis B Hepatitis C Hepatitis D • Autoimmune hepatitis • Drug induced (INH) • Wilson's disease • Alpha-1 antitrypsin deficiency |
|--|

to cirrhosis. The causes of chronic hepatitis are given in Table 2.21.

Chronic hepatitis is classified on the basis of its cause, its histological activity and its degree of progression.

Chronic Viral Hepatitis

Hepatitis B and C can progress to chronic phase in 1-2% and 80% cases respectively.

Clinical Features

The clinical features of chronic viral hepatitis are fatigue and persistent or intermittent jaundice. Patients may present with features of cirrhosis and its complications. Extrahepatic presentations like arthralgia and arthritis, immune complex glomerulonephritis, polyarteritis nodosa (hepatitis B) and essential cryoglobulinemia (hepatitis C) may occur.

Laboratory Features

High and fluctuating levels of serum aminotransferases are characteristic features. Serum bilirubin is only mildly raised whereas alkaline phosphatase is usually normal. Serum albumin is low. Prothrombin time is increased in severe cases.

Biopsy of the liver is required for assessment of histological activity to grade the chronic hepatitis as mild, moderate or severe.

Serological tests are needed to identify the viral types and their replication patterns.

Treatment

- Chronic hepatitis B:** The treatment of chronic hepatitis B is needed if there are markers of viral replication and ALT is elevated to levels of twice above normal. Different drugs used are interferon alpha, lamivudine or adefovir dipivoxil. Interferon alpha is given subcutaneously in dosage of 5 million units daily or 10

TABLE 2.22: Treatment of chronic hepatitis**Chronic hepatitis B***Interferon alpha*

5 million IU s/c daily × 16 weeks

10 million IU s/c thrice a week × 16 weeks

Pegylated interferon alpha (weekly)

Oral drugs

Lamivudine 100 mg OD

Adefovir dipivoxil 10 mg OD

Entecavir 0.5-1 mg OD

Telbivudine

Tenofovir

Chronic hepatitis C

PEG interferon alpha 2a 180 mcg weekly

or

PEG interferon alpha 2b 1.5mcg/kg weekly

Plus

Oral ribavirin 800-1200 mg/day

Autoimmune hepatitis

Prednisolone

Azathioprim

Drug induced chronic hepatitis

Withdrawal of causative drug

million units three times a week for 4 months. Lamivudine is used in daily oral dose of 100 mg. This is better tolerated. Adefovir is safe and effective and given orally in the dosage of 10 mg daily. (Table 2.22)

- b. *Chronic hepatitis C:* A combination of subcutaneous weekly pegylated interferon alpha (PEG-IFN) and daily oral ribavirin is the treatment of choice in chronic hepatitis C. The treatment is given for 6-12 months. The response to treatment varies according to the genotype of the HCV. The best response (up to 80%) is observed in genotype 2 and 3. (Table 2.22)
- c. Liver transplantation is indicated in advanced liver disease.

Prognosis

The course of the chronic hepatitis is variable. The sequelae of chronic hepatitis are cirrhosis, liver failure and hepatocellular carcinoma. Upto 40% patients with chronic hepatitis B and 30% with chronic hepatitis C will eventually develop cirrhosis in 5-30 years. The risk of development of hepatocellular carcinoma is 3-5% per year in patients with cirrhosis.

CIRRHOSIS OF LIVER

Cirrhosis of liver is the end result of the hepatocellular injury characterized by the presence of extensive fibrosis, regenerative nodules and loss of liver architecture.

Two of the commonest causes of cirrhosis are viral hepatitis (B, D and C) and prolonged excessive use of alcohol. Important causes of cirrhosis are given in Table 2.23.

Pathology

- The activation of stellate cells (fat storing cells, Ito cells) is the central event in the development of cirrhosis irrespective of the cause. The activated stellate cells transform into multifunctional cells upon interaction with hepatocytes, Kupffer cells and cytokines. The transformed cells form type I collagen leading to fibrosis.
- The cirrhosis can be micronodular typically in alcoholics where the regenerating nodules are small (<1 mm). The macronodular form is characterized by larger nodules and is seen in posthepatic or postnecrotic cirrhosis. However, either form can occur at different stages of disease.
- Clinical features of the cirrhosis are derived from the changes in the liver morphology as well as from liver cell dysfunction (Table 2.24).
 - Fibrosis and distorted vasculature may lead to portal hypertension and complications associated with it.
 - Hepatocellular dysfunction leads to jaundice, edema, coagulopathy and metabolic abnormalities.
 - Ascites and hepatic encephalopathy can result from both mechanisms.

Clinical Features

Cirrhotic patients may be asymptomatic and may be diagnosed incidentally.

- Mostly symptoms such as weakness, fatigue, weight loss, anorexia, nausea, vomiting and abdominal discomfort occur insidiously.

TABLE 2.23: Important causes of cirrhosis

- Viral hepatitis (postnecrotic)
- Alcohol (Laennec's cirrhosis)
- Autoimmune hepatitis
- Drug induced
- Biliary cirrhosis
- Hemochromatosis
- Wilson's disease
- Cardiac cirrhosis (prolonged CHF)
- Alpha-1 antitrypsin deficiency
- Cryptogenic (unknown etiology)

TABLE 2.24: Clinical features of cirrhosis

Features due to hepatocellular dysfunction

- Jaundice
- Ascites
- Hepatomegaly (initial phase)
- Spider nevi, palmar erythema
- Gynecomastia, testicular atrophy
- Menstrual abnormalities, breast atrophy
- Bleeding tendency
- Hepatic encephalopathy

Features due to portal hypertension

- Ascites
- Splenomegaly
- Variceal bleeding
- Hepatic encephalopathy

Others

- Parotid and lacrimal gland enlargement
- Clubbing
- Opaque nails (leukonychia)
- Dupuytren's contracture
- Skin pigmentation

- The liver is firm, nontender and nodular and is enlarged initially. The liver size reduces due to fibrosis as the disease progresses.
- Jaundice is generally absent or mild initially. It may become severe at later stages.
- Patient may present with features of *portal hypertension* mainly abdominal distension due to ascites and splenomegaly, hematemesis and melena due to variceal rupture or hepatic encephalopathy (Table 2.24).
- The signs of *chronic hepatic dysfunction* such as spider nevi, palmar erythema, gynecomastia, testicular atrophy and loss of hair may occur due to disturbances in hormonal metabolism. Females may have loss of libido, menstrual abnormalities and breast atrophy. Palmar erythema is redness of the thenar and hypothenar eminences which may also be found in normal persons and in hyperdynamic circulation. Spider nevi are dilated central arterioles with radiating small vessels (like spider) found mainly on the upper part of the body. They can also be seen in normal persons and also during pregnancy.
- Patients may have hemorrhagic manifestations such as epistaxis and increased menstrual flow.
- Other features include parotid gland and lacrimal gland enlargement, digital clubbing, Dupuytren's contracture and skin pigmentation. Complication of the cirrhosis is given in Table 2.25.

TABLE 2.25: Complications of cirrhosis

- Portal hypertension
 - Upper GI bleed
 - Ascites
 - Spontaneous bacterial peritonitis (SBP)
- Hepatic encephalopathy
- Hepatorenal syndrome
- Hepatocellular carcinoma
- Coagulopathy

Laboratory Featuresa. *Blood examination:*

- Aminotransferases (ALT, AST) are frequently elevated whereas a rise in the serum bilirubin and ALP may occur later.
- Serum albumin is low.
- PT is frequently prolonged.
- Anemia can occur due to bleeding, folate deficiency, marrow suppression or hypersplenism.
- Leukopenia and thrombocytopenia suggest hypersplenism.

b. *Imaging:* Ultrasonography is helpful in the evaluation of liver size and texture, ascites, portal hypertension and splenomegaly.c. *Endoscopy:* Upper gastrointestinal endoscopy is required to detect esophageal varices and to exclude other causes of upper gastrointestinal bleeding in the stomach and duodenum.d. *Liver biopsy:* Biopsy helps in the assessment of severity of the cirrhotic changes. Typical histological features may suggest the specific cause of the cirrhosis.**Management**

Management includes general management, treatment of specific cause, management of the complications and liver transplantation.

a. *General management:*

- The diet should contain an adequate amount of protein and calories. However, protein intake should be reduced in case of hepatic encephalopathy.
- Vitamin supplementation is helpful.
- Salt restriction is required in case of ascites.
- Medications which are hepatotoxic or metabolized in liver should be given with caution.

b. *Treatment of specific cause:* Alcohol abstinence is mandatory in alcoholic cirrhosis. Specific therapy is needed in hemochromatosis and Wilson's disease.

- c. *Management of complications:* (discussed later)
- d. *Liver transplantation:* The most common indication for liver transplantation is irreversible progressive chronic liver failure due to cirrhosis.

Prognosis

Overall the prognosis of cirrhosis is poor. Child Pugh's classification is used for the grading and the prognosis. The prognosis is favorable if the cause can be corrected. Around 25% of the patients survive for more than 5 years.

PORTAL HYPERTENSION

Portal hypertension is elevation of portal venous pressure above 10-12 mm Hg (normal 5-10 mm Hg). Portal hypertension results from (a) increased resistance to portal blood flow and (b) high portal blood flow. More commonly, portal hypertension occurs due to increased resistance.

The causes of portal hypertension are classified according to the site of obstruction in relation to hepatic sinusoids— (a) presinusoidal, (b) sinusoidal and (c) post-sinusoidal. Causes of portal hypertension are given in Table 2.26. Cirrhosis is the most common cause of portal hypertension. Portal vein obstruction is the next most common cause.

Clinical Features

The clinical features result from portal-systemic collateral formation and portal venous congestion. As there are no valves in the portal vein, retrograde blood flow from portal venous system (high pressure) to systemic venous circulation (low pressure) occurs during portal hypertension.

TABLE 2.26: Common causes of portal hypertension

Presinusoidal

Extrahepatic

- Portal vein obstruction (thrombosis, trauma, malignant diseases of pancreas)

Intrahepatic

- Schistosomiasis
- Sarcoidosis, vinyl chloride

Sinusoidal

Cirrhosis

Metastatic liver disease

Postsinusoidal

Intrahepatic

- Veno-occlusive disease

Extrahepatic

- Budd-Chiari syndrome (hepatic vein thrombosis)

The main clinical features are:

- a. hemorrhage from rupture of varices (hematemesis)
- b. ascites
- c. splenomegaly
- d. hypersplenism
- e. hepatic encephalopathy

Splenomegaly is the cardinal finding of portal hypertension.

Hypersplenism may give rise to thrombocytopenia and leukopenia. Anemia is rare.

Collateral formation occurs around the esophagus and stomach (esophageal varices and gastric varices), the rectum (hemorrhoids), retroperitoneal space and periumbilically in anterior abdominal wall. Collateral vessels in the anterior abdominal wall may be visible as tortuous vessels radiating from the umbilicus (caput medusae).

As a result of portal-systemic shunting, various toxic substances enter into systemic circulation without going to the liver. This is the basis for hepatic encephalopathy. *Musty odor of the breath (feter hepaticus)* due to the presence of mercaptans can occur in hepatic encephalopathy.

Investigations

- a. *Endoscopy:* Esophageal varices suggest the presence of portal hypertension. However, it does not detect the cause.
- b. *Ultrasonography:* It is useful in the evaluation of size and texture of the liver, splenomegaly, diameter and flow pattern of the portal vein, ascites and collateral circulation.
- c. *CT scan and MRI:* These are usually not needed. However, these are sensitive methods to detect collaterals.
- d. *Portal venous pressure:* Measurement of portal venous pressure is needed to confirm portal hypertension and to determine the level of obstruction. Portal venography helps in knowing the site and cause of the obstruction.

Management

The management includes—(a) reduction of pressure in the portal vein and (b) treatment of the complications.

- a. *Reduction in the portal venous pressure:* Propanolol and nadolol can be used to lower the portal venous pressure. Transjugular intrahepatic portosystemic shunt (TIPSS)

and surgical decompression procedures are other options to decrease portal venous pressure.

- b. *Treatment of complications:* Complications like upper GI bleed, ascites, spontaneous bacterial peritonitis (SBP), hepatic encephalopathy and hypersplenism are managed as given below.

UPPER GASTROINTESTINAL BLEEDING

Important causes of upper GI bleeding are given in Table 2.27. Peptic ulcer is the most common cause of upper GI bleeding. However, the most common cause of upper GI bleeding in the cirrhotic patient is rupture of esophageal varices. Bleeding in cirrhotic patients may also occur due to congestive gastropathy or rupture of gastric varices. The bleeding may be contributed by the presence of coagulopathy, thrombocytopenia, gastric erosions and drugs.

Variceal bleeding presents as painless hematemesis and melena. There may be circulatory shock if the bleeding is massive. GI bleeding can precipitate hepatic encephalopathy.

Upper GI endoscopy is the best method to detect the cause of bleeding. Endoscopy may also be used for therapeutic interventions.

Treatment of Variceal Hemorrhage

1. Urgent hospitalization in ICU
2. Restore the intravascular volume by the blood transfusion.
3. Fresh frozen plasma is given to correct coagulopathy.
4. Reduction of portal venous pressure is achieved by:
 - a. Vasopressin infusion (0.2-0.4 unit/min).
 - b. Somatostatin or its analogue octreotide (50-100 µg/hr).
 - c. TIPSS if bleeding does not respond to drugs and therapeutic endoscopic intervention.
 - d. Emergency portosystemic shunt surgery is rarely performed now.

TABLE 2.27: Causes of upper GI bleeding

- Peptic ulcer disease
- Variceal rupture
- Erosive gastritis (drugs, alcohol, severe illness/stress)
- Gastric malignancy
- Mallory-Weiss tear (laceration at gastroesophageal junction due to vomiting)

5. Local measures:

- a. *Endoscopic intervention:* It is most widely used as the first line of treatment. Bleeding is stopped in the majority of cases. Ligation of varices with bands and injection of sclerosing agent (sclerotherapy) are usual procedures.
- b. *Balloon tamponade* (with Sengstaken-Blakemore tube) is used when bleeding is massive and endoscopy is not available.
- c. *Esophageal transection* may rarely be needed as a last resort.

Prevention of Variceal Bleeding

1. Prevention of first hemorrhage (primary prevention) from varices is accomplished either (a) by the use of non-selective beta-blocker (propranolol) or (b) by prophylactic endoscopic banding.
2. *Prevention of recurrent bleeding:*
 - a. Propranolol and band ligation are useful in reducing recurrent variceal bleeding.
 - b. TIPSS or portal-systemic shunt surgery may be required in patients not responding to the above measures, however, the risk for hepatic encephalopathy is increased.

HEPATIC ENCEPHALOPATHY

Hepatic encephalopathy is a neuropsychiatric syndrome secondary to acute or chronic hepatic failure or due to portal-systemic shunt.

The toxic substances absorbed from the intestine are either not metabolized by the failing liver or bypass the liver through portal-systemic shunts. Thus, these harmful substances (mainly ammonia, mercaptans, short chain fatty acids and false neurotransmitters) reach the central nervous system and result in alteration in consciousness and other manifestations. The alteration in sensorium may range from mild confusion and disordered sleep patterns to deep coma (*hepatic coma*). Flapping tremor (*asterixis*) and *fetor hepaticus* are usually present.

Precipitating Factors

Certain factors may precipitate the onset of encephalopathy in otherwise stable cases of cirrhosis. These factors enhance

TABLE 2.28: Precipitating factors for hepatic encephalopathy

- Gastrointestinal bleeding
- Increased protein in the diet
- Constipation
- Uremia
- Hypokalemia and alkalosis
- Vigorous paracentesis
- Infections
- Drugs (diuretics, sedatives, hypnotics)
- Portal-systemic shunt (TIPSS or surgery)

the production of ammonia or other toxic nitrogenous substances or affect the CNS directly. The most common precipitating factor is upper GI bleeding. Other important factors are given in the Table 2.28.

Treatment

The treatment of hepatic encephalopathy includes following:

1. The precipitating factors should be identified and removed.
2. Protein should be withheld from the diet in severe cases or restricted to less than 20 g/day. However, the amount of protein in the diet is increased gradually once the patient improves. Vegetable proteins are preferred over animal proteins. Constipation should be avoided.
3. The nonabsorbable disaccharides (lactulose) act as osmotic laxative, favor conversion of ammonia to poorly absorbed ammonium ion and inhibit ammonia production by colonic bacteria. The usual dose is 15-30 ml thrice a day and it is adjusted so as to produce 2-4 soft stools daily.
4. Oral neomycin (0.5-1 g 6 hourly) can be used to reduce gut bacteria and ammonia production. Metronidazole (250 mg 8 hourly) can alternatively be used. Recently, nonabsorbable agent *rifaximin* (400 mg orally three times daily) has been found to be effective and safer than neomycin and metronidazole.
5. Sedatives and hypnotics should be avoided. However, short acting oxazepam may be given to agitated patients. Flumazenil, a benzodiazepine antagonist may be helpful in some cases.
6. Zinc supplementation may sometimes help.
7. Liver transplantation may be considered in nonresponding patients.

Management of Hemorrhagic Tendency in Liver Disease

- The administration of vitamin K (10 mg for 3 days subcutaneously) may be helpful in correction of coagulopathy in cases with vitamin K malabsorption.
- Fresh frozen plasma which contains all clotting factors, is indicated in active severe bleeding or prior to surgery.
- Platelet transfusion may be needed in thrombocytopenia with bleeding.

Hepatocellular Carcinoma

Patients with cirrhosis are monitored periodically for the development of hepatocellular carcinoma. This is done by the measurement of serum alpha fetoprotein level and imaging techniques.

Hepatorenal Syndrome

Renal failure occurring in the setting of cirrhosis and ascites without any evidence of specific cause of renal dysfunction is known as hepatorenal syndrome. Kidneys are intrinsically normal. The hepatorenal syndrome is believed to be due to altered systemic and renal blood flow. It has a very poor prognosis. Treatment is usually unsuccessful. Liver transplantation may be helpful in appropriate candidates.

ASCITES

Accumulation of excess fluid in the peritoneal cavity is called ascites. Normally there may be a small amount of fluid (<20 ml) in the peritoneal cavity. Ascites can be associated with normal or diseased peritoneum. The most common cause of ascites is cirrhosis with portal hypertension. Tuberculosis and malignancies are the other important causes. Table 2.29 shows causes of ascites.

Pathogenesis

Increased hydrostatic pressure in portal venous system, dilatation of splanchnic arterial system, hypoalbuminemia and reduced plasma oncotic pressure lead to loss of fluid from vascular compartment into peritoneal cavity. These changes cause reduction in the effective circulatory volume leading to increased activation of renin-angiotensin-aldosterone system, enhanced sympathetic activity, and

TABLE 2.29: Causes of ascites

<i>Normal peritoneum</i>
• Cirrhosis with portal hypertension
• Congestive heart failure (CHF)
• Hypoproteinemia (nephrotic syndrome, malnutrition)
• Budd-Chiari syndrome
• Pancreatitis
• Biliary ascites
• Chylous ascites (filariasis, trauma, tumor)
• Meig's syndrome
<i>Diseased peritoneum</i>
• Tuberculous peritonitis
• Bacterial peritonitis
• Malignancy (peritoneal, hepatic)

release of the antidiuretic hormone causing salt and water retention. This further contributes to the development of ascites.

Clinical Features

The main symptoms are distension of abdomen with or without discomfort or pain. Other symptoms may occur according to the cause of ascites. There may be fever in infective pathology while marked weight loss suggests malignancy. Patients should be questioned to ascertain if any risk factors of chronic liver disease such as viral hepatitis, alcohol use, transfusions and IV drug abuse exist.

Inspection: Inspection of abdomen may reveal distension with fullness in the flanks, everted and horizontal umbilicus, abdominal striae, prominent and tortuous veins and hernia.

Palpation: Fluid thrill is present in massive ascites (see Fig. 2.6).

Percussion: Shifting dullness is demonstrable on percussion in moderate ascites.

Other findings will be present depending on the underlying cause of ascites. For example, spider nevi, palmer erythema and gynecomastia suggest chronic liver disease. Splenomegaly is present in portal hypertension.

Investigations

a. **Ultrasonography:** Ultrasonography of abdomen can detect presence of minimal amount of fluid and is helpful when clinical signs are not present. It is also used for guiding paracentesis (drainage of fluid). It can also detect features of cirrhosis and portal hypertension and other findings like lymph nodes, masses, metastasis in liver.

CT scan can also be done for this purpose. Doppler ultrasound can detect obstruction in portal vein thrombosis and hepatic vein thrombosis.

b. **Paracentesis:** Abdominal paracentesis is performed as a routine investigation to determine the cause. It can be performed under ultrasound guidance. 50-100 ml of fluid is drained and examined for the color, total and differential cell count and malignant cells. Biochemical tests include protein and albumin levels, glucose, LDH and amylase. Culture, Gram stain and AFB staining may also be performed.

- In cirrhosis, the appearance of ascitic fluid is clear, straw colored. The fluid is milky white in chylous ascites while it is cloudy in infections. Hemorrhagic fluid may be seen due to trauma, tumor or tuberculosis.
 - The presence of more than 500 leukocytes/ μ L suggests inflammatory conditions. Polymorphonuclear count more than 250/ μ L is characteristic of bacterial peritonitis. Predominantly elevated lymphocytes suggest tuberculosis.
 - Based on the specific gravity and total protein concentration, ascitic fluid has traditionally been classified as transudative and exudative.
 - a. Transudative fluid (specific gravity <1.016 , protein <25 g/L) is typically found in cirrhosis and CHF.
 - b. Exudative fluid (specific gravity >1.016 , total protein >25 g/L) is seen in conditions with diseased peritoneum like tuberculosis, malignancy and bacterial peritonitis.
 - Serum-ascites albumin gradient (SAAG) is the best indicator to classify ascites into portal hypertensive and non-portal hypertensive causes. SAAG is the difference between the serum and ascitic fluid albumin levels. Transudative ascites is generally high gradient type and exudative ascites the low gradient type.
 - a. High gradient (>1.1 g/dL) suggests that ascites is due to portal hypertension (cirrhosis, CHF).
 - b. Low gradient (<1.1 g/dL) is seen in cases without portal hypertension.
- Laparoscopy and peritoneal biopsy may be needed in specific cases.

Management

The treatment of ascites in cirrhosis consists of the following steps:

- Salt in the diet is restricted to 2-3 g/day.
- Diuretics are given if there is no response to salt restriction. Spironolactone is initially used in a dosage of 100 mg/day which is increased to 400 mg/day. Frusemide (40-160 mg/day) is added in case the response is inadequate. The goal is to reduce the weight at a rate of not more than 1 kg/day in patients with anasarca and no more than 0.5 kg/day in patients with ascites alone. Vigorous diuresis may precipitate hepatic encephalopathy and renal failure.
- Therapeutic paracentesis is needed in massive ascites with respiratory compromise. A volume of 4-6 liters is removed. Intravenous albumin is given concomitantly to maintain intravascular volume.
- Portacaval shunt surgery or implantation of portocaval shunt (between abdominal cavity to superior vena cava) may be indicated in refractory ascites.
- TIPSS is an alternative to surgical shunting in cases with refractory ascites.

Treatment of underlying cause: Specific treatment is given in patients according to the cause. For example, antitubercular therapy is indicated in tuberculous ascites.

Spontaneous Bacterial Peritonitis

Spontaneous bacterial peritonitis (SBP) or primary bacterial peritonitis (PBP) is a common complication in patients with ascites due to cirrhosis. There is no apparent intra-abdominal source of infection. SBP must be differentiated from secondary bacterial peritonitis characterized by intra-abdominal infections due to appendicitis, perforated peptic ulcer or ruptured gallbladder.

- Etiology:** The most common organisms responsible for SBP are *E. coli*, *Klebsiella* and *Streptococcus pneumoniae*.
- Clinical features:** Clinical features include fever, abdominal pain, worsening or precipitation of hepatic encephalopathy or worsening of renal dysfunction. Abdominal tenderness and decreased bowel sounds are seen in many cases. Some patients may not have any suggestive symptoms or signs.
- Investigations:** Ascitic fluid is cloudy with high leukocyte counts. Polymorphonuclear (neutrophils) cell count of more than 250/ μ L strongly suggests SBP. Gram

stain may be positive. Culture of ascitic fluid confirms the diagnosis.

- Treatment:** Treatment of SBP includes commencing of antibiotic therapy. Third generation cephalosporins (cefotaxime 2 g IV 8 hourly or ceftriaxone 1-2 g IV daily) or quinolones (ciprofloxacin 500 mg IV 12 hourly) are given for 5 days.
- Prevention:** Recurrence of SBP is prevented by using norfloxacin (400 mg daily) or ciprofloxacin (750 mg weekly). Alternatively, a double strength tablet of trimethoprim-sulfamethoxazole may be given daily.

HEPATOMEGALY

Liver is examined when the patient is supine with relaxed abdominal wall. The fingers are placed in the right abdomen and slowly advanced upward as the patient inhales and exhales deeply.

The liver is normally not palpable; however, it may be palpable in thin individuals. It is palpable in the right upper abdomen when enlarged. In emphysema, liver may be palpable without being enlarged as it is pushed downward by the inflated lung. Hence, upper border of the liver should also be ascertained by percussion in order to measure the span of liver, which is normally 12-15 cm in height. Normally the upper liver dullness is at the level of fifth rib in the right mid-clavicular line. Important causes of hepatomegaly are given in Table 2.30.

TABLE 2.30: Causes of hepatomegaly

Vascular causes
• CHF
• Hepatic vein thrombosis
Inflammatory causes
• Hepatitis (viral, drugs, alcohol)
• Cirrhosis (early stage)
• Liver abscess (amoebic, pyemic)
Neoplasm
• Hepatoma
• Metastatic
• Lymphoma, leukemia
Granulomatous causes
• Tuberculosis
• Sarcoidosis
Infiltrative causes
• Fatty liver (alcohol, diabetes, toxin)
• Non-alcoholic steatohepatitis (NASH)
• Amyloidosis
• Hemochromatosis
• Storage disorders
Biliary tract obstruction

- The size of the liver is usually expressed in centimeters below the right costal margin.
- The consistency of the liver should be noted as soft, firm or hard.
- The surface may be smooth or nodular and edges may be sharp, regular or irregular.
- Liver may be nontender or tender to touch.
- Tender hepatomegaly may occur in CHF, acute hepatitis and liver abscess.
- The liver is soft, smooth and tender in CHF, and firm, regular in cirrhosis.
- Hard, irregular, nontender and nodular liver is detected in metastatic disease.

IMPLICATIONS ON DENTAL PRACTICE

1. Patients with gastric acid reflux may develop foul taste (dysgeusia), increased dental sensitivity, dental erosion and pulpitis. Such patients should be treated in semisupine position. Proton pump inhibitors and antacids are administered before the procedure.
2. Oral manifestations in peptic ulcer disease are rare. However, regurgitation of gastric acid may lead to dental erosion, particularly at the palatal aspect of maxillary teeth.
3. Use of aspirin and NSAIDs should be avoided in patients with history of peptic ulcer disease. Paracetamol can be used safely. Antacids may interfere with the absorption of antibiotics. Hence, patients are advised to take antibiotics 2 hours before or after the ingestion of antacids.
4. Patients with cirrhosis should be investigated for any hemostatic defects (coagulopathy, thrombocytopenia) before dental procedures. Tests recommended are complete blood count, PT, APTT, bleeding time and platelet count. Fresh frozen plasma and platelet infusion may be needed in patients with marked hemostatic defects.
5. Patients with obstructive jaundice may have bleeding tendency due to vitamin K deficiency. Such patients are given injectable vitamin K in order to correct bleeding tendency.
6. Neonate with obstructive jaundice may develop green teeth.
7. Sedative and tranquillizers should be avoided in patients

with cirrhosis as these agents may precipitate encephalopathy.

8. Most drugs are metabolized in the liver. Hence, caution should be taken in prescribing medications in patients with liver dysfunction.
9. Universal precautions must be taken while doing any procedure or intervention to prevent transmission of infection.
10. In case of exposure to hepatitis B positive blood (such as accidental needle prick or by sharp instruments), unvaccinated individuals should take immunoglobulin and vaccine as soon as possible after the exposure. Full course of vaccine should be completed.
11. All dentists should get vaccination against hepatitis B.
12. Invasive dental or oral surgical procedures may possibly increase the risk of SBP in patients with cirrhosis and ascites. Antibiotic prophylaxis with amoxicillin plus metronidazole may be recommended.

SELF ASSESSMENT

Multiple Choice Questions

1. Following are features of hemolytic jaundice except:

- A. Recurrent jaundice
- B. Mild jaundice
- C. Conjugated hyperbilirubinemia
- D. Increased urobilinogen in urine

2. Most of the bilirubin in normal persons is:

- A. Unconjugated
- B. Conjugated
- C. Equal amounts of conjugated and unconjugated
- D. Variable

3. Pruritus (itching) is a symptom of:

- A. Prehepatic jaundice
- B. Obstructive jaundice
- C. Both
- D. None

4. Causes of hepatocellular jaundice include all except:

- A. Viral hepatitis
- B. Drugs
- C. Alcohol
- D. Stone in common bile duct

5. Marked elevation of alkaline phosphatase is characteristic of:

- A. Hemolytic jaundice
- B. Hepatocellular jaundice
- C. Obstructive jaundice
- D. All of the above

6. In obstructive jaundice, urinary examination shows:

- A. No urobilinogen, no bilirubin
- B. Increased urobilinogen, increased bilirubin

- C. Increased urobilinogen, no bilirubin
D. No urobilinogen, increased bilirubin
7. In obstructive jaundice all are present except:
A. Pale stools
B. Pruritus
C. Vit K deficiency
D. Increased urobilinogen in urine
8. Which of the following is transmitted by feco-oral route:
A. Hepatitis B B. Hepatitis C
C. Hepatitis D D. Hepatitis E
9. Which of the following is not true for hepatitis A:
A. Feco-oral transmission
B. Progression to chronic stage
C. Vaccine against hepatitis A is available
D. Ig are used for passive immunization
10. Which of the following is not a feature of acute liver disease:
A. Increased ALT
B. Prolonged PT
C. Marked elevation in bilirubin
D. Low serum albumin
11. Vaccine is available for:
A. Hepatitis C B. Hepatitis E
C. Both D. None
12. Cirrhosis is a complication of all except:
A. Hepatitis B B. Hepatitis C
C. Hepatitis E D. Wilson's disease
13. Budd-Chiari syndrome results from the obstruction of:
A. Portal vein B. Biliary tract
C. Hepatic veins D. Hepatic artery
14. All the following contribute to hepatic encephalopathy except:
A. GI bleeding
B. Hypokalemia
C. Spontaneous bacterial peritonitis
D. Protein restriction in diet
15. Laennec's cirrhosis is associated with:
A. Hepatitis B B. Drugs
C. Chronic alcoholism D. Wilson's disease
16. The most common extraintestinal complication of amoebiasis is:
A. Lung abscess
B. Liver abscess
C. Brain abscess
D. Subdiaphragmatic abscess
17. Complications of peptic ulcer include all except:
A. Perforation
B. Hemorrhage
C. Gastric outlet obstruction
D. Gallstones
18. All the following organisms can cause bloody diarrhea except:
A. *Yersinia enterocolitica*
B. *Shigella*
C. *Giardia lamblia*
D. *Entamoeba histolytica*
19. Hepatitis B and C share all the following clinical features except:
A. Transmission through IV drug abuse
B. Transmission through sexual contact
C. Low mortality associated with acute disease
D. Development of chronic hepatitis in 70-80% patients
20. Patient vaccinated with hepatitis B is positive for:
A. Hepatitis B surface antigen (HBsAg)
B. Hepatitis B surface antibody (anti-HBs)
C. Both HBsAg and anti-HBs
D. Hepatitis B core antibody (anti-HBc) and anti-HBs
21. Kayser-Fleisher (KF) ring is seen in:
A. Wilson's disease
B. Hemochromatosis
C. Lipid storage disorders
D. Glycogen storage diseases
22. All the following are common causes of hematemesis except:
A. Esophageal varices B. Peptic ulcer
C. Pernicious anemia D. Mallory-Weiss tears
23. Stress ulcer may occur in:
A. Head injury B. Trauma
C. Burns D. All the above
24. Hunger pain is found in:
A. Duodenal ulcer B. Gastric ulcer
C. Gastric carcinoma D. Esophageal diseases
25. Jaundice is clinically detectable when serum bilirubin concentration exceeds:
A. 0.5-1 mg % B. 2-3 mg %
C. 5-8 mg % D. 9-12 mg %
26. Most common cause of hematemesis is:
A. Variceal bleed B. Peptic ulcer
C. Carcinoma stomach D. Gastric erosions
27. Tender hepatomegaly is seen in:
A. Acute viral hepatitis B. CHF
C. Amoebic liver abscess D. All
28. Portal hypertension can occur due to:
A. Portal vein thrombosis
B. Hepatic vein thrombosis
C. Cirrhosis
D. All
29. Gastroesophageal varices is complication of:
A. Gastric carcinoma
B. Esophageal carcinoma

- C. Portal hypertension
D. Varicose vein
- 30. Peptic ulcer is etiologically related to:**
A. *H. pylori* infection
B. Opioid analgesics
C. Increased consumption of tea, coffee and spicy food
D. All the above
- 31. Following is used to prevent NSAIDs induced peptic ulcer:**
A. Sucralfate B. PPI
C. H₂ blockers D. Steroid
- 32. *H. pylori* infection can cause:**
A. Gastritis
B. Peptic ulcer
C. Adenocarcinoma of stomach
D. All
- 33. Erosive and hemorrhagic gastritis can occur in:**
A. Alcohol
B. Patients with head injury
C. Patients with severe burns
D. All the above
- 34. Immediate treatment of percutaneous exposure (needle prick) in unvaccinated persons to hepatitis B positive blood is:**
A. Interferon and vaccine
B. Vaccine and lamivudine
C. Immunoglobulin and vaccine
D. Immunoglobulin and interferon
- 35. Following is not used in treatment of chronic hepatitis B:**
A. Interferon B. Lamivudine
C. Adefovir D. Ribavirin
- 36. Prevention of hepatitis C includes:**
A. Administration of vaccine
B. Administration of immunoglobulin
C. Both
D. None
- 37. Best indicator of acute hepatocellular dysfunction is:**
A. Increased ALT B. Increased ALP
C. Prolonged PT D. Decreased albumin
- 38. Black tarry stools usually suggest:**
A. Upper GI bleeding B. Lower GI bleeding
C. Amoebic dysentery D. Bacterial dysentery
- 39. Schilling test is used for the diagnosis of:**
A. Folate deficiency
B. Cobalamin deficiency
C. Protein malabsorption
D. Fat malabsorption
- 40. Vitamin B₁₂ absorption occurs in:**
A. Duodenum and jejunum
B. Large intestine
C. Stomach
D. Ileum
- 41. All the following can cause ascites with SAAG < 1.1 except:**
A. Tuberculosis B. Malignancy
C. Portal hypertension D. Bacterial peritonitis
- 42. Treatment of choice in bleeding due to hepatocellular dysfunction is:**
A. Whole blood transfusion
B. Fresh frozen plasma
C. Vitamin K administration
D. Packed red blood cell transfusion
- 43. Which of the following is used for prevention of recurrent variceal bleeding in patients with cirrhosis and portal hypertension:**
A. Amlodipine B. Atenolol
C. Propranolol D. Vitamin K
- 44. Treatment of choice in patients with acute watery diarrhea is:**
A. Antimotility agents B. Antibiotics
C. Rehydration therapy D. Antispasmodic agents

Fill in the Blanks

- Normal portal vein pressure is _____ mm Hg.
- Porto-systemic collaterals around umbilicus may be visible as _____.
- Normal bilirubin level is _____.
- Conjugation of bilirubin occurs in _____.
- Vitamin K dependent coagulation factors are _____.
- AST : ALT ratio of more than two is suggestive of _____.
- Coffee ground emesis or vomiting of dark, altered blood with clots is known as _____.
- Large, bulky stools which float in water are suggestive of _____.
- All coagulation factors except factor VIII are synthesized in _____.
- Sweet, musty odour present in patient of hepatic encephalopathy is known as _____.
- Shifting dullness is a sign of _____.
- Child-Pughs classification is used in assessing the prognosis in patients of _____.
- Stress ulcers in patients after head injury are known as _____ ulcers.
- Stress ulcers in patients with burns are known as _____ ulcers.
- Iron absorption occurs in _____.

Chapter 3

Hematological System

ANEMIA

DEFINITION: Anemia is defined as a decrease in the level of hemoglobin below normal for that age and sex. The normal hemoglobin level varies from 13-16 g/dL in the adult male and 12-15 g/dL in the female. The Table 3.1 shows the values of RBC count, hematocrit and hemoglobin, below which patient is diagnosed to have anemia.

Normal Hematopoiesis

- Hematopoiesis in adult life takes place in the marrow of flat bones such as ribs, vertebrae, pelvic bones, sternum, scapula and the proximal ends of long bones.
- All blood cells arise from a population of cells called hematopoietic stem cells (Fig. 3.1). About 2.5 billion red blood cells (RBCs), 1 billion granulocytes and 2.5 billion platelets per Kg of body weight are produced each day.
- The amount of hemoglobin is maintained by erythropoietin, the hormone secreted by renal peritubular cells.

TABLE 3.1: Values below which anemia is diagnosed

	Male	Female
RBC count (million/ μ L)	<4.5	<4.0
Hemoglobin level (g/dL)	<13	<12
Hematocrit %	<40	<37

- The average life span of RBCs is 90-120 days and around 1% of RBCs are destroyed in the spleen and replaced by the marrow everyday.

Clinical Presentations

The symptoms and signs of anemia depend on a number of factors. Important factors are:

- Severity of anemia
- Rapidity with which it develops
- Cause of anemia
- Presence of comorbid diseases

Patients, who develop anemia acutely, such as following massive bleeding, may be highly symptomatic whereas those with gradually developing anemia may have mild symptoms. This is due to the development of various compensatory mechanisms in order to improve tissue oxygenation.

The symptoms and signs of anemia are due to the anemia *per se* or due to conditions which have resulted in anemia. Following are symptoms and signs due to anemia itself.

General: Pallor, weakness, malaise, feverishness, tiredness, vertigo, tinnitus

Central nervous system: Lack of concentration, decreased memory, syncope, altered sensorium, seizures.

Cardiorespiratory: Palpitation, dyspnea, angina, cardiomegaly, congestive heart failure, flow murmur.

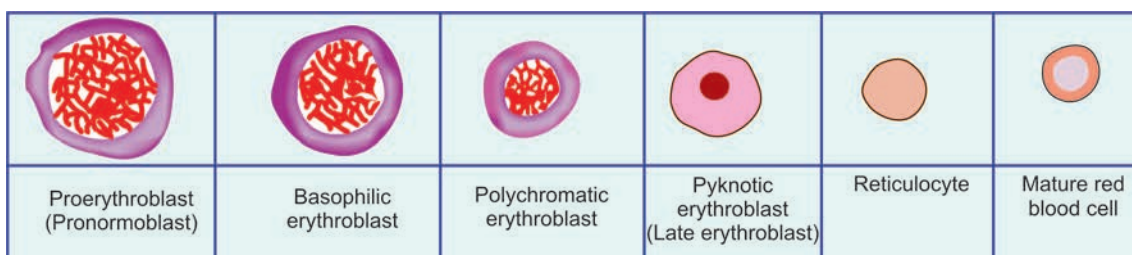


FIGURE 3.1: Normal erythropoiesis

Gastrointestinal: Anorexia, nausea, distaste, stomatitis, cheilosis, glossitis, dysphagia, malabsorption, hepatosplenomegaly.

Others: Loss of libido, impotence, menstrual abnormalities.

Classification

Anemia has been classified in various ways.

- Traditionally it is classified into three groups: (a) dyshematopoietic, (b) hemorrhagic and (c) hemolytic anemia.
- Another classification is based on the morphology of RBC (MCV, MCH, and MCHC) and is grouped into normocytic/microcytic/macrocytic and normochromic/hypochromic types (Table 3.2).
- However, the recent classification is based on reticulocyte index, which is a measure of RBC production. The reticulocyte index is increased (>2.5) due to increase in erythropoiesis as occurs in hemolytic and hemorrhagic anemias. A low reticulocyte index (<2) shows decreased marrow production or maturation defects during erythropoiesis (Table 3.3).

TABLE 3.2: Red blood cell indices (normal values)

1. Mean corpuscular volume (MCV)	90 ± 8 fL
2. Mean corpuscular hemoglobin (MCH)	30 ± 3 pg
3. Mean corpuscular hemoglobin concentration (MCHC)	$33 \pm 2\%$

TABLE 3.3: Classification of anemia

- Hypoproliferative (decreased production of red cells): normocytic normochromic
 - Bone marrow failure: Aplastic anemia
 - Bone marrow invasion (myelophthisis): Infiltration, fibrosis
 - Mild to moderate iron deficiency
 - Decreased stimulation by erythropoietin: Renal disease, metabolic defects, inflammation
- Maturation defect of RBC
 - Nuclear defect (defect in DNA synthesis): B_{12} deficiency, folic acid deficiency, drug toxicity, and refractory anemia (Macrocytic)
 - Cytoplasm defect (defect in hemoglobin synthesis): Severe iron deficiency, thalassemias, sideroblastic anemia, and lead poisoning (Microcytic)
- Hemolytic anemia
 - Intrinsic defects: Hereditary spherocytosis, G6PD deficiency, and sickle cell anemia (HbS)
 - Extrinsic defects: Immune mediated, malaria, micro-angiopathic anemia, and toxins
- Hemorrhagic anemia

TABLE 3.4: Investigations in anemic patients

- Hemoglobin, hematocrit
- RBC indices: MCV, MCH, MCHC, and Red cell distribution width (RDW)
- Reticulocyte count and index
- Total and differential leukocyte count
- Platelet count
- Cell morphology: size, color, anisocytosis, and poikilocytosis
- Iron status: serum iron level, total iron binding capacity (TIBC), ferritin, and marrow iron stain
- Serum vitamin B_{12} and folic acid levels
- Bone marrow aspiration and biopsy
- Liver function and renal function tests

Investigations

Tests are needed to assess the type of anemia and also to ascertain its cause. Table 3.4 shows the tests generally performed in anemic patients.

- Mean corpuscular volume (MCV) of less than 80 fL is called microcytosis and MCV of more than 100 fL is known as macrocytosis.
- The variation in red blood cell size and shape is known as anisocytosis and poikilocytosis respectively.
- Correction of reticulocyte count:** The reticulocyte count needs to be corrected according to the hematocrit. The corrected reticulocyte count is further adjusted if peripheral blood smear reveals prematurely released cells (polychromasia). This is known as reticulocyte index. Reticulocyte index is the main indicator of RBC production.

IRON DEFICIENCY ANEMIA (IDA)

IDA is the most common form of anemia prevalent worldwide.

Source and Daily Requirement

Iron is mainly available from a diet rich in meat, liver, beans, jaggery and green vegetables. Milk is poor source of iron. Daily iron requirement is 1 mg in males and 2 mg in females.

Metabolism

- Iron is absorbed in the proximal small intestine in the ferrous form. The absorption of iron is facilitated by the presence of acid in the stomach and vitamin C, while antacids, calcium, phosphates, and phytates decrease it.
- Ferrous iron is transported by metal transporter to enterocytes and stored in the form of ferritin. The iron

is transferred out of enterocytes by transporter ferroportin into the circulation.

- Iron released into the circulation binds to transferrin and is transported to sites of uses and storage. Hepatocytes, enterocytes and macrophages serve as storage reservoir of iron.
- Recently a liver hormone, hepcidin has been identified as a principal hormone involved in iron regulation. Hepcidin regulates the iron homeostasis by targeting at villus enterocytes, macrophages and hepatocytes. Thus, liver plays a central role in the regulation of iron absorption from the gut and in influencing the release of iron from the storage sites.

Causes of Iron Deficiency Anemia

The most common cause of IDA is blood loss. The blood loss can be due to hookworm infestation (*Ancylostoma duodenale* and *Necator americanus*), gastritis, peptic ulcer, esophageal varices, intestinal polyp, hemorrhoids, or heavy menstrual bleeding. Iron deficiency can occur during physiological states with high iron demand like adolescence and pregnancy. Decreased absorption due to gastrectomy, achlorhydria or diseases of the small intestine can also lead to IDA (Table 3.5).

Clinical Features

Besides having general symptoms of anemia, patients with IDA may specifically have pagophagia, i.e. craving for ice, cheilosis and spoon-shaped nails (koilonychia). Mild splenomegaly can occur in iron deficiency anemia, although it is uncommon. Additionally, patient may complain of dysphagia due to formation of post-cricoid web (Plummer Vinson or Patterson Kelly syndrome). *Plummer Vinson or Patterson Kelly syndrome includes:*

- Iron deficiency anemia

TABLE 3.5: Causes of iron deficiency anemia

1. Blood loss
a. Acute blood loss: accident and surgery
b. Chronic blood loss: gastritis, peptic ulcer, hookworm infestation, hemorrhoids, and menstrual loss
2. Increased demand
Infancy, adolescence, and pregnancy
3. Malabsorption
Post-gastrectomy, sprue, and Crohn's disease
4. Inadequate diet

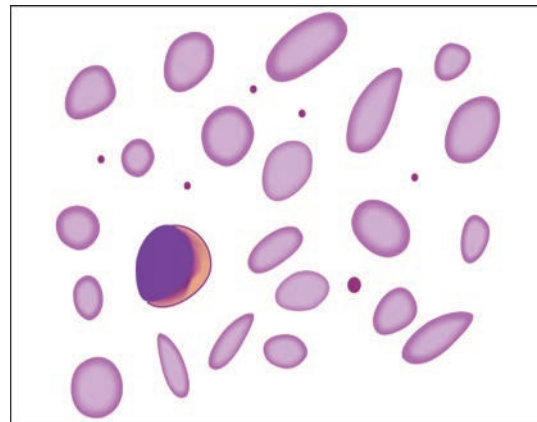


FIGURE 3.2: Microcytic hypochromic anemia

- Angular stomatitis
- Glossitis
- Dysphagia

Post cricoid web is a premalignant lesion and there is an increased risk of oral squamous cell cancer and esophageal cancer at post cricoid tissue web.

Investigations

Important investigations are peripheral blood examination, serum iron studies and bone marrow examination. (Fig. 3.2)

- The general blood picture is microcytic hypochromic.
- Serum iron and ferritin are low while total iron-binding capacity (TIBC) is increased. Transferrin saturation is below 16%.
- Bone marrow stains for iron reveal decreased or absent iron stores.
- Stool examination for parasites and occult blood is useful for detection of the cause of iron deficiency.
- Endoscopic and radiological examination of gastrointestinal tract is needed to detect the source of bleeding.

TABLE 3.6: Differential diagnosis of microcytic hypochromic anemia

Test	Iron deficiency	Inflammation	Thalassemia	Sideroblastic anemia
• Serum Iron	Low	Low	Normal/high	Normal/high
• TIBC	Increased	Decreased	Normal	Normal
• Percent saturation	Markedly low	Low	Normal	Normal
• Serum Ferritin	Low	Normal/high	High	High
• Hemoglobin pattern	Normal	Normal	Abnormal	Normal

Other conditions with microcytic hypochromic anemia should be differentiated from the iron deficiency anemia. These conditions are thalassemia, sideroblastic anemia, chronic inflammation and lead poisoning (Table 3.6)

Treatment

Oral Iron Therapy

The drug of choice is ferrous sulfate 200 mg thrice a day (elemental iron 60 mg thrice a day) orally taken in between meals. About 15-20% patients may develop intolerance to oral iron in the form of abdominal pain, nausea, vomiting, diarrhea or constipation. In such cases, the dose may be reduced or the salt is changed to ferrous gluconate, ferrous fumarate, sodium ironedite or carbonyl iron. The treatment with oral iron is usually given for a long duration and is sustained for 6-12 months even after normalization of hemoglobin.

Parenteral Iron Therapy

Intravenous iron therapy is indicated in following conditions:

- Patient is unable to tolerate oral iron
- Patient has malabsorption
- The needs for iron are relatively acute as in pregnancy or following bleed

Previously used iron compound, iron dextran has been associated with the risk of anaphylaxis which is almost never seen with newer preparations like sodium ferric gluconate and iron sucrose.

Red Blood Cell Transfusion

Red cell transfusion is indicated in patients with severe anemia where cardiorespiratory conditions warrant immediate intervention. In these situations, whole blood must be avoided since it may cause volume overload. Whole blood is needed if there is continued and excessive blood loss.

Anemia of Acute or Chronic Inflammation/Infection

Anemia of acute or chronic inflammation is one of the most common types of anemia which forms an important differential diagnosis of iron deficiency anemia because of similar general blood picture and low serum iron. This type of anemia is characterized by mild to moderate anemia

associated with inflammatory diseases such as rheumatoid arthritis, serious infections, carcinoma, and liver disease.

Mechanism

There is inadequate delivery of iron to the marrow despite normal or increased iron stores. There is an impaired marrow response to erythropoietin also. These effects are mediated by inflammatory cytokines (interleukin1, interferon γ , tumor necrosis factor) and hepcidin. Inflammation causes increased expression of hepcidin by the liver. Hepcidin suppresses iron absorption and release of iron from storage sites.

Investigations

Blood tests reveal hemoglobin levels of 8-10 g/dl, low serum iron, low total iron-binding capacity, decreased transferrin saturation, and a normal or increased serum ferritin. Bone marrow stain reveals adequate iron stores (Table 3.6).

Treatment

Treatment of underlying inflammatory disease usually corrects anemia. Iron therapy is not indicated.

MEGALOBLASTIC ANEMIA

Most important causes of megaloblastic anemia are vitamin B₁₂ deficiency (cobalamin) and folic acid deficiency. Vitamin B₁₂ and folic acid are essential for DNA synthesis and their deficiency may lead to impaired cellular proliferation and maturation. Rapidly dividing cells particularly those of hematopoietic and gastrointestinal tissues are most affected. A large number of erythroblasts in bone marrow fail to mature and are destroyed (ineffective erythropoiesis). Leukopenia and thrombocytopenia occur due to impaired proliferation and maturation of respective progenitor cells.

Cobalamin

- The daily requirement of vitamin B₁₂ (cobalamin) is about 2.5 μ g.
- Cobalamin is not synthesized in the human body and thus must be supplemented in diet. The only dietary source of cobalamin are animal products like meat and dairy foods.
- It is absorbed in the terminal ileum. The absorption is facilitated by the intrinsic factor produced by the parietal cells of the stomach.

- The cobalamin is largely stored in liver which may last upto 3-6 years.

Folic acid

- Folic acid (pteroylmonoglutamic acid) is found in foodstuff from both animal and plant sources. Dietary folic acid may be destroyed by cooking.
- The body stores are relatively small and may last for only a few months. The average daily requirement is 50-100 µg, although the demand may increase several fold during pregnancy.
- It is mainly absorbed in the jejunum.

Causes of Cobalamin and Folic Acid Deficiency

In tropical countries, megaloblastic anemia occurs mostly due to folic acid deficiency because of malnutrition and during pregnancy. Important causes of cobalamin and folic acid deficiency are given in Table 3.7.

Clinical Manifestations

The clinical manifestations are mainly due to the involvement of hematological, gastrointestinal and nervous systems.

- Patients are anemic (Fig. 3.3) and they may also have mild jaundice due to raised plasma unconjugated bilirubin. Purpura may rarely occur due to thrombocytopenia. Spleen may be enlarged.
- Anorexia, weight loss, diarrhea and smooth and beefy red tongue are important and significant gastrointestinal manifestations.

TABLE 3.7: Causes of cobalamin and folic acid deficiency

Causes of cobalamin deficiency

- Decreased intake: vegans
- Decreased absorption:
 - Intrinsic factor deficiency: pernicious anemia and postgastrectomy
 - Diseases of terminal ileum: sprue, Crohn's disease, and intestinal resection
 - Bacterial proliferation
 - Fish tapeworm (*Diphyllobothrium latum*) infestation

Causes of folic acid deficiency

- Decreased intake: unbalanced diet and alcoholism
- Increased demand: pregnancy, infancy, and hemolysis
- Decreased absorption: sprue, and celiac disease
- Drugs: phenytoin, methotrexate, pyrimethamine, and trimethoprim



FIGURE 3.3: Pallor visible over face

- Neurological manifestations such as paresthesia, ataxia, sensory-motor paraparesis (subacute combined degeneration), forgetfulness, psychosis are found in cobalamin deficiency and may occur even in the absence of anemia.

Investigations

- Complete blood count shows decrease in hemoglobin, leukopenia, and thrombocytopenia.
- Peripheral blood smear reveals macrocytosis (MCV >110 fL). Other causes of macrocytosis like hemolysis, liver disease, hypothyroidism and alcoholism must be ruled out. Hypersegmented neutrophils with a nucleus of six or more lobes in a single cell are highly characteristic of megaloblastic anemia (Fig. 3.4)

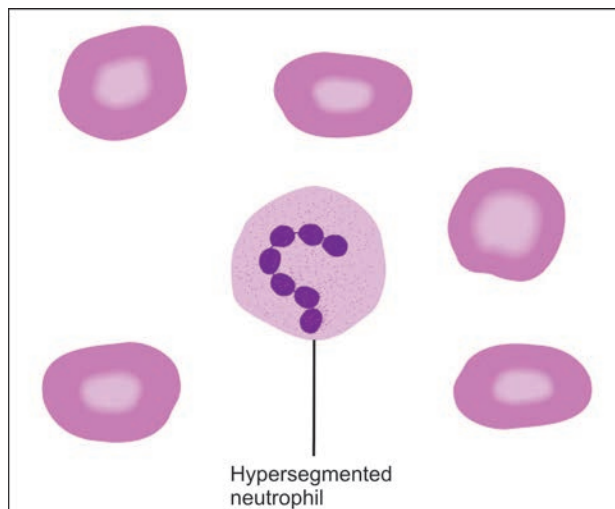


FIGURE 3.4: Hypersegmented neutrophil in megaloblastic anemia

- c. Lactate dehydrogenase (LDH 1) and unconjugated bilirubin are raised due to ineffective erythropoiesis.
- d. Bone marrow is hypercellular and reveals megaloblastic changes. Megaloblasts are abnormally large RBC precursors with nuclei less mature than would be expected from the cytoplasmic development.
- e. Serum cobalamin and folate levels are determined to detect the specific deficiency.
- f. Serum levels of both methylmalonic acid and homocysteine are increased in cobalamin deficiency while only homocysteine level is increased in folate deficiency.
- g. Schilling test is performed to detect the malabsorption of cobalamin. Auto-antibodies against intrinsic factor and parietal cells are found in *pernicious anemia*.

Treatment

Packed red cell transfusion is needed in case of severe anemia with cardiac symptoms. In addition, treatment is directed against the cause of the disease like antibiotics for bacterial overgrowth in the intestine.

Cobalamin deficiency: Parenteral therapy with intramuscular cyanocobalamin is preferred since deficiency is mostly due to malabsorption. The treatment begins with a dose of 1000 µg (1 mg) per week for 8 weeks followed by 1000 µg each month. The treatment is life long in case of pernicious anemia.

Folate deficiency: The usual dose of folic acid is 1 mg per day orally. However, higher dosage upto 5 mg daily may be needed in cases of malabsorption. Folinic acid is used in methotrexate-induced anemia.

APLASTIC ANEMIA

Aplastic anemia is characterized by pancytopenia (anemia, leukopenia and thrombocytopenia) and hypocellular bone marrow. In the majority of patients, the cause is not discernible (idiopathic) while the major known causes are drugs, radiation and viral infections (Table 3.8).

Pathogenesis

The pathogenesis of bone marrow aplasia is generally believed to be immune mediated. However, some genetic factors may predispose the individual to develop an abnormal T cell immune response to exogenous stimuli (drugs, viruses) and subsequent marrow failure.

TABLE 3.8: Types of aplastic anemia

a. <i>Idiopathic</i>
b. <i>Secondary</i>
Drugs: chloramphenicol, sulphonamides, indomethacin, gold, cytotoxic drugs, and anticonvulsants.
Radiation
Chemicals: DDT and benzene
Viruses: hepatitis viruses, parvovirus, HIV-1, Epstein-Barr virus
Pregnancy
Paroxysmal nocturnal hemoglobinuria (PNH)
c. <i>Inherited</i>
Fanconi's anemia

Clinical Features

- The common presentations are bleeding and symptoms of anemia.
- The excessive tendency to bleed is due to thrombocytopenia which may present as easy bruising, epistaxis, gum bleeding, heavy menstrual flow and petechiae (small pinpoint hemorrhage in skin and mucous membrane). Intracranial and retinal hemorrhages may also occur.
- Neutropenia may predispose patients to develop infections.
- Lymphadenopathy and splenomegaly are absent.

Investigations

- a. Peripheral blood smear shows normocytic or macrocytic anemia, decreased granulocyte and platelet count. Immature cells are absent. Reticulocytes are absent or few.
- b. The diagnosis is confirmed by the bone marrow aspiration and biopsy.
- c. Other investigations include viral markers, chromosomal studies, and tests for paroxysmal nocturnal hemoglobinuria (PNH). Chromosomal studies are done in children and younger adults to rule out inherited disorders like Fanconi's anemia.

Treatment

- a. **Bone marrow transplantation:** Allogeneic bone marrow transplantation (BMT) from HLA matched siblings is curative and the preferred mode of therapy in young patients (<40 years). However, this is limited by the high cost, nonavailability of matched donors and the significant morbidity and mortality. Moreover, this facility is available only at a few centers.

- b. *Immunosuppressive therapy*: Immunosuppressive agents such as anti-thymocyte globulin (ATG) or anti-lymphocyte globulin (ALG) along with cyclosporine is the treatment of choice for patients who cannot be given BMT.
- c. *Other drugs*: The role of anabolic steroids is not clear, though some patients may respond to them.
- d. *Supportive therapy*:
 - Severe anemia is managed with packed red cell transfusion.
 - Platelet concentrates are used to maintain platelet count at or more than 10,000/ μ L. Aspirin and NSAIDs which inhibit platelet function should be avoided.
 - Infections should be aggressively dealt with broad-spectrum antibiotics and antifungal agents. Granulocyte transfusion has also been used in overwhelming infections. Aseptic precautions must be observed to prevent infections.

HEMOLYTIC ANEMIA

The average life span of RBC is 90-120 days and around 1% of RBCs are destroyed in the spleen and replaced by the bone marrow every day. Hemolytic anemia results due to increased premature destruction of RBCs if the bone marrow is not able to replenish them adequately.

Hemolysis may be extravascular or less commonly intravascular. The hemoglobin, released following RBC breakdown is immediately bound to a plasma protein, haptoglobin. The hemoglobin also binds with albumin to form methemalbumin. In case of severe hemolysis the haptoglobin binding capacity of plasma is reduced. Hence, free hemoglobin passes through renal glomeruli and is converted to ferritin and hemosiderin in proximal tubules and passed in the urine (hemosiderinuria). Excess hemoglobin that is not absorbed by proximal tubules is passed as such in the urine (hemoglobinuria). Unconjugated bilirubin is raised due to increased hemolysis (prehepatic or hemolytic jaundice).

Hemolytic anemia can be acquired or hereditary. Hemolysis can occur due to defects in the RBC (intracorporeal or intrinsic defects) or due to causes other than in RBC (extracorporeal or extrinsic defects). The classification is given in Table 3.9.

TABLE 3.9: Classification of hemolytic anemias

Congenital (intrinsic)

1. Membrane defects: hereditary spherocytosis and hereditary elliptocytosis
2. Hemoglobinopathies:
 - a. Abnormal chain synthesis; thalassemia
 - b. Amino acid substitution: sickle cell disease (Hb S), HbC, HbD
3. Enzyme defects: Glucose-6-phosphate dehydrogenase and pyruvate kinase deficiency

Acquired (extrinsic)

1. Immune: Autoimmune hemolytic anemia (AIHA)
2. Non-immune:
 - a. Mechanical: Disseminated intravascular coagulation (DIC), toxemia of pregnancy, and artificial heart valve
 - b. Malarial and clostridium infections
 - c. PNH (intrinsic)*

* PNH is characterized by acquired RBC membrane defect (intrinsic).

Hemoglobinopathy

Normal hemoglobin (Hb) consists of heme and tetramer of globin polypeptide chains. Major adult Hb (HbA₁) has two alpha chains and two beta chains ($\alpha_2\beta_2$) and represents 98% of total Hb. During fetal life, fetal hemoglobin (HbF), containing two alpha chains and two gamma chains ($\alpha_2\gamma_2$), predominates. A small amount of HbA₂ ($\alpha_2\delta_2$) is also found in normal adults. The defect in Hb such as in thalassemia and sickle cell disease can result in hemolysis.

Thalassemias: Thalassemias are hereditary disorders characterized by reduced production of globin chains. Reduction in alpha chain synthesis leads to alpha thalassemia and that of beta chain leads to beta thalassemia. Hence in alpha thalassemia, excess beta chains form β_4 tetramer called HbH. In beta thalassemia, there is relative increase in the amount of HbF and HbA₂ as beta chains are missing and are substituted by gamma and delta chains respectively. Beta thalassemia major (Cooley's anemia) is a severe disease and manifests in childhood. This condition is transfusion dependent and generally fatal by 30 year of age. Beta thalassemia minor is mild disease, nontransfusion dependent, and patients can live full normal lives. In case of severe thalassemia, the expansion of bone marrow (due to extramedullary hematopoiesis) may cause bony deformities, pathological fractures and osteopenia.

Sickle cell anemia: Sickle cell anemia is caused by mutation in the beta globin gene that replaces the sixth amino acid

from glutamic acid to valine (Hb S, $\alpha_2\beta_2^{6\text{Glu-val}}$). Under hypoxic conditions, HbS polymerizes leading to sickling of the RBCs which are more liable to hemolysis.

Clinical Features

- The important clinical manifestations are anemia, recurrent mild jaundice and splenomegaly.
- Patients may have fever with chills during hemolysis.
- Other manifestations are gallstones (pigment stones) and red brown urine (hemoglobinuria).
- The family history may be positive in hereditary disorders like sickle cell disease, thalassemia and hereditary spherocytosis.
- A history of drugs and exposure to chemicals should also be obtained. This is important in patients with G6PD deficiency.
- Characteristic facial features due to marrow expansion and growth retardation may be seen in inherited disorders for example in thalassemia.

Investigations

- Peripheral blood smears reveal features of hemolysis like anisocytosis, poikilocytosis, macrocytosis and target cells. Characteristic RBC morphology is seen in specific causes (Table 3.10 and Fig. 3.5).
- Reticulocyte count is increased and is most useful in the diagnosis of hemolytic anemia.
- Bone marrow examination is generally not required.
- Serum unconjugated bilirubin, methemalbumin, and LDH are increased. Serum haptoglobin is decreased or absent (Table 3.11).

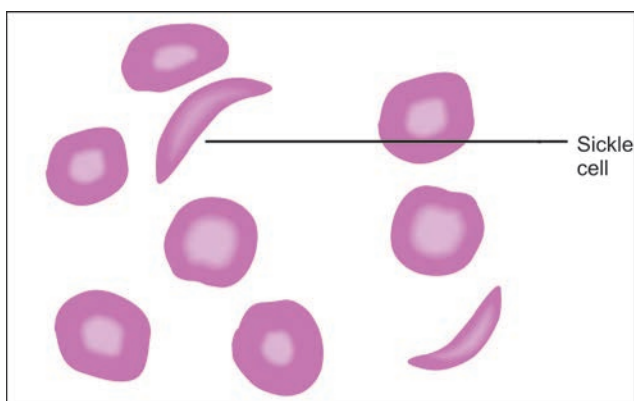


FIGURE 3.5: Sick cell anemia

TABLE 3.10: Red cell morphology in hemolytic anemias

Spherocytes	Hereditary spherocytosis and AIHA
Target cells	Thalassemia
Microcytes	Thalassemia
Schistocytes	DIC, eclampsia, and artificial heart valve
Sickled cells	Sickle cell syndrome

TABLE 3.11: Findings suggestive of hemolysis

<i>Increased RBC destruction:</i>	
a.	High serum indirect bilirubin
b.	Low serum haptoglobin
c.	High serum LDH
d.	Hemoglobinuria and hemosiderinuria (in intravascular hemolysis)
e.	Decreased RBC life span
<i>Increased RBC production:</i>	
a.	High reticulocyte count
b.	Nucleated RBCs in peripheral smear
c.	Erythroid hyperplasia in bone marrow

- Urine examination may reveal presence of hemosiderin or hemoglobin.
- Serum iron and ferritin levels are normal.
- Tests such as hemoglobin electrophoresis, osmotic fragility, Coombs' test, Glucose-6-phosphate dehydrogenase (G6PD) levels, sickling tests, and genetic studies are helpful in the diagnosis of the specific causes.
- Amniocentesis and chorionic villi biopsy are performed for prenatal diagnosis of inherited conditions like thalassemia and sickle cell disease.

Treatment

Supportive treatment in form of packed cell transfusion may be needed. However, repeated transfusion may lead to iron overload which in turn may require treatment with chelating agents, deferoxamine. Folic acid, 5 mg daily is given to meet the increased demand. Following are other modes of treatment in specific situations

- Bone marrow transplantation is curative in thalassemia and sickle cell disease.
- Splenectomy may be needed in hereditary spherocytosis and AIHA.
- Steroids and other immune suppressive drugs are given in immune-mediated anemias.
- Factors including infections, dehydration, hypoxia (in sickle cell disease) and drugs (primaquine, sulphonamides) that may precipitate hemolysis in persons with G6PD deficiency should be avoided.

ANEMIA OF ACUTE BLOOD LOSS

Acute blood loss due to trauma and surgery may lead to anemia. A loss of about 20% of blood volume (1 liter) can be tolerated; however, a loss of more than 40% can lead to shock.

Anemia may be apparent only after volume replacement in patients with acute blood loss.

Other findings include raised total leukocyte count and platelets, and presence of nucleated RBCs and immature leukocytes in the peripheral blood. Maximal rise in reticulocytes occur after 1 week. A rise in serum unconjugated bilirubin and fall in serum haptoglobin may occur in case of bleeding in an internal cavity.

The management includes treatment of the underlying conditions and the blood transfusion. Patient may need iron replacement.

LEUKEMIA

- Leukemia arises from malignant transformation of hematopoietic cells leading to increased number of white blood cells in blood and/or bone marrow.
- Depending on the onset and the course of the disease, leukemia is classified as acute or chronic.
- The course in acute leukemia is aggressive and the life span is short if not treated. The course in chronic leukemia is indolent and the survival period is much longer.
- Leukemia is also named as myeloid or lymphoid depending on the type of cell of origin.

In majority of cases the etiology of leukemia is not known. However, some are associated with ionizing radiation, drugs (alkylating agents), toxins (benzene) and viral infections. There may be a genetic predisposition for the development of leukemia.

Acute Leukemias

Acute leukemias are broadly classified into acute lymphoblastic leukemia (ALL) and acute myeloid leukemia (AML). Acute lymphoblastic leukemia is more common in children while acute myeloid leukemia is common in adults.

Classification

Acute leukemias are classified on the basis of cell of origin, morphology and immunophenotypic characteristics

TABLE 3.12: Classification of acute leukemias

Acute myeloid leukemia (French-American-British classification)
• M0 undifferentiated
• M1 Myeloblastic
• M2 Myeloblastic with differentiation
• M3 Promyelocytic
• M4 Myelo-monocytic
• M5 Monoblastic
• M6 Erythroleukemia
• M7 Megakaryoblastic
Acute lymphoblastic leukemia
• Pre B cell ALL
• T cell ALL
• B cell ALL

(Table 3.12). Newer World Health Organization (WHO) classification additionally includes cytogenetic and molecular features.

Clinical Features

There is uncontrolled proliferation of malignant hematopoietic cells which lose the capacity to differentiate and mature. This leads to accumulation of more and more malignant cells in the bone marrow at the expense of normal hematopoietic cells. There is a rapid onset of symptoms.

- Clinical features in acute leukemia arise chiefly due to marrow failure, presenting as anemia and recurrent infections due to neutropenia.
- Petechiae, gingival bleeding, epistaxis and menorrhagia may occur due to thrombocytopenia.
- Wide spread bleeding may occur due to disseminated intravascular coagulation (DIC) which is mainly present in patients with acute promyelocytic leukemia (AML-M3).
- Malignant cells may also infiltrate various organs leading to lymphadenopathy and hepato-splenomegaly.
- **Gum hyperplasia** may be seen in monocytic subtype (M4, M5) of AML (Fig. 3.6).
- Other manifestations are stomatitis, sternal tenderness, testicular enlargement and infiltration of skin and meninges.

Diagnosis

- Peripheral blood examination reveals the presence of blast cells with high, low or normal total leukocyte count (Fig. 3.7).



FIGURE 3.6: Gum hyperplasia in a AML-M5 patient

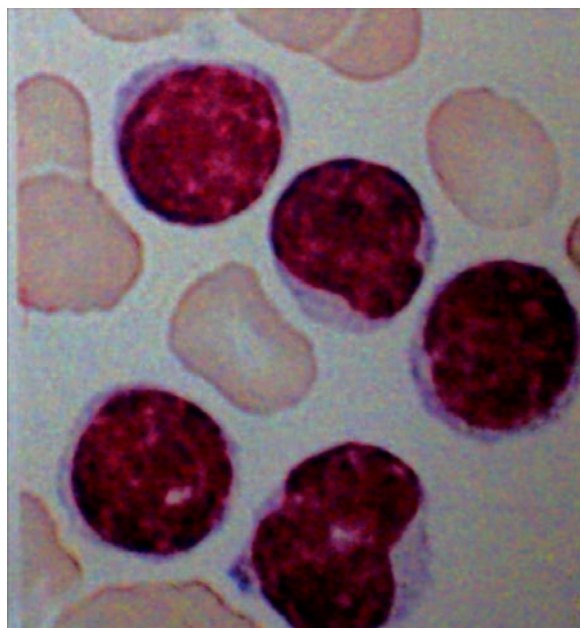


FIGURE 3.7: Acute myeloid leukemia

- There is also the evidence of anemia and thrombocytopenia.
- The bone marrow examination shows hypercellularity along with the presence of >20% leukemic blast cells. Presence of Auer rods in cytoplasm of blast cells is diagnostic of AML.
- Cytochemical staining, cytogenetics and immunophenotyping of the cells help in differentiating different types of leukemia.

Treatment

The management of acute leukemia consists of supportive and specific treatment.

- Supportive treatment:** Anemia is managed with infusion of red cell concentrate. Platelet transfusion is needed to treat bleeding manifestations and to maintain platelet count above 10,000-20,000/ μ L. The chance of life-threatening bleeding becomes more when platelet count is <10,000/ μ L. Infections are treated with parenteral broad spectrum antibiotics and antifungal drugs if required.
 - Specific treatment:** The objective of specific treatment is to eliminate leukemic cells without affecting the normal cells. However, the therapy may be associated with high morbidity and mortality. Hence, the decision to administer a specific therapy to a particular patient is based on the age, type of leukemia and the presence of other associated illnesses.
- **Chemotherapy:** In chemotherapy, a combination of various cytotoxic drugs is given under a standard protocol. The first step is to achieve remission (normal blood counts, normal bone marrow, and normal clinical status). The initial induction phase is followed by the consolidation phase and the maintenance phase. Table 3.13 shows drugs used in acute leukemias.
 - **Radiotherapy:** Cranial irradiation along with intrathecal methotrexate is given in ALL patients for CNS prophylaxis.

TABLE 3.13: Drugs commonly used to treat acute leukemias

Acute myeloid leukemia	
•	Daunorubicin
•	Cytosine arabinoside
•	Etoposide
•	All -trans-retinoic acid*
Acute Lymphoblastic leukemia	
•	Vincristine
•	Prednisolone
•	Daunorubicin
•	L-asparaginase
•	Methotrexate
•	Etoposide
•	Cytosine arabinoside
•	Mercaptopurine

*This is a vitamin A analogue, used in the treatment of AML-M3

- **Bone marrow transplantation:** Following initial remission of disease with chemotherapy, further treatment is given to cure the disease. The options are consolidation chemotherapy and bone marrow transplantation (BMT). Indications for BMT are:
 - Common (pre B) ALL in second remission
 - T and B cell ALL in first remission
 - AML in first remission.

Prognosis

The cure rate in AML is around 25-30% with chemotherapy and 50-60% with BMT. The cure rate of >90% can be achieved by chemotherapy in children with ALL (pre B type).

Chronic Myeloid Leukemia

Chronic myeloid leukemia (CML) is a type of myeloproliferative disorder. It occurs as a result of malignant transformation of pluripotent stem cell leading to accumulation of large number of immature leukocytes in the blood.

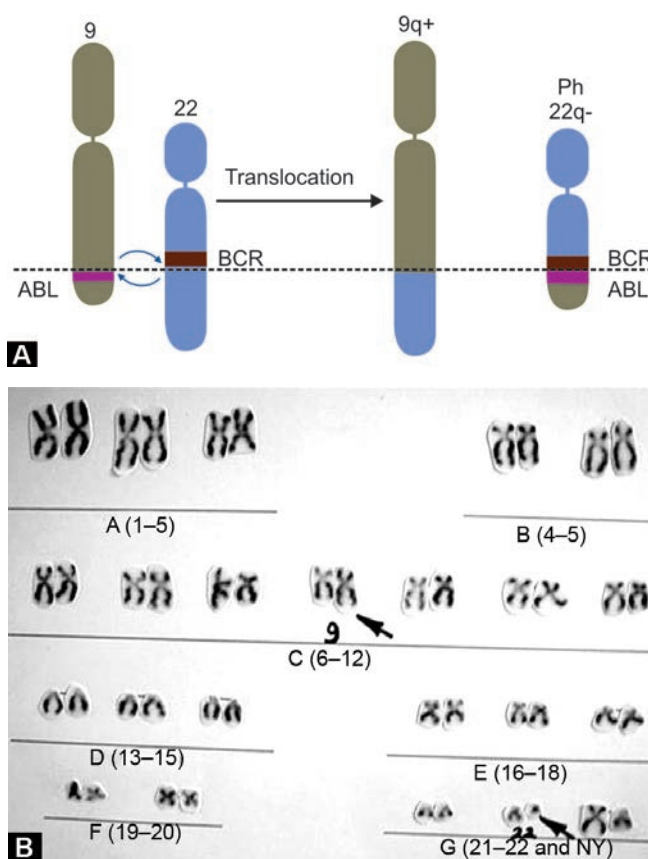
The underlying chromosomal abnormality in CML is the Philadelphia chromosome (short 22) which results due to the reciprocal translocation between chromosomes 9 and 22 (Figs 3.8A and B). The translocation leads to the formation of hybrid gene (bcr-abl) and synthesis of a 210 KD protein. This protein has intrinsic tyrosine kinase activity and is responsible for the initiation and maintenance of the malignant proliferation in CML.

Typically the course of CML consists of three phases.

1. The initial phase is the chronic phase which, without treatment, may last for 2-3 years.
2. This evolves into an accelerated phase which finally transforms into a terminal blast phase.
3. The blast phase is like acute leukemia, which is mostly myeloblastic but in about 20% cases, it can be lymphoblastic.

Clinical Manifestations

CML affects individuals in the third and fourth decade. The common presenting features are weakness, tiredness, weight loss and abdominal fullness due to splenomegaly. However, some patients are asymptomatic at the time of diagnosis. Seldom patients suffer from anemia and fever. Worsening anemia, splenomegaly, fever and bony pain may suggest



FIGURES 3.8A and B: (A) Philadelphia chromosome (B) Cytogenetic study showing short chromosome 22

the transformation into the accelerated phase. In addition to these, in blast phase, patient may also exhibit bleeding, severe infection and lymphadenopathy.

Investigations

- The blood examination reveals high total leukocyte count (generally in lacs), normocytic normochromic anemia and high platelet count.
- The presence of low platelet count denotes worsening of the disease.
- Large numbers of immature myeloid cells (metamyelocytes, myelocytes) are seen in the blood smear.
- Presence of 20% or more blast cells in bone marrow or peripheral blood is diagnostic of the blast phase of CML.
- The diagnosis of CML is confirmed by the demonstration of Philadelphia chromosome on cytogenetic analysis or the presence of bcr-abl fusion gene by molecular techniques.

Treatment

- At present the only known curative method of treatment of CML is allogeneic bone marrow transplantation (Allo BMT). However, most patients cannot avail this mode of treatment due to the high cost, lack of proper donor or the high risk involved in the procedure particularly if the age is above 40 years.
- Imatinib mesylate is the drug of choice for those who do not undergo Allo BMT. *Imatinib mesylate* is a new targeted drug which specifically inhibits the bcr-abl tyrosine kinase. This can result in molecular remission in around 60-70% patients. The usual dose is 400 mg oral daily. Newer tyrosine kinase inhibitor, *Dasatinib* is now available and is useful in patients who fail to respond to imatinib.
- The other useful agent is *interferon alpha* which is more toxic and less effective than imatinib.
- *Hydroxyurea* is used to control the white blood cell count and has almost replaced the *busulphan* used previously.
- The blast phase is managed as acute leukemia. Alternatively imatinib in larger dosage (600 mg daily) or dasatinib can be tried in blast phase.

Chronic Lymphocytic Leukemia

Chronic lymphocytic leukemia (CLL) is a malignancy of mainly B cell origin. It has slowly progressive course. The median age at presentation is 65 years.

CLL is characterized by the presence of large number of mature looking small lymphocytes in the blood smear and lymphoid organs like lymph nodes, liver and spleen. The patients are immunocompromised because B cells are less responsive to antigenic stimuli.

Clinical Manifestation

The onset is insidious. In many, the disease is asymptomatic and is diagnosed during incidental blood examination. The usual clinical features are painless lymphadenopathy, splenomegaly and anemia. Autoimmune hemolytic anemia and thrombocytopenia may also occur. Binet and Rai staging systems are used for the prognostic classification of the disease.

Investigations

- Examination of the blood reveals high lymphocyte count (WBC usually $>20000/\mu\text{l}$). Majority of the cells are mature lymphocytes.

- Bone marrow examination reveals infiltration with mature lymphocytes.
- Coombs test is positive in autoimmune hemolytic anemia.
- Thrombocytopenia may occur due to immune destruction or marrow failure.
- Serum immunoglobulin levels may be low which gives an idea about the degree of immunosuppression.
- Newer prognostic markers like CD38, ZAP 70 mutation and Vlg mutation are helpful in the management of CLL.

Treatment

Most patients do not require treatment in the early stage of the disease. Those with features of progressive disease such as symptomatic lymphadenopathy, anemia or thrombocytopenia need chemotherapy. The therapy of choice is fludarabine based chemotherapy (fludarabine alone or in combination with cyclophosphamide and/or anti-CD 20 monoclonal antibody) which provides better response and survival.

Alternatively oral chlorambucil may be given. Prednisolone is administered in cases of autoimmune hemolytic anemia.

LYMPHOMAS

Lymphomas are malignancies which primarily arise in lymphoid organs. Majority are of B cell origin. Clinically and histologically lymphomas are classified into non-Hodgkin's lymphoma (NHL) and Hodgkin's disease (HD).

Hodgkin's Disease

Hodgkin's disease (HD) is a lymphoid malignancy characterized by the presence of Reed-Sternberg cells of B cell origin (Fig. 3.9). The disease arises usually in single nodal area and successively spreads to contiguous lymph node areas. Extranodal involvement is rare.

Etiology: Etiology is generally unknown. Infection with HIV is a risk factor for developing HD. The etiological association with Epstein-Barr virus is not proven.

Clinical Features

There is a bimodal age distribution with one peak at 20-35 years and second peak in 50-70 year age group.

- The most common clinical presentation is painless lymphadenopathy, generally in neck, supraclavicular area and axillae (Fig. 3.10).

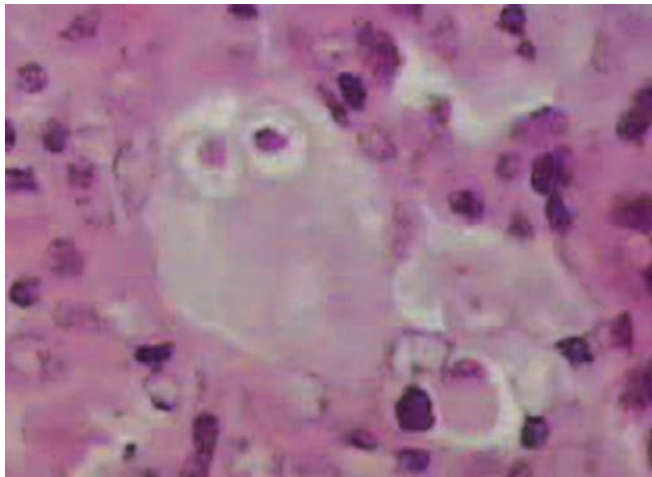


FIGURE 3.9: Reed-Sternberg cell in Hodgkin's lymphoma



FIGURE 3.10: Cervical lymphadenopathy

- A sizable number of patients have mediastinal lymphadenopathy.
- About one-third patients have B symptoms—fever, weight loss and night sweats.
- Unusual symptoms are generalized pruritus and pain in involved lymph nodes following alcohol ingestion.

Classification

HD is classified into four types depending on histological features (Table 3.14).

The most common type of HD in USA is nodular sclerosis whereas mixed cellularity is the most common type

TABLE 3.14: Classification of Hodgkin's disease

- Nodular sclerosis
- Mixed cellularity
- Lymphocyte-predominant
- Lymphocyte-depleted

of HD in India. Mixed cellularity HD occurs more commonly in elderly patients and in HIV positive individuals.

Investigations

- The diagnosis of HD is made by lymph node biopsy. Fine needle aspiration cytology (FNAC) may provide doubtful results. Presence of *Reed-Sternberg cells* (*Owl's eye appearance*) is characteristic of the disease (Fig. 3.9).
- Bone marrow examination, CT chest and abdomen are done for the staging purposes.
- PET scan (Positron emission tomography) is useful to document remission.
- Tests such as complete blood count, ESR, serum uric acid, liver function and renal function tests are helpful in planning and monitoring the treatment.
- Serum LDH is a useful prognostic marker, a high level being a bad prognostic sign.

Staging

The disease stage is determined after clinical evaluation and investigations. The Ann Arbor staging system is given in Table 3.15.

Treatment

Localized disease: The patients with localized disease (IA, IIA) are treated with 3 cycles of chemotherapy (ABVD-

TABLE 3.15: Ann Arbor staging system

Stage I:	Involvement of single lymph node region or lymphoid structure*.
Stage II:	Involvement of two or more lymph node regions on same side of the diaphragm.
Stage III:	Involvement of lymph node regions or lymphoid structures on both sides of the diaphragm.
Stage IV:	Disseminated disease with liver or bone marrow involvement.
A.	No systemic symptoms
B.	Weight loss >10 percent in 6 months, fever, and night sweats
*The lymphoid structures are defined as spleen, thymus, Waldeyer's ring, appendix and Peyer's patches	

doxorubicin, bleomycin, vinblastine and dacarbazine) followed by radiotherapy of involved nodal areas.

Extensive disease: Patients with B symptoms or extensive disease receive complete course (6-8 cycles) of ABVD chemotherapy.

Long-term cure can be achieved in >90% patients with localized disease and in 50-75% patients with extensive disease.

Nodular Lymphocyte Predominant Hodgkin's Disease

Nodular lymphocyte predominant Hodgkin's disease is a distinct entity which is different from classical HD. In this type, the Reed-Sternberg cells are rare. The course of the disease is chronic and relapsing and sometimes it transforms into diffuse large B cell NHL.

Non-Hodgkin's Lymphoma (NHL)

Etiology: A variety of etiological factors are associated with NHL. Viruses such as Epstein-Barr virus, HTLV-1, HIV, hepatitis C virus and Human herpes virus 8 are implicated in the occurrence of NHL. Infection with *Helicobacter pylori* is related with the genesis of gastric MALT (mucosa associated lymphoid tissue) lymphoma. NHL can occur in congenital and acquired immunodeficiency states, autoimmune disorders, and following exposure to chemicals, drugs, prior chemotherapy and radiation therapy.

Types: Clinically non-Hodgkin's lymphoma can be low grade or high grade type. It is also classified into B cell and T cell types. The most common low grade B cell lymphoma is follicular lymphoma while diffuse large cell lymphoma is most common high grade lymphoma. Table 3.16 shows some common types of NHL.

TABLE 3.16: Types of non-Hodgkin's lymphoma

B cell lymphoma

- Follicular lymphoma
- Small lymphocytic lymphoma
- Marginal zone lymphoma
- Mantle cell lymphoma
- Diffuse large B cell lymphoma
- Burkitt's lymphoma

T cell lymphoma

- Peripheral T cell lymphoma
- Anaplastic large cell lymphoma
- Mycosis fungoides/Sezary syndrome

Clinical Features

- The patients may present with localized or generalized painless lymphadenopathy, hepatosplenomegaly and constitutional symptoms such as fever, weight loss and night sweats.
- Extranodal presentations are more common in NHL than in HD. These include involvement of bone marrow, gut, lungs, testis, thyroid, skin and brain.
- Patients may present with superior vena cava obstruction, intestinal obstruction and spinal cord compression.
- Compared to HD, NHL is often disseminated at presentation.

Low-grade NHL is characterized by low proliferation rate, insidious onset of symptoms and slow progression. High grade NHLs have early onset of symptoms with aggressive course and are fatal if untreated.

Investigations

- The diagnosis of NHL is made by tissue biopsy (lymph node or extranodal tissue). Fine needle aspiration cytology (FNAC) may provide doubtful results.
- Immunophenotyping and cytogenetic analysis are done to further characterize the type of NHL.
- Bone marrow examination, CT chest and abdomen, and lumbar puncture are done for the staging purposes.
- Tests such as complete blood count, ESR, serum uric acid, liver function and renal function tests are helpful in planning and monitoring of the treatment.
- Serum LDH is a useful prognostic marker, high level is a bad prognostic sign.

Treatment

The disease stage is determined after clinical evaluation and investigations (see Table 3.15).

Treatment of low grade lymphoma: It depends on the stage of the disease and clinical status.

- Asymptomatic patients generally do not require therapy.
- Symptomatic localized (stage I) disease can be treated with radiotherapy.
- Symptomatic patients with extensive disease (stage III, IV) are treated with chlorambucil or combination chemotherapy (CVP; cyclophosphamide, vincristine, and prednisolone or CHOP; cyclophosphamide, doxorubicin, vincristine and prednisolone).

- Other newer effective agents are fludarabine, monoclonal antibody (anti-CD20) with or without radionuclides, and lymphoma vaccine.
- The relapse following therapy is common. Hence, low grade NHL is generally not curable. However, the cure is now possible in some cases with the use of radio-immunotherapy.

Treatment of high grade lymphoma:

- Localized high grade NHL (stage I and non-bulky stage II) is treated with three cycles of chemotherapy (CHOP with monoclonal antibody, anti-CD20) followed by radiotherapy.
- Extensive disease (bulky stage II, stage III, and stage IV) is treated with six to eight cycles of chemotherapy.
- Autologous bone marrow transplantation is indicated in relapsed cases.
- The cure rate is 60-90% in stage I and II and 30-40% in stage III and IV.

Burkitt's Lymphoma

This is a rare type of NHL. Epstein-Barr virus and HIV are associated with the development of Burkitt's lymphoma. This is the most rapidly progressive tumor. Most patients present with lymphadenopathy and abdominal mass. It has a tendency to metastasize to the CNS. Treatment should be initiated promptly with intensive chemotherapy. Prophylactic CNS therapy is also given. Seventy to eighty percent patients may be cured.

MULTIPLE MYELOMA

Multiple myeloma is a malignant disease arising from neoplastic transformation of plasma cells. Normally plasma cells produce polyclonal immunoglobulins in response to antigenic stimulation. In multiple myeloma, neoplastic plasma cells abnormally produce immunoglobulin of monoclonal origin called paraproteins. Most common type of paraprotein is of IgG type. In some cases, only light chains are produced which appear in urine (Bence Jones protein).

Clinical Features

Myeloma is a disease of older adults. Important clinical presentations are anemia, bone pain and infections. Bone pain is more common in back and ribs. Pathological fracture is common. Patient may also present with renal failure, spinal cord compression and neuropathy.



FIGURE 3.11: X-ray skull showing multiple osteolytic lesions

Investigations

There is normocytic normochromic anemia. Peripheral blood smear may reveal rouleau formation. ESR is high. Hallmark of the disease is raised levels of serum immunoglobulin and presence of monoclonal spike (M component) on electrophoresis. Hypercalcemia may also be present. X-ray shows lytic lesions in the skull (Fig. 3.11), ribs and long bones. Serum alkaline phosphatase is normal. Blood urea and serum creatinine may be raised. Urine examination may reveal proteinuria and Bence Jones proteins. Serum β_2 -microglobulin is a good prognostic marker.

Treatment

The treatment of multiple myeloma includes chemotherapy, radiotherapy, biphosphonates (pamidronate, zolidronate), and autologous peripheral stem cell transplantation. Newer therapeutic agents include thalidomide, linalidomide and proteasome inhibitors. Chemotherapeutic agents include melphalan, cyclophosphamide, doxorubicin, vincristine and dexamethasone.

NORMAL HEMOSTASIS

The blood remains fluid in the vessels while it clots at the site of breach or injury. The anticoagulant and procoagulant

mechanisms are intricately regulated so that no clotting occurs in flowing blood or at unwanted sites whereas there is an immediate platelet plug formation and fibrin deposition at the site of bleeding.

Exposure of blood to the subendothelial connective tissue following surgery, trauma or disease initiates primary and secondary hemostatic processes.

Primary Hemostasis

Within seconds of injury, the constriction of minor vessels like arterioles, capillaries and venules and the formation of platelet plug lead to the arrest of bleeding temporarily and allows time for fibrin deposition.

Platelet Plug Formation

Platelet plug formation requires three critical events, namely: (a) platelet adhesion, (b) granule release, and (c) platelet aggregation.

- Normally platelets do not adhere to the intact endothelium. This is mainly due to endothelial prostacyclin (PGI_2) which inhibits platelet activation. However, following injury, platelets adhere to the subendothelial collagen tissue through Gp Ia/IIa and Gp VI receptors. The von Willebrand factor (vWF) also helps in adhesion via linking platelet Gp Ib/IX receptor to the subendothelial collagen. (Fig. 3.12).
- Platelets become subsequently activated and release products like thromboxane A_2 (TXA_2), ADP, vWF and fibrinogen that promote platelet aggregation.
- The released ADP changes the conformation of platelet Gp IIb/IIIa complex. The fibrinogen binds with the GpIIb/IIIa receptor on the platelets, linking adjacent platelets into a hemostatic plug. The balance between

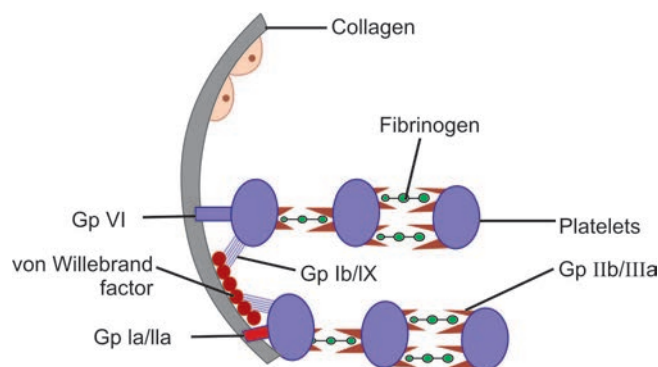


FIGURE 3.12: Platelet plug formation

the activity of TXA_2 and PGI_2 determines the rate and extent of platelet activation.

Secondary Hemostasis

Secondary hemostasis is important for bleed from larger vessels and involves activation of the plasma coagulation system as a result of which the platelet plug is stabilized by the fibrin deposition (clot formation). It requires several minutes and prevents recurrent bleeding hours or days after the initial injury.

Coagulation Mechanism

The process of coagulation involves specific plasma proteins (coagulation factors) and other substances like phospholipids, calcium and the tissue factor. Most coagulation factors are synthesized in the liver. Factors II (prothrombin), VII, IX (Christmas factor) and X (Stuart factor) require vitamin K for their synthesis.

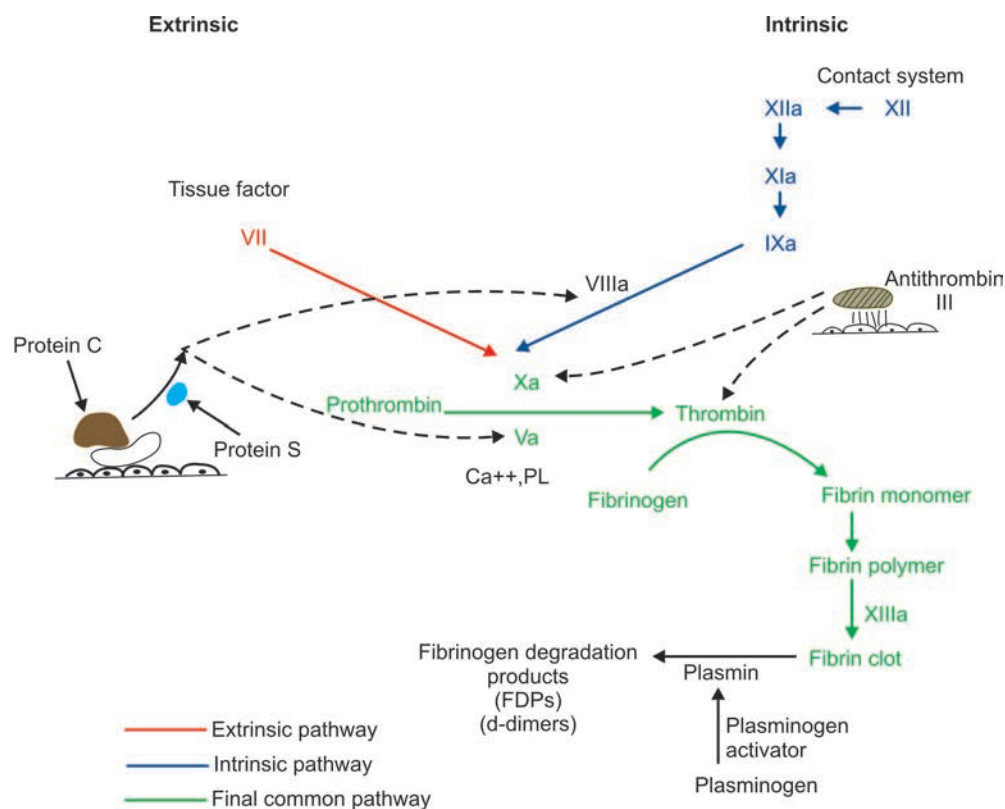
There are two pathways which initiate the process of coagulation: intrinsic and extrinsic (Fig. 3.13).

- The intrinsic pathway is activated by negatively charged surfaces and activated platelets. It involves coagulation factors XII (Hageman factor), XI, IX and VIII. The activity of the intrinsic process is assessed by determining the APTT (*activated partial thromboplastin time*).
- The extrinsic pathway is initiated by the tissue factor released from damaged cells. The tissue factor activates factor VII. This is measured by determining the PT (*prothrombin time*).

Both pathways converge to activate factor X which converts prothrombin to thrombin in the presence of factor V, calcium and phospholipids. Thrombin changes fibrinogen into fibrin monomer. The conversion of fibrinogen into fibrin can be measured by *thrombin time*. Fibrin monomers are crosslinked by activated factor XIII to form a stable fibrin polymer. An interpretation of different coagulation tests is given in Table 3.17.

Clot Lysis

The lysis of clot (fibrinolysis) and repair of vessel starts after the clot formation. Physiological fibrinolysis activators, tissue plasminogen activator (tPA) and urinary plasminogen activator (uPA or urokinase) are released from the endothelial cells and convert plasminogen into plasmin. Plasmin

**FIGURE 3.13**

degrades the fibrin polymer into small fragments, fibrin degradation products (FDPs) which are cleared by the monocyte-macrophage scavenger systems.

Coagulation Inhibitors

The formation of clot is limited to the site of injury only. This is tightly regulated by the coagulation inhibitors such as antithrombin III, protein C and protein S. These collectively maintain the fluidity of blood.

TABLE 3.17: Coagulation test abnormalities in various factor deficiencies

- | |
|--|
| a. Prolonged APTT: Factors XII*, XI, IX, VIII deficiency von Willebrand disease |
| b. Prolonged PT: Factor VII deficiency, early vitamin K deficiency |
| c. Prolonged APTT and PT: Factors II, V, X, late vitamin K deficiency, warfarin, heparin |
| d. Prolonged Thrombin time (TT): Heparin, afibrinogenemia and dysfibrinogenemia |
| e. Clot solubility in 5M urea: Factor XIII deficiency (in this PT, APTT, TT are normal) |

*no clinical bleeding occurs in Factor XII deficiency

Disorders of Hemostasis

Bleeding due to vessel or platelet disorders is generally localized to superficial sites like the skin and mucous membrane; it occurs immediately after trauma and is controlled by local measures. In contrast, bleeding due to coagulation disorders occurs in deeper tissues (subcutaneous, muscle, joint, and body cavities), hours or even days after the injury and is not affected by local therapy. The disorders of hemostasis can be classified into three types:

- Vessel wall abnormalities
 - Platelet disorders
 - Decreased platelet count (thrombocytopenia)
 - Platelet dysfunction
 - Coagulation disorders
 - Congenital: Hemophilia A and Hemophilia B
 - Acquired: Liver diseases, vitamin K deficiency, DIC and coagulation inhibitors (warfarin, heparin)
- Table 3.18 shows laboratory abnormalities in some common hemostatic disorders.

TABLE 3.18: Laboratory abnormalities in some common hemostatic disorders

Conditions	Bleeding time	Platelet count	PT	APTT	Others
Hemophilia A	Normal	Normal	Normal	Raised	Low factor VIII level
Hemophilia B	Normal	Normal	Normal	Raised	Low factor IX level
Von Willebrand's Disease	Prolonged	Normal	Normal	Raised	Decreased vWF level, abnormal platelet adhesion
Idiopathic thrombocytopenic purpura (ITP)	Prolonged	Low	Normal	Normal	Increased megakaryocytes on bone marrow examination
Disseminated intravascular coagulation (DIC)	Prolonged	Low	Raised	Raised	High FDP Low fibrinogen

Vessel Wall Abnormalities

The abnormalities of vessel wall, both congenital and acquired, may result in purpura (bleeding in the skin and mucous membrane). Platelet count, platelet function and bleeding time are normal. Important causes are senile purpura, Henoch-Schönlein purpura, vasculitis, paraproteinemia, scurvy and Ehlers Danlos syndrome.

PLATELET DISORDERS

The most common sites to observe bleeding in platelet disorders are skin and mucous membrane.

Collection of blood in skin is called purpura.

- Petechiae are small pinpoint hemorrhages into the dermis due to the leakage of red blood cells through capillaries.
- Ecchymoses are large subcutaneous collection of blood due to leakage from small arterioles or venules (common bruises).
- Deeper and palpable collection of blood is called hematoma.
- The bleeding in mucous membrane may present as gum bleeding, epistaxis, and menorrhagia.

The normal platelet count is 1,50,000 to 4,00,000/ μ L.

- The patient is usually asymptomatic if platelet count is more than 1,00,000/ μ L.
- Bleeding occurs only from severe trauma when the platelet count is 50,000 to 1,00,000/ μ L.
- There is a tendency to easy bruising and purpura after minor trauma at counts between 20,000 to 50,000/ μ L.
- Patients with counts lower than 20,000/ μ L bleed spontaneously and may have intracranial or internal bleeding.
- It is to be noted that bleeding does not always correlate with the platelet count; patient may bleed at a higher platelet count and vice versa.

The causes of thrombocytopenia are given in Table 3.19.

Idiopathic (Autoimmune) Thrombocytopenic Purpura (ITP)

This is an autoimmune disorder in which antibodies (IgG) are produced against platelet antigens (gp IIb/IIIa). The antibody-coated platelets are destroyed by phagocytic cells in the spleen.

Clinical Features

- In children, the onset is generally sudden (acute ITP). Purpura appears 2-3 weeks after viral infection. In most cases the condition is self-limiting.
- The onset is insidious in adults and the disease may persist for years (chronic ITP). Chronic ITP is more common in females.

The common presentations are skin and mucosal bleeding in the form of petechiae, ecchymosis, gum bleeding, epistaxis, and menorrhagia (Figs 3.14 and 3.15). Bleeding in gastrointestinal and genitourinary tract may also occur. Spleen is usually not palpable.

Diagnosis

- The peripheral blood examination reveals low platelet count.

TABLE 3.19: Causes of thrombocytopenia

1. Marrow disorder (decreased production)
 - a. Marrow failure – aplasia, hypoplasia
 - b. Infiltration – tumor, fibrosis
 - c. Vit B12/Folate deficiency
2. Splenic sequestration: splenic tumor and portal hypertension
3. Increased destruction of circulating platelets
 - a. Immune: ITP, SLE, drug induced, lymphoma, CLL, and HIV
 - b. Non-immune: DIC, sepsis, prosthetic valve, and TTP



FIGURE 3.14: Purpuric rashes on the face



FIGURE 3.15: Purpuric rashes on the upper limb

- Bone marrow examination shows normal morphology with increased megakaryocytes.
- Bleeding time is increased whereas clotting time, prothrombin time and APTT are normal.
- Tests are done to exclude the possibilities of HIV infection, SLE, viral hepatitis and infectious mononucleosis (EBV, CMV) since these conditions may be associated with thrombocytopenia.

Treatment

The treatment of ITP is indicated if platelet count is below 10000 to 20000/ μ L or there is extensive bleeding. The aim is to maintain the platelet count at or more than 20,000/ μ L.

- a. The initial treatment is with oral prednisolone 1-2 mg/kg/day. In most patients, platelet count improves within few days. The dose of prednisolone is tapered when the platelet count becomes normal.
- b. In case of severe and life-threatening bleeding, intravenous immunoglobulin (IV Ig) is given in the dosage of 1g/kg for 1-2 days.
- c. Splenectomy is indicated when there is no response to medical therapy or an unacceptably large dose of steroid is needed to maintain adequate platelet count. Most patients will respond to splenectomy which is curative in around 70-80% cases.
- d. Patients who fail to respond to steroid and splenectomy may be given danazol, vincristine, azathioprim, cyclophosphamide or cyclosporine. Monoclonal antibody (anti-CD20) has also been used in some cases with success.
- e. Platelet transfusion provides only a transient effect and is reserved only for cases with life-threatening bleeding.

Functional Platelet Disorders

Prolonged bleeding time despite normal platelet count indicates functional disorder of platelets. There may be defect in the platelet adhesion, aggregation or granule release. Important causes are given in Table 3.20. Commonly used drugs, aspirin and clopidogrel cause inhibition of platelet function. Aspirin irreversibly inhibits platelet cyclooxygenase I (COX -I) thus blocking synthesis of thromboxane A₂. Clopidogrel inhibits ADP-mediated platelet aggregation.

Renal Diseases

The bleeding in renal disorders is generally due to platelet dysfunction caused by low molecular weight products. Mild

TABLE 3.20: Causes of functional platelet disorders

Congenital	
•	Bernard-Soulier syndrome (defect in platelet receptor Gp Ib/IX)
•	Glanzmann's thrombasthenia (defect in platelet receptor Gp II b/IIIa)
•	von Willebrand's disease
•	Storage pool disorders
Acquired	
•	Uremia
•	Drug induced (aspirin, NSAIDs, and clopidogrel)
•	Myeloproliferative disorders

thrombocytopenia also contributes to bleeding. The treatment consists of platelet and red cell transfusion and dialysis. Desmopressin may also promote hemostasis in renal failure.

VON WILLEBRAND'S DISEASE

von Willebrand disease (vWD) is the most common inherited hemostatic disorder. It is mainly inherited in an autosomal dominant manner. von Willebrand factor (vWF) is a multimeric plasma glycoprotein synthesized by megakaryocytes and endothelial cells. It serves two major functions.

1. It is a carrier protein for factor VIII. Deficiency of the vWF leads to secondary reduction in the factor VIII level.
2. Platelets adhere to subendothelial collagen tissue through Gp Ia/IIa and Gp VI receptors. von Willebrand factor helps in adhesion via linking platelet Gp Ib/IX receptor with subendothelial collagen.

There are several subtypes of vWD.

1. The most common is type I which is characterized by mild to moderate decrease in the plasma vWF.
2. The vWF is qualitatively abnormal in type II vWDs, although its plasma level is normal.
3. Type III vWD is the most severe form where vWF is nearly absent.

Clinical Features

In most cases, mild symptoms in form of epistaxis, gum bleeding, menorrhagia and superficial bruises may be present. Gastrointestinal bleeding may also occur. Excessive hemorrhage may be observed after trauma, dental extraction or surgery. Unlike hemophilia, hemarthrosis (bleeding in joints) do not occur.

Investigations

- Platelet count is normal.
- Bleeding time and APTT are prolonged.
- The factor VIII activity is low.
- The plasma vWF level is reduced.
- The platelet adhesion activity of vWF as measured by ristocetin cofactor assay is reduced.

Treatment

- a. Mild bleeding episodes can be managed by giving desmopressin (DDAVP) which increases the vWF level.

- b. Cases with persistent and severe bleeding are managed with factor VIII concentrate that also contains considerable amount of vWF.
- c. Cryoprecipitate (rich in factor VIII and vWF) can be used alternatively.
- d. Antifibrinolytic agent (e.g. tranexamic acid) is useful as an adjunctive therapy during dental procedures. The dose of tranexamic acid is 25 mg/kg thrice a day for 5-7 days.

COAGULATION DISORDERS

Inherited Coagulation Disorders

In inherited coagulation disorders, only a single coagulation factor is involved (reduced or defective). The most common inherited coagulation abnormalities are factor VIII deficiency (hemophilia A) and factor IX deficiency (hemophilia B).

Hemophilia

Hemophilia is an X-linked recessive disorder characterized by the deficiency of factor VIII (Hemophilia A) or factor IX (Hemophilia B). The disease manifests in males while females are carriers (Fig. 3.16). Rarely it may also manifest in females in case the father is a hemophilic and mother is carrier or if the patient has Turner's syndrome (45 XO).

Clinical Manifestations

The clinical manifestation of hemophilia depends on the severity of deficiency of the factor VIII or IX.

The bleeding is spontaneous in severe cases (Factor activity <1%) while it may occur after minor trauma or surgery in moderate cases (level 1-5%).

- People with mild disease (factor activity 5-40%) rarely bleed spontaneously but can bleed following severe trauma or surgery.
- Severe cases generally manifest in early childhood when the child starts crawling or walking whereas mild and moderate cases may clinically manifest later in life.
- The bleeding occurs mainly in joints (hemarthrosis), muscles (hematoma), viscera or in retroperitoneum but it can involve any organ system.
- The most common joint involved is the knee joint (Fig. 3.17). Ankle, elbow and hip joints are also involved. Warmth, pain and swelling are initial presentations of

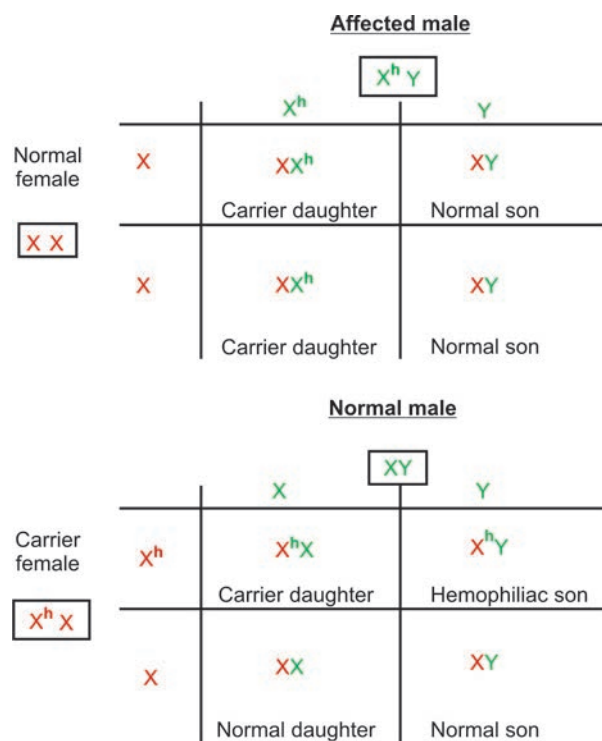


FIGURE 3.16: Genetics of hemophilia



FIGURE 3.17: Hemarthrosis of the knee

joint involvement. Recurrent bleeding in the joint can lead to synovial thickening, destruction of cartilage, fibrosis and deformity of joints.

- Calf and psoas muscles are most commonly affected with hematoma. Bleeding into iliopsoas muscle often causes femoral nerve palsy. Patients can also develop large calcified mass of blood with inflammation that may be mistaken as tumor (pseudotumor syndrome).
- Hematuria is also common in hemophilia.
- Intracranial and oropharyngeal bleeding may be fatal.

Investigations

- Activated partial thromboplastin time (APTT) is prolonged.
- Prothrombin time (PT), bleeding time and platelet count are within normal limits.
- Diagnosis is confirmed by the assay of factor VIII or IX.
- *Prenatal diagnosis* can be made by chorionic villi biopsy or amniocentesis.

Management

The management includes supportive and specific therapy.

Supportive Therapy

- The joint is immobilized with splint to reduce pain and bleeding.
- Ice cold packs are applied to the involved joints intermittently for few hours. This helps to arrest the bleeding.
- Physiotherapy is indicated once active bleeding stops to prevent joint deformity.
- Non-steroidal anti-inflammatory drugs and intramuscular injections should be avoided.

Specific Therapy

- Specific therapy of hemophilia A includes intravenous administration of factor VIII. The dose, frequency and duration of factor administration depend on the severity of the bleeding. A dose of 1 unit/kg factor VIII raises the activity of the factor by 2%. An increase to around 20-50% activity is generally needed to control bleeding. The chances of transmission of infections such as hepatitis, Parvovirus and HIV with uses of plasma derived factor can be eliminated by the use of recombinant factor.

- In case the factor is not available, cryoprecipitate which is rich in factor VIII, fibrinogen and vWF may be used.
- Mild hemophiliacs may benefit from desmopressin (DDAVP) which increases factor level to 2-3 folds.
- Antifibrinolytic agents like ϵ -amino caproic acid (EACA) and tranexamic acid can be used in minor dental and oral mucosal bleed. These agents inhibit plasminogen activator in oral tissue and stabilize the clot formation. EACA is given orally or IV in dosage of 100 mg/kg upto 10g initially followed by 50 mg/kg upto 5 g 6 hourly for 2-7 days. EACA can also be used as a mouthwash. Tranexamic acid can be given in dosage of 0.5-1 gm 8 hourly orally. The filling of carious tooth requires single infusion of factor VIII combined with antifibrinolytic agents for 3-4 days after the procedure. In case of major oral and periodontal surgery and extraction of permanent teeth, the patient needs hospitalization and regular infusion of factor VIII along with EACA.

Therapy of Hemophilia B

The specific treatment of hemophilia B is the intravenous administration of factor IX. Desmopressin and cryoprecipitate are not effective.

Acquired Coagulation Disorders

Acquired coagulation disorders are common and generally complex. There is deficiency of multiple coagulation factors. Most common acquired coagulation disorders are DIC, hemorrhagic diathesis due to liver disease, vitamin K deficiency and complications of anticoagulant therapy.

Liver Diseases

- Most of the coagulation factors are synthesized in liver. Hence, there may be deficiency of these factors in liver diseases.
- Vitamin K absorption is reduced in cholestatic liver diseases leading to decreased synthesis of vitamin K dependant factors (II, VII, IX, X).
- The bleeding in liver diseases is compounded by the presence of anatomical lesions like esophageal varices, gastritis and peptic ulcer disease.
- Thrombocytopenia can occur in patients having chronic liver disease with splenomegaly (*hypersplenism*).

Most patients have prolonged PT and APTT. Prolongation in PT correlates with bleeding risk. Treatment includes parenteral vitamin K and infusion of fresh frozen plasma as it contains all coagulation factors.

Protein C, protein S and antithrombin III are also synthesized in the liver. Deficiency of these coagulation inhibitors favours coagulation and can predispose the patients to develop DIC.

Disseminated Intravascular Coagulation (DIC)

Disseminated intravascular coagulation is characterized by widespread coagulation in the microcirculation leading to consumption of coagulation factors and platelets.

Mechanism: Endothelial damage in septicemia and other situations cause release of the endothelial tissue factors which activate the coagulation process. Thromboplastin released from placenta, damaged tissue and injured brain also activates the coagulation cascade. The formation of fibrin stimulates the fibrinolytic system generating fibrin degradation products (FDP). The deficiency of coagulation factors and platelets leads to hemorrhagic manifestation which is further exacerbated by the activation of the fibrinolytic system. The important causes of DIC are mentioned in Table 3.21.

Clinical Features

- Since DIC leads to both thrombosis and bleeding, the manifestations are either in the form of tissue infarction, digital ischemia and gangrene or bleeding at various sites. Spontaneous oozing from venipuncture sites or wounds is an important clue to the diagnosis of DIC.

TABLE 3.21: Causes of DIC

Infections
• Gram-negative bacterial infections
• Gram-positive bacterial infections
• Fungal infections
• Malaria
Obstetric
• Retained dead fetus
• Amniotic fluid embolism
• Abruptio placentae
Malignancies
• Lung, pancreas, prostate cancer
• Acute promyelocytic leukemia
Tissue injury
• Burns
• Head injury

- DIC can present as recurrent deep or superficial vein thrombosis, particularly in cancer patients.
- Microangiopathic hemolysis can lead to anemia.

Investigation

The DIC is suspected when there is a suggestive clinical situation (as mentioned in Table 3.21) along with following laboratory findings:

- Low platelet count
- Increased PT and APTT
- Low plasma fibrinogen level
- Elevated plasma FDP (d-Dimer assay)
- Schistocytes (fragmented RBC) on peripheral blood smear.

Management

This includes the following steps:

- Prompt management of underlying disorders such as:
 - antibiotics in septicemia
 - delivery of fetus in obstetric complications
- Patients with bleeding manifestations should receive fresh frozen plasma (FFP) to replace coagulation factors and platelet transfusion to correct thrombocytopenia.
- Heparin is indicated in patients where the predominant manifestation includes thrombosis and gangrene. Role of heparin in patients with bleeding manifestations is controversial. However, it may be given in such situations where the bleeding is not controlled despite adequate replacement with FFP and platelets.

Agranulocytosis

The mean blood neutrophil count is around 3600/ μ l. The important function of neutrophil is to prevent and contain bacterial and fungal infections. The decrease in neutrophil count is called neutropenia (<1500/ml) The neutrophil count below 500/ml is called severe neutropenia. Agranulocytosis term is used to describe a state of severe neutropenia or absence of circulating neutrophils.

Causes

Neutropenia can occur due to:

- decreased production
- increased destruction
- excessive peripheral pooling of neutrophils.

Important causes of neutropenia are given in Table 3.22.

TABLE 3.22: Important causes of neutropenia

Drug induced

- Anti-cancer drugs, antibiotics (sulphonamides, chloramphenicol), anticonvulsants (carbamazepine), and antithyroid drugs

Hematological diseases

- Aplastic anemia, acute leukemia, myelofibrosis, tumor invasion and megaloblastic anemia

Infections

- Tuberculosis, typhoid, kala azar, malaria, infectious mononucleosis, and HIV

Autoimmune

- Systemic lupus erythematosus, and Felty's syndrome
- Congenital*
- Cyclic neutropenia

Manifestations

The manifestations depend on the severity and duration of neutropenia and the type of underlying illness. Oral ulcers are usual in agranulocytosis. The patient develops fever and infections mainly of oropharynx, perirectal area, sinuses, lungs and skin. Common organisms responsible for infections are gramnegative enteric bacilli, *Pseudomonas* spp., *Staphylococcus* spp., *Candida* and *Aspergillus*.

A careful history of exposure to drugs, toxins and duration of illness should be asked. A careful examination of oropharynx and perirectal area should be done. The presence of lymphadenopathy and hepatosplenomegaly should also be noted.

Investigations

Peripheral blood examination reveals the degree of neutropenia. It can also reveal the presence of hematological malignancies. Bone marrow examination is necessary to diagnose aplasia, infiltration, fibrosis and hematological malignancies. Investigations for the presence of auto-antibodies are conducted as needed.

Management

Patients with neutropenia and fever must be admitted to the hospital. Sample for culture should be obtained and empirical parenteral broad spectrum antibiotics must be started immediately. Antifungal agents are to be given if fever does not respond to antibiotics. Neutrophil transfusion may help in severe refractory infections.

Prevention of infection is very crucial. Hand-washing is the best method for preventing the spread of infections.

Growth factors (G-CSF or GM-CSF) are used to reduce the chance and severity of neutropenia after chemotherapy and radiation therapy.

SPLENOMEGALY

Spleen is a reticuloendothelial organ which lies in the left upper quadrant of abdomen. The important functions of spleen are:

- Removal of old and defunct red blood cells from circulation
- Synthesis of antibodies in the white pulp
- Removal of antibody-coated bacteria and blood cells
- Spleen can form blood cells when the bone marrow is unable to meet the demand (extramedullary hematopoiesis).
- It also serves as a reservoir (store) for platelets and neutrophils.

Increase in the normal function can lead to splenomegaly. Causes of splenomegaly are given in Table 3.23.

The spleen is normally not palpable. If palpable, it is enlarged (Fig. 3.18). The direction of enlargement is towards right iliac fossa. The maximum diameter of spleen on ultrasonographic assessment is around 13 cm. Massive splenomegaly is defined as the spleen palpable more than 8 cm below costal margin. The common causes of massive splenomegaly are mentioned in Table 3.24.

TABLE 3.23: Causes of splenomegaly

Hyperplasia in response to infection

- Malaria and kala azar
- Infectious mononucleosis, cytomegalovirus infection, HIV infection, and viral hepatitis
- Endocarditis, tuberculosis, typhoid fever, and septicemia
- Histoplasmosis

Hyperplasia due to excessive function of red cell removal

- Spherocytosis, thalassemia, and early sickle cell disease

Hyperplasia in response to immune disorder

- Felty's syndrome, SLE, and sarcoidosis

Extramedullary hematopoiesis

- Myelofibrosis and marrow infiltration

Congestive splenomegaly

- Portal hypertension (cirrhosis, hepatic vein obstruction, portal vein thrombosis)

Infiltration of spleen

- Leukemias, lymphomas, Myeloproliferative disorders, amyloidosis, and storage diseases (Gaucher's disease, Niemann-Pick disease).

TABLE 3.24: Causes of massive splenomegaly

- Chronic myeloid leukemia
- Portal hypertension
- Myelofibrosis
- Malaria
- Kala azar

Clinical Manifestations

The symptoms due to splenomegaly are abdominal discomfort and pain. Massive splenomegaly may cause early satiety and abdominal bloating due to abdominal compression. Splenic infarction may result in severe pain radiating to the left shoulder. Repeated splenic infarction in sickle cell disease may lead to splenic destruction (autosplenectomy). Rupture of enlarged spleen can occur spontaneously or due to trauma and this may be fatal. Pancytopenia may occur in patients with splenomegaly (hypersplenism).

Diagnosis

A detailed history and clinical examination are helpful in knowing the underlying cause of splenomegaly. One should particularly look for the presence of fever, bleeding, lymphadenopathy and hepatomegaly. The following investigations can be helpful in making a diagnosis:

1. Complete blood count
2. Bone marrow examination
3. Imaging (ultrasonography, CT scan)
4. Screening for infections and autoimmune diseases
5. Endoscopy of the upper gastrointestinal tract.



FIGURE 3.18: Splenomegaly

LYMPHADENOPATHY

Enlarged lymph nodes may be localized or generalized. The Table 3.25 shows important causes of lymphadenopathy. Generalized lymphadenopathy is defined as the enlargement of three or more noncontiguous lymph node areas. Inguinal lymph nodes of upto 2 cm size and submandibular lymph nodes of upto 1 cm are generally considered normal.

- Lymphadenopathy may be the presenting symptom or it may be noticed incidentally when examined for other reasons.
- In more than two-third of the cases, the cause of lymphadenopathy is nonspecific or reactive.
- The enlargement of supraclavicular and scalene lymph nodes is always abnormal. The left supraclavicular node can be enlarged in metastasis from GI malignancy (Virchow's node, Fig 3.19).

TABLE 3.25: Causes of lymphadenopathy

Infections

- Viral: EBV, CMV, and HIV
- Bacterial: Streptococci, Staphylococci, and tuberculosis
- Spirochaetal: syphilis
- Fungal: histoplasmosis, and coccidioidomycosis
- Parasitic: filariasis, kala azar, and toxoplasmosis

Malignancies

- Primary: lymphomas, ALL, and CLL
- Metastasis: from various primary sites

Immunological diseases

- Rheumatoid arthritis, SLE, and drugs (phenytoin)

Miscellaneous

- Sarcoidosis and amyloidosis



FIGURE 3.19: Left supraclavicular lymphadenopathy (Virchow's node)



FIGURE 3.20: Cervical lymphadenopathy

- The most common site of localized lymphadenopathy is the neck (Fig. 3.20) which is generally due to upper respiratory infections, oral and dental lesions and viral illnesses (infectious mononucleosis and others). In India, tuberculosis is an important cause of lymphadenopathy.
- Metastasis from cancer of head and neck, lung and thyroid may also lead to cervical lymphadenopathy.
- Generalized lymphadenopathy may be found in viral infections (due to EBV, CMV, HIV), connective tissue disorders (SLE) and malignancies (ALL, CLL, lymphomas). Malignancy should be considered in elderly patients with lymphadenopathy.

Diagnosis

History: A detailed history of the presence of fever, weight loss, cough, sore throat, pain in lymph nodes, occupation, sexual behavior and drug intake is taken.

Examination:

- The lymph nodes are examined for the size, texture, tenderness, mobility and for the signs of inflammation over the node (Fig. 3.21).
 - Hard, fixed and nontender lymph nodes are metastatic.
 - Rubbery lymph nodes are found in lymphoma.



FIGURE 3.21: Examination of cervical lymph node

- Matted lymph nodes are characteristic of tuberculosis.
- b. The presence of skin lesions, hepatomegaly and splenomegaly should be noted.
- c. Careful oral, dental and throat examination is performed in cases with cervical lymphadenopathy.
- d. Abdominal and mediastinal lymphadenopathy may only be detected by imaging.

Investigations: The following investigations are helpful in making a diagnosis:

1. Complete blood count
2. Lymph node biopsy/FNAC
3. X-ray chest, ultrasonography, CT scan
4. Bone marrow examination
5. Screening for infections and autoimmune illnesses

IMPLICATIONS ON DENTAL PRACTICE

1. Elective oral surgical and periodontal procedures should be avoided in patients with severe anemia. Increased bleeding and impaired wound healing may occur in presence of anemia.
2. Glossitis, burning mouth, angular stomatitis and aphthous stomatitis can be early manifestations of iron, vitamin B₁₂ or folate deficiency. These may occur before the appearance of anemia.
3. Iron syrups may lead to staining of teeth. This can be prevented by using sodium ironedetate.
4. General anesthesia is not administered if Hb level is <10 g/dL. Hemolytic anemias, particularly sickle cell disease may pose difficult problem during general anesthesia.
5. Painful infarcts in the jaws in sickle cell disease may be mistaken for toothache or osteomyelitis.
6. Patients with splenectomy are more prone to infections. Hence, dental procedures in such patients should be done under prophylactic antibiotics.
7. Infection and bleeding pose major problem in the dental management in patients with aplastic anemia. Gingival bleeding may be controlled by the use of local hemostatic measures and use of systemic antifibrinolytic agents (tranexamic acid, aminocaproic acid). Oral infection can be reduced by use of chlorhexidine oral rinses.
8. Cyclosporin (used in patients with aplastic anemia) can result in gingival swelling.
9. Intramuscular injections and nerve block anesthesia are avoided in patients with thrombocytopenia. However, intraligamentary anesthesia can be used safely.
10. Oral ulcers, advanced periodontal disease, pericoronitis and pulpal infections can lead to life-threatening septicemia in patients with severe neutropenia. The patients should be given appropriate antibiotics and mouth rinses.
11. Oral bleeding, oral ulcers, gingival infiltrates, oral infections and cervical lymphadenopathy may be the presenting features of leukemias. Hence, the dentist may be the first clinician to suspect the disease.
12. Spontaneous gingival bleeding is common when platelet count is below 20,000/mm³.
13. A history of any bleeding manifestations (including family history) must be asked in patients undergoing dental procedures. In case of such history, investigations should be performed to diagnose or exclude hemostatic disorders.
14. A history of anticoagulant therapy must be noted. Non-surgical treatment can be carried out provided the PT/ INR is not grossly above the therapeutic range and trauma is minimal. Surgical treatment is not recommended for those who have INR >3.5. At an INR <3.5 where bleeding is expected (in procedures like multiple extractions, removal of wisdom teeth, full mouth or arch extraction) local measures should be used along with the reduction of INR to 2-3. Extensive flap surgery or multiple bony extraction may require an INR of <1.5.
15. The transfusion of blood and blood components should be done if absolutely necessary to avoid the risk of

transfusion-associated infections such as HIV, hepatitis B, hepatitis C, syphilis, malaria, cytomegalovirus and parvovirus.

SELF ASSESSMENT

Multiple Choice Questions

- 1. Raised PT and PTT are seen in:**
 - A. Thrombocytopenia
 - B. vWD
 - C. DIC
 - D. Hemophilia.
- 2. Which of the following manifestations is characteristic of hemophilia:**
 - A. Gum bleeding
 - B. Petechiae
 - C. Hemarthrosis
 - D. Epistaxis
- 3. Gum hypertrophy can be seen in:**
 - A. AML
 - B. CLL
 - C. CML
 - D. ALL
- 4. Gum bleeding can occur in:**
 - A. ITP
 - B. Acute leukemia
 - C. vWD
 - D. All of the above
- 5. Microcytic hypochromic anemia is seen in all except:**
 - A. Thalassemia
 - B. Iron deficiency
 - C. Folate deficiency
 - D. Pyridoxine deficiency
- 6. In hemolytic anemia the following can occur:**
 - A. Indirect hyperbilirubinemia
 - B. Splenomegaly
 - C. Gallstones
 - D. All of the above
- 7. Which is true in vWD:**
 - A. Normal PT
 - B. Normal APTT
 - C. Decreased platelet count
 - D. Raised plasma FDP
- 8. The most preferred way to assess RBC production is to measure:**
 - A. Serum ferritin
 - B. Reticulocyte count
 - C. RBC half life
 - D. Total RBC count
- 9. Following may be seen in megaloblastic anemia:**
 - A. Paresthesia
 - B. Macrocytosis
 - C. Smooth tongue
 - D. All of the above
- 10. Plummer Vinson syndrome is seen in:**
 - A. Iron deficiency
 - B. Esophageal carcinoma
 - C. Alcohol withdrawal
 - D. Lead poisoning
- 11. Following test is used to examine the intrinsic coagulation pathway:**
 - A. PT
 - B. PTT
 - C. Both of the above
 - D. None of the above
- 12. The bleeding due to oral anticoagulation therapy is managed by:**
 - A. Aspirin
 - B. Vitamin K
 - C. Factor VIII
 - D. Platelet transfusion
- 13. Most of the coagulation factors are synthesized by:**
 - A. Kidneys
 - B. Bone marrow
 - C. Liver
 - D. Capillary endothelial cells
- 14. Most of the bilirubin is formed in:**
 - A. Liver
 - B. Spleen
 - C. Gallbladder
 - D. Red blood cells
- 15. Platelet dysfunction can be found in following except:**
 - A. vWD
 - B. Uremia
 - C. Aplastic anemia
 - D. Aspirin therapy
- 16. The following can be found in aplastic anemia except:**
 - A. Thrombocytopenia
 - B. Splenomegaly
 - C. Low reticulocyte count
 - D. Infections
- 17. Which one of the following is seen in ITP:**
 - A. Raised bleeding time
 - B. Raised PT
 - C. Raised PTT
 - D. High plasma FDP
- 18. A 20-year girl develops excessive and prolonged bleeding following tooth extraction. Tests reveal raised PTT, normal PT, normal platelet count. The most probable diagnosis is:**
 - A. vWD
 - B. ITP
 - C. Hemophilia
 - D. DIC
- 19. Following are vit K dependent coagulation factors:**
 - A. VIII, IX, and X
 - B. II, VII, IX, X
 - C. II, V, VII, IX
 - D. V, VII, X, XII
- 20. The mechanism of action of tranexamic acid is:**
 - A. Anticoagulation
 - B. Antifibrinolysis
 - C. Enhancing platelet aggregation
 - D. All of the above
- 21. The following are found in DIC:**
 - A. Thrombocytopenia
 - B. Excessive bleeding
 - C. Thrombosis
 - D. All the above
- 22. Presence of gum bleed, infections, sternal tenderness and low platelet count suggests:**
 - A. ITP
 - B. Acute leukemia
 - C. Sickle cell disease
 - D. Aplastic anemia
- 23. The following may retard iron absorption except:**
 - A. Phytates
 - B. Calcium
 - C. Vitamin C
 - D. Achlorhydria
- 24. Following can be found in hemolysis except:**
 - A. Low plasma haptoglobin
 - B. Raised bilirubin
 - C. Hemoglobinuria
 - D. Low reticulocyte count
- 25. Which of the following is generally given in hemolytic anemia:**
 - A. Parenteral iron
 - B. Folic acid
 - C. Cobalamin
 - D. Vitamin E
- 26. Bleeding time is prolonged in:**
 - A. Thrombasthenia
 - B. Thrombocytopenia
 - C. Uremia
 - D. All of above

27. The following can be found in iron deficiency anemia except:

- A. Microcytosis B. Koilonychia
- C. Raised reticulocyte count
- D. Low plasma ferritin

28. The therapy of choice in bleeding secondary to liver dysfunction is:

- A. Cryoprecipitate B. Fresh frozen plasma
- C. Factor VIII D. Platelet transfusion

Fill in the Blanks

1. The most common type of leukemia in children is _____.
2. Philadelphia chromosome is present in patients of _____.
3. Oral anticoagulant therapy is monitored by _____ test.
4. PTT is used to monitor _____ therapy.
5. Koilonychia is seen in _____.
6. The plasma level of haptoglobin is _____ in hemolytic anemia.
7. Imatinib mesylate is used in the therapy of _____.
8. Spontaneous bleeding can occur in thrombocytopenia if platelet count falls below _____.
9. Reed-Sternberg's cells are seen in _____.
10. Coombs' test is positive in _____ anemia.
11. Most common inherited bleeding disorder is _____.
12. Hard, fixed and non-tender lymphadenopathy suggests a diagnosis of _____.
13. Virchow's node is _____.
14. Fish tapeworm can cause _____ deficiency anemia.
15. Vegetarians are prone to develop _____ deficiency.
16. Massive splenomegaly is defined as _____.
17. Localized lymphadenopathy most commonly occurs in _____ region.
18. Matted lymph nodes are seen in _____.
19. The daily dose of oral iron for treatment of iron deficiency anemia is _____.
20. The MCV is _____ in pernicious anemia.
21. The most common cause of microcytic hypochromic anemia is _____.
22. Normal absolute eosinophil count is _____.
23. The life span of RBC is _____.
24. Hypersegmented neutrophils are characteristically found in _____ anemia.
25. Schilling test is useful in the diagnosis of _____ anemia.

Chapter

4

Cardiovascular System

SYMPTOMS AND SIGNS OF CARDIOVASCULAR DISEASES

Symptoms

Following are important symptoms in patients with cardiac diseases:

- Dyspnea
- Chest pain
- Palpitation
- Oedema
- Syncope
- Fatigue

Dyspnea

Dyspnea is an abnormally uncomfortable awareness of breathing. Main causes of dyspnea are diseases of cardiovascular system and pulmonary system (see also chapter 5). In cardiovascular system this is mostly due to heart failure.

In heart failure elevated pulmonary venous and capillary pressure leads to pulmonary congestion. Pulmonary congestion leads to decreased lung compliance and increased airway resistance. Hence, an extra effort is needed to ventilate lungs. Other mechanisms of dyspnea are respiratory muscle fatigue and acidosis. Rapid, shallow breathing is characteristic of cardiac dyspnea.

Dyspnea is classified in four grades depending on severity, that is, level of exertion required to provoke its onset (Table 4.1). The breathlessness in cardiac patients may be on exertion, at rest or in the supine position (orthopnea).

Orthopnea: Orthopnea, breathlessness on lying flat, is due to redistribution of fluid from abdomen and lower extremities to chest in recumbent position (increased venous return) and elevation of diaphragm.

TABLE 4.1: The New York Heart Association (NYHA) grading of dyspnea

Grade 1	No breathlessness
Grade 2	Breathlessness on severe exertion
Grade 3	Breathlessness on mild exertion
Grade 4	Breathlessness at rest

Paroxysmal nocturnal dyspnea (PND): PND is a condition when patients may wake up in the night with complaints of severe breathlessness and wheezing. Patient is relieved by sitting upright or standing. Patient usually opens window or door for fresh air. PND is a variant of orthopnea and is due to same mechanisms. In advanced failure, patient may not be able to lie supine at all.

Palpitation

Palpitation is an unpleasant awareness of the beating of the heart. It may occur due to abnormalities in the rate, rhythm or force of contraction of the heart.

- Important causes of palpitation are atrial, junctional and ventricular tachyarrhythmias.
- Rapid irregular palpitation is due to atrial fibrillation.
- Other causes are extrasystole, high cardiac output states (anemia, thyrotoxicosis) and valvular regurgitation (mitral regurgitation, aortic regurgitation).

Chest Pain

Chest pain is the common presentation in the cardiac disease, although it can also occur in diseases of the lungs, chest wall or in anxiety states (Table 4.2).

- The chest pain due to ischemic heart disease typically is constricting, heavy or choking in character, occurs in retrosternal area, may radiate to left arm or shoulder and is provoked during exertion. The pain may be

accompanied by sweating, nervousness, breathlessness or marked anxiety (see Chapter 1: Table 1.1).

- The chest pain may also occur due to aortic dissection, pericarditis, pneumothorax or esophagitis.
- The pain in pericarditis is sharp central or to the left side of the chest which increases on deep inspiration and during coughing or postural changes.

Fatigue

Fatigue is an important symptom of cardiac failure. It is due to decreased oxygen delivery to muscles.

Edema

Edema in heart failure is a manifestation of systemic venous congestion and increased salt and fluid retention which occurs as compensatory mechanism. Edema generally presents first in the dependent parts of the body. This is usually bilateral and presents initially in feet, and is more marked at the end of the day (Fig. 4.1). However, edema may be localized in the sacral area in bed-ridden patients (see also Chapter 1).

Syncope

Syncope is defined as transient loss of sensorium due to decreased cerebral blood flow. There is generally a spontaneous recovery. Important cardiac causes of syncope are arrhythmias, aortic stenosis and massive myocardial



FIGURE 4.1: Bilateral pedal edema (pitting type)

infarction. A syncopal attack due to heart blocks is known as *Stokes-Adams attacks*. (see page 114)

Hemoptysis

Patients with mitral stenosis may have hemoptysis. Pink frothy sputum associated with sudden severe breathlessness and cough signifies pulmonary edema.

General Examination

The clinical examination in a cardiac patient should include a complete general examination together with a systemic cardiac examination. Other relevant systems are also examined for making a complete diagnosis.

A thorough examination is mandatory. One should specifically look for chest deformities, anemia, cyanosis, clubbing of fingers and toes, edema, hepatomegaly, cutaneous signs of infective endocarditis, arterial pulses, jugular venous pulsations and blood pressure.

Anemia

Anemia may be a precipitating factor in ischemic heart disease and heart failure. It can be present in infective endocarditis.

Cyanosis

Cyanosis is a bluish discoloration of the skin and mucous membrane caused by increased concentration of reduced hemoglobin (>4 g %) in superficial blood vessels. Cyanosis is looked for at lips, nailbeds, malar area, ear lobes and the mucous membrane of the oral cavity. Cardiac causes of cyanosis are cyanotic congenital heart disease (Fallot's

TABLE 4.2: Causes of chest pain

- | |
|-------------------------|
| a. Cardiac causes |
| • Myocardial infarction |
| • Pericarditis |
| • Dissection of aorta |
| b. Non-cardiac causes |
| • Functional |
| • Disease of thorax |
| Herpes zoster |
| Costochondritis |
| Myalgia |
| Pleurisy |
| Pneumothorax |
| Pulmonary embolism |
| c. Extrathoracic causes |
| • Esophagitis |
| • Peptic ulcer |
| • Cholecystitis |
| • Pancreatitis |

tetralogy, Eisenmenger's syndrome) and pulmonary edema. Cyanosis in congestive heart failure can be both central and peripheral types (see also Chapter 1).

Edema

Edema which pits on digital pressure is a feature of cardiac failure (Fig. 4.1). Later the edema may progress to anasarca.

Clubbing

Clubbing of fingers and toes is found in cyanotic heart disease and infective endocarditis (Fig. 4.2).

Peripheral Signs of Infective Endocarditis

Petechial hemorrhages, splinter hemorrhages in the nailbed, Osler's nodes (tender erythematous nodules at the finger pulp), Janeway lesions (painless red lesions of the palms) and Roth's spots (erythematous lesion on ocular fundi) should be looked for as an evidence of infective endocarditis.

Arterial Pulse

Arterial pulse should be examined for the rate, rhythm, volume, character and radio-femoral delay. The rate and rhythm are assessed by palpating radial artery (Fig. 4.3). The character of the pulse is better assessed by palpating the carotid artery (Fig. 4.4). The pulse rate is



FIGURE 4.2: Clubbing in cyanotic heart disease



FIGURE 4.3: Palpation of the radial artery



FIGURE 4.4: Palpation of the right carotid artery with the left thumb

determined by counting it for at least 30 seconds. The normal pulse rate varies from 60-100 per minute (for detail, see Chapter 1).

Blood Pressure

Blood pressure is measured in both arms and also in the lower limb. In coarctation of aorta, the arterial pressure in the upper limb is much higher than in lower limbs. The pulse pressure is wide in AR, anemia and pregnancy whereas it is narrow in AS. Hypotension may occur in cardiac failure (see Chapter 1).

Jugular Venous Pulse

A raised jugular venous pressure (JVP) is a sign of right heart failure. The JVP may rise during inspiration in constrictive pericarditis and cardiac tamponade (Kussmaul's sign). "a" wave is absent in atrial fibrillation whereas it is prominent in tricuspid stenosis (TS). Prominent "Y" descent is seen in tricuspid regurgitation (see Chapter 1).

Others

- *Inspiratory* crepitations over the lung bases are present in patients with heart failure. In patients with pulmonary edema, coarse crepitations are heard widely along with wheezing.
- *Tender hepatomegaly* may be present in cases with heart failure.
- *Splenomegaly* can be found in subacute infective endocarditis.
- *Ascites* may occur in severe heart failure.
- *Abdominal bruit* over renal area may be found in hypertension due to renal artery stenosis.

CARDIAC EXAMINATION

Inspection and Palpation

- The lowermost and outermost distinct cardiac pulsation is known as cardiac impulse or apex beat. It is normally situated in the fifth intercostal space just medial to mid clavicular line. The apex beat may be shifted inferiorly or laterally in cardiac enlargement. The apex beat is insignificant in cases of pericardial effusion and mitral stenosis (MS) whereas prominent in mitral regurgitation (MR), aortic regurgitation (AR) and aortic stenosis (AS). The apex beat is palpated to determine its location and character. It is "tapping" in character in MS while thrusting in AS and AR.

- Pulsations in left parasternal area and epigastrium may be seen in cases of right ventricular enlargement.
- Pulmonary component of second heart sound (P_2) is palpable in second left parasternal area in case of pulmonary hypertension.
- A thrill (palpable murmur) may be palpable in systolic phase (MR, AS, ventricular septal defect) or diastolic phase (MS, AR) in corresponding areas.

Auscultation

- The first heart sound (S_1) corresponds to the closure of mitral and tricuspid valves. It is loud in MS and muffled in MR.
- The second heart sound corresponds to the closure of aortic (A_2) and pulmonary valve (P_2). The pulmonary component of the second heart sound (P_2) is loud in pulmonary hypertension and soft in pulmonary stenosis. Aortic second sound (A_2) is soft in AS. There is a wide and fixed split of second heart sound in atrial septal defect (ASD).
- The presence of third heart sound (S_3) in children may be physiological. However, after the age of 40 yrs it may signify ventricular failure.
- The fourth heart sound (S_4) occurs due to vigorous atrial contraction with a noncompliant ventricle (as in hypertension, AS).
- Opening snap (OS) is audible due to forceful opening of the mitral valve in diastolic phase preceding the murmur and is heard in cases of noncalcific mitral stenosis.
- Ejection click may be heard due to opening of the stenosed aortic or pulmonary valve.

The murmurs are found because of turbulent flow within the heart and great vessels. An increased flow across the normal valve may also generate murmur (innocent or functional murmur). The murmurs are generally defined by characters such as site, loudness, quality, timing and radiation. Different murmurs found in heart diseases are mentioned in Tables 4.3 and 4.4.

TABLE 4.3: Auscultatory findings in rheumatic valvular heart disease

Mitral stenosis	Loud S_1 , Loud P_2	Opening snap	Mid-diastolic murmur
Mitral regurgitation	Soft S_1 , Loud P_2		Pansystolic murmur
Aortic stenosis	Soft A_2	Ejection click	Ejection systolic murmur
Aortic regurgitation	Soft A_2		Early diastolic murmur

TABLE 4.4: Types of murmur and their causes

Pansystolic	MR, TR, VSD
Ejection systolic	AS, PS
Early diastolic	AR, PR
Mid-diastolic	MS, TS, Austin Flint in AR
Continuous	PDA

INVESTIGATIONS

Chest X-ray

Chest X-ray is useful in detecting cardiomegaly (Fig. 4.5). Dilatation of individual chambers can also be recognized on X-ray of the chest. It may also reveal signs of increased pulmonary blood flow (pulmonary plethora) and pulmonary hypertension.

Electrocardiogram (ECG)

The electrical activity in the heart can be recorded at the surface with the help of electrodes and is reflected in waveforms (P, Q, R, S, T). The P wave denotes the atrial depolarization while QRS shows the ventricular depolarization. The T wave reflects the ventricular repolarization. A normal ECG pattern is shown in Figure 4.6. The ECG is helpful in diagnosing ischemic heart disease. It can also indicate the presence of cardiomegaly and arrhythmias. In stress ECG test, the recording is made while the patient is exercising on a treadmill. This is used to confirm the diagnosis of angina and to evaluate patients with IHD for the purpose of management.

Echocardiography

This very useful noninvasive procedure is employed to diagnose structural and functional abnormalities of the heart and the presence of cardiac vegetations. Doppler echocardiography can detect the abnormal direction of flow

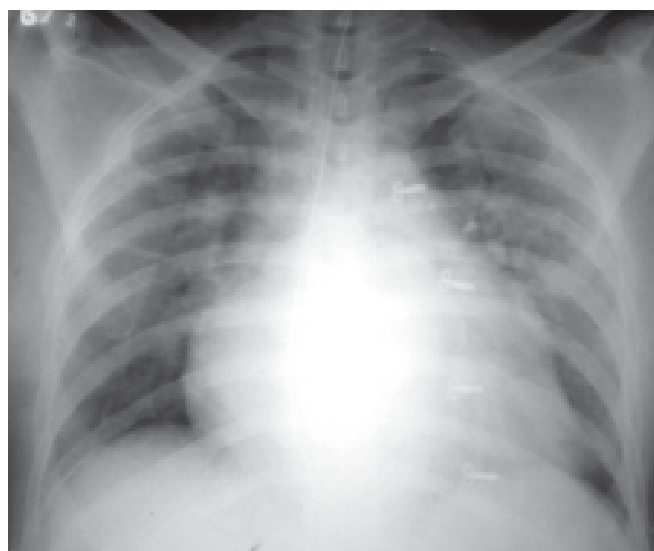


FIGURE 4.5: X-ray chest showing cardiomegaly

as in MR or AR and also the pressure gradient across the stenotic valve.

Transesophageal echocardiography is more sensitive method to detect smaller vegetations, prosthetic valve dysfunction and atrial septal defects.

Cardiac Catheterization

A catheter is passed in the heart via an artery or vein under fluoroscopic monitoring. It is mainly used to assess coronary artery disease, to evaluate valvular heart disease and to measure the chamber pressure.

Others

Myocardial perfusion imaging with radioactive thallium, magnetic resonance imaging (MRI) and positron emission tomography (PET) are other techniques employed in patients with cardiac diseases.

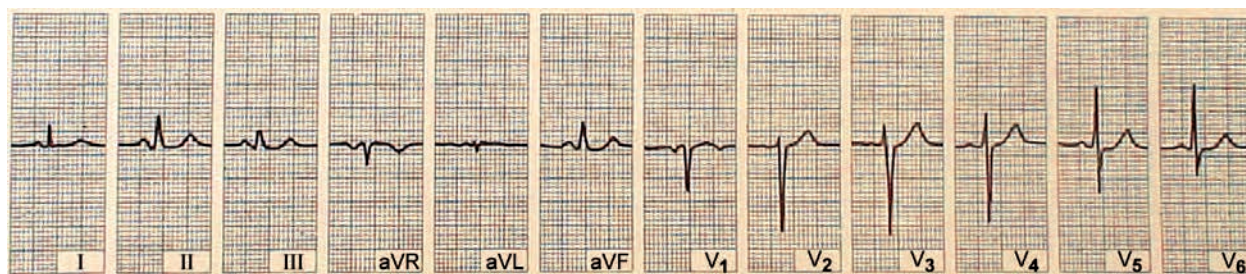


FIGURE 4.6: Normal electrocardiogram

ACUTE RHEUMATIC FEVER

Acute rheumatic fever (ARF) is a multisystem disorder which follows pharyngeal streptococcal infection. This is the commonest cause of acquired heart disease in childhood and adolescence. Acute rheumatic fever is still prevalent in developing countries, although it is less commonly seen in the industrialized world.

Acute rheumatic fever affects children, most commonly between 5 to 15 years of age. It occurs in about 3% individuals who develop pharyngeal infection with certain serotypes of group A. *streptococci*. Streptococcal antigens have cross reactivity with cardiac tissue; hence, anti-streptococcal antibodies mediate inflammatory reaction in the myocardium, endocardium and pericardium. Joint and skin tissues are also affected due to tissue cross reactivity.

Clinical Features

The manifestations of ARF are fever, lethargy, anorexia and symptoms caused due to involvement of heart, joints, skin and central nervous system. These manifestations occur generally 2-3 weeks after streptococcal pharyngitis. Its diagnosis is based on updated Jone's Criteria (Table 4.5). Presence of two or more major criteria or one major and at least two minor criteria plus evidence of previous streptococcal infection is required for the diagnosis of ARF.

Polyarthritis

- It is an early feature of ARF and occurs in about 75% of patients.
- There is painful inflammatory involvement of the large joints (ankles, knees, elbows) which are red, swollen and tender.

TABLE 4.5: The Jone's criteria for diagnosis of rheumatic fever

Major criteria	Minor criteria
Polyarthritis	Fever
Carditis	Arthralgia
Chorea	Raised ESR or CRP
Subcutaneous nodules	Leukocytosis
Erythema marginatum	Prolonged PR interval
And	
Positive throat culture for Streptococci or elevated or increasing streptococcal antibody titer	

- The pain and swelling in the involved joints subside or disappear as newer joints are affected (migratory polyarthritis).
- The response to salicylates is dramatic.
- The inflammation does not leave any joint deformity.

Carditis

- Involvement of pericardium, endocardium and myocardium (pancarditis) occurs in 40 to 60% of patients with ARF.
- The symptoms are chest pain, palpitation and breathlessness.
- Examination may reveal tachycardia, third heart sound, pericardial rub, murmur of mitral regurgitation and cardiomegaly. A mid-diastolic murmur (Carey Coombs' murmur) may be present due to mitral valvulitis.
- Fibrosis and adhesion of the valve may develop following healing of the valvulitis which may lead to stenosis and/or regurgitation (rheumatic valvular heart disease). The mitral valve is most commonly involved and the aortic valve is next most affected.
- The valvular involvement increases the risk of infective endocarditis.

Chorea

Chorea is a manifestation of central nervous system involvement in ARF. This is also known as Sydenham's chorea or St. Vitus dance. It is characterized by involuntary purposeless movement of hands, feet or face. Chorea is a late manifestation in the course of ARF and is more common in females.

Subcutaneous Nodules

This rare manifestation occurs in less than 10% cases of ARF. These are 0.5 to 2 cm, painless, firm nodules found over extensor surface of joints. The presence of subcutaneous nodules signifies underlying rheumatic heart disease.

Erythema Marginatum

This also occurs in less than 10% patients. These are evanescent red macules with pale center over the trunk and proximal extremities. Erythema marginatum may not be well appreciated in dark skinned people.

Investigations

- The blood examination reveals leukocytosis, raised ESR and CRP.
- Throat swab culture may be positive for group A streptococci.
- Anti-streptolysin O (ASO) titer is raised to >200 units in adults or >300 unit in children.
- Other antibodies like anti-deoxyribonuclease B and anti-hyaluronidase may also be helpful in the diagnosis.
- Chest radiograph, ECG, and echocardiography are helpful in the diagnosis of carditis.

Treatment

1. **Eradication of streptococci:** A 10 day course of erythromycin (250 mg 6 hourly oral) or oral penicillin V (500 mg twice daily) is administered to all patients to eradicate streptococcal infection. Alternatively, a single intramuscular injection of benzathine penicillin G (1.2 million units) may be given.
2. **Bedrest:** This is required in patients with severe carditis. This is also helpful in reducing joint pain. The duration of rest is guided by symptoms and markers of inflammation.
3. **Salicylates:** Aspirin is given in 6 divided dosages of 60 to 120 mg/kg per day to relieve the arthritis. This should be given for 4-6 weeks and then gradually tapered.
4. **Steroids:** Prednisolone (1-2 mg/kg/day) is given to patients with severe carditis with CHF or severe arthritis alone. As the patients improve, steroid should be tapered and salicylates added.

Secondary Prevention

This is used to prevent subsequent pharyngeal infection with group A streptococci. Following drugs may be used for secondary prevention of ARF:

- Oral penicillin V (250 mg twice daily)
- Oral sulphadiazine (1 g daily)
- Erythromycin (250 mg twice daily)
- Intramuscular benzathine penicillin G (1.2 million units) every three weeks

This is given for at least 5 years after the ARF. However, in patients with documented rheumatic heart disease or those at high risk of exposure, prophylaxis may be required indefinitely, preferably life long.

RHEUMATIC VALVULAR HEART DISEASE

About half of the patients who have rheumatic carditis will develop valvular disease. The mitral valve is most commonly involved, followed by the aortic valve. Other valves are involved rarely. Valvular defect may be stenotic or regurgitant. The patient may have single or multiple valvular involvements.

Mitral Stenosis (MS)

Mitral stenosis (MS) is more commonly found in females and its commonest cause is rheumatic heart disease. Rarely may it be congenital or degenerative (Table 4.6).

Pathophysiology

The size of the orifice of normal mitral valve is 4-6 cm². The orifice is progressively narrowed by fibrosis and calcification of valve leaflets, fusion of commissures and shortening of chordae tendineae. Mitral valve orifice less than 2 cm² is hemodynamically significant and becomes “critical” at <1 cm². The obstruction to the flow of blood through the stenosed mitral valve during diastole causes a rise in left atrial pressure. This leads to the hypertrophy and dilatation of the left atrium, pulmonary venous congestion and pulmonary arterial hypertension. Right ventricular failure may occur due to pulmonary hypertension. Reduced lung compliance causes breathlessness while decreased left ventricular filling leads to low cardiac output and fatigue. Atrial fibrillation (AF) commonly occurs because of left atrial dilatation. Patients, particularly with AF, are more susceptible to develop left atrial thrombus and systemic thromboembolism.

Clinical Manifestations

The early presentations of MS include breathlessness on exertion and fatigue. As the stenosis progresses, patients are dyspnic on rest and even have orthopnea and paroxysmal nocturnal dyspnea (PND). Acute pulmonary edema may also occur. There may be hemoptysis (due to rupture of pulmonary-bronchial venous connections secondary to pulmonary venous hypertension), edema of lower limbs,

TABLE 4.6: Causes of mitral stenosis

1. Rheumatic fever
2. Calcification/degeneration of mitral valve
3. Congenital mitral stenosis

palpitation and thromboembolic events (stroke, limb ischemia). Signs are given in Table 4.7.

Investigations

1. ECG may reveal the evidence of left atrial (LA) enlargement, right ventricular (RV) hypertrophy and atrial fibrillation (Fig. 4.7).
2. Chest X-ray shows findings suggestive of LA enlargement and pulmonary congestion.
3. Echocardiography is the most sensitive and specific of non-invasive methods to diagnose the valvular disease. It may reveal structural abnormalities of the valves, size of cardiac chambers, pulmonary artery pressure, ventricular dysfunction and presence of thrombi. Transesophageal echocardiography provides better information than transthoracic echocardiography.
4. Cardiac catheterization is used to assess associated valvular lesions and to detect coronary artery disease.

Treatment

Medical

Medical treatment of mitral stenosis consists of:

1. Restriction of physical activity.
2. Sodium restriction and diuretics are used in heart failure.

TABLE 4.7: Clinical signs of mitral stenosis

Malar flush (mitral facies)
Atrial fibrillation (irregularly irregular pulse)
Cardiac examination
Tapping apex beat
Diastolic thrill in mitral area
Auscultation
• Loud first heart sound
• Opening snap
• Mid-diastolic murmur in mitral area (low pitched, rumbling)
Signs of pulmonary hypertension
Loud P ₂
Right ventricular heave
Signs of raised pulmonary capillary pressure
Pulmonary crepts
Signs of right heart failure
Raised JVP
Tender hepatomegaly
Bilateral pitting pedal edema
Ascites and pleural effusion in severe cases



FIGURE 4.7: Atrial fibrillation

3. Digoxin is given to control ventricular rate in patients with AF. Beta blocker and calcium antagonists (verapamil, diltiazem) can also be used.
4. Oral anticoagulant (warfarin) is given to patients with a history of thromboembolic events or to those with AF (INR of 2.0-3.0).
5. Prophylaxis should be given to all patients to prevent rheumatic fever (see prophylaxis for rheumatic fever).
6. Prophylaxis for infective endocarditis should be given prior to procedures (see prophylaxis for endocarditis). *Recent guidelines recommend prophylaxis only if there is a prior history of endocarditis.*

Surgical

Surgical intervention is needed when patient remains symptomatic despite medical treatment or when mitral stenosis is severe.

1. **Mitral valvotomy**
 - a. Percutaneous balloon valvotomy is indicated when mitral valve is noncalcified and without regurgitation. The procedure involves the passing of catheter across the valve and inflation of the balloon to dilate the orifice.
 - b. Open valvotomy is carried out in patients where balloon valvotomy is not possible or in cases with restenosis. In this procedure the fusion of the valve is loosened and calcium deposit and thrombi are removed.
2. **Mitral valve replacement:** The mitral valve is replaced when there is critical MS (<1 cm² orifice size) and/or there is associated significant mitral regurgitation. Replacement is also done when the mitral valve is severely distorted and calcified.

Mitral Regurgitation (MR)

Mitral regurgitation (MR) is the backflow of blood from the left ventricle to the left atrium during systole. Rheumatic heart disease is the principal cause of MR. Other important causes are mitral valve prolapse, ischemic heart disease, dilated cardiomyopathy and connective tissue diseases like Marfan syndrome. Myocardial infarction, infective endocarditis and trauma can lead to *acute MR* (Table 4.8).

Pathophysiology

Mitral regurgitation leads to gradual dilatation of LA with little increase in pressure. However, chronic volume overload leads to left ventricular (LV) dilatation. Eventually the pressure in the LV and LA rises leading to breathlessness and pulmonary congestion. On the contrary, there is sudden elevation of LA pressure in *acute MR* causing severe pulmonary edema.

Clinical Manifestations

The symptoms of chronic MR are similar to that of MS and the main symptoms are fatigue, exertional dyspnea and orthopnea. Thromboembolic events are less common. Patients with acute severe MR commonly present with acute pulmonary edema. Clinical signs are listed in Table 4.9.

Investigations

1. ECG may reveal the evidence of LA and LV enlargement, RV hypertrophy and atrial fibrillation.
2. Chest X-ray shows findings suggestive of LA and LV enlargement and pulmonary congestion.
3. *Echocardiography is the most sensitive and specific of non-invasive methods to diagnose valvular disease. It may reveal structural abnormalities of the valves, size*

TABLE 4.8: Causes of mitral regurgitation (MR)

Chronic MR
Rheumatic fever
Mitral valve prolapse
Infective endocarditis (damage to valve)
Ischemic
Dilated cardiomyopathy (dilatation of ventricle/mitral valve ring)
Acute MR
Infective endocarditis
Myocardial infarction (papillary muscle rupture)
Trauma

TABLE 4.9: Clinical signs of mitral regurgitation

Atrial fibrillation (irregularly irregular pulse)
Cardiac examination
Hyperdynamic and shifted apex beat
Systolic thrill in mitral area
Auscultation
• Soft first heart sound
• Apical S ₃ (third heart sound)
• Pan systolic murmur in mitral area
Signs of pulmonary hypertension
Loud P ₂
Right ventricular heave
Signs of raised pulmonary capillary pressure
Pulmonary crepts
Signs of right heart failure
Raised JVP
Tender hepatomegaly
Bilateral pitting pedal edema
Ascites and pleural effusion in severe cases

of cardiac chambers, pulmonary artery pressure, ventricular dysfunction and presence of thrombi. *Doppler echocardiography* is needed to detect and estimate the MR. Transesophageal echocardiography provides better information than transthoracic echocardiography.

4. Cardiac catheterization is used to assess the severity of MR, to detect associated valvular lesions and coronary artery disease.

Treatment

Medical

Medical treatment of mitral regurgitation consists of:

1. Restriction of physical activities that cause fatigue and breathlessness.
2. Sodium restriction and diuretics are used to reduce pulmonary congestion.
3. Digoxin is given to control ventricular rate in patients with AF and to improve the ventricular systolic function.
4. Vasodilators like ACE inhibitors are given in chronic MR to reduce regurgitation and improve forward output. Intravenous nitroprusside or nitroglycerine is useful in *acute MR*.
5. Oral anticoagulant (warfarin) is given to patients with history of thromboembolic events or to those with AF.
6. Prophylaxis should be given to all patients to prevent rheumatic fever.

7. Prophylaxis for infective endocarditis should be given prior to procedures. *Recent guidelines recommend prophylaxis only if there is a prior history of endocarditis.*

Surgical

Mitral valve repair/Mitral valve replacement: Patients who are initially on medical therapy are watched for symptomatic worsening and for radiological evidence of progressive cardiac enlargement or deteriorating cardiac function. In this case, surgery is indicated. Surgery includes repair or replacement of the mitral valve.

Acute MR: Acute MR due to endocarditis, myocardial infarction and trauma often requires emergency surgery. Prior to surgery, patients are stabilized by vasodilators or intra-aortic balloon counter pulsation which reduces regurgitation by lowering systemic vascular resistance.

AORTIC STENOSIS (AS)

Aortic stenosis (AS) in *adults* may be due to (a) degeneration and calcification of a normal valve, (b) calcification of a bicuspid valve and (c) rheumatic aortic valve stenosis. Rheumatic AS is always accompanied with mitral valve involvement and presence of AR. Aortic stenosis in *children* is commonly congenital in origin (Table 4.10).

Pathophysiology

The obstruction to the left ventricular outflow causes pressure overload on the LV which subsequently leads to concentric LV hypertrophy. The increase in cardiac output during effort is limited by the valve stenosis. This can cause syncope and hypotension. Left ventricular failure occurs when it is unable to overcome the obstruction.

Ischemic symptoms (angina) may result due to increased oxygen demand by hypertrophied myocardial mass together

with decreased coronary supply. Angina may occur even in the absence of coronary artery disease.

Clinical Manifestations

The patients remain asymptomatic for many years. Symptoms occur when AS becomes severe (valve size $< 1 \text{ cm}^2$). Exertional dyspnea, angina pectoris and syncope are three cardinal symptoms of AS. Sudden death may occur. In the advanced stage, symptoms of LV failure such as orthopnea, PND and pulmonary edema may occur. Clinical signs are given in Table 4.11.

Investigations

1. ECG may show the evidence of LV hypertrophy and left bundle branch block.
2. Chest X-ray is generally normal. However, it may show post-stenotic dilatation of the ascending aorta and valvular calcification.
3. Echocardiography may reveal structural abnormalities of the valves including calcification, size of cardiac chambers, pulmonary artery pressure, ventricular dysfunction and presence of thrombi. Doppler echocardiography is needed to detect associated AR.
4. Cardiac catheterization is mainly required to accurately assess the severity of AR, LV dysfunction and to detect the presence of coronary artery disease.

TABLE 4.11: Clinical signs of aortic stenosis

General examination

Slow rising pulse with delayed peak (pulsus parvus et tardus)

Carotid thrill

Narrow pulse pressure

Cardiac findings

Inspection and palpation

- Forceful and sustained apex beat
- Systolic thrill at base of the heart radiating to carotid artery

Auscultation

- Soft or absent A_2
- S_4 at apex (S_3 if LVF)
- *Harsh ejection systolic murmur* best at aortic area and radiating toward carotid artery

Others

- Basal crackles (crepts) in lungs due to pulmonary congestion
- Signs of RV failure may be present in severe cases

TABLE 10: Causes of aortic stenosis (AS)

Degeneration/calcification of normal valve
Calcification of bicuspid aortic valve
Rheumatic fever
Congenital AS

Treatment

Medical

Medical treatment of aortic stenosis consists of:

1. Strenuous physical activities should be avoided.
2. Sodium restriction is advised in presence of CHF. Diuretics are used with caution to avoid hypotension.
3. Digoxin is given to improve the ventricular systolic function in patients with LVF.
4. Oral anticoagulant (warfarin) is given if there is AF.
5. Prophylaxis for infective endocarditis should be given prior to procedures. *Recent guidelines recommend prophylaxis only if there is a prior history of endocarditis.*
6. Statins have been shown to reduce the progression of degenerative calcific AS.

Surgical

1. Aortic valve replacement: All symptomatic adult patients with severe AS should be considered for aortic valve replacement.
2. Percutaneous balloon aortic valvuloplasty: it is useful in children with congenital non-calcific AS.

AORTIC REGURGITATION (AR)

- Aortic regurgitation (AR) is the backflow of blood from the aorta through an incompetent aortic valve into the left ventricle during diastole.
- Aortic regurgitation (AR) may be either due to valvular involvement or dilatation of aortic root or both.
- The important causes of valvular involvement are rheumatic disease, endocarditis, trauma and congenital bicuspid aortic valve. Marfan syndrome and severe hypertension can lead to aortic root dilatation and AR. Syphilis and ankylosing spondylitis can cause AR because of valvular involvement and aortic dilatation (Table 4.12).

Pathophysiology

Regurgitation causes volume overload leading to dilatation and hypertrophy of the LV. The stroke volume is increased to maintain the effective cardiac output. This is the basis for peripheral arterial signs in AR. As the LV function deteriorates, the effective forward output declines even during rest. Myocardial ischemia may occur even in the absence of concomitant coronary artery disease (CAD).

TABLE 4.12: Causes of aortic regurgitation

Valvular
• Rheumatic fever
• Endocarditis
• Trauma
• Congenital bicuspid aortic valve
• Syphilis
• Ankylosing spondylitis
Aortic root disease
• Aortic dissection
• Marfan syndrome
• Hypertension
• Syphilis
• Ankylosing spondylitis

Clinical Manifestations

- The patients remain asymptomatic for years. Palpitations, particularly on lying down are generally an early symptom. Later on, patients present with dyspnea on exertion followed by orthopnea and PND. Angina may also occur frequently in severe AR.
- The presentations in *acute severe AR* are pulmonary edema and/or cardiogenic shock (tachycardia, cold extremities, hypotension, cyanosis). The clinical signs are listed in Table 4.13.

Investigations

1. ECG shows the evidence of LV hypertrophy and ST-T changes.
2. Chest X-ray reveals cardiomegaly and aortic root dilatation. Features of pulmonary congestion may be present.
3. Echocardiography may reveal structural abnormalities of the valves, size of cardiac chambers, pulmonary artery pressure, ventricular dysfunction and presence of thrombi. Doppler echocardiography is needed to assess the severity of AR.
4. Cardiac catheterization is mainly required to detect the presence of coronary artery disease.

Treatment

Medical

Medical treatment of aortic regurgitation consists of:

1. Strenuous physical activities should be avoided.

TABLE 4.13: Clinical signs of aortic regurgitation**General examination**

Peripheral signs of AR

- Collapsing or water hammer pulse
- Dancing carotids (pulsatile and prominent carotid arteries)
- Quincke's sign (alternate flushing and paling of skin at root of nail on pressure)
- Pistol shot sound over femoral arteries (Traube's sign)
- Duroziez sign (to and fro murmur over femoral artery when it is compressed)
- de Musset's sign (head nodding with the pulse)

Blood pressure

- Increased pulse pressure (low diastolic pressure)
- Hill's sign: systolic BP in lower limbs is higher (>20 mm Hg) than in upper limbs.

Cardiac findings

Inspection and palpation

- Heaving and laterally displaced apex beat
- Diastolic thrill at left sternal border

Auscultation

- Soft or absent A₂
- S₃, (S₄ occasionally)
- High pitched blowing early diastolic murmur best at left sternal border
- Other murmurs include ejection systolic murmur at aortic area radiating to carotid artery and low pitch mid-diastolic murmur over mitral area (Austin flint murmur).

Others

- Basal crackles (crepts) in lungs due to pulmonary congestion.
- Signs of RV failure (edema and hepatomegaly) may be present in severe cases.

2. Sodium restriction and diuretics are advised in presence of CHF.
3. Digoxin and vasodilators (ACE inhibitors) are given to improve the ventricular systolic function in patients with LVF. Long-acting nifedipine has been found to delay the need for operation.
4. Oral anticoagulant (warfarin) is given if there is AF.
5. Prophylaxis for infective endocarditis should be given prior to procedures. *Recent guidelines recommend prophylaxis only if there is a prior history of endocarditis.*
6. Nitroprusside or ionotrope is given in acute AR to stabilize the patient before surgery.
7. Treatment of precipitating or underlying causes (syphilis, endocarditis) should be commenced.

Surgical

Aortic valve replacement and/or aortic root repair: All symptomatic patients with chronic AR should be considered for aortic valve replacement. However, surgery may also be advisable in asymptomatic patients who show evidence of progressive cardiomegaly or deteriorating LV function.

Acute AR: Repair of aortic root abnormalities and aortic valve replacement should urgently be performed in patient with acute AR. Patients may be stabilized by vasodilators or ionotropes before surgery.

INFECTIVE ENDOCARDITIS

Infective endocarditis is **microbial infection of endothelium of the heart**. The infection most commonly occurs at the **site of pre-existing endocardial damage in heart valves** (native or prosthesis) and **endocardium of the chambers**. Similar process involving **arterial endothelium in arteriovenous shunts, patent ductus arteriosus and coarctation of aorta** is known as **infective endarteritis**.

Pathology

The endothelium may get damaged due to **high pressure jet injury** as occurs in **valvular or congenital heart disease**. The **damaged surface of endothelium invites platelet adhesion and aggregation and fibrin deposition** (*nonbacterial thrombotic endocarditis*). Organisms which enter into the blood stream through mucosa, skin or sites of focal infection may **colonize the platelets-fibrin deposit** and form **vegetation**. These vegetations can **grow in size and cause obstruction** or can be dislodged as **emboli**. Locally, it can lead to **abscess formation and damage to the tissues like valves, chordae tendinae and myocardium**. Virulent organisms such as *S. aureus* may even colonize normal endothelium.

- Ventricular septal defect (VSD), mitral regurgitation, aortic stenosis, aortic regurgitation are particularly susceptible to endocarditis.
- Whereas conditions with low pressure shunt (atrial septal defect) are at low risk of developing endocarditis.
- Intravenous drug abusers are prone to develop right sided (tricuspid valve) endocarditis.

Endocarditis is classified into **acute and subacute types** depending on manifestations and their clinical course (Table 4.14).

TABLE 4.14: Acute and subacute infective endocarditis

Acute	Subacute
1. Rapid course	1. Indolent course
2. Rapid valve destruction and abscess formation	2. Slow structural damage
3. Metastatic spread common	3. Less common
4. Clubbing, splenomegaly and petechial hemorrhages not found	4. Clubbing, splenomegaly and petechial hemorrhages are seen
5. Fatal if not promptly treated	5. Prolonged course unless complicated by ruptured mycotic aneurysm or embolism
6. Causative organism: <i>Staph. aureus</i> , pneumococci, Beta hemolytic streptococci	6. <i>Strep. viridans</i> , enterococci, HACEK group

- The most common organism causing subacute endocarditis is *Viridans group of streptococci*. These are commensals in the upper respiratory tract and may enter the blood following brushing, chewing or dental procedures. Other organisms like *Enterococcus faecalis*, *S. bovis* may arise from urinary tract or bowel.
- *Staphylococcus aureus* is the most common cause of acute endocarditis. Prosthetic valve endocarditis is generally due to coagulase negative *Staphy. epidermidis*, a normal skin commensal.
- HACEK group of gram-negative bacteria (*Hemophilus*, *Actinobacillus*, *Cardiobacterium*, *Eikenella* and *Kingella*) can also cause native valve endocarditis.
- Other uncommon organisms which cause endocarditis are fungi (*Candida*), rickettsia, chlamydia and anaerobes.
- b. Meningitis, embolic infarcts and intracranial bleeding due to rupture of mycotic aneurysms are neurological presentations.
- c. Septic emboli may disseminate infection to distant organs such as skin, spleen, kidneys, bone and meninges. Embolic events may also be associated with infarction at various sites.
- d. Immune complex deposition can lead to glomerulonephritis and hematuria.

Investigations

1. **Blood culture:** The most important test is blood culture which tells about the organism and guides in antibiotic therapy. Venous blood samples are taken 30 minutes apart from three different sites. Aseptic technique is essential and the sample should be taken prior to antibiotic therapy except in severe cases. Serological tests may be useful if blood culture is negative.
2. **Echocardiography:** This can reveal the site and size of the vegetations, abscess formation, evidence of underlying heart disease and heart failure. **Trans-esophageal echocardiography** is a more sensitive method as compared to transthoracic echocardiography.
3. **Other findings:** These include normocytic normochromic anemia, leukocytosis, high ESR and CRP, microscopic hematuria, and proteinuria. Chest X-ray may show evidence of cardiomegaly and heart failure. Conduction defects may be observed on ECG recording.

Diagnostic criteria: **Duke's criteria** which are based on clinical, laboratory and echocardiographic findings are used for the diagnosis of endocarditis.

Clinical Manifestations

Endocarditis is suspected in any patient with cardiac disease who develops persistent fever. The manifestations are listed below;

1. **General:** Fever, weight loss, night sweats and weakness
2. **Cardiac:** New murmur, heart failure and heart blocks
3. **Extracardiac:**
 - a. Following findings may be present in IE
 - Anemia
 - Clubbing
 - Splenomegaly
 - Petechial hemorrhages
 - Osler nodes (painful lesions at fingertips)
 - Janeway's lesions (macular lesions over palm and soles)
 - Roth spots on fundus examination
 - Subconjunctival hemorrhages

Treatment

The treatment of infective endocarditis should be prompt and adequate. The principles of treatment are:

1. The **antibiotics** should be administered parenterally to achieve high serum concentration since the vegetation is avascular.
2. The therapy is generally of **prolonged duration**. The dose and duration should be meticulously adhered to ensure the proper response.
3. The antibiotics should preferably be **bactericidal**.
4. The selection of antibiotics should be based on culture reports and minimum inhibitory concentration (MIC) values. **Empirical therapy** may be initiated in acute severe cases after drawing blood samples for culture. The antibiotics are later changed, if necessary, based on sensitivity reports.

The list of antibiotics commonly used, their dosage and indications are given in Tables 4.15 and 4.16.

Empirical Therapy

Empirical therapy is given when the **culture is negative or before culture and sensitivity reports are available**. Ceftriaxone plus gentamicin is given in subacute native valve endocarditis. Vancomycin is added to the above regime in case of prosthetic valve endocarditis.

TABLE 4.15: The antibiotics commonly used in infective endocarditis

1. Penicillin G	2-4 million units IV 4 hrly
2. Gentamicin	1 mg/kg IV or IM 8 hrly
3. Ceftriaxone	2 g IV OD
4. Ampicillin/amoxycillin	2 g IV 4 hrly
5. Nafcillin or Oxacillin	2 g IV 4 hrly
6. Cefazolin	2 g IV 8 hrly
7. Vancomycin	15 mg/kg IV 12 hrly
8. Rifampicin	300 mg orally 8 hrly

Surgery

Surgery is often needed in patients with prosthetic valve endocarditis and fungal endocarditis who have (a) **heart failure** due to valve damage, (b) **no response to antibiotics**, (c) **large vegetations** and (d) **abscess formation**.

Prophylaxis

- Antibiotic prophylaxis prevents only a small number of cases of infective endocarditis. Bacteremia associated with routine daily activities like chewing brushing, and flossing is similar as with dental procedures like tooth extraction. It has been seen that cardiac tissue is exposed to bacteremia causing oral cavity organisms many times greater with routine daily activities than from dental procedures. As per American Heart Association recent guidelines antibiotic prophylaxis is now recommended only for patients at highest risk (Table 4.17).

TABLE 4.16: Common regimens used in IE

Organism	Sensitivity	Regimens
<i>Streptococci</i>	Penicillin susceptible (MIC <0.1 µg/ml)	Penicillin G (4 weeks) Penicillin G plus Gentamicin (2 weeks) Ceftriaxone (4 weeks) Vancomycin (4 weeks)
	Relatively penicillin resistant (MIC 0.1-0.5 µg/ml)	Penicillin G (4-6 weeks) plus gentamicin (2 weeks)
<i>Enterococci</i>	Moderately penicillin resistant (MIC 0.5-0.8 µg/ml).	Penicillin G plus Gentamicin (4-6 weeks) Penicillin G or Ampicillin or Vancomycin plus Gentamicin (4-6 weeks)
<i>Staphylococci</i>		
Native valve	Methicillin susceptible	Nafcillin or Oxacillin or Cefazolin (4-6 weeks) plus Gentamicin (3-5 days) Vancomycin (4-6 weeks)
	Methicillin resistant	Vancomycin (4-6 weeks)
Prosthetic valve	Methicillin susceptible	Nafcillin or Oxacillin (6-8 weeks) plus Gentamicin (2 weeks) plus Rifampicin (6-8 weeks)
	Methicillin resistant	Vancomycin (6-8 weeks) plus Gentamicin (2 weeks) plus Rifampicin (6-8 weeks)
<i>HACEK group</i>		Ceftriaxone (4 weeks) Ampicillin plus Gentamicin (4 weeks)
<i>Fungal</i>		Amphotericin B plus flucytosine

- Patients with **valvular and congenital heart disease** who are at high risk of endocarditis should receive prophylactic antibiotics (Table 4.18) before undergoing any procedure which may cause bacteremia. Antibiotic prophylaxis is not advised for other cardiac lesions.
- The patients must be advised to **maintain good dental and oral health**. In patients with poor oral hygiene, routine chewing and brushing of teeth may cause bacteremia. Hence, daily personal care and annual professional dental care is advised.
- **Tooth extraction, periodontal surgery, root canal therapy, scaling, removal of tartar and tooth implantation** are procedures that require antimicrobial prophylaxis (Tables 4.19 and 4.20). In addition, **respiratory, gastrointestinal and genitourinary procedures** also require prophylaxis.
- Patients with atrial septal defect, surgically repaired VSD and PDA, prior coronary artery bypass surgery and implanted pacemakers and defibrillators do not require prophylaxis (Table 4.21).

TABLE 4.17: Cardiac lesions (high risk) for which **antibiotic prophylaxis** is recommended before dental procedures

Previous episode of endocarditis
 Patients with prosthetic heart valves
 Cyanotic congenital heart diseases (unrepaired)
 Unrepaired congenital heart defects (VSD, PDA, Coarctation of aorta) or during first 6 months after repair
 Valvulopathy in post-cardiac transplant patients
 Patients with residual defects in incompletely repaired congenital heart disease

TABLE 4.19: Dental procedures for which endocarditis prophylaxis is advised in patients at high or moderate risk for endocarditis

1. Extractions
2. Periodontal procedures, cleaning causing gingival bleeding
3. Implant placement, reimplantation of avulsed teeth
4. Endodontic instrumentation (root canal) or surgery beyond the apex
5. Subgingival placement of antibiotic fibers or strips
6. Placement of orthodontic bands but not brackets
7. Intraligamentary injections (anesthetic)

TABLE 4.20: Oral procedures not requiring prophylaxis for infective endocarditis

- Local anesthetic agent (non-intraligamentary)
- Intracanal endodontic procedure
- Operative and prosthodontic procedures with or without retraction cord
- Adjustment of orthodontic appliance
- Placement of removable prosthodontic or orthodontic appliances
- Impression taking
- Exfoliation of primary teeth
- Oral radiography
- Placement of rubber dams
- Suture removal
- Fluoride treatment

HYPERTENSION

Elevated arterial pressure (hypertension) is one of the most important public health problems worldwide. A sizable proportion of such patients remain undiagnosed and around half of hypertensive patients are not adequately treated.

TABLE 4.18: Antibiotic regimens for high risk cardiac lesions

Drug	Dose	Route	Timing
Standard regimen			
• Amoxicillin	2.0 g	Oral	1 hour before procedure
Penicillin allergy			
• Clarithromycin or Azithromycin	500 mg	Oral	1 hour before procedure
• Cephalexin	2.0 g	Oral	1 hour before procedure
• Clindamycin	600 mg	Oral	1 hour before procedure
Patients unable to take orally			
• Ampicillin	2.0 g	IV or IM	1 hour before procedure
Patients with penicillin allergy and unable to take orally			
• Cefazolin or Ceftriaxone	1.0g	IV or IM	30 min before procedure
• Clindamycin	600 mg	IV or IM	1 hour before procedure
Cephalosporins should not be used in patients with Type 1 immediate hypersensitivity (anaphylaxis, urticaria, angioedema) to penicillin			

TABLE 4.21: Cardiac conditions where endocarditis prophylaxis is not recommended

- Atrial septal defect (ostium secundum)
- Mitral valve prolapse without regurgitation and/or thickening
- After surgical repair of ASD, VSD, or PDA
- Coronary artery bypass surgery (CABG) performed more than 6 weeks prior to treatment
- Physiological, functional heart murmurs
- Cardiac pacemakers
- Implanted defibrillators, previous rheumatic fever or Kawasaki disease without valvular dysfunction

There is no definite dividing line between normal and elevated arterial blood pressure. However, hypertension is defined as a level of blood pressure at which there is increased risk for target organ damage and the benefits of treatment outweighs the cost and hazards.

The seventh US Joint National Committee on Prevention, detection, evaluation, and treatment of high blood pressure (JNC VII) defined normal blood pressure as <120 mmHg systolic and <80 mmHg diastolic. The classification of blood pressure is given in Table 4.22.

- *Isolated systolic hypertension* is defined as systolic blood pressure of 140 mmHg or more together with diastolic pressure <90 mmHg.
- *Labile hypertensives* are those patients who some times have blood pressure in the hypertensive range.
- *Accelerated hypertension* is defined as significant recent increase in blood pressure over previous hypertensive level associated with rapidly progressive end organ damage.
- Presence of papilledema (swelling of optic disk) along with other features of accelerated hypertension signifies *malignant hypertension*.

TABLE 4.22: Classification of blood pressure for adults (>18 years)

Category	Systolic blood pressure (mm Hg)	Diastolic blood pressure (mm Hg)
Normal	<120	<80
Prehypertension	120-139	80-89
Hypertension		
Stage 1	140-159	90-99
Stage 2	>160	>100

Note: Blood pressure values are based on the average of two or more readings taken at each of two or more visits after an initial screening. When systolic and diastolic values fall into different categories, the higher category should be selected for classification.

Essential Hypertension

Patients with elevated arterial blood pressure without any identifiable cause are said to have essential or primary or idiopathic hypertension. More than 90% patients with hypertension belong to this category. Pathogenesis of essential hypertension is not clearly understood. However, this is thought to be multifactorial. Important environmental factors include high salt intake, heavy alcohol use, obesity and sedentary lifestyle. Genetic factors may also be responsible in some ethnic groups.

Secondary Hypertension

In secondary hypertension, a specific cause is identified. Important causes are given in Table 4.23. The most common cause of secondary hypertension is renal disease. Secondary hypertension should be suspected if onset of hypertension is at <35 years or >55 years of age or there is abrupt onset of severe hypertension at any age.

Effects of Hypertension on Target Organs

There is an increased chance of damage to the vascular bed of several organs mainly retina, heart, brain, kidneys and large arteries.

1. *Retinal:* Retinal changes depend on the severity of hypertension. These are classified into four grades (grade I-IV) based on the presence of arteriolar thickening, nicking, hemorrhages, exudates and papilledema.

TABLE 4.23: Important causes of secondary hypertension

Renal disorders
• Renovascular stenosis
• Parenchymal renal disease, particularly glomerulonephritis
• Polycystic kidney disease
Endocrinal disorders
• Pheochromocytoma
• Cushing's syndrome
• Primary hyperaldosteronism
• Acromegaly
• Hypo- and hyperthyroidism
Drugs
• Oral contraceptives
• Corticosteroids
• Sympathomimetic drugs
• Cyclosporine
Miscellaneous
• Toxemia of pregnancy
• Coarctation of aorta
• Raised intracranial pressure

2. **Cardiac:** These include left ventricular hypertrophy, pulmonary edema and high incidence of coronary artery disease.
3. **Central nervous system (CNS):** Stroke (infarction and hemorrhage), transient ischemic attacks (TIA), hypertensive encephalopathy (coma, seizures) are important effects on CNS.
4. **Renal:** Effect on kidneys consists of proteinuria and renal failure.
5. **Large vessels:** Hypertension leads to accelerated atherosclerosis and aneurysmal dilatation.

Clinical Features

- The majority of patients asymptomatic and is diagnosed on routine clinical examination. However, symptoms due to raised blood pressure are occipital headache, dizziness, palpitation and fatigue.
- Patient may also present with symptoms related to target organ damage like epistaxis, hematuria, blurred vision, TIA, angina and breathlessness.
- Symptoms pertaining to underlying cause may be present; for example, weight gain (Cushing's syndrome), weight loss (thyrotoxicosis), episodic headache, palpitation and sweating (pheochromocytoma).

History: History must include age, sex, occupation, lifestyle of patient along with history of smoking, diabetes mellitus, hyperlipidemia, alcohol and drug intake and presence of hypertension in family members.

Physical examination: Presence of truncal obesity (Cushing's disease), palpable kidneys (polycystic kidneys), radio-femoral delay (coarctation of aorta), abdominal bruit (renovascular) may help in identifying the secondary cause of hypertension. The signs of complications of hypertension such as heaving apex, fourth heart sound, loud aortic second heart sound, pulmonary crackles and retinal changes may be present.

Investigations

There are some investigations which should be done in all patients with hypertension. Tests are performed to know the cause of hypertension, target organ damage and to detect cardiovascular risk factors. Special tests are performed in others as indicated (Table 4.24).

TABLE 4.24: Investigations in patients with hypertension

Basic investigations in all patients
• Urine analysis for protein, blood and glucose
• Fasting blood sugar
• Serum creatinine and blood urea nitrogen
• Serum sodium and potassium
• Serum lipid profile
• Electrocardiography
Investigations in special group of patients
• Chest X-ray and echocardiography
• Renal ultrasonography and angiography
• Serum calcium and phosphate
• Thyroid stimulating hormone (TSH)
• Urinary cortisol and catecholamine
• Plasma renin activity and aldosterone

Management

The management of hypertension includes general measures and drug therapy.

General measures (lifestyle modification):

1. **Relief of stress:** Patients are advised to avoid unnecessary tension. Relaxation techniques may also be practised.
2. **Salt restriction:** Dietary restriction of sodium chloride upto only 5 g per day helps reduce blood pressure in some hypertensive patients. However, salt restriction potentiates the effect of almost all antihypertensive agents. Diet rich in potassium and calcium may also be useful in hypertensive patients.
3. **Weight reduction:** Caloric restriction in obese patients and weight reduction leads to decline in blood pressure.
4. **Control of risk factors:** Restriction of cholesterol and saturated fat in diet reduces the atherosclerotic complications in hypertensive patients. Alcohol intake should be reduced whereas smoking should be stopped. The blood sugar level should be tightly controlled in diabetics.
5. **Regular exercise:** Isotonic exercises like jogging and swimming lead to reduction in arterial pressure. Isometric exercises such as weight lifting should be avoided as these can increase arterial pressure. However, exercise should be planned under expert guidance.

Drug Therapy

Various classes of drugs used are given below. Commonly used drugs in each class are given in Table 4.25. Mechanism of action and side effects of antihypertensive drugs are given in Table 4.26.

TABLE 4.25: Commonly used antihypertensive agents

Dose range (daily)	
<i>Diuretics</i>	
Hydrochlorothiazide	Oral 12.5-25 mg
Furosemide	Oral/IV 20-320 mg
Torsemide	Oral 5-10 mg
Spironolactone	Oral 25-100 mg
<i>Beta blockers</i>	
Atenolol	Oral 25-100 mg
Metoprolol	Oral 50-300 mg
Carvedilol	Oral 12.5-50 mg
Bisoprolol	Oral 2.5-20 mg
Nevirilol	Oral 2.5-10 mg
<i>Calcium channel antagonist</i>	
Amlodipine	Oral 2.5-10 mg
Diltiazem	Oral 90-360 mg
Verapamil	Oral 80-480 mg
<i>ACE inhibitors</i>	
Enalapril	Oral 2.5-40 mg
Ramipril	Oral 1.25-20 mg
Lisinopril	Oral 5-40 mg
<i>ARBs</i>	
Losartan	Oral 25-100 mg
Valsartan	Oral 80-320 mg
Irbisartan	Oral 150-300 mg
Telmisartan	Oral 20-80 mg
Olmisartan	Oral 20-40 mg
<i>Alpha blockers</i>	
Prazosin	Oral 1-20 mg
Terazosin	Oral 1-20 mg
<i>Centrally acting agents</i>	
Clonidine	Oral 0.1-1.2 mg
Methyldopa	Oral 250-2000 mg
<i>Vasodilators</i>	
Hydralazine	Oral 40-300 mg

1. *Diuretics*: Thiazides are the most frequently used diuretics to treat hypertension. The side effects of thiazides such as hyperuricemia, hypertriglyceridemia and hypokalemia can be minimized by restricting the dose of hydrochlorothiazide to 25 mg per day. A combination of thiazide and potassium sparing agents (spironolactone, amiloride) may also be used. Loop diuretics are used hypertensive crisis and renal failure.
2. *ACE inhibitors and angiotensin receptor blockers (ARB)*: Angiotensin converting enzyme inhibitors (enalapril, lisinopril, ramipril) and ARB (losartan, valsartan) have good antihypertensive effect and cause significant reduction in death, acute MI and stroke. These agents also reduce the progression of nephropathy in type II diabetes. ARBs have lesser side effects like cough and angioedema than ACE inhibitors. These drugs are contraindicated in bilateral renal artery stenosis.
3. *Long acting calcium channel antagonist* (amlodipine) is particularly useful in elderly patients with isolated systolic hypertension. Other agents are less commonly used. These groups of drugs can cause pedal edema.
4. *Beta adrenergic antagonists*: These agents lower heart rate and cardiac output. The cardioselective agents which are preferred include atenolol and metoprolol. They are used with caution in patients with congestive heart failure, heart blocks and asthma.
5. *Other groups of drug* occasionally used in hypertension are:
 - a. alpha adrenergic receptor blockers: prazosin, doxazosin

TABLE 4.26: Mechanism of action and side effects of antihypertensive drugs

Drugs	Mechanism of action	Side effects
• Beta adrenergic antagonists	Blockage of sympathetic effects on heart, decreased heart rate and cardiac output	Heart failure, heart blocks, bradycardia, Raynaud's phenomenon, impotence, may precipitate bronchospasm
• Calcium channel blockers	Arterial vasodilatation	Pedal edema, flushing, headache, rash, tachycardia (nifedipine), gum hyperplasia
• ACE inhibitors	Inhibits ACE and blocks the production of angiotensin II leading to arterial and venous dilatation	Dry cough, angioneurotic edema, hypotension, taste disturbance, hyperkalemia
• ARBs	Block angiotensin receptors and cause vasodilatation	Angioedema, rash, hyperkalemia
• Diuretics	Cause natriuresis and decrease intravascular volume, may result in mild vasodilatation	Thiazide; hyperglycemia, hypercalcemia, hyperlipidemia, hypokalemia and hyponatremia Loop diuretics; hypocalcemia, hypokalemia and deafness K sparing; gynaecomastia, hyperkalemia

- b. centrally acting agents: clonidine, methyldopa
- c. vasodilators: hydralazine, minoxidil and
- d. agents with mixed alpha and beta adrenergic antagonist action: labetalol, carvedilol.

Treatment Guidelines

The treatment of hypertension depends on:

- a. The level of blood pressure
 - b. The presence of target organ damage
 - c. The presence of cardiovascular risk factors (see Table 4.30)
 - d. Presence of other disease
1. *Prehypertension*: Lifestyle modification should be advised. Drug therapy may be needed in case of presence of other cardiac risk factors or end organ damage.
 2. *Hypertension (Stage 1 and 2)*: In addition to lifestyle modification, drug therapy is indicated. The goal of therapy should be to achieve a blood pressure level of less than 140/90 mmHg (130/80 mmHg in diabetics). Thiazides are preferred initially as monotherapy. Other drugs (ACE inhibitors, ARBs, beta blockers or calcium channel blockers) may also be considered as monotherapy. However, combination therapy (preferably diuretic plus another agent) may be used where the blood pressure is not controlled by monotherapy. The choice of drug also depends on the presence of co-existing conditions (see Table 4.27).

Hypertensive urgencies is defined as substantial increases in blood pressure usually with systolic >220 mmHg or diastolic >120-130 mmHg. It also includes hypertension with disk edema, progressive end organ complications rather than damage and severe perioperative hypertension. In this situation, BP must be reduced in several hours. Parenteral therapy is usually not needed.

Hypertensive emergencies include hypertensive encephalopathy, hypertensive nephropathy, intracranial hemorrhage, unstable angina, myocardial infarction, acute left ventricular failure with pulmonary edema, aortic dissection and eclampsia. Malignant hypertension is characterized by nephropathy or encephalopathy with accompanying papilledema.

Control of blood pressure is vital for preventing ongoing end organ damage. Rapidly acting parenteral agents are used. Excessive and rapid decline in the BP should be avoided as it may lead to cerebral hypoperfusion and coronary insufficiency. The goal is to attain a decline of 20-25% in the mean arterial pressure or a diastolic pressure of 110 mmHg. Thereafter, the reduction in blood pressure is attained gradually over several days. The parenteral drugs used are nitroprusside, nitroglycerine, dioxide, esmolol, labetalol, enalaprilat and hydralazine. Intravenous frusemide is given as an adjunct. *Sublingual nifedipine should be avoided* in the acute management because it is associated with adverse cardiac events.

Surgery

Surgical intervention may be required in cases of pheochromocytoma and renal artery stenosis.

Hypertensive Crisis

Hypertensive crisis includes hypertensive urgencies and emergencies.

TABLE 4.27: Choice of antihypertensive drugs in various co-existing conditions

Diabetes mellitus	ACE inhibitors, ARBs
Coronary artery disease	Beta blockers, ACE inhibitors
Heart failure	ACE inhibitors, diuretics
Pregnancy	Methyldopa, hydralazine
Asthma	Calcium channel blockers
Prostatism	Alpha blockers
Elderly patients	Diuretics, amlodipine

ISCHEMIC HEART DISEASE / CORONARY ARTERY DISEASE

Ischemic heart disease (IHD)/Coronary artery disease (CAD) is the most common cause of premature death. The incidence of IHD is increasing worldwide. Ischemia is defined as the lack of oxygen due to reduced perfusion. IHD occurs when there is an imbalance between oxygen supply and oxygen demand. Clinical presentations of IHD are given in Table 4.28.

Causes

The most common cause of IHD is atherosclerosis of coronary arteries. Other less common causes of reduced coronary blood flow are spasm of coronary artery, vasculitis, emboli and congenital abnormalities of coronary artery (Table 4.29).

TABLE 4.28: Clinical presentations of coronary artery disease/IHD

1. Stable angina
2. Acute coronary syndromes (ACS)
 - a. Unstable angina
 - b. Non ST segment elevation myocardial infarction (NSTEMI)
 - c. ST segment elevation myocardial infarction (STEMI)
3. Heart failure
4. Arrhythmias
5. sudden cardiac death
6. Asymptomatic

TABLE 4.29: Causes of ischemic heart disease

Atherosclerosis of coronary arteries
 Spasm of coronary artery
 Vasculitis involving coronary artery
 Emboli
 Congenital anomalies of coronary arteries

Pathogenesis of Coronary Atherosclerosis

Atherosclerosis starts in second or third decade of life. The collection of lipoproteins within intimal layer of the coronary artery is the initial event. These lipoproteins undergo modification into oxidized LDL and subsequently are taken up by migrated monocytes in the intima (foam cells). Most foam cells die and result in formation of a lipid rich center. This necrotic area is surrounded by smooth muscle cells which lay down extracellular matrix that forms the bulk of the advance atheromatous lesion (atheromatous plaque or atheroma). Monocyte-mediated cytokines also play an important role in the evolution of the atheromatous plaque. The risk factors for atherosclerosis/cardiovascular disease are given in Table 4.30.

TABLE 4.30: Important risk factors for atherosclerosis

- Smoking
 - Hypertension
 - Lipid disorders (High LDL cholesterol, low HDL cholesterol)
 - Diabetes mellitus
 - Family history of IHD
 - Age (men >45 years, women >55 years)
 - Male sex
 - Sedentary lifestyle
 - Obesity
 - Atherogenic diet
- Newer risk factors*
- Homocysteine
 - Lipoprotein (a)

The ischemic heart disease can clinically manifest as:

- angina pectoris (stable or unstable)
- myocardial infarction
- arrhythmias
- heart failure
- sudden death

STABLE ANGINA

Stable angina results from transient myocardial ischemia arising because of imbalance between myocardial oxygen supply and demand. The reduced perfusion results from fixed 'stable' obstruction due to atheroma of the coronary arteries (Fig. 4.8).

Clinical features:

- The term angina, derived from Greek language means choking or strangulation.
- Stable angina is characterized by central chest pain or discomfort classically precipitated by exertion or stress and relieved by rest. The detailed characteristics of pain are given in Table 4.31.
- Diabetics and elderly may present with breathlessness, fatigue and faintness rather than chest pain. Symptoms other than chest pain are known as anginal equivalents.
- The physical signs are generally normal. The patient may have signs of cardiomegaly, mitral regurgitation, pulmonary edema and third or fourth heart sound (gallop rhythm).

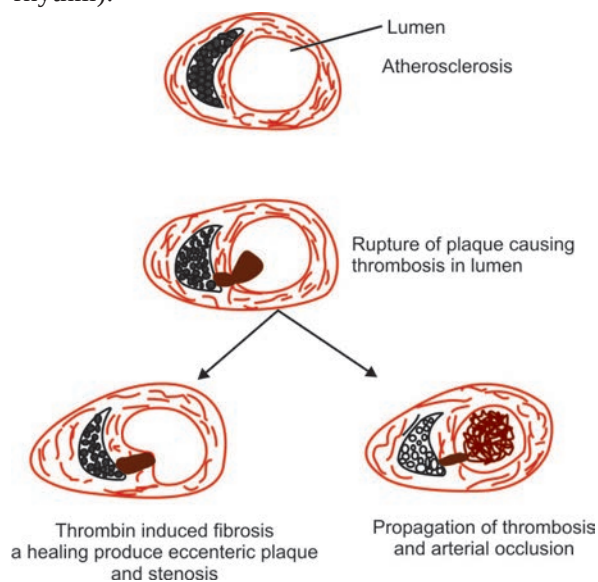
**FIGURE 4.8:** Atheromatous plaque

TABLE 4.31: Characteristics of chest pain in stable angina

Site of the pain	Retrosternal or precordial
Character of the pain	Squeezing, constricting, piercing, feeling of heaviness or pressure
Precipitating factors	Physical exertion, cold exposure, heavy meals, emotional stress, anemia, thyroid disease, vivid dreams (nocturnal angina)
Associated features	Feeling of impending death, breathlessness, apprehension, nausea, vomiting
Relieving factors	Rest, sublingual nitroglycerin
Radiation	Left shoulder, both arms, jaw, neck
Duration	Typically 2-10 minutes (>30 minutes suggests infarction)

- Evidences of risk factors like hypertension, diabetes mellitus and atherosclerosis (carotid bruit, diminished arterial pulse) may be present.

Investigations

Electrocardiogram (ECG)

The ECG changes are present only in around half of the cases. Reversible ST segment depression or elevation with or without T wave inversion during chest pain is specific finding in stable angina.

a. Stress Testing

- Exercise ECG:** This is used to diagnose IHD and to estimate the prognosis. The ECG is done on a treadmill protocol wherein the recording is made during and after exercise. The symptoms like chest pain, fatigue, breathlessness and changes in ST segment and blood pressure are noted.
- Myocardial perfusion scanning:** The uptake of radioactive thallium 201 or technetium 99m by myocardium is measured during and after exercise or pharmacological stress. This is used in patients who cannot exercise or the results of exercise ECG are inconclusive.
- Stress echocardiography:** This can be used alternatively to diagnose IHD in place of perfusion scanning. Exercise or pharmacological stress may cause the changes in myocardial contraction. The advantage is greater convenience, easily availability and the lower cost.

b. Coronary Arteriography

It provides information about the severity and extent of coronary artery disease and is useful when other tests

have failed to diagnose IHD. This is also indicated in patients who despite medical therapy are symptomatic and may need revascularization.

- Serum lipid profile and plasma glucose are done to diagnose risk factors.

Treatment

The treatment of stable angina includes:

- Medical treatment
- Coronary revascularization
- Identification and control of risk factors
- Education and reassurance.

Medical Treatment

- Antiplatelet agents:** Low dose aspirin (75-325 mg daily) reduces coronary events in patients of IHD. Aspirin inhibits platelet cyclo-oxygenase irreversibly. Clopidogrel 75 mg daily can be used in place of aspirin if patients develop GI bleeding, allergy, or dyspepsia due to aspirin.
- Nitrates:** Nitrates cause coronary vasodilatation, thus increasing the myocardial oxygen supply. They also reduce the myocardial oxygen demand by decreasing the preload and afterload on the heart. Nitroglycerine 0.4 to 0.6 mg is given sublingually for acute relief of the anginal pain. For long-term therapy, isosorbide dinitrate (10-20 mg 8 hrly) or isosorbide mononitrate (20-60 mg once or twice daily) is given orally. Nitroglycerine ointment or transdermal patches are also available.
- Beta blockers:** These reduce myocardial oxygen demand by decreasing heart rate, blood pressure and myocardial contractility. Atenolol (upto 100 mg daily), metoprolol (upto 200 mg daily) and Bisoprolol (5-10 mg daily) are commonly used beta blockers. These should be avoided in presence of bradycardia, severe bronchospasm, significant atrioventricular block and uncompensated heart failure.
- Calcium antagonists:** They reduce arterial pressure and myocardial contractility and cause coronary vasodilatation. These are useful in situations where beta blockers are contraindicated, ineffective or poorly tolerated. Commonly used agents are verapamil, diltiazem, sustained release nifedipine and amlodipine.
- Ranolazine:** This drug has been recently approved by FDA for use in chronic angina patients. This can be

used as first line agent or in patients who are symptomatic despite adequate antianginal therapy. Increase in intracellular sodium concentration in ischemic cardiac monocytes cause calcium overload via the $\text{Na}^+/\text{Ca}^{++}$ exchanger leading to contractile dysfunction and cellular injury. Ranolazine is piperazine derivative and it acts by inhibiting pathologically enhanced lateral inward sodium current thus preventing sodium overload of ischemic monocytes. This leads to decreased calcium overload via the $\text{Na}^+/\text{Ca}^{++}$ exchanger and attenuates the accompanying deleterious consequences. It thus minimizes diastolic tension and increases coronary nutrient flow as compression of intramyocardial arterioles is decreased. This drug can be classified as cytoprotective. This is given in the doses of 500-1000 mg per day in two divided doses.

- f. *Nicorandil*: This is potassium channel opener and given 20 mg BD. It leads reduction of free intracellular Ca^{++} ions through opening of potassium channels.

Coronary Revascularization

Percutaneous transluminal coronary angioplasty (PTCA): A wire is passed across coronary narrowing under radiographic control and the stenosed part is dilated with the balloon. A metallic stent is placed to maintain maximum dilatation and prevent restenosis. PTCA offers better control of symptoms than medical therapy in patients with chronic stable angina. However, there is no evidence that PTCA improves survival. The most common indication of PTCA in chronic stable angina is the presence of evidence of ischemia during stress test despite medical therapy.

Coronary artery bypass grafting (CABG): Internal mammary artery, radial artery or saphenous vein is used as a graft to bypass the obstructive lesion in the coronary artery. The indications for CABG are left main coronary stenosis, two vessel or three vessel disease with left ventricular dysfunction, and patients with diabetes mellitus.

Identification and Control of Risk Factors:

- a. Cessation of smoking
- b. Control of blood pressure
- c. Control of blood sugar
- d. Treatment of hyperlipidemia
- e. Control of weight
- f. Regular exercise (20 to 30 minutes daily)

UNSTABLE ANGINA

Presently, unstable angina, non-ST segment elevation myocardial infarction (NSTEMI) and myocardial infarction with ST segment elevation (STEMI) are included in a group called *acute coronary syndromes (ACS)*.

The diagnosis of unstable angina is considered when there is

- Ischemic chest pain on rest or with minimal exertion
- New onset angina
- Progression of angina of increasing frequency, severity, duration and at lower level of exertion.

The diagnosis of Non-ST segment elevation myocardial infarction (NSTEMI) is made when features of unstable angina are associated with elevation of cardiac markers of myocardial necrosis (Troponin T and I, Creatine kinase, CK-MB)

Pathophysiology of Unstable Angina

There is rupture or fissuring of coronary artery atheromatous plaque leading to formation of thrombus and acute reduction in the coronary blood flow. This may also be accompanied with coronary artery spasm. The angina due to vasospasm not associated with atheroma is termed as *Prinzmetal's angina*.

The obstruction in the coronary blood flow may progress, hence UA or NSTEMI may evolve into STEMI.

Investigations

- **ECG:** The ECG in patients with UA shows ST segment depression and T wave inversion without a rise in cardiac isoenzymes like creatine kinase MB (CK-MB) or troponin. Prinzmetal's angina is characterized by ST segment elevation.
- **Cardiac markers of necrosis:** NSTEMI is diagnosed when the features of UA are also associated with increased cardiac isoenzymes, i.e., markers of cardiac necrosis.
- **C-reactive protein (CRP):** CRP levels may aid in initial risk assessment in ACS patients. Increased levels of CRP in ACS are associated with poor prognosis and recurrent cardiac events.

Treatment

The mainstay of treatment of UA/NSTEMI includes following points. Various drugs used in ACS are given in Table 4.32.

TABLE: 4.32: Drugs used in acute coronary syndromes

1. Antiplatelet therapy
 - a. Oral
 - Aspirin
 - Clopidogrel
 - b. Intravenous
 - Glycoprotein IIB/IIIA inhibitors
 - Abciximab
 - Eptifibatide
 - Tirofiban
2. Anti-ischemic therapy/analgesia
 - a. Nitrates
 - Nitroglycerine
 - Isosorbide dinitrate
 - Isosorbide mononitrate
 - b. Beta blockers
 - Atenolol
 - Metoprolol
 - Esmolol
 - Bisoprolol
 - c. Calcium channel blockers
 - Amlodipine
 - Diltiazem
 - Verapamil
 - d. Morphine sulphate
3. Heparins (antithrombotics)
 - Unfractionated heparin
 - Low molecular weight heparin e.g. enoxaparin
4. Thrombolytic therapy (STEMI only)
 - Streptokinase
 - Alteplase (Recombinant tissue-plasminogen activator, rt-PA)
 - Tenecteplase (TNK-tPA)
 - Reteplase (r-PA)
5. Prevention of Ventricular remodelling
 - Ramipril
 - Captopril
 - Enalapril
6. Plaque stabilization therapy
 - HMG -coA reductase inhibitors
 - Atorvastatin
 - Simvastatin
 - Rosuvastatin

- a. Bed rest and supplemental oxygen
 - b. Antiplatelet therapy
 - c. Anti-ischemic therapy
 - d. Anticoagulant therapy
 - e. Risk stratification and revascularization
 - f. Control of risk factors.
- a. *Bed rest:* The patient should be hospitalized and put on bed rest. Continuous ECG monitoring is done and

supplemental oxygen is given. Morphine can be used for pain relief.

- b. *Antiplatelet therapy:* Aspirin and/or clopidogrel are given in the dosage as mentioned in the case of stable angina. Intravenous Gp IIB/IIIA inhibitors (Abciximab, Eptifibatide, Tirofiban) are indicated in high risk patients.
- c. *Anti-ischemic therapy:* It provides pain relief and prevents recurrence of chest pain.
 - *Nitrates* should be given sublingually or by buccal spray. If pain persists, intravenous nitroglycerine is initiated.
 - Initially beta blocker (metoprolol) is given intravenously. Later oral preparations of beta blocker (metoprolol, atenolol) are continued.
 - *Calcium antagonists* (verapamil, diltiazem) are added when there is inadequate response to beta blockers and nitrates. Calcium antagonists are also useful in cases where beta blockers are contraindicated.
 - *Statins (atorvastatin) and angiotensin converting enzyme (ACE) inhibitors* are given for long-term secondary prevention.
- d. *Anticoagulant therapy:* Heparin should be given either as intravenous unfractionated heparin (UFH) or subcutaneous low molecular weight heparin (LMWH). Treatment with UFH requires monitoring of PTT (partial thromboplastin time) for adjustment of the dose.
- e. *Risk stratification:* Based on clinical features, ECG findings and changes in cardiac enzymes, patients are stratified into low and high risk groups. High risk patients are subjected to coronary arteriography and subsequent PTCA or CABG. Low risk patients should undergo early stress test to ascertain the need for angiography.

ACUTE MYOCARDIAL INFARCTION OR STEMI

Acute MI results due to formation of occlusive coronary thrombus on ruptured atheromatous plaque. The prolonged ischemia leads to myocardial necrosis. The ischemic chest pain is the most common presenting symptom, which is generally

- severe, longer lasting (>30 minutes) and
- may not respond to sublingual nitroglycerine.

- Pain may be absent in diabetics, hypertensive, postoperative and elderly individuals (*silent MI*).
- Other clinical features include marked sweating, breathlessness, nausea, vomiting, syncope and collapse.
- Sudden death may occur due to ventricular arrhythmias or asystole.

General examination may reveal pallor, sweating, tachycardia or bradycardia, mild fever, hypotension, cold peripheries, raised jugular venous pressure and restlessness. Third or fourth heart sounds, murmur of mitral regurgitation, pericardial friction rub may be found on cardiac examination. The presence of crackles (crepts) in lungs suggests left ventricular failure.

Investigations

- Electrocardiography:** Initially ST segment is elevated and T wave may be tall. Later on, size of R wave
- Cardiac markers:** Intracellular cardiac enzymes and proteins are released into plasma due to myocardial necrosis. Serial estimation to detect the change in the levels of plasma CPK-MB and troponin T and troponin I are most useful for the diagnosis of MI. Troponin T and troponin I increase 3-12 hours after the onset of MI, peak at 24-48 hours, and return to baseline over 5-14 days. Serum CK-MB levels increase 3-12 hours after onset of MI, peak in 24 hours and return to baseline in 48-72 hours.
- Echocardiography:** This is helpful in the detection of wall motion abnormalities, left ventricular dysfunction

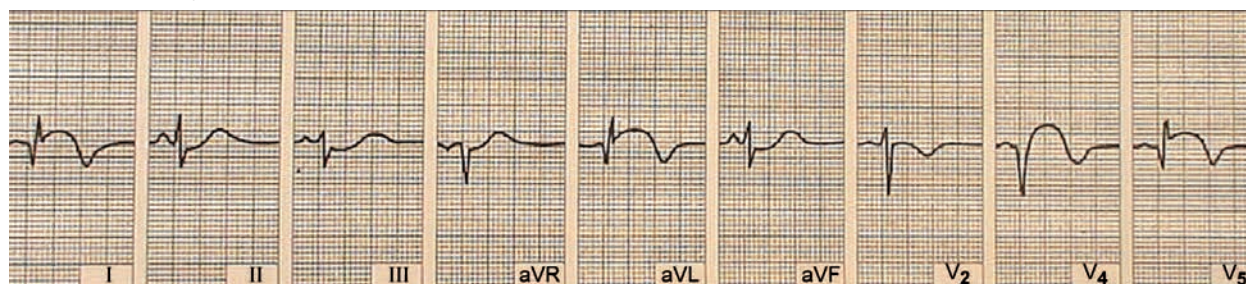


FIGURE 4.9: ECG of lateral wall myocardial infarction (early or hyperacute) showing ST segment elevation in leads I, aVL, V₄ and V₆

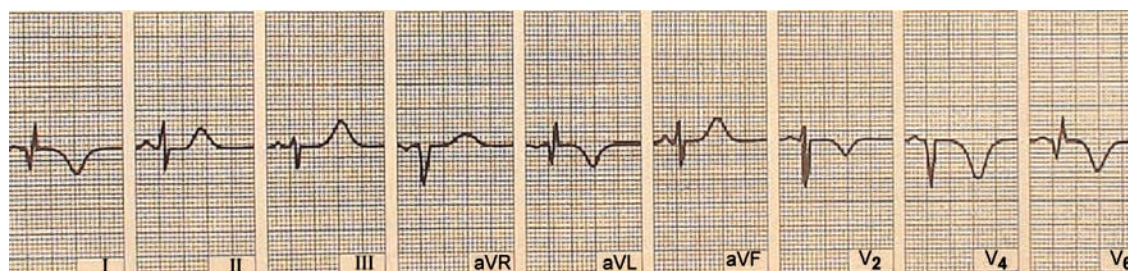


FIGURE 4.10: ECG of lateral wall myocardial infarction (recent) showing Q waves and prominent T wave inversion in leads I, aVL, V₄ and V₆

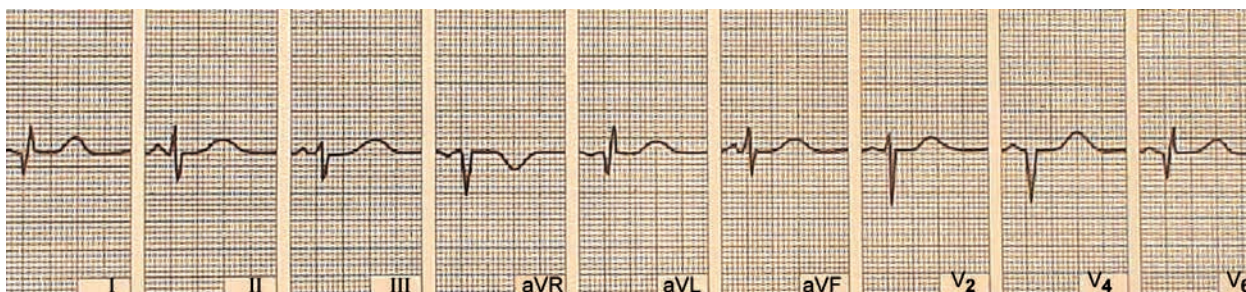


FIGURE 4.11: ECG of lateral wall myocardial infarction (old) showing Q waves in leads I, aVL, V₄ and V₆

and other complications such as pericardial effusion, mitral regurgitation, ventricular septal defect and left ventricular thrombus.

- d. *Other tests:* Chest X-ray may reveal pulmonary edema. Leukocytosis, raised ESR and CRP can also be present. Lipid profile, serum electrolytes, urea and creatinine are also measured.

Management

1. *Hospitalization:* Patient should be hospitalized in coronary care unit (CCU) with facilities of continuous ECG monitoring and defibrillator as promptly as possible since many patients die within the first 24 hours of onset of symptoms of MI and over half occur within first hour due to ventricular fibrillation. Patient should be confined to bedrest. The goal is a door-to-needle time of <30 min and door-to-balloon time of <90 min.
2. *Oxygen therapy:* Oxygen is administered for initial few hours and continued if hypoxemia is present.
3. *Antiplatelet agents:* Aspirin (nonenteric coated) in the dosage of 160-325 mg should be given to the patient to chew. Later aspirin is continued in the dosage of 75-162 mg orally daily. Clopidogrel (300 mg loading dose followed by 75 mg/day) can be used if there is allergy or intolerance to aspirin.
4. *Relief of pain:*
 - Sublingual nitroglycerine (0.4 mg) is given for pain relief. This may be given at interval of 5 minutes. Intravenous nitroglycerine is started if pain reoccurs or there is associated hypertension.
 - *Morphine* also can be used for pain relief (2-4 mg IV).
 - Intravenous beta blockers such as metoprolol are also helpful for pain relief followed by daily oral therapy.
5. *Reperfusion therapy:* This is given to restore the flow in the infarct related artery and to limit the infarct size.
 - a. Fibrinolytic agent like streptokinase or tPA (tissue plasminogen activator) is given through intravenous infusion. Alternatively, reteplase or tenecteplase can be administered as bolus (Bolus fibrinolytics). It should be administered as soon as possible. However, it can be useful if given within 6 hours of presentation. The contraindications are history of cerebrovascular accident, severe hypertension and

evidence of active internal bleeding. Bleeding is the most frequent side effect of fibrinolytic therapy. Streptokinase can result in hypersensitivity reaction.

- b. *Primary percutaneous intervention (PCI):* Primary percutaneous intervention (angioplasty, stenting) is expensive and available at select centers. When performed by experienced operators it appears to be more effective than fibrinolysis. This can also be performed in case of contraindication or failure of fibrinolytic therapy (rescue PCI).
6. *Anticoagulant therapy:* Unfractionated heparin is given as intravenous infusion to maintain the patency of infarct related artery. Later, oral anticoagulant is given for prolonged use where indicated.
7. *ACE inhibitors are given to prevent ventricular remodeling.* This is continued indefinitely in patients with hypertension or heart failure.
8. *Other measures:* Patients are on bed rest for at least 12 hours. Gradually, the activity is increased depending on the tolerance. Liquid diet is started when chest pain subsides. Later on, low fat, high fiber diet is prescribed. Laxatives are given to relieve constipation. Sedative (diazepam, lorazepam) is generally given for proper sleep.
9. *Risk stratification:* Patients are stratified into low and high risk groups depending on presence of persistent ischemia, heart failure and symptomatic ventricular arrhythmias. High risk patients are subjected to coronary angiography and subsequent PTCA or CABG. Low risk patients should undergo early stress test to ascertain the need for angiography.

Secondary prevention: The secondary prevention of MI includes long-term administration of aspirin or clopidogrel, beta blockers, ACE inhibitors, control of risk factors like cessation of smoking, statin therapy for hyperlipidemia, control of blood pressure and blood sugar, weight reduction in obese and regular exercise. Oral anticoagulants may be required in some cases.

Complications of MI

The complications include

1. *Arrhythmia:* Ventricular fibrillation is the major cause of early death in MI. Other common arrhythmias are atrial fibrillation and heart blocks.

2. Pump failure or cardiogenic shock where more than 40% myocardium is damaged. This is an important cause of inhospital mortality.
3. Recurrent ischemia or chest pain
4. Pericarditis, Dressler's syndrome
5. Thromboembolism
6. Ventricular remodeling
7. Left ventricular aneurysm
8. Mechanical complications like ventricular rupture and cardiac tamponade, mitral regurgitation and rupture of interventricular septum.

HEART FAILURE

Heart failure (HF) is a clinical syndrome in which the heart is unable to pump adequate amount of blood to meet the metabolic demands of the body. Abnormality in cardiac structure or function may be responsible for the inability of heart to pump blood adequately.

Causes

Heart failure generally occurs due to defect in the myocardial contractility. Ischemic heart disease and cardiomyopathy are the most common causes of HF. Causes of HF are listed in detail in Table 4.33.

Precipitating Factors

In certain situations, the load over an already burdened myocardium is suddenly increased leading to HF. Important precipitating causes are infection, arrhythmias, anemia, excessive physical exertion, emotional stress and fluid overload. Pregnancy, thyrotoxicosis and accelerated hypertension may also precipitate HF.

TABLE 4.33: Important causes of heart failure

- Ischemic heart disease
- Cardiomyopathy
- Myocarditis
- Valvular heart disease
- Hypertensive heart disease
- Congenital heart disease
- Infective endocarditis
- Pulmonary embolism
- Constrictive pericarditis

Pathophysiology

The pump failure leads to low cardiac output, which activates counter-regulatory neurohormonal mechanisms such as

- activation of renin angiotensin aldosterone system (RAAS)
- stimulation of sympathetic system
- increased vasopressin secretion

The net effects of these mechanisms are sodium and fluid retention and vasoconstriction. These tend to improve cardiac output; however, these are also responsible for various manifestations of heart failure due to systemic and pulmonary congestion. Natriuretic peptides are released from the stretched myocytes and mediate vasodilatation, diuresis and sodium loss.

Symptoms and Signs

Symptoms and signs in CHF are due to

- heart failure *per se*
- underlying causes of heart failure
- precipitating factor

Clinical features of heart failure are given in Table 4.34 and 4.35. Mechanisms underlying clinical features are given in Table 4.36.

Symptoms of HF mainly arise due to increased pulmonary and systemic venous and capillary pressure and reduced cardiac output. The important symptoms are dyspnea, orthopnea, paroxysmal nocturnal dyspnea, weakness and fatigue, edema and abdominal pain.

Dyspnea is the most important symptom and is multifactorial in origin. The most important mechanism of dyspnea is accumulation of interstitial and alveolar fluid due to increased pulmonary capillary pressure. Initially, the

TABLE: 4.34: Symptoms and signs of heart failure

Symptoms

- Dyspnea (breathlessness)
- Orthopnea
- Paroxysmal nocturnal dyspnea
- Fatigue and weakness
- Pink frothy sputum (Acute pulmonary edema)
- Right upper quadrant abdominal pain (Congestion of liver)
- Anorexia, nausea, early satiety (due to edema of intestine)
- Confusion, memory impairment, disorientation (reduced cerebral perfusion)

TABLE 4.35: Signs of heart failure

General physical examination
Tachycardia, Pulsus alternans
Elevated jugular venous pressure (JVP)
Bilateral, pitting pedal edema/sacral edema
Cyanosis of nail bed and lips
Cold extremities
Weight loss (cardiac cachexia)
Abdomen
Tender hepatomegaly
Ascites
Pulmonary
Pulmonary rales (crackles, crepitations)
Wheezing (cardiac asthma)
Pleural effusion
Cardiac
Cardiomegaly
Third heart sounds (S ₃)
Murmurs of mitral and tricuspid regurgitation

TABLE 4.36: Clinical features of heart failure

Due to increased preload
Increased systemic venous pressure (right heart failure)
• Raised JVP
• Tender hepatomegaly
• Edema
• Ascites, pleural effusion*
Increased pulmonary venous pressure (left heart failure)
• Dyspnea, orthopnea, PND
• Hemoptysis (pink frothy sputum)
• Central cyanosis
• Crackles on auscultation
• Pleural effusion*
Due to decreased cardiac output and increased afterload
• Cold clammy skin
• Peripheral cyanosis
• Fatigue
• Cardiac cachexia
Due to decreased cardiac contractility
• Ventricular dilatation (cardiomegaly)
• Third heart sound (S ₃)
• Pulsus alternans
Arrhythmias
• Sinus tachycardia
* Pleural effusion occurs when both pulmonary and systemic venous pressure are raised

dyspnea occurs only during exertion. Later on, as the disease progresses, the patient is dyspneic even on rest.

- Patients may feel dyspneic in the recumbent position (orthopnea) due to increased venous return to heart and elevation of the diaphragm. The dyspnea is usually relieved by sitting upright.

- There may be episodic severe breathlessness and cough which awaken the patient from sleep (*paroxysmal nocturnal dyspnea*).
- In its severe form, there is intense bronchospasm and wheeze, a condition known as *cardiac asthma*.
- *Acute pulmonary edema* is a severe form of cardiac asthma wherein the patient has alveolar edema and expectoration of blood-tinged sputum

Low cardiac output leads to poor effort tolerance, fatigue and weakness. Confusion, lack of concentration, memory impairment, headache and insomnia may occur particularly in elderly patients. There may also be a reduced urine output.

Patients may also have dependent edema in the lower limbs. Edema in the sacral region occurs in bed ridden patients. Congestion of liver and portal venous system leads to pain in the abdomen, nausea, anorexia and malabsorption. There may be marked weight loss in severe chronic HF (*cardiac cachexia*).

The findings on general physical examination in patients with HF include cold extremities, tachycardia, pulsus alternans, hypotension, cyanosis of lips and nail beds, increased jugular venous pressure, pitting edema and tender hepatomegaly. *Pulsus alternans* is regular pulse with regular alteration in pulse amplitude (alternate low and large volume pulse), and is a sign of severe heart failure.

Lung examination reveals inspiratory rales (crepitations) over the bases. In severe cases with pulmonary edema, coarse rales and wheezes all over lung fields are found (cardiac asthma). Third and fourth heart sounds are often heard. Sometimes, patients may exhibit signs of pleural effusion and/or ascites.

Types of Heart Failure

Heart failure has been classified in several ways:

- a. *Acute or chronic heart failure:* Acute HF occurs suddenly as in massive MI and acute infective endocarditis. Chronic HF presents slowly in conditions like valvular heart disease and cardiomyopathy. Chronic HF may be of the compensated or uncompensated type.
- b. *Right or left or biventricular heart failure:*
 - Left heart failure: Dysfunction of left ventricles (left ventricular failure) may lead to pulmonary congestion (increased pulmonary venous pressure) that may present as dyspnea, orthopnea, paroxysmal

nocturnal dyspnea and crackles in lungs (Table 4.37). The important causes of LVF are rheumatic valvular heart disease (AR, MR, AS), myocardial infarction, cardiomyopathy and hypertension.

- Right heart failure: Right ventricular failure (RVF) presents in the form of systemic congestion presenting as tender hepatomegaly, raised JVP, pedal edema (Table 4.38). RVF can occur secondary to LVF due to its effects on pulmonary circulation (raised pulmonary arterial pressure). Other important causes of RVF are cor pulmonale (right ventricular enlargement secondary to lung diseases), pulmonary embolism and pulmonary stenosis. Pulmonary congestion is uncommon. Subsequently both ventricular chambers may be affected (biventricular heart failure).
- c. *Forward or backward heart failure*: The cardiac output is primarily inadequate in forward failure whereas backward failure is characterized by marked salt and water retention causing systemic and pulmonary venous congestion.
- d. *Systolic or diastolic heart failure*: Heart failure due to impaired myocardial contraction is called systolic HF while failure of ventricle to relax and fill normally is called diastolic HF. The causes of diastolic HF are constrictive pericarditis, restrictive cardiomyopathy and ischemic heart disease.

TABLE 4.37: Clinical features of left heart failure

Symptoms
Dyspnea
Orthopnea
Paroxysmal nocturnal dyspnea (PND)
Acute pulmonary edema
Signs
Pulmonary rales (crackles, crepitations)
Third heart sound
Pulsus alternans
Pleural effusion

TABLE 4.38: Clinical features of right heart failure

Raised jugular venous pressure (JVP)
Tender hepatomegaly
Bilateral pitting pedal edema
Ascites
Pleural effusion

- e. *High output or low output heart failure*: High output heart failure occurs in patients with hyperthyroidism, anemia, pregnancy, AV fistula and beriberi. Low cardiac output failure occurs in Ischemic heart disease, hypertension, cardiomyopathy, valvular and pericardial disease.

Investigations

Investigations are carried out to detect the cause, nature and severity of the HF.

- Blood tests: The blood tests generally include hemoglobin, blood urea, electrolytes and thyroid function tests. X-ray chest may reveal cardiomegaly and pulmonary congestion.
- ECG: Electrocardiography may show arrhythmia and signs of chamber enlargement.
- Echocardiography: Echocardiography is very helpful in knowing about the cause of the HF, the type of cardiac dysfunction (systolic or diastolic) and the presence of valvular abnormalities.
- Brain natriuretic peptide (BNP): Recently measurement of *brain natriuretic peptide (BNP)* has been shown to be a highly sensitive and specific test for the diagnosis, prognosis and monitoring of the HF. High BNP level can differentiate cardiac asthma from the dyspnea of pulmonary origin.

Treatment

The treatment of HF includes:

- correction of the underlying cause
- removal of the precipitating factors
- control of congestive heart failure.

Correction of Precipitating Factors

Precipitating factors in patients with underlying heart disease should be searched and corrected. Some important factors are:

- Excessive salt and fluid intake
- Excessive activity
- Arrhythmias
- Infections
- Anemia
- Pulmonary embolism
- Noncompliance in therapy

Control of Heart Failure

Control of HF can be achieved by general measures and drug therapy.

1. General measures:

- Reduced physical exertion lowers myocardial oxygen demand and thus controls failure. Absolute bed rest may be needed in severe heart failure. In stable patients, regular isotonic exercises are advised.
- The salt intake is reduced to half of the normal or even less than 2 g per day in some cases.
- Weight loss in obese patients is also helpful in controlling heart failure.

2. Drug therapy: Drugs may help in HF by reducing preload and afterload and improving myocardial contractility. This can be achieved generally with the help of more than one class of drugs (Table 4.39).

- a. **Diuretics:** Diuretics lead to increased urinary sodium excretion and reduction in plasma volume. This lowers the preload to the heart.
- Thiazide diuretics can be used alone in mild HF and in combination with other class of diuretics in severe HF.
 - Loop diuretics are given in severe HF and in cases with renal impairment. Intravenous frusemide causes venodilatation and is useful in acute pulmonary edema.

TABLE 4.39: Drugs used in heart failure

1. <i>Diuretics</i>
Thiazide: chlorthiazide, metolazone
Loop diuretics: frusemide, bumetanide, torsemide
Potassium sparing: spironolactone, amiloride, triamterene
2. <i>Vasodilators</i>
Oral:
ACE inhibitors: ramipril, enalapril, lisinopril
ARBs : losartan, valsartan
Isosorbide dinitrate
Hydralazine
Parenteral:
Sodium nitroprusside
Nitroglycerine
Nesiritide (rBNP)
3. <i>Inotropic agents</i>
Digitalis
Sympathomimetic amines: dopamine, dobutamine
Phosphodiesterase inhibitors: amrinone, milrinone
4. <i>Beta blockers</i>
Metoprolol, carvedilol, bisoprolol

- Potassium sparing agents are often used in combination with other diuretics to minimize hypokalemia. Spironolactone in low dosage (25 mg daily) has been shown to improve morbidity and mortality in advanced HF.
- b. **Vasodilators:** Activation of renin-angiotensin-aldosterone system (RAAS), adrenergic nervous system and increased secretion of vasopressin cause vasoconstriction and salt and fluid retention. Vasodilators help in HF as they reduce preload and afterload through venodilatation and arterial dilatation respectively.
- ACE inhibitors improve exercise tolerance and reduce morbidity and mortality in patients with HF. They also prevent the onset of failure in patients with left ventricular dysfunction.
 - Angiotensin II receptor blockers (ARB) are equally effective as ACE inhibitors and have less incidence of cough.
 - Nitrates and hydralazine are also useful for chronic oral administration.
 - Intravenous nitroglycerine, sodium nitroprusside or nesiritide (recombinant BNP) is indicated in acute severe HF.
- c. **Inotropic agents:** They improve myocardial contractility. Various types of inotropic agents include:
- Digitalis: Digitalis relieves symptoms of HF and is particularly indicated in patients of HF with atrial flutter or fibrillation.
 - Sympathomimetic amines (dopamine, dobutamine): Sympathomimetic amines like intravenous infusion of dopamine or dobutamine are useful in acute severe HF.
 - Phosphodiesterase inhibitors (amrinone, milrinone): Amrinone and milrinone (phosphodiesterase inhibitors) also have vasodilator properties and are used alone or together with sympathomimetic amines in severe refractory HF.
- d. **Beta blockers:** Metoprolol, carvedilol and bisoprolol are shown to improve symptoms of HF and survival in patients with moderate to severe HF if added to other group of drugs. They also prevent arrhythmias.

These agents act by counteracting the adverse effects of enhanced adrenergic stimulation. Beta blockers are started in low dosage initially and the dose is increased gradually with care.

3. *Other measures:*

- **Anticoagulant therapy:** Anticoagulant therapy is given to patients with heart failure and atrial fibrillation or to patients with severe heart failure to prevent thromboembolism.
- **Antiarrhythmic therapy:** Amiodarone is given to control ventricular arrhythmias
- **Pacing:** Biventricular pacing resynchronizes the cardiac contraction and improves the performance in severe heart failure with conduction abnormalities
- **Circulatory support:** Mechanical circulatory support like intra-aortic balloon pump is helpful in intractable cases
- **Cardiac transplantation:** Cardiac transplantation is considered in patients with refractory severe end stage heart failure.

ACUTE PULMONARY EDEMA (CARDIOGENIC)

Acute pulmonary edema is characterized by the accumulation of fluid in the pulmonary interstitium and alveoli due to increase in the pulmonary venous and capillary hydrostatic pressure. The lungs become less compliant, airways resistance increases and the capillary-alveolar gas exchange is compromised.

Non-cardiac pulmonary edema is characterized by high venous and capillary permeability with low hydrostatic pressure.

Causes

Pulmonary edema can primarily be due to cardiac cause or of noncardiac etiology. The important cardiac causes of acute pulmonary edema are left ventricular failure (acute MI, acute MR and acute AR) and mitral stenosis. The noncardiac causes of pulmonary edema are listed in Table 4.40.

Clinical Presentations

There is rapid onset of dyspnea, cough, anxiety and restlessness. The dyspnea is more on lying down and sitting

TABLE 4.40: Noncardiogenic pulmonary edema

- | |
|---|
| 1. Acute respiratory distress syndrome (ARDS) |
| a. Infections |
| b. Inhaled toxins |
| c. Aspiration of gastric contents |
| d. Disseminated intravascular coagulation |
| e. Acute pancreatitis |
| f. Multiple trauma |
| g. Anaphylaxis |
| 2. High altitude |
| 3. Narcotic overdose |

upright may provide some relief. Additionally, patients may have excessive sweating and cyanosis. The sputum may be profuse, pink frothy or blood stained. The examination may reveal cold extremities, tachycardia and tachypnea with prominent use of accessory muscles of respiration, hypotension. There are extensive rales (crackles, crepitations) and wheezes on lung auscultation. Cardiac examination will reveal findings of underlying heart disease.

Investigations

- The chest X-ray may reveal cardiomegaly and signs of pulmonary edema (prominent interstitial and hilar shadows, Kerley B lines, pleural effusion).
- Echocardiography may help in differentiating cardiac from noncardiac causes.
- A rise in serum brain natriuretic peptide (BNP) level favors acute left ventricular failure as a cause of pulmonary edema.

Management of Cardiogenic Pulmonary Edema

Acute pulmonary edema is a life-threatening condition and needs prompt management.

Initial supports:

1. Oxygen is administered to improve hypoxia. In severe cases, positive pressure ventilation may be needed.
2. Patient is placed in sitting posture.

Drug therapy:

1. *Intravenous morphine* (2-4 mg) is given to relieve anxiety. It also reduces preload through venodilation and thus reduces dyspnea. This can be repeated if needed.
2. *Furosemide* (20-80 mg IV) offers rapid improvement in dyspnea by decreasing pulmonary congestion through

its venodilator effect prior to causing diuresis. Hence, it is the diuretic of choice in acute pulmonary edema.

3. *Intravenous nitroglycerine* is also helpful, particularly in cases of ischemic heart disease. Nitroglycerine decreases preload to heart as it is a venodilator. It is contraindicated if there is hypotension. Nitroprusside infusion is indicated in cases of acute pulmonary edema associated with hypertension.
4. *Inotropic agents* (dopamine, dobutamine) and phosphodiesterase inhibitors (amrinone, milrinone) stimulate myocardial contractility and are useful in patients with hypotension or shock.
5. Intravenous recombinant *BNP* is a potent vasodilator with diuretic properties. It is also effective in acute pulmonary edema.
6. *Hemodialysis* may be needed in patients with severe renal failure.

Correction of Precipitating Factors

The precipitating factors of acute pulmonary edema such as severe hypertension, IHD, arrhythmias, volume overload should be adequately managed.

CONGENITAL HEART DISEASE

Congenital heart disease (CHD) may present in early childhood or may remain asymptomatic till adult life. Its incidence is around 1% of live birth. Congenital abnormalities are due to multifactorial environmental and genetic causes. Rubella infection in mother is associated with patent ductus arteriosus (PDA) and pulmonary artery stenosis. Maternal lupus erythematosus is linked with congenital complete heart block. Use of drugs or toxins (alcohol) during pregnancy can also lead to CHD. Chromosomal abnormalities such as Turner's syndrome and Down's syndrome are associated with coarctation of aorta and septal defects respectively.

Classification

CHD may present with cyanosis or without cyanosis. Overall the most common CHD is ventricular septal defect (VSD) whereas the most common cyanotic heart disease is tetralogy of Fallot. This anomaly can occur in isolation or in combination. The classification of CHD is given in Table 4.41.

TABLE 4.41: The classification of congenital heart disease

1. <i>Acyanotic</i>
a. Left to right shunt
Ventricular septal defect (VSD)
Atrial septal defect (ASD)
Patent ductus arteriosus (PDA)
b. Without shunt
Bicuspid aortic valve
Coarctation of aorta
Congenital aortic stenosis
Pulmonary stenosis
Congenital mitral stenosis
2. <i>Cyanotic</i>
a. Increased pulmonary blood flow
Complete transposition of great arteries
Total anomalous pulmonary venous connection
b. Normal or decreased blood flow
Tetralogy of Fallot
Tricuspid atresia
Pulmonary atresia
3. <i>Others</i>
a. Dextrocardia
b. Congenital complete heart block

Ventricular Septal Defect (VSD)

VSD is the most common type of CHD. There is incomplete septation of the ventricles. The defect is most commonly present at the junction of membranous and muscular portion of the interventricular septum. VSD can occur alone or can be accompanied by other defects like aortic regurgitation.

The septal defect allows blood to shunt from left ventricle (high pressure) to right ventricle (low pressure). The right ventricular volume is increased as it receives blood from right atrium (RA) as well as from left ventricle (LV). Hence, the pulmonary flow is increased which in turn leads to an increase in the volume of left atrium (LA) and LV. The increased blood flow and obstructive structural changes in the pulmonary vascular bed result in the increase in pulmonary arterial pressure (pulmonary hypertension).

Pulmonary hypertension can in turn lead to elevated right ventricle (RV) pressure and reversal of shunt (right to left) as the pressure in RV exceeds that of LV. This condition is known as *Eisenmenger syndrome*. Central cyanosis and digital clubbing may also develop.

Clinical Features

The small VSD is generally asymptomatic whereas symptoms of heart failure (fatigue and dyspnea) occur in

large VSD. There is an increased incidence of respiratory infection. In advanced cases, cyanosis and digital clubbing appear due to reversal of shunt. There is a pansystolic murmur best heard at left sternal border in smaller VSD (*Maladie de Roger's* murmur). However, the murmur is softer in large VSD as the RV pressure is elevated.

Investigations

Echocardiography is used to detect the location and size of the VSD. The ECG may show biventricular hypertrophy and chest X-ray reveals pulmonary plethora (prominent pulmonary vascular markings).

Treatment

- Small defects should be watched as they may close spontaneously. Surgery is indicated in larger defects.
- Eisenmenger syndrome is a contraindication for surgery where the only option is heart-lung transplantation.
- In VSD, there is a moderate risk of developing infective endocarditis. Hence, endocarditis prophylaxis is mandatory before any procedure.

Atrial Septal Defect (ASD)

ASD occurs more commonly in females. The commonest type of ASD is *ostium secundum*. Other types are *ostium primum* and *sinus venosus*. Atrial septal defect may be accompanied by mitral stenosis (*Lutembacher's syndrome*).

A large amount of blood shunts from LA to RA. The right ventricular volume and pulmonary flow are increased which may lead to pulmonary hypertension. The pulmonary hypertension can in turn lead to elevated RV pressure and reversal of shunt (right to left) as pressure in RV exceeds that of LV. This condition is known as *Eisenmenger syndrome*.

Clinical Manifestations

ASD is generally asymptomatic in early life. However, there is an increased tendency for recurrent respiratory infection. Later during the fourth or fifth decade, it may present as atrial fibrillation, heart failure and Eisenmenger syndrome. Important clinical signs include wide and fixed splitting of second heart sound and a systolic flow murmur over the pulmonary area.

Investigations

Echocardiography is used to detect the location and size of the ASD. The ECG may show incomplete right bundle branch block. Chest X-ray reveals pulmonary plethora and enlargement of the heart.

Treatment

- Large ASD is closed surgically. Eisenmenger syndrome is a contraindication for surgery.
- Isolated secundum lesion has a low risk of developing infective endocarditis, hence prophylaxis is generally not needed.

Patent Ductus Arteriosus (PDA)

In fetal life, the blood from the pulmonary artery does not go to lungs but flows into the aorta through a communication, ductus arteriosus. However, the ductus closes soon after the birth as the blood from pulmonary artery goes to lungs. In some, the ductus fails to close, a condition known as PDA. This is more common in females and is sometimes associated with other congenital abnormalities.

In PDA, the blood flows from high pressure (aorta) to low pressure (pulmonary artery) side, resulting in left to right shunt. Pulmonary artery pressure rises with the rise in pulmonary vascular resistance. As the pressure in pulmonary artery exceeds that of aorta, right to left shunting occurs, the blood flow reverses (*Eisenmenger syndrome*). This leads to cyanosis, more in the feet than in upper part of the body, also called *differential cyanosis*.

Clinical Manifestations

Patients with small shunt are generally asymptomatic for years. However, there may be retarded growth and development in children with large shunt. With the onset of cardiac failure, the patients develop dyspnea. The arterial pulse is generally high volume. Cardiac finding includes a *continuous 'machinery' murmur* in the left second intercostal space often with a thrill. In advanced cases, P₂ becomes loud and the murmur is shortened. Patients develop cyanosis and features of right heart failure (*Eisenmenger syndrome*).

Investigations

The echocardiography is used to detect the location and size of the PDA. The ECG may show evidence of RV hypertrophy. Chest X-ray reveals pulmonary plethora and enlargement of heart.

Management

The large PDA is occluded with an implantable device in infancy before pulmonary hypertension develops. However,

surgery in small PDA is indicated to reduce the chance of endocarditis but may be delayed until later childhood.

A prostaglandin synthetase inhibitor (ibuprofen or indomethacin) may induce closure in the first week of life if there is no intrinsic abnormality of ductus.

Coarctation of Aorta

Congenital narrowing of aorta (coarctation) occurs most commonly just below the origin of left subclavian artery. “Acquired” coarctation of aorta may result due to trauma or arteritis (Takayasu’s disease). This is more common in males and may be associated with other congenital anomalies like bicuspid aortic valve and “berry aneurysms” of cerebral arteries.

Clinical Features

- If severe, it may cause heart failure in the newborn.
- In adults, headache may occur because of hypertension. Blood pressure is raised in upper extremities (in vessels above narrowing) but is normal or low in lower limbs.
- Lower extremities are less developed due to decreased blood supply. Weakness and cramps may occur in lower limbs. On examination, lower limbs are cold and arterial pulsations are weak. Femoral pulsations are delayed in comparison to radial pulse (radiofemoral delay).
- Systolic murmur may be present over the area of narrowing and beat is heard posteriorly.
- Bicuspid valve may cause systolic murmur in aortic area.
- Collateral vessels are well developed on chest and back (involving periscapular, intercostals and internal mammary arteries). Bruit may be heard over collaterals.
- Rupture of “berry aneurysm” may cause cerebral hemorrhage.
- Dissection of aorta and infective endocarditis are other complications.

Investigations

ECG may show LVH. Chest X-ray reveals dilated ascending aorta and indentation at the site of coarctation (the “3” sign). Notching of ribs due to collaterals may be seen. Echocardiography, CT and MRI are helpful in the diagnosis and evaluation of the severity of coarctation.

Management

Surgical correction of the coarctation is indicated in all except very mild cases. Balloon dilatation is used to correct postsurgical restenosis.

Tetralogy of Fallot

The tetralogy consists of following 4 components;

1. Ventricular septal defect
2. Pulmonary stenosis
3. Overriding of VSD by aorta
4. Right ventricular hypertrophy

This is the most common cause of cyanotic CHD. The pulmonary flow is reduced due to RV outflow obstruction. This results in shunting of desaturated blood from RV to LV across the VSD leading to cyanosis, clubbing and polycythemia. The cyanosis increases during feeding or crying while it may reduce in squatting posture. There is ejection systolic murmur in the pulmonary area. The complications include infective endocarditis, cerebral infarction or abscess and polycythemia.

Investigations

Electrocardiography shows RV hypertrophy. Chest X-ray reveals ‘boot shaped’ heart and small pulmonary artery. Echocardiography is diagnostic.

Management

- The treatment is corrective operation which is necessary in almost all cases. This includes correction of pulmonary stenosis and closure of VSD. In case of hypoplastic pulmonary artery, an initial palliative shunt between pulmonary artery and subclavian artery is created to facilitate definitive correction.
- Prophylaxis for infective endocarditis is indicated even after surgical correction.

SYNCOPE

Syncope is defined as a transient loss of consciousness due to diminished cerebral perfusion. Syncope is characterized by loss of postural control and spontaneous recovery. It may be preceded by symptoms of ‘presyncope’ such as lightheadedness, visual blurring, dizziness, sweating and nausea.

Causes of Syncope

The causes of syncope can be classified into:

1. Cardiovascular disorders
2. Vascular disorders
3. Cerebrovascular diseases

There are some situations which may resemble syncope (Table 4.42).

Cardiac Syncope

Arrhythmia is the most common cause of syncope due to cardiac cause. The syncope can occur in profound bradycardia or tachyarrhythmia. Under these conditions, the decreased stroke volume can lead to cerebral hypoperfusion. The syncope due to atrioventricular block is known as *Stokes-Adams attack*.

Vasovagal Syncope

- This is the most common cause of “common faint” in normal persons.
- The precipitating factors are hot or crowded environment, severe pain, extreme fatigue, prolonged standing, hunger and emotional situations.
- The vasovagal syncope occurs generally in the sitting and standing posture. Hence, it is essential that the patient

is immediately made to lie down (recumbent position) at the earliest.

- Venous pooling which may occur during prolonged standing or sitting posture reduces the filling of the ventricle. The underfilled ventricle vigorously contracts due to increased sympathetic activation which in turn stimulate myocardial mechanoreceptors and vagal afferent fibers. This causes vasodilation (due to sympathetic inhibition) and bradycardia (increased parasympathetic activity). Vasodilatation and bradycardia produce hypotension and syncope.

Postural Hypotension (Postural Syncope)

- Postural hypotension (orthostatic hypotension, OH) is a fall in systemic arterial pressure on assumption of upright posture. It is defined as a sustained drop in systolic (>20 mm Hg) or diastolic (> 10 mm Hg) blood pressure within 3 minutes of standing (see also Chapter 10).
- The common causes of postural (orthostatic) hypotension are defective postural reflexes and drugs. There is fall in the systemic arterial pressure on assumption of an upright posture.
- These defective postural reflexes are generally due to autonomic peripheral neuropathy such as in diabetes, parkinsonism and aging.
- Hypovolemia because of diuretic therapy, excessive sweating, diarrhea, or hemorrhage may also lead to postural syncope.
- Drugs that cause postural syncope mainly include vasodilators, diuretics and antidepressants.

TABLE 4.42: Causes of syncope

1. Cardiovascular disorders
Arrhythmias
Sinus bradycardia
Heart blocks
Ventricular and supraventricular tachycardia
Aortic stenosis
Hypertrophic obstructive cardiomyopathy
Left ventricular dysfunction
2. Disorders of vascular tone and blood volume
Vasovagal syncope
Postural hypotension
Carotid sinus sensitivity
Cough or micturition syncope
3. Cerebrovascular disease
Vertebrobasilar insufficiency
Basilar artery migraine
Conditions which resemble syncope
Hysterical fainting
Anxiety
Hypoglycemia
Seizures

Seizures and Syncope

The seizures may easily be confused with syncope. However, careful history and examination may reliably differentiate these situations. The important differences are listed in Table 4.43.

Investigations

- Tests like ECG, echocardiography, Holter ECG and electrophysiological studies may be needed to diagnose the cause of syncope.
- Upright tilt test is used to confirm the diagnosis of vasovagal syncope.
- Other tests like EEG, CT, MRI scan may be needed to diagnose any neurological cause.

TABLE 4.43: Differences between syncope and seizures

	Seizure	Syncope
Premonitory symptoms	None or aura	Lightheadedness, nausea, blurring of vision Sweating, palpitation
Precipitating factors	Usually none	Emotional stress, postural hypotension, prolonged standing
Posture	Any posture	Usually erect
Rapidity of onset	Immediate	Gradual
Period of unconsciousness	Prolonged (minutes)	Transient (seconds)
Facial appearance	Cyanosis and frothing	Pallor
Associated findings	Motor seizures, tongue biting, urinary incontinence	Motor movement uncommon and transient
Recovery	Prolonged headache, confusion, focal neurological signs	Rapid and uneventful

Treatment

The treatment depends upon the underlying cause. However, certain precautions are to be taken regardless of the cause.

Immediate Actions to be Taken During Syncope

- The patient should be placed in supine position with head tilted to the side to maximize cerebral blood flow and to avoid aspiration.
- Peripheral stimulation like sprinkling cold water over the face may help.
- Clothing should be loosened.
- The patient should not be allowed to rise again till weakness persists.

Instructions to the Patients

- Patients are advised to avoid situations that have caused the syncope.
- Patients should try to assume a recumbent position as soon as they feel premonitory symptoms.
- Patients who have recurrent syncope should avoid climbing ladders, swimming alone, driving or operating machines.

ARRHYTHMIA

Abnormalities in the **rhythm of heart rate** are known as arrhythmias. Cardiac arrhythmias result from abnormalities of **impulse generation, conduction** or both.

Arrhythmias may present with **palpitations, dyspnea, syncope, fatigue, light headedness, angina, exercise intolerance and skipped beat**. **Palpitations** are suggestive of

tachyarrhythmia. **Bradyarrhythmias** may present as **fatigue, light headedness, syncope and exercise intolerance**. A classification of arrhythmia is given in Table 4.44.

TABLE 4.44: Classification of arrhythmias

- A. Sinus rhythms
 1. Sinus arrhythmia
 2. Sinus bradycardia
 3. Sinus tachycardia
- B. Tachyarrhythmias
 1. Atrial Tachyarrhythmias
 - a. Atrial ectopic beats (extrasystoles, premature beats)
 - b. Atrial tachycardia
 - c. Atrial flutter (AF)
 - d. Atrial fibrillation (AFI)
 - e. Supraventricular tachycardia (SVT)
 - AV nodal re-entry tachycardia (AVNRT)
 - Atrioventricular re-entrant tachycardia (AVRT)
 - Junctional tachycardia
 - Atrial tachycardia.
 2. Ventricular tachyarrhythmias
 - a. Ventricular ectopic beats (extrasystoles, premature beats)
 - b. Ventricular tachycardia (VT)
 - c. Ventricular fibrillation (VF)
 - d. Torsades de pointes
- C. Bradyarrhythmias
 1. Sick sinus syndrome (SSS)
 2. Atrioventricular (AV) block
 - a. First-degree block
 - b. Second-degree block
 - Mobitz type I (Wenckebach) block
 - Mobitz type II block
 - c. Third-degree (complete) heart block
 3. Bundle branch block
 - a. Right bundle branch block (RBBB)
 - b. Left bundle branch block (LBBB)

Diagnostic tools for detecting arrhythmia are following:

- 12 lead **ECG**: An ECG during baseline and at the time of arrhythmia is initial investigation.
- Continuous ambulatory ECG monitoring**: Continuous ambulatory monitoring of ECG for 24-48 hours is useful for diagnosis of transient arrhythmias that occur with sufficient frequency. Correlation between symptoms and heart rate recordings aids in diagnosis of arrhythmia.
- Exercise ECG**: is helpful in diagnosis of exercise induced arrhythmia or to assess response to exercise.
- Event recorder**: They record ECG on activation by the patients when they feel symptoms. This can be kept by the patients for a month or more. These can also be implanted subcutaneously for up to 1-2 years and are useful for patients with infrequent arrhythmia or symptoms.
- Electrophysiological studies (EPS)**: EPS is an invasive procedure to induce arrhythmia to know their origin. It is particularly useful in **supraventricular arrhythmias (SVT)** and **ventricular arrhythmias (VT)**.

Anti-arrhythmic drugs

Anti-arrhythmic drugs are **classified into four classes based upon their actions** (Table 4.45 and 4.46).

TABLE 4.45: Classification of anti-arrhythmic drugs

Class I	Drugs that block sodium channels (membrane stabilizing effects)	
Ia	Prolong action potential:	Quinidine Procainamide Disopyramide
Ib	shorten action potential:	Lidocaine Mexiletine Phenytoin
Ic	no effect on action potential:	Flecainide Propafenone
Class II	Beta-blockers, slow AV conduction:	Atenolol Metoprolol Esmolol l-sotalol
Class III	drugs which prolong action potential:	Amiodarone d-Sotalol Ibutilide Dofetilide Bretylium
Class IV	slow calcium channel blockers:	Verapamil Diltiazem
Miscellaneous:		Adenosine Digoxin Atropine sulphate

TABLE 4.46: Intravenous dosage of some commonly used anti-arrhythmic drugs

- Adenosine: 6 mg IV as a rapid bolus followed by 12 mg after 1-2 min if needed
- Digoxin: 0.5 mg IV over 20 min followed by 0.25 or 0.125 mg increments to 1.5 mg in 24 hours
- Diltiazem: IV bolus of 0.25/kg mg over 2 min, repeat bolus of 0.35 mg/kg if required
- Verapamil: 5-10 mg over 2-3 min, can be repeated after 15-30 min
- Amiodarone: 150 mg IV over 10 min, maintain at 1 mg/min infusion for 6 hours
- Lidocaine: 1-2 mg/kg at 50 mg/min, maintain at 1-4 mg/min
- Ibutilide: 1 mg over 10 min, followed by second infusion of 1 mg
- Flecainide: 50 mg every 12h
- Propafenone: 150 mg over every 12h
- Metoprolol: 2.5-5 mg IV bolus over 2 min
- Esmolol: 0.5 mg/kg over 1 min

BRADYARRHYTHMIAS

Bradyarrhythmia is a rhythm **when heart rate is less than 60 beats per minute.**

Sinus Bradycardia

Sinus bradycardia is defined as **sinus rate of 60 beats/min with a normal P wave in ECG**. In healthy individuals particularly in athletes, heart rate may normally be less than 60/min.

Symptoms: It may cause **fatigue, confusion and syncope due to reduced cerebral perfusion.** It may also cause **breathlessness and angina on exertion.**

Causes: Causes of sinus bradycardia are **hypothyroidism, hypothermia, myocardial ischemia, primary sinus node disease, drugs** (digoxin, beta blockers, calcium channel blockers like diltiazem and verapamil, amiodarone).

Diagnosis: **ECG reveals rate of <60/min** with normal P wave. PR interval, QRS complexes, ST-T wave pattern are normal.

Treatment: In symptomatic patients, the treatment is directed towards the **underlying cause.** **Atropine (0.5 to 2.0) intravenously can be given in acute symptomatic patients.** **Cardiac pacing** (transcutaneous or transvenous) may be required.

Sick Sinus Syndrome

Sick sinus syndrome (SSS) is a group of rhythm disturbances that result due to **sinus node dysfunction**. It includes patients with sinus pause or sinus arrest, tachybrady syndrome and sinoatrial exit block.

Symptoms: Patients with SSS may present with **palpitation, dizziness, confusion, syncope because of pauses with sinoatrial or ventricular activity, intermittent tachycardia and bradycardia.**

Causes: Sick sinus syndrome occurs most commonly in **elderly patients and arises from intrinsic disease** (fibrosis, degenerative) of sinus node or cardiac conduction system. Sinus arrest is a condition when sinus node fails to generate an impulse intermittently.

Treatment: **Sinus pauses that result in ventricular asystole for 3 sec** are indications for **permanent pacing.**

Atrio-ventricular Block

AV block is condition when **atrial impulse is not conducted or conduction is delayed to ventricles.**

- a. **First-degree heart block** is due to delay of impulse conduction in AV node. ECG reveals prolonged PR interval (greater than 0.2 seconds). Important causes are increased vagal tone, antiarrhythmic drugs, ischemia, electrolytes disturbances and conduction system disease. It is usually asymptomatic and no treatment is required. However it may worsen heart failure in patients with pre-existing heart disease. In symptomatic patients, dual chamber pacemaker therapy may be considered.
- b. **Second-degree heart block** is a condition when some atrial impulses are not conducted to the ventricles. It is of two types:
 1. Mobitz type 1 (Wenckebach) block: Progressive delay in AV conduction occurs prior to blocked beat. ECG reveals progressive prolonged PR interval prior to blocked beat. It is benign and does not progress to complete heart block. Symptomatic patients are managed with IV atropine (0.5mg). Permanent pacemaker therapy is indicated when cause is untreatable.
 2. Mobitz type II AV block is characterized by abrupt AV conduction block without evidence of progressive

conduction delay. ECG reveals no change in PR interval prior to block or non-conducted P wave. Type II block may progress into complete heart block. Permanent pacemaker is indicated.

- c. **Complete (third-degree) heart block:** Complete (third-degree) heart block is a more advanced form in which transmission of atrial impulses to ventricles through AV node is completely blocked. Ventricle escape rhythm maintains a regular, slow ventricular rate of <45 beats/min. No increase in heart rate occurs with exercise.

Symptoms: Patient may be asymptomatic or may present with dyspnea and fatigue. Syncope may occur during transition partial heart block to complete heart block due to ventricular asystole, which may last for few seconds to minutes.

Causes: Causes may be congenital or acquired. Acquired causes of complete block are myocardial ischemia/infarction, drugs (digoxin, beta blockers), degeneration of conduction tissue, infiltrative diseases (amyloidosis, sarcoidosis), infectious diseases (Chaga's disease, Lyme disease) and rheumatological diseases.

Diagnosis: **Examination of patient reveals "cannon" a waves in jugular veins, varying intensity of first heart sound, changing systolic blood pressure level, slow and regular pulse rate of <45/min.**

Treatment: **Permanent pacing** is indicated if cause is not reversible.

Stokes-Adams Attacks

Patients with **ventricular asystole** may present with **recurrent syncope due to reduced cerebral perfusion**, this is known as Stokes-Adams attack. This occurs in patients with **complete heart block, Mobitz type II second degree block and sino-atrial disease.**

TACHYARRHYTHMIAS

Tachycardia is defined as a **heart rate of more than 100/min**. Tachycardias are classified as **supraventricular (SVT) and ventricular (VT)** depending on site of origin. They are also classified as **narrow complex** (QRS duration <120 ms) and **wide complex** (QRS duration >120ms).

A. Narrow complex tachycardias are of supraventricular origin.

1. *Sinus tachycardia* is defined as sinus rate of $>100/\text{min}$. Increased sympathetic activity due to exercise, anxiety, pregnancy and fever may cause tachycardia. Other cause are myocardial ischemia/infarction anemia, heart failure, thyrotoxicosis, phaeochromocytoma and drugs which increase sympathetic activity e.g., beta adrenoceptor antagonists like salbutamol, terbutaline. The ECG reveals rate of $100/\text{min}$ with normal P waves, PR interval and QRS complexes. Treatment is directed at the underlying process. Beta blockers may be useful in reducing rate particularly in the setting of myocardial ischemia.
2. Atrial tachycardia is defined as rate of more than $100/\text{min}$ with origin of electrical activity within the atrium, not in SA node. It is mostly seen in pulmonary diseases. Other causes are coronary artery disease, digitalis intoxication and acute alcohol ingestion. It produces a narrow complex tachycardia (rate $100\text{--}200/\text{min}$) with abnormal P wave morphology. Paroxysmal atrial tachycardia with atrio-ventricular block (PAT with block) is associated with digitalis intoxication. Multifocal atrial tachycardia (MAT) is seen in chronic obstructive pulmonary disease and heart failure. Treatment with beta blockers, calcium channel antagonists or digitalis is given to slow the ventricular rate. In digitalis intoxication, the digitalis is stopped and normal potassium levels are maintained. Digoxin antibodies with phenytoin and propranolol are used in refractory condition. Recurrent tachycardia or tachycardia not responding to drugs is treated by radiofrequency catheter ablation or surgical ablation.
3. Atrial flutter is characterized by single large re-entry circuit within the atrium. ECG reveals flutter waves (saw-toothed appearance) with atrial rate of around $300\text{ beats}/\text{min}$. It is commonly associated with 2:1, 3:1, 4:1 block so heart rate is usually less than the atrial rate ($75\text{--}150/\text{min}$). Block is increased by carotid sinus massage or intravenous adenosine and is helpful in diagnosis. Therapy includes (a) Restoration of sinus rhythm by direct current (DC) cardioversion, (b) digoxin, beta blockers or diltiazem may be

used to control ventricular rate, (c) amiodarone, propafenone or fleclainade are also helpful, (d) radiofrequency catheter ablation can be used for cure of atrial flutter in refractory or recurrent cases.

4. Atrial fibrillation (AF) is the most common sustained arrhythmia. Its prevalence increases with age (affecting 10% of those above 75 years). Conditions which increase atrial size or alter atrial conduction or disrupt atrial refractoriness nonuniformly are particularly susceptible to AF. It is due to multiple reentry circuits around the atria. During AF the atria beat rapidly in uncoordinated manner. Ventricles beat irregularly at a rate determined by conduction through AV node producing irregularly irregular pulse.

ECG reveals irregularly irregular rhythm (normal but irregular QRS complexes) with no well defined P waves; ventricular rate is $>100/\text{min}$ in untreated cases.

Symptoms of AF are palpitations, breathlessness, angina, syncope and light-headedness. It may precipitate or worsen heart failure in patients with ventricular dysfunction or structural heart disease. It may be asymptomatic and discovered on routine examination or ECG. Causes of atrial fibrillation are given in table 4.47.

Management of AF: Paroxysmal AF attacks are well tolerated and do not require treatment. In symptomatic patients beta blockers are used especially if it is associated with coronary artery disease, hypertension or heart failure. Class Ic drugs like flecainide or propafenone are also useful.

TABLE 4.47: Common causes of atrial fibrillation

Valvular heart disease (especially mitral valve disease)
Ischemic heart disease
Hypertension
Hyperthyroidism
Congenital heart disease
Acute alcohol ingestion
Pericardial disease
Pulmonary diseases
Drugs (theophylline etc.)
Idiopathic (Lone AF)

Radiofrequency ablation and overdrive atrial pacing are other modes of treatment in refractory cases. Management of persistent, AF includes:

- Rate control
- Prevention of thromboembolism
- Rhythm control.

Rate Control

Rate control of AF is done by agents which prolong conduction through AV node like beta blockers, calcium channel blockers (diltiazem, verapamil) and digoxin.

Prevention of Thromboembolism

Loss of atrial contraction and atrial dilatation produces stasis of blood and may precipitate thrombus formation in left atrium. Chronic warfarin ingestion in patients at risk for embolic stroke (cerebrovascular accident) is the most effective therapy for preventing stroke associated with AF. The INR is maintained between '2.0-3.0'.

In patients who have AF for >48 hours, anticoagulation with warfarin is recommended for at least 3 weeks before cardioversion.

Rhythm Control

Restoration of sinus rhythm can be achieved with electrical direct current (DC) cardioversion or chemical cardioversion (using antiarrhythmic drugs). If AF is present for <48 hours immediate DC cardioversion or chemical cardioversion (with intravenous ibutilide or flecainide or propafenone) can be done.

Supraventricular Tachycardias (SVT)

SVT is a generic term for paroxysmal, regular supraventricular tachyarrhythmia. The tachycardia may be due to AV nodal re-entry tachycardia (AVNRT), Atrioventricular re-entrant tachycardia (AVRT), junctional tachycardia, atrial tachycardia. The term is not generally used for atrial fibrillation (AF) and atrial flutter.

Supraventricular tachycardias (SVT) arise from the atria or the atrioventricular junction and are associated with a narrow QRS complex usually. They are due to re-entry circuit or automatic focus, the AV node is essential component of re-entry circuit.

AV nodal re-entry tachycardia (AVNRT): It is a regular tachycardia and may occur suddenly. Alcohol, tea, coffee

or exertion may precipitate or induce arrhythmia. It is due to re-entry in the right atrium and AV node.

ECG reveals normal regular QRS complexes at a rate of 140-240/min.

Symptoms are awareness of fast heart beat (palpitations), breathlessness or syncope. It may present with angina or heart failure if there is structural heart diseases. Polyuria may also occur because of release of atrial natriuretic peptide (ANP) due to increased atrial pressures.

Management

- Carotid sinus massage and other methods which increase vagal tone (Valsalva maneuver, facial immersion in cold water) may terminate the attack. If physical methods are not effective, intravenous adenosine, verapamil, diltiazem, flecainide or betablockers may terminate the attack.
- SVTs presenting with hemodynamic instability (hypotension, pulmonary edema) require emergency DC cardioversion.
- For recurrent attacks, prophylactic oral therapy with a verapamil, diltiazem, beta blockers or digoxin may be given. It can also be cured by catheter ablation techniques.

Atrioventricular re-entrant tachycardia (AVRT) and Wolf-Parkinson-White (WPW) Syndrome

This is due to long circuit involving atria, His bundle and ventricles. There is an abnormal connection which connects atria and ventricles (accessory pathway or bypass tract). In some cases this pathway conducts in the retrograde direction i.e., from the ventricles to atria resulting into normal ECG appearance. In some cases conduction takes place partly through AV node and partly through rapidly conducting accessory pathway during sinus rhythm. If the accessory pathway conducts from the atria to ventricles, the electrical impulses are conducted quickly and depolarize the part of ventricles abnormally (pre-excitation). Premature activation of ventricles produces a short PR interval, and a wide QRS complex with an initial slurred part called as 'delta wave'. Because AV nodes and bypass tracts have different conduction speeds and refractory periods, a re-entry circuit can develop, causing tachycardia. When these patients (with pre-excitation ECG) develop symptoms

due to tachycardia, this condition is known as Wolff-Parkinson-White (WPW) syndrome.

The ECG appearance may be indistinguishable from AVNRT.

Management

- Carotid sinus massage or intravenous adenosine may terminate the attack.
- If atrial fibrillation occurs, it is treated by DC cardioversion.
- In symptomatic patients, prophylactic therapy is indicated. Flecainide, propafenone or amiodarone is used for prophylaxis.
- Digoxin and verapamil are never used in atrial fibrillation associated with WPW syndrome because they shorten the refractory period of the accessory pathway and may allow a higher rate of conduction and may precipitate ventricular fibrillation.
- In symptomatic patients, catheter ablation of accessory pathway is curative.

Ventricular Tachyarrhythmias

Ventricular Premature Beats (VPBs)

Ventricular premature beats (VPBs) or ventricular extrasystole or ventricular ectopic beats (VEBs) is characterized by wide, bizarre QRS complexes that differ from normal beats in morphology and there is no preceding P wave.

- Every second or third beat if immature, the condition is known as ventricular bigeminy or trigeminy.
- VPC may disappear with exercise and rhythm may become normal if there is no cardiac disease.
- Patient may complain of skipped beat or may be asymptomatic.
- VPC occurring in patients with heart disease are associated with increased chances of sudden cardiac death due to ventricular fibrillation.

Treatment: In asymptomatic patients with no cardiac disease, no treatment is required. If frequent, the underlying cause (hypokalemia, hyperkalemia, hypomagnesemia, hyperthyroidism and heart diseases) should be treated. Beta blockers are agents of first choice in symptomatic patients with VPC.

Ventricular Tachycardia (VT)

It is defined as three or more consecutive VPCs. It has to be differentiated from SVT with aberrant conduction.

- The usual mechanism is re-entry or abnormal automaticity or triggered activity in ischemic tissue.
- It may be sustained (>30 sec) or non sustained (<30 sec). Patient may be asymptomatic or presents with syncope or symptoms due to reduced cerebral perfusion (dizziness) or dyspnea.
- VT is a frequent complication of acute myocardial infarction or cardiomyopathy. It may also occur in coronary artery disease, mitral valve prolapse, and myocarditis.
- VT is unfavorable sign in patients with heart disease and may cause hemodynamic compromise or degenerate into ventricular fibrillation.
- ECG shows tachycardia with a rate of >120/min with broad, bizarre QRS complexes.

Ventricular Fibrillation (VF)

It is characterized by rapid and irregular ventricular activation resulting into disorganized contraction of ventricle leading to hemodynamic collapse, cardiac arrest and sudden death.

- The patient is pulseless and becomes unconscious and respiration ceases.
- The ECG reveals irregular, rapid oscillations (250-400/min). The QRS complexes and T waves are not identifiable.

Treatment of VT and VF:

- Immediate DC cardioversion is the primary therapy for pulseless VT or VF.
- Intravenous antiarrhythmic agents may also be required if patient is resistant to defibrillation. In well tolerated VT, intravenous amiodarone or lidocaine is indicated. Electrolyte imbalance, hypoxemia, acidosis should be corrected because these can worsen the situation.
- Treatment of recurrent symptomatic ventricular arrhythmia includes administration of chronic antiarrhythmic agents (beta blockers, amiodarone)
- Nonpharmacologic treatment of VT/VF includes implantable cardiac defibrillator (ICDs) and radio-frequency catheter ablation to interrupt the arrhythmia focus or circuit.

- Torsades de pointes is a form of VT in which QRS morphology twists around the baseline. It may occur in hypokalemia, hypomagnesemia or after any drug that prolongs QT interval. It has poor prognosis.

IMPLICATIONS ON DENTAL PRACTICE

1. Patients with cardiac disease should receive dental treatment in minimal stressful environment. Anxiety, exertion and pain should be minimized.
2. The blood pressure may preferably be measured before any dental procedure to ensure that this is normal or controlled. A high blood pressure may result into precipitation of other related problems such as angina or heart failure.
3. Besides general examination, precordium should be auscultated to rule out any lesion in the heart.
4. Irregular pulse, engorged jugular veins and tachypnea may indicate the presence of cardiac disease.
5. Angina may present as pain in the mandible, teeth and other oral tissues.
6. A history of hypertension, ischemic heart disease or any other cardiac problem particularly congenital heart disease and drug intake (anticoagulant, aspirin) should be sought.
7. Epinephrine in the local anesthesia may raise the blood pressure and precipitate dysarrhythmias.
8. In patients with IHD, facilities for medical help, oxygen and nitroglycerine should be available.
9. General anesthesia should be avoided for at least three months in patients with recent onset angina.
10. Elective dental surgery should be deferred for 6 months following acute MI.
11. Prophylaxis for infective endocarditis is mandatory in cases where there is a risk.
12. Cardiac patients on anticoagulant drugs or aspirin are at increased risk of bleeding following dental procedures. Hence, these drugs should preferably be stopped a week before the procedure.
13. Calcium channel blockers may cause gingival swelling and lichenoid lesions in the oral cavity. ACE inhibitors can cause loss of taste, burning sensation in oral cavity, and angioedema. Dry mouth can result due to antihypertensive drugs such as diuretics, beta blockers and clonidine.
14. Sudden change in the posture from supine to standing following dental procedures may cause postural hypotension and syncope, particularly in patients using diuretics and calcium channel blockers.
15. Oral abnormalities such as enamel hypoplasia, delayed eruption of both dentitions, positional anomalies, bluish white “skimmed milk” appearance of teeth and vasodilatation in the pulps may be associated with cyanotic congenital heart disease.
16. Patient with left ventricular failure should be managed in partially reclining or erect position. Supine position may worsen dyspnea.

SELF ASSESSMENT

Multiple Choice Questions

1. Differential cyanosis can be seen in:
 - A. Bronchiectasis
 - B. PDA
 - C. ASD
 - D. Methemoglobinemia
2. “a” wave in JVP is absent in:
 - A. Complete heart block
 - B. Severe HF
 - C. AF
 - D. All the above
3. Major criteria for acute rheumatic fever include all except:
 - A. Pancarditis
 - B. Fever
 - C. Sydenham’s chorea
 - D. Polyarthritis
4. Following are features of right ventricular failure except:
 - A. Hepatomegaly
 - B. S3
 - C. Pulmonary congestion
 - D. Pedal edema
5. Most common congenital heart disease is:
 - A. ASD
 - B. VSD
 - C. Fallot’s tetralogy
 - D. Pulmonary stenosis
6. Most common congenital cyanotic heart disease is:
 - A. Fallot’s tetralogy
 - B. Eisenmenger syndrome
 - C. Tricuspid atresia
 - D. Pulmonary atresia
7. Clubbing may be found in all except:
 - A. Acute infective endocarditis
 - B. Fallot’s tetralogy
 - C. Bronchiectasis
 - D. Brochogenic carcinoma
8. Following features can be present in SABE except:
 - A. Splinter hemorrhages
 - B. Anemia
 - C. Erythema nodosum
 - D. Splenomegaly
9. Most common organism causing SABE is:
 - A. *S. aureus*
 - B. HACEK group
 - C. *Strep. viridans*
 - D. Enterococci
10. Following may be found in rheumatic AS except:
 - A. Water hammer pulse
 - B. Forceful apex beat
 - C. Syncope
 - D. Angina

11. Prophylaxis for infective endocarditis is not indicated in:

- A. Implanted pacemaker
- B. Secundum ASD
- C. CABG
- D. All the above

12. Following is the feature of ASD:

- A. Continuous murmur
- B. High chance of SABE
- C. Wide and fixed second heart sound
- D. Commonest type of congenital heart disease

13. Following can be found in mitral stenosis except:

- A. Loud S_1
- B. Mid-diastolic murmur at apex
- C. Opening snap
- D. Aortic ejection click

14. "T" wave in ECG is due to:

- A. Atrial depolarization
- B. Atrial repolarization
- C. Ventricular depolarization
- D. Ventricular repolarization

15. All are features of LVF except:

- A. Orthopnea
- B. Hepatomegaly
- C. Pulmonary edema
- D. S_3

16. Which antihypertensive drug is contraindicated in bradycardia:

- A. Amlodipine
- B. Atenolol
- C. Thiazide
- D. None

17. The most common form of hypertension is:

- A. Renovascular
- B. Essential hypertension
- C. Endocrinal
- D. Drug induced

18. Stokes-Adams attack is due to:

- A. AF
- B. Complete heart block
- C. Atrial extrasystole
- D. Severe hypertension

19. The preferred antihypertensive agent in diabetics is:

- A. Beta blocker
- B. ACE inhibitor
- C. Diuretics
- D. Centrally acting drugs

20. Most common cause of secondary hypertension is:

- A. Pheochromocytoma
- B. Eclampsia

C. Renal

D. Primary hyperaldosteronism

21. The diuretic preferred in acute pulmonary edema is:

- A. Loop diuretics
- B. Potassium sparing
- C. Thiazides
- D. All

Fill in the Blanks

1. Differential cyanosis is seen in _____.
2. Austin flint murmur is found in _____.
3. Carey-Coombs murmur is present in _____.
4. Reversal of shunt in VSD is known as _____.
5. Graham-Steel murmur is found in _____.
6. Maladie de Roger's murmur is found in _____.
7. Continuous machinery murmur is a feature of _____.
8. Most common type of ASD is _____.
9. Antihypertensive drug that can cause gum swelling is _____.
10. Brain natriuretic peptide (BNP) measurement is helpful in the diagnosis of _____.
11. Prehypertension is defined as Systolic BP _____ and diastolic _____.
12. Patients with coronary artery bypass surgery require prophylaxis for infective endocarditis (Yes/No)
13. Janeway's lesions are seen in _____.
14. Berry's aneurysm is a feature of _____.
15. Mycotic aneurysm is found in _____.
16. Hill sign is present in _____.
17. Wide fixed splitting of second heart sound is found in _____.
18. First heart sound is _____ in mitral stenosis.
19. Crackles are feature of _____ ventricular failure (right or left).
20. Drugs used for the prophylaxis of rheumatic fever are 1 _____, 2 _____ and 3 _____.

Chapter 5

Respiratory Diseases

Diseases of respiratory system are not uncommon. The respiratory tract may be primarily involved or it may be affected secondary to diseases of other systems. A thorough history and detailed clinical examination are mandatory in order to make a proper diagnosis.

The respiratory system is customarily divided into the upper and lower respiratory tract.

- The upper respiratory tract includes the nose, nasopharynx and the larynx.
- The lower respiratory tract includes trachea, bronchi and further structures.

The right lung is divided into three lobes: upper, middle, and lower while the left lung has two lobes: upper and lower. Each lobe is further divided into bronchopulmonary segments. Upper and middle lobe on the right side and upper lobe on the left side occupy most of the area of the chest anteriorly (Fig. 5.1).

SYMPTOMS

The important respiratory symptoms are:

- Dyspnea
- Cough
- Sputum

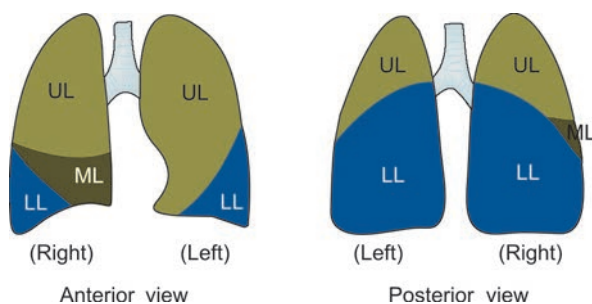


FIGURE 5.1: Anterior and posterior aspects of the lung

- Hemoptysis
- Chest pain.

Dyspnea

Dyspnea is defined as an abnormally uncomfortable awareness of breathing. The pathophysiology of dyspnea is multifactorial and mainly includes:

- a. hypoxia
- b. hypercapnia
- c. altered lung and chest wall compliance
- d. increased respiratory effort and respiratory muscle weakness

The dyspnea can be acute or chronic.

- The respiratory causes of acute dyspnea are severe asthma, acute exacerbation of COPD, pneumonia, pneumothorax, pulmonary embolism, foreign body inhalation, laryngeal edema and ARDS (acute respiratory distress syndrome).
- Important respiratory causes of chronic dyspnea are COPD and interstitial lung disease. Other causes of dyspnea are given in Table 5.1.

Dyspnea can occur only during exertion or it may be present even at rest, depending on the severity of disease. The dyspnea which occurs mainly at the night (*nocturnal dyspnea*) may be due to asthma, sleep apnea, gastroesophageal reflux or left ventricular dysfunction. Episodic dyspnea typically occurs in asthma (see also page 131).

Cough

Cough is one of the most important symptoms in respiratory diseases. It may be acute (less than 3 weeks) or chronic (Table 5.2)

The cough may be *dry* or associated with sputum (*productive cough*).

- Dry cough can occur in bronchial asthma, early phase of pneumonia and interstitial lung diseases. Episodic dry cough particularly in the night may be an early symptom of asthma.

TABLE 5.1: Important causes of dyspnea

Acute dyspnea	Chronic dyspnea
a. Respiratory causes • Asthma • Acute exacerbation of COPD • Pneumonia • Pneumothorax • Pulmonary embolism • ARDS • Foreign body inhalation • Laryngeal edema	a. Respiratory causes • COPD • Interstitial lung disease • Pleural effusion
b. Cardiac causes • Acute pulmonary edema	b. Cardiac causes • Chronic heart failure
c. Other causes • Metabolic acidosis (uremia, diabetic ketoacidosis) • Psychogenic	c. Other causes • Anemia • Obesity

TABLE 5.2: Causes of cough

Acute cough	Chronic cough
Upper respiratory tract infection	Chronic bronchitis
Pneumonia	Post-nasal drip
Aspiration	Gastroesophageal reflux
Pulmonary edema	Asthma
Pulmonary embolism	Drugs (ACE inhibitors)

- Common causes of productive cough are bronchiectasis, chronic bronchitis, pneumonia, lung abscess and tuberculosis.

TABLE 5.3: Characteristics of cough

Site of origin	Character of cough	Common causes
Pharynx	Persistent, throat pain	Pharyngitis, post nasal drip
Larynx	Harsh, painful, persistent. May have stridor	Laryngitis, tumour, whooping cough, vocal cord palsy (bovine cough)
Trachea	Persistent, painful, stridor	Tracheitis, tumour
Bronchi	Dry or productive, wheeze	Bronchitis, Bronchial asthma

Contd...

Contd...

	May have hemoptysis	Bronchogenic carcinoma
Lung parenchyma	dry initially followed by productive cough, rusty sputum Dry, distressing Often nocturnal, pink frothy sputum	Pneumonia Interstitial fibrosis Pulmonary edema

The cough may have different characteristics based on the sites involved. These are given in table 5.3.

Sputum

Sputum production is a common symptom in respiratory diseases. Its characteristics such as amount, colour and consistency are helpful in making the diagnosis.

- Large amount of sputum is produced in bronchiectasis and lung abscess while scanty thick sputum is found in asthma.
- Sputum is mucoid, scanty and thick in chronic bronchitis.
- Purulent yellow or green sputum signifies infection.
- Foul smelling sputum is present in anaerobic infection.
- Pink frothy sputum suggests pulmonary edema.
- Patients with bronchiectasis or lung abscess may bring out large amount of sputum in a particular posture which forms the basis of '*postural drainage*' in the management.

Hemoptysis

The coughing of blood in the sputum is called hemoptysis. The blood may be coughed up alone (frank hemoptysis) or there may be streaking of sputum with blood. Expectoration of 200 to 600 ml blood in 24 hours is defined as massive hemoptysis. Hemoptysis should be differentiated from bleeding from other sites such as gums, nose and stomach. The differentiating features between hemoptysis and

TABLE 5.4: Differences between hemoptysis and hematemesis

Hemoptysis	Hematemesis
1. Bright red appearance	1. Coffee ground or dark red
2. Presence of cough or other respiratory symptoms	2. Presence of vomiting or other abdominal symptoms
3. Alkaline pH	3. Acidic pH
4. Melena generally not found	4. Associated with black tarry stool (melena)

TABLE 5.5: Causes of hemoptysis*Respiratory*

- Chronic bronchitis
- Tuberculosis
- Bronchogenic carcinoma
- Bronchiectasis
- Pulmonary infarction
- Pneumonia

Cardiac

- Mitral stenosis
- Left ventricular failure

Others

- Goodpasture's syndrome
- Wegener's granulomatosis
- Hemorrhagic disorders
- Anticoagulant therapy

hematemesis (bleeding from gastrointestinal tract) are given in Table 5.4.

Common causes of hemoptysis are chronic bronchitis, pulmonary tuberculosis, and bronchogenic carcinoma. Other causes are given in Table 5.5.

Wheezing

Wheeze is a high pitched continuous musical sound heard due to narrowing of airways. This may be noticed by the patient or relatives or others as in case of asthma.

A low pitched inspiratory sound heard in the obstruction of larynx or trachea is known as 'stridor'.

Chest Pain

Chest pain in respiratory diseases is mainly due to the involvement of parietal pleura. The pain is sharp "stabbing" which increases on deep inspiration and coughing (pleuritic chest pain). The causes of pleuritic pain are conditions with

TABLE 5.6: Differences between pain of pleural and cardiac origin

Pleural origin	Cardiac origin
Site is usually lateral, axilla or back	Central, retrosternal
Character of pain is stabbing, cutting or tearing	Heaviness or constriction type
Referred to abdomen or back	Referred to neck or left upper limb
Aggravated by respiratory movement or cough	Aggravated by exertion, emotions
Relieved by rest or decreased movement of chest	Relieved by rest or nitroglycerine
Examination reveals pleural rub localized to the site of pain	Pericardial rub may be present

pleural inflammation and malignancies involving pleura. The pain due to pleural inflammation may occur in pneumonia and pulmonary infarction if parietal pleura is involved. Chest pain due to cardiac diseases should be differentiated from that of pleural origin. Important differentiating points are given in Table 5.6.

EXAMINATION

General

The general examination in the patients with respiratory disease is of great importance. Particularly one should look for the physique, cyanosis, digital clubbing, elevation of JVP, edema and lymph node enlargement.

Physique: Weight loss may be apparent in cases of tuberculosis, chronic suppurative lung diseases and malignancies of the lung or pleura.

Cyanosis: Patients with respiratory failure may have central cyanosis. This may occur acutely in extensive pneumonia, pulmonary edema and massive pulmonary embolism. Chronically cyanosis is present in COPD. The presence of polycythemia in COPD may lead to cyanosis even in mild hypoxemia.

Clubbing: Digital clubbing may be found in patients with chronic suppurative lung diseases (bronchiectasis, lung abscess, empyema), primary and metastatic lung cancers, mesothelioma, chronic interstitial lung disease and advanced pulmonary tuberculosis. The presence of digital clubbing along with swelling above the wrist and ankle due to periostitis of long bones (*hypertrophic pulmonary osteoarthropathy*) is particularly associated with bronchogenic carcinoma. Clubbing develops rapidly in patients with lung abscess and bronchogenic carcinoma. (see Table 1.6)

Edema: Pedal edema can be found in patients with respiratory disease and it indicates the presence of cor pulmonale (right ventricular enlargement secondary to pulmonary disease) and right ventricular failure.

Swelling over the face and neck may also occur in superior vena cava obstruction due to enlarged lymph nodes or malignant mass.

Lymph node enlargement: The enlargement of lymph nodes, mainly axillary and cervical may indicate tuberculosis or carcinoma lung. Other groups of lymph nodes like mediastinal, hilar and paratracheal can only be detected by imaging.

Respiratory System Examination

The respiratory system, like other systems, should be examined under four headings: inspection, palpation, percussion and auscultation.

Inspection (Table 5.7)

Shape and symmetry of chest:

- The shape of the chest is normally bilaterally symmetrical and transverse diameter is more than anteroposterior diameter (elliptical). Over-inflated *barrel shaped* chest (AP diameter is equal or more than transverse diameter) can be present in COPD. Other deformities of the chest like *kyphosis* (forward bending) or *scoliosis* (lateral bending) should be noted. Deformities such as *pigeon chest* (marked bulging of sternum) and *Harrison's sulcus* (a horizontal groove at the lower part of chest) can rarely be seen.
- Tracheal shift can be assessed by looking for the prominence of clavicular end of sternocleidomastoid muscle which becomes more prominent on the side of shift (*Trail sign*).
- Localized bulging in the chest may occur in aneurysm, empyema or cardiac hypertrophy whereas localized recession (flattening) can be seen in pleural or lung fibrosis.

Movements: The movement of the chest is normally equal on both sides. The movement is decreased on the side of disease. Drawing of intercostal spaces with inspiration indicates severe upper airway obstruction. Use of accessory muscles of respiration (intercostals, sternocleidomastoid, etc.) denotes significant pulmonary impairment.

Rate, rhythm and depth of respiration:

- The rate, rhythm and depth of the respiration should also be noted. The normal respiratory rate is around 12-16 per minute. An increase in the respiratory rate is known as tachypnea.
- Rapid and deep respiration (*Kussmaul's breathing*) is present in metabolic acidosis whereas rapid shallow breathing is a feature of restrictive lung disease.

TABLE 5.7: Inspection of chest

Shape and symmetry
Movements
Use of accessory muscles
Rate, rhythm and depth of respiration
Trail sign
Others: Venous prominence, scar, sinuses

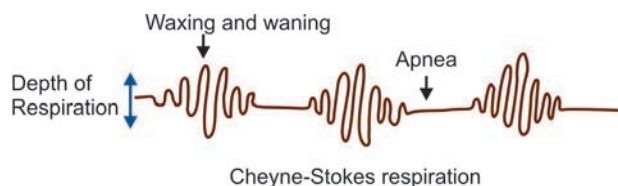


FIGURE 5.2: Cheyne-Stokes respiration

- *Cheyne-Stokes* respiration is characterized by cyclical waxing and waning of the rate and depth of the respiration intervened with a short period of apnea (Fig. 5.2). It is observed in narcotic overdose and severe left ventricular failure.

Palpation

- The *trachea* and the *apex beat* are palpated to find out the shift of the mediastinum. The shift of the mediastinum to the side of the disease occurs in fibrosis and collapse. Mediastinum is pushed to the opposite side in cases of pleural effusion and pneumothorax.
- The *expansion of the chest* is measured and compared on both sides.
- *Tactile vocal fremitus* is detected with the flat of the hand placed on the chest wall. This occurs due to the transmission of sound waves produced in the larynx and proximal airways to the chest wall. The vocal fremitus is diminished in pleural effusion and pneumothorax and increased in consolidation (pneumonia).
- Palpation can also detect tenderness which may occur in rib fracture and costochondritis.

Percussion

Normally the percussion note over the chest is resonant.

- It is *impaired* or *dull* in fibrosis, collapse, consolidation and pleural thickening
- The note is '*stony dull*' in pleural effusion.
- A *hyperresonant* note is found in pneumothorax and emphysema.
- Normal liver dullness is masked in emphysema.

Auscultation

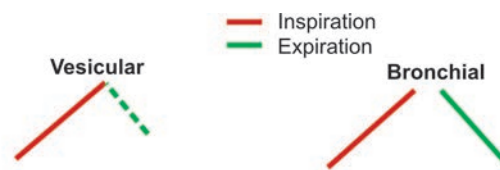
Type of breath sound: Normal breath sounds heard over the chest wall are called *vesicular*. This is a rustling sound heard throughout the phase of inspiration and during initial phase of expiration. There is no pause in-between. The intensity of the vesicular breath sounds is reduced in cases with

emphysema, pleural effusion or thick chest wall. Individuals with very thin chest wall may have vesicular breath sounds with increased intensity. Vesicular breath sounds with prolonged expiration are heard in asthma and COPD.

The *bronchial* breath sounds are higher pitched hissing sounds typically heard over trachea in normal individuals. The presence of bronchial breath sounds over other areas of the chest is abnormal. These are classically heard over areas of consolidation and cavity. These sounds are hollow in character, louder during expiration and last for most of the expiratory phase. There is a gap between inspiratory and expiratory phases (Fig. 5.3).

Added sounds: Added sounds include wheezes (rhonchi), crackles (crepitations or rales) and pleural rub.

- **Wheezes** are high pitched continuous musical sounds associated with narrowing of airways. Diffuse wheezes (polyphonic) are heard in cases of bronchial asthma and COPD whereas localized wheezes (monophonic) are generated at narrowed bronchus due to tumor or foreign body.
- **Crackles** are bubbling or clicking brief interrupted sounds produced by the explosive opening of small airways due to sudden changes in gas pressure. Fine crackles are heard in interstitial fibrosis and early pulmonary edema while coarse crackles are heard



1. Expiration shorter and softer
2. Inspiration longer and louder
3. Rustling
4. No gap between inspiration and expiration

1. Expiration and Inspiration of same duration and intensity
2. Hollow
3. Gap between inspiration and expiration

Harsh vesicular with prolonged expiration



1. Expiration prolonged
2. Present in asthma and COPD
3. Wheeze may be present

FIGURE 5.3: Breath sounds

in pneumonia, bronchiectasis, chronic bronchitis and late pulmonary edema.

- **Pleural rub** is a leathery rubbing sound found in cases with pleural inflammation.

Vocal resonance: In consolidation, spoken or whispered sounds are auscultated louder and more clearly over the chest wall. These are called *bronchophony* and *whispering pectoriloquy* respectively. The voice may sound nasal or bleating over the level of pleural effusion (*aegophony*).

TABLE 5.8: Chest examination findings in common conditions

Condition	Inspection/palpation	Percussion	Auscultation
Consolidation (Pneumonia)	• Reduced chest wall movement over the affected area • No mediastinal shift • Increased vocal fremitus	• Dull note	• Bronchial breath sounds • Crackles • Increased vocal resonance • Whispering pectoriloquy present
Pleural effusion	• Reduced movements over affected side • Mediastinal shift to opposite side • Decreased vocal fremitus	• Stony dull note	• Diminished or absent breath sounds • Pleural rub rarely (above effusion) • Aegophony
Pneumothorax	• Reduced movements over affected side • Mediastinal shift to opposite side • Decreased vocal fremitus	• Hyper-resonant	• Diminished or absent breath sounds
Emphysema	• Reduced movements bilaterally • Barrel shaped chest • No mediastinal shift • Decreased vocal fremitus	• Normal or hyper-resonant liver dullness masked	• Diminished vesicular breath sounds with prolonged expiration
Chronic bronchitis	• Movement normal or symmetrically reduced • No mediastinal shift • Normal vocal fremitus	• Normal note	• Wheezes • Vesicular breath sounds with prolonged expiration • Wheezes • Coarse crackles

The chest examination findings in common respiratory disease are given in Table 5.8.

INVESTIGATIONS

Chest X-ray: Chest radiograph is an important initial investigation in patients with respiratory diseases. Consolidation, pleural effusion, pneumothorax, cavity and infiltrates can be reliably diagnosed.

Posteroanterior (PA) view is the standard view. Lateral view provides additional information in certain conditions like minimal pleural effusion (lateral decubitus view) and mediastinal lesions. Comparison with old films is sometimes helpful to detect new lesions or any change in size.

Sputum examination: Examination of the sputum provides useful information. Gram staining, AFB (Acid Fast Bacilli) smear and culture/sensitivity of the sputum are needed to diagnose pulmonary infections. Malignant cells can be seen in sputum in cases of bronchogenic carcinoma. Nebulization with hypertonic saline can be used to obtain sputum samples. Sample can also be obtained by bronchoscopy or trans-tracheal aspiration.

Computed tomography: This is useful in the diagnosis and staging of lung and pleural malignancies. High resolution CT is particularly helpful in the diagnosis of bronchiectasis, interstitial fibrosis and pulmonary embolism.

Radioisotope imaging: Combined ventilation-perfusion scanning is useful in the diagnosis of pulmonary thromboembolism.

Pulmonary angiography is the definitive method of diagnosing pulmonary emboli.

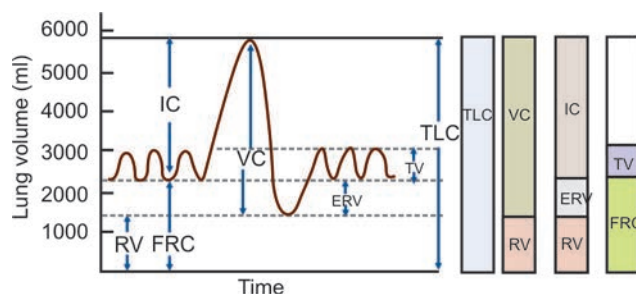
Arterial blood gas (ABG) analysis: Partial pressures of oxygen (PaCO_2), carbon dioxide (PaO_2) and pH can be measured in arterial blood sample. This is helpful in the assessment of the degree and type of respiratory failure.

Bronchoscopy: Trachea or bronchi are directly visualized by flexible or rigid bronchoscope. Samples from airways can also be obtained by bronchial washings, aspiration or biopsy. Foreign body can be removed by bronchoscopy (Therapeutic bronchoscopy).

Pleural biopsy and aspiration: These are helpful in the detection of the cause of pleural effusion.

Spirometry: Lung volumes are measured by spirometry (Fig. 5.4). This is most useful in differentiating obstructive from restrictive lung diseases.

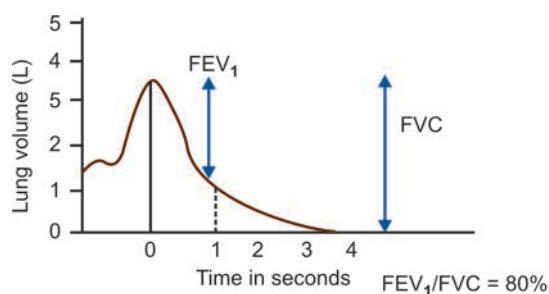
- Forced expiratory volume (FEV_1) is the amount of gas exhaled in the first second of this maneuver which is around 80 % of the FVC in normal subjects (Figs 5.5A and B).
- Forced vital capacity (FVC) is the total amount of gas which can be forcefully exhaled following a maximal inhalation.
- In obstructive lung diseases such as COPD, asthma, and bronchiectasis, the FEV_1/FVC ratio is reduced.



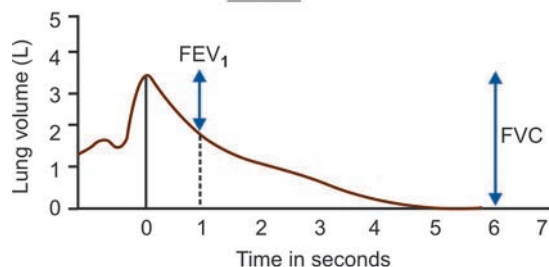
Lung volumes

RV = Residual volume, FRC = Functional residual capacity
ERV = Expiratory reserve volume, IC = Inspiratory capacity
TLC = Total lung capacity, TV = Tidal volume,
FRC = Functional residual capacity

FIGURE 5.4: Lung volumes



Normal



Obstructive lung disease

$\text{FEV}_1/\text{FVC} = 52\%$

FIGURES 5.5A and B: Spirometry in (A) normal person (B) patient with obstructive lung disease

- The ratio is normal or raised in restrictive lung diseases such as pulmonary fibrosis.
- *Peak expiratory flow rate* (PEFR) can be measured by a portable meter and is helpful in the diagnosis and management of patients with asthma.

PNEUMONIA

Pneumonia is defined as the infection of the lung parenchyma (alveoli and distal airways) and interstitium of the lung.

Etiology

Pneumonia is more common at extremes of age and in winter months.

- The risk factors for developing pneumonia are smoking, alcohol use, immunocompromised states (HIV disease, end stage renal disease), diabetes mellitus, congestive heart failure, COPD and malignancies.
- Pneumonia is broadly divided into two types: Community acquired pneumonia (CAP) and Health care associated pneumonia (HCAP).
- HCAP is defined as new episodes of pneumonia occurring at least two days after admission or initiation of ventilatory support. It is subcategorized as hospital acquired pneumonia (HAP) and ventilator associated pneumonia (VAP).

Route of Infection

The most common route of infection is microaspiration of oropharyngeal secretions colonized with pathogenic microorganisms. Other routes are gross aspiration, aerosolization, hematogenous spread from a distant site and spread from adjacent tissues.

Organisms

More than 100 organisms including viruses, bacteria, fungi and parasites are reported to cause pneumonia. Common organisms responsible for CAP and HCAP are listed in Table 5.9 and 5.10, respectively.

By far the most common organism causing CAP is *S. pneumoniae* (>50%). The etiological organism may vary according to the age of the patient and the specific clinical situations. *P. carinii* occurs in HIV infection while anaerobic and gram-negative bacilli (mixed etiology) can cause pneumonia due to aspiration in patients with stroke, epilepsy,

TABLE 5.9: Common pathogens causing CAP

Bacteria

Streptococcus pneumoniae
Mycoplasma pneumoniae
Hemophilus influenzae
Chlamydia pneumoniae
Staphylococcus aureus
Legionella sps.

Viruses

Influenza
 Adeno viruses
 Respiratory syncytial viruses

TABLE 5.10: Common pathogens causing HCAP

Non-MDR pathogens	MDR pathogens
<i>Streptococcus pneumoniae</i>	<i>Pseudomonas aeruginosa</i>
<i>Hemophilus influenzae</i>	Methicillin resistant <i>S. aureus</i>
Methicillin sensitive <i>S. aureus</i>	<i>Acinetobacter</i> spp.
Enterobacteriaceae (<i>E. coli</i> , <i>klebsiella</i>)	Enterobacteriaceae (antibiotic resistant)

and dental caries. *Pseudomonas* infection is common in case of bronchiectasis.

Pathology

Pathologically, pneumonia is classified into four types

- Lobar pneumonia:** This type is typically seen in pneumococcal pneumonia. Generally, entire lobe of the lung is involved.
 - The first stage is known as *congestion*, which occurs within 24 hours. The lung parenchyma is highly vascular and edematous with plenty of bacteria and scanty neutrophils.
 - Second stage (*red hepatization*) is characterized by the presence of numerous erythrocytes, neutrophils, desquamated epithelial cells and fibrin in alveoli resulting in a red and airless lung, and a consistency similar to that of liver.
 - In the third stage (*grey hepatization*), lung parenchyma becomes grey, dry and friable.
 - Finally, in the *stage of resolution*, the exudates are digested and removed by scavenger cells or coughed out.
- Bronchopneumonia:** This type is most commonly seen in HCAP. There is neutrophilic exudate in bronchi and bronchioles with peripheral spread of infection to alveoli.

One or several lobes of the lung may be involved. Generally lower and posterior segments of lobes are affected.

- c. *Interstitial pneumonia*: This is seen in viral and pneumocystis jiroveci pneumonia. There is predominant involvement of the interstitium, alveolar wall and connective tissue around the bronchovascular tree. The pattern may be patchy or diffuse.
- d. *Miliary pneumonia*: Hematogenous spread of pathogens to the lung may result into diffuse and discrete 2-3 mm lesions resembling millet seeds.

Clinical Manifestations

The onset may be sudden or insidious, and the disease may be mild or severe. Fever with chills or rigors, cough (dry or productive), pleuritic chest pain, breathlessness are typical manifestations of pneumonia. Other symptoms are headache, nausea, vomiting, diarrhea, altered sensorium, myalgia and arthralgia. The signs of lobar pneumonia are depicted in Table 5.11.

The severity of pneumonia is indicated by respiratory rate >30 per minute, pulse rate >125 per minute and blood pressure less than 90 mmHg. Other markers of severity are altered mentation, hypoxia ($\text{PaO}_2 < 60 \text{ mmHg}$), hyponatremia, acidosis and azotemia. Pneumonia due to certain organisms like *P. aeruginosa*, *Klebsiella spp*, *E. coli*, *S. aureus* are associated with high mortality rate. The CURB-65 criteria (Table 5.12) are helpful in deciding whether patient should be treated on out-patient basis or hospitalized. Patients with two or more criteria need hospitalization.

TABLE 5.11: Respiratory signs in lobar pneumonia

Inspection and palpation:

- Tachypnea
- Decreased movement of the chest on the side of disease
- No shift of trachea or apex beat (no mediastinal shift)
- Increased vocal fremitus

Percussion:

- Impaired or dull note

Auscultation:

- Bronchial breath sounds
- whispering pectoriloquy
- Increased vocal resonance
- Aegophony
- crackles (crepts)
- pleural rub

TABLE 5.12: CURB-65 criteria

C	confusion
U	blood urea nitrogen >20 mg/dL
R	respiratory rate >30 min
B	blood pressure systolic <90 mm Hg or diastolic <60 mm Hg
65	age >65 years

Complications

The complications of pneumonia are given in Table 5.13.

Investigations

1. *Radiological*: Chest X-ray shows homogeneous opacity localized to the lobe or segment (lobar consolidation) or diffuse infiltrates in bronchopneumonia (Fig. 5.6). Other findings may include pleural effusion, lung abscess and hilar lymphadenopathy. Pneumatoceles may be seen in *S. aureus* pneumonia.

TABLE 5.13: Complications of pneumonia

1. *Pulmonary*
 - Parapneumonic pleural effusion
 - Empyema
 - Suppurative pneumonia/lung abscess
 - ARDS
 - Pneumothorax (in *S. aureus* pneumonia)
2. *Extrapulmonary*
 - Hepatitis, pericarditis, meningoencephalitis
 - Multiorgan failure
 - Ectopic abscess formation



FIGURE 5.6: X-ray chest showing consolidation in the right lung

2. *Sputum or pulmonary secretion examination:* Gram staining and culture are helpful in identifying the causative organisms and finding out the sensitivity pattern.
3. *Blood tests:* These include blood culture, arterial blood gas analysis, and total and differential leukocyte counts. Neutrophilic leukocytosis is commonly found in bacterial pneumonia while leukopenia may indicate viral etiology.
4. *Serological tests:* The detection of antibodies may be helpful in the diagnosis of Chlamydia, Mycoplasma, and Legionella pneumonia.
5. *Other tests:* Polymerase chain reaction (PCR) based tests and specialized culture tests may be needed in some cases.

Treatment

1. *General principles:* The patients are managed as out-patient or in-patient (hospitalized) depending upon the severity of the disease. The severity is generally assessed using CURB 65 criteria. Other criteria used are PORT (Pneumonia Patient Outcomes Research Team) and CPIS (Clinical Pulmonary Infection Score).
2. *Supportive therapy:* Oxygen therapy is given to all hypoxemic patients. Those who do not respond may need mechanical ventilation. Intravenous fluids may also be required in hospitalized patients.
3. *Antibiotic therapy:* The antibiotics should be started as early as possible. If possible, the specimen should be sent for culture and sensitivity before starting antibiotics. The etiology is commonly unknown; hence, the initial choice of antibiotics is empirical. However, this may be modified once culture reports are available.
 - a. In out-patients with CAP, the preferred empirical antibiotic options are macrolide or doxycycline or fluoroquinolones (with activity against *S. pneumoniae*) (Table 5.14). A 10-14 days therapy is generally required.
 - b. Empirical antibiotic options for hospitalized patients with CAP are following. These are given intravenously. The duration of treatment is 10-14 days.
 1. Beta lactum (ceftriaxone 1-2 gm/day or Cefotaxime 2 gm 6 hrly) plus a macrolide or fluoroquinolones
 2. Beta lactam/beta lactamase inhibitor (ampicillin/sulbactam or piperacillin/tazobactam) plus macrolide or fluoroquinolones

TABLE 5.14: Empirical antibiotic options in out-patients with CAP

1. <i>Macrolides</i>	
Clarithromycin	500 mg orally twice a day
Azithromycin	500 mg orally on day 1 then 250 mg daily × 4 days
2. <i>Doxycycline</i>	100 mg orally twice a day
3. <i>Fluoroquinolones</i>	
Gatifloxacin	400 mg orally once a day
Levofloxacin	500 mg orally once a day
Moxifloxacin	400 mg orally once a day
4. <i>Alternative regimens</i>	
Erythromycin	500 mg orally 4 times a day
Amoxicillin-clavulinate	625 mg orally thrice a day
Amoxicillin	500-1000 mg orally thrice a day
Cefpodoxime	200 mg orally twice a day
Cefuroxime axetil	500 mg orally twice a day

3. Patients with aspiration pneumonia should receive fluoroquinolones plus clindamycin or metronidazole.
4. In pseudomonas infection (as in patients with underlying bronchiectasis), a combination of fluoroquinolones with carbapenem or cefipime.

LUNG ABSCESS

Suppurative pneumonia is characterized by destruction of lung parenchyma by the inflammatory process generally leading to micro-abscess formation. When a large collection of pus occurs within the lung parenchyma lined by chronic inflammatory tissue, the condition is known as lung abscess.

The most commonly involved areas are dependent segments of the lungs: *posterior segments of the upper lobes and superior segments of the lower lobes* in supine position.

Risk Factors

Aspiration of oropharyngeal secretion is the most important cause of lung abscess. Predisposing factors for aspiration include:

- Altered consciousness due to any cause (alcohol, drugs, seizures, stroke and general anesthesia)
- Impaired swallowing due to esophageal disorders (achalasia, gastroesophageal reflux)
- Vocal cord palsy.
- *Poor oral hygiene and gross oral sepsis is also associated with greater risk of pulmonary infection due to aspiration.*

Causative Organisms

Mostly the lung abscess is caused by mixed bacterial flora, both aerobic and anaerobic.

- Common aerobic pathogens responsible for lung abscess are *S. pneumoniae*, *S. aureus*, *S. pyogenes*, *H. influenzae*, *E. coli* and *Pseudomonas*.
- Anaerobic bacteria such as bacteroides, *Prevotella*, *Fusobacterium* and *Peptostreptococcus* species are generally responsible in cases with poor dental hygiene or oral sepsis.

Clinical Features

The presentation is generally insidious in anaerobic infection while more acute presentation is typical of aerobic infection. Important symptoms are high remittent fever, weight loss, malaise, night sweats, cough with large amount of purulent sputum. The foul smelling sputum suggests anaerobic infection. Chest pain may occur due to pleural involvement.

General examination may reveal clubbing. Respiratory examination shows signs of consolidation or cavity such as bronchial breath sound and coarse crackles.

Diagnosis

- The chest X-ray may show, in early stages, a homogenous opacity due to consolidation. This may cavitate and present as large cavity with air fluid level (Fig. 5.7). Other causes of cavitary lung disease should be excluded (Table 5.15).



FIGURE 5.7: X-ray chest showing lung abscesses with horizontal air fluid level

TABLE 5.15: Causes of cavitary lung disease

1. Lung abscess
2. Tuberculosis
3. Fungal infections
4. Cavitating carcinoma
5. Pulmonary infarction
6. Wegener's granulomatosis

- Sputum or pulmonary secretion obtained by bronchoscopy is cultured for both aerobic and anaerobic organisms.
- CT thorax can be helpful to exclude the presence of tumor or other underlying causes in patients not responding to treatment.

Treatment

Medical

- The antibiotic therapy should be chosen according to the organism isolated.
- The therapy is generally given for prolonged period (6-8 weeks) until the radiographic resolution of the cavity.
- Commonly, a combination of penicillin (amoxicillin) and metronidazole is effective. However, with the emergence of beta lactamase producing organisms, clindamycin (150-300 mg 6 hrly) is now considered standard therapy. Beta lactam/beta lactamase inhibitors may also be used.
- Postural drainage is also helpful in the management of lung abscess.

Surgical

Surgical intervention like percutaneous drainage or lobectomy is needed in case there is no response to medical treatment or there is a large abscess.

BRONCHIECTASIS

Abnormal and permanent dilatation of bronchi is called bronchiectasis. There is inflammation of the medium sized airways leading to the destruction of the wall. The inflammatory cells, chiefly neutrophils are primary mediators and they release proteolytic enzymes leading to destruction and fibrosis of bronchial structures. Repeated obstruction and infections are important factors. Common causes of bronchiectasis are suppurative pneumonia and tuberculosis. Other important causes of bronchiectasis are given in Table 5.16.

TABLE 5.16: Important causes of bronchiectasis

Congenital	
•	Kartagener's syndrome*
•	Cystic fibrosis
Acquired	
•	Suppurative pneumonia
•	Tuberculosis
•	Post-viral infection (in children, adenovirus, influenza virus)
•	Foreign body
•	Bronchial tumor
•	Allergic bronchopulmonary aspergillosis (ABPA)

* Kartagener's syndrome includes bronchiectasis, sinusitis and situs inversus.

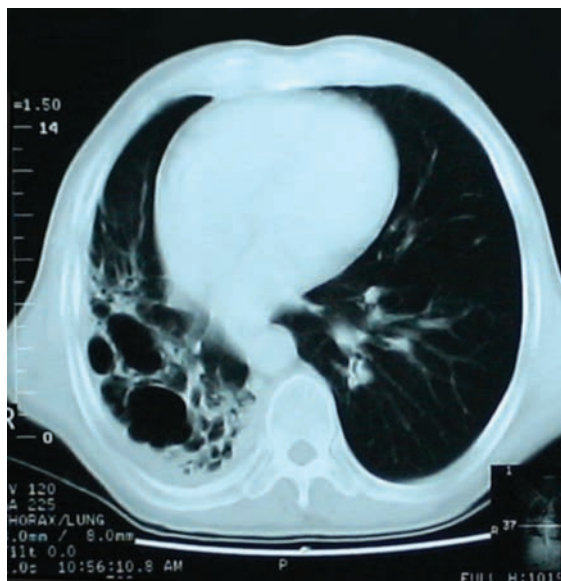
Clinical Manifestations

Chronic persistent or intermittent cough with large amount of purulent sputum is the main symptom. Patient may also have fever, weight loss and hemoptysis. Recurrent lung infection may lead to pneumonia and pleurisy.

Clinical examination reveals digital clubbing and anemia. Lung signs are non-specific and include basal coarse crackles and wheezes. There may be signs of right heart failure (cor pulmonale).

Investigations

- Chest X-ray may be normal or it may show small cystic spaces.
- High resolution CT scan (HRCT scan) is the test of choice to make the diagnosis (Fig. 5.8).

**FIGURE 5.8:** CT scan showing bronchiectasis

- Sputum examination and culture guide in choosing the antibiotics.

Treatment

- The main steps in the management of bronchiectasis are antibiotics, postural drainage and inhaled bronchodilators.
- Surgery is indicated in cases of massive hemoptysis and localized bronchiectasis which fail to respond to adequate medical therapy.

Complications

The complications of bronchiectasis are amyloidosis, cor pulmonale and abscess at distant site.

BRONCHIAL ASTHMA

Asthma is a disease of the airways in which there is a chronic inflammation and increased responsiveness to a wide variety of stimuli, leading to reversible airway obstruction. Patients generally have paroxysms of cough, wheeze and dyspnea, interspersed with symptom-free periods.

Pathophysiology

Genetic and environmental factors play a role in the pathogenesis of bronchial asthma.

- Early onset (extrinsic) asthma begins in childhood and a family history of atopy and other allergic disorders such as allergic rhinitis and eczema is usually present. Skin test to antigens is positive and serum levels of IgE are raised. The symptoms resolve in about 80 % of children as they grow.
- Late onset (intrinsic) asthma starts in adulthood in non-atopic individuals with no family history of allergic disorders. Skin test is negative and serum levels of IgE are normal. This type of presentation is found in minority of asthmatic patients (10%).

Atopic individuals are sensitized after an exposure to allergen and develop IgE antibodies. Subsequent exposure to the allergen causes a two-phase bronchoconstrictor response. Early reaction (type I) occurs because of rapid interaction of allergen with mast cells via IgE dependant mechanism. It results in the release of preformed mediators like histamine and leukotrienes which cause bronchoconstriction. Late reaction (type II) is characterized by T-

cell mediated influx of inflammatory cells, chiefly eosinophils. It leads to chronic inflammatory reaction in the bronchial wall.

The pathological changes are;

- A. Bronchial smooth muscle hypertrophy and hyperplasia
- B. Hyperplasia of mucous glands
- C. Mucosal edema with infiltration of granulocytes
- D. Hyperplasia of mucosal and submucosal vessels
- E. Thickening of basement membrane
- F. Infiltration with eosinophils, lymphocytes, mast cells, neutrophils.

A wide variety of stimuli can provoke bronchospasm in an asthmatic. Stimuli can be allergic or non-allergic (Table 5.17).

TABLE 5.17: Stimuli that can provoke asthma

Allergic

- Inhaled allergens : pollen, house-dust, feathers, animal danders, fungal spores
- Ingested allergens : fish, eggs, milk, nuts, strawberries, yeast, wheat
- Additives in food: monosodium glutamate (ajinomoto), metabisulphite preservative used in beers, wine and preserved food
- Occupational: grain-dust, wood-dust
- Environmental: cold air
- Physiochemical agents: gases, fumes and smoke
- Drugs: aspirin, NSAIDs, beta-blockers

Non-allergic

- Infections : respiratory viral infections
- Exercise
- Emotional outburst

Clinical Features

The symptoms of bronchial asthma may be episodic or persistent.

- *Episodic asthma* is characterized by paroxysm of dyspnea, cough and wheezing which occurs more commonly in children and young adults. The attacks may be mild or severe and may last for hours, days or rarely weeks. Between episodes, the patients are usually asymptomatic.
- In older non-atopic patients, the asthma is chronic and persistent. Symptoms are worse in the early morning. This can be confused with the COPD.

Examination

- The examination may reveal tachypnea, tachycardia and involvement of accessory respiratory muscles.

- The breath sound is harsh vesicular with prolonged expiration.
- Prominent wheeze is audible in both phases of respiration.

Depending on the symptoms, signs, frequency of exacerbations and PEF rate, the asthma is classified as mild, moderate or severe.

Acute severe asthma: This is a severe life-threatening attack of asthma, previously known as *status asthmaticus*. The patient may additionally have tachycardia, pulsus paradoxus, cyanosis and active accessory respiratory muscles. The air entry is drastically reduced (silent chest on auscultation). The patient may become confused or drowsy.

Investigations

Pulmonary function tests: (see also page 134)

- Demonstration of reversible air flow obstruction is the hallmark of the diagnosis.
- In bronchial asthma, FEV_1 , FEV_1/VC ratio and peak expiratory flow rate (PEFR) are reduced.
- An improvement of at least 15% in FEV_1 or PEFR following administration of bronchodilator is diagnostic of bronchial asthma. PEFR is generally used for long-term home monitoring (Fig. 5.9).
- In asymptomatic individuals, exercise, histamine or methacholine can be used to provoke bronchospasm for the diagnosis of asthma.



FIGURE 5.9: Peak expiratory flowmeter

Chest X-ray: This is usually normal. However, it may appear hyperinflated. This is also helpful in ruling out other conditions.

Arterial blood gas analysis: Initially PaCO_2 is low. A normal or high PaCO_2 is a sign of impending respiratory failure. $\text{PaO}_2 < 60$ mmHg signifies severe bronchoconstriction and respiratory failure.

Others: Peripheral blood may show eosinophilia. Sputum may also contain an increased number of eosinophils.

Management

The overall aim of management should be

- To abolish or minimize symptoms
- To attain the best possible PEF

- To prevent exacerbation
- To normalize exercise capacity

Avoidance of factors: The identification of allergens is possible only in a few cases. Wherever possible, the exposure to such agents must be avoided.

Desensitization: Desensitization by repeated administration of allergen has not been proven effective in most studies.

Drug therapy: The drugs used in asthma can be grouped as:

- Quick relievers: which inhibit smooth muscle contraction and cause bronchodilatation (Table 5.18)
- Long-term control medications: which prevent or reverse inflammation (Table 5.19).

TABLE 5.18: Bronchodilator drugs used in asthma

Group	Route of administration	Major side effects
Adrenergic (Beta 2 receptor agonist)		
Epinephrine	subcut. injection	tremor, tachycardia
Salbutamol, albuterol, terbutaline	oral and inhaled	
Salmeterol, formoterol (long acting)	inhaled	
Methylxanthines		
Theophylline	oral and IM/IV	nausea, vomiting
Aminophylline	IV	arrhythmia, seizures
Anticholinergics		
Ipratropium bromide		
Tiotropium	inhaled	no major side effects

TABLE 5.19: Anti-inflammatory drugs used in asthma

	Route of administration	Major side effects
Steroids		
Systemic		
Methyl prednisolone	oral and IV	long-term side effects: adrenal suppression, cataract, growth retardation, osteoporosis, hyperglycemia, hypertension
Prednisolone	oral	
Inhaled		
Beclomethasone	inhaled	dysphonia, thrush, long-term use may cause systemic side effects as above
Budesonide		
Fluticasone		
Mast cell stabilizers	inhaled	
Cromolyn sodium		
Nedocromil		
Leukotriene modifiers	oral	hepatic toxicity, drug interaction
Montelukast		
Zafirlukast		
Zileuton		

Note: Long acting inhaled adrenergic agents (salmeterol, formoterol) and theophylline also have anti-inflammatory properties.

Chronic Persistent Asthma

A stepwise approach is chosen according the severity of disease. Once the disease is controlled, a step down therapy is attempted.

Step 1: Inhaled short acting adrenergic agents (salbutamol, terbutaline) are used as needed for minor symptoms.

Step 2: If the symptoms are not controlled with occasional inhaled adrenergic drugs as mentioned above, low dose of inhaled steroid is added. Mast cell stabilizers can be used prophylactically before exposure to allergens, exercise or cold air.

Step 3: The dose of inhaled steroid is escalated. An inhaled long acting adrenergic agent such as salmeterol or formoterol can be added. Sustained release theophylline may also be used orally.

Step 4: In addition to drugs used in step 3, leukotriene modifiers and inhaled ipratropium bromide are tried.

Step 5: Regular oral steroid in the lowest effective dosage is added to step 4 regimen to control symptoms.

Acute Severe Asthma

1. *Hospitalization and oxygen therapy:* The patient should be hospitalized for urgent management. Arterial blood gas analysis, chest X-ray and ECG should be done to assess severity and to rule out other causes. A high concentration oxygen inhalation is given to maintain $\text{PaO}_2 > 60 \text{ mmHg}$.

2. Repeated dosage of short-acting adrenergic drugs (salbutamol 2.5-5 mg or terbutaline 5-10 mg) is given through a nebulizer at an interval of 20-30 minutes. The PEFR should be recorded frequently to assess the response. Metered dose inhaler with a spacer device can alternatively be used. Ipratropium bromide can be added to get additional bronchodilator effect.
3. *Systemic steroids:* This is essentially needed in all cases of severe asthma. Prednisolone (30-60 mg daily) or methylprednisolone is given orally or intravenously.
4. Others:
 - a. Mechanical ventilation is needed in patients with coma, respiratory arrest, exhaustion and deteriorating blood gases despite adequate treatment.
 - b. IV fluid is administered to prevent dehydration.
 - c. Antibiotics are not used routinely. These are only given if infection is present.
 - d. Opiates, sedatives or tranquillizers are contraindicated.

Patient Education and Monitoring of Therapy

- The patients are educated about the nature of disease and its treatment.
- They should also be trained to recognize the severity of their disease and monitor the response to therapy with the use of peak flow meter.
- Patients should also be demonstrated the proper use of inhalation devices such as metered dose inhalers (pressurized aerosol system), rotahaler (dry powder system) and nebulizers (Figs. 5.10A to C).



FIGURES 5.10A to C: Inhalation devices, nebulizer and patient on nebulizer therapy

- Use of inhaler therapy should be encouraged as it is effective in lower dosage together with a rapid onset of action and has fewer side effects.

Complications

The main complications of bronchial asthma include pneumothorax, respiratory failure and cor pulmonale.

CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

COPD is characterized by chronic airflow obstruction which is not fully reversible. It includes chronic bronchitis and emphysema. Other causes of pulmonary obstruction (*asthma, bronchiectasis, cystic fibrosis, bronchiolitis*) are not classified as COPD.

1. *Chronic bronchitis* is defined as a condition with a history of cough and sputum on most days for at least 3 consecutive months for more than 2 successive years (provided other causes of chronic productive cough are excluded). Recently, chronic bronchitis has been described as clinically defined condition with chronic cough and sputum.
2. *Emphysema* is defined as irreversible destruction and enlargement of airspaces distal to terminal bronchioles. Pure forms of these two conditions are rare and in the majority of cases they coexist.

Etiological Factors

- *Cigarette smoking* is the most important etiological factor for COPD. There is a positive correlation between the intensity of smoking (number of cigarettes and duration of smoking) and the development of COPD. However, individual susceptibility varies since only 15 % smokers develop clinically significant COPD.
- *Air pollution* in the environment may also be associated with the development of COPD.
- Other factors are occupational exposure to dusts, bronchial hyper-responsiveness and low birth weight.
- Severe alpha-1 antitrypsin deficiency is a proven genetic risk factor for the development of emphysema.

Pathophysiology

The important pathological changes in the airways are inflammation, hypertrophy of the mucous secreting glands

and increase in the number of goblet cells. This leads to decreased mucociliary clearance. The mechanical obstruction in the airflow and loss of elastic recoil of lungs result in air flow limitation.

The emphysema in smokers is generally centriacinar which involves mostly the upper lobes of the lung (centriacinar emphysema). In alpha-1 antitrypsin deficiency, the lower lobes of lung are predominantly affected (panacinar emphysema).

The persistent hypoxia and pulmonary vascular changes cause pulmonary hypertension, right ventricular hypertrophy (cor pulmonale) and failure.

Clinical Features

The important symptoms are cough, sputum production and exertional dyspnea. Initially, the cough and expectoration are more marked in winters but later these may occur throughout the year. Sputum is mucoid, scanty and thick. There may be streaking of sputum with blood; massive hemoptysis is rare. Increased production of purulent sputum and fever indicates bacterial infection. As the disease advances, dyspnea becomes more severe and occurs even during rest.

In mild cases, the physical examination may be normal.

- Chest examination reveals vesicular breath sound with prolonged expiratory phase and generalized wheeze. The crackles (crepitations) may be heard mainly in the lower zones.
- The patient with predominant emphysema shows barrel shaped chest (increased AP diameter), loss of cardiac and liver dullness and diminished vesicular breath sound.
- The use of accessory respiratory muscles, breathing with pursed lips and lack of cyanosis characterize emphysema (pink puffers).
- Cyanosis and right heart failure are more common and seen earlier in the course of disease in patients with predominant chronic bronchitis (blue bloaters).

The signs are also depicted in Table 5.20.

Investigations

1. *Pulmonary function tests*: (see also page 131) Spirometric demonstration of airway obstruction without significant reversal is the hallmark of the diagnosis of COPD.

TABLE 5.20: Clinical signs in COPD**General**

- Cyanosis
- Signs of right heart failure
Pedal edema, raised JVP, hepatomegaly, ascites
- Signs of CO₂ retention
Flapping tremor (asterixis), bounding pulse
obtundation
- Pursled lip breathing
- Wasting and weight loss

Respiratory

- Use of accessory muscles of respiration (scalene, sternocleidomastoid)
- Indrawing of intercostal muscles during inspiration (Hoover's sign)
- Increased AP diameter of the chest
- Loss of cardiac dullness and liver dullness
- Diminished vascular breath sounds
- Prolonged expiratory phase
- Generalized wheeze
- Crackles mostly at bases

Significant reversibility of the obstruction (as seen in asthma) is defined as 15% or more increase in FEV₁ after two puffs of beta adrenergic agonist. An abnormal FEV₁ (< 80% predicted) with an FEV₁/VC ratio of < 70% strongly suggests COPD. A normal FEV₁ excludes the diagnosis of COPD. Lung volumes (total lung capacity and residual volume) are increased due to air trapping.

2. *Chest X-ray* may demonstrate bullae, hypertranslucent lung fields, flattened diaphragms and prominent pulmonary arterial shadows. The X-ray is also helpful in excluding other conditions presenting with chronic cough and dyspnea. CT is used to diagnose emphysema and to detect its severity.
3. *Arterial blood gas analysis* reveals hypoxemia (low PaO₂) and hypercarbia (high PaCO₂). Diffusing capacity of the lung for carbon monoxide (DLco) is markedly low in emphysema.
4. *Serum alpha-1 antitrypsin* level should be measured in young patients with family history of COPD.

Complications

The complications of COPD include, (a) pneumothorax due to rupture of bullae, (b) respiratory failure (type II) and (c) cor pulmonale.

Treatment

Treatment of COPD can be described as of (a) chronic stable phase, and (b) acute exacerbation

Chronic Phase

Most forms of therapy are directed to relieve the symptoms and to decrease the frequency and severity of acute exacerbations. These measures do not alter the natural course of the disease. However, cessation of smoking, oxygen therapy in chronically hypoxic patients and lung volume reduction surgery (LVRS) in selected patients are known to influence the natural history of COPD.

- Smoking cessation:* All patients with COPD should be encouraged to stop smoking since it improves the rate of decline in the pulmonary function. The patients should be educated and adequately counseled. Bupropion or nicotine replacement therapy may be employed to enhance the chance of smoking cessation.
- Oxygen therapy:* Long-term oxygen therapy at home is required in patients with PaO₂ <55 mmHg or in patients with PaO₂ 55-60 mmHg who have pulmonary hypertension, heart failure or polycythemia. Oxygen is given in low concentration (2 liters/min) for at least 15 hours a day.
- Bronchodilators:* Bronchodilators are given for symptomatic relief. Anticholinergic agents (ipratropium bromide, tiotropium) are used in inhaler form. Beta agonists like salbutamol (short acting) or salmeterol and formoterol (long acting) are also useful. The addition of beta agonists with anticholinergic agents provides incremental response. Oral theophylline may result in improved exercise tolerance and quality of life in moderate to severe COPD.
- Corticosteroids:* Inhaled corticosteroids have been shown to decrease the frequency of acute exacerbations of COPD. Low dose inhaled steroids are thus indicated in severe COPD. However, chronic use of oral corticosteroid is not recommended in view of side effects.
- Antibiotics:* Respiratory infections must be treated promptly as they may precipitate acute exacerbation. The most common bacteria responsible for infection are *S. pneumoniae* and *H. influenzae*. The antibiotics of choice are amoxicillin, amoxicillin-clavulanate combination or clarithromycin given for at least 5-10 days. Influenza vaccine is recommended annually.

- f. *Pulmonary rehabilitation*: Exercise and pulmonary rehabilitation programs improve dyspnea and quality of life.
- g. *Surgery*:
 - Lung volume reduction surgery (LVRS) may be indicated in patients with emphysema. It improves pulmonary functions and symptoms and also provides mortality benefit.
 - Surgical resection is also required in patients with large bullae.
 - Severe COPD is an important indication for lung transplantation.

Acute Exacerbation

The acute exacerbation of COPD presents as fever, increased breathlessness and cough, production of increased volume of purulent sputum and some times as right heart failure. The patients may have cyanosis and impaired sensorium.

Acute exacerbation of COPD should be differentiated from other conditions presenting as acute onset dyspnea (Table 5.21).

Chest X-ray, arterial blood gas analysis, ECG, blood counts, urea and electrolytes and sputum culture are generally performed to assess severity of the exacerbation and to exclude other causes.

Most patients require hospitalization. The main steps in the treatment include:

1. Oxygen therapy
2. Antibiotics are given to treat respiratory infections.
3. Bronchodilators: The dose and frequency of bronchodilators are escalated. These are administered initially by nebulization. Intravenous infusion of aminophylline is needed if there is no response.
4. Short course of oral corticosteroid hastens recovery and reduces chances of subsequent relapses.

TABLE 5.21: Differential diagnosis of acute exacerbation of COPD

Pneumonia
Pneumothorax
Left ventricular failure
Pulmonary embolism
Acute severe asthma
Airways obstruction

5. Diuretics are given in case of raised jugular venous pressure (JVP) and edema.
6. Mechanical ventilatory support (invasive and non-invasive) may be required in patients with severe respiratory distress, severe hypoxia and hypercapnia.

PULMONARY EOSINOPHILIA

Pulmonary eosinophilia or pulmonary infiltrates with eosinophilia (PIE) are distinct syndromes characterized by the presence of pulmonary infiltrate and eosinophilia (peripheral eosinophil count $>500/\mu\text{L}$). The most common cause of PIE in tropical countries is parasitic infestation and ABPA (allergic bronchopulmonary aspergillosis) in developed countries. Other causes of PIE are listed in Table 5.22.

Tropical Eosinophilia

- Tropical eosinophilia is caused by filarial infection.
- The manifestations are due to allergic and inflammatory reactions elicited due to clearance of microfilaria by the lungs.
- The main presenting features are paroxysmal dyspnea, wheeze, cough and low grade fever.
- Peripheral blood examination shows eosinophilia ($>3000/\mu\text{L}$) and X-ray chest generally reveals diffuse miliary opacities. Total serum IgE level and anti-filarial antibody titer are specifically elevated.
- The treatment includes diethylcarbamazine (DEC) in dosage of 4-6 mg/kg body weight daily for 14 days.

Acute eosinophilic pulmonary infiltrate may also occur due to passage of helminthes larvae through lungs (Loeffler's Syndrome).

TABLE 5.22: Causes of pulmonary infiltrates with eosinophilia (PIE)

1. *Parasitic infestation*
 - Ascaris, microfilaria, ankylostoma, toxocara, strongyloides
2. *ABPA*
3. *Drug induced*
 - Nitrofurantoin, penicillin, sulphonamide, PAS, phenyl butazone
4. *Conditions with unknown etiology*
 - Hypereosinophilic syndromes
 - Churg-Strauss syndrome (multisystem vasculitis)
 - Eosinophilic pneumonias

Allergic Bronchopulmonary Aspergillosis (ABPA)

- *Aspergillus fumigatus* is the most common cause of ABPA.
- The features include asthma, raised serum IgE, skin reaction to *A. fumigatus*, evidence of peripheral eosinophilia ($>1000/\mu\text{L}$) and pulmonary infiltrates.
- The sputum culture may be positive for the *A. fumigatus*.
- The treatment requires long-term use of glucocorticoids.

Hypereosinophilic Syndrome

The diagnosis depends on the fulfilment of following criteria;

- Eosinophilia ($>1500/\mu\text{L}$) for at least 6 months
- Lack of evidence of any detectable cause
- Sign and symptoms of multisystem organ dysfunction, such as involvement of heart, nervous system, lungs and liver

The therapy includes the use of glucocorticoids and/or hydroxyurea.

PULMONARY EMBOLISM

Pulmonary embolism is an important cause of mortality in hospitalized patients. The most common source of emboli is proximal leg and pelvic deep vein thrombosis (DVT). Table 5.23 shows important predisposing conditions for DVT. Other causes of pulmonary embolism are shown in Table 5.24.

Clinical Syndromes

The pulmonary embolism (PE) can result in different syndromes depending upon the site, size and speed of onset of embolism. Dyspnea is the most frequent symptom and tachypnea is the most frequent sign.

TABLE 5.23: Predisposing factors for DVT

- Genetic
 - Factor V leiden
 - Prothombin gene mutation
 - Protein C or protein S deficiency
- Acquired
 - Smoking
 - Oral contraceptive
 - Pregnancy
 - Surgery and trauma
 - Long air travel
 - Malignancies
 - Anti-phospholipids antibody syndrome

TABLE 5.24: Causes of pulmonary embolism

- Deep vein thrombosis (in lower limb)
- Non-thrombotic causes
 - Amniotic fluid embolism (tear of placental margin, fetal membrane leak)
 - Air embolism
 - Fat embolism (long bone fractures, blunt trauma)
 - Tumor embolism (choriocarcinoma)
 - Septic emboli (Infective endocarditis affecting tricuspid and pulmonary valves)

Acute massive pulmonary embolism:

- This is due to the obstruction of main or proximal pulmonary artery leading to decreased cardiac output and right ventricular dilatation.
- Presentations are sudden onset of severe dyspnea, apprehension, central chest pain, faintness and syncope.
- Signs include tachycardia, tachypnea, cyanosis, hypotension, increased JVP, right sided S_3 , and wide split P_2 .

Acute small/medium pulmonary embolism:

- There is obstruction of segmental small pulmonary artery which often leads to peripheral pulmonary infarction.
- Usual symptoms are dyspnea, pleuritic chest pain, fever, cough and hemoptysis.
- Signs may include tachycardia, tachypnea, pleural rub, and features of pleural effusion.
- Clinical features may be more severe if there is underlying cardiopulmonary disease.

Chronic pulmonary embolism:

- Chronic occlusion of pulmonary microvasculature is generally asymptomatic initially. Later, it may produce pulmonary hypertension and right heart failure.
- This can present as chronic exertional dyspnea.
- Signs of pulmonary hypertension (loud P_2) and right ventricular failure (raised JVP, RV heave, pedal edema) can be found.

Differential Diagnosis

The PE should be differentiated from conditions which present with acute dyspnea or chest pain as shown in Table 5.25.

Investigations

- Chest X-ray:** Generally the X-ray chest is normal in PE but it is helpful in ruling out other conditions like

TABLE 5.25: Differential diagnosis of pulmonary embolism**Cardiac diseases:**

- Acute myocardial infarction
- Acute left ventricular failure
- Pericardial tamponade
- Aortic dissection

Pulmonary diseases:

- Pneumothorax
- Pneumonia
- Exacerbation of asthma or COPD
- Primary pulmonary hypertension

Others:

- Anxiety

pneumothorax, pneumonia and heart failure. The findings in PE include peripheral wedge shaped opacity (due to pulmonary infarction), local infiltrates, pleural effusion and pulmonary oligemia. *A normal X-ray chest along with acute dyspnea and hypoxemia suggests the diagnosis of PE.*

- Electrocardiography:** The ECG findings are common. These could be in form of non-specific ST-T changes or more specifically $S_1Q_3T_3$ pattern or T wave inversion in V_1 to V_4 .
- Arterial blood gas analysis:** The values may reveal low PaO_2 , normal to low $PaCO_2$ (type I respiratory failure) and metabolic acidosis in severe cases.
- D-dimer test:** D-dimer is a degradation product released into circulation when fibrin is thrombolized by endogenous plasmin. The plasma level of D-dimer is raised in >90 % cases of PE. However, it can also be high in other conditions like pneumonia, MI and sepsis. Hence, a raised value is not diagnostic of PE. *The low value of D-dimer excludes the diagnosis of PE.*
- Imaging:**
 - Spiral CT of the chest with IV contrast has good sensitivity and specificity for the diagnosis of PE and is now the investigation of choice in patients with acute dyspnea.
 - Magnetic resonance (MR) imaging with contrast is also increasingly being used for the diagnosis of PE.
 - Ventilation perfusion (V/Q) lung scanning, a popular investigation in past, is now less commonly used. However, it may be useful when performed within 24 hours of presentation and in patients with no previous cardiopulmonary disease such as COPD.

- Dopplar ultrasonography:** This is done to detect deep vein thrombosis in the leg veins. It has now replaced venography.
- Echocardiography:** This is useful to exclude other conditions such as MI, pericardial tamponade and dissection of aorta. Echocardiography can diagnose right ventricular dysfunction and may detect major central PE.
- Pulmonary angiography:** This is the “gold standard” method for the definitive diagnosis of PE. However, in the current era, the CT has replaced pulmonary angiography as the former is less invasive.

Treatment

- Supportive measures:** Important supportive measures include:
 - Relief from pain by non-steroidal anti-inflammatory drugs or opiates
 - Oxygen therapy to maintain oxygen saturation over 90%
 - Dobutamine in patients having severe right heart failure and cardiogenic shock
- Anticoagulation:** Anticoagulation prevents additional thrombus formation. The already formed clot is eventually lysed by endogenous fibrinolytic mechanisms.
- Heparins:** Low molecular weight heparin (LMWH), such as enoxaparin is given subcutaneously twice a day. Alternatively unfractionated heparin can be used intravenously (bolus of 5000-10,000 IU followed by a continuous infusion at the rate of 1000 IU /hr) . The dose of heparin is adjusted according to activated partial thromboplastin time (aPTT). The LMWH is preferred over unfractionated heparin because of ease of administration and no need for monitoring.
- Oral anticoagulants:** Oral anticoagulant (warfarin) is also started simultaneously. The heparin is stopped after 5 days while the warfarin is continued. The dose of warfarin is adjusted according to prothrombin time and international normalized ratio (PT/INR).
- Duration and complications:** The duration of anticoagulation therapy is generally upto 3 months if cause of DVT is reversible. However, patients with history of previous embolism or having underlying prothrombotic conditions should be given oral

anticoagulant life long. The most common complication of anticoagulation therapy is hemorrhage.

- c. **Thrombolytic therapy:** Thrombolysis is indicated in acute massive pulmonary embolism with right ventricular dysfunction or hypotension. Thrombolytic agent used is recombinant tissue plasminogen activator (r-tPA) or streptokinase.
- d. **Embolectomy:** Open surgical or catheter embolectomy is indicated in patients with massive PE where thrombolysis is contraindicated because of high risk of intracranial hemorrhage.
- e. **Inferior vena caval filters:** A filter is placed in inferior vena cava to prevent the passage of emboli from pelvic and lower limb veins to the lungs. This is done in patients with recurrent emboli despite adequate anticoagulation or where anticoagulation is contraindicated because of bleeding.

Prevention of PE

The preventive measures include early mobilization after surgery, adequate hydration and prophylactic use of LMWH or warfarin in patients undergoing major surgery.

TUBERCULOSIS

Epidemiology: Tuberculosis (TB) is among the oldest infections in humans. Around one-third of the world population is infected with *M. tuberculosis*. It is still a significant cause of morbidity and mortality in the developing countries. There is an increase in the incidence of tuberculosis because of HIV infection.

Causative organism: It is caused by bacteria belonging to *Mycobacterium tuberculosis* complex.

- The most common agent of human disease is *M. tuberculosis*.
- *M. bovis*, an important cause of infection in those who consume unpasteurized milk, is now uncommon.

Transmission: Most commonly the infection is transmitted from infected patients to other persons through droplet nuclei released by coughing, sneezing or speaking. Other rare routes of transmission are ingestion, through skin and transplacental.

Risk factors: The risk of acquiring infection is increased by factors like poverty and overcrowding. Patients with

diabetes, silicosis, alcoholism or immunocompromised states are at a greater risk of acquiring tuberculosis. Health workers are also at increased risk as they may be exposed to TB patients.

Primary Tuberculosis

Primary tuberculosis occurs due to initial infection in lungs and occasionally in tonsils or intestine. The primary lesion in lungs, tonsils and intestine is almost always accompanied by the lymph node involvement such as mediastinal, cervical and mesenteric groups respectively.

Pathology: The macrophages ingest bacilli and eventually are killed through cellular immune responses leading to tissue necrosis and caseation. Activated monocytes turn into epithelioid cells and form granuloma at the periphery of caseation. In the majority of cases lesions heal by calcification. In some the healing is incomplete and the bacilli may disseminate to different organs through the blood stream. Lesions in these organs may develop years later.

Tuberculosis can be pulmonary, or extrapulmonary or both. Pulmonary tuberculosis is more common than extrapulmonary. However, in patients with HIV disease, extrapulmonary form of tuberculosis is more commonly seen.

Pulmonary Tuberculosis

Pulmonary tuberculosis can be classified into:

- a. Primary pulmonary tuberculosis
- b. Post-primary pulmonary tuberculosis (adult type, reactivation or secondary tuberculosis)

Primary Pulmonary Tuberculosis

The bacilli enter the lung parenchyma and cause a peripheral parenchymal lesion. The bacilli eventually travel to the mediastinal lymph nodes. This is known as *primary complex*.

- Middle and lower lobes of the lung are usually involved.
- In most (80-90%), the primary complex heals within 4-6 weeks. The healed calcified peripheral parenchymal lesion is known as Ghon's lesion (Fig. 5.11).
- It usually affects children and remains asymptomatic in most. However, in some, particularly in immunocompromised individuals, the disease may progress in the following forms:
 - a. The parenchymal lesion may enlarge and cavitate (progressive primary tuberculosis).



FIGURE 5.11: X-ray chest showing Ghon's lesion

- The disease may involve pleura and result into pleural effusion.
- Enlarged lymph node may compress bronchi and result in collapse (epituberculosis).
- Hematogenous dissemination is common and generally asymptomatic. However, occasionally it may lead to meningitis or miliary tuberculosis.
- Manifestations due to hypersensitivity reaction may occur in form of erythema nodosum or phlyctenular conjunctivitis.

Post-primary Pulmonary Tuberculosis

This is also known as adult-type, reactivation or secondary tuberculosis. It usually results from reactivation of latent infection and is localized to the upper lobes as growth of the mycobacteria is favored by high oxygen concentration. Parenchymal involvement may be in the form of small infiltrates, pneumonia, collapse, extensive cavitary lesions or miliary lesions. The lesion may remit spontaneously or in some, it may progress to chronic fibrosis.

Clinical features: The patients initially present with symptoms like cough, malaise, loss of appetite, loss of weight, low grade fever with evening rise, night sweats and hemoptysis. The chest examination may be normal or may reveal inspiratory crackles, particularly after cough, and bronchial breathing over large cavities.

Other uncommon presentations may be in the form of pleural effusion, spontaneous pneumothorax and pyrexia of unknown origin (PUO).

Extrapulmonary Tuberculosis

Virtually any organ may be involved due to hematogenous spread of the infection. The occurrence of extrapulmonary form of tuberculosis has become more common with emergence of HIV disease. The most common extrapulmonary site involved is lymph nodes. Table 5.26 shows different forms of extrapulmonary tuberculosis.

TABLE 5.26: Forms of extrapulmonary tuberculosis

Tuberculous lymphadenitis	Painless enlargement of lymph nodes generally cervical, may be matted, may form sinus (scrofula) and collar stud abscess
Pleural tuberculosis	Pleural effusion, empyema, pneumothorax
Gastrointestinal	Most commonly ileocecal involvement, may present as abdominal pain, diarrhea, intestinal obstruction, ascites
Genitourinary	Hematuria, frequency, sterile pyuria, epididymitis, prostatitis, tubo-ovarian abscess, infertility
Pericardial	Pericardial effusion, constrictive pericarditis
Central nervous System	Meningitis, tuberculoma
Bone and joints	Spinal tuberculosis, arthritis

Miliary Tuberculosis

The miliary tuberculosis is a severe form of tuberculosis that results from hematogenous spread of tuberculous bacilli. This may be a form of primary tuberculosis or may occur due to reactivation of old foci. The lesions are characterized by granuloma (1-2 mm) that resembles millet seeds.

Clinical features: Besides constitutional symptoms like fever, weight loss and anorexia, patients may have hepatosplenomegaly and lymphadenopathy. The fundus examination may show choroidal tubercles.

Investigations: Chest X-ray shows miliary shadows (Fig. 5.12). The hematological features include anemia, leukopenia or leukemoid reaction. The tuberculin skin test (PPD test) is negative in half of the patients. Liver or bone marrow biopsy may be required in some cases for the diagnosis.

HIV and Tuberculosis

Tuberculosis is the most common opportunistic infection in HIV infected individuals in India. The disease may appear at any stage of HIV disease.



FIGURE 5.12: X-ray chest showing miliary tuberculosis

Clinical presentation: The presentation is typical (upper lobe infiltrates and cavity) in early stages when the immunity is only partially compromised. However, in late stages, the presentation can be like primary tuberculosis. The extrapulmonary involvement is more common in HIV infected persons than in HIV negatives. *M. avium* complex (MAC) infection may occur when CD4 count becomes less than 50/cmm.

Diagnosis: The diagnosis of tuberculosis in HIV infected patients becomes difficult because of atypical clinical and radiological features. Moreover, the PPD skin test and sputum smear for AFB (Acid Fast Bacilli) are negative in most cases.

Investigations

1. **Demonstration of AFB:** The diagnosis of tuberculosis is based on the demonstration of AFB in the smear of the sputum or in other specimens such as tissue biopsy materials or body fluids. *Ziehl-Neelsen* or *Auramine fluorescence* staining is usually done for this purpose. If the patient is not passing sputum, nebulization with hypertonic saline can be used to induce sputum expectoration. Samples can also be obtained by gastric lavage (in children), bronchoalveolar lavage or transbronchial biopsy.
2. **Culture methods:** The culture provides confirmation of the diagnosis by the isolation and identification of *M.*

tuberculosis from the specimen. Drug sensitivity can also be tested. The growth on solid media (Löwenstein-Jensen media) is slow and may take 4-8 weeks. However, the time required for culture confirmation is shorter (2-3 weeks) when liquid media (BACTEC) is used.

3. **Molecular methods:** Nucleic acid amplification method provides diagnosis in hours but sensitivity is lower than culture and the cost is high. It is useful in cases with AFB negative pulmonary and extrapulmonary tuberculosis.
4. **Radiological tests:** The typical findings include infiltration of upper lobe with fibrosis and/or cavity



FIGURE 5.13: X-ray chest showing right sided infiltration and fibrosis of lung



FIGURE 5.14: X-ray chest showing right sided massive pleural effusion with mediastinal shift

(Fig. 5.13). However, any type of radiographic pattern such as consolidation, collapse, pleural effusion or miliary can be seen in tuberculosis (Fig. 5.14).

5. **Skin test:** Purified protein derivative (PPD) skin test (using Mantoux method or Heaf method) is positive (a) in persons infected with *M. tuberculosis*, (b) in persons sensitized by non-tuberculous mycobacteria and (c) in those who have received BCG vaccination. A positive skin test does not tell about the active disease. A positive test in those who have not received BCG may suggest the diagnosis of tuberculosis. The test may be negative in miliary tuberculosis and in immunocompromised patients with tuberculosis.
6. **Histopathological tests:** The FNAC (fine needle aspiration cytology) or biopsy specimens from the involved tissue may typically reveal caseous granuloma. AFB can be demonstrated in tissue specimens. The tissue can be cultured to demonstrate mycobacteria.
7. **Other tests:**
 - The hematological findings include anemia, raised ESR and C reactive protein.
 - The fluid (pleural, pericardial, peritoneal) is exudative in nature.
 - Liver biopsy and bone marrow biopsy specimens may also be examined for the evidence of granuloma and AFB.

Treatment

The main aims of treatment of tuberculosis are:

1. To cure the patients of tuberculosis
2. To decrease transmission of tuberculosis to others
3. To prevent relapse
4. To prevent morbidity and mortality from active tuberculosis
5. To prevent late effects of tuberculosis.

Anti-tubercular Drugs

The anti-tubercular drugs are bactericidal or bacteriostatic in nature. The bactericidal drugs are used to rapidly reduce the number of viable organisms and render patients non-infectious. These also have sterilizing activity (kill all bacilli), hence, they prevent relapse. Bacteriostatic drugs are used along with the bactericidal drugs to prevent emergence of resistance.

First line drugs: The first line drugs are rifampicin (R), isoniazid (H), pyrazinamide (Z) and ethambutol (E). Isoniazid and rifampicin are active against all population of TB bacilli while pyrazinamide and streptomycin are active against certain population of TB bacilli. In addition, pyrazinamide is active against bacilli in acid environment inside macrophages and has good CSF penetration. Streptomycin is particularly active against extracellular bacilli. The dosage and side effects of anti-tubercular drugs are given in Table 5.27.

TABLE 5.27: Anti-tubercular drugs and their dosage

Drugs	Mode of action	Daily dose mg/kg	Intermittent (thrice weekly) mg/kg	Common side effects
Isoniazid (H)	Bactericidal	5	10	Hypersensitivity, neuropathy, hepatitis
Rifampicin (R)	Bactericidal	10	10	Hepatitis, thrombocytopenia, flu like symptoms, hypersensitivity, drug interactions
Pyrazinamide (Z)	Bactericidal	25	35	Gout, hepatitis, hypersensitivity
Ethambutol (E)	Bacteriostatic	15	30	Optic neuritis, hypersensitivity, colour blindness for green
Streptomycin (S)	Bactericidal	15	15	Vestibular toxicity, deafness, hypersensitivity, renal toxicity

Second line drugs: The Second line drugs (Table 5.28) are less efficacious and more toxic. These are used when first line drugs fail.

Treatment Regimen

The treatment consists of the initial phase and the continuation phase. The schedule is daily or intermittent

(twice or thrice weekly). The response to therapy is monitored through clinical improvement, X-ray chest and culture or smear examination.

Initial phase: The aim is to rapidly kill the bacilli, resolve symptoms and bring out sputum conversion (AFB negative) so that the patient becomes non-infectious. Generally, a combination of 3-4 drugs is used for 2-3 months.

TABLE 5.28: Second line anti-tubercular drugs

Oral agents
• Ethionamide
• Cycloserine
• PAS
• Fluoroquinolones: Gatifloxacin, Levofloxacin, Moxifloxacin, Ofloxacin
Injectable agents
• Streptomycin (S)
• Kanamycin
• Amikacin
• Capreomycin
Other drugs
• Clofazimine
• Thioacetazone
• Amoxicillin-clavulanic acid
• Linezolid

Continuation phase: The purpose is to eliminate the remaining bacilli from the lesion (sterilizing effect) so that relapse may not occur. This is generally given for 4-6 months.

The most commonly used regimen consists of a 2 month initial phase of RHZE followed by a 4 month continuation phase of HR (2RHZE/4HR). The drugs can also be given thrice weekly (2H₃R₃Z₃E₃/4H₃R₃).

Other Drugs/Therapy

- The supplementation of pyridoxine (10-25 mg/day) prevents isoniazid related neuropathy. It is given in those who are at high risk of pyridoxine deficiency such as

alcoholics, elderly, malnourished, diabetics, pregnant ladies and HIV patients.

- Corticosteroids are useful if given with antitubercular drugs in seriously ill patients (miliary TB), and meningeal, pericardial, pleural or ureteric diseases to prevent adhesions.
- Surgery may be needed in patients with massive and recurrent hemoptysis, constrictive pericarditis, lymph node suppuration, empyema and spinal cord compression.

National Guidelines

Patients with tuberculosis are grouped into four categories and the treatment regimens are chosen accordingly (Table 5.29).

Drug Resistant Tuberculosis

Resistance to TB drugs develops because of spontaneous point mutation in the mycobacterium genome. It may be primary (patients infected with resistant bacilli and have not received TB drugs earlier) or secondary/acquired (resistance develops during treatment). Factors which favor the development of resistance are inadequate regimen, incomplete dosage or duration. Multidrug resistant (MDR) TB is characterized by the resistance to more than one anti-TB drugs or resistance to both isoniazid and rifampicin. The treatment of MDR TB needs second line drugs (Table 5.28) for a longer duration. Recently XDR TB (extensive drug resistant TB) has been reported where MDR strains are also resistant to at least the fluoroquinolones and one or

TABLE 5.29: National guidelines for treatment of tuberculosis

TB treatment category	TB patient	Initial phase (daily or intermittent)	Continuation phase
I	New smear positive pulmonary TB; new smear negative pulmonary TB with extensive parenchymal involvement; new cases of severe form of extrapulmonary TB	2 RHZE (RHZS)	4 HR or 4 H ₃ R ₃ or 6 HE
II	Sputum smear positive: relapse, treatment failure or treatment after interruption	2 RHZES plus 1 RHZE	5 HRE or 5H ₃ R ₃ E ₃
III	Smear negative pulmonary TB (other than category I); new less severe form of extrapulmonary TB	2 RHZ	4 HR or 4 H ₃ R ₃ or 6 HE
IV	Chronic case (sputum positive after supervised treatment), most likely multidrug resistant (MDR)	Refer to specialized center for second line drugs	NA

Note: Less severe form of extrapulmonary TB includes lymph nodes, unilateral pleural effusion, bone excluding spine, peripheral joint or skin involvement. Severe form of extrapulmonary TB includes meningitis, miliary TB, pericarditis, peritonitis, bilateral or extensive pleural effusion, spinal, genitourinary, intestinal TB.

more of the injectable drugs: amikacin, kanamycin or capreomycin.

Directly Observed Treatment, Short Course (DOTS)

Patients swallow the drugs before a health personnel or some other responsible person. This ensures that the TB patient takes the right drugs, in the right dosage at the right intervals. DOTS helps to improve cure rate and to reduce the chance of drug resistance.

Complications of Pulmonary Tuberculosis

The complications of TB are given in Table 5.30.

Vaccination

Most countries recommend the use of *Bacillus Calmette Guérin (BCG)* vaccination. This contains attenuated strain of *M. bovis*.

- Intradermal route is recommended for the administration of the vaccine at the lower deltoid area.
- In India, it is recommended to be administered to infants after birth or at the first contact of the infant with the health worker. The BCG vaccination is contraindicated in children with symptomatic HIV disease.
- Ulceration at injection site and regional lymphadenopathy may occur in 1-10 % of vaccinated persons.
- The vaccination does not prevent infection or reactivation. However, it leads to inhibition of lymphohematogenous spread of bacilli. Thus, highest efficacy of the BCG vaccine is in children with those forms of disease which involve hematogenous spread such as meningitis, miliary and disseminated TB. The vaccine provides about 70 % protection for more than 10 years.

TABLE 5.30: Complications of pulmonary tuberculosis

Pulmonary complications

- Massive hemoptysis
- Aspergilloma
- Bronchiectasis
- Cor pulmonale
- Pleural calcification
- Bronchopleural fistula

Extrapulmonary complications

- Amyloidosis
- Laryngitis
- Enteritis
- Empyema necessitans

Chemoprophylaxis

- Chemoprophylaxis has been applied in (a) tuberculin negative individuals to prevent occurrence of infection in high risk situations (b) treating tuberculin positive cases to prevent disease.
- In India where a large majority of the population is tuberculin positive, it may be an expensive proposition. Hence, chemoprophylaxis has been recommended in all known recent tuberculin converters or in household contacts of newly diagnosed active cases regardless of tuberculin positivity.
- Isoniazid is given for 6 months or isoniazid plus rifampicin for 3 months or rifampicin alone for 4 months.

RESPIRATORY FAILURE

Respiratory failure is defined as a condition in which lung function is inadequate to meet the metabolic requirement of the individual. The dysfunction can be in one or more components of the respiratory system such as airways, respiratory muscles, alveolar units and pulmonary vessels.

Respiratory failure manifests as hypoxemia ($\text{PaO}_2 < 60$ mmHg) and/or hypercapnia ($\text{PaCO}_2 > 50$ mmHg).

Clinical features of respiratory failure consist of those of underlying disease combined with manifestations of hypoxemia and hypercapnia (Table 5.31).

Type I Respiratory Failure (Hypoxemic Respiratory Failure)

Type I Respiratory Failure may be acute, chronic or acute on chronic.

1. **Acute:** This is characterized by low PaO_2 and normal or low PaCO_2 . The important causes are pulmonary edema, pneumonia, ARDS, pneumothorax, pulmonary embolism and acute asthma. The management includes:
 - a. high concentration oxygen therapy

TABLE 5.31: Clinical features of hypoxemia and hypercapnia

Hypoxemia

Symptoms	Dyspnea
Signs	Cyanosis, confusion, tachypnea, tachycardia, arrhythmias, flapping tremors (asterixis)

Hypercapnia

Symptoms	Dyspnea, headache
Signs	Impaired sensorium, hypertension, tachycardia, tachypnea, papilledema, asterixis

- b. prompt treatment of underlying disorders
 - c. mechanical ventilation if no improvement
2. **Chronic:** There is low PaO_2 and normal PaCO_2 . The important causes are emphysema and lung fibrosis. The treatment is directed toward underlying conditions.

Type II Respiratory Failure (Hypercapnic Respiratory Failure)

Type II Respiratory Failure may be acute, chronic or acute on chronic.

- a. **Acute:** There is hypoxia and hypercapnia. The latter can result into respiratory acidosis. Causes include:
- diminished respiratory drive (drugs, brainstem lesions)
 - impaired respiratory muscles function (myasthenia gravis, myopathies, Guillain-Barré syndrome)
 - asphyxia (inhaled foreign body, laryngeal edema)
 - acute severe asthma.

The treatment is aimed at reversal of precipitating events. The patient may need tracheostomy or mechanical ventilation. Respiratory stimulant (doxapram infusion) may be helpful in case the facilities for mechanical ventilation are not available.

- b. **Chronic:** This is characterized by the presence of hypoxemia, hypercapnea with near normal pH due to compensatory rise in bicarbonate (HCO_3). The most important cause is COPD. The treatment is directed toward the cause. A controlled long-term oxygen therapy may be helpful.
- c. **Acute on chronic:** Any precipitating event such as acute infection, airways obstruction and cardiac failure can lead to acute deterioration in patients with chronic respiratory disease. The treatment consists of control of precipitating events and controlled low concentration oxygen therapy.

Type III Respiratory Failure

Type III Respiratory Failure type of respiratory failure can occur as a result of lung collapse, generally during perioperative period. That is why this is also called as *perioperative respiratory failure*. It can be avoided by frequent change in position and chest physiotherapy.

Type IV Respiratory Failure

Type IV Respiratory Failure occurs due to hypoperfusion of respiratory muscles in patients with shock. Respiratory failure occurs due to increased demand by respiratory

muscles. The distribution of cardiac output to respiratory muscle is increased. However, respiratory failure may occur if the demand is not met with by the increased supply. Intubation and mechanical ventilation are helpful in the management of such condition.

PLEURAL DISEASES

Pleura consists of two layers. The parietal pleura lines the inner surface of the chest wall while visceral pleura covers the lung surface. There is a space between both layers of pleura which normally contains very thin layer of fluid (5-15 ml).

Pleurisy

Pleurisy is a term described when pleura is involved in any disease process. The inflammation in the pleura is also known as pleuritis.

- Common causes of pleurisy are viral infections, pneumonia, tuberculosis and malignancy.
- The most important symptom of pleurisy is sharp and stabbing chest pain that worsens by coughing, sneezing or deep breathing (pleuritic pain).
- The clinical examination reveals pleural rub and other features of primary underlying disease. The disappearance of rub and chest pain suggests recovery or the development of pleural effusion.
- The management of pleurisy includes treatment of the primary cause and the symptomatic control of pain. Analgesics and anti-inflammatory drugs are used for pain relief. Codeine is helpful in suppressing the cough and chest pain.

Pleural Effusion

Pleural effusion is defined as excessive accumulation of fluid in the pleural space. The collection of grossly purulent

TABLE 5.32: Common causes of pleural effusion

Transudative

- Congestive heart failure
- Cirrhosis with ascites
- Nephrotic syndrome

Exudative

- Tuberculosis
- Pneumonia
- Malignancy
- Pulmonary embolism
- Connective tissue diseases
- Pancreatitis

fluid is known as empyema. Presence of blood and lymphatic fluid in pleural space is called hemothorax and chylothorax respectively. Important causes of pleural effusion are pneumonia, tuberculosis, cardiac failure, cirrhosis and malignancy. Other causes are mentioned in Table 5.32.

Clinical Features

Pleuritic chest pain often precedes the onset of effusion. The most important symptom of pleural effusion is dyspnea and the severity depends on the rate and size of the accumulation of fluid. Examination reveals diminished movements on the side of the effusion, mediastinal shift to the opposite side and stony dull note on percussion. Auscultation of the chest shows reduced or absent breath sounds.

Investigations

Chest X-ray reveals the presence of fluid and the evidence of underlying disease in the lungs (Fig. 5.14). Ultrasound is helpful to confirm the presence of fluid and to guide the aspiration. CT chest is occasionally required to diagnose the underlying pathology.

The aspiration of fluid confirms the diagnosis of pleural effusion. The biochemical, microbiological and cytological examination of the pleural fluid is helpful in knowing the underlying cause. The measurement of lactate dehydrogenase (LDH) and protein in the pleural fluid helps in differentiating transudative from exudative effusion. The biopsy of the pleura may be needed to diagnose the underlying cause.

Management

- The aspiration of fluid may be required to relieve the dyspnea in cases with large effusion.
- The underlying causes should be treated.
- Recurrent accumulation of pleural fluid in malignancy can be managed by the injection of sclerosing agents in the pleural cavity after the complete evacuation of fluid. The sclerosing agents cause pleural inflammation and adhesion leading to obliteration of the pleural space (pleurodesis).
- Empyema requires intercostal tube drainage and sometimes surgical intervention.

PNEUMOTHORAX

Presence of air in the pleural space is known as pneumothorax. Incidence of pneumothorax is highest in tall young males, smokers and those with apical subpleural blebs.

Classification

- Spontaneous pneumothorax: It can be without evidence of any overt lung disease (Primary) or due to underlying lung disease such as COPD, tuberculosis (Secondary).
- Traumatic pneumothorax: It may be iatrogenic or due to trauma to the chest wall.

Types of Pneumothorax

- Close type: The communication between pleural space and lung seals off as the lung deflates and it does not reopen.
- Open type: The communication between pleural space and lung fails to seal off and air continues to transfer freely.
- Tension pneumothorax: The communication between pleural space and lung works as a one way valve that allows air to enter the pleural space during inspiration and coughing but prevents its escape.

Clinical Features

Most important symptoms of pneumothorax are sudden onset breathlessness and unilateral chest pain. Clinical signs are given in Table 5.33.

Investigations

Chest X-ray usually shows collapsed lung which has sharply defined edge with complete translucency without lung markings surrounding it. The shift of mediastinum to opposite side is seen (Fig. 5.15).

TABLE 5.33: Signs of pneumothorax

Tachycardia, cyanosis, hypotension
Decreased movement of chest wall on the side of lesion
Mediastinal shift to opposite side
Hyperresonant percussion note
Decreased or absent breath sound
Decreased vocal fremitus and decreased vocal resonance



FIGURE 5.15: Right sided pneumothorax

Management

- Primary spontaneous pneumothorax: It can be initially managed by simple aspiration. Thoracoscopy with stapling of blebs is indicated if lung expansion does not occur or patient develops recurrent pneumothorax.
- Secondary spontaneous pneumothorax: It is treated with tube thoracostomy with underwater seal drainage. Alternatively, thoracoscopy with stapling of blebs can be performed.
- Traumatic pneumothorax: It should be treated with tube thoracostomy. Two tubes are needed in case of hemo-pneumothorax, one to be placed in upper part to drain air and another in lower part to drain blood.
- Tension pneumothorax: This is a medical emergency. A large bore needle should be inserted into pleural space through second anterior intercostals space. Later thoracostomy should be done and the needle removed.
- Supportive treatment: This includes oxygen inhalation and management of shock.

INTERSTITIAL LUNG DISEASE

Interstitial lung disease (ILD) is a group of diseases involving parenchyma of the lung: the alveoli, alveolar epithelium, capillary endothelium and spaces between these structures.

Classification

Interstitial lung disease is classified into two main types depending on the underlying histopathological changes.

- Predominant inflammation and fibrosis associated
- Predominant granulomatous reaction associated

Pathogenesis

There are inflammatory changes in the air spaces and alveolar walls initially which gradually involve interstitium and vessels. Eventually, chronic inflammation leads to fibrosis. In granulomatous type, lymphocytes, macrophages and epithelial cells organize into discrete granuloma in the lung parenchyma.

Clinical features

Progressive dyspnea is the most common presenting symptom whereas tachypnea and basal end inspiratory crackles is the important sign. Other symptoms and signs are given in the Table 5.34.

Common causes of ILD

Important causes of ILD are

- Idiopathic interstitial pneumonia
- Occupational: fumes, gases, asbestos, organic and inorganic dust
- Sarcoidosis
- Connective tissue diseases: SLE, rheumatoid arthritis
- Drugs and others

Investigations

- Chest X-ray: The findings are nonspecific. Most common findings are bibasilar reticular pattern, nodular, and mixed pattern (Fig. 5.16).

TABLE 5.34: Clinical features of ILD

Symptoms:

Progressive dyspnea
Dry cough (or minimal sputum)
Fatigue, weight loss
Chest pain, hemoptysis (uncommon)

Signs:

Tachypnea
Cyanosis, clubbing (rare)
Bilateral basal end inspiratory crackles
Wheezes (occasional)
Signs of cor pulmonale



FIGURE 5.16: Interstitial lung disease (X-ray showing reticulonodular shadows)

- b. CT thorax: High resolution CT scan confirms the diagnosis and can detect the disease in early stages.
- c. Pulmonary function tests: It reveals restrictive pattern. FEV₁ and FVC are reduced; FEV₁/FVC is usually normal or increased. Total lung capacity, functional residual capacity and residual volume are decreased.
- d. Arterial blood gas analysis may reveal hypoxia and respiratory alkalosis.
- e. Bronchoalveolar lavage (BAL): This may be helpful in the diagnosis of sarcoidosis.
- f. Others: ANA, rheumatoid factor, serum ACE level

Treatment

The main goal of treatment is aggressive suppression of inflammation. This can be achieved by:

- a. Corticosteroids
- b. Immunosuppressive drugs: cyclophosphamide, azathioprim

LUNG CANCERS (BRONCHOGENIC CARCINOMA)

Lung cancer is the leading cause of death due to cancers worldwide. Smoking is causally associated with lung cancer; however, it can occur in non-smokers also.

Etiology of Lung Cancers

The most common cause of lung cancer is cigarette smoking. Various carcinogens and tumor promoters are inhaled during smoking.

- a. In active smokers, the relative risk of developing lung cancer is 13 fold while it is 1.5 fold in passive smokers.
- b. The chance of developing cancer is even more in case a smoker has COPD.
- c. The chance of developing lung cancer depends on the amount and duration of cigarette smoking (cigarette pack years).
- d. The risk of lung cancers reduces following smoking cessation.

Apart from smoking, thoracic radiation therapy increases the incidence of lung cancer.

Classification

The classification of lung cancers is mainly based on the cell types. Four major cell types constitute most of the primary lung cancers and these are:

1. Squamous (epidermoid) carcinoma
2. Small cell (oat cell) carcinoma
3. Adenocarcinoma
4. Large cell carcinoma

The most common type of lung cancer worldwide is adenocarcinoma. Adenocarcinoma is also common in non-smokers, females and young patients. Other types include undifferentiated carcinoma, carcinoids and bronchial gland tumors.

Clinical Features

The symptoms and signs of lung cancers occur due to various factors/mechanisms, which are given below.

- a. *The presence of local tumor* growth may lead to cough, hemoptysis, wheeze, stridor, dyspnea. Peripherally present growth may also cause chest pain due to pleural involvement.
- b. *Invasion or obstruction of nearby structures* can cause dyspnea (tracheal obstruction), dysphagia (esophageal compression), hoarseness of voice (recurrent laryngeal nerve palsy) and Horner's syndrome (enophthalmos, ptosis, miosis and ipsilateral loss of sweating over face). Local invasion of structures such as 8th cervical and 1st

and 2nd thoracic nerves and destruction of first and second ribs by tumor in the apex of lung with shoulder pain is known as Pancoast's syndrome. Regional spread of tumor can cause superior vena caval obstruction (Superior vena cava syndrome). Lymphatic obstruction may result in pleural effusion.

- c. Extrathoracic metastasis can occur at numerous sites like brain, bones, bone marrow, liver and lymph nodes giving rise to various symptoms and signs.
- d. Paraneoplastic syndromes occur due to biologically active hormones secreted by the tumor. However, in many cases, the mechanism is unknown. Some of the important paraneoplastic syndromes found in lung cancers are hypercalcemia, hyponatremia, hypokalemia, hypertrophic pulmonary osteoarthropathy, peripheral neuropathies, cerebellar degeneration, venous thrombosis and Eaton Lambert's syndrome.

Investigations

- a. X-ray chest reveals mass lesion which is generally central in squamous cell and small cell carcinoma and peripheral in adenocarcinoma and large cell carcinoma. Cavitory lesion can be seen in some cases of squamous cell and large cell carcinoma. Other findings are mediastinal lymphadenopathy, bone erosions and pleural effusion (Fig. 5.17).

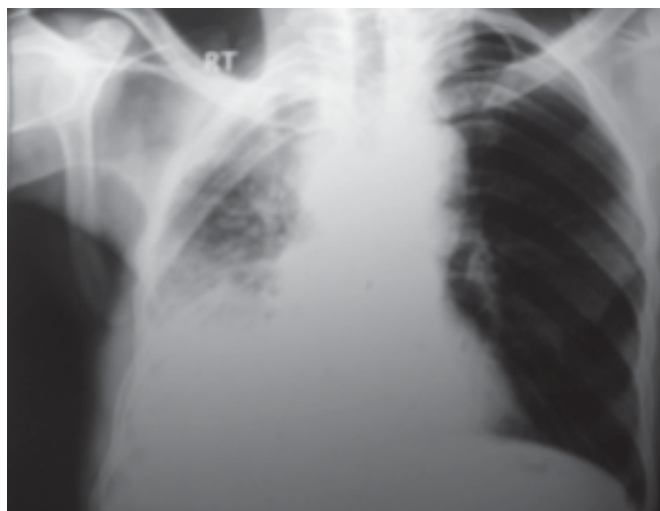


FIGURE 5.17: Carcinoma right lung (X-ray showing collapse of lung with destruction of ribs)

- b. CT scan helps in detecting the small size tumour and local and distant metastasis. These findings are helpful in staging the disease.
- c. Fiberoptic bronchoscopy is useful in cases of endobronchial growth where tumour tissue can be obtained for histological diagnosis.
- d. Other investigations include sputum examination for presence of malignant cells and biopsy from sites like enlarged lymph nodes, soft tissue mass, lytic bone lesion or pleural lesion. Positron emission tomography scan (PET scan) is especially useful in assessing mediastinal and solitary pulmonary nodules.

Management

Following modalities of treatment are applied in patients with bronchogenic carcinoma:

- a. Surgery
- b. Chemotherapy
- c. Radiotherapy
- d. Palliative therapy

Surgery: Surgery offers best hope in patients with early stage disease (stage I and II), where the 5 year survival rates are more than 75% and 55% respectively. However, surgery is not possible in majority of cases due to extensive spread of disease or presence of severe cardiac or renal disease or other comorbid illnesses.

Chemotherapy: The combination chemotherapy drugs can increase median survival in patients with small cell carcinoma, whereas, in general, chemotherapy is less effective in non-small cell bronchogenic carcinoma. The drugs used in combination chemotherapy are cisplatin, etoposide, cyclophosphamide, vincristine and doxorubicin.

Radiotherapy: The important role of radiotherapy in bronchogenic carcinoma is in the palliation of distressing complications such as superior vena caval obstruction, recurrent hemoptysis and pain due to chest wall infiltration or skeletal metastatic deposits. Radiotherapy when used along with chemotherapy in small cell carcinoma can improve the survival. It can also be helpful in patients with localized disease where surgery is not indicated due to various reasons.

Palliative therapy: This includes removal of obstruction of airways by endobronchial stenting or laser treatment

through fiberoptic bronchoscopy. Effective pain management is also important in patients with advanced cancer.

IMPLICATIONS ON DENTAL PRACTICE

1. Elective dental care is deferred in acute respiratory infections including common cold, sinusitis, pneumonia and acute bronchitis.
2. General anesthesia should be avoided in presence of respiratory infections.
3. Prolonged use of corticosteroid inhalers may lead to increased incidence of oral candidiasis. Patients are advised to rinse mouth with water after using inhalers.
4. Use of decongestant and antihistaminics may cause oral dryness.
5. Toothache may occur due to maxillary sinusitis. It should be differentiated from odontogenic infections. Pain in more than one tooth in the same maxillary quadrant suggests sinus infection.
6. Mouth breathing due to chronic sinusitis may lead to oral dryness and oral diseases such as gingivitis.
7. Poor oral health predisposes to the development of pneumonia due to aspiration of salivary secretion containing both aerobic and anaerobic bacteria. Colonization of dental plaque and oral mucosa with pathogens is more common among hospitalized patients. Hence, regular use of chlorhexidine gluconate mouth washes is recommended in hospitalized patients.
8. Inhalation of tooth or fragments or rarely endodontic instruments can cause lung abscess.
9. Numerous dental products and materials such as toothpaste, tooth enamel dust, methyl methacrylate may exacerbate the asthma.
10. Elective dental procedure should only be done when asthma is well controlled. Patients are asked to bring the asthma medication with them.
11. The history of allergy to aspirin or NSAIDs should always be asked in asthmatic patients as these agents may precipitate asthmatic attacks.
12. Use of beta blockers may preferably be avoided in asthmatic patient.
13. If the patient develops acute asthmatic attack during dental procedure, the procedure is stopped and all intraoral devices are removed. The patient is placed in comfortable position and the patency of airways is ensured. Oxygen is given and beta 2 agonists are administered through nebulization. Medical assistance may be called for.
14. Drug interactions should be taken in mind. For example, macrolide antibiotics (erythromycin, clarithromycin) may increase the level of theophylline. Use of epinephrine containing local analgesics may precipitate arrhythmias if patient is taking theophylline or beta agonists.
15. Aminoglycoside group of drugs (streptomycin, amikacin) can enhance the activity of neuromuscular blocking drugs and increase the weakness in myasthenic patients.
16. Long-term use of beta 2 agonists may cause reduced salivary flow and increased incidence of caries and candidiasis. Fluoride supplement is advised for all asthmatic patients particularly those taking beta 2 agonists. Ipratropium can also cause dry mouth.
17. There is association between oral infections and exacerbation of COPD. Therefore maintenance of good oral hygiene is must in these patients.
18. Patients with COPD are better treated in upright position as they may become more breathless if laid flat. Patients may not be able to tolerate rubber dam.
19. Special caution is needed during dental procedures in patients with pulmonary embolism who are on oral anticoagulants. In such patient, prothrombin time and INR are estimated before the procedure. A level of INR <2.5 is safe for dental care including simple extraction.
20. Tuberculosis is unlikely to be transmitted to the dental staff unless the patient is having active pulmonary tuberculosis or the dental staff is immunocompromised. Hence, dental treatment is best deferred until the active tuberculosis has been treated.
21. Patient with tuberculosis and HIV infection can have oral lesions in the form of chronic oral ulcerations and candidiasis.
22. Mycobacteria are resistant to many disinfectants. Hence, heat sterilization should be used.
23. The drug interaction with anti-tubercular agents must be considered while prescribing the dental treatment.
24. Rifampicin can cause red saliva.

SELF ASSESSMENT

Multiple Choice Questions

1. **Following part of lung is mostly involved in lung abscess:**
 - A. Middle lobe of the lung
 - B. Posterior segment of lower lobe
 - C. Superior segment of lower lobe and posterior of upper lobe
 - D. Posterior segment of upper lobe and superior of lower lobe
2. **Following are causes of cavity in lung except:**
 - A. Lung abscess
 - B. Tuberculosis
 - C. COPD
 - D. Wegener's granulomatosis
3. **Which is not a clinical manifestation of lung abscess:**
 - A. Clubbing
 - B. Chest pain
 - C. Copious amount of sputum
 - D. Episodic dyspnea with wheeze
4. **Kussmaul's breathing is seen in:**
 - A. Metabolic alkalosis
 - B. Respiratory alkalosis
 - C. Metabolic acidosis
 - D. Respiratory acidosis
5. **Polycythemia is seen in:**
 - A. Emphysema
 - B. Pleural effusion
 - C. Bronchial asthma
 - D. Lung abscess
6. **Stony dull note on percussion is present in:**
 - A. Pleural effusion
 - B. Pneumothorax
 - C. Pulmonary fibrosis
 - D. Bronchial asthma
7. **Following is the sign of consolidation:**
 - A. Bronchial breath sound
 - B. Increased vocal fremitus
 - C. Whispering pectoriloquy
 - D. All of above
8. **Most common source of pulmonary embolism:**
 - A. Splenic vein
 - B. Arteriovenous fistula
 - C. Leg and pelvic veins
 - D. Subclavian vein
9. **Peripheral neuropathy can be caused by:**
 - A. Isoniazid
 - B. Rifampicin
 - C. Ethambutol
 - D. Streptomycin
10. **Hepatitis can be caused by:**
 - A. Rifampicin
 - B. Isoniazid
 - C. Pyrazinamide
 - D. All of the above
11. **Significant improvement in FEV₁ after bronchodilator inhalation is seen in:**
 - A. COPD
 - B. Bronchiectasis
 - C. Bronchial asthma
 - D. None of the above
12. **Following indicates severe asthma except:**
 - A. Pulsus paradoxus
 - B. Cyanosis
 - C. PaO₂ >80 mm Hg
 - D. Silent chest
13. **Clubbing can be found in all except:**
 - A. Lung abscess
 - B. Lung cancer
 - C. Pneumonia
 - D. Bronchiectasis
14. **Hemoptysis can be present in:**
 - A. Pulmonary tuberculosis
 - B. Chronic bronchitis
 - C. Mitral stenosis
 - D. All the above
15. **Tropical pulmonary eosinophilia is associated with:**
 - A. Aspergillus infection
 - B. Nitrofurantoin
 - C. Filaria
 - D. Malaria
16. **Fibro-cavitary lesion is feature of:**
 - A. Primary tuberculosis
 - B. Post-primary tuberculosis
 - C. Miliary tuberculosis
 - D. All of above
17. **Following can be used in the treatment of pulmonary edema except:**
 - A. Morphine
 - B. Frusemide
 - C. Beta blockers
 - D. Nitrates
18. **The most common symptom in pulmonary embolism is:**
 - A. Chest pain
 - B. Fever with chills
 - C. Dyspnea
 - D. Hemoptysis

19. Following is preferred for the definitive diagnosis of acute pulmonary embolism:

- A. D-dimer test
- B. Arterial blood gas analysis
- C. Ventilation perfusion scan
- D. Spiral CT

20. Following can be found in pulmonary infarction:

- A. Dyspnea
- B. Pleuritic chest pain
- C. Hemoptysis
- D. All the above

21. Following are the causes of type I respiratory failure except:

- A. Pneumonia
- B. Pulmonary embolism
- C. Respiratory muscle paralysis
- D. Pulmonary edema

22. Domiciliary oxygen therapy is indicated in:

- A. Pneumonia
- B. COPD
- C. Pulmonary eosinophilia
- D. None of the above

23. Following drugs are used in patients with eosinophilia except:

- A. Prednisolone
- B. Hydroxyurea
- C. Diethyl carbamazine
- D. Nitrofurantoin

24. Hyper-resonant note and absent breath sounds are features of:

- A. Pleural effusion
- B. Pneumothorax
- C. Pneumonia
- D. Pulmonary fibrosis

25. "Blue bloater" is associated with:

- A. Emphysema
- B. Chronic bronchitis
- C. Pulmonary fibrosis
- D. Bronchial asthma

Fill in the Blanks

1. Enlargement of _____ secondary to lung disease is called cor pulmonale.
2. Foul smelling sputum suggests infection due to _____ organisms.
3. Pink frothy sputum can be present in _____.
4. Most common cause of community acquired pneumonia is _____.
5. Most common cause of hospital acquired pneumonia is _____.
6. Barrel shaped chest is seen in _____.
7. FEV₁/FVC ratio is _____ in COPD.
8. Interrupted brief explosive sounds on lung auscultation is known as _____.
9. Continuous musical adventitious lung sound is called _____.
10. Inspiratory sound heard in case of tracheal obstruction is called _____.
11. Healed calcified parenchymal lesion in primary tuberculosis is called as _____ lesion.
12. Cyclical waxing and waning of respiration intervened with a short period of apnea is known as _____ breathing.
13. The most common cause of chronic type II respiratory failure is _____.
14. The drug of choice in tropical pulmonary eosinophilia is _____.
15. Pink puffers are associated with the disease _____.
16. Pleural rub is a sign of _____.
17. Nasal or bleating sounds on lung auscultation are called _____.
18. Normal partial pressure of oxygen in arterial blood (PaO₂) is _____.
19. Normal partial pressure of carbon dioxide in arterial blood (PaCO₂) is _____.
20. Normal arterial blood pH is _____.

Chapter 6

Renal Diseases

RENAL SYNDROMES

Structure and Function

The functional unit of the kidney is the nephron. There are about one million nephrons in each kidney. Nephrons are mainly composed of glomeruli and tubules. The important function of glomeruli is ultrafiltration whereas those of tubules are selective reabsorption and maintenance of fluid, electrolytes and acid-base balance. The important physiological functions of kidneys are given in Table 6.1.

The possible sites of injury may be the glomeruli, the tubules, the interstitium or the vasculature. Hence manifestations in renal disorders may vary according to the predominant site/structures involved, constituting distinct combination of symptoms, signs and laboratory abnormalities. These are known as **renal syndromes** (Table 6.2)

The renal syndromes can present with one or more of the following features;

- Alteration in urine volume:** The urine output in 24 hours is normally 2-3 liters. It may be decreased (oliguria/anuria) or increased (polyuria). *Anuria* means there is no urine output or <50 ml per 24 hours. A urine output of <400 ml/24 hours is defined as *oliguria*. When the urine output is more than 3 liters in 24 hours, it is known as *polyuria*.
- Abnormalities of urinary sediments:** The presence of >2-5 RBC's /high power field is defined as *hematuria*.

TABLE 6.1: Important functions of kidneys

- Maintenance of water and electrolyte balance
- Maintenance of acid-base homeostasis
- Synthesis of erythropoietin
- Vitamin D metabolism (hydroxylation of 25-hydroxy-cholecalciferol to the active form, 1,25-dihydroxycholecalciferol)
- Secretion of renin

TABLE 6.2: Renal syndromes

- Acute renal failure (ARF)
- Chronic renal failure (CRF)
- Nephritic syndrome
- Nephrotic syndrome
- Asymptomatic urinary abnormalities (hematuria, proteinuria, pyuria)
- Urinary tract infection (UTI)
- Renal tubule defects
- Hypertension
- Nephrolithiasis (renal stones)
- Urinary tract obstruction

Hematuria, if visible by naked eye is called macroscopic or gross hematuria and if detected by dipstick test or microscopy, known as microscopic hematuria. However a WBC count of <3/high power field in females and <1/high power field in males may be regarded as normal. The presence of WBCs more than the above normal values for males and females is defined as *pyuria*. Normal urine may also contain variety of casts, particularly hyaline casts and crystals.

- Abnormal excretion of serum proteins (proteinuria):** Proteins of low molecular weight are normally filtered at the glomeruli. These proteins are reabsorbed by the tubules. Hence, a normal person excretes less than 150 mg protein and less than 30 mg albumin daily. *Proteinuria* is defined as excessive protein excretion in the urine (>150 mg in 24 hours). The excretion of 30-300 mg albumin per 24 hour (20-200 µg/min) is known as *microalbuminuria*. Microalbuminuria is an early marker of glomerular disease in diabetes. *Massive proteinuria* (>3.5 g/day) is the hallmark of nephrotic syndrome. The dipstick method measures mostly albumin in urine. The sulfosalicylic acid method is used to detect total protein concentration and other types of protein in urine such as light chains (Bence Jones proteins).

- d. *Decline in glomerular filtration rate (GFR)*: This manifests as retention of nitrogenous waste products such as urea and creatinine. This is known as *azotemia*.
- e. Hypertension and edema
- f. Electrolyte and acid-base abnormalities
- g. Fever, abdominal pain and dysuria.

INVESTIGATIONS IN RENAL DISORDERS

- a. *Complete urine analysis*: Urine is examined for amount, specific gravity, pH, colour, protein, glucose, red blood cells, leukocytes, casts, and crystals.
- b. *Serum biochemistry*: The estimation of blood urea, serum creatinine, blood sugar and electrolytes (mainly Na, K) are important.
- c. *Estimation of glomerular filtration rate (GFR)*: Creatinine clearance is the usual method of estimation of GFR in clinical practice. GFR can be estimated by using the Cockcroft-Gault formula for creatinine clearance.

$$\text{Creatinine clearance (ml/min)} = \frac{(140 - \text{age}) \text{ weight in kg}}{72 \text{ serum creatinine (mg/dL)}}$$

(in females, the value is multiplied by 0.85)

However, GFR can be measured more accurately by radionuclide studies using ^{125}I -iothalamate, diethylene triamine-penta-acetic acid labeled with technetium ($^{99\text{m}}\text{Tc}$ -DTPA), or dimercaptosuccinic acid labeled with technetium ($^{99\text{m}}\text{Tc}$ -DMSA).

- d. Imaging studies include plain X-rays, ultrasound, intravenous urography (IVU), CT scan, MRI, renal arteriography and renal venography.
- e. *Renal biopsy*: Light microscopy, electron microscopy and immunofluorescence of the biopsy specimen are helpful in the diagnosis, prognosis and treatment of various renal disorders.
- f. Other investigations may be required in specific situations to know the etiology of renal diseases such as antinuclear antibody (ANA) in SLE, serum protein electrophoresis in myeloma and serum IgA level in IgA nephropathy.

NEPHROTIC SYNDROME

Nephrotic syndrome is characterized by massive proteinuria (>3.5 g per 1.73 m² body surface area per 24 hr), hypoalbuminemia, edema, hyperlipidemia and hypercoagulability.

Nephrotic syndrome results from glomerular involvement. Massive proteinuria is the most important feature of nephrotic syndrome.

Causes

Nephrotic syndrome can be due to primary glomerular disease or secondary to systemic diseases. Important causes of nephrotic syndrome are idiopathic, diabetes mellitus, autoimmune disorders, drugs, and amyloidosis. Causes of nephrotic syndrome are given in Table 6.3.

Pathophysiology

The main component of nephrotic syndrome is proteinuria which occurs due to an alteration in the glomerular filtration. Other features such as hypoalbuminemia and edema are secondary to loss of protein in urine.

- Increased catabolism of protein in the kidney and an inadequate synthesis by liver also contribute to lower levels of serum albumin.
- The edema is due to low plasma oncotic pressure resulting in leakage of fluid into the interstitium (underfilling hypothesis) and/or primary salt and water retention by kidneys.
- Escape of fluid into the interstitium causes low intravascular volume which results into activation of renin-angiotensin-aldosterone axis and the sympathetic system, increased secretion of vasopressin (ADH), and decreased secretion of atrial natriuretic peptide. The net result is renal salt and water retention and increased intravascular volume and further leakage of fluid into interstitial space.

TABLE 6.3: Causes of nephrotic syndrome

Primary renal disorders (idiopathic)

Minimal change disease
Membranous glomerulopathy
Focal and segmental glomerulosclerosis
Membranoproliferative glomerulonephritis

Systemic diseases and others

Diabetes mellitus
Amyloidosis
Infections (HIV, hepatitis B, hepatitis C, plasmodium malariae)
Drugs (gold, penicillamine, NSAIDs, captopril)
Autoimmune (systemic lupus erythematosus, rheumatoid arthritis)
Malignancies (Hodgkin's disease, carcinoma breast, colon, and lung)

- The fall in oncotic pressure due to hypoalbuminemia leads to compensatory increase in synthesis of lipids by liver. There is also an altered metabolism of lipids. Both these mechanisms result in hyperlipidemia.
- Hypercoagulability is due to increased urinary loss of antithrombin III and altered levels of protein C and S.
- Infections may occur more commonly due to increased urinary loss of immunoglobulins.
- Vitamin D deficiency that occurs due to enhanced excretion of cholecalciferol binding protein may cause hypocalcemia and secondary hyperparathyroidism.
- Loss of transferrin in the urine may lead to iron-unresponsive microcytic hypochromic anemia.

Clinical Features

Edema is the main presenting feature of nephrotic syndrome. In children, edema is often seen only in the face (Fig. 6.1). However in adults, initially it may be present in the dependant parts (lower extremities) but later on it may become generalized edema known as anasarca (Figs 6.2A and B). In the morning, the face and upper limb may be more affected. Fluid collection can present as pleural fluid, ascites, or pulmonary edema. Fever may occur due to infection. Other uncommon features are arterial and venous thrombosis, pulmonary embolism, and renal vein thrombosis. Urine out put is generally normal. Hypertension and hematuria are rare.



FIGURE 6.1: Facial edema in nephrotic syndrome



FIGURE 6.2A: Anasarca in nephrotic syndrome - facial and upper limb edema



FIGURE 6.2B: Anasarca in nephrotic syndrome pedal edema)

Investigations

(a) Urine analysis

- The urine is examined for the presence of protein. Presence of protein in urine ($>3.5\text{gm}/24\text{ hours}$) is the hallmark of nephrotic syndrome.
- Microscopic examination may reveal lipid casts.
- Hematuria is rare.

(b) Blood Examination

- The blood examination reveals low serum albumin (<3 g/dL) and the presence of hyperlipidemia.
- Blood urea level and serum creatinine are generally normal.
- Other tests are required in certain settings to know the cause of nephrotic syndrome such as blood sugar, antinuclear antibody (ANA), rheumatoid factor, and HIV.

(c) Renal biopsy

- Renal biopsy is generally needed to know the type of primary renal disease, particularly in adults. Light microscopy, immunofluorescence, and electron microscopy of biopsy specimen are helpful in knowing the type of the lesion.
- The minimal change disease is characterized by normal findings on light microscopy and foot process fusion on electron microscopy.
- The thickening of glomerular basement membrane (GBM) and subepithelial deposits of IgG and C3 are features of membranous nephropathy.
- Minimal change disease is the most common histological type found in children whereas membranous glomerulopathy is most common in adults with nephrotic syndrome.

Treatment**General Measures**

1. **Control of edema:** The salt in diet is restricted to 1-2 gm per day. Diuretics (thiazide or loop diuretics or both) are used judiciously because these can lead to electrolyte imbalance and hypotension.
2. **Reduction of proteinuria:** ACE inhibitors and ARBs are useful in reducing proteinuria in patients. The daily protein loss is compensated by adequate dietary protein intake.
3. **Management of hyperlipidemia:** This is managed with the lipid lowering drugs such as statins (simvastatin, atorvastatin, and rosuvastatin). Dietary modification and exercise are also helpful.
4. **Management of hypercoagulable states:** Anticoagulation is needed in patients with the evidence of thrombosis.
5. **Others:** Vitamin D (1,25-dihydroxycholecalciferol) supplementation is required in patients with evidence

of vitamin D deficiency. Despite having evidence of iron deficiency anemia, patients with nephrotic syndrome may not respond to iron supplementation. This is due to loss of iron binding protein *transferrin* in the urine.

Specific Measures

1. **Steroid:** Immunosuppressive therapy is required in primary renal causes of nephrotic syndrome and in some secondary diseases such as SLE. Initially oral prednisolone is given in the dosage of 1-2 mg/kg/day. Once response is obtained (monitored by urinary protein excretion) it is tapered gradually.
2. **Cytotoxic/immunosuppressive agents:** In patients (a) who are steroid dependent, (b) who are steroid resistant, or (c) who frequently undergo relapse, treatment with cytotoxic or other immunosuppressive drugs is indicated. Common agents used are:
 - Cyclophosphamide (2 mg/kg/day)
 - Chlorambucil (0.1 – 0.2 mg/kg/day)
 - Cyclosporin (5 mg/kg/day)
 - Mycophenolate mofetil.

Prognosis

Remission occurs in 90% children and 50% adults in minimal change disease. About 20-30% patients with membranous nephropathy develop chronic renal failure.

NEPHRITIC SYNDROME

Nephritic syndrome arises due to acute glomerular inflammation (glomerulonephritis, GN) and is characterized by sudden onset (days to weeks) of:

- Oliguria
- Edema
- Hypertension
- Hematuria
- Subnephrotic proteinuria (<3 g/day)
- Worsening renal function

Pathophysiology

The renal blood flow and glomerular filtration rate (GFR) are reduced because of obstruction of glomerular capillaries by inflammatory cells and glomerular cell proliferation. Impaired GFR and increased reabsorption of salt and water

TABLE 6.4: Causes of nephritic syndrome

1. *Idiopathic*
Proliferative glomerulonephritis (focal or diffuse)
Rapidly progressive glomerulonephritis (RPGN)
2. *Post-infectious*
Streptococci
Bacterial endocarditis
Hepatitis B
Malaria
3. *Multisystem disorders*
SLE (lupus nephritis)
Henoch-Schönlein purpura
Goodpasture's syndrome
Wegener's granulomatosis

by tubules result in edema and hypertension. Injury to the glomerular capillary wall leads to the appearance of dysmorphic RBC, red blood cell cast and protein in the urine. Hematuria is often macroscopic.

Clinical Features

The onset is often sudden. The presenting features are decreased urine output and edema. A urine output less than 400 ml per day is defined as **oliguria** while less than 50-100 ml per day is called **anuria**. Hypertension and hematuria are often present. Patients also have generalized symptoms such as anorexia, nausea, vomiting, headache and malaise.

Causes

The important causes of nephritic syndrome are given in Table 6.4.

The nephritic syndrome is classified on the basis of pathological features into three broad categories. The types of nephritic syndrome along with the serological markers are described in Table 6.5. The most common type is immune complex glomerulonephritis.

Investigations

(a) Urine examination

The urine examination reveals dysmorphic red blood cells, red blood cell cast, and proteinuria. 24 hours urine output is measured which is low.

(b) Blood Examination

- Blood urea and creatinine are raised.
- Other tests which help in diagnosing different types of GN are serum complement levels (C3), anti-GBM antibody, ANCA (anti-neutrophil cytoplasmic antibody), antinuclear antibody (ANA), and ASO titers.
- Tests like serum electrolytes, arterial blood gases, and complete blood count may be needed.

(c) Renal biopsy

The biopsy of kidney is the gold standard for the diagnosis, treatment plan and the prognosis of the disease. Light microscopy, immunofluorescence, and electron microscopy of biopsy specimen are helpful in distinguishing the major types of acute nephritis.

Treatment

General Measures

1. *Control of edema and volume overload:* The salt in diet is restricted to 1-2 g per day. Water intake is restricted to the urine output plus 500 ml/day. Diuretics (thiazide or loop diuretics or both) are used carefully.
2. *Control of hypertension:* The blood pressure is controlled with antihypertensive agents. This helps in decreasing proteinuria and slowing the progression of the disease. The target BP should be 130/80 mm Hg.
3. *To minimize protein loss:* ACE inhibitors and ARBs are useful in reducing proteinuria. However, these agents can cause hyperkalemia and a rise in creatinine; hence

TABLE 6.5: Types of nephritic syndrome

Pathogenic type	Serological features	Immunofluorescence	Important causes
1. Immune complex GN	Low C3	Granular deposition of Ig and C3	- Idiopathic - Post-infectious - SLE
2. Anti-GBM disease	Anti-GBM antibody +	Linear deposition of Ig and C3	- Anti-GBM disease - Goodpasture's syndrome
3. Pauci-immune GN	ANCA+	Absent Ig and C3	- Wegener's granulomatosis

serum potassium and serum creatinine should be monitored.

4. **Dialysis:** Dialysis may be needed to control hypervolemia and uremia.

Specific Measures

- Corticosteroids (prednisolone, methylprednisolone) and other immunosuppressive drugs (cyclophosphamide, azathioprine, cyclosporine and mycophenolate mofetil) are occasionally used for the control of glomerular inflammation, however their role is debatable.
- Antibiotic is given in post-streptococcal glomerulonephritis.
- Patients who do not respond to immunosuppressive therapy and progress to end stage renal disease (ESRD) require dialysis or renal transplantation.

Prognosis

The course of the disease varies according to the cause and histopathological types. Majority of patients with post-streptococcal nephritic syndrome remit spontaneously and few patients (<5%) may develop rapidly progressive glomerulonephritis (RPGN) or ESRD.

ACUTE RENAL FAILURE (ARF)

Acute renal failure is defined as sudden decline in renal ability to maintain fluid and electrolyte homeostasis and to excrete nitrogenous wastes. Clinically ARF manifests as rising blood urea and serum creatinine levels or decreased urine output over hours or days. Acute renal failure may be oliguric or nonoliguric.

TABLE 6.6: Causes of acute renal failure

1. <i>Prerenal failure</i>
Hypotension or volume contraction (blood or fluid loss)
Heart failure
2. <i>Intrinsic renal failure</i>
Acute tubular necrosis
Glomerulonephritis
Interstitial nephritis
Reno-vascular diseases
3. <i>Postrenal failure (Obstructive)</i>
Ureteric obstruction (stone, clot, tumor, external compression)
Bladder outlet obstruction (prostate hypertrophy/carcinoma, stone, clot)

Important causes of ARF are classified in the Table 6.6. The most common type of ARF is prerenal failure. This is a response to renal hypoperfusion in situations like hypotension and volume depletion. However, prolonged and severe hypoperfusion may lead to renal parenchymal injury, a type of intrinsic renal failure (acute tubular necrosis).

Clinical Manifestations

The clinical manifestations of ARF are often due to azotemia (retention of nitrogenous waste products such as urea and creatinine) and are nonspecific. The clinical features also depend on the underlying cause of renal failure.

- Usual symptoms are nausea, vomiting, malaise, and anorexia.
- Cardiac manifestations include pulmonary edema due to fluid retention, pericardial effusion and arrhythmias.
- Encephalopathic features such as drowsiness, confusion, asterixis, seizures and coma may occur.
- There may be a bleeding tendency due to platelet dysfunction and altered coagulation.
- Features of hyperkalemia and metabolic acidosis are common.
- Anemia occurs due to blood loss or decreased RBC production.
- Infection is a serious complication and the most common cause of death in ARF patients.

Management

The management of ARF is generally aimed at maintaining fluid, electrolyte and acid-base balance. This requires a regular fluid intake and output recording and frequent estimation of urea, creatinine and electrolytes. The recovery of renal function occurs in most patients with ARF.

1. **Fluid management:** Intravenous fluid replacement is done in patients with hypovolemia in prerenal ARF. In other types of ARF, fluid intake is restricted according to the urine output (500 ml plus urine output in last 24 hours). Diuretics are used in cases with volume overload.
2. **Diet management:** Adequate caloric intake should be maintained to avoid excessive catabolism. Protein intake is restricted upto 0.6 gm/kg/day. Salt, potassium and phosphorus intake should also be restricted.

3. *Management of metabolic complications:* Hyperkalemia is managed with the use of calcium, insulin, bicarbonate and glucose. Sodium bicarbonate is given to control acidosis. Hypocalcemia and hyperphosphatemia require prompt management.
4. *Adjustment of drug dosage:* The dosage of the drugs is adjusted according to the severity of renal failure. Nephrotoxic drugs should be avoided.
5. *Management of anemia:* Severe anemia is managed by blood transfusion. Erythropoietin, though useful in CRF is not helpful in ARF.
6. *Bleeding complications:* Bleeding can be controlled by the use of desmopressin or estrogen. Regular use of antacids reduces the incidence of gastrointestinal hemorrhage.
7. *Control of infections:* Aseptic precautions should be practised to avoid infections. Prompt use of antimicrobial agent is necessary to manage infection.
8. *Dialysis:* Majority of patients improve on conservative management. However, the dialysis is indicated in following situations:
 - refractory hyperkalemia
 - acidosis
 - volume overload
 - uremic complications (encephalopathy, pericarditis and seizures).
 Peritoneal dialysis and hemodialysis are equally effective.

TABLE 6.7: Important features of CRF

Chronic features (>3 months) of uremia
Hypertension
Anemia
Renal osteodystrophy
Isothenuria (fix low specific gravity of urine)
Broad cast in the urine
Small contracted kidneys (on ultrasonography)

TABLE 6.8: Newer classification of CRF (CKD)

Stage	Description	GFR (ml/min/ 1.73m ²)
1	kidney damage with normal/increased GFR	>90
2	kidney damage with mildly decreased GFR	60-89
3	Moderately decreased GFR	30-59
4	Severely decreased GFR	15-29
5	kidney failure	<15

9. *Removal of obstruction:* Measures to remove obstruction in the urinary tract should be undertaken.

CHRONIC RENAL FAILURE (CRF)

Chronic renal failure is defined as irreversible loss of renal function which develops over a period of months to years. Eventually clinical symptoms and signs ensue due to severe loss of metabolic, excretory and endocrinal functions of the kidneys (**uremia**). Blood urea and serum creatinine are raised. Important features of CRF are given in Table 6.7. Now the term “chronic renal disease (CRD)” or “chronic kidney disease (CKD)” are preferred over CRF.

End stage renal disease (ESRD) is a severe stage of CRF (GFR 10-15 ml/min), in which the patient is permanently dependant on renal replacement therapy to avoid life-threatening uremia.

Newer classification of CRF (CKD): According to the estimated GFR, chronic kidney disease is classified as per guidelines issued by the National Kidney Foundation, USA. (Table 6.8)

Causes of CRF

Diabetes mellitus and hypertension are most common causes of CRF. Other important causes are glomerulonephritis, polycystic renal disease, tubulointerstitial diseases, and obstructive nephropathies.

Management

1. *Diet management:* Adequate caloric intake should be maintained to avoid excessive catabolism. Protein intake is restricted up to 0.6 g/kg/day to reduce accumulation of nitrogenous waste products and to slow down the progression of renal failure. Fluid, salt, potassium and phosphorus intake should also be restricted.
2. *Management of metabolic complications:* Hyperkalemia is managed with the use of calcium, insulin, bicarbonate and glucose. Sodium bicarbonate is given to control acidosis.
3. *Management of renal osteodystrophy:* Hyperphosphatemia is management by restriction of phosphate (milk, cheese, eggs) in the diet and use of phosphate binding drugs such as calcium carbonate and aluminium hydroxide gel. Supplementation of vitamin D (1-alpha hydroxyl vitamin D3 or 1,25-dihydroxy vitamin D3) and calcium is needed to correct hypocalcemia.



FIGURE 6.3: Patient on hemodialysis

4. *Management of proteinuria:* ACE inhibitors and ARBs are helpful in reducing proteinuria and slowing the progression of renal damage.
5. *Control of blood pressure:* Hypertension must be controlled rigorously with a target of blood pressure less than 130/80 mmHg. In CKD patients with diabetes mellitus or proteinuria greater than 1 gm/24 hrs, BP should preferably be reduced to 125/75 mm Hg.
6. *Adjustment of drug dosage:* The dosage of the drugs is adjusted according the severity of renal failure. Nephrotoxic drugs should be avoided.
7. *Management of anemia:* Erythropoietin (subcutaneous or intravenous) is useful in the management of anemia. Iron stores must be adequate to ensure response to erythropoietin. Intravenous iron (Ferric gluconate or iron sucrose) may be needed to replenish iron stores. Supplementation of vitamin B12 and folic acid is also required. Anemia not responding to erythropoietin is corrected by blood transfusion.
8. *Bleeding complications:* Bleeding can be controlled by the use of desmopressin or estrogen. Regular uses of antacids reduce the incidence of gastrointestinal hemorrhage.
9. *Renal replacement therapy:* This is indicated in ESRD patients. Different forms of replacement therapy are:
 - (a) Peritoneal dialysis
 - (b) Hemodialysis (Fig. 6.3)
 - (c) Renal transplantation

Dialysis is indicated in refractory hyperkalemia, acidosis or volume overload and uremic complications (encephalopathy, pericarditis and seizures).

IMPLICATIONS ON DENTAL PRACTICE

1. Poor oral health is common in patients with chronic renal failure.
2. Elective dental care is avoided in patients with acute renal failure.
3. Elective dental procedures are better tolerated on non-dialysis days.
4. Blood pressure measurement is advised at every visit.
5. Complete blood count, platelet count and bleeding time should preferably be done in uremic patients before dental procedures.
6. Peritoneal dialysis is generally not a contra-indication for dental procedures.
7. Hemorrhagic complications are more common during dental treatment in patients with uremia. This can present as gingival bleed or hematoma formation following periodontal procedures. Hence, appropriate care must be taken and hemostatic agents (tranexamic acid, DDAVP) may be used.
8. Uremic patients are immunocompromised. Hence, these patients need meticulous oral hygiene and frequent check up to maintain oral health to reduce the incidence of infection from dental origin.
9. The incidence of hepatitis B, hepatitis C and HIV is more in patients on hemodialysis. Screening and appropriate precautions should be exercised.
10. Nephrotoxic drugs must be avoided. The dosage and frequency of administration of medications should be adjusted according to renal function.
11. The dietary restriction in the patients must be kept in mind before giving any advice to the patient regarding diet.
12. Renal patients who are on long-term immunosuppressive therapy should be given prophylactic antibiotics to avoid infections.
13. Cyclosporine therapy may cause gingival hyperplasia.

SELF ASSESSMENT

Multiple Choice Questions

1. **Most common type of acute renal failure is:**
 - A. Renovascular
 - B. Prerenal
 - C. Postrenal
 - D. Drug induced
2. **The nephrotic syndrome in adult is mostly due to:**
 - A. Minimal change disease
 - B. Membranous glomerulonephritis

- C. Goodpasture's disease
D. Wegner's granulomatosis
3. **Dysmorphic red blood cells in urine are found in:**
A. Bladder cancer B. Renal stone
C. Glomerulonephritis D. Prostatism
4. **Prerenal ARF can occur in:**
A. Massive blood loss
B. Severe diarrhea
C. Heart failure
D. All of the above
5. **Which of the following is usually not a feature of minimal change disease:**
A. Massive proteinuria B. Hypoproteinemia
C. Oliguria D. Hyperlipidemia
6. **The most important abnormality in ARF is:**
A. Ketosis
B. Metabolic acidosis
C. Metabolic alkalosis
D. All of the above
7. **The most important electrolyte disturbance in renal failure is:**
A. Hyponatremia B. Hypernatremia
C. Hypokalemia D. **Hyperkalemia**
8. **Amount of protein normally excreted in urine per day is:**
A. 1 gm B. 500 mg
C. **150 mg** D. No protein is present
9. **Microalbuminuria is defined as excretion of following amount of albumin daily:**
A. Less than 30 mg B. **30-300 mg**
C. 300-650 mg D. More than 650 mg
10. **Following is the commonest systemic cause of nephrotic syndrome:**
A. SLE B. Diabetes mellitus
C. Leukemia D. Gold therapy
11. **The commonest cause of death in ARF is:**
A. **Infections** B. Pulmonary edema
C. Hypertension D. Hypocalcemia
12. **Following feature is specific for chronic renal failure:**
A. Hyperkalemia B. Acidosis
C. Oliguria D. **Small contracted kidneys**
13. **Following are the features of CRF:**
A. Hypercalcemia and hyperphosphatemia
B. Hypocalcemia and hypophosphatemia
C. Hypocalcemia and hyperphosphatemia
D. Hypercalcemia and hypophosphatemia
14. **The key component of nephrotic syndrome is:**
A. Anasarca B. Hypoalbuminemia
C. Massive proteinuria D. Hyperlipidemia
15. **The following is not seen in nephritic syndrome:**
A. Hematuria B. Oliguria
C. Proteinuria D. **None of the above**
16. **Insulin is used in the treatment of:**
A. **Hyperkalemia** B. Hypercalcemia
C. Hypernatremia D. Hypermagnesemia
17. **Following are present in case of nephrotic syndrome:**
A. Massive proteinuria B. Hyperlipidemia
C. Renal vein thrombosis D. **All of the above**
18. **Most common cause of nephrotic syndrome in children is:**
A. Minimal change disease
B. Membranous glomerulopathy
C. Diabetes mellitus
D. Amyloidosis

Fill in the Blanks

- The amount of proteinuria in nephrotic syndrome is more than _____ per day.
- The oliguria is defined as urine output less than _____ per day.
- The retention of nitrogenous waste products due to renal insufficiency is called _____.
- Normal level of blood urea is _____.
- Normal serum creatinine level is _____.
- Normal level of serum total protein and albumin is _____.
- In end stage renal disease, the GFR is _____/min

Chapter

7

Nervous System

EXAMINATION OF CRANIAL NERVES

A basic knowledge of cranial nerves is desirable for dental practitioners since patients may present with a number of symptoms pertaining to the oromaxillary area. There are twelve cranial nerves. These are named as follows:

- I Olfactory
- II Optic
- III Oculomotor
- IV Trochlear
- V Trigeminal
- VI Abducent
- VII Facial
- VIII Vestibulocochlear
- IX Glossopharyngeal
- X Vagus
- XI Accessory
- XII Hypoglossal

Olfactory Nerve

Olfactory nerve, which carries the sensation of smell, arises in the olfactory mucous membrane from receptor nerve cells, situated in upper part of the nasal cavity. The nerve fibres pass through the cribriform plate of the ethmoid bone and enter the olfactory bulb in the cranial cavity. Olfactory tract connects the olfactory bulb to the olfactory area of the cerebral cortex.

The presentation of olfactory nerve damage may be lack of smell sense (anosmia) or altered sense of smell (parosmia). Hallucination of smell may occur as aura of temporal lobe seizure. Important causes of anosmia are given in Table 7.1.

TABLE 7.1: Important causes of anosmia

Sinusitis and other local nasal disorders

Head injury

Subfrontal meningioma

Bacterial meningitis

Common substances such as clove oil, peppermint oil, soap or fruits are used to test the sense of smell. Each nostril is tested separately. However, local causes like sinusitis etc. should be ruled out before contributing the dysfunction due to the neurological cause.

Optic Nerve

The optic nerve is the nerve of sight. It is composed of the axons of the cells in the ganglionic layer of the retina. The optic nerve emerges from the back of eyeball and passes through optic canal to enter in the cranial cavity where it unites with the opposite optic nerve to form the optic chiasma. From the optic chiasma fibers run in optic tract. The fibers from the medial half of retina cross the midline in optic chiasma and enter the optic tract of opposite side, while the fibers from the lateral half of each retina pass in the optic tract of the same side. Most of the fibers in the optic tract end by synapsing with nerve cells in lateral geniculate body. The axons of nerve cells in lateral geniculate body pass as the optic radiation to the visual cortex of the cerebral hemisphere.

Few fibres from the optic tract pass to pretectal nucleus and superior colliculus and are concerned with light reflexes.

Following are different components of optic nerve testing:

- Visual acuity
- Visual fields

- Color vision
- Pupillary examination

Visual Acuity

Visual acuity is measured by Snellen's chart. Patient is placed at a distance of 6 meters and asked to read letters on the chart. This tests the distant vision. Normal eye can read down the letters upto seventh line from top (visual acuity 6/6). In case of very poor visual acuity (1/60) when patient is not able to read the top letters from a distance of 1 meter, other tests are performed. These are counting fingers (CF), hand movements (HM), perception of light (PL) or no perception of light (no PL).

Near vision is tested with the help of test types of varying sizes from the ordinary reading distance. The near vision is recorded as the smallest type patient can read comfortably. Smallest print is N5.

Visual Field

Different methods to test visual field are (a) confrontation test using a finger (b) red pin confrontation test (c) perimetry. When the size of test object is 5 mm, the field of vision is 100 degrees laterally, 60 degrees medially, 60 degrees superiorly and 75 degrees inferiorly.

Scotoma is known as a zone of loss of vision in the visual field. The scotoma can be central or paracentral. Important cause of unilateral central scotoma is optic neuritis. Unilateral paracentral scotomas are due to vascular disease (retinal embolism or retinal artery branch occlusion). Bilateral paracentral scotomas are due to vitamin B₁₂ deficiency or alcoholism. Glaucoma can cause comma shaped defect in visual field (paracentral 'arcuate' scotoma) due to damage of nerve fibres in retina or optic nerve.

Loss of vision in one half of visual field is known as Hemianopia. When vision is lost in same half of visual fields in both eyes it is known as homonymous. Right homonymous hemianopia means right half of the field in both eyes is affected. Quadrantanopia is a term used to describe blindness in one quadrant of visual field. Loss of vision in outer half of field (temporal halves) in both eyes is known as bitemporal hemianopia. Binasal hemianopia is loss of visual field in inner or nasal halves of visual fields. Hemianopia may occur due to lesions in optic nerve or chiasma (incongruous) or post-geniculate lesions (congruous). Bitemporal hemianopia is due to damage of fibres in optic chiasma derived from the nasal part of each

eye due to pituitary tumors, inflammatory and traumatic lesions extending out of sella turcica. Binasal hemianopia is due to bilateral lesion of outer aspect of optic chiasma.

Color Vision

Colour vision is tested using Ishihara's pseudoisochromatic plates consisting of multicolor dots outlining certain digits. Most common color vision anomaly is red green deficiency, which is inherited as sex-linked recessive condition. Acquired causes of defective color vision are macular and optic nerve disease.

Pupillary Examination

The pupillary light reflex consists of optic nerve fibres as afferent component and parasympathetic fibres in oculomotor nerve as efferent. The light shown in one eye results in constriction of pupil in the same eye (direct pupillary reaction) as well as in other eye (consensual pupillary reaction). The absence of direct pupillary light reaction but the presence of consensual reaction indicates the lesion of the optic nerve in that eye. This is performed using swinging light test (swinging torch test).

Accommodation Reflex

Pupillary accommodation reflex includes miosis (pupils become smaller) and convergence of the eyes on accommodating for the near objects. This is impaired in the lesions of third nerve and in autonomic neuropathies.

Argyll Robertson pupil: The pupil is small and irregular. Accommodation reflex is present while there is no light reflex. This is seen in neurosyphilis.

Adie pupil (tonic pupil): There is absence or delayed pupillary constriction to light or to accommodation. It is of little clinical significance.

Horner's syndrome: This is due to paralysis of cervical sympathetic nerve. Sympathetic nerve fibres originating in the lower cervical and upper thoracic region of the spinal cord pass through the ophthalmic division of the Fifth cranial nerve to the eye. Sympathetic activity mediates pupillary dilatation and elevation of the upper lid through the contraction of smooth muscles of levator palpebrae superioris. Features of Horner's syndrome are:

- Pupillary constriction
- Slight drooping of the upper lid (ptosis)

- Absence of papillary dilatation in the dark
- Loss of cilio-spinal reflex
- Impaired sweating of the face on the side of lesion

Oculomotor Nerve

The oculomotor nerve after emerging from the anterior surface of mid-brain passes forward between the posterior cerebral and superior cerebellar arteries and continues in the middle cranial fossa in the lateral wall of cavernous sinus. It then passes through superior orbital fissure to enter the orbital cavity. The oculomotor nerve supplies the following muscles of the eye:

- Levator palpebrae superioris
- Superior rectus
- Inferior rectus
- Inferior oblique

It also supplies the ciliary muscles and constrictor pupillae muscles of the iris.

The functions of the oculomotor nerve are

- Lifting the upper eyelid
- Turning the eye upward, downward and medially
- Constriction of pupil
- Accommodation of the eye

The test is done by asking patient to look in different directions. Horizontal and outward movement of eye ball is called abduction whereas horizontal inward movement is termed adduction. Vertical upward movement is termed elevation and downward is termed depression. Superior and inferior recti act as elevator and depressor alone when the eye is in abduction. Inferior and superior oblique act as elevator and depressor respectively when eye is in adduction.

Lower motor neuron lesion (infranuclear lesion) of III, IV and VI cranial nerves leads to weakness of individual or group of muscles. While upper motor neuron lesions (supranuclear lesion) lead to paralysis of conjugate movement of the eye.

The infranuclear lesion of the oculomotor nerve may lead to following abnormalities:

- No movement of eyeball in medial and upward direction, movement is possible in lateral direction and slightly downwards
- Lateral and downward displacement of the eye
- Dilatation of the pupil
- Loss of accommodation reflex of the pupil.

Trochlear Nerve

The trochlear nerve leaves the midbrain and passes forward in middle cranial fossa in the lateral wall of the cavernous sinus. It then passes through superior orbital fissure to enter the orbital cavity. It supplies the superior oblique muscle of the eyeball and helps in turning the eyeball downward and laterally.

Infranuclear lesions of the trochlear nerve may result in impaired downward movement when eyeball is in medial position.

Trigeminal Nerve

Trigeminal nerve is the largest cranial nerve. It has a sensory root and a small motor root. It leaves the anterior side of pons and passes through the posterior cranial fossa to reach the apex of petrous temporal bone in the middle cranial fossa. Here the sensory root forms the trigeminal ganglion. The three divisions namely ophthalmic (V1), maxillary (V2) and mandibular (V3) arise from the ganglion.

- The ophthalmic division is sensory and supplies skin of forehead and scalp as far as the vertex, the conjunctiva of the eye, cornea, the upper eyelid, lacrimal gland, the mucous membranes of the paranasal sinuses, the nasal cavity and the skin of nose as far as the tip.
- The maxillary division is also sensory and it passes through foramen rotundum to supply the skin overlying the maxilla, upper lip, the lower eye lid and its conjunctival surface, the teeth of upper jaw, the maxillary sinus, the palate, and the mucous membrane of the nose.
- The mandibular division is both motor and sensory. These roots pass through the foramen ovale. The sensory fibers supply the skin of the lower part of cheek, the skin over the mandible, the lower lip and the side of the head. It also supplies the teeth of lower jaw, temporomandibular joint, mucous membrane of the cheek, the floor of the mouth and the anterior part of the tongue.

The motor fibers supply:

- Muscles of masticulation
- Mylohyoid muscle
- Anterior belly of the digastric muscle
- Tensor veli palatine of the soft palate
- Tensor tympani of the middle ear

The sensory component of the trigeminal nerve is examined by testing the touch, pain and cold sensations in

the areas supplied. The lesion of the whole trigeminal nerve leads to loss of sensations in the skin and mucous membrane of the face and nasopharynx. Lacrimal and salivary secretions may be diminished. *Trophic ulcers may develop in the mouth, nose and cornea. Lack of salivary secretion may also alter taste perception.*

The motor component of trigeminal nerve is tested by asking the patient to clench the teeth. Normally temporal and masseter muscles become prominent which can be easily palpable. Muscles do not become prominent on the paralyzed side. While opening the mouth, jaw deviates to the paralyzed side. This is because of uninhibited action of lateral pterygoid muscles which push the jaw to the paralyzed side.

Corneal reflex: The corneal reflex may be absent either due to lesion of Vth nerve (afferent component of the reflex) or due to facial nerve (efferent component of the reflex). This is tested by touching the lateral edge of cornea with a fine cotton wisp. Normally this leads to the blinking of the eyes. Alternatively it can be tested by lightly blowing puff of air into each cornea.

Abducens Nerve

The Abducens nerve emerges between the pons and the medulla oblongata and passes forward in the middle cranial fossa along with internal carotid artery through the cavernous sinus. It then passes through superior orbital fissure to enter the orbital cavity.

It supplies the lateral rectus muscle and its function is to turn the eyeball laterally.

Infranuclear lesions of the sixth nerve may result in inability to move eye outward (Figure 7.1) with diplopia (double vision) and convergent squint because of unopposed action of medial rectus.

Facial Nerve

Facial Nerve emerges as two roots between the pons and the medulla oblongata and passes laterally in the posterior cranial fossa with the VIII nerve and enters the internal acoustic meatus in the petrous part of temporal bone. The nerve then enters the facial canal and laterally through the inner ear. It emerges from the canal through the stylomastoid foramen. The nerve then passes through the parotid gland and divides into branches which innervate the muscles of the face and scalp except the *levator palpebrae superioris*. It also supplies the platysma, stylohyoid, posterior belly of the digastric and the stapedius muscle of the middle ear.



FIGURE 7.1: Right VI nerve palsy

The facial nerve controls facial expression, salivation and lacrimation. It also carries the taste sensation from the anterior two-third of the tongue, floor of the mouth and the palate. The taste fibers pass from the lingual nerve (a branch of fifth cranial nerve) into the chorda tympani, to join the facial nerve above stylomastoid foramina. The sensory component of facial nerve carries sensations from the part of the skin of external ear. Examination of the facial nerve is described under 'facial nerve palsy'. Innervation of tongue is given in Table 7.2.

TABLE 7.2: Innervation of the tongue

Sensory	Taste
Anterior two third - V nerve	Anterior two third - VII nerve
Posterior one third - IX nerve	Posterior one third - IX nerve

Taste sensation examination: Patient is asked about the loss or change in the taste. The examination is done by applying solutions one by one on the surface of protruded tongue. The solutions generally used for different taste modalities are sugar (sweet), common salt (salt), citric acid (sour) and quinine (bitter). The patient is asked to indicate the nature of taste before withdrawing tongue inside the mouth. This is to differentiate the impairment of taste sensation in anterior or posterior parts of the tongue.

Ageusia is defined as loss of taste sensation.

Vestibulocochlear Nerve

Vestibulocochlear nerve consists of vestibular and cochlear fibers. These fibers leave the brain between the pons and the medulla oblongata. After crossing the posterior cranial fossa it enters the internal acoustic meatus with the facial

nerve. The vestibular fibers originate from the vestibule and semicircular canals and are concerned with the sense of position and movements of head. The cochlear fibers originate from the cochlea to the internal ear and are concerned with hearing.

Important symptoms of vestibular dysfunction are vertigo, giddiness and unsteadiness. In vertigo, external objects seem to move around the patient. Auditory manifestations include tinnitus, hyperacusis, and auditory hallucination and delusion. Tinnitus means persistent ringing in the ears. In hyperacusis, there is excessive sensitivity to sound and it occurs due to paralysis of stapedius muscle.

Tests of Hearing

- Hearing is tested by producing whispered and conventional voice at varying distances.
- Hearing may also be tested using vibrating tuning fork (512 Hz). Rinne test and Weber test are used to diagnose sensoryneural and conductive hearing loss.
- Audiometry is used to measure the extent of hearing loss and to locate the site of lesion.

Tests of Vestibular Functions

- Eye movements are observed in all gazes to find out nystagmus.
- Positional testing
- Caloric test

Glossopharyngeal Nerve

Glossopharyngeal nerve has both motor and sensory components. It emerges from the anterior surface of the medulla oblongata and passes laterally in the posterior cranial fossa. It then passes through the jugular foramen and descends through the upper part of neck to the tongue. The motor fibers supply the stylopharyngeus muscle, parasympathetic fibers supply the parotid gland, and sensory fibers carry general sensations and taste sensation from the posterior part of the tongue and pharynx. It also innervates carotid body and carotid sinus.

The function of this nerve is to assist swallowing and promoting salivation.

The loss of taste sensation in the posterior part of the tongue occurs if the glossopharyngeal nerve is damaged.

Vagus Nerve

The Vagus Nerve has both motor and sensory components. It emerges from the anterior surface of the medulla oblongata and passes laterally in the posterior cranial fossa. It then passes through the jugular foramen and descends through the neck along with carotid arteries and internal jugular vein within the carotid sheath. After passing through the mediastinum, it pierces the diaphragm and terminates in the abdomen.

Vagus nerve innervates the heart and great vessels, the larynx, trachea, bronchi, lungs and alimentary canal (from pharynx to splenic flexure of colon). It also supplies liver and pancreas and other glands associated with alimentary canal.

Accessory Nerve

The Accessory nerve is a motor nerve and has cranial root and a spinal root. The cranial root emerges from the anterior surface of the medulla oblongata and runs laterally in the posterior cranial fossa. The spinal root arises from nerve cells in the anterior column of the upper five segments of the cervical part of the spinal cord. The nerve ascends through the spinal cord and enters the skull through the foramen magnum and joins the cranial root.

The two roots leave the cranium through the jugular foramen and separate. The cranial root is distributed to muscles of the soft palate, pharynx through branches of vagus nerve (pharyngeal plexus). It also supplies the muscles of larynx except the cricothyroid muscle. The spinal root supplies the trapezius and sternocleidomastoid muscles.

Examination of IX, X, and XI cranial nerves

Due to common innervations of various structures by IX, X and XI cranial nerves, symptoms are generally common and overlapping.

PALATAL PARALYSIS

Unilateral paralysis of palate may not lead to symptoms or symptoms are very minimal. However, bilateral paralysis of palate may produce following:

- There is regurgitation of fluid during swallowing through the nose. This is due to paralysis of soft palate.
- Patient is not able to pronounce the words which require closure of nasopharynx. For example, 'egg' and 'rub' are pronounced as 'eng' and 'rum' respectively.

- Dysphagia (difficulty in swallowing) for solid foods may occur in lower motor neuron lesion and markedly for liquids in upper motor neuron lesion.

Examination of palatal movement is performed by asking the patient to say 'ah'. In case of bilateral paralysis there is no movement of the palate. Palate will remain flat and immobile on the side of paralysis and median raphe is pulled toward the other side in case of unilateral lesion.

Palatal reflex: It may be impaired in the lesion of IX or X cranial nerves. Normally there is reflex contraction of the palate when the back of the pharynx is tickled.

Laryngeal paralysis: The innervation of larynx is given in Table 7.3.

- Bilateral damage of superior laryngeal branch of vagus nerve leads to hoarse voice.
- Unilateral paralysis of recurrent laryngeal nerve (branch of vagus nerve) leads to blurred speech and bovine cough (patient cannot cough clearly and effectively). Bilateral paralysis may cause stridor or respiratory obstruction.

Laryngoscopic examination reveals characteristic features of vocal cord paralysis.

Examination of spinal part of accessory nerve: Trapezius and sternomastoid muscles are tested for the evidence of lesion in the spinal part of the XI nerve. Patient is asked to shrug the shoulder against resistance for testing trapezius muscle. There is weakness in the rotation of chin toward the opposite side in sternomastoid paralysis.

Hypoglossal Nerve

The hypoglossal nerve is a motor nerve. It emerges from the anterior surface of the medulla oblongata and passes laterally in the posterior cranial fossa. It then passes through the hypoglossal canal and runs downward and forward in the neck to the tongue. The hypoglossal nerve innervates the muscles of the tongue except the palatoglossus muscle and controls the movements and shape of the tongue.

TABLE 7.3: Innervation of larynx

Superior laryngeal branch	• Sensory: larynx above the level of vocal cords • Motor: cricothyroid muscle
Recurrent laryngeal branch	• Sensory: larynx below the level of vocal cords • Motor: all laryngeal muscles except cricothyroid

The hypoglossal nerve is examined by asking the patient to protrude the tongue, move the tongue from side to side and to lick each cheek with the tongue. In unilateral lesion, the tongue deviates to the paralyzed side on protrusion. Presence of wasting or fasciculations on the tongue indicates lower motor neuron lesion of the hypoglossal nerve.

EXAMINATION OF A COMATOSE PATIENT

Consciousness: It is a state when patient is aware of self and the environment. The person is able to respond to internal as well external changes/needs.

Sleep: This is a normal variation in consciousness. Patient can be aroused to a normal state of consciousness either spontaneously or by external stimuli from the state of sleep.

Confusional state: It is an altered state of consciousness characterized by clouded alertness and disorientation.

Delerium is a confusional state with marked agitation.

Drowsiness: It is like light sleep, which cannot be easily aroused and the alertness can only be maintained for brief period. Once awakened, patient is usually disoriented and tends to fall asleep despite verbal stimulation.

Stupor: It is a condition where arousal can occur only by vigorous stimuli and arousal is brief and incomplete.

Coma: It is a state of persistent loss of consciousness, where patient is unarousable and unable to respond to external events or inner needs like hunger. The coma indicates disorders of arousal mechanisms in the brain.

Pathophysiology of Coma

Alertness is maintained by proper functioning of reticular activating system (RAS) that contains brain stem, ascending projection to cortex and the cortex. Hence, coma occurs due to:

- Effect on RAS and its projections
- Involvement of both cerebral hemispheres
- Suppression of reticulo-cerebral functions by drugs, toxins or metabolic factors such as hypoglycaemia, uremia, anoxia or hepatic failure.

Unilateral lesions like stroke and tumor generally do not alter consciousness unless they produce mass effect to compress opposite hemisphere or brain stem. Subtentorial lesions produce coma by compressing the brain stem.

Causes: They can be classified into metabolic or diffuse causes and structural causes. Structural causes can be

TABLE 7.4: Important causes of coma**Metabolic causes:**

Hypoglycemia
 Hypoxemia or hypercapnoea
 Hypothermia
 Hypercalcemia
 Hyponatremia or hyponatremia
 Liver failure
 Renal failure
 Diabetic ketoacidosis
 CNS infection/inflammation
 Drugs and toxins
 Sepsis
 Post-ictal state or subclinical seizure
 Hypertensive encephalopathy
 Hypothyroidism

Head injury**Structural lesions:**

Stroke
 Brain abscess
 Brain hematoma
 Hydrocephalus
 Brain tumour
 Venous thrombosis

supratentorial or infratentorial. Important causes of coma are given in Table 7.4.

Diagnosis

Early diagnosis of underlying cause is of vital importance. The diagnosis is based on history, general physical and neurological examination and the laboratory investigations.

History

The following points must be taken to ascertain the cause of coma:

- History of trauma
- Ingestion of drugs or toxins
- Presence of chronic disease of liver, kidneys, heart or lungs
- Diabetes
- Circumstances and preceding symptoms like fever, seizures, headache, vomiting etc.

General Physical Examination

- Presence of fever suggests a systemic infection, meningitis, encephalitis or brain stem fever.
- Hypothermia can be observed in alcoholism, barbiturate poisoning, hypothyroidism, hypoglycaemia and circulatory failure.

- Severe hypertension indicates hypertensive encephalopathy or cerebral hemorrhage.
- Hypotension indicates sepsis, heart failure, internal hemorrhage, alcoholism and barbiturate poisoning.

Neurological Examination

The level of consciousness can be assessed by Glasgow coma scale. The score ranges from 3 (unresponsive) to 15 (normal). (Table 7.5)

The type of neurological presentation may indicate the underlying cause:

- Hemiplegia (weakness of half of body) indicates stroke
- Myoclonus; metabolic encephalopathy
- Asterix (flapping tremor); metabolic encephalopathy; drug intoxication
- Decorticate rigidity (flexion of elbow and wrist with supination of arm); lesion rostral to midbrain
- Decerebrate rigidity (extension of elbow and wrist with pronation of arm); lesion in midbrain or caudal diencephalon
- Unilaterally dilated pupil signifies third cranial nerve damage whereas bilaterally dilated or unresponsive pupils suggests severe midbrain lesion, atropine or methyl alcohol intoxication, and severe anoxic encephalopathy. Bilaterally small or pin point pupils are seen in narcotic overdose, metabolic encephalopathy and thalamic or pontine lesions.

TABLE 7.5: The Glasgow Coma Scale**Eye opening:**

Spontaneous	4
To voice	3
To pain	2
None	1

Best verbal response:

Oriented	5
Confused	4
Inappropriate words	3
Unintelligible sounds	2
None	1

Best motor response:

Obeys command	6
Localizes pain	5
Normal withdrawal	4
Flexor response	3
Extensor response	2
None	1

- g. Conjugate horizontal deviation of eye balls on one side indicates damage to pons on other side or frontal lobe lesion on the same side (eyes look toward a hemispheric lesion and away from a brain stem lesion). Eyes turned down and inward indicates midbrain lesion.
- h. Moving the head from side to side and observing eye movement: movement of eyes opposite to the direction of head movement is called Doll's eye phenomenon (Oculocephalic test). This is suppressed in awake patient while is present in comatose patient with intact brain stem oculomotor function.
- i. Caloric stimulation test: the external auditory canal is irrigated with cold water, which leads to tonic deviation of eye balls to the side of cold water irrigation and nystagmus in the opposite side. Loss of this reflex suggests brain stem damage.
- j. Cheyne stokes breathing is seen in bilateral hemispheric damage whereas Kussmaul's breathing suggests metabolic acidosis.
- k. Presence of herniation must be recognized and managed promptly otherwise it may lead to irreparable brain damage and death. Herniation occurs due to shift in brain tissue due to edema or mass lesions.

Investigations

The following laboratory investigations are helpful in making the diagnosis of a patient with coma.

- a. Complete blood count
- b. Blood sugar, blood urea, serum sodium and potassium, serum calcium
- c. Liver function tests, serum ammonia level, serum ethanol
- d. Thyroid function tests
- e. Imaging of brain (CT, MRI)
- f. Cerebrospinal Fluid (CSF) examination
- g. Electroencephalography (EEG)

Management

Patient should be admitted in intensive care unit. Following are important steps in the management of coma:

General

- a. Maintenance of airway, breathing and circulation
- b. Maintain normal body temperature

- c. Administration of oxygen
- d. Mechanical ventilation

Medications

- a. Administration of IV thiamine 100 mg
- b. Administration of dextrose 50 ml (50%)
- c. Rapid correction of causative factors like hypotension, hypoglycaemia, hypoxia, hypercalcemia, hypercapnea and hyperthermia
- d. IV naloxone for opiate intoxication and flumazenil for benzodiazepine intoxication
- e. Intravenous antibiotics in case of meningitis, septicaemia
- f. Management of increased intracranial pressure (a) IV mannitol, (b) hyperventilation (c) dexamethasone

Operative Management

- a. Surgical evacuation of hematoma or hemorrhage
- b. Shunt for hydrocephalus
- c. Surgery for structural lesion

Brain Death

A state of cessation of cerebral function while somatic function is maintained by artificial means and the heart continues to pump. Three essential elements are required for clinical diagnosis of brain death.

- a. Widespread **cortical** destruction reflected by deep coma and unresponsiveness to all stimuli including visual, auditory or painful stimuli. There is no spontaneous movement or any motor response to noxious stimuli.
 - b. **Brain stem** damage shown by absence of papillary response, loss of oculo-vestibular, corneal or gag reflexes
 - c. Destruction of **medulla** manifested by complete apnoea
- Ancillary diagnostic tests, although not required in most cases may be helpful in diagnosing brain death. These tests are;
- a. EEG: absence of brain derived electrical activity
 - b. Radionuclide brain scan: absence of cerebral blood flow
 - c. Somato-sensory-evoked potentials: absence of cortical and subcortical responses with intact peripheral nerve responses
 - d. Conventional angiography: absent cerebral blood flow

EPILEPSY

Epilepsy is defined as any disorder characterized by recurrent seizures due to chronic underlying process. A single seizure is not epilepsy.

Seizure: This is a transient clinical event due to abnormal paroxysmal excessive discharges from a group of central nervous system neurons. The seizure may present in form of motor, sensory, autonomic or psychic manifestations.

Convulsion: The motor form of seizure is called convulsion.

Classification

Seizures may be of two major types (Table 7.6).

- a. **Partial seizure:** In this condition seizure activity is restricted to localized part of one cerebral hemisphere.
- b. **Generalized seizure:** It involves both cerebral hemispheres diffusely and simultaneously.

Partial Seizures

Simple partial seizures are those where consciousness is preserved whereas it is impaired in complex partial seizures. Partial seizures may spread diffusely throughout the cortex and result in secondary generalized seizures. The manifestations depend on the area of the brain involved and may be in the form of motor, sensory, autonomic or psychic symptoms.

Simple Partial Seizures

Various presentations of simple partial seizures are:

- a. **Focal motor:** The jerky movements of limbs or other parts of the body depending upon the area of the motor cortex involved.
- b. **Somatosensory:** Localized paresthesias if sensory cortex is involved.

- c. **Special sensory:** Involvement of visual, auditory, olfactory and gustatory regions of the brain can lead to light flashes, buzzing, unusual odor (burning rubber) or epigastric sensations.
- d. **Autonomic:** Flushing, sweating and piloerection are due to autonomic involvement.
- e. **Psychic:** Illusions, hallucinations, affective disturbances and *déjà vu* can be the manifestations of simple partial seizures.

Partial motor seizures may present with additional characteristic features like:

- a. Abnormal movements beginning in a restricted region like fingers may spread to involve larger part of the limb. This is known as **Jacksonian march**.
- b. Weakness in the involved parts following motor seizures may occur for minutes or last for hours. This is called **Todd's paralysis**.
- c. Rarely focal motor seizures may continue for hours or days (**epilepsia partialis continua**).

Complex Partial Seizures

Complex partial seizures are characterized by focal seizure activity accompanied with impaired consciousness.

- They arise generally from the temporal or frontal lobes.
- The complex partial seizure begins with the aura (in form of simple partial seizure) and is the same (stereotypic) every time in a particular patient.
- The patient stares blankly (blackouts) and has involuntary automatic behavioral movements (**automatism**) such as chewing, lip smacking, emotion display or running.
- The patients do not respond to visual or verbal commands during the seizure and become drowsy following seizures.
- There is amnesia about the recollection of ictal (seizure) phase.

Primary Generalized Seizures

Tonic-Clonic Seizures

Tonic-clonic seizures (grand mal, primary generalized seizures) may occur abruptly without warning or there may be some premonitory symptoms. There is sudden loss of consciousness and the patient may fall on the ground and sustain injuries.

Tonic phase: The initial phase is characterized by increased muscle tone throughout the body. Cyanosis,

TABLE 7.6: Classification of seizures

1. Partial seizures
Simple partial seizures
Complex partial seizures
Partial seizures with secondary generalization
2. Primarily generalized seizures
Absence (petit mal)
Tonic clonic (grand mal)
Myoclonic
Tonic
Atonic
3. Unclassified seizures
Infantile spasm
Neonatal seizures
4. Status epilepticus

impaired respiration, pooling of secretions in the oral cavity occurs because of tonic contractions of muscles of respiration. Increased tone in laryngeal muscles may cause 'ictal cry'. Tongue bite may occur due to contraction of jaw muscles. Increased sympathetic manifestations such as tachycardia, hypertension and dilatation of the pupils may also occur.

Clonic phase: Tonic phase lasts for a minute and is followed by 'clonic phase' characterized by jerky movements for a few minutes. Subsequently, the patient goes in a state of muscular flaccidity and unresponsiveness (post-ictal phase). There may be bladder or bowel incontinence.

Patient gradually regains consciousness over minutes to hours and may have postictal headache, confusion, muscle ache and fatigue for hours.

The presence of aura suggests partial seizure with secondary generalization.

Absence Seizures (Petit mal)

- This occurs in childhood and ceases after 20 years of age.
- This is characterized by brief lapses of sensorium with loss of postural control.
- These are too subtle to be noticed or referred to as day dreaming.
- The attacks are much briefer (seconds) and more frequent than complex partial seizures.
- There is no post-ictal confusion.
- There may be mild motor movements like blinking of eyes, chewing and hand clonus.
- This is associated with characteristic EEG pattern.

Causes of Seizures

The etiology depends on the age of the patient. A list of causes of seizures is given in Table 7.7.

- Hypoxia, metabolic derangements, congenital defects and birth trauma are important causes of seizures in neonates.
- Febrile seizures are more common in early childhood.
- Head trauma is common cause of seizures in young adults.
- In older people, cerebrovascular disease, tumors and degenerative disorders are important cause of seizures.
- Metabolic and electrolyte imbalance, drugs, and systemic illnesses can cause seizures at any age.

Investigations

- EEG:** This may help in the diagnosis and classification of seizure disorders. Characteristic EEG changes in petit mal seizures (3 Hertz spike and wave activity) differentiate it from complex partial seizures.
- Brain imaging:** CT or MRI is indicated in patients with focal seizures, focal neurological signs or if age of onset is more than 20 years (Fig. 7.2).
- Metabolic:** Serum electrolytes, urea, calcium, blood glucose and liver function tests are done to rule out any metabolic cause.

TABLE 7.7: Causes of seizures

Idiopathic
Genetic
Neurofibroma
Inborn errors of metabolism
Trauma
Birth trauma
Head injury
Metabolic
Alcohol withdrawal
Hypoglycemia
Hypocalcemia
Hyponatremia
Hypoxia
Renal failure
Liver failure
Intracranial space occupying lesions
Tuberculoma
Neurocysticercosis
Brain abscess
Brain tumor
Cerebrovascular diseases
Hemorrhage
Emboli
Infections
Encephalitis
Meningitis
HIV
Toxoplasmosis
Inflammatory
SLE
Sarcoidosis
Degenerative
Alzheimer's disease
Drugs
Lignocaine
Quinolones
Penicillins
Theophylline
Chloroquine, Mefloquine
Psychotropic agents

5. *Other tests:* Blood count, ESR, X-ray chest, CSF examination, ANA and other specific tests may be needed to diagnose infective or inflammatory causes.

Differential Diagnosis

The seizures should be differentiated from transient ischemic attacks (TIA), syncope (see page 100) and pseudoseizures. Pseudoseizures do not occur during sleep and are not associated with unconsciousness, cyanosis, incontinence and tongue bite. There are no post-ictal headache, confusion and focal neurological deficit. EEG changes and high serum prolactin level are features of “true” seizures and not found in pseudoseizures. Pseudoseizures may occur in hysterical reactions and malingering and may be precipitated by emotional stress.

Management

General Precautions

- The patient should refrain from working with dangerous equipments and should avoid swimming, fishing, or cycling.
- He should also avoid any activity such as driving where loss of consciousness is dangerous. The patient should avoid working near fire or at a height.



FIGURE 7.2: CT head showing solitary ring lesion (tuberculoma)

- These precautions should be taken until a good control of seizures is obtained.

Immediate Care of Seizures

- The patient is shifted to a safer place, away from danger (water, fire and machine).
- The patient is turned to the semi-prone position to prevent aspiration.
- The patient should not be left alone until full recovery from seizures as there may be drowsiness and confusion in the post-ictal stage or the seizures may reoccur.
- Nothing should be given by mouth until the patient has fully recovered.
- If convulsions continue for a prolonged duration (>5 minutes) or reoccur without the patient regaining consciousness, hospitalization is a must.

Treatment of the Underlying Condition

If the cause of seizures is a metabolic abnormality (electrolyte abnormality or glucose abnormality) it should be corrected effectively. In case of drug induced seizures, the offending drug should be withdrawn. Structural CNS lesions like brain tumor, abscess and vascular malformation must be treated appropriately. The antiepileptic treatment may not be required once the underlying condition has been well treated.

Avoidance of Precipitating Factors

Seizures may be precipitated by specific trigger factors such as sleep deprivation, alcohol withdrawal, mental stress, physical exhaustion, flickering lights (TV or monitor), music, loud sounds and drug abuse. Such situations should be avoided.

Anti-epileptic Drug Therapy

Drug treatment to prevent seizures is indicated in patients with recurrent seizures of unknown etiology or when known cause of seizures cannot be reversed. Choice of medicine depends on the type of seizures (Tables 7.8 and 7.9).

- Treatment should be initiated with a single drug.
- Dose of the drug should be gradually increased until seizures are controlled or side effects appear. If seizures are not controlled by a single drug, a second drug is added while the first drug is gradually withdrawn. In

most patients, seizures can be controlled by a single drug. In some, a combination of two or more drugs is required to control the seizures.

- If seizures are refractory to medical treatment, the patient may benefit from surgical interventions.

Monitoring

Patient should be monitored for the side effects of the medication. Blood counts, liver function tests, renal function tests are done at regular intervals. Serum level of drug can be measured to guide appropriate dosage and to check the compliance.

Duration of Therapy

Treatment should be continued until there have been no seizures for at least 2-3 years. The dosage of the drugs should be tapered and withdrawn gradually over 2-3 months.

Status Epilepticus

Status epilepticus refers to continuous seizure activity or intermittent seizures with impaired consciousness in interictal period. This can be convulsive (tonic clonic) or nonconvulsive type. Practically seizure activity lasting for more than 5 minutes should be managed as status epilepticus.

Causes

Abrupt drug (antiepileptic drugs) withdrawal or noncompliance is the most common cause of the status epilepticus. Other causes are metabolic disorders, intracranial infections and structural lesions of the brain.

TABLE 7.9: The choice of anti-epileptic drugs

Type of epilepsy	Drug(s) of choice
Partial or secondary generalized seizures	Carbamazepine Phenytoin Valproic acid
Primary generalized seizures	Valproic acid Lamotrigine
Absence seizures	Valproic acid Ethosuximide
Myoclonic	Valproic acid Clonazepam

Management

The status epilepticus should be treated as emergency. The patient should be hospitalized.

1. Airway is maintained and oxygen is administered.
2. Intravenous line is started, a blood sample is withdrawn for laboratory analysis and intravenous dextrose (50% dextrose 25-50 ml) is promptly given. The initial laboratory tests include glucose, electrolytes, calcium, urea, creatinine, liver transaminases and complete blood count.
3. Intravenous diazepam (10 mg) or lorazepam (4 mg) is given slowly in 2 minutes. This can be repeated once after 15 minutes if seizures are not controlled.
4. If seizures continue beyond 30 minutes, intravenous phenytoin (20 mg/kg) at a rate not more than 50 mg/min is given. Alternatively, fosphenytoin can be given. This may be repeated in a dosage of 5-10 mg/kg. One should monitor for cardiac arrhythmia and hypotension.
5. Phenobarbitone (IV 20 mg/kg) at a rate not more than 50 mg/min is given if seizures are still uncontrolled. An additional dose of 5-10 mg/kg may be repeated if needed.

TABLE 7.8: The anti-epileptic drugs

Drug	Daily dose (adult)	Important side effects
Phenytoin	300-400 mg	Ataxia, diplopia, drowsiness, gingival hyperplasia, folate deficiency, osteomalacia
Carbamazepine	200-2000 mg	Ataxia, diplopia, drowsiness, blood dyscrasia, hepatotoxicity, hyponatremia
Valproic acid	400-2000 mg	Rashes, liver damage, drowsiness, blood dyscrasia
Phenobarbital	60-180 mg	Ataxia, confusion, drowsiness, rash,
Clonazepam	1-12 mg	Ataxia, drowsiness
Gabapentin	300-2400 mg	Ataxia, drowsiness
Lamotrigine	25-500 mg	Ataxia, drowsiness, diplopia, rash
Ethosuximide	500-1500 mg	Ataxia, rash, blood dyscrasia
Primidone	250-1000 mg	Ataxia, confusion, drowsiness, rash, megaloblastic anemia
Topiramate	200-600 mg	Ataxia, confusion, drowsiness, renal stones
Tiagabine	32-56 mg	Ataxia, confusion, drowsiness, paraesthesia, depression
Oxycarbazepine	900-2400 mg	As for carbamazepine
Levetiracetam	1000-3000 mg	Sedation, fatigue, psychosis
Zonisamide	200-400 mg	Sedation, headache, psychosis, renal stone

Respiratory depression and hypotension are important adverse events.

6. Uncontrolled seizures are finally managed by general anesthesia and neuromuscular blockade along with ventilatory support. Anesthetic agents used are midazolam, propofol and pentobarbitone.
7. Once the status is controlled, long-term anti-epileptic medications are started.
8. The underlying cause is identified if any and treated accordingly.

Complications

Important complications of status epilepticus are cardio-respiratory dysfunction, hyperthermia, rhabdomyolysis and irreversible neurological damage.

MENINGITIS

Meningitis means inflammatory changes in the meninges and the subarachnoid space. Involvement of brain parenchyma along with meninges is known as meningoencephalitis.

Causes: Meningitis is generally due to infective causes. These could be viruses, bacteria or fungi. The noninfective causes of meningitis are SLE, sarcoidosis, or neoplastic invasion.

Meningitis can be classified into three main categories:

- acute purulent (pyogenic, septic, bacterial)
- viral (aseptic)
- chronic meningitis (tuberculous, fungal).

Acute Bacterial Meningitis

The organisms responsible for acute bacterial meningitis are variable and may differ with the age of the patients. The most common organisms in adults are *S. pneumoniae* and *N. meningitidis*. A list of various organisms responsible for acute meningitis is given in Table 7.10.

Bacterial meningitis is either secondary to bacteremic illness or due to spread of infection from adjacent structures like ear, sinus, nose or fractured skull. Old age, alcoholism, diabetes, splenectomy are additional risk factors. Meningococcal meningitis results from air borne infection that initially leads to nasopharyngeal colonization.

Pathology

- Pia-arachnoid is congested and infiltrated with inflammatory cells.

- The pus may form in the subarachnoid space which may lead to adhesions.
- Adhesions may result in obstruction to CSF flow (hydrocephalus) and cranial nerve damage.
- Cerebral infarction may occur due to obliterative endarteritis.
- Cerebral edema can lead to raised intracranial pressure (ICP) and coma.

Clinical Manifestations

The classical triad is fever, headache and neck rigidity. The majority of patients develop impairment in sensorium. Nausea, vomiting and photophobia are also common. Patients with meningococcal meningitis may develop rash and adrenocortical failure with shock (**Waterhouse Friderichsen syndrome**). Other features are seizures and signs of raised ICP (Table 7.11).

Signs of Meningeal Irritation

- **Neck rigidity** is the pathognomonic sign of meningeal irritation (Figs 7.3A and B). This is demonstrated as an increased resistance to passive flexion of the neck. Other causes of neck rigidity are diseases of the cervical spine, posterior fossa tumor and subarachnoid hemorrhage.
- **Kernig's sign** is elicited when the patient is lying supine. The thigh is flexed on the abdomen with the knee flexed

TABLE 7.10: Causes of bacterial meningitis

Neonates

Gram-negative bacilli
Group B streptococci
Listeria monocytogenes

Children

H. influenzae
S. pneumoniae
N. meningitidis

Adults

S. pneumoniae
N. meningitidis
H. influenzae
Listeria monocytogenes

Elderly

S. pneumoniae
N. meningitidis
Listeria monocytogenes
Gram-negative bacilli

Posttraumatic/post-surgical

S. aureus



FIGURE 7.3A and B: Neck rigidity (A) Supine (B) Lateral position



FIGURE 7.4: Elicitation of Kernig's sign

(Fig. 7.4). Passive extension of the knee causes pain and spasm of the hamstrings. This is due to irritation of meninges in the lower spine.

- **Brudzinski's sign** is elicited when the patient is in a supine position. Passive flexion of the neck results in a spontaneous flexion of hips and knees.

Investigations

CSF examination: Diagnosis is made by lumbar puncture and CSF examination. CT scan is performed to rule out the presence of intracranial space-occupying lesion prior to lumbar puncture particularly in cases with signs of increased ICP, focal neurological signs and seizures. Lumbar puncture in such cases may lead to fatal brain stem herniation.

The CSF is examined for pressure, cell count, protein, glucose, smear stain for bacteria and culture. PCR may rarely be needed. The findings of CSF examination of normal and patients with meningitis are given in Table 7.12.

Blood examination: There is generally leukocytosis. Blood culture may be helpful in the detection of organism.

Treatment

Antimicrobial Treatment

Acute bacterial meningitis is an emergency. Antimicrobial treatment should be started as soon as possible. Prognosis depends on early initiation of antimicrobial treatment. Drugs are given intravenously and for prolonged period. The dosage of antimicrobials used in treatment of bacterial meningitis is given in Table 7.13.

When lumbar puncture cannot be done or attempt to obtain CSF has failed, antimicrobials are given intravenously

TABLE 7.11: Signs of meningitis

Signs of meningeal irritation
Neck rigidity
Kernig's sign
Brudzinski's sign
General
Skin rashes (meningococcal)
Shock (meningococcal)
Signs of raised ICP
Altered consciousness
Papilledema
Sixth nerve palsy
Dilated pupil
Cushing's reflex (bradycardia, hypertension)
Others
Focal neurological signs

TABLE 7.12: CSF examination findings

	Normal	Bacterial	Tubercular	Viral
Color	Clear, water like	Turbid	Straw color, Cobweb formation	Clear
Cell count	0-5 cells/mm ³	200-20,000	100-5,000	10-2,000
Cell type	All lymphocytes	Mainly polymorphs	Mostly lymphocytes (polymorphs predominate in acute stage)	Mostly lymphocytes
Protein	15-45 mg/dL	High	High	Mildly increased
Glucose	50-70% of blood glucose	Low	Low	Normal
Other findings	—	Gram stain	AFB stain, mycobacterial culture positive	Viral isolation, PCR culture positive

after obtaining a blood sample for culture. However, CSF should be obtained as soon as possible.

Empirical treatment is started before the CSF culture and gram stain report is available. It is directed against the most common microorganisms present in a particular age group (Table 7.14). Antimicrobials used against specific organisms in meningitis are given in Table 7.15.

- Ceftriaxone or cefotaxime provides adequate coverage against *S. pneumoniae*, *H. influenzae*, Group *B streptococci* and *N. meningitides*.
- Vancomycin is added to cover cephalosporin resistant *S. pneumoniae*.
- Ampicillin is added to cover *L. monocytogenes* in patients less than 3 months of age or more than 55 years and also in immunocompromised patients.

TABLE 7.13: Drugs and their dosage used in bacterial meningitis

Agent	Daily adult dose, frequency of administration
Ceftriaxone	4 g/d, 12 h
Cefotaxime	12 g/d, 4 h
Ceftazidime	6 g/d, 8 h
Ampicillin	12 g/d, 4h
Penicillin G	20-24 million units/d, 4 h
Vancomycin	2 g/d, 12 h
Cefepime	6 g/d, 8 h
Metronidazole	1.5-2 g/d, 8 h

TABLE 7.14: Empirical choice of antibiotics in bacterial meningitis

Neonates and infants < 3 months	Ampicillin + Ceftriaxone or cefotaxime
Children and adults	Ceftriaxone or cefotaxime + vancomycin
Adults > 55 years	Ampicillin + ceftriaxone or cefotaxime+ vancomycin
Hospital-acquired meningitis	Ampicillin + ceftazidime + vancomycin
Posttraumatic or postsurgical, Immunocompromised patients	

- Ceftazidime is active against *P. aeruginosa* and is preferred over ceftriaxone or cefotaxime in hospital-acquired meningitis, posttraumatic and postsurgical meningitis.

Duration of Therapy

The duration of antibiotic therapy depends on the type of organism. A one-week therapy is given in case of *H. influenzae* and *N. meningitides* infection. *S. pneumoniae* needs antibiotic therapy for two weeks whereas *L. monocytogenes* and gram-negative bacilli infections require a 3 weeks therapy.

Adjunctive Therapy

The use of dexamethasone has decreased morbidity and mortality in pneumococcal meningitis in adults. This is also beneficial in pneumococcal and *H. influenzae* meningitis in infants and children. Ten milligrams of dexamethasone is given 15-20 minutes before or concurrent with the first dose of antibiotic. Therapy is continued at the same dosage every 6 hours for 4 days.

TABLE 7.15: Choice of antibiotics for specific microorganism

Organisms	Antibiotics
1. <i>N. meningitides</i>	
Penicillin sensitive	Penicillin G or ampicillin
Penicillin resistant	Ceftriaxone or cefotaxime
2. <i>S. pneumoniae</i>	
Penicillin sensitive	Penicillin G
Penicillin resistant	Ceftriaxone or cefotaxime
	Vancomycin
3. Gram-negative bacilli	Ceftriaxone or cefotaxime
4. <i>H. influenzae</i>	Ceftriaxone or cefotaxime
5. <i>Pseudomonas spp</i>	Ceftazidime
6. <i>S. arueus</i>	
Methicillin sensitive	Naficillin
Methicillin resistant	Vancomycin
7. <i>Listeria monocytogenes</i>	Ampicillin + gentamicin
8. Anaerobic organisms	Metronidazole

Supportive Therapy

Patients with raised (ICP) are managed in intensive care unit. Intravenous mannitol, hyperventilation and elevation of patient's head to 30 degree are measures to reduce ICP.

Prevention of Meningococcal Infection

- Persons in close contact with patients of meningococcal meningitis should be given oral rifampicin for 2 days (adult: 600 mg 12 hrly, children: 10 mg/kg 12 hrly). Alternatively, adults can be given single dose of ciprofloxacin (750 mg), azithromycin (500 mg) or intramuscular ceftriaxone (250 mg).
- Vaccines are available against meningococci of group A and C and *H. influenza*.

Complications

Sensorineural hearing loss, decreased intellectual functions, seizures and gait disturbances are the main complications of bacterial meningitis.

Viral Meningitis

Involvement of meninges may occur in viral infections which primarily affect other organs. Most common viruses are enteroviruses (coxsackie, echo, polio virus). Others are mumps, measles, influenza, herpes zoster, herpes simplex, HIV, EBV and hepatitis.

- Headache is the main feature.
- Neck rigidity is generally present. Kernig's and Brudzinski's signs are usually absent.
- Seizures and focal neurological deficit generally do not occur.
- Sensorium may be mildly altered.
- Other features are fever, myalgia, malaise, anorexia, vomiting and abdominal pain.
- CSF examination reveals a rise in cell count, mostly lymphocytes, slightly increased protein and normal sugar (Table 7.12).

Treatment is supportive. However, acyclovir may be useful in severe cases of herpes simplex, varicella zoster or EBV meningitis. The course is generally benign and self limiting.

Chronic Meningitis

The patients with chronic meningitis present insidiously. Common pathogens are *Mycobacterium tuberculosis*,

atypical mycobacteria, fungi (*Cryptococcus*, *histoplasma*) and spirochaetes (*Treponema pallidum*).

Tuberculous Meningitis

Meningeal involvement by the *Mycobacterium tuberculosis* may occur through hematogenous spread in cases of primary or post-primary pulmonary disease. The rupture of the brain tubercle into the subarachnoid space may also cause meningitis. The meninges, especially at the base are covered with exudates and tubercles.

Clinical Features

The onset of symptoms is insidious; in some it may be acute. Patients have headache, vomiting, low grade fever and alteration in sensorium. Signs of meningeal irritation are present. Other signs are cranial nerve (oculo-motor) palsies, papilledema and focal neurological deficit. Hydrocephalus is a common complication.

Investigation

- CSF examination
 - The CSF is straw coloured clear but when allowed to stand, a fine clot (spider web) is formed.
 - There is high lymphocyte count, high protein and low glucose.
 - In acute cases, polymorphs may predominate.
 - The AFB stain may be positive. Culture for AFB is positive in 80% cases (Table 7.12).
- CT or MRI brain may show meningeal enhancement or hydrocephalous.

Treatment

Anti-tubercular treatment should be started as soon as possible. Addition of steroids for initial few weeks increases survival and reduces complications. Mortality is high if treatment is delayed.

Complications

Complications like hydrocephalous, focal deficits and cranial nerve palsies may occur. These are seen more commonly in cases where treatment is delayed.

Fungal Meningitis

Fungal meningitis usually occurs in the immunocompromised individuals. Its incidence has gone up due to emergence of

HIV disease. Cryptococcal meningitis is an important opportunistic infection in HIV patients. Biochemical and cytological findings in CSF are similar to that of tuberculous meningitis. The diagnosis can be confirmed by microscopy (India ink preparation for *Cryptococcus*) and specific serological tests. The treatment of cryptococcal meningitis is amphotericin B followed by oral fluconazole.

HEADACHE

Headache is one of the commonest symptoms. In most cases it is benign in nature. Occasionally, it may be the manifestation of serious illness such as brain tumor, intracranial hemorrhage, meningitis and temporal arteritis.

Pathological Basis of Headache

- The distension, stretching and irritation of the pain sensitive intracranial structures such as dura mater and proximal parts of the blood vessels cause headache.
- Brain parenchyma, cerebral ventricles, choroids plexus and arteries over the cerebral convexities are insensitive to pain.
- Inflammation and trauma to cranial and cervical muscles and irritation of cranial and spinal nerves may also lead to headache.

Classification

The headache is classified as primary and secondary (Table 7.16). Secondary headache has underlying cause and the clinical features vary accordingly. The mechanism of primary headache is poorly known and it is probably due to disturbances in serotonergic neurotransmission.

Approach to a Case of Headache

A detailed history is important in the case of headache. The onset, severity and the site of headache are important.

- Acute severe headache may suggest subarachnoid hemorrhage or meningitis.
- Hemicranial headache is a feature of migraine.
- Inquiry should be made about precipitating factors like alcohol, emotional stress, foods or medications.
- Chewing movements may exacerbate the pain due to temporomandibular joint disease, trigeminal neuralgia and glossopharyngeal neuralgia.

Cranial CT or MRI is indicated if the onset of headache is new at middle age, progressive headache that disturbs sleep, or is associated with neurological symptoms or signs.

Migraine

Migraine is a common cause of headache and it is more common in females. It is defined as episodes of unilateral throbbing headache, nausea and vomiting and/or symptoms of neurological dysfunctions. Following are different types of migraine:

- Classical migraine:** Headache is characteristically associated with premonitory sensory, motor or visual symptoms (aura).
- Common migraine:** There is headache without aura. It is most frequent type of migraine.
- Migraine equivalent:** Rarely migraine can present with focal neurological deficit without headache.
- Complicated migraine:** Migraine with transient focal neurological features or that leaves a persistent neurological deficit is called complicated migraine.

TABLE 7.16: Causes of headache

Primary headache syndromes

Migraine
Tension headache
Cluster headache
Benign paroxysmal headache

Secondary headache

Intracranial causes
Subdural and intracerebral hematoma
Subarachnoid hemorrhage
Brain abscess, meningitis, encephalitis
Obstructive hydrocephalus
Vasculitis
Benign intracranial hypertension (pseudotumor cerebri)
Cerebral ischemia or infarction
Extracranial causes
Giant cell arteritis (temporal arteritis)
Sinusitis
Glaucoma
Optic neuritis
Dental diseases
Temporomandibular joint disease
Disease of cervical spine
Systemic causes
Fever, hypoxia, hypercapnia, hypertension, anemia, allergy
Drugs (nitrates)
Depression

The headache typically starts with nonspecific prodromal symptoms like malaise and irritation followed by an aura of a focal neurological event. There is severe throbbing hemicranial headache with nausea, vomiting, photophobia and phonophobia. The patient prefers to be in quiet and darkened room and to go to sleep. The aggravating factors for the headache are menses, red wine, hunger, lack or excess of sleep, alcohol, emotional or physical stress, perfumes and oral contraceptive pills. The deactivators or relieving factors are sleep, pregnancy and triptans. Family history of migraine is often present.

The most common aura is visual, which is in the form of scotomas, hallucinations and fortification spectra. The latter is *pathognomonic* for migraine and is characterized by silvery zig zag lines marching across the visual fields for 20 to 25 minutes.

Variants of Migraine

Basilar migraine: The neurological symptoms are in the form of vertigo, diplopia, and dysarthria. These symptoms are followed by headache and impaired sensorium. Transient blindness may also occur, particularly in young females.

Ophthalmoplegic migraine: This is characterized by pain around the eye, nausea, vomiting and diplopia. Diplopia is due to transient external ophthalmoplegia mainly involving third cranial nerve and rarely sixth nerve.

Facial migraine (carotidynia, lower-half headache): The pain occurs repeatedly in jaw, neck and periorbital area. There may be an associated throbbing headache. The pain is accompanied with tenderness and prominent pulsations of cervical carotid artery. There may be soft tissue swelling over the artery. The condition is more common in older patients and it is generally precipitated by **dental trauma**.

Pathophysiology of Migraine

The mechanism of migraine is multifactorial and complex. It has definite genetic predisposition and is influenced by strong environmental components. Headache and related symptoms are caused by dilatation of the extracranial vessels whereas focal neurological dysfunctions are due to intracranial vasoconstriction. Neurotransmitters such as serotonin and dopamine are implicated in the genesis of the migraine. Other possible mechanisms are activation of dorsal raphe neurons in upper brainstem.

Treatment

Avoidance of Trigger Factors

The trigger factors are identified and best avoided. Alcohol, red wine, chocolate are avoided. Adequate sleep should be ensured. Meals should not be missed.

Treatment of Acute Attack

- Rest in a quite darkened room is helpful.
- Analgesics (aspirin, paracetamol and other NSAIDs) are usually effective if taken early at the onset of headache. This is often combined with dopamine antagonists (metoclopramide, domperidone, prochlorperazine).
- Severe attacks may be treated with triptans (sumatriptan, zolmitriptan) and ergotamine. These are 5-HT (serotonin) agonists and are potent constrictors of extracranial vessels. Triptans can be given orally, subcutaneously or intranasally. The usual dose of sumatriptan is 6 mg subcutaneously repeated at one hour (not more than two doses per day) or 25-100 mg orally which may be repeated at 2 hours (maximum 200 mg/day). These are contraindicated in patients with coronary artery disease, uncontrolled hypertension and migraine with neurological deficits.
- Ergotamine is best effective when taken early during prodromal phase. The initial dose is 1-2 mg orally which may be repeated at every 30 minutes to a maximum of 10 mg per day or 16 mg per week. It is contraindicated in coronary artery disease and pregnancy. This is given orally often in combination with 100 mg caffeine. Dihydroergotamine (DHE) can be given parenterally (1-2 mg IM, SC) at the onset of the headache. Intravenous DHE (1 mg) along with prochlorperazine (5-10 mg) is effective during severe attacks.
- Narcotic analgesics (meperidine) can be used in severe acute attacks. However, this should not be used for chronic headaches to prevent addiction and tolerance.

Prophylactic Therapy

This is necessary if migrainous headache occurs more than 3 times a month. The drugs are taken daily. It may take 2-6 weeks before the drugs are effective. A prolonged treatment is required. Once the response is obtained, the dose can be tapered and withdrawn. The common drugs used for prophylaxis are given in Table 7.17. Most patients respond to low dose amitriptyline, propranolol or valproate.

TABLE 7.17: Drugs used for prophylactic therapy of migraine

Drug	Dosage (daily)
Propranolol	80-320 mg
Amitriptyline	10-50 mg (at night)
Sodium valproate	300-1000 mg
Cyproheptadine	4-16 mg
Methysergide	4-8 mg
Verapamil	80-240 mg

Tension Headache

This is a very common type of headache which is experienced by the majority of the population at some time or other.

- The headache is constant, generalized and may continue for weeks or months.
- It is commonly described as a dull, tight, pressure or band like sensation.
- Pain is less noticeable when the patient is busy and becomes worse at the end of the day.
- Emotional stress, noise and fatigue may precipitate the headache.
- Unlike migraine, there is no photophobia, nausea, vomiting or focal neurological symptoms.

This responds poorly to analgesics. Management of underlying anxiety or depression often helps. Relaxation techniques like massage, hot baths and biofeedback are also helpful.

Cluster Headache (Migrainous Neuralgia)

- It is common in middle-aged males and there is no family history.
- The pain occurs periodically at a specific time of the day, generally in the early morning. The pain is severe and lasts for 30-90 minutes. There is unilateral periorbital pain associated with nasal congestion, lacrimation, rhinorrhea or redness of the eye.
- Horner's syndrome may occur during the attack.
- Patients may remain asymptomatic for weeks or months before another bout of headache (cluster) occurs.
- Alcohol triggers the attack.

Treatment of acute attack with oral drugs is unsatisfactory. Inhalation of 100% oxygen (7 L/min for 15 minutes) is the most effective modality of treatment for acute attack. Subcutaneous sumatriptan (6 mg) may be effective.

Preventive therapy is generally effective. The drugs used are propranolol, amitriptyline, valproate, verapamil, cyproheptadine. Lithium carbonate and prednisolone are also effective.

Headache Due to Raised Intracranial Pressure

Headache due to raised intracranial pressure is worse in the morning upon waking and improves as the patient becomes upright. It may be associated with vomiting. It may worsen on bending forward, coughing or straining. The headache is relieved by simple analgesics.

Intracranial mass lesions and hydrocephalus are main causes of raised intracranial pressure. It is rare for mass lesion to present with severe headache alone; seizure or focal neurological deficits are usual presenting features.

FACIAL PAIN

The pain in the facial area may be due to various causes. Most cases are due to dental problems, trigeminal neuralgia, post-herpetic neuralgia and atypical facial pain. *Provocation of pain due to hot, cold or sweet foods is typical of dental origin. The stimulation by cold stimulus repeatedly induces dental pain whereas pain cannot be repeatedly induced in case of neuralgias.* A list of important causes is given in Table 7.18.

Trigeminal Neuralgia

Trigeminal neuralgia (*tic douloureux*) occurs in middle-aged or elderly patients, more commonly in females. Pain occurs in the facial region supplied by second and third divisions of the trigeminal nerve.

TABLE 7.18: Causes of facial pain

1. Trigeminal neuralgia
2. Atypical facial pain
3. Postherpetic neuralgia
4. Glossopharyngeal neuralgia
5. Other causes:
 - Glaucoma
 - Sinusitis
 - Mastoiditis
 - Dental problems
 - Temporomandibular joint disease
 - Jaw pain due to angina pectoris
 - Giant cell arteritis

Pathogenesis

There is demyelination of the trigeminal nerve root at the point of their entry into the pons. The demyelination is commonly due to the compression mainly by aberrant blood vessels (superior cerebellar artery, tortuous vein). In a few patients it may occur due to multiple sclerosis or brainstem tumor.

Clinical Features

Trigeminal neuralgia is characterized by episodes of sudden, severe, sharp lancinating pain in the area of lips, gums, cheek and chin. The pain lasts for a few seconds or sometimes persists for a minute or two. The pain is so severe that patients may flinch as if a motor tic (hence called tic douloureux). The pain is triggered by chewing, speaking, smiling and touching trigger zones in the trigeminal territory. There is a tendency for remissions and relapses.

Examination does not reveal any neurological sign. Presence of neurological signs and sensory loss suggest multiple sclerosis or posterior fossa tumor.

Investigations

CT and MRI may be needed to detect the underlying cause such as multiple sclerosis and tumors. High resolution MR angiography is helpful in visualizing aberrant vessels.

Treatment

Medical: The drug of choice is carbamazepine (upto 1200 mg/day). If not controlled, other drugs such as phenytoin (300-400 mg daily) or gabapentin (upto 2400 mg/day) are tried. Baclofen (10-20 mg thrice a day) may be used alone or in combination with carbamazepine or phenytoin.

Surgical: This is tried when medical treatment fails.

- The most common therapy is heat lesion of the trigeminal ganglion or nerve (*radiofrequency thermal rhizotomy*).
- Injection of *alcohol* in Meckel's cave has been practiced in the past. Excessive damage of the nerve may lead to loss of sensation in the face (anesthesia dolorosa).
- *Posterior craniotomy and microvascular decompression* is quite effective and generally done in younger patients.
- *Gamma radiosurgery* to the trigeminal root is a recent noninvasive effective tool to treat trigeminal neuralgia.

Atypical Facial Pain

Atypical facial pain occurs usually in middle-aged depressed females. It is a constant pain often burning type which is centered on the maxilla, but may spread to the rest of the face on the affected side. Sometimes it may involve other side of the face, neck or back of the head. Tricyclic antidepressant, analgesics, carbamazepine or phenytoin may be tried. The response to therapy is poor.

Glossopharyngeal Neuralgia

Glossopharyngeal nerve (IX cranial nerve) supplies taste sensation to the posterior one-third of the tongue and sensation to the posterior pharynx (along with X nerve). Glossopharyngeal neuralgia is similar in quality to trigeminal neuralgia but is less common. Pain occurs on one side of the throat in the area of tonsillar fossa which may radiate to the ear. In some the pain may remain localized in the ear. The pain may be initiated by swallowing, coughing, chewing or yawning. Pain may be accompanied by syncope in some cases. Sensory and motor examination is normal. No cause is demonstrable in most cases; however, multiple sclerosis may be responsible for the pain in some.

Medical treatment is the same as for trigeminal neuralgia. Carbamazepine is the drug of choice. Microvascular decompression and partial rhizotomy can be done in refractory cases.

Post-herpetic Neuralgia

Post-herpetic neuralgia is the complication of *herpes zoster* (shingles) infection. It occurs more commonly in elderly patients when first division of trigeminal nerve is involved. Pain is continuous and burning in character throughout the affected territory of trigeminal nerve. Even a light touch may precipitate the pain. Pain may last for years. Simple analgesics may be helpful. Phenytoin (300 mg/day), carbamazepine (upto 1200 mg/day), gabapentin (upto 3200 mg/day) may be tried if analgesics fail to control the pain. Tricyclic agents (amitryptiline) alone or along with phenothiazine are often effective in severe cases.

FACIAL NERVE PALSY

The facial nerve (VII cranial nerve) arises from the pons, passes through the facial canal and exits from the skull through stylomastoid foramina. It then passes through the

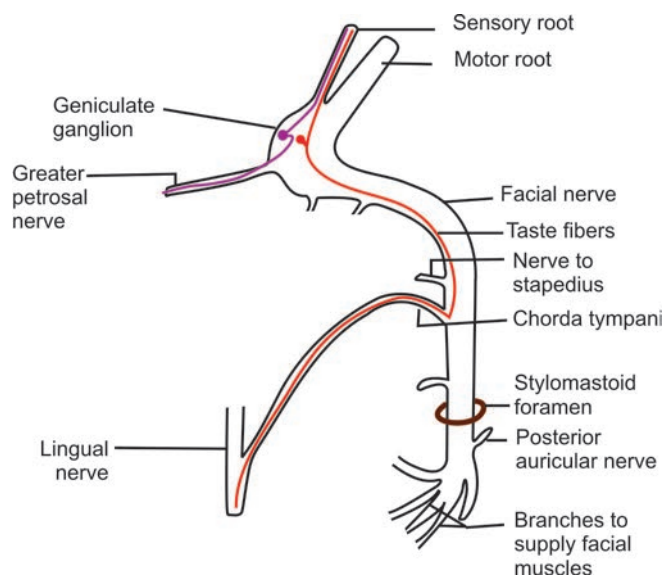


FIGURE 7.5: Pathway of VII cranial nerve

parotid gland and subsequently divides into branches (Fig. 7.5). It provides motor innervation to all muscles of the facial expression and the stapedius. Through its branch, the chorda tympani, it carries taste sensation from the anterior two-third of the tongue. It also carries cutaneous impulses from the anterior wall of the external auditory canal. (see also 'examination of the cranial nerve')

Features of Facial Nerve Palsy

The findings of motor lesion of unilateral facial nerve are as follows:

- There is loss of facial expression at the side of the palsy.
- Furrows of the forehead are absent.
- The eye is more widely open and the eyelids do not close completely on the side of lesion.
- There is drooping of the corner of the mouth. When the patient is asked to smile or show his teeth, the angle of the mouth at the side of the lesion does not move (Fig. 7.6).
- There is loss of nasolabial fold on the affected side (Fig. 7.6).
- On attempted closure of the eye lids, the eye ball on the paralyzed side rolls up (Bell's phenomenon). It is a normal phenomenon which is preserved and can be visualized in facial nerve paralysis.
- Food collects between teeth and the lips.
- Saliva dribbles from the angle of the mouth on the paralyzed side.



FIGURE 7.6: Facial nerve palsy (There is loss of nasolabial fold and drooping of angle of mouth on right side)

- The patient is unable to whistle or blow.
- If the nerve to stapedius is damaged, there is hyperacusis (sensitivity to loud sounds).

There is loss of taste in the anterior two-third of the tongue on the same side. There is no sensory loss over the face.

Causes of Facial Nerve Palsy

The involvement of facial nerve at its origin in the pons or at any site throughout its course may lead to infranuclear facial palsy (lower motor neuron type). The causes of the lesion at different sites are given in Table 7.19. Important causes of supranuclear palsy are cerebral thrombosis, cerebral embolism, cerebral hemorrhage and brain tumor.

Melkersson-Rosenthal syndrome is triad of recurrent facial nerve palsy, recurrent facial edema and plication of tongue.

TABLE 7.19: Causes of infranuclear facial nerve palsy

Pons:	Infarction, tumor, multiple sclerosis
Cerebellopontine angle:	Acoustic neuroma
Temporal bone:	Chronic suppurative otitis media (CSOM), cholesteatoma, Bell's palsy, Ramsay Hunt syndrome, dermoid, carotid body tumor
Outside skull:	Parotid lesions, trauma, lymph node swelling
Others:	Guillain-Barré syndrome, sarcoidosis (uveo-parotid fever, Heerfordt syndrome), diabetes mellitus, leprosy, lyme disease, Melkersson-Rosenthal syndrome

Ramsay Hunt syndrome is a complication of herpes zoster infection of geniculate ganglion associated with facial palsy and vesicular eruption in external auditory canal and pharynx.

Facial diplegia is bilateral facial infranuclear palsy seen in Guillain-Barré syndrome and sarcoidosis.

Localization of the Site of Lesion in Infranuclear Facial Palsy

1. Lesion outside the stylomastoid foramina:

- If the lesion of the facial nerve is outside the stylomastoid foramen, only motor manifestations are present.
- Taste is not involved as chorda tympani joins the facial nerve in the facial canal before the facial nerve emerges from the stylomastoid foramen.
- There is no hyperacusis.

2. Lesion in the temporal bone:

- The lesion of the facial nerve in facial canal near the middle ear may cause hyperacusis in addition to loss of taste.
- Other cranial nerves (auditory and vestibular) are also involved if the lesion is near the internal auditory meatus.

3. Lesion in the pons: In the pontine lesion, sixth cranial nerve may also be involved and there may be hemiplegia of the opposite side (crossed hemiplegia).

If the recovery of motor function is incomplete, aberrant regeneration can cause synkinesis, i.e., movement of one muscle can cause movement of another or all muscles; for example, closure of eye can cause deviation of the angle of the mouth. Lacrimal gland fibers may join with fibers of other muscles so that while eating, tears may also flow (*crocodile tears*). Orbicularis oculi fibers may join with orbicularis oris resulting in the closure of eyelids when the mouth is opened (*jaw winking*).

Supranuclear Facial Palsy

The damage of corticonuclear fibers from motor cortex to facial nucleus may cause supranuclear facial palsy (upper motor neuron type). The infranuclear facial paralysis must be differentiated from supranuclear type of facial palsy. (Table 7.20)

In supranuclear facial palsy, the upper part of the face (frontalis, orbicularis oculi) is involved to a lesser extent than the lower part of the face whereas whole of the face is

TABLE 7.20: Features of supranuclear and infranuclear facial palsy

Supranuclear facial palsy	Infranuclear facial palsy
1. Upper motor neuron lesion	1. Lower motor neuron lesion
2. Upper part of face relatively spared	2. Whole face involved
3. May be associated with aphasia	3. Aphasia not present
4. Usually accompanied with hemiplegia on same side (un-crossed hemiplegia)	4. May be accompanied with hemiplegia on opposite side (crossed hemiplegia)
5. Taste sensation not involved	5. Taste sensation may be involved
6. Emotional facial movement preserved	6. Emotional facial movement not preserved

equally involved in infranuclear facial palsy. This is because upper part of the face is innervated by both motor cortices whereas the lower part of the face is innervated by the opposite hemisphere only.

- Supranuclear facial palsy is often associated with paralysis of the arm and leg of the same side or aphasia.
- The taste sensation in the anterior two-third of the tongue is not involved in supranuclear type.
- Emotional facial movements are also preserved in supranuclear type.
- In supranuclear type, the lesion is on the opposite side of the palsy whereas it is on the same side of the palsy in infranuclear type.
- Important causes of supranuclear palsy are cerebral thrombosis, cerebral embolism, cerebral hemorrhage and brain tumor.

Bell's Palsy

This is the most common type of facial palsy. It is a type of idiopathic infranuclear (lower motor neuron type) facial palsy, the cause of which is not known. The site of involvement is facial canal. Bell's palsy has been associated with reactivation of herpes simplex type I infection, but its causal role is not established.

Clinical Features

Onset is abrupt or subacute and the maximum weakness occurs in 48 hours. Pain around the ear may occur prior to or along with the weakness. Motor manifestations are the

same as described earlier. Bell's phenomenon is present. There is loss of taste in the anterior two-thirds of the tongue on the same side. Hyperacusis occurs on the side of lesion if nerve to stapedius is also involved. There may be diminished salivation and tear secretion.

Investigations

MRI may show swelling and enhancement of geniculate ganglion and facial nerve. EMG has prognostic value.

Treatment

Symptomatic: This includes massage of the facial muscles and protection of the eye during sleep to prevent corneal damage. A lubricating eye drop is used to avoid dryness.

Specific: Administration of glucocorticoids with or without acyclovir improves the outcome. The dose of prednisolone is 60-80 mg daily for 5 days and then tapered over the next 5 days. Acyclovir is given in a dosage of 400 mg 5 times daily for 10 days. Combination of prednisolone with acyclovir is more effective than prednisolone alone.

Prognosis

Over 80% of the patients recover completely in a few weeks time. Patients with complete paralysis have a less favorable prognosis than those with incomplete paralysis.

CEREBROVASCULAR DISEASES

Cerebrovascular diseases mainly include ischemic stroke, hemorrhagic stroke and vascular anomalies such as intracranial aneurysms and arteriovenous malformation. The incidence of cerebrovascular diseases increases with age.

- **Stroke or cerebrovascular accident** are characterized by abrupt onset of symptoms and neurologic deficits attributable to focal vascular cause.
- Ischemia may be transient and there may be recovery from the neurological deficit within few hours but always within 24 hours. This is known as *Transient ischemic attack (TIA)*. TIAs are usually embolic. TIA is a risk factor for future stroke.
- A reversible ischemic neurological deficit (RIND) is similar to TIA persisting for >24 hours but resolves within a week.
- In some patients, the deficit continues to worsen after about 6 hours of the onset (**stroke-in-evolution**).

Causes

Stroke may be ischemic (thrombotic and embolic) or hemorrhagic. Important causes of stroke are given in Table 7.21. Risk factors for stroke are given in Table 7.22.

Clinical Manifestations

There is an acute onset of neurological dysfunction. The presentation is variable and depends upon the vessel involved. Patients generally present with headache, sudden loss of consciousness and seizures.

TABLE 7.21: Causes of stroke

Ischemic stroke	
Thrombosis	
Lacunar infarction	
Large vessel thrombosis	
Embolic	
From artery	
Carotid bifurcation	
Cardioembolic	
Atrial fibrillation	
Myocardial infarction	
Infective endocarditis	
Valvular lesions	
Others	
Hypercoagulable states	
Vasculitis	
Meningitis	
Hemorrhagic	
Hypertension	
Trauma	
Anticoagulant therapy	
Aneurysm	
AV malformation	
Blood dyscrasias	
Brain tumor	

TABLE 7.22: Risk factors for stroke

Hypertension	
Heart diseases (AF, CHF, IE)	
Diabetes mellitus	
Smoking	
Hyperlipidemia	
Age (old age)	
Gender (male)	
Previous vascular event	
(MI, stroke, peripheral embolism)	
AF = Atrial fibrillation, CHF = Congestive heart failure, IE = Infective endocarditis, MI = Myocardial infarction	

- Involvement in the carotid artery area (anterior circulation) leads to hemiplegia (weakness of upper and lower limbs of one side) and cortical sensory loss on the opposite side of the involved hemisphere.
- Aphasia (loss of speech) may occur in left hemispheric involvement.
- Vertebrobasilar strokes (posterior circulation) produce unilateral or bilateral motor sensory deficits of cranial nerves and brain stem signs. Symptoms like vertigo, nausea, vomiting, dizziness, dysarthria, ataxia are more common in posterior circulation strokes.
- Sudden onset of severe headache and neck rigidity suggests subarachnoid hemorrhage.

Investigations

- CT or MRI head is performed to diagnose the type (ischemic or hemorrhagic) and site of lesion (Fig. 7.7).
- Other tests done are carotid Doppler (to diagnose carotid stenosis), echocardiography (to rule out cardiac lesions) and CSF analysis.
- Complete blood count, platelet count, PT, APTT and electrolytes are also performed.
- Tests such as blood sugar, lipid profile and ECG are done to find out the presence of risk factors.
- Specific tests may be needed to diagnose an uncommon underlying cause.

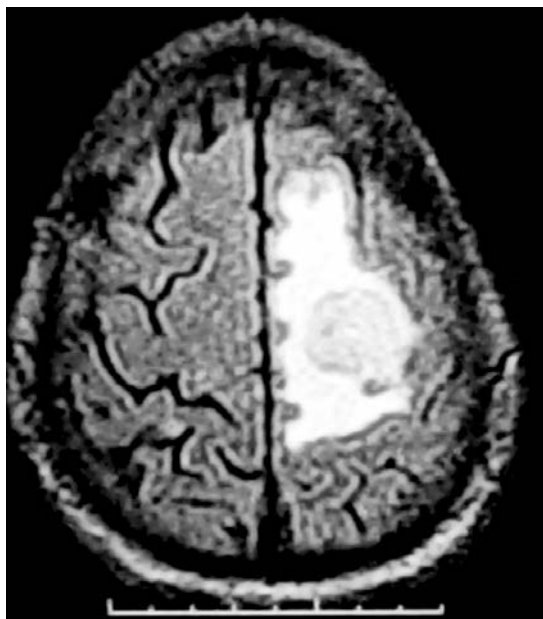


FIGURE 7.7: MRI brain showing intracerebral hemorrhage

Treatment

Following are the main components of management of stroke:

1. **Medical support:** The patient should be urgently hospitalized. The medical management includes control of blood pressure, maintenance of fluid and electrolytes and control of intracranial pressure (with IV mannitol).
2. **Thrombolysis:** This is indicated in early ischemic stroke within 3 hours of onset of symptoms. Recombinant tissue plasminogen activator (rTPA) is used for thrombolysis.
3. Aspirin in doses of 160-325 mg daily is indicated for acute and long-term management of ischemic stroke.
4. Anticoagulation is required to prevent recurrent embolic strokes. Warfarin is used for chronic anticoagulation and the target INR should be 2-3 (in case of prosthetic valve 2.5-3.5).
5. Carotid endarterectomy is indicated in patients with >70% carotid stenosis. It decreases the risk of stroke and death.
6. Modification of risk factors includes the control of blood pressure, blood sugar, serum lipids and cessation of smoking.
7. Physiotherapy and rehabilitation.

IMPLICATIONS ON DENTAL PRACTICE

1. In patients of epilepsy, fixed prosthesis is placed rather than removable appliances because latter may dislodge during the seizure.
2. Gingival growth occurs in patients taking phenytoin.
3. There is no need to increase the dose of anticonvulsant therapy in well-controlled epilepsy prior to dental treatment. Routine use of sedation is not indicated.
4. Seizure may occur during the dental procedures. Hence, one should be aware of the management of patients during seizures.
5. Drugs may precipitate seizures or may interfere with the anticonvulsant drugs. Avoidance or the dose adjustment of the drugs may be required.
6. Accumulation of food debris in the vestibule and the plaque on the teeth may occur on the affected side in cases of facial palsy. Saliva may trickle from the angle of mouth and lead to angular stomatitis.
7. There is loss of taste in the anterior two-third of the tongue on the same side and there may be diminished salivation and tear secretion in Bell's palsy.

8. CSF leak (CSF rhinorrhea) and recurrent bacterial meningitis may occur as a result of maxillofacial fracture involving the cribriform plate of the ethmoid.
9. Patients with maxillofacial injuries should be given prophylactic antibiotics because of risk of bacterial meningitis.
10. Patients with trigeminal neuralgia may develop oral complications as they may be reluctant to brush their teeth.
11. Migrainous neuralgia must be differentiated from pain of dental origin by careful history and examination.
12. Elective dental care should be deferred for 6 months in a patient with stroke since there is a risk of developing another stroke.
13. Stroke should be considered in patients who develop sudden loss of consciousness or neurological deficits during dental surgery.

SELF ASSESSMENT

Multiple Choice Questions

1. Which is true in Bell's palsy:
 - A. Is lower motor neuron lesion
 - B. Is always bilateral
 - C. May be caused by a cerebellopontine angle tumor
 - D. All the above
2. Sodium valproate is an:
 - A. Antiepileptic drug
 - B. Antihypertensive drug
 - C. Antithyroid drug
 - D. Antidiabetic drug
3. Most effective drug for trigeminal neuralgia is:
 - A. Carbamazepine
 - B. Clonazepam
 - C. Phenobarbitone
 - D. Ibuprofen
4. Commonest cause of facial nerve palsy, proximal to its branching in parotid is:
 - A. Bell's palsy
 - B. Ramsay-Hunt syndrome
 - C. Acoustic neuroma
 - D. Brainstem diseases
5. CSF glucose levels are normal in following type of meningitis:
 - A. Tubercular
 - B. Bacterial (pyogenic)
 - C. Viral
 - D. Fungal
6. Marked increase in polymorphs in CSF is a feature of which type of meningitis:
 - A. Pyogenic
 - B. Tubercular
 - C. Viral
 - D. Fungal
7. Which of the following is a cause of upper motor neuron facial palsy:
 - A. Bell's palsy
 - B. Sarcoidosis
 - C. Chronic suppurative otitis media (CSOM)
 - D. Stroke (Cerebrovascular accident)
8. In status epilepticus, the initial drug to be given is:
 - A. Phenytoin
 - B. Diazepam
 - C. Phenobarbitone
 - D. Carbamazepine
9. Drug of choice for partial seizures is:
 - A. Phenytoin
 - B. Clonazepam
 - C. Carbamazepine
 - D. Gabapentin
10. Which one is not true about trigeminal neuralgia:
 - A. Mainly occurs in middle aged and elderly
 - B. Pain lasts only for few seconds
 - C. Ophthalmic division of trigeminal nerve is most commonly involved
 - D. Neurological examination is normal (no motor or sensory deficit)
11. Probable site of lesion in Bell's palsy is:
 - A. Within parotid gland
 - B. Brainstem
 - C. Cerebellopontine angle
 - D. Within facial canal
12. Which one is not true about Bell's palsy:
 - A. Presence of hyperacusis
 - B. Taste sensation is impaired
 - C. Spontaneous recovery is common
 - D. Treatment of choice is carbamazepine and acyclovir
13. Side effects of phenytoin do not include:
 - A. Osteomalacia
 - B. Gum hypertrophy
 - C. Folate deficiency
 - D. Blindness
14. Following is not seen in upper motor neuron facial lesion:
 - A. Loss of nasolabial fold
 - B. Inability to blow or whistle
 - C. Loss of wrinkles over forehead
 - D. Dribbling of saliva on the side of palsy
15. Ramsay-Hunt syndrome is characterized by all except:
 - A. Presence of vesicles over external auditory canal
 - B. Ipsilateral facial nerve palsy
 - C. Cause is herpes simplex of geniculate ganglion
 - D. Lower motor type of lesion
16. Which of the following opportunistic meningeal infection is commonly seen in AIDS:
 - A. Cryptococcus
 - B. Candida
 - C. Aspergillus
 - D. Histoplasma
17. Drugs useful in treatment of status epilepticus are all except:
 - A. Phenytoin
 - B. Phenobarbitone
 - C. Clonazepam
 - D. Midazolam

18. Tongue bite, incontinence of urine or stools, cry and cyanosis are features of:
 A. Petit mal seizures B. Grand mal seizures
 C. Myoclonic seizures D. Partial seizures
19. Jacksonian march is seen in:
 A. Simple partial seizures
 B. Petit mal seizures
 C. Grand mal seizures
 D. Complex partial seizures
20. Neck rigidity, altered sensorium and skin rashes suggest:
 A. Meningococcal meningitis
 B. Pneumococcal meningitis
 C. *H. influenzae* meningitis
 D. Tubercular meningitis
21. Neck rigidity is present in:
 A. Posterior fossa tumor
 B. Meningitis
 C. Subarachnoid hemorrhage
 D. All the above
22. Gum hypertrophy can occur with the use of:
 A. Calcium channel blockers
 B. Phenytoin
 C. Cyclosporine
 D. All of the above
23. The consciousness is not impaired in:
 A. Grand mal seizures
 B. Status epilepticus
 C. Simple partial seizures
 D. Complex partial seizures
24. All are features of common migraine except:
 A. Unilateral headache
 B. Visual symptoms
 C. Vomiting
 D. Premonitory symptoms (Aura)
25. Which one is not true about "Cluster headache":
 A. It is common in males
 B. Pain occurs periodically at a specific time of the day
 C. Severe pain lasting about 24-48 hours
 D. Pain is associated with nasal congestion and lacrimation
26. Which one is not effective in acute attack of migraine:
 A. Sumatriptan
 B. Paracetamol
 C. Amitriptyline
 D. Ergotamine
27. Most common cause of pyogenic meningitis in children is:
 A. *S. pneumoniae*
 B. *N. meningitidis*
 C. *H. influenzae*
 D. *Listeria monocytogenes*

Fill in the Blanks

- Weakness of the involved part following motor seizure is called _____.
- Tic douloureux is _____.
- Anesthesia dolorosa is a complication of _____.
- Most frequent type of migraine is _____.
- The cell count in the normal CSF is _____.
- Spider web is formed in the CSF of patients with _____ meningitis.
- India ink preparation is used to diagnose _____ infection.
- Hyperacusis is due to paralysis of _____ muscle.
- Taste sensation from anterior two-third of the tongue is carried by _____ nerve.
- Complex partial seizures generally arise from _____ lobes.
- Rifampicin is used for the prophylaxis of _____.
- Waterhouse-Friedreichsen syndrome is a complication of _____.
- Most effective modality of treatment in cluster headache is _____.
- Sumatriptan is contraindicated in patients with _____ and _____.
- Recurrent meningitis can result from fracture of _____.

Chapter 8

Endocrine and Metabolic Disorders

THYROID DISORDERS

Anatomy of thyroid gland: The thyroid gland is situated in the neck anterior to the thyroid cartilage. It weighs around 15-20g in the adult. The thyroid gland has two lobes, the right and the left, joined by an isthmus.

Physiology: It secretes mainly thyroxine (T_4) and a small amount of triiodothyronine (T_3).

- T_3 is the active form of the hormone. Most of the T_4 is converted into T_3 in peripheral tissues such as liver, muscles and kidneys.
- Thyroid hormones are carried in plasma, predominantly in the bound form with a plasma protein (thyroid binding globulin, TBG), while only a small amount circulates unbound (free hormone).
- Only the free form can enter cells and exerts its metabolic action.

Regulation of thyroid hormone secretion: The production of thyroid hormones is stimulated by the thyroid stimulating hormone (thyrotropin, TSH), released by the anterior pituitary in response to the hypothalamic thyrotropin releasing hormone (TRH). There is a negative feedback effect of thyroid hormone on the pituitary gland so that TSH secretion is suppressed if the level of thyroid hormone is raised or *vice versa* (Fig. 8.1).

Hypothyroidism

Hypothyroidism is more common in females. It can be due to either:

- Primary disorders of the thyroid gland (primary hypothyroidism) or
- Decreased TSH secretion by the pituitary gland (secondary hypothyroidism)

Causes are given in Table 8.1. The most common cause of hypothyroidism worldwide is iodine deficiency. However,

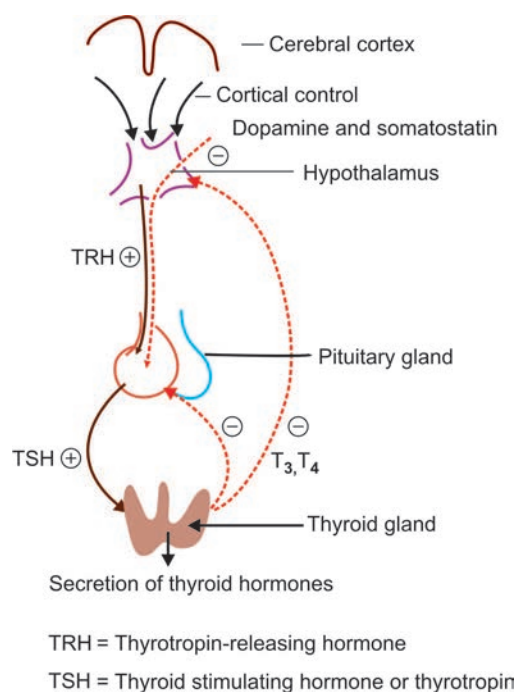


FIGURE 8.1: Feedback control of thyroid hormone

TABLE 8.1: Causes of hypothyroidism

Primary hypothyroidism

- Spontaneous atrophic
- Post-thyroidectomy
- Following ^{131}I administration for hyperthyroidism
- Congenital
- Goitrous
 - Hoshimoto's thyroiditis
 - Iodine deficiency
 - Drug induced (lithium, amiodarone, methimazole)
 - Genetic enzyme defects (dyshormonogenesis)

Secondary hypothyroidism

- Destruction of pituitary gland
 - Post-surgery
 - Post-radiation
 - Tumor
- Disorders of hypothalamus

in areas where iodine deficiency is not present, the important causes are autoimmune thyroiditis and iatrogenic.

Autoimmune hypothyroidism is initially associated with goiter (Hoshimoto's thyroiditis) which later becomes atrophic (atrophic thyroiditis). It is also associated with other autoimmune disorders.

Clinical Features

Table 8.2 shows important symptoms and signs in hypothyroidism.

Common symptoms in hypothyroidism are a dry coarse skin, weakness, tiredness, cold intolerance, puffy face, edema in hands and feet, constipation, weight gain and a hoarse voice (Fig. 8.2). Prolonged hypothyroidism results in the deposition of hydrophilic mucopolysaccharides in tissues. This causes non-pitting edema (myxedema).

TABLE 8.2: Clinical features of hypothyroidism

General

Weakness, tiredness
Cold intolerance
Dry coarse skin
Pallor
Hair loss
Puffy face, hand, and feet
Myxedema
Weight gain, poor appetite
Hypothermia
Goiter
Hoarse voice

Gastrointestinal

Constipation
Large tongue

Reproductive

Menorrhagia
Amenorrhea
Infertility
Galactorrhea

Nervous system

Poor memory
Poor concentration
Carpal tunnel syndrome
Delayed relaxation of deep reflexes

Cardiovascular

Bradycardia
Hypertension
Ischemic heart disease
Pericardial effusion



FIGURE 8.2A: Enlarged tongue in hypothyroidism



FIGURE 8.2B: Facial puffiness and pallor in hypothyroidism

Investigations

- Thyroid function tests:
 - High serum TSH is the earliest and most sensitive indicator of primary hypothyroidism.
 - The serum TSH level is low in secondary hypothyroidism.
 - Serum T_4 is low.
 - Serum T_3 levels is low but is not reliable for the diagnosis of hypothyroidism.
 - High serum TSH and normal serum T_4 indicates subclinical or mild hypothyroidism.
- High titer of antibodies against thyroperoxidase (TPO) and thyroglobulin are found in patients with Hashimoto's thyroiditis and atrophic thyroiditis.

- c. Other findings are anemia (normocytic or macrocytic), high serum cholesterol and triglycerides, increased serum creatine kinase and LDH, hyponatremia, and low voltage ECG.

Treatment

- Hypothyroidism should be treated with oral levothyroxine. The treatment is generally started with low dose (50-100µg daily) and gradually increased.
- In elderly, a smaller starting dose (25 µg) is preferred to avoid cardiac side effects.
- TSH is monitored periodically and the dosage adjustment of thyroxine is made accordingly.
- The ideal goal is to maintain TSH at the lower half of the reference range.
- Treatment is needed lifelong.

Myxedema Coma

This is a rare presentation of hypothyroidism. The clinical features are altered sensorium, hypothermia, hypoglycemia, hypoventilation, hypoxia, hypercapnia, hyponatremia, and hypotension. Convulsions may occur. Coma is induced by infection, cold exposure and systemic illness. This is more commonly seen in the elderly. It has a high mortality rate despite intensive treatment.

Treatment

- The treatment includes intravenous administration of levothyroxine, 400 µg as a loading dose followed by 50-100 µg daily.
- If parenteral T_4 is not available, T_3 can be given “intravenously” in the dosage of 10-25 µg, 8 hourly. T_4 can also be given through nasogastric tube.
- Supportive measures include external warming, use of broad spectrum antibiotics, and correction of hypoglycemia and hyponatremia.
- As there is impaired adrenal reserve in severe hypothyroidism, 50 mg-100 mg hydrocortisone is also given every 6 hourly.

Congenital Hypothyroidism

Congenital hypothyroidism occurs in about 1 in every 4000 newborn. Thyroid gland dysgenesis is the most common cause of neonatal hypothyroidism (85%). Dyshormono-

TABLE 8.3: Clinical features of congenital hypothyroidism

Feeding problems
Prolonged jaundice
Hypotonia
Enlarged tongue
Delayed physical growth and development
Mental retardation
Short stature
Umbilical hernia
Other features as seen in adult patients with hypothyroidism



FIGURE 8.3: Cretinism

genesis and TSH-receptor antibody mediated hypothyroidism are other causes.

Clinical features: The infants appear normal at birth. Only 10% can be diagnosed clinically. Important clinical features are given in Table 8.3. Cretinism is severe hypothyroidism beginning in infancy. This is marked by mental retardation (Fig. 8.3). Permanent neurological damage occurs if treatment is delayed. Congenital malformations are also common in congenital hypothyroidism.

Diagnosis: This is made by the estimation of serum TSH levels and T_4 levels. Serum T_4 is low while TSH is high. In developed countries, routine screening of the TSH levels is performed in the newborn.

Treatment: It includes administration of levothyroxine. The dose is adjusted according to the TSH levels.

Hyperthyroidism

A state of excessive thyroid hormone due to hyperfunction of the thyroid gland is called hyperthyroidism. Thyrotoxicosis is defined as clinical manifestations due to excessive thyroid hormones. The main causes of hyperthyroidism are Graves' disease, toxic multi-nodular goiter and

TABLE 8.4: Causes of thyrotoxicosis**Primary hyperthyroidism**

Graves' disease
 Multinodular goiter
 Toxic adenoma (solitary nodule)
 Subacute thyroiditis (de Quervain's)
 Iodide induced (Jod-Basedow disease)
 Drugs – amiodarone
 Radiographic contrast media
 Excessive iodine ingestion
 Struma ovarii (ovarian teratoma producing thyroid hormone)
 Functioning thyroid carcinoma metastasis

Secondary hyperthyroidism

TSH secreting pituitary tumor
 Pregnancy and trophoblastic tumors

Thyrotoxicosis without hyperthyroidism

Thyrotoxicosis factitia (ingestion of excess thyroid hormone)

toxic adenomas (Figs 8.4A and B). Other causes are given in Table 8.4.

Graves' Disease

Graves' disease (Basedow's disease) is the most common cause of thyrotoxicosis (60-80%).

- It is more common in females and occurs generally between 20-40 years of age.
- Graves' disease is an autoimmune disorder with genetic predisposition. The antibodies (TSH-R antibodies) bind TSH receptors on thyroid follicular cells and stimulate thyroid hormone production and goiter formation.
- It can be accompanied by other autoimmune diseases like pernicious anemia, myasthenia gravis and diabetes mellitus.

**TABLE 8.5:** Clinical features of thyrotoxicosis**Symptoms**

Weight loss with increased appetite
 Heat intolerance and sweating
 Nervousness and restlessness
 Palpitation
 Diarrhea
 Oligomenorrhea, amenorrhea
 Muscle cramps and weakness
 Anginal chest pain

Signs

Tachycardia, atrial fibrillation
 Fine finger tremors
 Moist warm skin
 Goiter
 Hyper-reflexia
 Lid lag and lid retraction
 Clubbing
 Bruit over thyroid gland*
 Ophthalmopathy*
 Dermopathy*

*Signs present in Graves' disease.

- Graves' disease is characterized by diffuse enlargement of the thyroid gland, infiltrative ophthalmopathy (exophthalmos) and pretibial myxedema (dermopathy).

Clinical Features

- Important manifestations are weight loss with increased appetite, sweating, palpitation, tremors, and nervousness. The signs are tachycardia, atrial fibrillation, lid retraction, wide palpebral fissure, and exophthalmos. Other clinical features of thyrotoxicosis are given in Table 8.5.
- Ophthalmopathy is present in 20-50% patients of Graves' disease. It may precede the development of

**FIGURES 8.4A and B:** Enlarged thyroid gland

thyrotoxicosis or may develop after successful treatment of hyperthyroidism in Graves' disease. It usually consists of chemosis, scleral injection, periorbital edema and proptosis. Proptosis may cause corneal drying and damage. In severe cases exophthalmos, diplopia and optic nerve compression may occur.

- c. Dermopathy occurs in about 5% of patients with Graves' disease. Purple or pink patches over anterior and lateral aspect of the leg (pretibial myxedema) are commonly seen.
- d. Thyroid acropachy is unusual feature of Graves' disease and manifests as digital clubbing and swelling of fingers and toes.

Laboratory Findings in Hyperthyroidism

- Serum total and unbound (free) T_3 and T_4 are increased and TSH level is suppressed. Serum TSH is the best initial diagnostic test; normal TSH levels exclude clinical hyperthyroidism. In some cases, only T_3 levels are raised whereas T_4 is normal (T_3 toxicosis).
- TSH-R antibodies levels are increased in about 75 % of cases of Graves' disease.
- ESR may be increased in subacute thyroiditis.
- The uptake of radioactive iodine by thyroid is high in Graves' disease and toxic nodular goiter whereas it is low in subacute thyroiditis.
- Ultrasonography of thyroid gland reveals diffuse enlargement of thyroid gland. It helps in differentiating Graves' disease from nodular goiter.

Treatment

The hyperthyroidism of Graves' disease is treated by antithyroid drugs, radioactive iodine (^{131}I) or subtotal thyroidectomy. The choice of treatment depends on the cause and severity of hyperthyroidism, the patient's age and clinical situation.

- a. **Antithyroid Drugs:** The most commonly used drugs are **carbimazole**, **methimazole** and **propylthiouracil**.
 - These drugs reduce the synthesis of thyroid hormones by inhibiting the iodination of tyrosine. These drugs also reduce the thyroid antibody levels.
 - These are used in young adults and in patients with mild thyrotoxicosis and small goiter.
 - The drugs are given (carbimazole 5-40 mg / day, propylthiouracil 100-200 mg every 6-8 h) for prolonged periods of about 1-2 years. Relapse occurs

in about 50% patients with hyperthyroidism after stopping treatment.

- Rash, fever, arthralgia are common side effects. Agranulocytosis is an uncommon and serious side effect.

Propranolol is used to control adrenergic symptoms (tachycardia, tremor, sweating and anxiety) that occur due to hyperthyroidism of any origin. It has no effect on thyroid hormone synthesis and secretion and the usual dose is 20-40 mg every 6 hours.

- b. **Radioactive Iodine:** (^{131}I); causes progressive destruction of thyroid cells. It can be used as initial treatment or for relapses after "stopping treatment" antithyroid drugs or surgery. This treatment is contraindicated in pregnancy and breast feeding. The majority of patients develop hypothyroidism following radioactive iodine therapy. Hence, a long-term follow-up with measurement of thyroid hormones and TSH is necessary.
- c. **Thyroid Surgery (Subtotal Thyroidectomy):** This is indicated in cases of relapse after antithyroid drugs and in young males with large goiter or severe hyperthyroidism. This is also preferred in pregnant women. Before surgery, the patient is made euthyroid by antithyroid drugs. Potassium iodide is then added 1-2 weeks before surgery. Both drugs are discontinued after surgery. Complications of surgery are recurrent laryngeal nerve palsy and hypoparathyroidism. However, recurrence of hyperthyroidism or development of hypothyroidism may occur.

Thyroid Crisis

This is a severe form of thyrotoxicosis which can occur during stressful illnesses, thyroid surgery or radioactive iodine administration. It manifests as delirium, tachycardia, vomiting, diarrhea, and high fever. In elderly, heart failure may occur. The mortality rate is high.

Treatment

- The patients should be rehydrated and given broad spectrum antibiotics.
- Propranolol is given orally or intravenously.
- Iodate sodium (500 mg orally per day) is very effective in bringing the T_3 levels to normal. This is a radiographic contrast medium which inhibits the release of thyroid hormones and peripheral conversion of T_4 to T_3 .

- The patient is also given carbimazole 40-60 mg daily which is continued at a maintenance dose.

CALCIUM METABOLISM

The total amount of calcium is about 2 % of the body weight. Most of it (99%) is in the bones.

- The normal total serum calcium level is 9-10.5 mg/dL (2.2-2.6 mmol/L). Half of this is present in free form (ionized calcium) and the remainder is bound with proteins mainly albumin. The total serum calcium level is low in conditions in which hypoalbuminemia exists, however, free calcium level is normal.
- The ionized calcium is responsible for the physiological functions of the calcium such as nerve function and muscle contraction.

Regulation of Calcium Metabolism

The calcium metabolism is regulated chiefly by the parathyroid hormone (parathormone) and vitamin D.

Parathyroid hormone: Serum calcium level is principal regulator of parathyroid hormone release. Low serum calcium level stimulates parathyroid hormone secretion. Parathyroid hormone maintains serum calcium level by the following mechanisms:

- a. It promotes resorption of calcium from bones
- b. It promotes resorption of calcium from renal tubules
- c. It stimulates the synthesis of 1, 25-dihydroxy-cholecalciferol by the kidneys and thus indirectly promotes the absorption of calcium from the intestine.

Vitamin D: It enhances the absorption of calcium and phosphate from the gut.

Hypocalcemia

Hypocalcemia is defined as serum calcium level below 9 mg/dL with normal serum albumin level or an ionized calcium of less than 4.2 mg/dL. The important causes of hypocalcemia are chronic renal failure, vitamin D deficiency, and hypoparathyroidism. The causes of hypocalcemia are given in Table 8.6.

- Calcium chelators (citrates) present in the transfused blood may lead to hypocalcemia following multiple blood transfusions.

TABLE 8.6: Causes of hypocalcemia

Chronic renal failure
Vitamin D deficiency
Decreased intake
Decreased exposure to sun light
Malabsorption
Decreased production of active forms
Hypoparathyroidism
Pseudohypoparathyroidism
Multiple blood transfusions
Hyperphosphatemia
Hypomagnesemia
Acute pancreatitis
Low serum albumin (free calcium is normal)

- Hypomagnesemia causes decreased secretion of PTH and inhibits the action of vitamin D and PTH on bones leading to hypocalcemia.
- Pseudohypoparathyroidism is characterized by tissue resistance to the action of parathyroid hormone with normal or increased secretion of parathyroid hormone.

Manifestations

Hypocalcemia causes increased excitability of peripheral nerves leading to tetany. Triad of manifestations of hypocalcemia is carpopedal spasm, convulsions and laryngeal spasm. Other features are muscle spasm, perioral and limb parasthesia.

In carpopedal spasm (Fig.8.5), there is flexion of metacarpophalangeal joints, extension of interphalangeal joints of fingers and thumb and apposition of thumb (**main d' accoucheur**).

Prolonged hypocalcemia as in hypoparathyroidism may cause cataract, basal ganglia calcification, raised intracranial pressure, papilledema, and psychosis.

Tetany which is not obvious (*Latent tetany*) can be detected by eliciting **Trousseau's sign** and **Chvostek's sign**.



FIGURE 8.5: Elicitation of Trousseau's sign

- Trousseau's sign is the appearance of carpal spasm within 3 minutes when sphygmomanometer cuff on the upper arm is inflated more than systolic blood pressure (Fig. 8.5).
- Chvostek's sign: Contraction of facial muscles in response to tapping over the branches of facial nerve as they emerge from the parotid gland is called Chvostek's sign.

Investigations

- Serum calcium is low.
- Total serum calcium is normal but ionized calcium is low in alkalosis. In hypoalbuminemia, total serum calcium is low but ionized calcium is normal (Table 8.7).
- Serum phosphorus is elevated in most of the causes of hypocalcemia except in vitamin D deficiency where it is low.
- Serum parathyroid hormone level is elevated except in hypoparathyroidism and magnesium deficiency (Table 8.7).
- Serum magnesium is measured to rule out hypomagnesemia.
- The ECG may show prolongation of QT interval. Arrhythmias may occur.

Treatment

Treatment of Severe Symptomatic Hypocalcemia: Calcium gluconate 2g, equivalent to 180 mg elemental calcium (20 ml of 10% calcium gluconate intravenously in 10-15 minutes) should be given. This should be followed by infusion of 60 ml of 10% calcium gluconate in 500 ml dextrose water slowly in 4-6 hours. Subsequently the infusion rate should be adjusted to maintain serum calcium

TABLE 8.7: Laboratory findings in hypocalcemia

	Total serum calcium	Ionized serum calcium	Serum phosphate	Serum PTH
Hypoalbuminemia	Low	Normal	Normal	Normal
Alkalosis	Normal	Low	Normal	Normal
Osteomalacia (vit D deficiency)	Low	Low	Low	High
Chronic renal failure	Low	Low	High	High
Hypoparathyroidism	Low	Low	High	Low
Pseudohypoparathyroidism	Low	Low	High	High

level between 8-9 mg/dL. Magnesium should also be corrected if low.

The underlying cause should be treated and long term therapy started.

Long-term Treatment:

- Oral calcium supplement is given in the dosage of 1-2g elemental calcium daily. The preferred salt is calcium carbonate which is the least expensive and is well tolerated. The calcium when given with food is well absorbed. The goal is to maintain serum calcium level between 8-8.5 mg/dL. At this level, the symptoms of hypocalcemia are avoided and the chance of hypercalciuria is minimal.
- The vitamin D is supplemented in the dosage of 400-1000 units per day. The dose of active form of vitamin D (1,25-dihydroxycholecalciferol, calcitriol) is 0.25-0.5 µg daily.

Hypercalcemia

Hypercalcemia is defined as serum calcium level above 10.5 mg/dL with normal serum albumin or an ionized calcium of more than 5.2 mg/dL. Important causes of hypercalcemia are given in Table 8.8.

- Primary hyperparathyroidism and malignancy account for 90 % of all the case of hypercalcemia.
- Hypercalcemia in malignancy may occur due to (a) metastasis in bone, (b) increased bone resorption due to increased osteoclast activating factor (OAF), and (c) production of PTH related peptide (PTHrP).
- Sarcoidosis may cause hypercalcemia by increased production of vitamin D₃ by granulomatous tissue.

TABLE 8.8: Important causes of hypercalcemia

Primary hyperparathyroidism

Adenoma
Hyperplasia
Carcinoma

Malignancy

Tumors producing PTH related proteins (malignancy of lung, ovary, kidney)
Hematological malignancies (Myeloma, lymphoma, leukemia)

Other causes

Sarcoidosis
Vitamin D excess
Hyperthyroidism
Lithium and thiazide use
Milk alkali syndrome
Immobilization

- Increased bone turn over in hyperthyroidism may lead to hypercalcemia.
- Prolonged immobilization may cause hypercalcemia due to continuing bone resorption in the absence of normal postural stimuli for bone formation.
- Milk alkali syndrome is due to ingestion of large amount of calcium and absorbable antacids such as milk or calcium carbonate.
- e. Oral glucocorticoids are effective in hypercalcemia due to hematological malignancies, sarcoidosis and vitamin D toxicosis.
- f. Hemodialysis may be needed in cases of renal failure and heart failure.
- g. The underlying cause should be treated. Parathyroidectomy is done in primary hyperthyroidism.

Clinical Features

The symptoms generally occur if serum calcium level is more than 12 mg/dL. Symptoms are more marked if hypercalcemia develops rapidly.

- The gastrointestinal manifestations are anorexia, nausea, vomiting, peptic ulcer and constipation.
- Renal symptoms include polyuria, polydipsia, renal colic and nephrolithiasis.
- Neurological presentations are confusion, depression, drowsiness, stupor and coma.
- Ectopic calcification may occur in soft tissue if serum calcium level is more than 13 mg/dL.

Investigations

Serum calcium and serum PTH levels are measured.

- High PTH level is present in primary hyperparathyroidism while it is low in malignancies where parathyroid related protein (PTHrP) is raised.
- Other tests are done to detect the presence of malignancies if suspected.
- Measurement of thyroid hormones and vitamin D levels may be required.
- ECG findings include short QT interval and ventricular arrhythmias.

Treatment

- a. Restoration of extracellular fluid volume is done with 0.9 % saline. Three to four liters of fluid may be needed in the first 24 hours.
- b. Saline diuresis is induced by giving an infusion of saline which promotes the excretion of calcium. Frusemide may be added in case of renal impairment or heart failure.
- c. Bisphosphonate (60-90 mg pamidronate or zoledronate 4 mg) is given intravenously. It inhibits bone resorption. This is the drug of choice in malignancies.
- d. Calcitonin (IM or SC) may be given particularly when there is renal failure. It inhibits bone resorption and promotes calcium excretion.

PARATHYROID DISORDERS

There are four parathyroid glands situated posterior to the thyroid gland. These produce parathyroid hormone (PTH) which is a peptide comprising of 84 amino acids. The secretion of PTH is regulated by the ionized calcium levels in serum. The important physiological roles of PTH are:

- a. It promotes resorption of calcium from bones
- b. It promotes resorption of calcium from renal tubules
- c. It stimulates the synthesis of 1, 25-dihydroxy-cholecalciferol by the kidneys, thus indirectly promoting the absorption of calcium from the intestine.
- d. It inhibits the absorption of phosphate by the renal tubules.

Hypoparathyroidism

Important causes of hypoparathyroidism are post-thyroidectomy, after parathyroid gland surgery, hemochromatosis and hypomagnesemia.

- Hypoparathyroidism can also be hereditary such as in **Di George syndrome**. This is characterized by hypocalcemia, cardiovascular and facial anomalies.
- Hypoparathyroidism can be a manifestation of polyglandular autoimmunity type I in which other features are candidiasis, Addison's diseases and vitiligo.
- Pseudohypoparathyroidism is a condition in which the production of PTH is normal, but there is tissue resistance to PTH. This is associated with structural abnormalities such as short stature and short fourth metacarpal.
- Pseudohypoparathyroidism is characterized by the presence of structural abnormalities as seen in pseudohypoparathyroidism but normal calcium and PTH levels.

Clinical Features

The manifestations of hypoparathyroidism are due to hypocalcemia (see hypocalcemia).

Investigations

- The serum calcium is low, serum phosphate is high, and serum alkaline phosphatase is normal.
- The PTH levels are low. It is normal or high in pseudohypoparathyroidism.
- The magnesium level is measured to rule out hypomagnesemia.

Treatment

Treatment of hypoparathyroidism includes supplementation of calcium and vitamin D.

Hyperparathyroidism

Primary hyperparathyroidism is caused by hypersecretion of PTH. In majority of cases, this is due to autonomous hypersecretion of PTH. Primary hyperparathyroidism (adenoma or hyperplasia) may be familial and part of multiple endocrine neoplasia (MEN types 1, 2a).

Secondary hyperparathyroidism is characterized by the hypersecretion of PTH due to stimulation by hypocalcemia (Table 8.9). There is hyperplasia of all parathyroid glands.

In **tertiary hyperparathyroidism** hyperplastic parathyroid glands (as in case of secondary hyperparathyroidism) may result in adenoma formation and autonomous PTH secretion.

Clinical Manifestations

Majority of patients may be asymptomatic. However, symptoms are generally due to hypercalcemia (see hypercalcemia). Bone resorption occurs due to excessive PTH activity. This may lead to demineralization, pathological fractures and generalized cystic bone lesions (**osteitis**

fibrosa cystica). This can also present as “**brown tumors**” (cysts of the jaw).

Investigations

- In primary hyperparathyroidism serum PTH level is elevated, serum calcium is high and phosphate is low. Serum alkaline phosphatase is raised if bone disease is present.
- In secondary hyperparathyroidism, PTH level is elevated. Serum calcium is low and phosphate is high.
- Bone X-ray may show demineralization, subperiosteal erosions, resorption of terminal phalanges and loss of lamina dura of the teeth. Skull X-ray may reveal pepper-pot appearance.
- Radioimaging and ultrasound neck are needed to localize and diagnose parathyroid tumors.

Treatment

Parathyroidectomy is recommended for patients with symptomatic hyperparathyroidism. Medical management of hypercalcemia is described elsewhere (see hypercalcemia).

PITUITARY GLAND

Anatomy: The pituitary gland is situated in the sella turcica. The gland is connected to the hypothalamus with pituitary stalk or infundibulum. The pituitary gland has two lobes, anterior and posterior. Portal vessels carry blood from the hypothalamus to the anterior lobe while the posterior lobe receives nerve fibers from the hypothalamus.

Physiology: The anterior lobe of pituitary gland secretes growth hormone (GH), prolactin (PRL), adrenocorticotrophic hormone (ACTH), thyroid stimulating hormone (TSH), follicle stimulating hormone (FSH) and luteinizing hormone (LH). The secretion of these hormones is controlled by the hypothalamus. The hypothalamus stimulates or inhibits the secretion of anterior pituitary hormones through the release of substances in the portal vessels (Fig. 8.6).

Anti-diuretic hormone (ADH) and oxytocin are synthesized in the hypothalamus and transported through the nerve axons to the posterior pituitary.

Hypopituitarism

Hypopituitarism may be caused by hypothalamic dysfunction or pituitary disease. There may be single or multiple hormonal deficiencies.

TABLE 8.9: Types of hyperparathyroidism

Primary hyperparathyroidism
Single adenoma (80%)
Hyperplasia of two or more glands
Carcinoma
Secondary hyperparathyroidism
Chronic renal failure
Malabsorption
Osteomalacia and rickets
Tertiary hyperparathyroidism

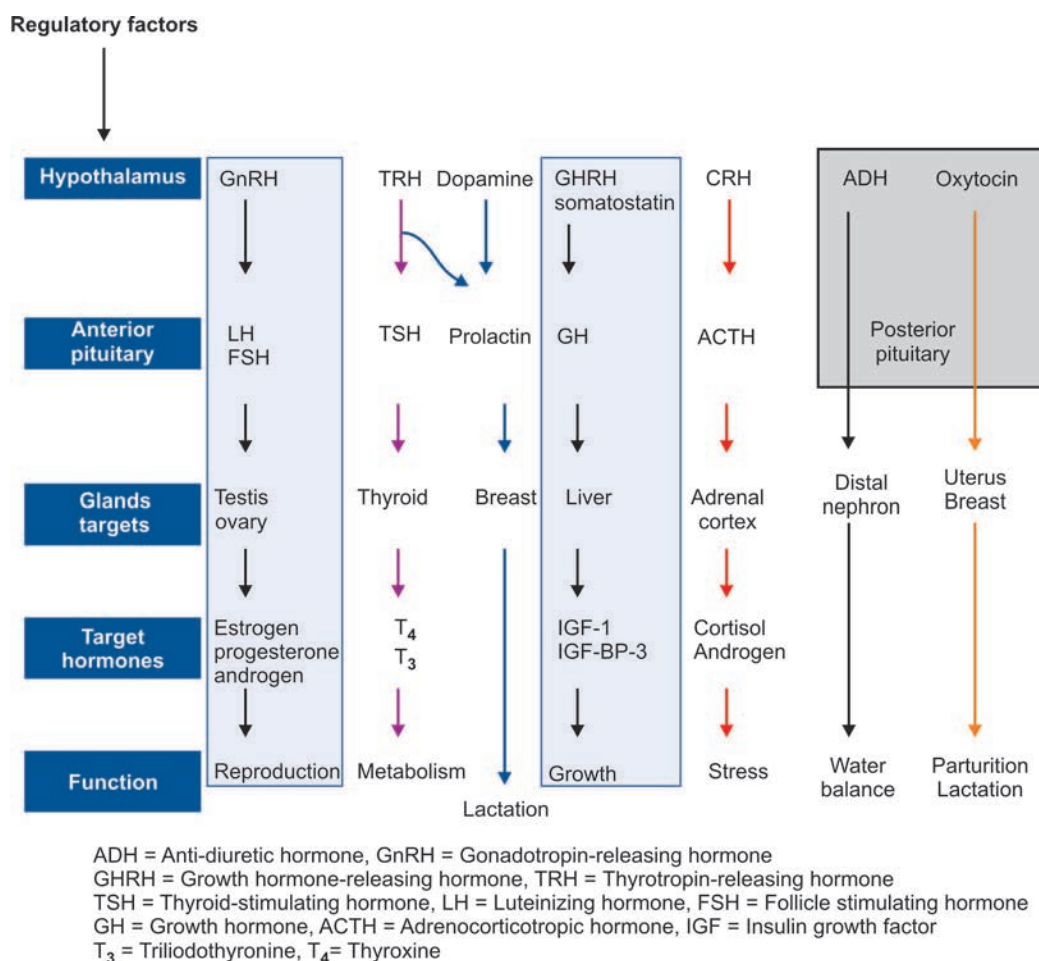


FIGURE 8.6: Anterior pituitary hormones

Important causes of hypopituitarism are pituitary adenoma, granulomas, apoplexy, metastatic carcinoma and brain tumors such as craniopharyngioma and meningioma. Pituitary tumor can be a part of MEN (multiple endrocrinal neoplasia type I).

Other causes are trauma, surgery, radiation, encephalitis, hemochromatosis and stroke.

Clinical Features

The clinical manifestations are variable and depend upon the deficient hormones and the underlying cause. Panhypopituitarism means absence of all anterior pituitary hormones. With progressive lesions of the pituitary, the deficiency of GH occurs first and TSH deficiency occurs in the last. Important clinical features are given in Table 8.10. Other manifestations are due to the underlying cause. For example, tumor may produce headache, visual field defects, and diplopia.

Investigations

- Visualization of the pituitary region is best done with the help of MRI.
- The levels of pituitary hormones are low in the serum.
- The levels of hormones produced by target glands on which pituitary hormones act are also low.
- Stimulation tests are performed to detect the deficiencies of these hormones.

TABLE 8.10: Clinical features of hypopituitarism

Hormone	Clinical features
GH deficiency	Short stature, lethargy, asthenia, small heart, central obesity
Gonadotropins (FSH, LH) deficiency	Hair loss, decreased libido, amenorrhea, infertility, osteopenia
ACTH deficiency	Weakness, fatigue, weight loss, hypotension, pallor
TSH deficiency	Fatigue, weakness, weight gain, hyperlipidemia, cold intolerance

Treatment

- a. *Removal of Underlying Cause:* Trans-sphenoidal surgery of pituitary tumor can correct hypopituitarism. Radiation therapy with gamma knife or X-ray may also be helpful.
- b. *Substitution Therapy:* Substitution therapy with hormones may be needed life long.
 1. Cortisol replacement is done by giving hydrocortisone or prednisolone in case of ACTH deficiency. Mineralocorticoid replacement is not required.
 2. Levothyroxine (25 to 300 µg /day) is given to maintain serum T4 to the upper limit of the reference range. TSH level is not helpful for monitoring the therapy.
 3. Human growth hormone is given subcutaneously to correct its deficiency.
 4. Sex hormone replacement is indicated if there is gonadotropin deficiency.

Acromegaly

Acromegaly is caused by an excess of the growth hormone secreted by pituitary adenoma. In most cases the adenoma is more than 1 cm in diameter (macroadenoma).

Clinical Features

If the growth hormone excess occurs early in life, i.e. before the fusion of epiphyses, tall stature and gigantism will result. In adult life, after closure of epiphyses, acromegaly results.

- The most common symptoms are headache and sweating.
- Head size increases and mandible becomes more prominent. It leads to prognathism and malocclusion.
- Tooth spacing is widened.
- The tongue is hypertrophied (macroglossia).
- Hands are enlarged and feet are widened.
- There may be hypertension, cardiomegaly, and diabetes mellitus.
- The expansion of macroadenoma may lead to local complications including hypopituitarism.

Investigations

- The serum GH is measured during oral glucose tolerance test. In normal subjects, plasma GH is suppressed to below 2 mU/L. Failure of suppression of GH level or paradoxical rise suggests acromegaly.

- The assessment of other pituitary functions is also done.
- Serum insulin like growth factor I (IGF-I) is elevated.
- The plasma glucose is high.
- MRI demonstrates pituitary tumor. X-ray of skull may reveal enlarged sella.

Treatment

- a. *Surgical:* Trans-sphenoidal surgery is the first line of treatment. External radiotherapy (gamma knife, heavy particle radiation) is given to patients not responding to surgery.
- b. *Medical:* Somatostatin analogues (octreotide, lanreotide) are used to treat patients with persistent acromegaly after surgery. Growth hormone receptor antagonist (pegvisomant) can be used to block the effects of GH.

Addison's Disease

Adrenal cortex secretes three major classes of steroids:

- a. Glucocorticoids (cortisol)
- b. Mineralocorticoids (aldosterone)
- c. Adrenal androgens

Adrenal medulla secretes catecholamines. Hypothalamic-pituitary-adrenal axis controls the secretion of glucocorticoids and adrenal androgens through adrenocorticotrophic hormone (ACTH). Mineralocorticoid secretion is under control of rennin-angiotensin-aldosterone system.

Adrenal disorders mainly include (a) hypofunction (adrenal insufficiency), and (b) hyperfunction (Cushing syndrome).

Adrenal insufficiency may result from either due to disease of adrenal glands (primary adrenal failure, Addison's disease) or due to adrenocorticotrophic hormone (ACTH) deficiency caused by disorders of pituitary gland or hypothalamus (secondary adrenal failure). Secondary adrenal failure may also occur due to chronic glucocorticoid therapy that suppresses ACTH secretion. In primary adrenal failure, there is inadequate secretion of both cortisol and aldosterone with increased level of ACTH, whereas in secondary adrenal failure, there is deficiency of cortisol alone with decreased ACTH level.

Etiology

The most common cause of Addison's disease is autoimmune adrenalitis. This may be associated with other

TABLE 8.11: Causes of Addison's disease

Autoimmune
Tuberculosis
Histoplasmosis HIV/AIDS
Bilateral adrenalectomy
Intraadrenal hemorrhage (Waterhouse-Friedrichsen syndrome)
Amyloidosis Hemochromatosis
Metastatic carcinoma

autoimmune disorders like hypothyroidism. Other important causes of tuberculosis, HIV/AIDS, and histoplasmosis (Table 8.11)

Clinical Features

The clinical features are generally non-specific. Important symptoms include anorexia, nausea, vomiting, weight loss, weakness and fatigue. Patient may have postural hypotension and hyponatremia. Although presentation is usually chronic, patient may present acutely as shock triggered by surgery, illness, or injury. Other features are hyperpigmentation due to increased secretion of ACTH (see Fig. 8.8), volume depletion and hyperkalemia.

Diagnosis

Tests for primary adrenal insufficiency:

- Random plasma cortisol level is usually low. Its level may be normal.
- ACTH stimulation test: Cortisol level fails to increase in response to exogenous ACTH.
- ACTH level: This is high in Addison's disease
- Plasma rennin activity is high with low or low-normal aldosterone level.
- There may be hyponatremia and hyperkalemia.

Tests to ascertain the cause of adrenal failure:

- Measurement of antibodies against steroid secreting cells
- Tests for tuberculosis
- CT/MRI to identify metastatic malignancy
- Elisa for HIV

Treatment

a. Immediate treatment:

In case of adrenal crisis, patient should be admitted and managed urgently. Inj. Hydrocortisone 100 mg I.V. is administered 8 hrly. Intravenous fluid (dextrose saline) is infused until hypotension is corrected. Patient is later

maintained on oral prednisolone. Precipitating cause should be treated.

b. Maintenance therapy:

All patients will require cortisol replacement with prednisolone. The dose of prednisolone should be kept to a minimum to avoid the side effects. Most will need aldosterone replacement therapy with fludrocortisone. Usually a dose of 7.5-15 mg/day of prednisolone is required. However, a higher dose may be needed in case of intercurrent stress, severe illness, or surgery. The maintenance dose of fludrocortisone is generally 0.05 to 0.2 mg daily. Salt intake is increased.

c. An identification tag is provided.

Cushing's Disease

Cushing's disease is due to hypersecretion of ACTH by the pituitary gland and it is the most common cause of Cushing's syndrome. **Cushing's syndrome** is defined as a state of cortisol excess due to any cause.

ACTH causes increased production of corticosteroids by the adrenal cortex. Cushing's disease is usually because of benign small pituitary adenoma (micro adenoma).

Clinical Features

The manifestations include central obesity, thin extremities, plethoric "moon face", and "buffalo hump". Other features are purple striae over abdomen and thigh, easy bruisability, acne, hypertension, osteoporosis, muscle wasting, hyperglycemia, impaired wound healing and psychosis. The tendency to infections is increased.

Investigations

- Overnight dexamethasone suppression test, 24 hour urinary free cortisol measurement and plasma ACTH level are some important tests performed to diagnose Cushing's disease.
- Cushing's disease must be differentiated from other causes of hypercortisolism such as exogenous glucocorticoid, adrenal tumor secreting cortisol independent of ACTH, and ectopic production of ACTH by non-pituitary tumors (small lung cell carcinoma).
- MRI is needed to detect pituitary tumor.

Treatment

Patients are treated by trans-sphenoidal resection of pituitary adenoma. Medical treatment with drugs that inhibit corticosteroid synthesis is given before surgery. The drugs are metyrapone, aminoglutethimide and ketoconazole. Patients who fail to respond to pituitary surgery undergo pituitary irradiation or bilateral adrenalectomy.

Diabetes Insipidus

This disorder is characterized by increased thirst and the passage of an increased amount of dilute urine. It is caused by the deficiency of antidiuretic hormone (vasopressin, ADH) or by resistance to the effects of ADH.

Diabetes insipidus can be classified into two types;

- (a) central diabetes insipidus (deficient ADH)
- (b) nephrogenic diabetes insipidus where ADH secretion is normal but renal tubules are unresponsive to ADH.

Causes of diabetes insipidus are given in Table 8.12.

Clinical Manifestations

The patient has excessive thirst. The volume of ingested fluid may reach up to 20 liters a day (polydipsia). The patient passes large amount of dilute urine (polyuria). If the fluid intake is not maintained as in an unconscious patient, the problems of hypernatremia and dehydration may occur.

Investigations

- The plasma osmolality is increased and the urine is dilute (low specific gravity and low osmolality).
- Water deprivation test is used for the diagnosis of diabetes insipidus.

TABLE 8.12: Important causes of diabetes insipidus

Central

Hypothalamic or pituitary stalk lesion
Tumor, head injury, surgery, meningitis, encephalitis

Genetic

Idiopathic

Nephrogenic

Genetic

X-linked recessive

Acquired

Metabolic - hypokalemia, hypercalcemia.

Drugs - demeclocycline, lithium, methicillin

Heavy metals

Others - pyelonephritis, amyloidosis, sickle cell anemia

- In central type of diabetes insipidus, serum ADH is low and the urine osmolality improves on desmopressin (DDAVP) administration.
- On the contrary, serum ADH level is normal and there is no response to DDAVP in nephrogenic diabetes insipidus.
- MRI is useful in locating the hypothalamic or pituitary tumor.

Treatment

- a. DDAVP, an analogue of ADH is the drug of choice in central type of diabetes insipidus. This is usually given intranasally.
- b. Thiazide, amiloride and indomethacin are helpful in nephrogenic diabetes insipidus.

DIABETES MELLITUS

Diabetes mellitus (DM) is a metabolic disorder characterized by hyperglycemia due to absolute or relative deficiency of insulin.

Classification

Two broad categories of DM are type 1 and type 2. Other types include gestational diabetes mellitus, and diabetes due to drugs, pancreatic diseases and genetic syndromes (Table 8.13). The prevalence of DM has risen over past decades. Type 2 DM is more common.

TABLE 8.13: Classification of diabetes mellitus

1. **Type 1 diabetes mellitus**
 - A. Immune mediated
 - B. Idiopathic
2. **Type 2 diabetes mellitus**
3. **Other specific types:**
 - a. Genetic defects of beta cell function
 - b. Genetic defects in insulin action
 - c. Diseases of pancreas: pancreatitis, neoplasia, hemochromatosis, pancreatectomy
 - d. Endocrinopathies: acromegaly, Cushing's syndrome, hyperthyroidism pheochromocytoma, glucagonoma
 - e. Drugs and chemicals: thiazides, corticosteroids, phenytoin, thyroid hormone
 - f. Infections: congenital rubella, coxsackie, cytomegalo virus
 - g. Uncommon forms of immunemediated diabetes
 - h. Genetic syndromes: Down's syndrome, Klinefelter's syndrome, Turner's syndrome
4. **Gestational diabetes mellitus**

Type 1 Diabetes Mellitus

This form of diabetes is immune mediated in 90% of cases (type 1 A). It occurs most commonly in children and young adults.

In this disorder, there is virtually no circulating insulin. Pancreatic beta cells fail to respond to insulinogenic stimuli. Exogenous insulin is therefore required to control hyperglycemia and prevent ketosis.

Type 1A diabetes is a T-cell mediated autoimmune disease and may be associated with other autoimmune diseases like thyroid disease, Addison's disease, vitiligo and pernicious anemia. Autoimmune process is triggered by infectious or environmental stimuli in genetically susceptible individuals. Possible triggers are viral infections (coxsackie, rubella), bovine milk protein and nitrosuria compounds.

These patients have normal beta cell mass at birth which gradually declines over months or years due to immunologic destruction. Pancreatic islets are infiltrated with lymphocytes (insulinitis).

In many patients immunologic markers appear before diabetes manifests clinically. Immunological markers are islet cell antibodies (ICA), insulin antibodies (IAA), antibodies against glutamic acid decarboxylase (GAD) and tyrosine phosphatase (IA-2). These antibodies can be used for screening siblings of the patients.

HLA DR3 and/or HLA DR4 haplotypes are associated with the type 1 DM. Risk of developing DM is increased 10 fold in the relatives of type 1 DM patients.

Type 2 Diabetes Mellitus

Type 2 diabetes mellitus is a disease of middle aged and elderly. It is polygenic and multifactorial in origin involving genetic and environmental factors. There are three basic pathophysiological abnormalities in type 2 DM:

- a. Impaired insulin secretion
- b. Insulin resistance (inability of insulin to act on target tissues mainly liver and muscles)
- c. Increased hepatic glucose production.

Most of the patients are obese and insulin resistant. Insulin resistance increases with age, sedentary lifestyle, and abdominal-visceral obesity.

In addition, there is impairment of beta cell function. Diabetes occurs only when insulin secretion is not adequate to meet the requirement. There is hyperglycemia but

endogenous insulin prevents the development of ketoacidosis. Hyperglycemia worsens insulin resistance and beta cell response to glucose (glucose toxicity) and both improve when glucose is normalized with the treatment.

Many biologic products are secreted by adipocytes that modulate insulin secretion, insulin action, and body weight. Leptin and adiponectin improve insulin sensitivity while tumor necrosis factor alpha and resistin interfere with insulin action.

In the early stages, glucose remains normal, despite insulin resistance, because beta cells compensate by increasing insulin secretion. Gradually there is decline in the insulin secretion and overt hyperglycemia occurs. There is also an increased output of glucose from liver which contributes to fasting hyperglycemia.

Genetic factors are important in the development of type 2 diabetes. The concordance rate in twins is 100%. Several genes are involved. Individuals with family history of diabetes are at increased risk of developing diabetes, if both parents have diabetes, the risk is 40%.

Clinical Features

The classical symptoms are polyuria, thirst (polydipsia), nocturia, and rapid weight loss despite good appetite. These are seen in type 1 DM.

Polyuria is due to osmotic diuresis secondary to hyperglycemia. Thirst is due to hyperosmolality of the plasma. Weight loss is due to depletion of glycogen and triglycerides, increased lipolysis and increased protein catabolism and loss of muscle mass.

Other symptoms are tiredness, fatigue, and irritability. Blurring of vision, frequent infections, and slow healing of wounds may also be present. The infections include bacterial and fungal infections of skin and genitalia.

The patients with type 2 DM may be asymptomatic when they are diagnosed during routine laboratory tests. The patients are usually obese. They may present with the features of chronic neurological, cardiovascular, ophthalmic and renal complications (Table 8.14). Chronic skin infections, generalized pruritus and fungal vaginitis are common.

Females who have unexplained fetal loss or deliver large babies should be screened for DM.

The patients may present with acute complications such as diabetic ketoacidosis (DKA), hyperglycemic hyperosmolar state (HSS) or features of chronic complications.

TABLE 8.14: Clinical features of type 1 and type 2 diabetes mellitus

	Type 1	Type 2
Onset (age)	<30 years	>30 years
Body habitus	Lean	Obese
Ketoacidosis	Common	Rare
Family history	Uncommon	Yes
Autoantibodies	Present	No
Other autoimmune diseases	Yes	No
Complications at diagnosis	No	May be present
Insulin therapy	Must	May be needed
Polyuria, polydipsia, polyphagia, weight loss	Common	Less common

Diagnostic Criteria

Fasting plasma glucose of < 100 mg/dL is considered normal. **Impaired fasting glucose (IFG)** is defined as fasting plasma glucose levels between 100 mg/dL and 126 mg/dL. **Impaired glucose tolerance (IGT)** is defined as plasma glucose levels between 140mg/dL and 200 mg/dL, 2 hours after 75 g glucose load. Patients with IFG or IGT are at substantial risk for developing type 2 DM and cardiovascular disease.

Diagnosis of DM is made if any of the following criteria is present:

- If plasma glucose is > **126 mg/dL** after an overnight fast (this should be confirmed by a repeat test)
- If random plasma glucose is > **200 mg/dL** in presence of symptoms of DM
- If oral glucose tolerance test shows plasma glucose of > **200 mg/dL at 2 hours** after 75 g glucose load.

Investigations

Urine Test

The urine is tested for the presence of glucose, ketones, and protein. Presence of albumin (>30 mg/dL) in the urine suggests nephropathy.

Blood Tests

The plasma glucose tests are performed for the diagnosis of DM (see above). The fasting plasma glucose is done after overnight fast (at least 8 hours fast). The random test is defined as without regard to time since the last meal.

Oral glucose tolerance test (OGTT) is performed by giving 75 g glucose dissolved in water and measuring plasma

glucose after 2 hours. The persons should be on unrestricted carbohydrate diet for 3 days before the test.

The random blood sugar is performed first. If the random plasma glucose level is increased but not diagnostic (<200 mg/dL), fasting plasma glucose is measured. When the fasting plasma glucose level is <126 mg/dL, OGTT is performed.

The criteria for the diagnosis of DM in pregnancy (gestational diabetes) are different and more stringent.

Glycated Hemoglobin (HbA1) Measurement

HbA1 is formed by the non-enzymatic condensation of glucose molecules with free amino groups of hemoglobin. It provides an accurate measure of glucose control over a period of 8-12 weeks. The major form of glycated Hb is HbA_{1c} which normally comprises of 2-4 % of total Hb. The measurement of HbA_{1c} should be made at 3-4 months interval and the therapy is adjusted accordingly. However, this is not used for the diagnosis of the DM.

Serum Fructosamine Measurement

Serum fructosamine is formed by non-enzymatic glycation of serum proteins, mainly albumin. Its level reflects the glycemic control for the preceding 2 weeks. This is particularly useful for the assessment of the glycemic control in diabetic women.

Self Monitoring of Blood Glucose

Glucometers are available for the measurement of capillary blood glucose. The test can be performed by the patients themselves at home.

Continuous Blood Glucose Monitoring System

The blood glucose can be monitored continuously throughout 72 hours with the help of a subcutaneously placed sensor. This is useful in the detection of asymptomatic hypoglycemia.

Other Tests

Serum lipids are routinely measured. Patients with type 2 DM may have dyslipidemia characterized by high triglycerides, low HDL cholesterol, and presence of small dense LDL particles. These patients are more susceptible to atherosclerosis.

Type 1 diabetics have raised triglycerides and LDL cholesterol but normal HDL cholesterol.

Blood urea, serum creatinine, electrolytes, liver function tests are also performed.

Complications of Diabetes

The complications of DM may be classified as acute or chronic. The complications are listed in Table 8.15.

Mechanisms of diabetic complications are complex. However, increased metabolism of glucose to sorbitol via the polyol pathway, glycation of structural proteins and deposition of advanced glycation end products in various tissues, and free radical mediated damage are some of the key pathogenetic mechanisms. Hyperglycemia leads to increased formation of diacylglycerol and activation of

protein kinase C. Various functional abnormalities in diabetes have been linked to the protein kinase C activity.

Diabetic Ketoacidosis

This is a serious complication seen more frequently in type 1 DM and less frequently in type 2 DM. (Diabetic Ketoacidosis—DKA) is caused by insulin deficiency often in association with stress and activation of counter-regulatory hormones. The stress conditions include infections, trauma, myocardial infarction and surgery.

The biochemical abnormalities are hyperglycemia, ketosis and metabolic acidosis. Hyperglycemia leads to osmotic diuresis, dehydration and electrolyte loss. Insulin deficiency and elevated glucagon, growth hormone and catecholamines lead to increased lipolysis from adipose tissue and increased synthesis of ketone bodies in the liver.

Clinical Features

The usual clinical manifestations are nausea, vomiting and abdominal pain. Patients may have respiratory distress, shock, confusion and coma. Examination reveals the presence of dehydration, rapid and deep breathing (Kussmaul's breathing), and fruity breath odor. There may be tachycardia and hypotension.

Investigations

Plasma glucose is very high. Arterial blood gas measurement shows metabolic acidosis. Urine examination reveals positive ketone test. Hyponatremia, hyperkalemia, azotemia and hyperosmolality are the other findings. Urine culture, blood culture, blood counts and X-ray of chest should be done to detect the evidence of infections. ECG is performed to rule out cardiac events.

Management

This is the medical emergency. The patient is hospitalized. The essential components of treatment are;

1. Intravenous saline is initially given to correct dehydration. Once blood glucose level falls to 250 mg/dL, the fluid is changed to 5% dextrose solution.
2. Intravenous infusion of regular insulin is administered until the ketoacidosis is corrected.
3. Potassium and phosphate are replaced as needed.
4. Sodium bicarbonate is generally not required. However, it is given if blood pH is less than 7.0.
5. Antibiotics are given to treat infections.

TABLE 8.15: Complications of diabetes mellitus

Acute complications

- Diabetic ketoacidosis (DKA)
- Hyperglycemic hyperosmolar state (HHS)

Chronic complications

Microvascular

- Retinopathy
- Neuropathy
- Nephropathy

Macrovascular

- Coronary artery disease
- Cerebrovascular disease
- Peripheral vascular disease

Non-vascular

- Gastrointestinal (gastroparesis, diarrhea)
- Genitourinary
- Dermatologic
- Infectious
- Cataract and glaucoma

Miscellaneous (multiple etiology)

- Diabetic foot (Fig. 8.7)
- Erectile dysfunction



FIGURE 8.7: Diabetic foot

6. The electrolytes, blood glucose, blood urea, serum creatinine, arterial blood gases are periodically monitored.

Hyperglycemic Hyperosmolar State (Non-ketotic Hyperosmolar Diabetic Coma, HHS)

This is characterized by severe hyperglycemia, hyperosmolality and dehydration without ketoacidosis. Ketoacidosis is generally absent because residual insulin secretion is adequate to inhibit lipolysis and ketogenesis.

It is a complication of type 2 DM. The precipitating factors include stress, infections, drug non-compliance, stroke and alcohol. The onset is insidious. The usual symptoms are polyuria, weight loss, drowsiness and altered sensorium. Severe dehydration, hypotension and tachycardia are the usual clinical findings. Thromboembolic complications are common.

Laboratory findings include marked hyperglycemia (>600 mg/dL), plasma osmolality greater than 350 mOsm/L, absence of ketonemia, pH >7.3 and prerenal azotemia. Lactic acidosis may occur in some patients.

Treatment includes fluid replacement, insulin therapy, and management of electrolytes especially potassium and phosphates. The requirement for insulin is less in HHS than in DKA.

Chronic Complications

Ocular Complications

These are important causes of blindness in the diabetics in developed countries. Diabetic retinopathy is an important ocular complication of DM. This generally develops after 10 years of the disease and almost all patients have retinopathy after 20 years of the disease. Duration of the disease and degree of the glycemic control are important predictors of this complication.

The diabetic retinopathy has two stages non-proliferative (background or simple retinopathy) and proliferative retinopathy. Simple retinopathy is characterized by microaneurysms, retinal hemorrhages, exudates and cotton wool spots. New vessel formation occurs in proliferative retinopathy.

Other ocular complications are macular edema, premature cataract and glaucoma.

Prevention is the most desired way to control the complications. Strict control of blood sugar and blood

pressure prevents the development or slow the progression of retinopathy. Regular eye examination is recommended. The treatment of proliferative retinopathy and macular edema is laser photocoagulation.

Diabetic Nephropathy

This is the leading cause of end stage renal disease (ESRD). About 30-40% patients with type 1 DM and 15-20% patients with type 2 DM develop nephropathy after 20 years. The early manifestation of nephropathy is the onset of microalbuminuria. Microalbuminuria is defined as the urinary excretion of 30-300 mg albumin per day. It progresses to overt proteinuria (>300 mg /day) in few years.

Intensive control of blood sugar and blood pressure is an effective way to reduce the progression of microalbuminuria to overt nephropathy. Angiotensin converting enzyme (ACE) inhibitors or angiotensin II receptor blockers (ARB) reduces the progression of overt nephropathy. Dietary protein restriction may be beneficial in some patients.

Diabetic Neuropathy

This may be sensory, motor, autonomic or combined. It may manifest as polyneuropathy or mononeuropathy. Neuropathy occurs in 50% cases with long standing DM. The presence of neuropathy is a risk factor for foot trauma, ulceration and arthropathy. Cranial nerves may also be involved. The most commonly affected nerves are 3rd and 6th cranial nerves. Autonomic neuropathy may manifest as postural hypotension, resting tachycardia, nocturnal diarrhea, impotency and urinary bladder dysfunctions. Hyperhidrosis or anhidrosis may occur.

The neuropathy may be asymptomatic. Annual examination of lower extremities for the touch and vibration sensations is recommended.

Painful neuropathy is treated with tricyclic antidepressant (amitriptyline) or anticonvulsants (carbamazepine, gabapentin, and phenytoin). Blood glucose should be controlled strictly. Alcohol should be avoided and vitamins (B12, B6, and folate) are replaced if deficient.

Macrovascular Complications

Various risk factors for the macrovascular disease include insulin resistance, hyperglycemia, micro-albuminuria, hypertension, hyper-lipidemia, smoking, and obesity. Optimal glycemic control, control of blood pressure to less than

130/80 mmHg (125/75 mmHg in patients with proteinuria), treatment of dyslipidemia (target LDL level <100 mg/dL, Triglyceride <150 mg/dL), cessation of smoking and weight reduction are ways to minimize the macrovascular complications.

Management of Diabetes

This includes diet modification, exercise, education and medications.

Ideal goals of the glycemic control in patients with DM are as follows. However, these goals are individualized.

1. Average preprandial glucose values 90-130 mg/dL
2. Bedtime glucose value of 100-140 mg/dL
3. Peak postprandial glucose less than 180 mg/dL
4. The level of HbA_{1c} less than 7%

The management of patients with DM includes:

- a. Glycemic control
- b. Treatment of associated conditions like hypertension, dyslipidemia, obesity and cardiovascular disease
- c. Detection and management of diabetes related complications.

Diet Modification

The diet is modified in such a way that it helps in maintenance of ideal body weight and provides essential nutrients. The caloric intake is optimized. Protein and fat should provide 15-20% and less than 30% of the total caloric intake respectively. Saturated fat should provide less than 10 % of the total caloric intake. Cholesterol intake is restricted to less than 300 mg per day. Intake of dietary fibers is increased. Carbohydrate intake is individualized based on glycemic control, plasma lipids and body weight. Use of sweeteners is acceptable.

Exercise

It provides multiple benefits. It improves insulin sensitivity, lowers plasma glucose, reduces blood pressure and lowers cardiovascular risk. Other benefits are weight reduction and maintenance of muscle mass.

Education

The education of patients on different aspects of the disease is very important. They should be educated about the diet,

exercise, self monitoring of blood glucose and urinary glucose and ketones. Patients should be warned about the symptoms and dangers of hypoglycemia. They are trained in foot and skin care and about insulin injections.

Pharmacological Management

If glycemic control is not achieved by non-pharmacological means such as diet control, and increased physical activities in 3-4 weeks, pharmacological therapy is indicated. Pharmacological therapy consists of oral glucose lowering agents and insulin.

Glucose Lowering Agents

Oral glucose lowering agents are generally indicated in type 2 DM since residual pancreatic function is required for their glucose lowering effects. Insulin is ultimately required in advanced stage of type 2 DM. Oral glucose lowering agents are classified into various groups (Table 8.16).

The treatment should be started with lower dosage and increased slowly to the optimal dose in several days or weeks.

Some of these agents are available in extended release forms. Hence, these need to be administered less frequently. *Insulin secretagogues*: These agents stimulate insulin secretion by pancreatic beta cells. Important side effects are weight gain and hypoglycemia. Sulfonylurea should be given 30 to 60 minutes before meals. Non-sulfonylureas are given with each meal to reduce meal related increase in blood glucose. These should not be given if a patient is observing fast.

Biguanides: Biguanides reduce hepatic glucose production and improve glucose utilization in peripheral tissues. These are taken with food. They also improve lipid profile and promote weight loss. Side effects are diarrhea, anorexia, nausea and lactic acidosis. They do not result in hypoglycemia. Metformin is contra-indicated in renal insufficiency, heart failure, liver disease, severe hypoxia and other conditions associated with the acidosis.

Alpha glucosidase inhibitors: They reduce post-prandial hyperglycemia by delaying glucose absorption. These are given with the first bite during meal. Side effects are diarrhea and flatulence. These agents can also be used in type 1 DM.

Thiazolidinediones: Thiazolidinediones enhance insulin sensitivity in muscle, adipose tissue and liver, hence reduce

TABLE 8.16: Glucose lowering agents (Anti-diabetic drugs)

Drugs	Daily dose
ORAL	
Insulin secretagogues	
<i>Sulfonylureas</i>	
a. First generation:	
Chlorpropamide	100-500 mg, single dose
Tolbutamide	500-2000 mg in 2-3 divided doses
Tolazamide	100-1000, in 1 or 2 divided doses
b. Second generation:	
Glyburide	1.25-20 mg in 1 or 2 divided doses
Glipizide	2.5-40 mg in 1 or 2 divided doses
Gliclazide	40-320 mg in 2 divided doses
Glimepiride	1-8 mg, single dose
Glibenclamide	2.5-10 mg in 1 or 2 divided doses
<i>Non-sulfonylureas:</i>	
Repaglinide	1-16 mg in 2-4 divided doses
Nateglinide	180-360 mg in 3 divided doses
Biguanides:	
Metformin	500-2000 mg in 2-3 divided doses
Alpha glucosidase inhibitors	
Acarbose	75-300 mg in 3 divided doses
Miglitol	75-300 mg in 3 divided doses
Thiazolidinediones	
Rosiglitazone	2-8 mg in 1 or 2 divided doses
Pioglitazone	15-45 mg, single dose
Incretins	
<i>DDP-IV inhibitors</i>	
Sitagliptin	100 mg OD
Vildagliptin	50 mg BD
PARENTERAL	
Insulin (see Table 8.16)	
<i>Incretins</i>	
GLP-1 receptor agonist	
Exenatide	5-10 mcg BD
<i>Amylin agonist</i>	
Pramlintide	

insulin resistance. These can cause hepatotoxicity and fluid retention hence, are not given in cases of liver disease and congestive heart failure.

Incretins: The glucose dependent insulinotropic polypeptide (GIP) and glucagon-like peptide (GLP-1) are two major incretin hormones released in response to ingestion of meal. In the fasting stage, their levels are low but after eating a meal their secretion increases rapidly.

- GIP is a peptide (42 amino acid) secreted by endocrine K cells in proximal gastro intestinal tract
- GLP-1 is a peptide (30-31 amino acid) secreted from L cells in the distal gastro intestinal tract

They mediate their actions by binding and activation of receptors (GLP-1R and GIP-R) located in several tissues, including alpha and beta cells in the pancreas.

GIP and GLP-1 stimulate glucose dependent insulin release from beta cells. They contribute 60-70% of the total post-prandial insulin response in healthy individuals (Incretin effect). This incretin effect is markedly decreased in diabetic patients due to low levels of post-parandial incretin hormones. GLP-1 also suppresses hepatic glucose output by inhibiting glucagon response from alpha cells in a glucose dependent manner.

After secretion these are rapidly metabolized by the enzyme dipeptidyl peptidase (DPP-IV). The plasma half-life of these incretins is short (2-5 minutes). Following degradation, the metabolites are eliminated from the kidneys.

Agents that either have similar action as GLP-1 or increase GLP-1 activity are now available for use in diabetes. Two major strategies have been developed to increase the effects of incretins:

1. **Incretin mimetics (GLP-1 analogues, GLP-1 receptor antagonists):** These are injectable agents that mimic the action of GLP-1 on glucose control. Exenatide is a 39 amino acid peptide and exhibits same glucoregulatory effects as the human incretin hormone GLP-1. It is resistant to endogenous DDP-IV, thus have prolonged GLP-1 like action. It improves glycemic control by reducing glucose concentration by following mechanisms:
 - a. Beta cell effect-enhancement of glucose dependent insulin secretion and restoration of first phase insulin response.
 - b. Suppression of elevated glucagon secretion
 - c. Slowing of gastric emptying
 - d. Reduction of food intake

Use of Exenatide is associated with weight loss. It is indicated as an adjunct therapy to improve glycemic control in patients with type 2 who are taking metformin, sulfonylurea or combination of both but have not achieved glycemic control.

It is given in doses of 5-10µg subcutaneously twice daily. Major side effects include nausea, vomiting, diarrhea and hypoglycemia.

2. **Incretin enhancers (DDP-IV inhibitors):** These are new class of anti-diabetic drugs and are orally

effective agents able to inhibit DDP-IV enzyme. By inhibiting the DPP-IV, the endogenous levels of incretins (GLP-1) are increased. It is indicated as an adjunct therapy to improve glycemic control in patients with type 2 DM who are taking metformin, sulfonylurea or combination of both but have not achieved glycemic control. Trials are under going to evaluate their use as first line agents (monotherapy) in diabetes. These can be used in combination with metformin, sulfonylurea and insulin. They are weight neutral having no effect on body weight. Major side effects include nasopharyngitis, dizziness, headache, diarrhea, constipation and hypoglycemia.

These include:

- Sitagliptin is used orally in doses of 100 mg daily.
- Vildagliptin is used orally in doses of 50 mg twice daily.

Insulin

Human insulin is produced through recombinant DNA technology. Bovine and porcine insulins are now rarely used. Insulins are divided into various groups based on the onset and duration of action (Table 8.17). Insulin is usually given subcutaneously. However, regular insulin can also be given intravenously. Insulin lispro, aspart, glulisine, glargine and detemir are genetically modified insulin analogues.

TABLE 8.17: Types of insulin

Insulin preparations	Onset (in h)	Peak (in h)	Duration (in h)
Ultrashort acting			
Insulin lispro, insulin aspart, insulin glulisine	<0.25	0.5-1.5	3-4
Short acting			
Regular insulin	0.5-1	2-3	6-8
Intermediate acting			
NPH (neutral protamine hagedorn)	2-4	6-12	10-16
Lente insulin	3-4	6-12	12-18
Long acting			
Ultralente insulin	4-6	10-16	18-20
PZI (protamine zinc insulin)	3-8	14-24	24-36
Insulin glargine			
Insulin detemir	4	—	24
Combination			
70/30, 70% NPH and 30% regular			
50/50, 50% NPH and 50% regular			
75/25, 75% protamine lispro and 25 % lispro			

Insulin Regimens

Various insulin regimens are followed in order to achieve the uniform glycemic control. Basal insulin requirement is fulfilled by long acting or intermediate acting insulin. These are often combined with the regular, lispro or aspart insulin to mimic physiological release of insulin with meals. The preparation containing lispro or aspart insulin is injected just before or after the meals while those with the regular insulin is given 30-45 minutes prior to meal. Generally total dose of insulin required is 0.5-1.0 unit/kg/day which is divided into multiple doses. 40-50 % of the daily requirement is given as basal insulin. Commonly used regimen is twice daily injection of intermediate insulin mixed with short acting insulin. Two-third of the total dose is given before the morning meal and remaining one third before the evening meal.

Continuous subcutaneous insulin infusion (CSII) is another regimen delivered by programmable pump. Insulin pen injector devices are also available which cause lesser pain. Inhaled insulin is also being tried to determine its efficacy and side effects.

Complications of insulin therapy include hypoglycemia, insulin allergy, insulin resistance due to antibodies and lipodystrophy at injection sites.

Management Steps in Diabetic Patients

The steps in the management of diabetic patients depend upon the type of the DM and the specific needs of the patients.

Type 1 DM: The treatment of type 1 DM is lifelong insulin replacement. However, alpha glucosidase inhibitors (acarbose, miglitol) can be given to reduce carbohydrate absorption.

Diet modifications, physical exercise and education are also important for the proper control of blood sugar.

Type 2 DM: In obese type 2 patients, weight reduction should be achieved by diet modification and increased physical activity. Metformin is the first line therapy in obese patients. If this is inadequate, sulfonylurea, thiazolidinediones or alpha glucosidase inhibitors are added alone or in combination. If the combination of these drugs fails to control blood glucose levels, insulin therapy is instituted.

Weight maintenance diet is prescribed to non-obese patients. Sulfonylurea is the first line therapy. If the blood

glucose is not controlled by the maximum dose of sulfonylurea, other agents like biguanide or thiazolidinediones or both are added. If the combination of these drugs fails, insulin is indicated.

Other indications of insulin in type 2 DM are (a) as a initial therapy in patients with severe weight loss, (b) patients with hepatic or renal disease where oral agents are contraindicated, (c) patients who are acutely ill (DKA, HHS) and (d) during pregnancy.

Hypoglycemia

Hypoglycemia means low blood sugar levels, generally <50 mg/dL. However, the threshold may vary widely from patient to patient. This is the most common complication of insulin therapy. Hypoglycemia can also occur due to sulfonylurea therapy. Risk factors for the development of hypoglycemia include skipped meal, delay in the meal, unusual physical exertion, alcohol ingestion and drug overdose. Hypoglycemia can also occur in other conditions such as insulinomas (beta islet tumor), liver disease, adrenal insufficiency and hypopituitarism.

Clinical Presentations

Manifestations of hypoglycemia are commonly due to activation of autonomic nervous system and secondary to glucose deprivation of the brain. Autonomic symptoms are sweating, trembling, palpitation, hunger and anxiety. Neurological manifestations are impaired concentration, irritation, confusion, seizures and coma. Nausea, headache and tiredness are non-specific symptoms.

Diagnosis

The diagnosis is made by the measurement of plasma or capillary blood glucose, which is low.

Treatment

The treatment depends upon the severity of hypoglycemia and consciousness level of the patient.

- If the patient is able to swallow, rapidly absorbable carbohydrates can be administered orally (glucose and sugar). Alternatively milk, fruit, candy bars, or biscuits may be given to patients with mild hypoglycemia.
- If the patient is unable to swallow or is in severe hypoglycemia, intravenous glucose (20-50 ml of 50% dextrose) is given initially. This is followed by an infusion

of 10% dextrose to maintain blood glucose above 100 mg/dL.

- Glucagon injection (1 mg SC or IM) may be given in severe hypoglycemia if intravenous access can not be established promptly.
- Prevention of hypoglycemia includes proper education regarding causes and symptoms of hypoglycemia and proper adjustment in medication and diet.

IMPLICATIONS ON DENTAL PRACTICE

1. Oral conditions seen in diabetic patients include burning mouth, altered wound healing and increased incidence of infections. Xerostomia may result from diabetes *per se* or from medications.
2. Diabetes is a risk factor for the prevalence and severity of gingivitis and periodontitis. Periodontal infection may adversely affect glycemic control. Proper dental treatment may have favorable effects on glycemic control.
3. Diabetes may be associated with the increased incidence of oral candidiasis. Severe diabetes with ketoacidosis may predispose patient to mucormycosis of the paranasal sinuses and nose.
4. Blood sugar should be adequately controlled before any surgery.
5. The normal pattern of food intake may be disturbed by dental disease or dental treatment. This can interfere with diabetic control.
6. Most common emergency in diabetic patients during dental treatment is hypoglycemia. A proper history about the time of meals and anti-diabetic medications must be recorded. Easily absorbable carbohydrates (glucose, sugar, candies, chocolates, and fruit juices) should be available for emergency use.
7. Patients with Cushing's syndrome may develop complications following dental procedures. These include excessive bleeding, poor wound healing and increased susceptibility to infections.
8. Patients with Cushing's syndrome have higher incidence of Candida and other fungal infections.
9. Brown or black pigmentation of the oral mucosa (Fig. 8.7) is seen in patients with Addison's disease (adrenal insufficiency). This condition must be suspected if there is hypotension, weakness, nausea, anorexia, vomiting, abdominal pain and weight loss.



FIGURE 8.8: Pigmentation of oral mucosa (Addison's disease)

10. In hyperthyroidism, elevated systolic blood pressure may require caution. Longer duration of local pressure is needed to stop bleeding in such cases. Patients may be difficult to deal as they may be anxious and irritable.
11. Deposition of subcutaneous mucopolysaccharides in hypothyroidism may lead to impaired ability of small vessels to contract when cut. This may result in excessive bleeding. Hence, local pressure is required for longer period to control bleeding.
12. There is also a delayed wound healing and increased risk of infection due to poor healing in hypothyroidism.
13. Use of sedatives, opioid analgesics and tranquillizers may precipitate myxedema coma in patients with hypothyroidism.
14. Well controlled hyperthyroidism and hypothyroidism do not pose any additional risk for dental procedures.
15. Dryness of the mouth may be present in diabetes insipidus.
16. Brown tumor of the jaw should be differentiated from giant cell granuloma of the jaw.
17. Enamel hypoplasia, delayed eruption and chronic mucocutaneous candidiasis may occur in idiopathic congenital hypoparathyroidism.

SELF ASSESSMENT

Multiple Choice Questions

1. **Excess of cortisol causes:**
 - A. Conn's syndrome
 - B. Cushing's syndrome
 - C. Diabetes insipidus
 - D. Acromegaly
2. **Signs of hypoglycemia include the following except:**
 - A. Sweating
 - B. Mental confusion
 - C. Tachycardia
 - D. Hyperpyrexia
3. **Which of the following is associated with tetany:**
 - A. Chvostick's sign
 - B. Kussmaul's sign
 - C. Corrigan sign
 - D. Lock Jaw
4. **Features not associated with type 2 diabetes mellitus:**
 - A. Onset after 30 years
 - B. Obesity
 - C. Ketoacidosis very common
 - D. Insulin resistance
5. **Initial drug preferred in type 2 obese diabetics:**
 - A. Gliclazide
 - B. Repaglinide
 - C. Metformin
 - D. Acarbose
6. **Treatment of choice for type 1 diabetics:**
 - A. Metformin
 - B. Insulin
 - C. Acarbose
 - D. Glipizide
7. **Earliest feature in diabetic nephropathy is:**
 - A. Hematuria
 - B. Raised serum creatinine
 - C. Microalbuminuria
 - D. Overt proteinuria
8. **All of the following hormones are secreted by anterior pituitary except:**
 - A. ACTH
 - B. TSH
 - C. ADH
 - D. FSH
9. **Bone pain, bone cysts, fracture and renal stones are found in:**
 - A. Multiple myeloma
 - B. Hyperparathyroidism
 - C. Osteomalacia
 - D. Marfan's syndrome
10. **In alkalosis, following is true:**
 - A. Normal total serum and normal ionic calcium
 - B. Low total serum and normal ionic calcium
 - C. Normal total serum and low ionic calcium
 - D. Low total serum and low ionic calcium
11. **Tetany may occur in:**
 - A. Alkalosis
 - B. Hyperparathyroidism
 - C. Myxedema
 - D. Hyperglycemia
12. **Following are features of diabetes insipidus except:**
 - A. Polyurea
 - B. Polydipsia

- C. Polyphagia
- D. High plasma osmolality

13. Most common cause of thyrotoxicosis is:

- A. Graves' disease
- B. Multinodular goiter
- C. TSH secreting pituitary tumor
- D. Toxic adenoma

14. Pretibial myxedema is seen in:

- A. Hypothyroidism
- B. Graves' disease
- C. Hashimoto's disease
- D. All of above

15. Which one of the following is the early feature of hypothyroidism:

- A. Low T_3
- B. High TSH
- C. Low T_4
- D. Low TSH

16. In hypoalbuminemia, following is true:

- A. Low total serum calcium
- B. Low ionized calcium
- C. Low total serum calcium and low ionized calcium
- D. Low total serum calcium and normal ionized calcium

17. Following is seen in hypocalcemia:

- A. Convulsions
- B. Laryngeal spasm
- C. Perioral parasthesia
- D. All of the above

18. The common causes of hypocalcemia include all except:

- A. Chronic renal failure
- B. Hypoparathyroidism
- C. Vitamin D deficiency
- D. High serum albumin

19. Hypercalcemia associated with malignancy is most often mediated by:

- A. Parathyroid hormone
- B. Parathyroid related protein (PTHrP)
- C. Interleukin-6
- D. Calcitonin

20. Hyperthyroidism is treated by:

- A. Carbimazole
- B. Radioactive iodine
- C. Subtotal thyroidectomy
- D. All of the above

21. Glycated Hb (HbA1) provides measure of glucose control over a period of:

- A. Within 6 hours
- B. Within one week
- C. Within 1-3 weeks
- D. Within 8-12 weeks

22. Which one is not a feature of hyperglycemic hyper-osmolar state:

- A. Severe hyperglycemia
- B. Marked dehydration

- C. High incidence of thromboembolic complications
- D. Presence of ketosis

23. Which of the following drug is not a insulin secretagogue:

- A. Glipizide
- B. Glimperide
- C. Repaglinide
- D. Rosiglitazone

24. Hypoglycemia is not a complication of:

- A. Glipizide
- B. Glimperide
- C. Metformin
- D. Repaglinide

25. Drug which inhibits absorption of glucose from intestine is:

- A. Gliclazide
- B. Pioglitazone
- C. Acarbose
- D. Nateglinide

Fill in the Blanks

1. Moon face and buffalo hump are seen in _____.
2. Active form of thyroid hormone is called _____.
3. Trousseau's sign is present in _____.
4. Normal level of total serum calcium is _____.
5. Carpopedal spasm (main d' accoucheur) is a feature of _____.
6. Goiter, bruit over thyroid gland, ophthalmopathy and dermapathy is seen in thyrotoxicosis due to _____.
7. Condition when the production of PTH is normal but there is tissue resistance to PTH is _____.
8. Most common cause of primary hyperparathyroidism is _____.
9. Basal ganglion calcification and cataract formation are seen in _____.
10. Fasting plasma glucose more than _____ is diagnostic of diabetes mellitus.
11. Impaired fasting plasma glucose is between _____ and _____ mg/dL.
12. Rapid and deep breathing in diabetic ketoacidosis is known as _____.
13. Microalbuminuria is defined as excretion of _____ to _____ albumin daily.
14. The drug of choice in hypertension with diabetes mellitus is _____.
15. The insulin which can be given intravenously is _____.

Chapter 9 Infections

MEASLES (RUBEOLA)

Measles (Rubeola) is an acute paramyxoviral infection. It is a cause of major morbidity and mortality in children worldwide.

- *Mode of transmission:* Measles is a highly infectious disease. It spreads through infected droplets. Patients in pre-eruptive and catarrhal phase are most infectious.
- The incubation period is about 7-14 days.
- Illness confers permanent immunity.

Clinical Features

- The illness starts with high fever, dry cough, sore throat, malaise and redness of conjunctivae.
- After about 2 days of fever, tiny white spots surrounded by red margin appear on buccal mucosa. These '*Koplik's spots*' are pathognomonic of measles.
- Rashes appear over the face and behind the ears 4 days after the onset of fever. These *non-pruritic erythematous maculopapular rashes* descend and involve the whole body. The rashes fade in order of appearance.
- Other features include pharyngeal edema, red tongue, generalized lymphadenopathy, and splenomegaly.
- The disease is more severe and prolonged in adults. The clinical features may be atypical.

Complications

- *Neurological:* Such complications are rare. Encephalitis can occur after 3-7 days of the appearance of rashes. Other complications are aseptic meningitis and transverse myelitis. Subacute sclerosing panencephalitis (SSPE) is a very late complication which occurs years after the initial infection.

- *Respiratory:* Pneumonia or bronchopneumonia can occur due to the measles virus or due to secondary bacterial infection. This is the most common complication of measles in adults. However, otitis media is the most common complication in young children.
- *Gastrointestinal:* Possible complications are gastroenteritis, hepatitis and ileocolitis.

Laboratory Diagnosis

- The blood examination shows leukopenia. Leukocytosis suggests the presence of bacterial infection.
- A specific diagnosis can quickly be made by the immunofluorescence techniques which detect the presence of viral antigens in the respiratory secretion.
- RT-PCR can detect viral RNA.
- Specific IgM antibodies suggest recent infection.

The important differential diagnosis includes drug reaction, infectious mononucleosis and mycoplasma infection.

Management

The management of measles is mainly symptomatic and supportive.

1. The patients should be isolated for the week following onset of rashes and put on bedrest. Vitamin A administration has been shown to reduce morbidity in children.
2. Antibiotics should not be used routinely. They are given only when signs of superinfection exist. Immunoglobulin can attenuate the disease and can be given in non-immune or immunocompromised patients.
3. Antiviral drug ribavirin may be considered for use in immunocompromised patients.
4. SSPE is managed symptomatically.

Prevention

Live attenuated measles vaccine should be given at the age of 9 months to all children (Table 9.1). However a combined measles, mumps and rubella (MMR) vaccine can be given at the age of 12-15 months. MMR booster is given at 4-12 years of age. This provides life long protection.

Post-exposure prophylaxis is indicated for immuno-compromised persons within 6 days of contact. A dose of immune globulin 0.25 ml/kg is given in adults. Vaccine and immune globulin should not be given concurrently.

MUMPS

Mumps is a systemic viral infection caused by paramyxovirus.

- It spreads through respiratory droplets or saliva.
- The multiplication of the virus in the upper respiratory tract epithelium is followed by a phase of viremia. Subsequently various organs such as salivary glands, pancreas, meninges and gonads are involved.
- The incubation period is 14-21 days.
- The patients become infective one day before the appearance of symptoms and remain so for a week.

Clinical Features

The prodromal symptoms like fever, malaise and myalgia appear first.

This is followed by bilateral tender parotid swelling generally within next 24 hours. Both parotids may not be involved simultaneously in some. Orifice of Stensen's duct may be red and swollen. Other salivary glands (submaxillary, sublingual) are less commonly affected. There can be pain during eating, swallowing or talking. The glandular swelling disappears within a week.

Complications

Common complications are meningitis, orchitis or oophoritis and pancreatitis. Other complications are shown in Table 9.2.

- Aseptic meningitis may occur in both children and adults. Presence of stiff neck, headache and drowsiness suggests meningitis. Cerebrospinal fluid (CSF) findings show mild pleocytosis (mostly lymphocytes), and raised protein. Unlike other viral meningitis the sugar level in CSF may be low. The meningitis is self-limiting.
- Generally, unilateral orchitis occurs in postpubertal males. This is the most common complication among

TABLE 9.1: Universal immunization schedule*

At birth	BCG, oral polio
At 6 weeks	BCG (if not given at birth), DPT and oral polio
At 10 weeks	DPT and oral polio
At 14 weeks	DPT and oral polio
At 9 months	Measles
At 12-15 months	MMR
At 16-24 months	DPT and oral polio
At 5-6 years	DT
At 10, 16 years	Tetanus toxoid
DPT: diphtheria, pertussis, tetanus	
MMR: measles, mumps, rubella	
DT : diphtheria, tetanus	
*Hepatitis B vaccination is also included in the schedule in some states.	

postpubertal males. There is pain, tenderness and swelling of the testicle. Testicular atrophy may occur but sterility is rare. Lower abdominal pain in females suggests oophoritis.

- Mumps is the leading cause of pancreatitis in children. This presents with upper abdominal pain, nausea and vomiting.

Investigations

- Serum amylase is raised either due to parotitis or pancreatitis.
- There may be relative lymphocytosis in the peripheral blood.
- The diagnosis can be confirmed by isolating the virus from saliva, throat, CSF or urine during first few days of illness.
- Demonstration of specific IgM antibody or four-fold rise in IgG antibody titer in acute and convalescent sera also help in the diagnosis.

TABLE 9.2: Complications of mumps

Common
Aseptic meningitis
Orchitis/oophoritis
Pancreatitis
Rare
Hepatitis
Myocarditis
Thyroditis
Nephritis
Arthritis
Thrombocytopenia
Encephalitis
Guillain-Barré syndrome

TABLE 9.3: Causes of parotid swelling

Unilateral	
	Parotid cyst
	Parotid tumor
	Stone in the parotid duct
Bilateral	
<i>Infectious</i>	
	Viral: mumps, Coxsackie, HIV, influenza, EBV
	Bacterial: <i>S. aureus</i> , Streptococci, Gram-negative bacteria
<i>Non-infectious</i>	
	Sarcoidosis
	Sjögren's syndrome
	Diabetes mellitus
	Cirrhosis
	Drugs (propyl thiouracil, phenothiazines)

Differential Diagnosis

The parotitis in mumps should be differentiated from other causes of parotid swelling. The Table 9.3 shows other causes of parotid enlargement.

Treatment

The patient should be isolated until the swelling subsides. The patient needs bedrest during the febrile period. The treatment of parotitis, meningitis and pancreatitis is purely symptomatic. Analgesics and cold or warm compresses are helpful in relieving parotid pain.

The role of prednisolone in the management of orchitis is not fully established. However, it may reduce the discomfort of orchitis. Interferon alpha has been shown to be helpful in some cases of orchitis.

Prevention

The infection confers life long immunity. The primary prevention of mumps is done by the administration of live attenuated vaccine. MMR vaccine is given during 12-15 month of age (Table 9.1). The vaccine provides life long immunity. The vaccine can also be given to susceptible adults who have not had mumps. However, this is contraindicated in pregnant women and immunocompromised individuals.

RUBELLA (GERMAN MEASLES)

Rubella or German measles is a systemic viral infection caused by togavirus.

- The infection spreads through droplets.

- Incubation period is 14-21 days.
- Patients are infective from up to a week before to two weeks after the onset of the rash.
- Skin, joint and placenta are mainly involved.

Clinical Features

The main presenting features are fever, lymphadenopathy and rash. In many, the infection is subclinical.

- The fever is mild and occurs on the first day of the rash.
- There may be coryza and conjunctivitis.
- Rash is maculopapular and nonconfluent which begins on the face and then spreads downwards. Petechial rash may appear on the soft palate (Forschheimer's spots).
- Posterior auricular, cervical and suboccipital lymphadenopathy are typically present.

The most devastating effect of rubella is on the fetus if infection occurs in pregnant women during first trimester and early second trimester.

- Spontaneous abortion may occur.
- *Congenital rubella*: Classical signs of congenital rubella are cataract, heart disease (patent ductus arteriosus, pulmonic stenosis), deafness and other defects such as mental retardation, autism, microcephaly and retinopathy.

Complications

Polyarthrits occurs in about 25% adult female patients. Other complications are bleeding due to thrombocytopenia and vascular damage, encephalitis and hepatitis.

Investigations

The diagnosis is made by the demonstration of virus or specific antibodies. The presence of IgM antibody or four-fold rise in IgG antibodies in paired acute and convalescent sera is diagnostic.

Treatment

There is no specific therapy. The fever and joint pain are managed symptomatically.

Prevention

The rubella vaccine (in form of MMR) is given to children at 12-15 months of age (Table 9.1). A second dose is given at 4 years.

CHICKENPOX (VARICELLA) AND HERPES ZOSTER (SHINGLES)

Varicella-zoster virus is a human herpes virus 3. It causes two important clinical entities in humans chickenpox (varicella) and herpes zoster (shingles).

Chickenpox

It is a highly contagious disease. The spread of infection is through inhalation of infected droplets (aerosol) or through contact with the skin lesions. Incubation period is 10-21 days.

Clinical Features

Chickenpox is generally a disease of children, although it can also occur in adults. The disease is mild in children and more severe in adults, pregnant women and immunocompromised individuals.

- The initial manifestations are fever and malaise followed by the appearance of rash. Rashes appear first over mucous membrane and then spread in centripetal distribution over skin (more in the trunk, sparse in limbs). Initial lesions are pruritics and maculopapular which eventually evolve into vesicles, pustules and finally crusts (Figs 9.1A and B). Successive new crops of lesions appear so that all stages of rashes are seen simultaneously (*pleomorphic rash*).
- Patient is infective till crusts slough and separate (7-14 days).
- Intense itching leads to infection due to scratching of lesions.
- After the primary infection, the virus remains dormant in nervous tissue. Reactivation later in life may lead to herpes zoster, more common in elderly.

Complications

- a. The most common complication of chickenpox is secondary infection of the skin lesions by *Streptococcus pyogenes* or *Staphylococcus aureus*.
- b. Acute cerebellar ataxia and meningitis may occur after 3 weeks of onset of rash. It is generally self-limiting. Encephalitis, Guillain-Barré syndrome are other rare neurological complications.
- c. Varicella pneumonia is a serious complication of varicella. It is more common in adults especially pregnant



FIGURES 9.1A and B: (A) Pleomorphic rash over abdomen in chickenpox (B) Rash over face

women and immunocompromised patients. It manifests as tachypnea, cough, dyspnea and fever.

- d. Hepatitis can occur in small number of patients.
- e. Other complications include myocarditis, nephritis and arthritis.
- f. Infection in females during first trimester of pregnancy can lead to congenital malformations in fetus (*congenital varicella*). Mortality rate in neonates is high if they get infection within five days of delivery (*perinatal varicella*).
- g. *Reye's syndrome* (fatty liver with encephalopathy) may complicate children with varicella who receive aspirin.

Diagnosis

Diagnosis is usually clinical from the typical appearance of the lesions.

- The confirmation of infection is made by isolation of varicella on tissue culture or by detection of viral DNA by PCR.
- Serological diagnosis by the demonstration of rising titers of antibody is also helpful.
- Multinucleated giant cells can be demonstrated by Tzanck smear.

Treatment

1. *General:*
 - a. The patient is isolated till the complete sloughing of crusts.
 - b. Personal hygiene should be maintained to prevent skin infection. Nails should be closely cropped.
 - c. Antipruritic agents, tepid water baths and wet compresses can be used to ease pruritus.
 - d. Aspirin should be avoided in children.
2. *Antiviral Therapy:* Acyclovir, 800 mg orally five times a day for 5-7 days is given within 24 hours of onset of rashes and is of benefit particularly in immunocompromised host and complicated varicella.

Prevention

- a. Live attenuated varicella vaccine can be given to children >1 year of age who have not had chickenpox. A single dose is adequate. Seronegative adults should receive two doses 1-2 months apart.
- b. Varicella-zoster immunoglobulin (VZIg) is given to susceptible individuals (HIV positive, pregnant ladies, premature infants, neonates born to mothers who develop varicella in perinatal period) within 4 days of exposure to an active varicella case.
- c. Susceptible individuals who are not eligible for VZIg or who have had exposure more than 4 days before should receive acyclovir. Acyclovir is initiated in these patients after seventh day of exposure.

Herpes Zoster

Herpes Zoster is due to reactivation of latent varicella-zoster virus from the dorsal root ganglion. Its incidence is highest in the sixth decade, although it can occur at all ages.



FIGURES 9.2: Rash over thoracic dermatome in herpes zoster

Clinical Features

Severe pain occurs initially and is followed by the appearance of rash after 2-3 days.

- The rashes are initially erythematous maculopapular which evolve into vesicles and pustules (Figs 9.2A and B). These occur along the distribution of nerve root (dermatome). Thoracic and lumbar roots are more commonly involved (Fig. 9.3). The rash usually subsides in 1-2 weeks. However, it may take 4 weeks for the skin to become normal.
- In immunosuppressed patients (such as HIV/AIDS), the rash may be multidermatomal and occur recurrently (Fig. 9.4).
- Lesions may appear on the face, mouth, tongue and eye if branches of the trigeminal nerve are involved.
- *Ramsay Hunt syndrome:* Ipsilateral facial palsy, loss of taste over the anterior two-third of tongue and painful



FIGURE 9.3: Rashes over thoracic and lumbar roots



FIGURE 9.4: Multidermatomal herpes zoster in an immunocompromised patient

vesicles in the external auditory canal can occur if geniculate ganglion is involved.

Complications

- Most common complication is post-herpetic neuralgia; pain persists for 1-6 months or more following healing of the rash. This is more common in elderly.
- Other complications are meningitis and encephalitis.

Treatment

Antiviral therapy (acyclovir, valacyclovir, famciclovir) is helpful as it accelerates the healing and reduces the pain during attack. It also reduces the incidence and severity of post-herpetic neuralgia. The dose of acyclovir is 800 mg

five times a day for 7-10 days and of valacyclovir is 1000 mg three times a day for 5-7 days. The dose of famciclovir is 500 mg three times a day for 7 days.

Post-herpetic neuralgia is treated with analgesics, amitriptyline, gabapentin and lidocaine patches.

HERPES SIMPLEX

Herpes viruses are associated with various human diseases. Eight types of herpes viruses may cause disease in humans (Table 9.4). Primary infection with herpes virus may be subclinical and is followed by latent stage. Each type of virus may affect particular organ. Immunodeficiency or impaired immunity later in life is associated with recurrences or relapses due to reactivation of virus.

Herpes simplex (HSV)/*Herpesvirus hominis*

HSV-1 and HSV-2 infection is associated with mucocutaneous lesions. HSV-1 causes mainly oral lesions while HSV-2 causes lesions in genital area predominantly.

Source of Infection

The source of infection is patients suffering from primary infection or recurrences. Asymptomatic shedding of virus occurs after primary infection or shedding may occur during symptomatic recurrences and lead to transmission.

Clinical Presentation

Primary infection may be asymptomatic or may cause gingivostomatitis or pharyngitis. Systemic manifestations are more common during primary infection and patient may present with fever, myalgias, weakness and cervical lymphadenopathy. HSV-1 disease mainly involves mouth

TABLE 9.4: Human Herpes viruses

Herpes simplex (HSV)/ <i>Herpesvirus hominis</i>	
HSV Type 1	
HSV Type 2	
Varicella zoster virus (VZV)-	causes chickenpox and shingles
Type 3	(Herpes zoster)
Epstein-Barr virus (EBV)-	causes infectious mononucleosis
Type 4	
Cytomegalovirus (CMV)-Type5	
Human herpes virus (HHV-6)	causes exanthema subitum
Human herpes virus (HHV-7)	causes disease in immunocompromised patients.
Human herpes virus (HHV-8)	associated with Kaposi's sarcoma



FIGURE 9.5: Herpes simplex labialis

and oral cavity (herpes labialis, Fig.9.5). Lesions may occur on gingiva, soft palate, tongue and lips. Lesions in oral area are characterized by vesicles, which occur in groups. Vesicles later ulcerate and epithelialize in few weeks.

Recurrence: After primary infection, virus remains in latent stage in ganglion of sensory nerves. Stress, infection, radiotherapy, chemotherapy, sunlight, drugs and impaired immunity may lead to reactivation of virus and recurrences.

- Recurrences usually are associated with milder disease, few lesions and heal more rapidly as compared to primary infection.
- Recurrent herpes labialis is more commonly a manifestation of reactivation.
- Ulcerations on lips, oral mucosa or facial skin may occur due to reactivation of HSV from the trigeminal ganglia.
- Patient may develop oral-labial lesions few days after procedures like *trigeminal root decompression and dental extraction*.
- Ulceration in oral mucosa is difficult to differentiate from aphthous ulcers, traumatic ulcers and drug-induced ulcers.

In immunosuppressed individuals, infection is more severe and extends into deeper tissues. Lesions are characterized by severe pain, inability to eat, bleeding and necrosis. Differential diagnosis includes bacterial infections, fungal infections, drug-induced ulcers and trauma.

Other Manifestations

Bell's palsy: HSV-1 has also been associated with Bell's palsy (infranuclear VII nerve palsy). See chapter 7.

Genital herpes: HSV-1 may also cause vulvovaginitis and cervicitis in females, balanitis in males. Bilateral lesions of external genitalia occur. HSV may also cause rectal infection (proctitis) and perianal infection.

- Primary genital herpes may be associated with systemic manifestations like fever, headache, weakness, myalgia and inguinal adenopathy.
- Lesions occur in form of painful grouped vesicles or pustules in the genital and perianal regions that ulcerate. Discharge (vaginal, urethral), dysuria and itching may occur.
- First episode of HSV-1 and HSV-2 are similar clinically, however, recurrent genital herpes is more common with HSV-2 infection. HSV-2 infection increases the risk of acquiring HIV infection. In HIV infection recurrences of HSV-2 occur more frequently.

Eye involvement: HSV infection can cause keratitis, keratoconjunctivitis and chorioretinitis. Keratitis presents as acute pain, blurred vision, chemosis and dendritic ulcers of cornea. HSV infection of eye can lead to blindness.

Congenital herpes: Transmission of infection occurs during delivery or during intrauterine stage. Disseminated and potentially fatal infection may occur in neonates (congenital herpes). It is characterized by visceral and CNS involvement. Maternal infection during pregnancy is associated with highest risk of transmission.

Skin: HSV infection in patients with eczema may result in severe infection involving large areas of skin and oral lesions may spread to visceral organs (eczema herpeticum). Infection of the finger (herpetic whitlow) may occur as a complication of HSV oral or genital infection or through occupational infection. Vesicular or pustular lesion of fingertip occurs with edema, redness and tenderness. Lesion may be difficult to differentiate from pyogenic bacterial infection.

Nervous system: HSV infection can cause encephalitis, the most serious complication of herpes simplex infection. Recurrent benign lymphocytic meningitis (Mollaret's meningitis) is another CNS complication of HSV infection.

Diagnosis

Diagnosis is usually clinical but to differentiate from other vesicular eruptions laboratory tests may be needed.

- Multinucleated giant cells or intranuclear inclusion bodies can be demonstrated in scrapings from the lesion and staining with Giemsa's stain (Tzanck preparation) or Papnicolaou's stain. These findings may be present in other viral infection like VZV infection.
- Confirmation of HSV can be done by detection of HSV DNA by PCR, culture from vesicular fluid and direct fluorescent antibody staining of scraped lesions.
- Serology is useful in primary infections.
- CSF-PCR is useful in HSV encephalitis.

Treatment

In immunocompetent persons, therapy is usually not required but if initiated early it may ameliorate or shorten the duration of symptoms. Antiviral treatment is effective if started in initial stages (first 48 hours). Severe manifestations require antiviral therapy irrespective of time of presentation.

Serious manifestations (CNS infections, visceral HSV infections, neonatal infection, disseminated infections) require intravenous administration of acyclovir (10 mg/kg 8 hourly for 10 days).

Acyclovir, valacyclovir and famciclovir are used for treatment and these agents inhibit the viral DNA polymerase. Treatment of mucocutaneous herpes is given in Table 9.5. For eye infections, various topical preparations are available (idoxuridine, vidarabine and cidofovir).

Prevention

- Use of barrier contraception during sexual activity (condoms) decreases the rate of transmission of HSV infection.
- Valacyclovir 500 mg once daily reduces chances of transmission of infection between sexual partners.
- Isolation and use of gloves, gowns and handwashing decrease spread of infection from patients with mucocutaneous, genital and disseminated infection to hospital staff and other patients

INFECTIOUS MONONUCLEOSIS

Infectious mononucleosis is mostly caused by Epstein-Barr virus (EBV), a human herpes virus (HHV-4) and is

TABLE 9.5: Treatment of herpes simplex infection

Primary HSV	
• Acyclovir	200 mg five times daily for 7-14 days
• Valacyclovir	1 g twice daily for 7-14 days
• Famciclovir	250 mg three times daily for 7-14 days
• Acyclovir	IV 5 mg/kg 8 h for 5 days in Severe primary infection
Recurrent HSV	
Oro-labial	
• Famciclovir	1500 mg single dose or 750 mg b.d. for 1 day
• Valacyclovir	2 g single dose or 2 g b.d. for 1 day
Genital	
• Acyclovir	800 mg tid for 2 days or 200 mg five times for 5 days
• Valacyclovir	500 mg b.d. for 3 days
• Famciclovir	1500 mg single dose or 750 mg bid for 1 day or 125 mg b.d. for 5 days
Disease suppression	
• Acyclovir	400 mg b.d.
• Valacyclovir	500 mg - 1 g daily
• Famciclovir	250 mg b.d. or 500 mg once daily

characterized by fever, lymphadenopathy, pharyngeal inflammation, and atypical lymphocytes on peripheral blood smear examination.

EBV is also associated with many tumors including nasopharyngeal carcinoma, Burkitt's lymphoma, Hodgkin's disease and B cell lymphoma in immunosuppressed persons.

Source of Infection

Its mode of transmission is saliva, contact with oral secretions of patients or asymptomatic carriers. About 90% of asymptomatic seropositive persons have virus in their saliva. It is also transmitted through blood transfusion and bone marrow transplantation. The incubation period is 4-6 weeks.

Pathogenesis

The virus infects epithelial cells of the oral cavity and salivary glands and is shed in salivary secretions. Lymphocytes in tonsils may be infected directly or B lymphocytes may get infected after contact with epithelial cells. The virus then spreads through blood stream. Lymphoid tissue is enlarged because of proliferation and expansion of B cells and reactive T cells. Activated B cells produce antibodies against EBV antigens and host cells. CD4 + T cells decrease in number and CD8+ T cells increase

leading to inverse CD4⁺/CD8⁺ cells ratio. B cells form the reservoir for EBV in the body.

Clinical Features

Infection may occur at any age.

- a. In developing countries infection occurs most commonly in early childhood. Infection is usually subclinical and most patients are asymptomatic. By adulthood majority of population has antibodies against EBV.
- b. In developed countries infection occurs during adolescence and young adults (10-36 years). In this situation many patients have typical infectious mononucleosis (IM).
 - It presents with nonspecific symptoms like fever, myalgia, weakness and malaise.
 - Pharyngitis with or without tonsillitis is usually present.
 - Lymphadenopathy, mainly posterior cervical group of lymph nodes are involved but generalized lymphadenopathy may also be seen. Lymph nodes are tender, mobile and discrete.
 - Splenomegaly may be seen in second or third week of illness. Maculopapular rash is present in 5-15% of cases. If ampicillin is given, most patients (>90%) will develop rash.

Complications

Most cases are self-limiting but complications may be seen in some patients (Table 9.6).

- Complications may be due to CNS involvement (meningitis, encephalitis, cranial nerve palsies mostly VII nerve, transverse myelitis and polyneuritis).
- Upper airway obstruction occurs because of enlargement of lymphoid tissue in tonsils and adenoids and is an indication for hospitalization.
- Coombs positive hemolytic anemia may occur because of formation of antibodies against the red cell antigen. Granulocytopenia and thrombocytopenia may also occur.

Laboratory Findings

- a. After a week of illness, WBC count is elevated with lymphocytosis and atypical lymphocytes. Atypical lymphocytes are larger than normal lymphocytes with plenty of cytoplasm, vacuoles and dark chromatin with indented cell membrane. For diagnosis at least 20% of

TABLE 9.6: Complications of Infectious mononucleosis

Central nervous system complications

- Meningitis
- Encephalitis
- Cranial nerve palsies
- Acute transverse myelitis
- Polyneuritis

Airway obstruction

Splenic rupture

Bacterial superinfection

Hematological complications

- Autoimmune hemolytic anemia
- Granulocytopenia
- Thrombocytopenia
- Red cell aplasia
- Pancytopenia

Rare complications

- Hepatitis
- Myocarditis
- Pericarditis
- Interstitial nephritis
- Interstitial pneumonia

peripheral lymphocytes must be atypical. Granulocytopenia, thrombocytopenia may be seen.

- b. Serum levels of aminotransferases and alkaline phosphatase are usually elevated. Bilirubin levels may be raised in some cases.
- c. Presence of heterophile antibodies in serum is an important test for the diagnosis. Tests for heterophile antibodies are positive in about 40% of patients in first week of illness but they become positive in almost 90% of cases during the third week. Heterophile antibodies when adsorbed in guinea pig kidney, agglutinate erythrocytes of other species e.g. sheep, horse or cow. Paul-Bunnell test or slide test (Monospot test) is used to demonstrate presence of heterophile antibodies.
- d. Elevation in other antibodies directed at various EBV antigens may also be useful in diagnosis patients who don't have heterophile antibodies and patients with atypical presentation. These include
 - IgM and IgG antibodies against viral capsid antigen (anti-VCA)
 - Antibodies to early antigen (anti-EA-D and anti-EA-R)
 - Antibodies to EBV nuclear antigen (anti-EBNA)
- e. Detection of EBV DNA by polymerase chain reaction is useful in demonstrating virus in EBV-associated malignancies and for diagnosis in immunocompromised individuals when serology is not helpful.

Differential Diagnosis

Most of the cases of IM are due to EBV. Few percentages of cases may be caused by cytomegalovirus (CMV). IM caused by CMV is less severe and heterophile antibody test is negative. Other less common causes of IM are acute Toxoplasmosis, acute HIV infection, HHV-6, hepatitis virus and hypersensitivity drug reactions. Exudative pharyngitis and heterophile antibodies are not seen in other causes of IM.

Treatment

Treatment is usually symptomatic as majority of patients will recover.

- Rest and avoidance of excessive exertion are helpful.
- Acetaminophen and other nonsteroidal anti-inflammatory drugs (NSAIDs) are used for symptomatic relief. Warm saline gargles and throat irrigation also provides relief.
- Acyclovir therapy is not effective and is not recommended.
- Glucocorticoids (prednisolone) are used in airway obstruction, autoimmune hemolytic anemia, and severe thrombocytopenia. Glucocorticoid therapy is not recommended in uncomplicated cases.
- Ampicillin, amoxicillin and other penicillins should be avoided because they can precipitate rash in patients of IM.
- Interferon alpha and antibodies to CD20 have been useful in some cases.
- Patients should avoid sports activities for few weeks until splenomegaly has resolved because of the danger of splenic rupture.

Other EBV-Associated Diseases

- a. EBV-associated lymphoproliferative disease may occur in immunodeficient patients including AIDS patients, post-transplant patients and patients with severe combined immunodeficiency. In such patients it may cause B cell hyperplasia or B cell lymphoma.
- b. Duncan's disease is a X-linked recessive lymphoproliferative syndrome seen in young boys who develop fatal lymphoproliferative syndrome after EBV infection.
- c. *Oral hairy leukoplakia* is seen in HIV infection and is characterized by raised, corrugated lesions on tongue; EBV-DNA can be detected from these lesions.

- d. It is associated with Burkitt's lymphoma (90% of cases in Africa and about 15% of cases in USA). EBV-DNA can be detected from tumor tissue and these patients have high titre of antibodies to EBV.
- e. Nasopharyngeal carcinoma is also associated with EBV infection. EBV-DNA levels in plasma correlates with poor prognosis and EBV-DNA levels in such patients are used to monitor patients.
- f. EBV infection is also associated with Hodgkin's lymphoma (especially mixed cellularity type). Viral antigen and EBV-DNA can be demonstrated in Reed-Sternberg cells in such patients and patients have high levels of antibodies to EBV.

DIPHTHERIA

Diphtheria although eradicated from most of the developed world still occurs in developing countries.

- This is caused by aerobic gram-positive bacteria *Corynebacterium diphtheriae*.
- The infection occurs mostly in the upper respiratory tract via droplets which spread from the cases or carriers. Occasionally bacteria may primarily infect conjunctiva, genital tract or skin wounds.
- The organism *does not invade the tissue*. The manifestations occur due to damage of heart muscle and nervous system by the *soluble bacterial exotoxin* absorbed at the site of infection.
- The incubation period is 2-4 days.

Clinical Manifestations

The presentations may be predominantly nasal, laryngeal, pharyngeal or cutaneous.

- The pharyngeal form is the most common one. The initial presentations are usually mild fever, sore throat and membrane formation over tonsils. The membrane is grayish, firm and adherent, surrounded by a zone of inflammation. The membrane often extends beyond the margin of tonsils unlike the exudates in staphylococcal pharyngitis. Swelling of neck may occur along with tender lymphadenopathy (*bull-neck*).
- Husky voice, high-pitched cough and respiratory obstruction may occur due to laryngeal involvement.
- Patients may have epistaxis and nasal obstruction.
- Death may occur due to acute circulatory failure (Table 9.7).

TABLE 9.7: Clinical manifestations of diphtheria

Mild fever, sore throat
Tachycardia
"Pseudomembrane" over tonsils, larynx, nasal cavity
Hoarseness of voice
Respiratory obstruction
Acute circulatory failure

Complications

Myocarditis
Palatal and laryngeal paralysis
Peripheral neuropathy

Complications

The important complications are respiratory obstruction, myocarditis and neurological involvement.

- Manifestations of diphtheric myocarditis include arrhythmia, conduction defects and cardiomyopathy.
- Neurological complications appear 3-5 weeks after the onset of diphtheria and are in the form of palatal palsy, paralysis of accommodation and polyneuritis.

Diagnosis

A definitive diagnosis of diphtheria is made on the basis of typical clinical presentations including 'pseudomembrane' and isolation of *C. diphtheriae* from the lesions. Diphtheric pseudomembrane should be distinguished from other pharyngeal exudates due to staphylococcal infection, infectious mononucleosis and candida.

Treatment

The patient is managed in isolation.

- The most important step is the prompt administration of antitoxin since the antitoxin if given late cannot neutralize the toxin which is already bound to the tissues. The dose of antitoxin is given in Table 9.8. A history of previous horse serum administration and any allergic reaction must be sought. A small test dose is given to detect the allergy. Injectable adrenaline must be ready so as to be used in case anaphylaxis occurs.
- Antibiotic (penicillin, amoxicillin or erythromycin for 10-14 days) is given to eliminate *C. diphtheriae* so that the spread of infection to others is prevented. The dose of erythromycin is 500 mg 6 hourly and that of amoxicillin is 500 mg 8 hourly. Clarythromycin and azithromycin are equally effective.

TABLE 9.8: Dose of diphtheria antitoxin

Mild cases	4000-8000 units IM
Moderate cases	16000-40000 units IM
Severe cases	up to 100000 units IV

- An urgent tracheostomy may be needed in case of respiratory obstruction.
- Following recovery, the patient should be immunized with diphtheria toxoid.
- The patient is isolated till three daily consecutive cultures are negative for the organism.

Prophylaxis

Active immunization for diphtheria is a part of immunization program in children. A combined vaccine DPT is used (Table 9.1). Susceptible contact person should be given booster dose of toxoid and a course of antibiotic.

ENTERIC FEVER (TYPHOID FEVER)

Enteric fever (typhoid fever) is a systemic disease caused by *Salmonella typhi* and *Salmonella paratyphi*. Humans are the only host for *S. typhi* and *S. paratyphi*.

- In most, the infection results from consumption of contaminated food or water.
- The sources of infection are acutely infected patients or chronic carriers.
- Incubation period is 5-14 days.

Pathophysiology

Organisms enter the intestinal epithelial cells and subsequently invade and multiply in macrophages in Peyer's patches, mesenteric lymph nodes and the spleen. This is followed by a phase of bacteremia. The organisms may spread to the lungs, gallbladder, kidneys and central nervous system. Finally, bacteria localize mainly in the lymphoid tissues of the distal small intestine. The typical lesion occurs in the Peyer's patches in small intestine which become swollen, ulcerate and heal. Bleeding and perforation may occur.

Clinical Features

The onset is insidious.

- Initially patient presents with prodromal symptoms such as malaise, headache, cough, sore throat, abdominal pain

and constipation followed by the appearance of fever. Children may present with diarrhea and vomitings.

- Fever typically rises in stepwise fashion for few days and then attains plateau. It disappears over 7-10 days in uncomplicated cases.
- There may be relative bradycardia.
- Spleen becomes palpable in second week.
- Rashes (rose spots) may appear on the back and upper abdomen at the end of first week. These are pink maculopapular rashes that fade on pressure and disappear in 3-4 days.

About 5% of patients with enteric fever become long-term asymptomatic carriers who continue to shed bacteria in stool or urine. Women, patients with gallstones or gastrointestinal malignancies are more likely to become carriers.

Complications

Important complications are intestinal perforation and/or intestinal hemorrhage which occur in untreated cases in third and fourth week. Presence of tachycardia, hypotension, abdominal pain and tenderness, dark or fresh blood in stool and leukocytosis suggest these complications. Other complications are shown in Table 9.9.

Investigations

- Blood culture is positive for *S. typhi* or *S. paratyphi*. The yield is higher in first week of illness (90% in untreated cases) but decreases to 50% by third week.
 - Culture of bone marrow may be positive when blood cultures are negative.
 - Stool, urine, rose spots and gastrointestinal secretions can also reveal positive culture. Stool culture is positive during the third week of infection.

TABLE 9.9: Complications of enteric fever

Intestinal perforation
Intestinal hemorrhage
Others
Osteomyelitis
Arthritis
Cholecystitis
Meningitis
Myocarditis
Nephritis
Parotitis
Psychosis

- White blood cell counts are normal but leukopenia can occur. Leukocytosis may occur in children or when infection is complicated by intestinal perforation or hemorrhage.
- Serological tests are available but these are not reliable because of high false-positive and false-negative rates. Widal test detects antibodies to the causative organisms and is usually positive by the second week. Administration of antibiotics or previous typhoid vaccination may interfere with the interpretation of Widal test.

Treatment

Treatment includes administration of antibiotics parenterally or orally depending on patient's condition. Effective antibiotics are fluoroquinolones, third generation cephalosporins (ceftriaxone, cefotaxime, cefixime and cefpodoxime) and azithromycin (Table 9.10). Ampicillin, amoxycillin, chloramphenicol and trimethoprim-sulfamethoxazole (TMP-SMX) are not used now because of resistance. Quinolone resistance has also been reported. Dexamethasone is indicated in severe cases (altered sensorium or shock) as a single dose of 3 mg/kg followed by 8 doses of 1 mg/kg every 6 hourly.

Carriers are treated with oral antibiotics (amoxycillin, ciprofloxacin, TMP-SMX) for 6 weeks.

Prevention

- Immunization is indicated in close household contacts, travelers to endemic area and during epidemics. Multiple dose oral vaccines or single dose parenteral vaccines are available. These provide protection for 5 and 3 years respectively.
- Other preventive measures are proper waste disposal and protection of food and water from contamination.
- Carriers should not work as food handlers.

TABLE 9.10: Usual antibiotics for enteric fever

Drug	Dose	Duration
Ciprofloxacin	500-750 mg twice daily	10-14 days
Ofloxacin	300-400 mg twice daily	10-14 days
Levofloxacin	500 mg daily	10-14 days
Ceftriaxone	1-2 g IV or IM daily	10-14 days
Azithromycin	1 g once daily	5 days

GONORRHEA

Gonorrhea is caused by *Neisseria gonorrhoeae*, a gram-negative diplococcus. It is transmitted through sexual activity (vaginal, oral, anal sex). Sexually active young population is the most commonly affected one. The incubation period is 2-10 days.

Clinical Features

Infection may be asymptomatic, or they may present as the following:

- In males, urethra is most commonly involved. Dysuria and purulent urethral discharge are initial symptoms. It may progress to involve the prostate, epididymis and periurethral glands. Chronic infection may lead to urethral stricture and prostatitis. In homosexual men, rectal infection may occur which is either asymptomatic or presents with discharge, discomfort or rectal bleeding.
- In females, the infection may be asymptomatic in 50% cases. Urethral infection may present as dysuria, discharge or increased frequency. Vaginitis or cervicitis may occur and present as vaginal discharge. Infection of Bartholin's glands and periurethral glands is also common. Rectum may be involved due to anal sex or spread from urogenital infection. Uterus and fallopian tubes may be involved resulting into salpingitis. Chronic salpingitis may lead to scarring of fallopian tubes and sterility. Pelvic inflammatory disease (PID) refers to mixed infection with gonococci, chlamydia and anaerobes. It may present as lower abdominal pain, discharge, pain during coitus and vaginal bleeding.
- Pharyngeal gonorrhea may be asymptomatic or present as sore throat and occurs due to oral sex.
- Eye involvement occurs in the form of conjunctivitis due to direct inoculation of gonococci. The purulent conjunctivitis and edema of eyelids are found in neonates (ophthalmia neonatorum). It may lead to panophthalmitis. Urgent treatment is required to prevent corneal damage and loss of vision.
- Dissemination of gonococci through bloodstream from primary site may cause bacteremia and systemic infection. Intermittent fever, arthritis, tenosynovitis and skin involvement are common. Rash may be maculopapular or hemorrhagic. Endocarditis and meningitis are rare presentations.

Differential Diagnosis

Gonorrhea should be differentiated from non-gonococcal urethritis or vaginitis due to *Chlamydia trachomatis*, trichomoniasis, candidiasis, bacterial vaginosis.

Investigations

- Smear examination of the discharge may show gram-negative diplococci in polymorphonuclear leukocytes.
- Diagnosis is confirmed by culture.
- Ligase chain reaction (LCR) assay may detect both *N. gonorrhoeae* and *Chlamydia trachomatis* in urethral or cervical swab and urine.

Treatment

- Antimicrobial is started as soon as possible. A single dose is curative in uncomplicated infection (Table 9.11). Cefixime and ceftriaxone are first line agents because of fluoroquinolones resistance. In patients with sensitivity to beta lactams, spectinomycin should be used. Doxycycline (100 mg twice daily for 7 days) or azithromycin (1 g orally) is generally added to cover other coinfections like Chlamydia.
- Complicated infection requires prolonged therapy. PID is treated with antimicrobials effective against gonococci, chlamydia and anaerobes. Complicated gonorrhea (salpingitis, prostatitis, arthritis, bacteremia) is treated by any of the following:
 - Penicillin G (10 million units intravenously daily for 5 days)
 - Ceftriaxone (1 g intravenously daily for 5 days)
 - Fluoroquinolones (ciprofloxacin 500 mg twice daily orally for 5 days or levofloxacin 500 mg daily for 5 days).
- All sexual partners and patient should be treated and tested for syphilis and human immunodeficiency virus (HIV).

TABLE 9.11: Treatment of uncomplicated gonorrhea

Cefixime	400 mg	Orally
Ceftriaxone	125 mg	Intramuscularly
Ciprofloxacin	500 mg	Orally
Ofloxacin	400 mg	Orally
Levofloxacin	250 mg	Orally
Spectinomycin	2 g	Intramuscularly

Prevention

The preventive methods include education, and adoption of mechanical measures like condoms. Sexual partners of the patient should be treated. The patient is advised to abstain from sex until therapy is completed. Effective drugs in therapeutic dosage if taken within 24 hours of exposure may abort an infection.

SYPHILIS

Syphilis is sexually transmitted chronic systemic infection caused by *Treponema pallidum*. It is transmitted during sexual contact through abrasions in skin or mucous membranes. It can be transferred from mother to fetus via placenta (congenital syphilis).

Transmission may also occur by blood transfusion, kissing and percutaneous injury.

Clinical Manifestations

The clinical features of syphilis are broadly classified as early and late syphilis.

- a. Early syphilis includes primary syphilis, secondary syphilis and latent syphilis.
- b. Late syphilis includes ‘Late’ latent syphilis and tertiary syphilis (benign tertiary syphilis, cardiovascular and neurosyphilis).

Early Syphilis

Primary Syphilis

Primary syphilis is the first stage in the natural history of the disease.

- The incubation period is 10-90 days (average 2-4 weeks).
- Typical lesion is a chancre, which develops at the site of the inoculation, usually located on the penis in heterosexual males. It begins as a painless papule that ulcerates and becomes indurated.
- The inguinal lymph nodes are enlarged, rubbery, discrete and non-tender. It heals without treatment in 2-6 weeks; a scar may form, particularly if secondary bacterial infection has occurred.
- In homosexual male, lesion can occur in rectum or anal canal. In female patients, cervix and labia are common primary sites. Atypical primary sites of chancre include tongue, tonsils, nipple and fingertips.

The differential diagnosis of primary chancre is chancroid (painful nonindurated ulcer), lymphogranuloma venereum, genital herpes and neoplasm.

Secondary Syphilis

Secondary stage occurs 6-8 weeks after the development of chancre. The manifestations are due to the dissemination of *Treponema pallidum* to distant sites. Main features are fever, rash and lymphadenopathy.

- Rashes are non-pruritic and symmetrical. The palm and soles are generally involved. Rashes may be in the form of macular, papular, pustular or follicular or a combination of these but not vesicular.
- Mucosal patches and ulcers may occur in the oral cavity, and genitalia. These are highly infectious. In the oral cavity the lips, oral mucosa, tongue, palate and pharynx may be involved. Differential diagnosis of oral ulcers in secondary syphilis is given in Table 9.12.
- “Weeping” papules over warm and moist areas like vulva and perianal region are called *candyloma lata*.
- Lymphadenopathy is generalized, non-tender, discrete and rubbery.
- Other features are meningitis, cranial nerve palsies, uveitis, hepatitis, gastritis, periosteitis, glomerulonephritis. alopecia (moth-eaten appearance) may also occur.

Latent Syphilis

Latent syphilis is characterized by positive serological tests for syphilis, absence of clinical manifestations and normal CSF.

- a. The latent phase during first year of infection is known as *early latent syphilis* when the patient is sexually infectious.
- b. *Late latent syphilis* begins after 1 year of infection in untreated patients. This stage is sexually non-infectious.

TABLE 9.12: Differential diagnosis of oral lesions in secondary syphilis

Aphthous ulcers
Ulcerative stomatitis
Herpes simplex
Infectious mononucleosis
Primary HIV infection
Agranulocytosis
Stevens-Johnson syndrome
Behcet syndrome
Reiter’s syndrome

However, transmission of infection through blood or placenta may occur in both early and late latent syphilis. Latent syphilis may persist throughout life or could end in the development of late syphilis (tertiary syphilis) in 30 to 40% cases.

Tertiary Syphilis

Benign Tertiary Syphilis

Benign Tertiary Syphilis may develop between 3-10 years of infection and can involve skin, mucous membrane, bone, muscle or viscera. Chronic granulomatous lesion (gumma) is the characteristic feature. Gumma can be single or multiple and most commonly involved sites are the skin, bone, mouth, upper respiratory tract, liver and the stomach. Upper respiratory gumma can lead to perforation of the palate or nasal septum. Gummatous tongue may heal to develop leukoplakia which is premalignant.

Other causes of granulomatous lesions such as tuberculosis, sarcoidosis, leprosy and fungal infection must be considered in the differential diagnosis of gumma.

Cardiovascular Syphilis

Cardiovascular Syphilis may present after many years of infection. Aorta, coronary ostia and aortic valve may be affected leading to aortic aneurysm, angina and aortic regurgitation.

Neurosyphilis

Neurosyphilis takes years to develop. Neurosyphilis may be asymptomatic or symptomatic.

- Asymptomatic neurosyphilis is characterized by CSF abnormalities in the absence of clinical signs.
- Symptomatic neurosyphilis may present as:
 - a. Meningovascular syphilis
 - b. General paresis of insane
 - c. Tabes dorsalis

Some important clinical features are Argyll Robertson pupils and Charcot's joints. *Argyll Robertson pupil* is small irregular pupil which reacts to accommodation but not to light.

Congenital Syphilis

Transmission of infection from untreated mother to fetus transplacentally can lead to congenital syphilis. Untreated

maternal syphilis can lead to still birth, premature birth and neonatal death. A child may be born with syphilitic features, same as in secondary syphilis, or may develop manifestations during first few weeks of life.

- The features of early syphilis are rhinitis (snuffles), maculopapular rash, condyloma lata, mucous patches, periosteitis, lymphadenopathy, hepatosplenomegaly, anemia and thrombocytopenia. This stage is infectious.
- Some children remain in latent phase and develop features of congenital syphilis or residual stigmata after 2 years. Late congenital syphilis presents as benign tertiary syphilis, interstitial keratitis, deafness, neurosyphilis, and Clutton's joint (bilateral knee effusion).
- Stigmata include *Hutchinson's teeth*, *mulberry molars*, frontal bossing, saddle nose, *maxillary hypoplasia*, high arched palate, rhagades and saber tibia.
 - a. Hutchinson's teeth are widely spaced peg-shaped notched upper central incisors.
 - b. Mulberry molars are sixth year molars with poorly developed cusps.
 - c. Rhagades are linear scars at angle of the mouth, nose and anus following secondary bacterial infection of the rashes.

Investigations

T. pallidum cannot be cultured *in vitro* hence diagnosis depends on (a) Microscopic examination, (b) serological tests and (c) other tests.

- a. *Dark-field microscopic* examination may show motile spirochaetes from lesions in infectious syphilis. Fluid from chancre or moist cutaneous lesions in secondary syphilis or congenital syphilis is used for this purpose. Organism is not found in late syphilis.

The direct fluorescent antibody *T. pallidum* test (DFA-TP) is also used to detect the pallidum in fixed smears prepared from lesions. PCR test is also used in some centers.

- b. *Serological tests*: Tests used in the diagnosis of syphilis are given in Table 9.13.

Non-treponemal tests measure the ability of serum to flocculate cardiolipin-lecithin-cholesterol antigen complex. These tests are used for primary screening and are positive 4 weeks after the primary infection, secondary syphilis and at birth in congenital syphilis.

TABLE 9.13: Serological tests for syphilis**Non-treponemal tests**

Venereal diseases research laboratory test (VDRL)

Rapid plasma reagin test (RPR)

Treponemal antibody tests (specific)

Fluorescent treponemal antibody-absorbed test (FTA-ABS)

T. pallidum hemagglutination test (TPHA)Agglutination assays for antibodies to *T. pallidum*.

False positive reactions may occur in other diseases and diagnosis must be confirmed by specific treponemal antibody test. The titer correlates with disease activity and can be used to monitor the response to therapy.

Specific tests are used to confirm the diagnosis as these are negative in non-treponemal infections. These are positive in primary and secondary syphilis.

- c. *Other tests:* CSF examination is required in late syphilis such as in benign tertiary syphilis, cardiovascular syphilis and neurosyphilis. It should also be examined in seropositive patients with neurological signs. Biopsy may be required to diagnose gumma. Chest X-ray, ECG and echocardiography are done to detect cardiovascular syphilis.

Treatment

Penicillin G is the drug of choice in all stages of syphilis. Alternatively tetracycline, doxycycline, cephalosporin, and azithromycin can be used in penicillin allergic patients (Table 9.14).

- a. *Treatment of early syphilis (primary, secondary and early latent)* Single dose of benzathine penicillin G is the drug of choice. Alternatively, two weeks course of tetracycline or doxycycline can be given in penicillin sensitive patients. Ceftriaxone (for 8-10 days) or azithromycin (single dose) are also effective.

TABLE 9.14: Drugs used in syphilis

Benzathine penicillin G	2.4 million units	IM, single dose
Aqueous penicillin G	18-24 million unit daily	IV, divided doses
Procaine penicillin G	2.4 million unit daily	IM, divided doses
Probenecid	500 mg q.i.d.	Oral
Doxycycline	100 mg b.d.	Oral
Tetracycline	500 mg q.i.d.	Oral
Azithromycin	2 g	Oral single dose
Ceftriaxone	1 g daily	IM/IV

- b. *Treatment of late syphilis (late latent, cardiovascular, benign tertiary):* Lumbar puncture is done to examine CSF. If CSF is abnormal, patient is treated for Neurosyphilis.

- Patients with normal CSF are treated with benzathine penicillin weekly for 3 weeks. Alternatively, doxycycline or tetracycline is given for 4 weeks.
 - Neurosyphilis (asymptomatic or symptomatic) is treated with aqueous penicillin G for 10-14 days. Probenecid is added if procaine penicillin G is given.
- c. The only drug indicated in neurosyphilis, pregnancy with syphilis and HIV coinfecting with syphilis is aqueous penicillin G. Patients allergic to penicillin are treated after desensitization.
- d. *Jarisch Herxheimer reaction:* This is a mild acute febrile reaction which occurs following initiation of treatment. The features are fever, chills, myalgia, headache, tachycardia and hypotension. The reaction occurs due to release of toxic products after massive destruction of spirochaetes. Treatment is symptomatic. The symptoms generally resolve within 24 hours.

Prevention

Use of condom is helpful in prevention. Following exposure to infectious syphilis, azithromycin 1 gm single dose or procaine penicillin G 2.4 million units IM may be used to prevent infection.

AMOEBIASIS

Amoebiasis is an infection caused by the intestinal protozoan *Entamoeba histolytica*. The infection is spread by cysts which are present in water and uncooked food contaminated by human feces. About 90% of infections are asymptomatic.

Pathogenesis

- The trophozoites are released from cysts in the intestine and invade the mucous membrane of the large bowel, chiefly the cecum. They can produce flask-shaped ulcers surrounded by healthy mucosa. These ulcers can bleed or lead to perforation.
- Rarely, intestinal infection may lead to formation of mass lesion (*amoeboma*).
- The trophozoites may reach the liver through the portal vein and cause parenchymal destruction leading to the

formation of *amoebic liver abscess*. The abscess is usually found in right lobe of the liver.

- Other extraintestinal sites such as genitourinary tract, skin and cerebrum may become involved rarely.

Clinical Manifestations

- In endemic areas, around one-third of the population is asymptomatic passers of amoebic cysts.
- The usual symptom is frequent loose stools containing mucus with or without blood. The stool has an offensive odor. This is known as *amoebic dysentery*. There may also be mild abdominal pain.
- On palpation, the colon particularly cecum is tender. Cecal involvement may mimic acute appendicitis.
- The acute diarrheal presentation with blood in stool mimics bacillary dysentery or ulcerative colitis.
- The presentation may be severe in elderly and in immunocompromised individuals.

Amoebic Liver Abscess

- The symptoms are fever with chills, malaise and pain in right upper abdomen and right shoulder.
- The liver is enlarged and tender. Jaundice is rare.
- The large abscess may rupture through the diaphragm into the lung. The contents of the abscess thus may be coughed out. Occasionally the abscess may rupture into the pleural cavity, peritoneal cavity or pericardial sac.
- The diagnosis is suspected on clinical grounds and confirmed by ultrasonography.

Investigations

1. Fresh stool examination reveals the presence of trophozoites or amoebic cysts. Trophozoites are diagnostic of acute dysentery. The stool is positive for heme (occult blood).
2. Sigmoidoscopy may reveal typical flask-shaped ulcers. Biopsy of the margin of the ulcer may show the presence of trophozoites.
3. X-ray chest may reveal raised diaphragm in cases of liver abscess.
4. Ultrasonography confirms the diagnosis of liver abscess (Figs 9.6 and 9.7). Amoebic liver abscess is generally solitary. Presence of multiple abscesses indicates pyogenic liver abscess.

5. Aspiration of abscess: The aspirated fluid is pink to chocolate brown color (*anchovy sauce*) and may contain free amoebae.
6. Immunofluorescence techniques show the presence of antibodies in the serum of about 95% patients with amoebic liver abscess and 60% with amoebic dysentery.
7. Blood tests reveal mild leukocytosis, normal liver transaminases and raised alkaline phosphatase.

Management

- The drug of choice is metronidazole in dosage of 750-800 mg 8 hourly for 5-10 days. The common side effects of metronidazole are metallic taste and disulfiram like reaction. Longer acting nitroimidazoles (tinidazole,

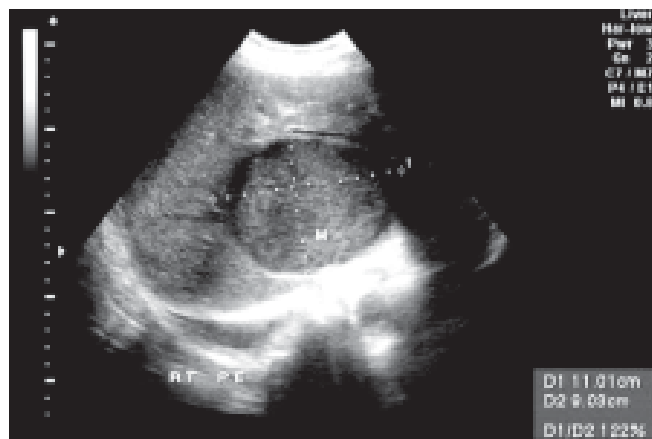


FIGURE 9.6: Ultrasound showing solitary abscess (Amoebic liver abscess)

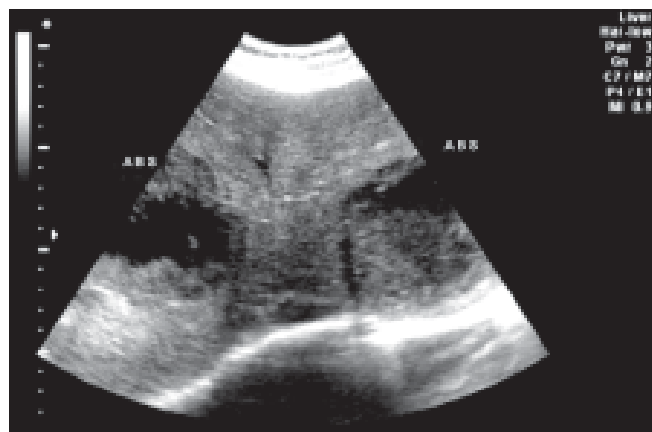


FIGURE 9.7: Ultrasound showing multiple abscesses (pyogenic liver abscess)

- Asymptomatic cyst passers or carriers can be given luminal agents like diloxanide furoate (500 mg 8 hourly for 10 days) or iodoquinol (650 mg 8 hourly for 20 days).
- The amoebic liver abscess is treated with medications as mentioned above. The aspiration of the amoebic liver abscess is indicated if it is large and threatens to burst or does not respond promptly (within 72 hr) to drug therapy. Aspiration may also be needed to differentiate it from pyogenic liver abscess especially in case of multiple abscesses.

Malaria is a parasitic infection caused by *Plasmodium vivax*, *P. falciparum*, *P. ovale* and *P. malariae*. Globally around 1.5-3 million people die each year due to malaria, mostly due to *P. falciparum* infection. Currently malaria is appearing in the areas from where it had disappeared.

Malaria is transmitted from human to human through bite of infected female *Anopheles* mosquitoes. There is no animal reservoir for human malaria. Other rare routes of transmission are transplacental and through blood transfusion.

The mosquito is infected with the sexual form of parasite (gametocyte) when it feeds on human blood. These gametocytes develop into sporozoites in the salivary gland of the mosquito within 7-20 days.

Merozoites leave the liver and infect erythrocytes. In the erythrocytes, merozoites go through the stages of ring form, trophozoite and schizont (Fig. 9.8). The rupture of

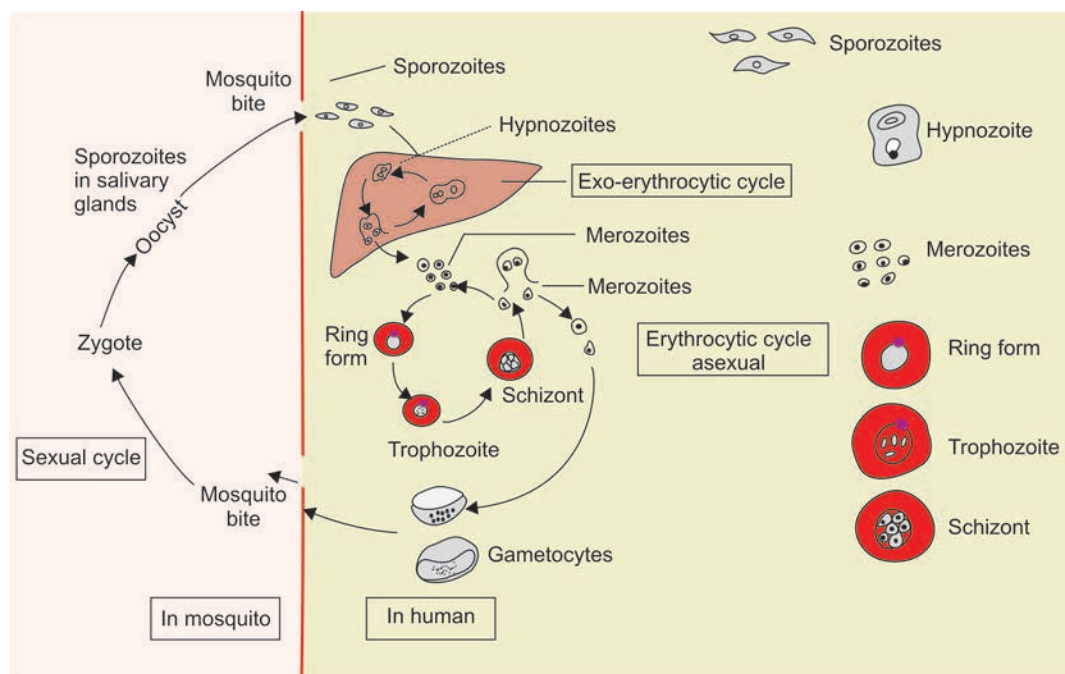


FIGURE 9.8: Life cycle of malarial parasite

schizont causes release of plenty of newly produced merozoites in the blood (asexual erythrocytic stage). This event is accompanied with the febrile episode.

The newly released merozoites invade erythrocytes to repeat the erythrocytic cycle. However, some merozoites develop into gametocytes (sexual form) that can transmit malaria.

Recurrent malaria can occur in all four types of plasmodium infections due to the failure of treatment to eradicate all infected erythrocytes (*recrudescence*).

Pathogenesis

P. falciparum invades erythrocytes of all ages whereas *P. vivax* and *P. ovale* preferably invade reticulocytes and *P. malariae* normoblasts. Parasitemia is most severe in *P. falciparum* infection; more than 2% of all erythrocytes can be infected. That is why the hemolysis is most severe in *P. falciparum* infection. Erythrocytes infected with *P. falciparum* adhere to capillary endothelium (cytoadherence), agglutinate with other infected erythrocytes and form rosettes with uninfected erythrocytes. This leads to ischemia and dysfunction of various organs such as brain, liver, kidney, lungs and the gut.

Clinical Features

The approximate incubation period is 12 days in case of *P. falciparum*, 14 days for *P. vivax* and *P. ovale* and 30 days for *P. malariae*.

The typical clinical feature is fever with chills and rigors. This may be associated with nonspecific symptoms such as headache, fatigue, myalgia, arthralgia and abdominal symptoms like anorexia, nausea, vomiting, diarrhea and pain. Initially, there is shaking chills for few hours (cold stage) followed by high fever (hot stage) and marked sweating (sweating stage). The fever is intermittent which may occur every other day in *P. vivax* and *P. ovale* infection (*tertian malaria*), every third day in *P. malariae* (*quartan malaria*). *P. falciparum* has no specific pattern of fever.

The patients may have anemia and mild jaundice due to hemolysis. Splenomegaly is usual while mild hepatomegaly may also occur.

Severe Falciparum Malaria

Severe falciparum malaria is most dangerous form of malaria and is responsible for most of the deaths. Cerebral malaria is the most serious complication. It is characterized by confusion, coma, convulsions and neurological signs. Severe intravascular hemolysis may lead to hemoglobinuria (*black water fever*). Other complications of falciparum malaria are given in Table 9.15. Complications are more frequent in children and pregnant females.

Chronic Complications of Malaria

Tropical splenomegaly (hyperactive malarial splenomegaly): Abnormal immune response to repeated attacks of malaria may result in massive splenomegaly, hepatomegaly and raised titers of IgM and malarial antibodies.

Nephrotic syndrome: Chronic and repeated infections with *P. malariae* may cause immune-mediated injury to renal glomeruli and cause nephrotic syndrome.

Investigations

- Thick and thin blood films stained with Giemsa and Wright's stain are examined for the presence of sexual and asexual forms of the parasites. The presence of asexual forms is important for the diagnosis of the infection. Sexual forms appear late and may persist even after the treatment of the infection.
- Antibodies are detectable 8-10 days after the onset of symptoms and persist for years. Hence, serological tests are neither helpful in the diagnosis of initial acute attack nor in the differentiation of current or past infections.

TABLE 9.15: Complications of falciparum malaria

Cerebral malaria
Acute noncardiogenic pulmonary edema (ARDS)
Acute tubular necrosis and renal failure
Hyperpyrexia
Hypoglycemia
Severe anemia
Metabolic acidosis
Bleeding and disseminated intravascular coagulation (DIC)
Hypotension and shock
Convulsions

- Antigen detection methods (ELISA and dipstick format), polymerase chain reaction and indirect fluorescent antibody tests are commercially available for the diagnosis.
- Normocytic normochromic anemia is usually present. Laboratory findings in severe falciparum malaria include altered hepatic and renal function tests, acidosis and hypoglycemia.

Treatment

Several drugs are available for the treatment of malaria. The drugs are given in Table 9.16.

- Only primaquine acts against hypnozoites and prevent relapse in *P. vivax* and *P. ovale* (radical cure). The gametocytes of *P. falciparum* are killed by primaquine. Primaquine is contraindicated in patients with G6PD deficiency.
- Chloroquine, quinine, artesunate, pyrimethamine and mefloquine act on blood schizonts and gametocytes of *P. vivax*, *P. ovale* and *P. malariae*.
- Chloroquine is the drug of choice in uncomplicated malaria except in chloroquine resistant *P. falciparum* infection. Primaquine is added in the case of *P. vivax* and *P. ovale* infection to destroy the hypnozoites and to prevent relapse.
- In severe attacks, parenteral therapy with quinine is started and switched to oral as soon as possible. Primaquine is added in case of *P. vivax* and *P. ovale* infections. Parenteral artesunate, artemether or chloroquine are other alternatives and can be used in place of quinine.

Treatment of Resistant Malaria

- The resistance to chloroquine is usually observed in *P. falciparum* malaria. The treatment is started with oral quinine sulfate plus doxycycline, tetracycline or clindamycin. WHO recommends the use of combination therapy with artemether plus lumefantrine for treatment of resistant falciparum malaria in most regions. Alternative drugs are mefloquine, artesunate followed by mefloquine and halofantrine.
- Severely ill patients who cannot take medication orally are given IV quinine plus doxycycline, tetracycline or clindamycin. Oral treatment is started as soon as possible. Alternative to quinine is parenteral artesunate or artemether followed preferably by mefloquine.

TABLE 9.16: Drugs used in malaria

Drug	Adult dose and side effects
Chloroquine	1g salt (600 mg base) initially, then 0.5 g (300 mg base) at 12, 24 and 36 hours. Parenteral dose: IV infusion; 10 mg/kg base in 8 hours then 15 mg/kg in next 24 hours or seven IM doses of 3.5 mg base/kg every 6 hour. Side effects: Nausea, postural hypotension, distaste Cardiac arrhythmia and severe hypotension if given IV
Quinine	Oral: 10 mg salt/kg thrice a day for 7 days plus Doxycycline 100 mg twice a day for 7 days or tetracycline 500 mg 4 times a day for 7 days or clindamycin 900 mg 3 times a day for 7 days. Parenteral dose: IV infusion 20 mg of salt/kg over 4 hours followed by 10 mg/kg infused over 2-8 hours every 8 hours. Start oral therapy when possible. Side effects: Nausea, vomiting, tinnitus, hearing loss (cinchonism), hypotension if given IV
Mefloquine	Oral: 750 mg followed by 500 mg after 12 hours. Side effects: Convulsion, neuropsychiatric symptoms
Sulfadoxine/ pyrimethamine	25/1.25 mg/kg single oral dose (3 tablets of 500/25 mg) Side effects: Hypersensitivity, megaloblastic anemia
Primaquine	15 mg base daily for 14 days Side effects: Hemolysis (in G6PD deficiency patients) nausea, diarrhea, methemoglobinemia
Artesunate	Oral : 4 mg/kg followed by 2 mg/kg per day for 4 days Parenteral: 2.4 mg/kg IV followed by 1.2 mg/kg at 12 and 24 hours then once daily Side effects: Anaphylaxis, urticaria, fever, lowers reticulocyte count
Artemether	Oral dose same as artesunate Parenteral: 3.2 mg/kg IM followed by 1.6 mg/kg daily Side effects same as artesunate

Cerebral Malaria

Patients are hospitalized promptly. The management includes appropriate antimalarial chemotherapy, detection and treatment of fluid, electrolyte, acid-base abnormalities and other complications.

Prophylaxis

Chemoprophylaxis should begin one week before entering the malarious area and should be given till 4 weeks after leaving it. The drug of choice for prophylaxis is chloroquine (150 mg base 2 tablets weekly). In regions with chloroquine resistant malaria, proguanil (100 mg once a day) plus atovaquone (250 mg), mefloquine (250 mg weekly) or doxycycline (200 mg daily) can be used.

Other preventive measures are wearing long sleeve shirts and trouser, using mosquito repellents and nets and using screens in houses.

HUMAN IMMUNODEFICIENCY SYNDROME/ ACQUIRED IMMUNODEFICIENCY SYNDROME (HIV/AIDS)

HIV/AIDS is a global pandemic. It was first detected in USA in 1981 whereas the first case of HIV infection was diagnosed in India in 1986. Now almost all countries of the world have it.

- According to a recent estimate around 30-40 million people are infected with HIV and 4 million people have AIDS worldwide.
- India is next to South Africa in having maximum number of HIV positive people. According to an estimate, there are around 2.3 million HIV positive individuals in India. Worst affected states are Tamil Nadu, Maharashtra, Andhra Pradesh, Karnataka, Manipur and Nagaland.

Mode of Transmission

- The most common mode of transmission of HIV in India as well as worldwide is heterosexual contact. In India, around 85-90% patients have received infection through unsafe sex with commercial sex workers (CSW), mostly in metropolitan cities. In Mumbai, about half of the CSW are HIV positive.
- Other modes of transmission are through contaminated blood or blood products, contaminated syringes (as in IV drug abusers) and from infected mother to fetus/newborn baby (Table 9.17).
- The transmission of HIV from mother to child occurs mostly during labour although it can happen during any time in pregnancy or after birth, through breastfeed.
- Stool, saliva, sweat, urine have not shown to transmit infection although viral particles have been demonstrated in these fluids.

TABLE 9.17: Modes of transmission of HIV

1. Sexual route (heterosexual, homosexual)
2. Parenteral route
 - a. blood and blood products
 - b. injection drug users
 - c. occupational injury
3. Mother to child transmission

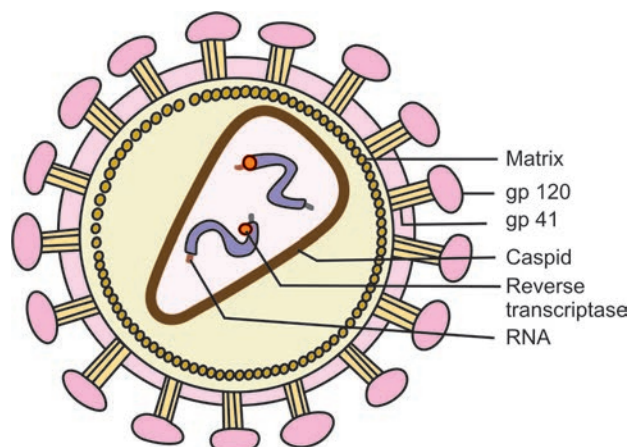


FIGURE 9.9: Human immunodeficiency virus

Causative Organism

Most cases of AIDS (98%) are due to HIV 1 while HIV 2 is responsible in only few cases particularly in Africa. The course of HIV 2 is less aggressive. Human immunodeficiency virus is a retrovirus belonging to lentivirus family (Fig. 9.9). It is a RNA virus. HIV 1 is subdivided into three groups, M (major), O (outlier) and N. The most common group is M, which is further classified into nine subtypes or clades, lettered A, B, C, D, F, G, H, J, and K. Subtype C is prevalent in India whereas subtype B is common in North America and Europe.

Pathogenesis

HIV infects CD₄ positive immune cells chiefly T helper lymphocytes, but other cells like B lymphocytes, monocytes, dendritic cells and microglial cells are also infected. A glycoprotein gp 120 present on the surface membrane of the virus has high affinity with the CD₄ molecule present on the surface of immune cells. Certain coreceptors such as CXCR4 (on lymphocytes) and CCR5 (on monocytes) are also needed for the binding of the virus over cell surface. Following binding with the lymphocytes, HIV enters the cell (Fig. 9.10). Its genomic RNA is released in the cytoplasm and converted into viral DNA by a unique enzyme *reverse transcriptase*. Viral DNA is integrated into the host cell DNA and captures the genetic machinery of the host cell. This leads to the rapid production of viral genome which attains the shape of full virus with the help of protease enzymes.

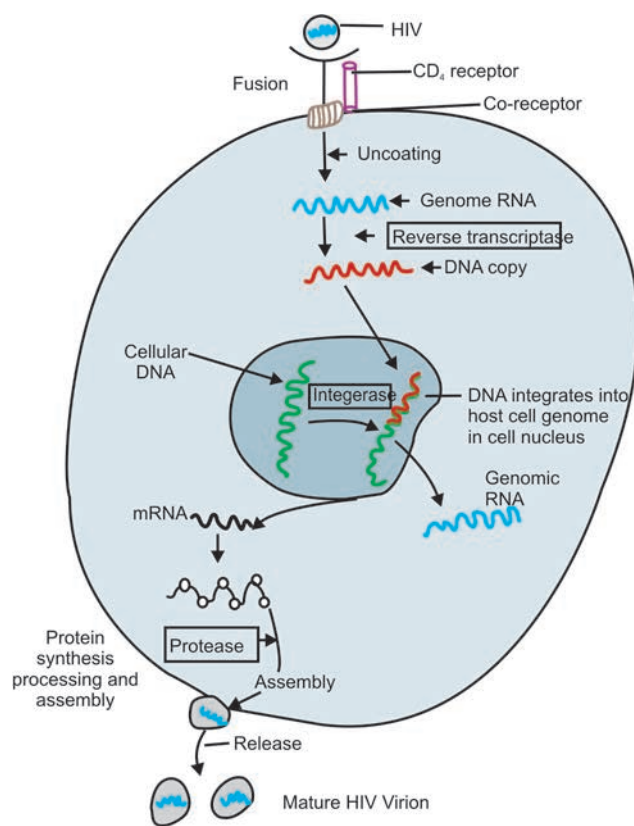


FIGURE 9.10: Pathogenesis of HIV disease

Clinical Stages

The course of HIV disease follows various stages (Table 9.18).

Acute HIV Syndrome

About half to two-third of the patients infected with HIV develop prodromal symptoms 3-6 weeks after infection. This stage is known as acute HIV syndrome and is characterized by fever, myalgia, arthralgia, rashes, lymphadenopathy, thrombocytopenia and neurological syndromes. In some this phase goes unnoticed. The symptoms spontaneously resolve within 3-4 weeks. The plasma viral load is very high during this stage.

TABLE 9.18: Clinical stages of HIV disease

1. Acute HIV syndrome
2. Asymptomatic stage
3. Symptomatic stage
 - a. mildly symptomatic disease
 - b. advanced disease (AIDS)

Asymptomatic Stage

Following resolution of the acute HIV syndrome, infected individuals remain asymptomatic (clinical latency) for 2-8 years, although HIV continues to multiply and damage the immune system of the patient. There is a progressive decline in the CD₄ count.

Mildly Symptomatic Disease

There is mild immune deficiency leading to manifestations, although these are not AIDS defining illnesses. The manifestations are oral or vaginal candidiasis, hairy cell leukoplakia, herpes zoster, bacillary angiomatosis, pelvic inflammatory disease and idiopathic thrombocytopenic purpura.

Acquired Immunodeficiency Syndrome

As the disease progresses, there is a significant decrease in the CD₄ count.

- Constitutional symptoms include unexplained weight loss (>10%), fever (>1 month) and chronic diarrhea. The patient may become severely cachexic (Fig. 9.11). Persistent generalized lymphadenopathy may occur.
- A number of opportunistic infections may occur when CD₄ count falls below 200/mm³.
- Important opportunistic infections are tuberculosis (pulmonary and extrapulmonary), oro-esophageal



FIGURE 9.11: Severely cachexic AIDS patient

candidiasis (Fig. 9.12), cryptococcal infection, toxoplasmosis, cytomegalovirus and *Pneumocystis carinii* pneumonia.

- Tuberculosis is the most common opportunistic infection in HIV patients in India.
- Patients may also have malignancies like Kaposi sarcoma and lymphoma.
- Diffuse skin eruption due to *Molluscum contagiosum* may be seen in patients with advanced HIV disease (Fig. 9.13).

Diagnosis

- The diagnosis of HIV infection is generally based on the detection of antibodies against viral proteins in the serum of the patients. The tests are ELISA and western blot. The former is more sensitive and is used for the screening whereas the later is more specific and is needed to confirm the diagnosis if the ELISA test is inconclusive.
- According to NACO guidelines, two different ELISA in symptomatic and three different ELISA in asymptomatic are required to be positive for the diagnosis of HIV infection.



FIGURE 9.12: Oral candidiasis



FIGURE 9.13: Molluscum contagiosum

- ELISA test may be negative during the first 3-6 weeks of infection when detectable amount of antibodies may not have formed. During this 'window period' tests based on detection of antigens (p24) or viral RNA through RT-PCR (reverse transcriptase-polymerase chain reaction) are recommended.
- The CD₄ count, measured by flow cytometry tells about the current immune status of the patient and guides us to start prophylactic therapy and anti-retroviral treatment. Viral RNA load suggests the prognosis of the patient.
- Patients are investigated for the evidence of co-infection with hepatitis B, and hepatitis C and positivity for syphilis (VDRL test).
- Investigations are also performed, if need be, to detect the presence of opportunistic infections.
- Besides these, complete blood count, liver function tests, blood urea, serum amylase, blood sugar and lipid profile are done as the baseline before initiation of anti-retroviral therapy.

Management of HIV Infection

The management of HIV disease starts the day the patient is diagnosed. The initial assessment includes a detailed history, mainly about occupation, mode of infection, fever, weight loss, loose motions, cough, dyspnea, rashes, lymphadenopathy, neurological problems, oral ulcer. The examination must include a detailed general and physical examination. Following are the main components of the management:

1. Counseling
 2. Prevention of spread of infection
 3. Prophylaxis of opportunistic infections (OI)
 4. Treatment of opportunistic infections
 5. Anti-retroviral therapy.
1. *Counseling*: Initially patient needs counseling so as to be able to cope up with stress. He is also counseled about the nature of disease and possibilities of the benefits and the risks of the treatment. He should also be convinced as regards regular follow-up visits to the doctor for initial assessment, prophylaxis for opportunistic infections and anti-retroviral therapy.
 2. *Prevention of spread of infection*: HIV positive individuals should also be advised to refrain from donating blood and unsafe sex so that they may not spread HIV infection to others.
 3. *The initiation of primary prophylactic therapy of OI* depends on the CD₄ count. A recommended schedule is given in Table 9.19.
 4. *Management of opportunistic infection*: Prompt diagnosis of OI and effective treatment has resulted into improved survival of HIV patients. The control of OI also retards the progression of HIV. Some infections are to be treated lifelong.

The co-infection of tuberculosis with HIV poses special problem in India and other developing countries. The presence of tuberculosis adversely affects the prognosis of HIV infection and the presence of HIV infection in tuberculosis leads to problem in the diagnosis as the manifestations are more commonly atypical. HIV positive patients are 15 times more prone to have reactivation of tuberculosis than their HIV negative counterparts.

TABLE 9.19: Primary prophylaxis of opportunistic infections

Infections	Indications	Medications
<i>Pneumocystis carinii</i>	CD ₄ count <200/mm ³	Cotrimoxazole, dapsone, pentamidine
Toxoplasmosis	CD ₄ count <100/mm ³	Co-trimoxazole, pyrimethamine plus dapsone
Cytomegalovirus	CD ₄ count <50/mm ³	Gancyclovir
<i>Mycobacterium avium</i> complex (MAC)	CD ₄ count <50/mm ³	Azithromycin, clarithromycin, rifabutin

Treatment of tuberculosis is same in HIV positive individuals as in HIV negatives. However, special precaution should be taken regarding the interaction between rifampicin and anti-retroviral drugs. In such cases, rifabutin is preferred over rifampicin.

5. *Anti-retroviral therapy (ART)*: The availability of anti-retroviral drugs has transformed the prognosis in AIDS patients. Now AIDS is regarded as chronically manageable disease rather than an essentially fatal disease.
 - The anti-retroviral therapy reduces viral load, improves the immune status of the patients (increases CD₄ count) and thus decreases the incidence of opportunistic infections.
 - The decision to start ART is based upon the clinical stage of disease, CD₄ count and HIV viral load. The criteria may differ from country to country.
 - In India, according to recent NACO guidelines (April 2009), the anti-retroviral therapy is recommended in symptomatic AIDS patients or in those with CD₄ count <250/mm³. The ART should also be initiated if patient has CD₄ count between 250-350 and WHO clinical stage II or III. (see NACO website).

Limitations of ART

- ART is not curative. Only 70-80% patients respond to ART.
- The treatment needs to be given life long.
- There is also the problem of drug resistance.
- The anti-retroviral drugs are not free from toxic effects. These can lead to significant complications like gastrointestinal intolerance, bone marrow failure, pancreatitis, lipodystrophy (Fig. 9.14), myopathy, neuropathy and increased cardiovascular mortality. Anemia is a common side effect of zidovudine (AZT) while neuropathy may occur with stavudine therapy.

Anti-retroviral Drugs and Regimens

More than 20 drugs are approved for therapy. Most drugs are directed against viral enzymes such as reverse transcriptase and protease. Recent drugs like entry inhibitors and integrase inhibitors are also being tried. Anti-retroviral drugs are listed in Table 9.20.



FIGURE 9.14: Lipodystrophy

Triple drug therapy is the gold standard. Mono or dual drug combination is contraindicated since these are often ineffective and lead to the emergence of resistance.

The combinations are:

2 NRTI plus 1 NNRTI

2 NRTI plus 1 PI (boosted with ritonavir)

3 NRTI

TABLE 9.20: Anti-retroviral drugs

Reverse transcriptase inhibitors(RTI)

Nucleoside RTI (NRTI)

Zidovudine
Stavudine
Lamivudine
Didanosine
Emtricitabine
Abacavir

Non-nucleoside RTI (NNRTI)

Nevirapine
Efavirenz
Delavirdine

Nucleotide RTI

Tenofovir

Protease inhibitors (PI)

Saquinavir
Ritonavir
Indinavir
Nelfinavir
Atazanavir
Lopinavir
Fosamprenavir

Entry inhibitors

Enfuvirtide (T20)
(fusion inhibitor)

The response to therapy is monitored clinically, immunologically and by viral load reduction. In patients who respond, there is a rise in CD₄ count and decrease in the plasma viral load.

Occupational Exposure and Post-exposure Prophylaxis (PEP)

There is a minimal but definite risk of infection to the health care worker following exposure to patient's blood and body fluids. To minimize this, certain precautions and steps are to be taken. This is known as post-exposure prophylaxis (PEP). The chance of transmission of infection depends on the severity of exposure, the stage of the disease in the patient and measures taken after exposure. The guidelines for PEP are given in Figure 9.15 (NACO guidelines).

Majority of occupational infections occur percutaneously through needle sticks and other instruments contaminated with blood. The mucocutaneous transmission of HIV may also occur. Risk of infection following percutaneous exposure to contaminated blood is about 0.3% and after mucous membrane exposure 0.09%.

Following needle stick injury, the part should be thoroughly washed with soap and water. The injured finger should not by reflex be put in the mouth. Open wounds should be irrigated with saline. Antiseptic agents can be applied but caustic agents are avoided. Mucous membrane is flushed with copious amount of water. Eyes are irrigated with clean water or sterile saline.

The need for PEP is decided according to the NACO guidelines (Fig. 9.15).

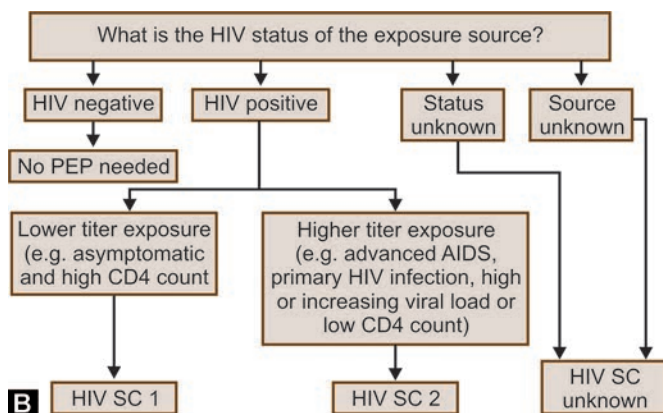
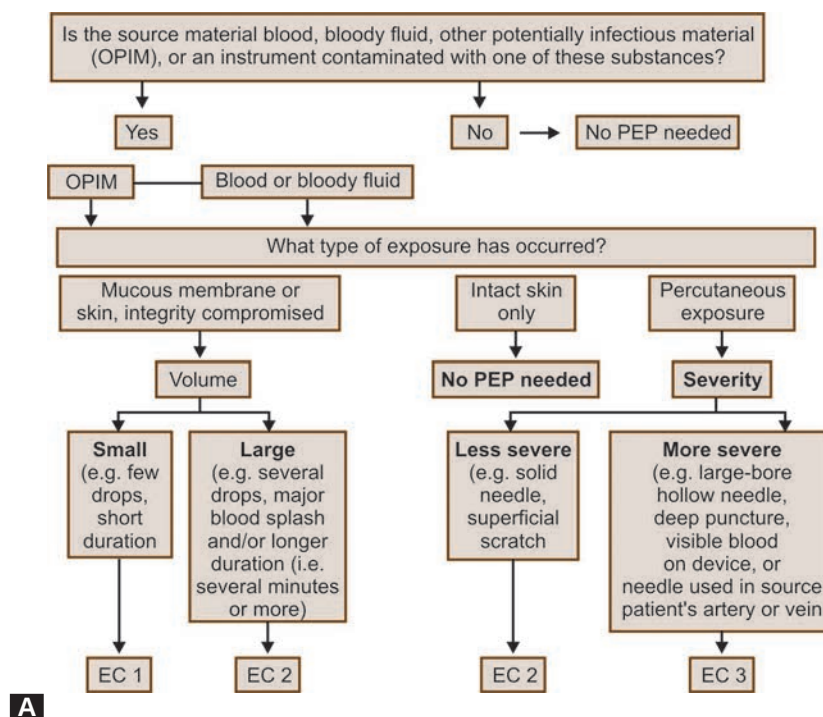
Combination of drugs (two or three drugs) is started as soon as possible, ideally within one to two hours of exposure. The drugs are zidovudine (300 mg BD), lamivudine (150 mg BD) and PI (nelfinavir 1250 mg BD or boosted lopinavir or indinavir). The treatment is given for 4 weeks (Table 9.21).

Universal precaution must be taken to protect health care worker from HIV patients.

Prevention of HIV Infection

The prevention of HIV infection can be achieved by taking the following measures:

- To practice safe sex: Use of condom can effectively reduce the chance of infection.



Determinants of PEP Recommendation

EC	HIV SC	PEP Recommendation
1	1	PEP may not be warranted.
1	2	Consider basic regimen (Negligible risk for HIV transmission)
2	1	Basic regimen (increased risk for HIV transmission?)
2	2	Expanded regimen (increased risk for HIV transmission)
3	1 or 2	Expanded regimen (increased risk for HIV transmission)
2/3	Unknown	Consider basic regimen

FIGURES 9.15A to C: NACO guidelines for PEP

- Use of sterilized syringes and needles and instruments for every injection, immunization and surgery.
- Screening of all blood and blood products for HIV infection.

Mother to Child Transmission of HIV Infection

The success story of ART in the prevention of transmission of HIV infection from mother to fetus/newborn has instilled

hopes. The zidovudine (300 mg twice a day) is given to HIV positive pregnant mothers from the 10th to 12th week of pregnancy or immediately after the diagnosis, if done later. During labor, AZT is given intravenously. The newborn is given syrup AZT 1mg/kg three times a day for 6 weeks. As a result of this, the chances of transmission are reduced by one-third, from 26-30% to 8%. Recent trials have reported the efficacy of single dose of nevirapine (200 mg) given at the time of labor.



FIGURE 9.16: Diffuse infiltrative lymphocytosis syndrome in HIV patient

TABLE 9.21: Regimens for PEP

Basic regimen
Zidovudine 300 mg twice a day
plus
Lamivudine 150 mg twice a day
Expanded regimen
Basic regimen
plus
Indinavir 800 mg thrice a day or Boosted indinavir (Indinavir 800 plus ritonavir 100 mg BD)
or
Boosted Lopinavir (400 mg plus ritonavir 100 mg BD)
or
Nelfinavir 1250 mg twice a day

Oral Manifestations in HIV Disease

Oropharyngeal lesions are common in HIV infection and are mostly due to secondary infections (Table 9.22).

- The presence of *oral candidiasis* (thrush) may be an indicator of underlying HIV disease. Thrush appears as white, cheesy exudates on erythematous mucosa in the posterior oropharynx. Diagnosis is based on the examination of scraping for fungal elements.
- Palatal, glossal or gingival ulcer may also result from *cryptococcal disease* or *histoplasmosis*.

TABLE 9.22: Oral manifestations in HIV infection

Infections

Fungal

- Candidiasis
- Histoplasmosis
- Cryptococcosis

Viral

- Oral hairy leukoplakia (EBV)
- Herpes simplex
- Cytomegalovirus
- Varicella zoster
- Papilloma virus

Bacterial

- Periodontal infections
- Necrotizing ulcerative gingivitis
- Necrotizing stomatitis

Neoplasms

- Kaposi sarcoma
- Lymphoma

- Idiopathic oral aphthous ulcers**
- HIV salivary gland disease (DILS)**
- Cervical lymphadenopathy**

- Oral hairy leukoplakia* is a white corrugated lesion presumed due to EBV. It occurs mostly on the lateral borders of the tongue but may involve adjacent buccal mucosa. It does not transform into malignancy. The presence of this lesion is highly suggestive of concurrent HIV infection and is more prevalent in advanced disease. Diagnosis is clinical and usually no treatment is required as it does not cause any symptoms.
- Herpes simplex* may manifest as recurrent single or multiple oral or labial ulcers. Oropharyngeal ulcers can also be due to *Cytomegalovirus* (CMV). *Herpes zoster* can lead to ulcers in the trigeminal nerve distribution.
- Papilloma virus* infection can present as pink or white single or multiple warts in the oral cavity. These are usually asymptomatic. The treatment is surgery, laser excision, cryotherapy or coagulation.
- Periodontal infections may become rapidly destructive which are difficult to manage. Anaerobic bacteria play a dominant role. Aggressive necrosis and ulceration of free gingival margin (*necrotizing ulcerative gingivitis*) may occur. The dissemination of this process to adjoining oral mucosa and palate leads to *necrotizing stomatitis*. Extension of ulcer can lead to bone destruction, tooth loosening and loss. These lesions are associated with pain, fever, gingival bleeding and foul breath. The

treatment includes mouthwashes with povidine-iodine followed by chlorhexidine. Local debridement of necrotic material may be needed. Antibiotics (clindamycin, metronidazole, amoxicillin-clavulanate) are given in severe infection.

- g. Neoplasms of the oral cavity may occur. The mouth is the common site for *Kaposi's sarcoma*. The lesions are red or violet, flat, raised and nodular found usually over hard palate. Bulky lesions may ulcerate and bleed. They may cause pain and swallowing problems. Diagnosis is made by biopsy.
- h. *Lymphoma* may manifest as nodules, ulcers or diffuse swelling in the oral cavity.
- i. Recurrent painful, single or multiple *aphthous ulcers* may occur in HIV disease. Large destructive ulcers are seen in advanced HIV disease. The treatment includes application of local anesthetic and steroids. In case of larger lesions, biopsy is done to rule out infectious or neoplastic cause. Thalidomide (200 mg/day orally) for 4 weeks is effective treatment for large ulcers. Systemic steroids can also be tried.
- j. *Diffuse infiltrative CD₈ lymphocytosis syndrome (DILS)* may present as bilateral salivary gland enlargement, usually parotid (Fig. 9.16) and xerostomia. Xerophthalmia can also occur. *Sjögren syndrome* and lymphoma may develop in these patients.

IMPLICATIONS ON DENTAL PRACTICE

1. HIV patients are usually able to tolerate all types of dental procedures. No dental modification is needed for patients based on their HIV status. Patients with HIV are not more susceptible to complications after dental treatment regardless of CD₄ cell count. Routine use of antibiotic prophylaxis is not recommended in HIV patients. However antibiotics are added in case of neutropenia (count <750/mm³)
2. HIV patients may have xerostomia particularly due to anti-retroviral medications. Xerostomia may also occur due to HIV associated disease of salivary glands. Hence, this should be taken into consideration while planning dental procedures like restorative procedures and fabricating fixed or removable prosthodontics.
3. The drug interactions are common with anti-retroviral agents. This should be considered when prescribing any medicines to such patients.

4. The practice of universal precaution must be maintained. In case of accidental exposure such as needle prick or injury by sharp instruments, PEP must be immediately followed.
5. Patients with HIV disease may present for the first time to dental surgeon with oral manifestations. Most common of these manifestations is oro-esophageal candidiasis.
6. Gonorrhea may present uncommonly with oral manifestations like inflamed tonsils, lymphadenitis, and painful oral and pharyngeal ulcers.
7. Patient with late syphilis may present with leukoplakia which may become malignant.

SELF ASSESSMENT

Multiple Choice Questions

1. HIV subtype predominantly found in India is:
 - A. Clade E
 - B. Clade C
 - C. Clade B
 - D. All
2. Western blot test for HIV infection is based on the detection of following in the serum of patients:
 - A. Antigen
 - B. Antibody
 - C. Viral RNA
 - D. Viral particles
3. Estimated number of HIV infected people in the world:
 - A. 4 million
 - B. 40 million
 - C. 10 billion
 - D. 1 billion
4. Universal work precaution is needed while caring for:
 - A. Symptomatic AIDS patients
 - B. All patients with HIV/AIDS
 - C. All patients whether HIV positive or negative
 - D. Only during surgery
5. HIV can infect:
 - A. CD₄ T lymphocytes only
 - B. CD₄ and CD₈ T lymphocytes
 - C. All CD₄ bearing cells
 - D. All cells arising from bone marrow
6. Most common opportunistic infection in AIDS patients in India is:
 - A. Candidiasis
 - B. Amoebiasis
 - C. Tuberculosis
 - D. Hepatitis
7. Mother to child transmission of HIV occur mostly:
 - A. During labor
 - B. During throughout pregnancy
 - C. Through breastfeeding
 - D. During last trimester of pregnancy
8. The risk of transmission of HIV through needle injury is:
 - A. 2-10%
 - B. 0.3%

- C. 10-30%
D. 0.09%
9. Following malignancies are increasingly associated with HIV infection except:
A. Kaposi sarcoma
B. Hodgkin's disease
C. Carcinoma breast
D. Primary CNS lymphoma
10. Following are at increased risk of getting HIV except:
A. Truck driver
B. IV drug abuser
C. Who frequently donate blood
D. Those having STD
11. Following are anti-HIV drugs except:
A. Ritonavir B. Acyclovir
C. Saquinavir D. Efavirenz
12. Following test is used for diagnosis of HIV infection during window period:
A. Western blot B. p24
C. B₂ microglobulin D. CD₄
13. Diagnosis of tuberculosis in HIV positive patients may be difficult because:
A. Tuberculosis is not common in HIV patients
B. Presentation of tuberculosis may be atypical
C. Bacterial strain is different
D. All the above
14. Following drug is contraindicated in pregnancy:
A. AZT (Zidovudine)
B. Stavudine
C. Efavirenz
D. All the above
15. Following percentage of HIV patients respond to ART:
A. 10-20% B. 60-70%
C. >90% D. 30-40%
16. Anemia is important side effect of:
A. Lamivudine B. Efavirenz
C. AZT (Zidovudine) D. All of the above
17. Which is true for ART:
A. ART is least effective
B. ART is very effective in all patients
C. ART is accompanied with increased cardiovascular mortality
D. All the above
18. Following opportunistic infection may occur even at CD₄ count >500/cmm:
A. CMV
B. Herpes zoster
C. Mycobacterium avium intercellulare
D. PCP
19. In resource poor settings, if CD₄ facility is not available, following can be used as parameter for starting ART:
A. Eosinophilia
B. Total lymphocyte count
C. ESR
D. PPD test
20. Following is true during latent period of HIV disease:
- | | Elisa | p24 | Symptoms |
|----|-------|-----|----------|
| A. | + | +++ | - |
| B. | - | - | + |
| C. | + | +/- | - |
| D. | - | ++ | - |
21. Following drugs are used in herpes zoster except:
A. Acyclovir B. Valacyclovir
C. Indinavir D. Famcyclovir
22. Which one of the following is the most common complication of mumps in postpubertal males?
A. Myocarditis B. Orchitis
C. Nephritis D. Parotid abscess
23. Koplik's spots are seen in:
A. Mumps B. Measles
C. Rubella D. Chickenpox
24. Following are features of rashes in chickenpox except:
A. Pleomorphic rash
B. Rashes are pruritic
C. First appears in extremities
D. Rashes are infective till crusts slough
25. Most common complication of chickenpox is:
A. Encephalitis
B. Meningitis
C. Viral pneumonia
D. Secondary infection of skin lesion
26. Subacute sclerosing panencephalitis (SSPE) is associated with:
A. Mumps B. Measles
C. HIV D. Japanese encephalitis
27. Following are features of congenital syphilis except:
A. Mulberry molars
B. Maxillary hyperplasia
C. High arched palate
D. Rhagades
28. Following drugs are used in syphilis except:
A. Penicillin B. Doxycycline
C. Azithromycin D. Cotrimoxazole
29. Jarisch Herxheimer reaction can occur following initiation of treatment of:
A. Plasmodium falciparum
B. HIV disease
C. Syphilis
D. Gonorrhea

30. The longest incubation period is found in which type of malaria:

- A. *P. falciparum* B. *P. ovale*
C. *P. vivax* D. *P. malariae*

31. Primaquin is given for radical cure in:

- A. *P. vivax* and *P. falciparum*
B. *P. falciparum* and *P. ovale*
C. *P. vivax* and *P. ovale*
D. *P. malariae* and *P. ovale*

32. Following agents can be used for chemoprophylaxis of malaria except:

- A. Chloroquine B. Mefloquine
C. Doxycycline D. Quinine

33. Most common site involved in amoebiasis is:

- A. Jejunum B. Rectum
C. Caecum D. Gastro-esophageal junction

34. Most common organ involved in extra-intestinal amoebiasis is:

- A. Brain B. Lung
C. Liver D. Pancreas

35. Palatal palsy is a complication of:

- A. Mumps
B. Diphtheria
C. Syphilis
D. Rubella

36. Following may occur through blood transfusion except:

- A. Syphilis B. HIV
C. Gonorrhea D. Malaria

Fill in the Blanks

1. Forchheimer's spots are seen in _____ infection.
2. Hutchinson's teeth are seen in _____.
3. Gumma occurs in _____ stage of syphilis.
4. Drug of choice in primary syphilis is _____.
5. Syphilis is caused by _____.
6. Cerebral malaria is caused by Plasmodium _____.
7. Tropical splenomegaly is associated with _____.
8. Amoebiasis is caused by _____.
9. Anchovy sauce pus is a feature of _____.
10. "Bull neck" may occur in _____ infection.

Chapter 10

Medical Emergencies in Dental Practice

Patients who come for treatment for their problems to dental surgeon may also be suffering from other systemic disorders. Some of these disorders may have direct relevance to the dental illness while others may not. However patients may develop serious life threatening problems during or just after the dental treatment. Emergency situations may arise more commonly when invasive or painful procedures are performed or medically compromised patients are treated. Hence the knowledge of these disorders is necessary in order to avoid or manage emergency situations. A list of important medical emergencies that may occur during dental treatment/ procedure is given in the Table 10.1. A general guideline should be followed (Table 10.2).

TABLE 10.1: Important medical emergencies in dental practice

- General: syncope, postural hypotension
- Cardiovascular: angina, myocardial infarction, cardiac arrest, heart failure, severe hypertension
- Respiratory: asthmatic attack, hyperventilation, airways obstruction
- Neurological: seizures, stroke
- Allergic: anaphylaxis
- Endocrinal: hypoglycemia, adrenal crisis
- Hematological: excessive bleeding

TABLE 10.2: General precautions and guidelines

- Patient should lay supine while receiving local anesthesia
- Ensure that diabetic patient has taken meal timely
- Always have an assistant
- Assistant should be well versed with techniques such as venepuncture, CPR, monitoring vitals and maintaining airways
- Emergency kit should be readily accessible and periodically checked
- Watch carefully for any change in behaviour, breathing pattern, wheeze, sensorium in patient while on procedure
- Contact phone number of physician/emergency department/ ambulance

The dental clinic should be well equipped in order to deal with any emergency. A list of paraphernalia and drugs which must be available in the clinic is given in Table 10.3.

History: A quick but discrete history must be taken to know about the past and present illness in the patient and also about the medication he/she is taking. In addition to this, a proper counseling is needed so that the patient understands the nature of treatment he is going to receive and the side effects expected if any. This will allay fear from the mind of the patient and build trust and confidence. This will also ensure full cooperation on the part of the patient. Such steps are very helpful in avoiding complications like

TABLE 10.3: List of drugs and equipments to deal with medical emergencies

Drugs

- Inj adrenaline (0.5 ml ampules of 1:1000 solution)
- Inj chlorphenamine
- Inj hydrocortisone sodium succinate (100-200 mg)
- Inj glucagon 1 mg
- Inj diazepam
- IV dextrose (20%, 50%)
- glucose powder
- glyceryl trinitrate 0.3-1 mg,
- aspirin
- salbutamol solution for nublizing

Equipments

- oropharyngeal airway
- oxygen supply kit
- suction machine
- ambubag and airways
- tourniquet
- laryngoscope
- sphygmomanometer
- stethoscope
- IV cannula
- disposable syringes and needles
- IV drip sets and DNS bottles
- nebulizer

Contact numbers of physician/emergency department/ambulance

TABLE 10.4: History taking before dental treatment/procedure

- Any problem during previous dental treatment
- History of hypertension, angina or any heart disease, seizures, fainting, asthma, syncope, drug allergy, renal disease, diabetes, peptic ulcer, chronic bronchitis, bleeding diathesis, anemia, jaundice, pregnancy, risk factors for HIV,
- Details of present medications

TABLE 10.5: Clinical examination

General

- Pallor
- Pulse
- blood pressure
- jaundice
- edema
- goiter
- lymphadenopathy
- temperature
- skin

Systemic

- higher functions
- heart sounds
- respiratory
- liver
- spleen

exacerbations of bronchial asthma, vasovagal syncope, surge in blood pressure, intense tachycardia or excessive bleeding. A scheme of history taking is given in Table 10.4.

Examination: A quick but thorough clinical examination should be performed before any dental treatment/procedure. This includes general and important systemic examination. A list is given in the Table 10.5

POSTURAL HYPOTENSION (ORTHOSTATIC HYPOTENSION)

Postural hypotension (orthostatic hypotension, OH) is a fall in systemic arterial pressure on assumption of upright posture.

- It is defined as a sustained drop in systolic (>20 mmHg) or diastolic (>10 mmHg) blood pressure within 3 minutes of standing.
- Symptoms of postural hypotension are dimming or loss of vision, light headedness, diaphoresis, nausea, pallor and weakness. Syncope may result if cerebral perfusion is impaired (see Chapter 4).
- Postural hypotension is generally due to defective postural reflexes, hypovolemia or drugs. A history of medications, previous postural syncope, diabetes and

TABLE 10.6: Important causes of postural hypotension

- Defective postural reflexes: Diabetes, aging, parkinsonism
- Drugs: vasodilators, beta blockers, diuretics, antidepressants
- Hypovolemia: diarrhea, excessive sweating, hemorrhage, diuretics

TABLE 10.7: Measures to avoid Postural Hypotension

- To avoid prolonged standing
- To avoid sudden change in posture to sitting or standing
- Movement of legs prior to standing
- Plenty of fluids and salt intake prior to the procedure

causes precipitating hypovolemia must be obtained before the procedure (Table 10.6)

Management

- a. Patients who have underlying causes of postural hypotension should be advised to attain upright position slowly. Preferably they may be asked to sit down first for sometimes before standing (Table 10.7).
- b. Plenty of fluids should be given to prevent hypovolemia.
- c. These patients should not be made to remain in standing position for long.
- d. Medications that aggravate the problem should be discontinued.
- e. Movement of legs prior to rising may facilitate the venous return and help avoid OH.
- f. Upon development of initial symptoms of OH, patient should immediately be laid supine. He is given water to drink and advised to attain sitting posture slowly as described earlier.

HYPERTENSIVE CRISIS (See also Chapter 4)

A sudden rise in blood pressure may occur during the treatment at dental clinic. This may happen due to skipping in the dose of antihypertensive medication or due to fear and anxiety.

History should be taken about the hypertension and its medications and whether the patient has taken the dose timely. The patient is counseled about the nature of treatment/procedure in order to allay fear and anxiety. Blood pressure is measured before the procedure.

Management

- a. Excessive and rapid decline in the BP should be avoided as it may lead to cerebral hypoperfusion and coronary insufficiency.

- b. In patients with malignant hypertension without encephalopathy, it is advisable to bring down BP over hours. This is done with frequent oral dosing with captopril, labetalol or clonidine.
- c. Parenteral labetalol and nicardipine are effective agents for lowering down the blood pressure in patients with hypertensive encephalopathy.
- d. Sublingual nifedipine should be avoided in the acute management because it is associated with adverse cardiac events.

ACUTE PULMONARY EDEMA

The symptoms of acute pulmonary edema are rapid onset dyspnea, cough, anxiety and restlessness. Patient cannot remain supine and prefers to sit upright for some relief. The sputum may be profuse, pink and frothy or blood stained. Patient has cold extremities, cyanosis, excessive sweating, tachypnea and tachycardia. Lungs are full with crackles and wheezes.

Precipitating factors for acute pulmonary edema are stress and exertion in a cardiac patient, arrhythmia, myocardial ischemia or hypertensive crisis.

An adequate medical history about the presence of cardiac disease should be taken. One should make sure that patient is regularly taking medications. The patient should be properly counseled prior to the procedure. The procedure is done in cordial and comfortable environment. The management is done as mentioned in Table 10.8. (see also chapter 4).

TABLE 10.8: Management of acute pulmonary edema

- Patient is put in sitting posture
- Stoppage of dental procedure
- Reassurance
- Oxygen is administered to improve hypoxia
- IV furosemide (20-80 mg) gives rapid relief
- Call for medical help
- Monitor vital signs
- IV morphine (2-4 mg) can be given
- If no relief, shift the patient to medical emergency

CHEST PAIN

Patients with ischemic heart disease may develop retrosternal chest pain due to angina. Occasionally it may develop for the first time during procedure in patient not previously known to be a cardiac patient. Hence, a history

TABLE 10.9: Management of chest pain (angina)

Reassure the patient
Defer further dental treatment
Glyceryl trinitrate 0.5 mg sublingual
Give aspirin
Summon assistance

should be taken for the presence of heart disease in all patients and also about the medications. The immediate management is done as shown in the Table 10.9.

ASTHMATIC ATTACK

The asthmatic attack may get precipitated before, during or after dental procedure. Important precipitating factors are anxiety, infection, allergy or drugs. The symptoms are acute dyspnea, wheeze, dry cough, tightness in the chest, anxiety, diaphoresis, cyanosis and confusion. (see also Chapter 5) In case the patient is a known case of bronchial asthma, one should make sure that he has properly taken medications as prescribed. In order to avoid the precipitation of acute attack in asthmatic patient, one should:

- Avoid using materials/drugs with allergic potentials
- Assure and calm the patient
- Ensure that patient has taken medicines for asthma as prescribed
- Avoid numerous dental products which may exacerbate asthma such as some toothpastes, tooth enamel dust and methacrylate.

The patient is urgently managed as discussed in Table 10.10. However, if recovery does not occur, he should be shifted to medical emergency department.

AIRWAYS OBSTRUCTION

During dental procedures, objects such as fragments of tooth, filling materials or segments of bone may find their way into the posterior part of the oral cavity, pharynx and

TABLE 10.10: Management of acute asthmatic attack

Stop the procedure
All dental materials removed from patient's mouth
Reassure the patient
Do not lay patient supine, place in comfortable posture
Administer oxygen
Nebulize with salbutamol (2.5-5 mg)
Inject hydrocortisone IV 200 mg
Call an ambulance in case the condition does not settle

TABLE 10.11: Prevention of airways obstruction

Use of rubber dams wherever possible
Adequate use of oral packings
Proper positioning of patient
Effective uses of dental assistance and suction

larynx. In conscious patient, airway obstruction is less common due to cough reflex. The symptoms are violent respiratory efforts, feel of choking and deepening cyanosis. Preventive measures should be taken to avoid such events (Table 10.11).

If the foreign body is swallowed, its location should be confirmed with the help of radiologists. Tips of broken instruments, burs or dental restorations may have pointed or jagged edges that might injure the gut mucosa. The patient is advised to eat soft bulk diet and to keep watch for evacuation of foreign body. A follow-up X-ray must be done to ensure its elimination.

If the foreign body has been aspirated, certain maneuvers are performed and patient is referred to thoracic surgeon. Different maneuvers are:

- Make patient stand up and use abdominal thrusts until foreign body is expelled (Heimlich maneuver).
- A Trendelenberg position (legs raised) can allow the gravity to move the object close to the oral cavity wherefrom it can be easily removed.
- Use laryngoscope and pharyngeal suction.
- If obstruction persists, insert a cannula of 10-12 IV size through cricothyroid ligament (cricothyrotomy).
- Arrange transfer to hospital for specialized care.

HYPERVENTILATION

Patient may develop hyperventilation (rapid breathing) during dental procedure. This is commonly due to pain, fear and anxiety (Table 10.12). Hyperventilation may result into hypocarbia (low PaCO_2) and alkalosis. Symptoms due to hyperventilation are tingling sensation around mouth, fingers and toes, muscle tremors, tetanic spasms of fingers and hand, light headedness and altered sensorium. The problem is self-limiting and does not generally require specific treatment (Table 10.13).

SEIZURES

Seizure is a transient clinical event in the form of motor, sensory, autonomic or psychic manifestations due to abnormal excessive paroxysmal discharges from a group of

central nervous system neurons. Precipitating factors for seizures are given in Table 10.14. (see also Chapter 7)

Management

The patient should be shifted to a safe place, away from danger (water, fire, machine). The patient is made to lie in semi-prone position in order to prevent aspiration (Table 10.15). Nothing should be given orally until patient has fully recovered. Oral cavity and airways are cleaned and cleared preferably with a piece of gauze (finger should not be inserted into mouth to avoid bite injury).

Proper history of any seizure disorder must be asked from the patient. If patient has seizure disorder, one should make sure he is taking the medication regularly. Precipitating factors for seizures must be avoided.

TABLE 10.12: Causes of hyperventilation

- Psychogenic or anxiety
- Pulmonary diseases (pneumonia, asthma, embolism)
- Cardiac diseases (CHF, hypotension)
- Metabolic acidosis (diabetic, renal, lactic)
- Drug induced (salicylates, methylxanthines)
- Fever, pain, sepsis

TABLE 10.13: Management of hyperventilation

- Stop the procedure
- Assure the patient
- Put the patient in comfortable position
- Generally the problem is self-limiting
- Patients may be asked to breathe in a paper bag
- May give mild sedative in case of intense anxiety
- Exclude other causes of hyperventilation

TABLE 10.14: Precipitating factors for seizures

- Fatigue
- Skipped meal
- Flashes of light (TV, monitor)
- Loud music/sound
- Sleep deprivation
- Alcohol use or withdrawal
- Physical and emotional stress
- Missing the dose of medication

TABLE 10.15: Emergency management of seizures

- Patient is put semi-prone
- Airways patency ensured
- Nothing oral till full recovery
- Protect from injury
- If seizure does not terminate spontaneously, give IV diazepam 10-20 mg slowly and may repeat if no recovery in 5 minutes
- Shift to medical emergency department

STROKE (CEREBROVASCULAR ACCIDENT)

Stroke means sudden neurological deficit due to focal vascular cause. The incidence of stroke increases with age. The causes and risk factors for stroke are described in chapter 7. Important risk factors are hypertension, heart disease, diabetes mellitus, old age, smoking and previous vascular events. Patient develops sudden loss of unconsciousness with paralysis, generally hemiplegia and facial paralysis. The initial management is done in the dental clinic before shifting the patient to the emergency room (Table 10.16).

TABLE 10.16: Initial management of stroke in dental clinic

- Stop dental procedure
- Lay the patient supine with head turned to one side
- Do not give anything orally
- Ensure patent airways
- Monitor vitals
- Give oxygen
- Inject 20-50 ml 50% dextrose (in case there is hypoglycemia)
- Arrange for shifting to emergency room

ANAPHYLAXIS

Anaphylaxis is acute type 1 hypersensitive reaction presenting as sudden hypotension, bronchospasm and urticarial skin reaction. Patient may develop respiratory arrest, cardiac arrhythmia and cardiac arrest. Thorough history about any such reaction in the past must be asked. The etiology, mechanism and management are discussed in detail in Chapter 12.

In order to avoid this complication, intradermal sensitivity test should be performed, particularly in case of xylocaine, penicillin, streptomycin, anti-serum and radiographic contrast media. Fainting after injection is common and can be confused with anaphylaxis. The management is summarized in Table 10.17.

TABLE 10.17: Management of anaphylaxis

- Patient is laid supine with legs raised
- Give 0.5 ml of 1:1000 adrenaline intramuscularly, may be repeated at every 10 min.
- Inject 10-20 mg chlorpheniramine IV
- Inject 100-200 mg hydrocortisone hemisuccinate IV
- Ventilate with oxygen
- Rapid IV saline infusion
- Be prepared for CPR
- Call ambulance, hospitalize the patient

HYPOGLYCEMIA

Hypoglycemia is defined as low blood sugar generally less than 50 mg/dL. (see also Chapter 8). However, the levels of blood sugar at which patients develop symptoms greatly vary. It is one of the most common emergencies in diabetic patients during dental treatment. Important symptoms of hypoglycemia are hunger, sweating and tiredness. Other manifestations are listed in Table 10.18.

Hypoglycemia is a complication of insulin or oral hypoglycemic therapy. Risk factors for the development of hypoglycemia include overdose of anti-diabetic medication, skipped meal and delay in the meal, unusual physical exertion and alcohol ingestion. Hypoglycemia can also complicate liver disease, adrenal insufficiency and hypopituitarism.

History prior to the dental treatment/procedure must include whether patient has diabetes. The timing of the intake of the anti-diabetic medication and the meal last taken should be asked. Patient should also be inquired for the presence of any symptoms suggestive of hypoglycemia. In case of any such symptoms, patient is given sugar, glucose or candy to eat. Blood may preferably be drawn before the administration of glucose to allow documentation of low plasma glucose. Patient is managed as mentioned in the Table 10.19.

Prevention of hypoglycemia includes proper education regarding the causes and symptoms of hypoglycemia and proper adjustment in medication and diet.

ADRENAL CRISIS

Adrenal crisis is acute adrenal insufficiency. This should be suspected if patient collapses, becomes pale, develops

TABLE 10.18: Symptoms of hypoglycemia

- Clinical settings wherein hypoglycemia is known to occur
- Hunger, sweating, anxiety, irritation
- Palpitation, trembling, nausea, headache
- Confusion, lack of concentration
- Seizures, coma

TABLE 10.19: Management of hypoglycemia

Conscious patient: Give rapidly absorbable carbohydrates orally like glucose or sugar. Milk, fruit, candy bars or biscuits may also be given.
If patient is unable to swallow or severe hypoglycemia: IV glucose (20-50 ml of 50% dextrose) followed by infusion of 10% dextrose Inj glucagon may be given for immediate effect.

hypotension and there is history of prolonged uses of corticosteroid with a period of withdrawal. The adrenal crisis is precipitated in certain situations (Table 10.20). The patient should be put supine with legs raised and injection hydrocortisone hemisuccinate 200 mg IV is immediately given. Medical specialist/endocrinologist is consulted.

EXCESSIVE BLEEDING

Excessive bleeding during dental procedures is not uncommon. In most cases, it may be related to the procedure itself or as a consequence of anxiety. However, in some cases, when it is due to underlying hemostatic disorders, may be life threatening. Hemostatic disorders commonly seen in practice are listed in the Table 10.21. A proper history of undue bleeding, purpura, ecchymosis, hemarthrosis and intake of aspirin or anticoagulants must be taken before the dental procedure.

If patient is a known case of hemostatic disorder, proper management must be done before the procedure (see Table 10.22 and Chapter 3). In case the patient bleeds excessively and prior diagnosis is not known, appropriate blood samples for the investigations must be drawn before administration of any drug/factors. Initial investigations include tests listed in Table 10.23. Other tests can be done later.

For detailed description of hemostatic disorders, readers are advised to see chapter on hematological disorders (Chapter 3).

DENTAL PROCEDURES IN PREGNANT WOMEN

Elective procedures may be deferred during pregnancy; however, acute and emergent dental problems do arise and should be treated. It is advisable to consult the obstetrician before any intervention. Following points should be kept in mind:

- Local anesthesia is preferred over general anesthesia.
- Prolonged procedures may preferably be avoided.

TABLE 10.20: Precipitating factors for adrenal crisis

- Infection
- Trauma
- General anesthesia
- Steroid withdrawal

TABLE 10.21: Common hemostatic disorders

- Hemophilia A
- Hemophilia B
- Idiopathic (autoimmune) thrombocytopenic purpura
- Von Willebrand disease
- Drugs (aspirin, anticoagulants)
- Liver disease

TABLE 10.22: Steps to prevent excessive bleeding

- Stop aspirin/antiplatelet drugs 4-7 days before the procedure
- Maintain platelet count (50,000-1,00,000/cmm) in thrombocytopenic patients
- Avoid NSAIDs in thrombocytopenic patient
- Administer FFP in patient with CLD if prothrombin time is highly raised
- Maintain adequate factor activity in hemophilic patients with administration of factors (factor VIII or IX) an hour before the procedure
- In case of mild hemophilia or vVWD, tranexamic acid or DDAVP may be used

TABLE 10.23: Initial investigations in patients with excessive bleeding

- Complete blood count (including platelet count)
- Prothrombin time
- Activated partial thromboplastin time
- Bleeding time

- Pregnant patients may have gingival hyperplasia which does not need any treatment. The gum swelling resolves after the delivery.
- Pre-eclamsia (hypertension, edema, proteinuria) should be excluded before any procedure.
- Only drugs safe in pregnancy should be prescribed.

Chapter

11

Critical Care

SHOCK

Shock is the clinical syndrome when the oxygen delivery fails to meet the metabolic requirements of the tissues. It is due to reduced tissue perfusion leading to decreased delivery of oxygen and metabolic substrates to the tissues.

Cellular dysfunction caused by reduced perfusion leads to release of inflammatory mediators, which further cause changes in the microvasculature leading to more reduced perfusion and ultimately multiple organ failure and death if not treated.

Shock is usually accompanied by low blood pressure (hypotension) and circulatory failure. Mean arterial pressure (diastolic blood pressure plus one-third of pulse pressure) is usually less than 60 mmHg.

Classification

Classification of shock is given in Table 11.1. Hypovolemic shock is the most common form of shock.

Hypovolemic Shock

Hypovolemic shock is caused by any condition causing reduction in blood volume. Hypovolemic shock is characterized by low cardiac output, low central venous pressure (JVP) and sympathetically mediated increased systemic vascular resistance.

TABLE 11.1: Classification of shock

Hypovolemic
Cardiogenic
Obstructive
Neurogenic
Septic
Anaphylactic
Traumatic
Hypoadrenal

It may be caused by:

- hemorrhage (external or internal)
- loss of fluid from gastrointestinal tract
- urinary loss or insensible loss (skin etc.)
- sequestration of fluid.

Mild hypovolemia (loss of <20% blood volume) is characterized by few clinical signs like anxiety and tachycardia.

Moderate hypovolemia is characterized by loss of 20-40% of blood volume and is characterized by tachycardia, tachypnea, and postural hypotension. Blood pressure may be normal in supine position but when patient is made to stand for few minutes, blood pressure may fall and patient may complain of dizziness, light headedness and syncope in standing position. Postural hypotension (orthostatic hypotension) is defined as > 20 mmHg fall in systolic blood pressure and >10% fall in diastolic blood pressure within 3 minutes of standing.

Severe hypovolemia is defined as loss of >40% blood volume. Clinical features of severe hypovolemia are:

- Hypotension: Systolic blood pressure <90 mmHg or mean arterial pressure of <60 mmHg
- Tachycardia: heart rate >100/min
- Tachypnea: increased respiratory rate
- Oliguria: decreased urinary output of <30 ml/hr
- Signs of reduced cerebral perfusion: agitation, confusion, drowsiness and coma
- Cold and clammy extremities
- Reduced central venous pressure
- Multiple organ failure: Acute lung injury and adult respiratory distress syndrome (ARDS), acute renal failure, liver dysfunction, reduced cerebral perfusion (coma), disseminated intravascular coagulation (DIC).

Diagnosis

Diagnosis is made by history of blood loss or fluid loss and signs of hypovolemic shock mentioned above. Occult blood loss, mainly in gastrointestinal tract and internal hemorrhage, should be ruled out when patient is having postural hypotension and other features of hypovolemic shock but there is no history of blood or volume loss.

Measurement of hemoglobin and hematocrit is misleading as these values may be normal in acute blood loss despite significant blood loss.

Management

- Patient with shock should be admitted in intensive care unit (ICU).
- Assessment of airway, breathing and circulation is done.
- Blood pressure, pulse rate, respiratory rate, urinary output, arterial oxygen saturation and mental status are continuously monitored.
- Aim is to restore tissue perfusion, maintain cardiac output and oxygen saturation so that there is adequate delivery of oxygen and substrates to the tissues.
- Volume resuscitation to maintain intravascular blood volume and measures to control ongoing losses.
 - Isotonic saline or Ringer's lactate are given through rapid intravenous infusion.
 - Blood transfusion or packed cell transfusion may be required when there is continuing blood loss and hemoglobin is lower than 10 g/dL.
 - Infusion of inotropic agents such as dopamine, dobutamine or vasopressin may be required to maintain adequate cardiac output.
- Supplemental oxygen is also required to maintain respiratory function. If arterial oxygen saturation is not improving, endotracheal intubation may be required.

Neurogenic Shock

Neurogenic shock is caused by traumatic high spinal cord injury, effects of spinal anesthesia or head injury.

- It is due to loss of sympathetic tone which results in impaired vasomotor tone causing dilatation of arterioles and veins. Dilatation of vessels leads to venous pooling causing decreased venous return and cardiac output.
- Peripheral examination reveals warm extremities because of venodilation (extremities are cold in

hypovolemic and cardiogenic shock due to vasoconstriction) and other features of shock.

- Vagal stimulation which causes increase in parasympathetic tone may result in hypotension, bradycardia and syncope (shock like features). It may mimic neurogenic shock. Pain due to any cause, fright or gastric dilatation may cause reflux vagal stimulation.

Management

Treatment of neurogenic shock includes correction of hypovolemia and drugs such as norepinephrine, phenylephrine which increase sympathetic tone and cause increase in blood pressure.

Obstructive Shock

This is characterized by acute reduction in cardiac output and development of shock. Important causes of obstructive shock are given in Table 11.2.

In cardiac tamponade there is inadequate diastolic filling because of compression of heart which becomes less compliant. Tension pneumothorax, diaphragmatic hernia and positive pressure ventilation causes increase in intrathoracic pressure which causes compression and decreased venous return.

Diagnosis depends on clinical features, X-ray chest and echocardiography.

Management depends on prompt recognition of cause and its treatment (Table 11.3)

Hypoadrenal Shock (see also Chapter 8)

There is increased secretion of cortisol from adrenal glands in stress due to any cause like illness, trauma and major

TABLE 11.2: Causes of obstructive shock

Cardiac tamponade (pericardial effusion, constrictive pericarditis)
Tension pneumothorax (air in pleural space)
Massive pulmonary embolism
Diaphragmatic hernia (herniation of abdominal viscera in thoracic cavity)
Mechanical ventilation (excessive positive pressure ventilation)

TABLE 11.3: Treatment of obstructive shock

Cardiac tamponade	Pericardiocentesis
Tension pneumothorax	Chest tube placement
Massive pulmonary embolism	Thrombolytic therapy, surgical embolectomy.

TABLE 11.4: Causes of adrenocortical insufficiency

Steroid therapy (chronic administration of high dose steroids)
Atrophy of adrenal glands
Tuberculosis, amyloidosis, metastatic disease of adrenal glands
Bilateral hemorrhage in adrenal glands

surgery. This response is impaired in certain clinical settings leading to decreased secretion of cortisol from adrenal gland and impaired host response to stress. Causes of adrenal insufficiency are given in Table 11.4.

Shock due to hypoadrenal state is characterized by hypovolemia, decreased systemic vascular resistance and reduced cardiac output.

Diagnosis

Adrenal insufficiency can be diagnosed by ACTH stimulation test.

Treatment

Hydrocortisone (100 mg every 4-6h) or dexamethasone (4 mg) is given intravenously, along with volume replacement and intravenous vasopressors.

Septic Shock

Septic shock and anaphylactic shock are classified under distributive shock. Sepsis is the most common cause of distributive shock. It is characterized by hypotension due to decreased vascular tone (reduced systemic vascular resistance), inadequate cardiac output and normal circulatory volume.

- Sepsis is defined as SIRS (systemic inflammatory response syndrome) with proven or suspected infectious etiology.
- SIRS may be due to infectious or non-infectious etiology and is characterized by fever or hypothermia, leukocytosis or leucopenia, tachypnea and tachycardia.
- Severe sepsis is sepsis with one or more organ dysfunction.
- Septic shock is defined as sepsis with hypotension i.e. mean arterial blood pressure <70 mmHg or Arterial systolic blood pressure <90 mmHg for at least 1 hour despite adequate fluid resuscitation. Septic shock is usually caused by inflammation due to bacterial infection. It is caused mainly by gram negative bacteria (Table 11.5).

- SIRS caused by inflammation due to non-infectious causes like pancreatitis, crush injuries and drugs (salicylates) may also lead to shock.

Clinical Presentation

Patients present with fever, chills, hypotension, altered mental status and features of end organ failure due to hypoperfusion. Systolic blood pressure is 90 mmHg or mean arterial blood pressure <70 mmHg. In some patients clinical features may point toward specific etiology. Generalized erythroderma may be present in toxic shock syndrome, *Staph. aureus* or *Strep. pyogenes* infection. *N. meningitidis* infection causes hemorrhagic or petechial rashes over skin.

Septic shock is also characterized by hypoperfusion of organs and may lead to dysfunction or failure of many organs if not treated promptly.

- Restlessness, confusion, delirium and coma may result from encephalopathy as a result of hypoperfusion of the brain.
- ARDS may result from increased permeability of pulmonary capillaries and pulmonary edema.
- Acute renal failure may occur due to renal hypoperfusion. Prolonged hypoperfusion or toxic injury may cause acute tubular necrosis.
- Increased levels of serum bilirubin and hepatic enzymes (AST, ALT) suggest hepatic dysfunction.
- DIC is frequently present in severe septicemia.

TABLE 11.5: Causes of septic shock/SIRS

Septic
Infection
Gram negative bacteria
• <i>E. coli</i>
• <i>Klebsiella</i>
• <i>Proteus</i>
• <i>Pseudomonas</i>
Gram positive cocci
Gram negative anaerobes
• <i>Bacteriodes</i>
SIRS (systemic inflammatory response syndrome)
Pancreatitis
Crush injuries
Drugs (salicylates)

Investigations

- Blood culture may reveal causative organism.
- Culture of material from local infected foci should also be performed to know causative organism and antibiotic sensitivity.
- TLC reveals increased WBC count and DLC reveals polymorphonuclear leukocytosis.
- Increased levels of serum urea and creatinine suggest renal dysfunction.
- Increased bilirubin levels and hepatic enzymes (AST, ALT) are usually found.
- Low platelet count, increased prothombin time (PT) and partial thromboplastin time (aPTT) suggest DIC.
- Arterial blood gas analysis may show hypoxia and metabolic acidosis.
- Chest X-ray may show features of ARDS.

Management

As in all types of shock, initial treatment consists of basic life support (airway, breathing and circulation). Airway maintenance, oxygen inhalation, ventilatory support, intravenous access and fluid resuscitation are important measures in management.

- Antimicrobial therapy should be started as soon as possible after sending blood sample and other specimen for culture. Empirical therapy is started with broad spectrum antibiotics to cover gram negative and gram positive organisms (Table 11.6). Empirical therapy

TABLE 11.6: Antimicrobial regimens used in septic shock

Group 1		
• Ceftriaxone	2g	od
• Ticarcillin-clavulanate	3.1g	q 4-6h
• Piperacillin-tazobactam	3.375g	q 4-6 h
Group 2		
• Imepenem-cilastatin	0.5g	q 6 h
• Meropenem	1g	q 8 h
• Cefipime	2 g	q 12 h
Group 3		
• Gentamycin	5-7 mg/kg	q 24 h
• Tobramycin	5-7 mg/kg	q 24 h
Group 4		
• Ciprofloxacin	400 mg	q 12h
• Levofloxacin	500-750 mg	q 12 h
• Clindamycin	600 mg	q 8h
Group 5		
Vancomycin	15 mg/kg	q 12 h

consists of one antibiotic from group 1 or group 2 plus one from group 3 (e.g. Ceftriaxone plus gentamycin, Meropenem plus tobramycin). If the patient is allergic to betalactam antibiotics, one antibiotic from group 3 plus clindamycin is given. Vancomycin is added if infection due to methicillin resistant (*Staph. aureus*) (MRSA) is suspected. Antibiotics are changed, if required, after culture and sensitivity reports are available. Maximum recommended doses of antimicrobial agents are given intravenously.

- Fluid administration is usually required for correction of hypotension and it is monitored by measuring central venous pressure (CVP). Inotropic or vasopressor agents are given if hypotension does not respond to fluid therapy.
- Septic shock may be associated with relative adrenal insufficiency and corticosteroids like hydrocortisone (50 mg 6 h for 5-7 days) are associated with improved survival.
- Drotrecogin alfa (activated human protein C) administration has been shown to reduce mortality in severe sepsis and shock. It has anti-inflammatory, antithrombotic properties and is given as continuous infusion of 24 mcg/kg body weight/hr for 96 hours.

Cardiogenic Shock

Shock as a result of impaired cardiac function is known as cardiogenic shock (CS). It is most commonly seen after acute myocardial infarction causing pump failure. Other causes of cardiogenic shock are given in Table 11.7.

CS secondary to acute myocardial infarction is characterized by

- Hypotension (systolic blood pressure < 90 mmHg, mean arterial pressure < 60 mmHg),

TABLE 11.7: Causes of cardiogenic shock

Pump failure
• Acute myocardial infarction
• Cardiomyopathy
Acute myocarditis
Arrhythmias
Valvular lesions
• Acute severe aortic or mitral regurgitation
• Critical mitral or aortic stenosis
Rupture of ventricular septum or free ventricular wall
Pericardial tamponade
Right ventricular failure due to pulmonary embolism

- Decreased cardiac index
- Elevated intracardiac pressure (pulmonary capillary wedge pressure >18 mmHg) and
- Increased peripheral vascular resistance.

In cardiogenic shock, CVP is usually raised (it is low in hypovolemic shock). Depressed myocardial contractility, usually due to ischemia, results in low cardiac output and hypotension. A vicious cycle is formed as hypotension results in hypoperfusion and ischemia of myocardium and further depression of myocardial contractility. Compensatory reflex arteriolar vasoconstriction leads to reduced blood flow to kidneys, muscle, abdominal viscera and skin while vasodilatation of cerebral and coronary arteries occurs in order to maintain circulation in these territories. Vasoconstriction not only helps in elevating blood pressure but also causes tissue hypoxia and acidosis.

Clinical Features

Shock (acute circulatory failure, low output state) is a clinical syndrome characterized by

- Arterial hypotension (systolic blood pressure <90 mmHg, mean arterial pressure <60 mmHg)
- Weak and rapid pulse
- Cold extremities and cyanosis
- Oliguria (low urine output <30 ml/h) and
- Features of altered mentation (agitation, confusion, drowsiness and coma).

In addition, features of underlying cardiac disease are present.

Most patients with acute myocardial infarction present with severe chest pain, dyspnea, anxiety and sweating. Cardiac findings include S3 gallop (third heart sound) and systolic murmurs. Raised jugular venous pressure (JVP) and pulmonary rales are audible in patients with left ventricular failure and cardiogenic shock.

Investigations

Chest X-ray, electrocardiogram and echocardiography are investigations which help in diagnosis. Echocardiogram should be done to find etiology of CS. Pulmonary artery catheterization helps in measurement of filling pressures and cardiac output and helps in diagnosis and proper management (volume replacement and use of pressor agents).

Management

- The aim of therapy is to increase blood pressure sufficient to maintain systemic and coronary perfusion and maintain volume status to ensure proper ventricular function and preventing volume overload.
- Oxygen inhalation is given to maintain PaO₂ of >60 mmHg. Endotracheal intubation and mechanical ventilation may be required. It reduces work of breathing and increases cardiac pressure which may improve cardiac output.
- Hematocrit is maintained >30%, fluid replacement is done to maintain adequate preload and ventricular function. Excessive fluids may cause pulmonary edema.
- Pressors are administered only after adequate fluid resuscitation.
- Drugs are used to increase blood pressure and cardiac output in patients of CS. The dose is titrated to maintain blood pressure of >90 mmHg or mean arterial pressure of >60 mmHg.
 - Dopamine is usually used in CS. It has variable effects according to dosage. At low dose (2 mcg/kg/min) it dilates renal vascular bed and increases glomerular filtration rate because of dopaminergic effects. At moderate dosage (2-10 mcg/kg/min) it stimulates beta adrenergic receptors and increases myocardial contractility (positive inotropic effect) and heart rate (positive chronotropic effect). At higher dosage (10 mcg/kg/min) it produces vasoconstriction because of alpha adrenergic stimulation.
 - Dobutamine produces greater inotropic effects as compared to dopamine. It is a synthetic sympathomimetic amine and is given in dosage of 2-20 mcg/kg/min. It also reduces afterload because of its vasodilating effects. Amrinone or milrinone is given with dobutamine.
 - Norepinephrine is a potent vasoconstrictor and inotropic agent and is given in dosage of 2-10 mcg/min. It is reserved for patients with CS and refractory hypotension. It is also useful in distributive or neurogenic shock.
- Aortic counterpulsation: Intraaortic balloon pumping, a mechanical circulatory assist device, augments arterial diastolic pressure and cardiac output. It is used in patients

not responding to medical treatment or in acute valvular insufficiency/VSD. It is inserted percutaneously and the balloon inflates during diastole and deflates during systole, reducing afterload and improving cardiac output. It also increases coronary blood flow during diastole.

- g. Definitive treatment is directed toward underlying cause. This can be in form of angioplasty, coronary artery bypass grafting, valvular replacement or cardiac transplantation. The rapid establishment of bloodflow in infarct-related artery is essential in acute myocardial infarction.

ACUTE RESPIRATORY DISTRESS SYNDROME (ARDS)

ARDS is most severe form of acute lung injury (ALI) characterized by acute hypoxemic respiratory failure following systemic or pulmonary insult without heart failure. ARDS is a physiological and radiographic syndrome rather than a specific disease.

Causes

Various causes which cause direct or indirect lung injury cause ARDS. The causes are given in Table 11.8.

Pathophysiology

The mechanism of lung injury varies with the etiology, however, the proinflammatory cytokines play a central role in the pathogenesis of lung injury. Damage occurs in the

alveolar epithelial cells as well as capillary endothelial cells. This causes increased vascular permeability and decreased production of surfactant. There is accumulation of protein rich fluid and inflammatory cells in the alveoli and interstitium leading to pulmonary edema, alveolar collapse and hypoxemia.

Clinical Features

The usual symptom is rapid onset dyspnea which follows the underlying causative event within 12-48 hours. Examination reveals tachypnea, labored breathing and intercostal retraction. Auscultation of lungs shows bilateral crackles (crepts). Most patients with ARDS have multiorgan failure involving kidneys, liver, gut, CNS and cardiovascular system.

Diagnosis

Diagnosis is based on following criteria:

- Acute onset of severe dyspnea
- $\text{PaO}_2/\text{FiO}_2 < 200$ mmHg
- Bilateral infiltrates on chest X-ray
- Pulmonary capillary wedge pressure (PCWP) < 18 mmHg or no clinical evidence of heart failure

Differential Diagnosis

Cardiogenic pulmonary edema must be excluded. (see Chapter 4).

Investigations

Important investigations include:

- X-ray chest PA view
- Arterial blood gases
- Hemogram
- Blood sugar
- Blood urea/Serum creatinine
- Liver function tests, serum amylase/lipase

Treatment

General measures include:

- Recognition and treatment of underlying cause
- Minimizing invasive procedures
- Prophylaxis for venous thromboembolism
- Management of nosocomial infection

TABLE 11.8: Causes of ARDS

Systemic insult (Indirect lung injury)

- Sepsis
- Multiple blood transfusions
- Severe trauma
- Burns
- Pancreatitis

Pulmonary insult (Direct lung injury)

- Pneumonia
- Aspiration of gastric contents
- Miliary tuberculosis
- Oxygen toxicity
- Pulmonary contusion
- Near drowning
- Toxic inhalation
- High altitude

Mechanical Ventilation

Mechanical ventilation may be needed in patient with respiratory failure. Positive pressure mechanical ventilation with lowest level of PEEP and supplemental oxygen required to maintain the PaO₂ above 60 mmHg or SaO₂ above 90% is used. Volume cycle ventilation with small tidal volumes have been shown to reduce mortality over standard forms of mechanical ventilation.

Currently, use of corticosteroids is not recommended as no mortality benefit has been demonstrated with steroids. Surfactants and inhaled nitric oxide have also been tried.

Prognosis

Mortality is 40-60% on treatment. It is around 90% if ARDS is accompanied by sepsis.

CARDIOVASCULAR COLLAPSE

Cardiovascular collapse is defined as a loss of effective blood flow due to acute dysfunction of heart or peripheral vasculature. Cardiovascular collapse may be caused by:

- Cardiac arrest: which requires intervention to prevent death.
- Other causes which are transient and non-life-threatening with spontaneous return of consciousness. These include syncope (vasovagal, postural hypotension, neurocardiogenic syncope) or transient severe bradycardia.

CARDIAC ARREST

Cardiac arrest is defined as abrupt cessation of cardiac pump function. This may be reversible by prompt intervention or can lead to death if no intervention is done. There is aberration in the underlying electrical activity leading to following conditions. Any of these conditions may be responsible for cardiac arrest.

- Ventricular fibrillation (VF) –responsible for 50-80% of arrests
- Asystole
- Persistent bradyarrhythmias
- Pulseless electrical activity (PEA)
- Pulseless sustained ventricular tachycardia (VT)

TABLE 11. 9: Causes of sudden cardiac death/cardiac arrest

Coronary artery disease
Hypertrophic cardiomyopathy
Aortic stenosis and pulmonary stenosis
Cyanotic congenital heart disease
Atrial myxoma
WPW syndrome
Electrolyte abnormalities
Hypoxia
Acidosis
Prolonged QT interval
Brugada syndrome

Sudden Cardiac Death (SCD)

SCD is defined as unexpected non-traumatic death in clinically well or stable patient who dies within 1 hour of onset of symptoms. The causative rhythm in most cases is VF.

Causes of cardiac arrest and are given in Table 11.9.

Management of Cardiac Arrest

Following are important steps in the management of cardiac arrest. (see cardiopulmonary resuscitation below)

- The initial response and basic life support
- Defibrillation
- Advanced life support
- Postresuscitation care
- Long-term management

CARDIOPULMONARY RESUSCITATION

All health care providers must know the procedures of basic life support (BLS) and advanced cardiac life support (ACLS). When cardiac arrest occurs, BLS must be started immediately. The goal of resuscitation is to maintain cerebral perfusion until cardiopulmonary function is restored. The longer the period of cardiopulmonary arrest, the lower is the chance of restoring healthy life. Irreversible cerebral damage may occur after 3 minutes of anoxia. Ventricular arrhythmias are responsible for majority of cardiac arrests, hence defibrillation facilities should be available.

Following sequence should be followed during adult resuscitation:

- Assessment of unresponsiveness
- Activation of emergency medical services

3. BLS until defibrillation is available
4. Defibrillation if indicated
5. Intubation
6. Administration of appropriate medications

BLS

Airway, breathing and circulation are the essentials of BLS. In an unconscious patient, the following is recommended:

1. Determine the responsiveness by gently shaking the patient and shouting into ear. Shaking of patient's head and neck should be avoided if trauma to these areas is suspected.
2. If no response is obtained, activate the emergency medical services system. If breathing problem is the likely cause of unconsciousness (drowning, choking, alcohol intoxication, trauma), resuscitation should be performed before going for help.
3. Patient should be laid on firm, flat surface. If the patient is moved, he should be moved as a single unit so that head, neck and body moves simultaneously.
4. *Airway*: The patient's mouth should be opened, debris (e.g. mucus and blood) should be removed and dentures should be left in place. Loose or ill-fitting dentures should be removed. Open the patient's airway by flexing the neck and extending the head. The head tilt-chin lift should be performed if neck injury is not suspected. If a cervical injury is suspected, use the jaw thrust maneuver to limit the potential for spinal cord injury.
5. *Breathing*: Assess for the presence of respiration once the airway is open by the look, listen and feel method. An open airway may be all that is necessary for spontaneous respiration to resume and continue. One should try to feel and hear breath sounds by placing an ear or cheek near to the victim's mouth. Breathing can also be assessed by observing chest and abdominal movements (rise and fall). If the spontaneous breathing is not present, give two slow breaths (1.5 to 2 seconds per breath), taking a breath after each ventilation. Each ventilation should be performed with sufficient volume to make the patient's chest rise. Rapid or high pressure breath may result in gastric distention. If the chest does not rise, the patients head should be repositioned and ventilation attempted again.
6. *Circulation*: Cardiac arrest is accompanied by circulatory collapse. Assess circulation by palpating for pulses and checking for signs of perfusion including swallowing or breathing for more than occasional gasps. Absence of carotid pulse for 10 seconds confirms circulatory collapse.
 - a. External chest compressions should be initiated at a rate of 80-100 per minute in the absence of carotid pulse .
 - b. The patients should be lying on firm surface.
 - c. The heel of one hand is placed over the lower half of the victim's sternum and the heel of the second hand is placed over the first with fingers interlocked. Chest compressions are performed with the heel of one hand on the back of the other hand. Hand position should be one inch cephalad to the patient's xiphoid process. With shoulder directly above the hands and elbows in locked position, the patient's sternum should be compressed below to 1.5 to 2 inches.
 - d. The adequacy of compressions should be assessed periodically by having another rescuer palpate the carotid pulse.
 - e. Respiratory and circulatory support is continued by providing two effective breaths for every 15 cardiac compressions. Fifteen chest compressions should be performed followed by two ventilations. Once the patient is intubated, ventilation can be performed at a rate of 12 to 15 per minute without pausing for compressions
- Stop BLS for five seconds at the end of first minute and every 1-2 minutes thereafter to determine whether patient has resumed spontaneous breathing or circulation. If spontaneous pulse has returned check the BP and continue ventilation as needed. BLS should not be withheld for more than five seconds except to intubate or defibrillate the patient. Attempts at intubation should not exceed 30 seconds before CPR is resumed.
- If an unconscious patient cannot be ventilated after two attempts at positioning the head and chin and a laryngoscope is unavailable, 6 to 10 abdominal thrusts should be performed. Careful technique

should be used in proper hand position, to avoid damage to the internal organs. After this maneuver is performed, debris should be swept from the patient's mouth with a finger, and then one should attempt to ventilate the patient. Cricothyrotomy and transtracheal ventilation are rarely necessary for ventilating the patient.

Advanced Cardiac Life Support (ACLS)

ACLS is an extension of BLS and is implemented by a team. Early designation of a team leader facilitates coordination of the efforts. The team leader is responsible for ensuring proper implementation of ACLS. The leader should ascertain that BLS is being properly performed and that defibrillation occurs early when indicated. After this, the leader should ensure the following:

- Proper endotracheal intubation and ventilation
- Placement of functioning IV catheter
- Administration of appropriate cardiovascular medicines
- Diagnosis and treatment of the cause of the arrest

The team leader should examine the patient and determine the events preceding the arrest, obtain a relevant history including current medications and treatments, and obtain appropriate tests (electrolytes, hematocrit, arterial blood gas analysis, ECG, radiographs, etc.) to determine and treat the cause of the arrest. The patient's rhythm and pulse should be assessed after every intervention. The team leader also should determine when termination of resuscitative efforts is warranted.

- a. ACLS consists of ECG monitoring, endotracheal intubation and setting up an intravenous line in a large peripheral vein or central vein.
- b. Immediate therapy includes defibrillation, oxygen and administration of cardioactive drugs.
- c. As soon as possible the cardiac rhythm should be established. If the ECG reveals ventricular arrhythmia or there is doubt to the nature of the rhythm (e.g. Fine VF may be confused with asystole) immediate defibrillation is performed.
- d. If initial defibrillation attempts are unsuccessful, victim is intubated and setting of intravenous line is performed while the circulation is supported by external chest compressions. If there is any difficulty in intubating the patient, ventilation should be continued by means of an airway, a ventilating bag and oxygen.

- e. Intravenous epinephrine results in vasoconstriction and increases the proportion of cardiac output delivered to brain.

Sudden unexpected cardiac arrest (Table 11.10) occurs due to:

- a. Ventricular fibrillation or pulseless (VF)/VT.
- b. Asystole and pulseless electrical activity also known as electro mechanical dissociation(non-VF/VT).

Majority of deaths are due to VF and rapid VT. Small proportion of deaths occurs due to pulseless electrical activity and remainders are due to asystole.

- VF and pulseless VT is treated with immediate defibrillation, CPR and drugs. Intravenous amiodarone is the first line drug in the refractory VF/pulseless VT.
- Asystole is more difficult to treat but the heart may respond to atropine or epinephrine (adrenaline). Recently vasopressin has been shown to be successful. If there is any sign of slow electromechanical activity (e.g. bradycardia with a weak pulse), emergency pacing should be done.
- Pulseless electrical activity may be due to several potential reversible causes (Table 11.11). It carries a very poor prognosis. Effective treatment involves addressing the underlying cause.

TABLE 11.10: Causes of unexpected cardiac arrest

Cardiac arrhythmias (e.g. ventricular fibrillation)
Sudden pump failure (e.g. acute myocardial infarction)
Acute circulatory obstruction (e.g. acute pulmonary embolism)
Cardiovascular rupture (e.g. myocardial rupture, aortic dissection)
Vasomotor collapse (e.g. pulmonary hypertension)

TABLE 11.11: Most frequent causes of pulseless electrical activity

Hypovolemia
Hypoxemia
Hypokalemia/hyperkalemia
Hypothermia
Metabolic acidosis
Cardiac tamponade
Tension pneumothorax
Pulmonary thromboembolism
Acute coronary syndrome (coronary thrombosis)
Toxins

Chapter 12

Anaphylaxis and Drug Allergy

ANAPHYLAXIS

Anaphylaxis is a type I IgE mediated rapidly developing systemic allergic reaction. Exposure to an antigen results in the formation of specific IgE which binds with the mast cells. Once IgE is bound to the surface of mast cells, the individual is primed to develop type I hypersensitivity reaction. Repeat exposure to same antigen in the sensitized individual can lead to the activation of IgE bound mast cells and release of a number of mediators such as histamine, serotonin, leukotriens and prostaglandins (Fig. 12.1). These mediators cause bronchoconstriction, vasodilatation, fluid exudation and smooth muscle contraction which are responsible for clinical features of the anaphylactic reaction (Table 12.1).

Anaphylactoid reaction: Certain substances can cause activation of mast cells directly (non-IgE mediated) and lead to similar clinical picture. This is known as *anaphylactoid reaction*. Radio-contrast media are the most common cause of anaphylactoid reaction.

TABLE 12.1: Mediators and their actions in anaphylaxis

Primary mediators	
Histamine	Vasodilatation, increased vascular permeability, bronchospasm, increased mucus secretions
Chemotactic factors (ECF-A, NCF-A)	Chemotaxis of eosinophils and neutrophils
Proteases	Generate other inflammatory mediators
Secondary mediators	
Leukotrienes (LT)	
LTC ₄ and LTD ₄	Vasoactive and spasmogenic
LTB ₄	Chemotaxis of neutrophils, eosinophils and monocytes
Prostaglandins (PGD ₂)	Bronchospasm and increased mucus productions
Platelet activating factor (PAF)	Platelet aggregation, histamine release

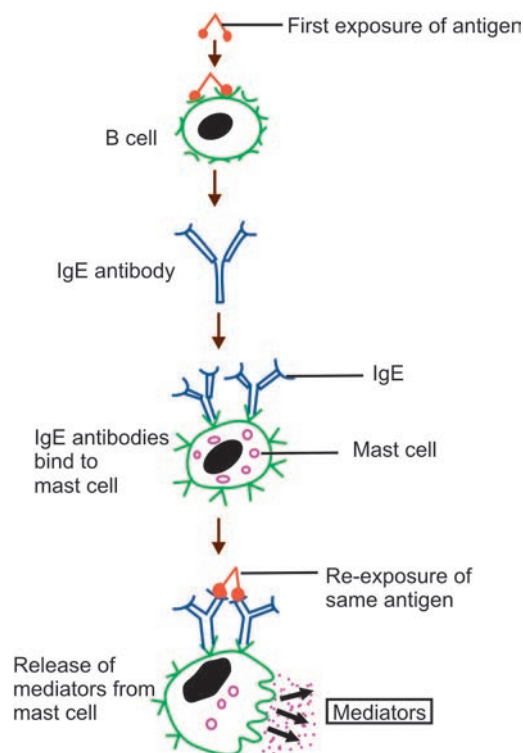


FIGURE 12.1: Mechanism of anaphylaxis

Causes

Important causes of anaphylactic reaction are drugs, foodstuffs and insect venoms. These are listed in Table 12.2.

Clinical Manifestations

The manifestation may be mild, very severe or life threatening. Type I reaction can occur as systemic disorder (anaphylaxis) or local reaction (urticaria and/or angioedema). Serious manifestations occur within seconds to minutes after exposure to antigen. Sometimes, the reaction is delayed by a few hours. Important clinical features are given

TABLE 12.2: Causes of anaphylactic reaction

Drugs	
Antibiotics:	penicillins, cephalosporins, nitrofurantoin, quinolones
Amphotericin B	
Local anesthetics:	procaine, lidocaine
Muscle relaxants:	gallamine, suxamethonium, pancuro-nium
Vitamins:	thiamine, folic acid
Diagnostic agents:	contrast media
Horse serum:	Antithymocyte globulin, antidiaphtheric serum
Enzymes:	Streptokinase, chymotrypsin, L-asparginase
Foods	
Milk, eggs, sea food, nuts, grains, beans, gelatin in capsules	
Pollen extracts	
Non-pollen extracts: dust mites, animal danders	
Insect bites: Honeybee, wasps, Hymenoptera	
Latex rubber products	
Vaccine preservative: Thiomerosal	

in Table 12.3. Respiratory involvement is the most common cause of death.

- a. Skin lesions in the form of urticaria or angioedema may occur. In mild cases, this could be the only manifestation.
 - Urticaria is a raised flat-topped well demarcated skin lesion with surrounding erythema (Figs 2A and B). Individual lesions usually last from minutes to hours. They are localized or disseminated and are pruritic.

**FIGURES 12.2A and B:** Urticaria**TABLE 12.3:** Clinical features of anaphylaxis

Skin	Pruritus, urticaria, angioedema, flushing
Respiratory	Nasal blockade, rhinorrhea, cough, laryngeal edema, breathlessness, hoarseness, stridor, wheezing, cyanosis
Cardiovascular	Tachycardia, palpitation, hypotension, shock, arrhythmia, cardiac arrest
Gastrointestinal	Abdominal pain, nausea, vomiting, diarrhea

- Angioedema is a deeper cutaneous process causing areas of localized nonpitting swelling. It most often involves lips, tongue and eyelids (Figs 12.3A and B).
- b. Respiratory distress may occur due to laryngeal edema, laryngospasm or bronchospasm. Patient may have a feeling of tightness of chest, stridor and wheezing.
 - c. Hypotension and shock can occur. Patients have cold extremities, decreased urinary output, peripheral cyanosis and altered sensorium.
 - d. Gastrointestinal manifestations include abdominal cramps, nausea, vomiting and diarrhea.

Diagnosis

The diagnosis of anaphylactic reaction is clinical, based on the history of exposure to the triggering agent and appearance of symptoms and signs within minutes of exposure. An elevated serum tryptase during an episode confirms the diagnosis of anaphylactic reaction.



FIGURES 12.3A and B: Angioedema

Treatment

Prompt recognition and management of anaphylactic reactions is a must as death may occur within minutes. Upon any suspicion of anaphylaxis, airway, breathing and circulation are assessed.

- The drug of choice is adrenaline (epinephrine).* This is given in a dose of 0.3-0.5 mg (0.3-0.5 ml of 1:1000 solution) intramuscularly or subcutaneously. Repeated injections can be given at 20 minutes intervals if necessary. Patients who are already being treated with beta adrenergic blocker drugs are refractory to epinephrine. A higher dose of adrenalin is required for the desired effect. Glucagon is also beneficial in such patients. Epinephrine causes bronchodilatation and vasoconstriction and prevents further mast cell degradation. This can also be given sublingually, intravenously or via an endotracheal tube in patients with hypotension and airway compromise.
- Airway management should be a priority. Hundred percent oxygen should be administered. Endotracheal intubation may be necessary. If laryngeal edema does not respond to epinephrine, tracheostomy may be required. Inhalation of β_2 adrenergic agonist (terbutaline, salbutamol) and intravenous aminophylline are helpful in bronchospasm.
- Intravenous fluids are given to maintain intravascular volume. The amount of fluid to be given is adjusted according to the blood pressure and urine output.

Vasopressor drugs (dopamine, norepinephrine) are used if the patient remains hypotensive.

- H_2 receptor antagonists (diphenhydramine, promethazine) are useful in relieving skin symptoms (pruritus, urticaria and angioedema) and abdominal cramps. Addition of H_2 receptor antagonists (ranitidine) may provide better effects.
- Glucocorticoids have no immediate significant effect but may reduce prolonged reactions or relapses. Hydrocortisone (200 mg) or methylprednisolone (125 mg) can be given intravenously.

The patient is observed for 24 hours as the late phase reactions can cause reappearance of symptoms.

Prevention of Recurrent Anaphylaxis

- Identification and avoidance of the offending agent is crucial for the prevention of anaphylactic reaction.
- Self-administered epinephrine therapy is advocated for those who develop recurrent anaphylaxis to a food or Hymenoptera sting.
- Venom immunotherapy is tried in patients with anaphylaxis to hymenoptera sting.

Treatment of Urticaria and Angioedema

These represent mild type I hypersensitivity reactions.

- Ideal treatment is identification and avoidance of specific cause, such as medications, cosmetics, foods.
- Antihistaminics (H_1 blocking agents) are given for the control of symptoms. Second generation antihistaminics

like fexofenadine, cetirizine, levocetirizine, rupatadine, loratidine, desloratidine are used as first line agents. Classic antihistaminics such as hydroxyzine is added as a evening dose for better control of the lesions or in refractory cases. H₂ blocking agents (ranitidine) may be added in refractory cases.

3. Oral corticosteroids are given to patients with severe symptoms who have not responded to antihistaminics.
4. Use of ACE inhibitors and ARBs can be associated with the onset of angioedema. They should be discontinued if angioedema develops.

DRUG ALLERGY

Drug reactions are a common medical problem. The basis of drug reaction is idiosyncratic, toxic or immunological.

Immunological reactions can be of various types and occur with relatively low dose. These occur usually after re-exposure to the drug in an individual who has already been sensitized by the previous exposure. Various types of immunological drug reactions are given in Table 12.4.

Skin test predicts immediate hypersensitivity reaction. If the test is negative, immediate hypersensitivity reaction will not develop, but delayed non-IgE-mediated reaction may still develop in these cases.

The allergic reactions to drugs can manifest through organ specific syndromes. Important syndromes are as follows:

TABLE 12.4: Drug reactions (immune-mediated)

Type of reaction	Mechanism	Features
Anaphylactic	IgE-mediated activation of mast cells and release of mediators	Anaphylaxis Urticaria Angioedema
Cytotoxic	IgG and IgM antibodies against cell antigens and complement activation	Autoimmune hemolytic anemia, Drug-induced thrombocytopenia, Interstitial nephritis
Immune complex	Immune complex deposition and complement activation	Serum sickness* Vasculitis
Cell mediated	T cell activation against cell surface antigen	Photosensitivity Contact dermatitis

* Rash, fever, arthralgia, lymphadenopathy

Cutaneous Reactions

Cutaneous reactions are most common manifestations of drug allergy.

- a. Urticaria: It is commonly seen with penicillin, cephalosporin and sulfonamides
- b. Fixed drug eruptions: These are discrete macular or bullous non-pruritic lesions which occur at the same place each time the drug is administered. Drugs commonly implicated are tetracycline, sulfonamide and penicillin.
- c. Photodermatitis: There are bright erythematous eruption or eczematoid lesions in area exposed to ultraviolet light (sun exposure). Doxycycline, coal tar derivatives and psoralens are important drugs causing photodermatitis.
- d. Contact dermatitis: Local application of neomycin and PABA can lead to dermatitis.
- e. Febrile mucocutaneous syndromes: These are erythema multiforme, Stevens-Johnson syndrome (SJ syndrome), and toxic epidermal necrolysis (TEN or Lyell syndrome). Common drugs implicated are rifampicin, phenobarbitone, phenytoin, trimethoprim-sulfamethoxazole. SJ syndrome is a severe form of erythema multiforme and presents as popular, urticarial, vasicular or purpuric lesions involving two or more mucosal surfaces (Fig. 12.4). Lyell syndrome presents as epithelial bullae and subsequent desquamation (Fig. 12.5).

Erythema multiforme is characterized by target lesion on skin. Sloughing of the dermis and mucous membrane



FIGURE 12.4: Stevens-Johnson syndrome



FIGURE 12.5: Toxic epidermal necrolysis

occurs in SJS, (<10% of total body surface area involved) and TEN (if >30% body surface is involved). Involvement of 10-30% of the body surface area is known as SJS-TEN overlap. The offending drug which has resulted in the above reactions should not be given again. Future skin testing with the offending drug is absolutely contraindicated.

Hepatic Syndromes

Hepatic syndromes can present as hepatocellular damage (phenytoin, halothane) and cholestatic jaundice (phenothiazine, erythromycin and azathioprine).

Renal Syndromes

Renal syndromes may present as interstitial nephritis (fever, pyuria, hematuria, proteinuria, rash and eosinophilia). Drugs implicated are methicillin, sulfonamides, and cephalosporins.

Pulmonary Syndromes

Pulmonary syndromes presents as pulmonary infiltration (nitrofurantoin, gold) and bronchospasm (aspirin, NSAIDs).

Management of Drug Allergy

A careful history is taken for any drug allergy. In case of history of drug reaction, the name of the offending drug and the date of the reaction should be noted. Patients may lose their sensitivity to a drug over time. Prompt appearance of symptoms of drug reaction suggests IgE-mediated anaphylactic reaction. In other types of immune-mediated reactions, symptoms may develop several days after the completion of course of therapy.

Drugs which have caused anaphylaxis should be avoided. However, if no alternative drug is available, skin testing is performed. In case skin test is positive, desensitization is performed. There may be cross reactivity within the same group of antibiotics. A high degree of cross reactivity exists between penicillin, carbapenem and imipenem. Cross reactivity between penicillin and cephalosporin is uncertain, however, latter is avoided if anaphylaxis had occurred due to penicillin.

Treatment

1. Discontinuation of suspected drugs
2. If the reaction is mild (skin rash) and the offending drug needs to be given essentially, the drug may not be discontinued. The reaction is treated symptomatically.
3. For treatment of anaphylaxis, urticaria and angioedema, see page 258.

IMPLICATIONS ON DENTAL PRACTICE

1. Swelling of the lips and floor of the mouth may occur in angioedema.
2. Dry mouth and drowsiness may occur in patients taking antihistaminics.
3. Emergency kit should always be available in order to deal with any allergic reactions during dental procedures.
4. Latex and rubber products can cause allergic reactions. These are now increasingly seen as the use of gloves has become more common with the advent of HIV/AIDS.
5. Patients should be in supine position when injections are given. Fainting after injection is common and can be confused with anaphylaxis.

SELF ASSESSMENT

Multiple Choice Questions

1. **The drug of choice in a patient with an acute allergic reaction involving bronchospasm and hypotension is:**
 - A. Aminophylline
 - B. Epinephrine
 - C. Dexamethasone
 - D. Chlorphenaramine maleate
2. **The blood cell, responsible for humoral immunity are:**
 - A. T-cells
 - B. B-cells
 - C. Both T and B cells
 - D. No cells only antibodies

3. Type I hypersensitivity is mediated by which of the following immunoglobulins:
A. Ig A B. Ig M
C. Ig D D. Ig E
4. The first drug of choice to be administered in anaphylaxis is:
A. Atropine B. Adrenaline
C. Hydrocortisone D. Cetirizine
5. A positive tuberculin test is an example of which hypersensitivity reaction:
A. Type I B. Type II
C. Type III D. Type IV
6. Most abundant immunoglobulin is:
A. IgG B. IgM
C. IgD D. IgA
7. Which of the following is not true about urticaria:
A. Skin lesions B. Raised flat-topped lesions
C. Pruritus is present D. No response to antihistaminics
8. Angioedema is commonly seen with the use of:
A. ACE-inhibitors B. Calcium channel blockers
C. Beta blockers D. Diuretics
9. Anaphylactoid reaction is:
A. IgE mediated
B. IgG mediated
C. Non-immunoglobulin-mediated release of mast cells mediators
D. None of the above
10. Treatment of urticaria includes following except:
A. Ranitidine B. Amantadine
C. Fexofenadine D. Hydroxyzine

Chapter 13 Nutrition

DIET AND NUTRITION

Around 40 to 50 nutrients are required by human body which are supplied through the food we eat. These are broadly divided into macronutrients and micronutrients. Macronutrients consist of proteins, fats and carbohydrates. They form the main bulk of food. Micronutrients are vitamins and minerals. They are required in a very small amount.

Energy

Body requires energy for normal function, growth and repair. This is provided by the oxidation of proteins, fats and carbohydrates. Oxidation of one gram of protein or carbohydrate provides 4 kcal (kilo-calorie) and one gram of fat provides 9 kcal.

The energy requirement mainly depends upon basal energy expenditure (BEE) and physical activity. It also depends upon the energy requirement for metabolizing food (thermic effect of food). Average energy requirement varies according to the age, sex, body composition and physical activity of the individual. An adult male requires 2800-3800 kcal per day and an adult female requires 1800-2900 kcal per day depending upon the type of work (light, moderate or heavy). Requirement increases in pregnancy (+300 kcal) and lactation (+500 kcal).

Protein

Protein is needed for growth and for maintenance of body structure and functions. About 10-15% of the total calories should be provided by the proteins in a healthy diet. Essential amino acids cannot be synthesized in the body and thus should be supplied through food. There are nine essential amino acids namely histidine, isoleucine, leucine, lysine,

methionine, phenylalanine, threonine, tryptophan and valine. Usual protein requirement in adults is 1g/kg per day. The protein requirement increases during growth, pregnancy, lactation and convalescence. The biological value of animal proteins is higher than that of proteins from vegetable sources.

Fat

Fats are concentrated sources of energy. It should provide 30% of the total calorie demand. Dietary fats are chiefly composed of fatty acids and cholesterol. Fatty acids are either saturated or unsaturated. Unsaturated fatty acids are further classified into mono-unsaturated fatty acids (MUFA) and polyunsaturated fatty acids (PUFA). Saturated fats are derived from animal foods and are solid at room temperature. Unsaturated fats are found in vegetable oils. Essential fatty acids, mainly linoleic acid cannot be synthesized in the body and thus is needed in the diet. Linoleic acid is required for the synthesis of other essential fatty acids, linolenic acid and arachidonic acid.

Deficiency of essential fatty acids leads to rough and dry skin (phrenoderma), hair loss and poor wound healing.

Carbohydrate

Dietary carbohydrates mainly consist of simple sugars, complex carbohydrates (starches) and indigestible carbohydrates (dietary fibers). At least 55% of the total calories should come from carbohydrates. Diets high in fibers are associated with lower incidence of digestive and cardiovascular diseases.

Vitamins

Vitamins are organic compounds required by the body for variety of essential metabolic functions. These are grouped

TABLE 13.1: Recommended dietary allowances (RDA) of vitamins

Vitamin	Adult Male	Adult Female	Pregnancy	Lactation
Vitamin A (g/d)	900	700	750	1200
Vitamin C (mg/d)	90	75	85	120
Vitamin D (g/d)*	5	5	10	10
Vitamin E (ng/d)	15	15	15	19
Vitamin K (g/d)	120	90	90	90
Thiamine (mg/d)	1.2	1.2	1.4	1.5
Riboflavin (mg/d)	1.6	1.4	1.6	1.6
Niacin (mg/d)	18	16	18	18
Vitamin B ₆ (mg/d)	2.0	2.0	2.5	2.5
Folate (g/d)	200	200	400	300
Vitamin B ₁₂ (g/d)	1	1	1.5	1.5
Pantothenic acid (mg/d)	5	5	6	7
Biotin (g/d)	30	30	30	35
Choline (mg/d)	550	425	450	550

*as calciferol (1 g is equal to 40 IU)

as water soluble and fat soluble vitamins. Fat soluble vitamins are vitamin A, D, E and K. Water soluble vitamins are thiamin (B₁), riboflavin (B₂), niacin (B₃), pyridoxine (B₆), cobalamin (B₁₂), folate, pantothenic acid, biotin and L-ascorbic acid (vitamin C). Recommended dietary allowances (RDA) of vitamins are given in Table 13.1.

Minerals

Body requires major minerals, electrolytes and trace elements. These are needed for growth, repair and vital body functions. Major minerals are calcium, magnesium and phosphorus. Electrolytes include sodium, potassium and chloride. Trace elements are iron, zinc, copper, manganese, molybdenum, fluoride, iodide, cobalt, chromium and selenium.

Balanced Diet

A balanced diet is defined as a diet which provides adequate amount of energy, amino acids, carbohydrates, fats, vitamins and minerals to maintain health and general wellbeing. The diet should also provide extra amount of nutrients to meet the demands during short periods of deficiency.

Recommended Dietary Allowance (RDA)

Nutritional requirements are generally expressed in the form of RDA. This is the average daily dietary intake that meets the nutrient requirements in healthy persons of a specific sex, age and physiological condition (pregnancy, lactation).

PROTEIN ENERGY MALNUTRITION (PEM)

PEM is still a major health problem in the developing countries including India. It occurs mainly in childhood and is an important cause of morbidity and mortality.

PEM occurs due to absolute or relative deficiency of energy and protein. It may be due to inadequate food intake or secondary to illnesses such as diarrhea, respiratory infections, measles and intestinal worms. Factor contributing to PEM are poverty, poor environmental conditions, large family size, poor maternal health, failure of lactation, premature termination of breastfeeding and adverse customs.

Classification

PEM is classified in various ways depending on weight, height, severity and relative contribution of energy or protein deficit. The commonly used classification is Wellcome Trust Classification which is given in Table 13.2. Another classification (WHO classification) describes PEM under different nomenclature normal, stunting and wasting.

Clinical Manifestations

The clinical manifestations of PEM range from mild growth retardation and weight loss to distinct clinical forms like kwashiorkor and marasmus. The weight to height ratio is low. There is loss of subcutaneous fat, wasting of muscles and edema. Features of specific vitamin deficiency may also be found. Oral manifestations of malnutrition are given in Table 13.3.

TABLE 13.2: Classification of PEM

Kwashiorkor	60-80% of expected weight with edema
Under nutrition	60-80% of expected weight without edema
Marasmus	Less than 60% of expected weight without edema
Marasmic Kwashiorkor	Less than 60% of expected weight with edema

TABLE 13.3: Oral manifestations in malnutrition

Findings	Deficiency of nutrients
Glossitis	Niacin, riboflavin, Vitamin B ₁₂ , folate, pyridoxine
Bleeding gums	Vitamin C, riboflavin
Cheilosis and angular stomatitis	Riboflavin, niacin
Atrophic lingual papillae	Niacin, riboflavin, iron, folate, Vitamin B ₁₂
Hypogeusia	Vitamin A, zinc
Tongue fissuring	Niacin

Kwashiorkor occurs due to protein deficiency while the energy intake is adequate. The child develops dependant edema, ascites, anasarca, 'flaky paint' dermatosis and fatty liver. *Marasmus* is caused by the deficiency of both, energy and proteins. Patient typically develops weight loss, wasting and cachexia. Body fat stores are depleted and muscle mass decreases. There may be intermediate clinical forms having features of both *kwashiorkor* and *marasmus*.

Kwashiorkor like PEM may develop in adults following trauma, burn and sepsis. *Marasmus* like PEM may result from chronic obstructive pulmonary disease, congestive heart failure, cancer and AIDS.

Laboratory Assessments

- Laboratory assessment includes anthropometric measurements for subcutaneous fat and skeletal muscles (mid-arm circumference and thickness of the triceps skin fold of the posterior mid-upper arm).
- Serum levels of protein, minerals, vitamins and assessment of immune functions are important laboratory tests.

Treatment

Treatment of PEM is a slow process. Patients fed rapidly may develop complications like hypokalemia, hypophosphatemia and congestive heart failure due to volume overload.

- Fluid and electrolyte abnormalities and infections are controlled.
- The treatment is directed toward repletion of protein, energy, vitamins and micronutrients.
- The supplementation can be done orally or parenterally.

VITAMINS

Vitamins are organic compounds required by the body for variety of essential metabolic functions. These are grouped as water soluble and fat soluble vitamins. **Fat soluble** vitamins are vitamin A, D, E and K. **Water soluble** vitamins are thiamin (B_1), riboflavin (B_2), niacin (B_3), pyridoxine (B_6), cobalamin (B_{12}), folate, pantothenic acid, biotin and L-ascorbic acid (vitamin C).

Vitamins are not synthesized at all or are inadequately synthesized in the body; hence they are required in the diet. They are generally needed in very small quantities for essential functions. Multiple vitamin deficiency is more

common than single vitamin deficiency. Body stores for vitamins B_{12} and vitamin A are large and persist for more than an year after being on a deficient diet while folate and thiamine stores are short lived and deplete within weeks. Daily requirement of vitamins are given in Table 13.1.

Most deficiencies are associated with malnutrition, malabsorption, alcoholism, medications, total parenteral nutrition, hemodialysis, food faddism or inborn errors of metabolism.

Vitamin deficiency develops gradually and the symptoms are nonspecific in the early stages. The physical findings appear late. Subclinical deficiency of vitamins is common particularly in the geriatric age group and can be recognized by laboratory testing.

Excess vitamins can also cause disease. Some vitamins are also used as drugs, e.g. niacin for hyperlipidemia and vitamin A for cystic acne and skin wrinkles.

Thiamine (Vitamin B_1)

Functions

- Thiamine pyrophosphate (TPP) acts as a coenzyme in the decarboxylation of alpha-ketoacids and branched-chain amino acids and thus is a source of energy production.
- Thiamine pyrophosphate acts as a coenzyme for a transketolase reaction that mediates conversion of hexose and pentose phosphates.
- It plays a role in peripheral nerve conduction.
- It is a coenzyme for the enzyme alcohol dehydrogenase.

Sources

Important sources of thiamine are yeast, pork, legumes, beef, whole grains and nuts. Milled and polished rice is a poor source of vitamin B_1 . Hence, deficiency of vitamin B_1 is more commonly seen in those with polished rice based diet. Tea and coffee contain thiaminases that destroy the thiamine.

Deficiency

The deficiency of thiamine most commonly occurs due to poor intake and alcoholism. The deficiency presents with nonspecific symptoms in the early stages such as anorexia and irritability. Prolonged deficiency of thiamine causes beriberi. Beriberi may be wet or dry.

- a. Wet beriberi is characterized by the presence of cardiomegaly, tachycardia, congestive heart failure and peripheral edema.
- b. Dry beriberi presents with symmetrical sensorimotor neuropathy.

Alcoholics with chronic thiamine deficiency may also have central nervous system manifestations. These include Wernicke's encephalopathy (nystagmus, ophthalmoplegia, cerebellar ataxia and mental impairment) and Korsakoff's syndrome (amnesia, confabulation and impaired learning).

Diagnosis

In most cases, the clinical response to thiamine therapy supports the diagnosis. However, biochemical tests can also be performed to confirm the deficiency. These include an assay of transketolase activity before and after addition of thiamine pyrophosphate and measurement of thiamine levels in blood and serum.

Treatment

Thiamine in a dose of 100 mg/day parenterally is given for 7 days in acute cases. This is followed by 10 mg/day orally till complete recovery. Anaphylaxis has been reported with the parenteral use of thiamine.

Benfotiamine is a newer derivative of thiamine. It is lipid soluble and hence is absorbed much better than water soluble thiamine hydrochloride. It activates the enzyme transketolase ten times more than thiamine.

Riboflavin (Vitamin B₂)

Functions

Riboflavin plays an important role in the oxidation-reduction reactions and is a cofactor in a number of enzymes involved in energy metabolism.

Sources

Milk and other dairy products, eggs and green leafy vegetables are important sources. Other sources are cereals, legumes, broccoli and fish.

Deficiency

Riboflavin deficiency is usually due to poor dietary intake and is commonly associated with multiple vitamin

deficiencies. People who consume rice as staple diet are more prone to riboflavin deficiency. The clinical manifestations are mucocutaneous lesions of the mouth and skin such as cheilosis, angular stomatitis, magenta tongue and seborrheic dermatitis. Corneal vascularization and anemia may also occur.

Diagnosis

The treatment is usually started empirically. However, the diagnosis can be confirmed by measurement of red blood cell and urinary riboflavin concentrations or red blood cell glutathione reductase activity.

Treatment

Riboflavin deficiency is easily treated with oral preparations of the vitamin with the dosage of 5-15 mg/day. It is given till the clinical findings are resolved.

Niacin (Vitamin B₃)

Functions

Niacin refers to nicotinic acid, nicotinamide and their derivatives. It serves as a precursor of nicotinamide adenine dinucleotide (NAD) and nicotinamide adenine dinucleotide phosphate (NADP) which are involved in many oxidation-reduction reactions. Unlike other vitamins, niacin is synthesized in the body from the amino acid tryptophan, but to a very small extent.

Sources

Important sources of niacin are milk, eggs, beans and meat. The bioavailability of niacin from cereals especially maize is low.

Deficiency

The deficiency of niacin leads to pellagra. This is more common in people who eat corn-based diet. Pellagra can also occur in alcoholics and in inborn errors of tryptophan metabolism (Hartnup's disease).

Pellagra is characterized by dermatitis, diarrhea and dementia. There are dark, dry and scaly lesions in sun exposed areas such as back of the hands, lower legs, face and neck (Casal's necklace). Other features are anorexia, irritability, *bright red glossitis*, stomatitis and weight loss.

Depression, seizures and psychosis may also occur. Advanced pellagra can result in death.

Diagnosis

Pellagra is generally diagnosed clinically. However, measurement of niacin metabolites in the urine can be performed to confirm the diagnosis.

Treatment

Oral nicotinamide or nicotinic acid (10-150 mg daily) effectively treats pellagra.

Toxicity

Toxicity may occur at high dosage of nicotinic acid used to treat hyperlipidemia. Symptoms include flushing, skin dryness and itching. Others are gastric irritation, hepatotoxicity, hyperglycemia, gout and macular edema.

Vitamin B₆ (Pyridoxine)

Functions

Vitamin B₆ refers to a group of closely associated substances including pyridoxine, pyridoxal, pyridoxamine and their 5-phosphate esters. Pyridoxal-5-phosphate is a cofactor for a number of enzymes involved in amino acid metabolism. Vitamin B₆ is also involved in the metabolism of fat, carbohydrates and several vitamins including the conversion of tryptophan to niacin. It is also involved in heme synthesis.

Sources

Milk, liver, meat, legumes, whole grain cereals, nuts and vegetables are good sources of B₆.

Deficiency

Deficiency of vitamin B₆ may occur in association with various drugs (isoniazid, cycloserine, oral contraceptives) or alcoholism. Glossitis, cheilosis, weakness are usual features of the vitamin B₆ deficiency. Severe deficiency results in the peripheral neuropathy and personality changes including depression and irritability. Microcytic hypochromic anemia and seizures may also occur.

Diagnosis

A low level of pyridoxal phosphate in blood is diagnostic of vitamin B₆ deficiency.

Treatment

Vitamin B₆ is given orally in a dosage of 10-20 mg/day. However, higher doses (100-200 mg/day) are needed in cases with medication-related vitamin deficiency. It is also indicated in the prevention of isoniazid-induced neuropathy.

Toxicity

High doses of vitamin B₆ can cause sensory neuro-pathy and dermatitis.

Folate and Vitamin B₁₂

These vitamins are discussed in Chapter 3.

Biotin

This is a water soluble vitamin which plays a role in gluconeogenesis and fatty acid synthesis. Dietary sources of biotin are liver, beans, yeast and egg yolk. Significant amounts are also formed by the bacteria of the gut. In adults, the deficiency of biotin results into depression, hallucination, paresthesia, anorexia. Seborrhic rash may occur over face. The deficiency of biotin in infants can lead to lethargicness, hypotonia and alopecia. Treatment is supplementation of biotin 10 mg /day.

Pantothenic Acid

Dietary sources of pantothenic acid are liver, yeast, egg yolk and vegetables. It plays a role in fatty acid metabolism and in the synthesis of cholesterol and steroid hormones. Deficiency symptoms are nonspecific and are in the form of muscle cramps, paresthesia (burning feet syndrome) and depression.

Choline

Choline is a precursor of acetylcholine, phospholipids and betane. It is also necessary for the integrity of cell membrane. Deficiency can lead to fatty liver and elevated transaminases levels.

Vitamin C (L-ascorbic Acid)

Functions

Vitamin C is a potent antioxidant which is involved in many oxidation-reduction reactions. It is also required for the synthesis of collagen and conversion of dopamine to norepinephrine. Vitamin C promotes iron absorption because

it reduces ferric iron in diet to ferrous form which is better absorbed. It is also involved in drug metabolism, wound healing and carnitine biosynthesis.

Sources

Amla, citrus fruits, guava, green vegetables, potatoes and tomatoes are good sources of vitamin C. The requirement for vitamin C increases under stress, smoking and in those on hemodialysis.

Deficiency

Deficiency may occur in poor, elderly and chronic alcoholics. Vitamin C deficiency causes scurvy in which there is impaired formation of mature connective tissue. The patients may have petechiae, ecchymosis, swollen and bleeding gums, hemarthrosis, anemia and poor wound healing. In children bone growth is impaired. *Periodontal signs do not develop in edentulous patients.*

Diagnosis

Diagnosis is made clinically and confirmed by the low level of ascorbic acid in plasma and leukocytes.

Treatment

The usual dose of vitamin C is 200-500 mg/day orally. Studies have reported that high intake of vitamin C may protect against certain cancers. Vitamin C in the dosage of 1-2 g/day may decrease the duration and symptoms of upper respiratory infection.

Toxicity

More than 2 g/day can cause gastric irritation and diarrhea. Higher doses can lead to hyperuricemia and raised levels of alanine amino transferase (ALT) and lactate dehydrogenase (LDH). It can also lead to false negative test for fecal occult blood and disturbs the urinary glucose estimation.

Vitamin E

Functions

Vitamin E is a collective name for naturally occurring tocopherols. α tocopherol is the most potent of all tocopherols. It is a powerful antioxidant. Vitamin C, glutathione and various enzymes maintain vitamin E in the reduced state.

Sources

Rich sources of vitamin E are sunflower oil, safflower oil and wheat germ oil. It is also found in meat, nuts, cereals, fruits and vegetables.

Deficiency

Dietary deficiency of vitamin E is rare. Deficiency can occur in chronic and severe malabsorption, abetalipoproteinemia, and in children with chronic cholestatic liver disease or cystic fibrosis.

Manifestations of vitamin E deficiency include hemolytic anemia, areflexia, gait disturbances, decreased vibration and position sense and ophthalmoplegia. Myopathy and retinopathy may also occur. It also causes posterior column and spinocerebellar symptoms.

Diagnosis

Low blood levels of α tocopherol is diagnostic of vitamin E deficiency.

Treatment

Symptomatic vitamin E deficiency is treated with 800-1200 mg of α tocopherol daily. Higher doses are required in abetalipoproteinemia. In higher doses, vitamin E protects from intraventricular hemorrhage of prematurity and oxygen-induced retrolental fibroplasias and bronchopulmonary dysplasia.

Toxicity

High dose of vitamin E (>800 mg/day) may interfere with vitamin K metabolism, hence it should not be given to patients taking warfarin. Higher doses can also cause nausea, flatulence and diarrhea.

Vitamin A

Vitamin A (retinol) is a high molecular weight alcohol, synthesized from plant carotenoids. The oxidized metabolites of retinol, retinaldehyde and retinoic acid are also biologically active compounds. Retinaldehyde is required for normal vision and retinoic acid is necessary for growth and cell differentiation. Vitamin A also plays a role in iron metabolism and humoral and cellular immunity.

Sources

Liver and fish have preformed vitamin A. Green leafy vegetable, green and yellow fruits (papaya, mango, pumpkin) and carrots are good sources of β -carotene also known as provitamin A. Milk is a poor source of vitamin A. Fortified food (vanaspati and milk) are also sources for vitamin A.

Deficiency

Vitamin A deficiency is one of the most common vitamin deficiency syndromes and is a common cause of blindness.

Night blindness is the earliest symptom. Conjunctival xerosis, Bitot's spots, corneal xerosis and keratomalacia are other ocular signs of vitamin A deficiency. Xerophthalmia refers to all ocular manifestations of vitamin A deficiency. Keratomalacia is ulceration and necrosis of cornea which may perforate.

Extraocular manifestations include hyperkeratotic skin lesions, anorexia and growth retardation.

Diagnosis

Measurement of serum retinol is diagnostic. Abnormalities of dark adaptation are strongly suggestive of vitamin A deficiency.

Treatment

Ocular changes are treated with 30 mg (100,000 IU) vitamin A intramuscularly or 60 mg (200,000 IU) orally. The dose is repeated every six months in endemic areas.

Toxicity

Acute toxicity has been reported after excess consumption of polar bear liver or intake of more than 150 mg vitamin A. Presentations include increased intracranial pressure, vertigo, diplopia, seizures and exfoliative dermatitis.

Chronic toxicity (ingestion of 15 mg/day for several months) manifests as dry skin, cheilosis, glossitis, alopecia, bone pain, hypercalcemia and increased intracranial pressure.

High dose of carotenoids may cause yellowing of skin (palm and soles) but not the sclera.

Vitamin K

Vitamin K exists in two natural forms, vitamin K I (phylloquinone) and vitamin K II (menaquinone). Vitamin K II is synthesized by intestinal bacteria and vitamin K I is found in animal and plant sources. Menadione (K III) is a synthetic provitamin which is converted to vitamin K II by the liver.

Vitamin K plays an important role in the coagulation process by acting as a cofactor for the post translational gamma carboxylation of factor II, VII, IX and X, protein C and S. These modified proteins are able to bind with platelets in a calcium dependent reaction and participate in the coagulation process more efficiently. Without gamma carboxylation, these factors do not function efficiently. Warfarin type drugs prevent the conversion of vitamin K to its active hydroquinone form.

Sources

Dietary source of vitamin K are green leafy vegetables, fruits, milk, butter, liver and coffee. It is also present in olive oil and soyabean oil. Vitamin K II is endogenously produced by gut bacteria.

Deficiency

Deficiency of vitamin K occurs because of poor diet, malabsorption and broad spectrum antibiotics which suppress colonic flora. Body stores of vitamin K are small, hence deficiency can occur in as little as 1 week. Newborns are particularly susceptible to vitamin K deficiency because of low stores, low levels of vitamin K in the breast milk, lack of intestinal flora and liver immaturity.

Deficiency of vitamin K may present as bleeding from any site.

Diagnosis

Prothrombin time (PT) is prolonged in mild deficiency. PT and APTT both are elevated in advanced cases; however, PT is elevated to a greater extent. The vitamin K level can be measured by chromatography.

Treatment

The deficiency is treated with 10 mg vitamin K parenterally. Patients with chronic malabsorption should receive 1-2 mg

parenterally every week. Newborns are given 1 mg vitamin K intramuscularly at the time of birth as a prophylaxis.

Toxicity

Toxic effects of vitamin K are hemolysis and hyperbilirubinemia which are reported after parenteral use of menadione.

Vitamin D

Vitamin D exists in two forms: ergocalciferol (vitamin D₂) and cholecalciferol (vitamin D₃). Ergocalciferol is derived from plants (the ergot fungus found especially on rye) and cholecalciferol is synthesized in the skin from 7-dehydrocholesterol under the influence of ultraviolet rays present in sunlight or is synthetically produced. Vitamin D is stored largely in fat depots. Cholecalciferol is converted to active forms by hydroxylation in the liver and subsequently in kidneys. In the liver, it is converted to 25-hydroxycholecalciferol (25[OH] D₃) and in kidney, to most potent metabolite 1, 25-dihydroxycholecalciferol (1,25[OH]₂ D₃) or calcitriol.

The main function of vitamin D is to increase absorption of calcium and phosphate from the intestine. It also increases the renal tubular resorption of phosphate. Vitamin D promotes bone mineralization. It may also have other systemic effects since vitamin D receptors are found in many body tissues.

Vitamin D is actually a hormone and if sufficient exposure of sunlight is given to the skin it is not required in the diet. It was classified as vitamins when much was not known about it.

Sources

Vitamin D is derived from both sunlight and food. Liver, butter, cheese, egg and fish are important sources of vitamin D. Fish liver oil is the richest source of vitamin D. Other sources are food artificially fortified with vitamin D such as vanaspati and milk (milk as such is a poor source).

Deficiency

Deficiency can occur due to inadequate exposure to sunlight, lack of dietary intake, malabsorption, impaired

vitamin D activation (in renal disorders) and resistance to the effects of vitamin D (defective receptors). Drugs such as phenytoin, rifampicin and barbiturates can lead to increased metabolism of vitamin D in liver, thus they can cause vitamin D deficiency.

Vitamin D deficiency leads to rickets in children and osteomalacia in adults. *Rickets* is characterized by growth failure, bony deformities, muscular hypotonia, tetany and convulsion. Bony deformities include curved legs, pigeon chest, Harrison sulcus, rickety rosary, kyphoscoliosis and deformed pelvis. Defective skeleton mineralization, bone fractures and proximal myopathy are found in *osteomalacia*.

Diagnosis

Serum calcium and phosphate levels are low. There is a compensatory increase in PTH level. Serum concentration of vitamin D is low.

Treatment

Treatment involves replacement of vitamin D and calcium. Vitamin D can be replaced in various forms like ergocalciferol, 1- α -hydroxy vitamin D₃ (α -calcidol) and 1,25-dihydroxy vitamin D₃ (calcitriol). The dose of calcitriol is 0.25 to 0.5 μ g daily.

Toxicity

Excessive intake of vitamin D may cause hypercalcemia which present as anorexia, nausea, vomiting, thirst, polyurea and drowsiness. Cardiac arrhythmias, renal failure and coma may occur.

IMPLICATIONS ON DENTAL PRACTICE

1. Burning mouth syndrome, mouth ulcers, glossitis, gum bleeding and angular stomatitis may be the first presentation of nutritional deficiency.
2. Undernutrition in children can result in retarded tooth eruption.
3. Undernutrition can lead to immunodeficiency which predisposes to the development of oral ulcerations, necrotizing gingivitis and gangrenous stomatitis.
4. Dysphagia may be a manifestation of iron deficiency.

SELF ASSESSMENT

Multiple Choice Questions

- Alcoholics are prone to develop deficiency of:
 - Vitamin D
 - Vitamin B₁
 - Vitamin C
 - Vitamin E
- Vitamin B₁₂ deficiency can give rise to all of the following except:
 - Myelopathy
 - Optic atrophy
 - Peripheral neuropathy
 - Myopathy
- All the following are true about manifestations of vitamin E deficiency except:
 - Hemolytic anemia
 - Posterior column abnormalities
 - Cerebellar ataxia
 - Autonomic dysfunction
- Following is true about Vitamin A:
 - Water soluble
 - Deficiency causes impaired vision
 - Maintains normal plasma calcium levels
 - Is required for formation of plasma clotting factors
- The final metabolically active form of vitamin D is synthesized in:
 - Kidney
 - Liver
 - Bone
 - Skin
- Glossitis and angular stomatitis are features of all the vitamins except:
 - Niacin
 - Pyridoxine
 - Riboflavin
 - Ascorbic acid
- All the following are causes of bleeding gums except:
 - Acute myeloblastic leukemia
 - Phenytoin therapy
 - Scurvy
 - Candidiasis
- Deficiency of vitamin D can lead to:
 - Hypercalcemia
 - Decreased absorption of calcium from intestine
 - Decreased renal excretion of calcium
 - Increased renal calcium excretion
- The syndrome of angular oral fissure, corneal vascularization and glossitis is due to:
 - Pyridoxine deficiency
 - Niacin deficiency
 - Riboflavin deficiency
 - Xerophthalmia (vitamin A deficiency)
- Sensory neuropathy can occur with high doses of which of the following vitamin:
 - Thiamine
 - Pyridoxine
 - Niacin
 - Vitamin B₁₂
- Which of the following is used to treat hyperlipidemia:
 - Vitamin A
 - Vitamin D
 - Niacin
 - Vitamin E
- Increased intracranial pressure can occur with excess doses of:
 - Vitamin A
 - Vitamin K
 - Vitamin E
 - All the above
- Consumption of polar bear liver can cause acute toxicity of:
 - Vitamin A
 - Vitamin C
 - Vitamin K
 - Folic acid
- Following is used in hyperhomocysteinemia:
 - Folic acid
 - Pyridoxine
 - Riboflavin
 - Ascorbic acid
- Which of the following is a cardiac risk factor?
 - Hyperhomocysteinemia
 - Excess niacin levels
 - Increased HDL cholesterol
 - All of the above
- Yellow discoloration of palm and soles can occur with high doses of:
 - Provitamin A
 - Vitamin K
 - Vitamin D
 - Vitamin C

Fill in the Blanks

- Diarrhea, dementia and dermatitis are due to _____ deficiency.
- Wernicke's encephalopathy is due to the deficiency of _____ vitamin.
- Night blindness is a feature of _____ deficiency.
- Bitot's spot is seen in _____.
- Osteomalacia is due to deficiency of _____.
- Vitamin _____ is required for synthesis of clotting factors.
- Subacute combined degeneration of spinal cord can occur in _____ deficiency.
- Deficiency of vitamin B₁ is more common with _____ based diet.
- Corneal vascularization can occur in _____ deficiency.
- _____ is used to prevent isoniazid induced neuropathy.
- People who eat corn based diet are prone to develop _____ deficiency.

Chapter 14

Preoperative Evaluation

Patients must be thoroughly assessed preoperatively for their fitness for anesthesia and surgery. This is important because there is a risk of morbidity and mortality in patients undergoing surgery. The mortality increases with age as geriatric population is at increased risk of cardiac events. Cardiac causes are major causes of death in postoperative period (First 48 hours). Death in the later period is mainly due to sepsis, pneumonia, cardiac arrest, pulmonary embolism and renal failure.

Identification of high risk patients requires a proper history taking and a focused clinical examination. Evaluation of such patients is important so that their suitability for surgery may be ascertained. American Society of Anesthesiologists (ASA) have classified the surgical risk on the basis of patient's functional status, comorbid diseases and the type of surgery. Head and neck surgery is stratified as intermediate risk (<5% mortality) and superficial procedures as low risk surgery (<1% mortality).

CARDIOVASCULAR EVALUATION

Cardiac risks of surgery include myocardial infarction and other cardiac events which may be fatal. The risk for cardiac complications depends upon functional capacity, age of the patient, type of surgery and presence of comorbid conditions. The predictor of cardiac ischemia in postoperative period include left ventricular hypertrophy on ECG, history of hypertension, definite coronary artery disease, diabetes mellitus and use of digoxin.

During preoperative cardiac evaluation one should particularly look for the evidence of ischemic heart disease, valvular heart disease, congestive heart failure, arrhythmias and hypertension.

Routine surgery must be delayed for at least 6 months after a myocardial infarction.

Blood pressure should be well controlled before surgery.

PULMONARY EVALUATION

Pulmonary complications in the postoperative period include pneumonia, hypoventilation and atelectasis. Perioperative pulmonary function tests such as FEV₁ are indicated in patients with suspected lung disease, COPD and in smokers.

FEV₁ is the best predictor of the surgical risk. Chest X-ray is also recommended in suspected lung disease. Arterial blood gas analysis is generally not needed, however, it is recommended in patients with severe COPD. Patient should be advised to stop smoking weeks before the surgery. The dose of bronchodilators must be optimized.

HEMATOLOGICAL EVALUATION

Patients should be evaluated for the presence of hemorrhagic diatheses. A history of use of aspirin and other NSAIDs, warfarin or heparin must be obtained. Aspirin should be discontinued one week prior to elective procedure. Warfarin is advised to be discontinued at least five days before the surgery. However, if anticoagulation is required, low molecular weight heparin (LMWH) may be given. This is discontinued 12-24 hours prior to surgery and restarted postoperatively. An INR of <1.5 is considered safe for most surgeries. Patients with risk of venous thromboembolism require prophylactic administration of LMWH and early mobilization.

ENDOCRINAL EVALUATION

A moderate control of blood sugar (<150 mg/dL) is desirable in perioperative period. Patient on oral hypoglycemic agent (OHA) should hold the medication on the morning of

surgery (48 hours before in case of metformin). Subsequently, if needed, insulin is given to control blood sugar.

Surgery is contraindicated in uncontrolled hyperthyroidism until the patient is euthyroid.

Adrenal insufficiency may occur in patients who have received long-term steroid therapy. These patients require corticosteroids in perioperative period.

NUTRITIONAL EVALUATION

Patients with malnutrition and hypoalbuminemia have higher postoperative morbidity and mortality. They require preoperative nutritional intervention.

EVALUATION OF OTHER ORGANS

Patients with chronic liver disease have high risk for anesthesia (delayed clearance of anesthesia and prolonged intubation). PT and APTT tests are mandatory prior to surgery.

Patients with renal failure should be monitored for fluid and electrolyte imbalance, hypertension, hypotension and

risks of drug-induced nephro-toxicity. Patient with end-stage renal disease may need hemodialysis prior to surgery and in postoperative period.

Elective surgery is absolutely contraindicated in patients with severe acute infections.

PREOPERATIVE LABORATORY EVALUATION

Certain routine tests are recommended before elective surgery (Table 14.1). However, specific tests may be required based on the clinical conditions of the patient.

TABLE 14.1: Preoperative laboratory tests

- Complete blood count (Hb, TLC, DLC, and Platelet count)
- Prothrombin time (PT) and activated partial thromboplastin time (APTT)
- Urine analysis
- Electrolytes
- Biochemistry (Blood sugar, blood urea, serum creatinine, liver function tests)
- Chest radiograph
- Electrocardiography

Test Paper

1. Which of the following clinical features is indicative of left ventricular failure:
A. Neck vein distension
B. Ascites
C. Orthopnea
D. Edema
2. Which of the following is a sign of aortic stenosis:
A. Diastolic rumble
B. An opening snap
C. Loud second heart sound
D. Harsh systolic murmur
3. All the following are cardiac risk factors except:
A. Hypertension
B. Family history of CAD
C. High levels of HDL
D. Diabetes mellitus
4. All of the following features are seen in patients with pulmonary embolism except:
A. Cyanosis B. Hypoxia
C. Bradycardia D. Right heart failure
5. A reduced one second forced expiratory volume to forced vital capacity is present in all the following except:
A. Allergic asthma B. Chronic bronchitis
C. Emphysema D. Lung abscess
6. Match the following:
A. VSD 1. Absent femoral pulses
B. ASD 2. Continuous murmur
C. PDA 3. Fixed, wide second heart sound
D. Co-arcuation of aorta 4. Holosystolic murmur
7. Match the following:
A. AS 1. Loud first heart sound
B. MS 2. Parvus et tardus
C. MR 3. Austin Flint murmur
D. AR 4. Pansystolic murmur
8. Match the following:
A. Pulsus alternans 1. Cardiac tamponade
B. Pulsus paradoxus 2. LVF
C. Water-Hammer pulse 3. AF
D. Irregularly irregular pulse 4. AR
9. Following are the features of iron deficiency anemia except:
A. Low serum ferritin
B. Hypochromic microcytic
C. Absent marrow iron on biopsy stain
D. Decreased total iron binding capacity
10. All are true regarding idiopathic thrombocytopenic purpura except:
A. Steroids are used in treatment
B. Splenomegaly is usually present
C. Bone marrow reveals increased megakaryocytes
D. Splenectomy can be effective therapy
11. All of the following patients may have DIC except:
A. Patients with gram negative infections
B. Patients with multiple trauma
C. Patients with administered thrombolytic agents
D. Patients with retained dead fetus
12. Hemolysis is indicated by all of the following except:
A. Increased LDH levels
B. Shortened red blood cell survival
C. Decreased number of reticulocytes
D. Reduced serum haptoglobin
13. All the following can cause inflammatory bloody diarrhea except:
A. *Entamoeba histolytica*
B. *Giardia lamblia*
C. *Campylobacter jejuni*
D. *Shigella* species
14. Following is not true regarding Plummer-Vinson syndrome:
A. Esophageal webs are present
B. Webs may be associated with iron deficiency anemia

- C. Also known as Paterson-Kelly syndrome
D. Lesions are not premalignant
15. **Therapy of ascites includes all the following except:**
A. Diuretics
B. Paracentesis
C. Unrestricted salt intake
D. Bedrest
16. **Rifampicin is used in:**
A. Treatment of tuberculosis
B. Prevention of meningococcal meningitis
C. Treatment of staphylococcal endocarditis
D. All the above
17. **Match the following:**
- | | |
|----------------------------|-----------------------------|
| A. Cough | 1. Calcium channel blockers |
| B. Gingival hyperplasia | 2. Beta blockers |
| C. Precipitation of asthma | 3. ACE inhibitors |
| D. Hypokalemia | 4. Diuretics |
18. **Limb edema can occur with the use of following drug:**
A. Beta blockers
B. Calcium channel blockers
C. Diuretics
D. Nitrates
19. **Following is a cause of secondary hypertension:**
A. Eclampsia
B. Renovascular
C. Coarctation of aorta
D. All the above
20. **Which of the following is given to prevent isoniazid induced neuropathy:**
A. Steroids
B. Folic acid
C. Vitamin B₁₂
D. Pyridoxine
21. **Match the following:**
- | | |
|---------------------------|----------------------------------|
| A. Rheumatic fever | 1. <i>H. influenzae</i> |
| B. Subacute endocarditis | 2. <i>S aureus</i> |
| C. Nosocomial pneumonia | 3. Group A streptococci |
| D. Meningitis in children | 4. <i>Streptococcus viridans</i> |
22. **Match the following:**
- | | |
|-----------------|---------------------|
| A. Rifampicin | 1. Optic neuritis |
| B. Ethambutol | 2. Ototoxicity |
| C. Streptomycin | 3. Neuropathy |
| D. Isoniazid | 4. Thrombocytopenia |
23. **The most common organism for lobar pneumonia is:**
A. *S. aureus*
B. *S. pneumoniae*
C. *Mycoplasma pneumoniae*
D. Gram negative bacteria
24. **Which of the following should be avoided in treatment of tuberculosis in mild to moderate renal failure:**
A. Rifampicin
B. Pyrazinamide
C. Isoniazid
D. Streptomycin
25. **All the following can cause hypercalcemia except:**
A. Sarcoidosis
B. Multiple myeloma
C. Tumor lysis syndrome
D. Prolonged immobilization
26. **A 52-year-old female with type 2 diabetes is found to have 24-hr urinary proteinuria of 280 mg. Which of the following drug should be used to retard progression of renal disease:**
A. Aspirin
B. Thiazide
C. Ramipril
D. Amiloride
27. **In a patient of *Listeria meningitis* who is allergic to penicillin, the antimicrobial of choice is:**
A. Gentamicin
B. Ceftriaxone
C. Trimethoprim-sulphamethoxazole
D. Vancomycin
28. **All the following can cause megakaryocytic thrombocytopenia except:**
A. Aplastic anemia
B. Systemic lupus erythematosus
C. Disseminated intravascular coagulation
D. Idiopathic thrombocytopenic purpura
29. **The following test may be abnormal in disseminated intravascular coagulation except:**
A. D-dimer level
B. Clot solubility
C. Prothrombin time (PT)
D. Activated partial thromboplastin time (APTT)
30. **A patient of acute leukemia is admitted with febrile neutropenia. On day four of being treated with broad spectrum antibiotics, his fever increases. Which of the following should be most appropriate next step in the management:**
A. Add antiviral therapy
B. Add antifungal therapy
C. Add cotrimoxazole
D. Continue chemotherapy
31. **A 28-year-old man is noted to have blood pressure of 180/104 mmHg. He has prominent ejection click in aortic area and murmurs heard over the ribs on both sides anteriorly and posteriorly. In addition, the pulses in the lower extremities are feeble and he complains of mild claudication with exertion. The most likely diagnosis is:**
A. Cardiomyopathy
B. Atrial septal defect
C. Coarctation of aorta
D. Aortic stenosis

32. All the following are true about therapy for tuberculosis **except**:
- Hyperuricemia is a side effect of pyrazinamide
 - Red-green color impairment is a sign of ethambutol toxicity
 - Ethambutol accumulates in renal failure
 - 'Flu like syndrome' is usually seen in people taking rifampicin on daily basis
33. Which of the following organisms, when isolated in blood, requires the synergistic activity of penicillin plus an aminoglycoside for appropriate therapy:
- Enterococcus faecalis*
 - S. aureus*
 - Streptococcus pneumoniae*
 - Bacteriodes fragilis*
34. All the following can cause DIC **except**:
- Amniotic fluid embolism
 - Intrauterine fetal death
 - Diabetes mellitus
 - Abruptio placentae
35. Ebstein-Barr virus is associated with:
- Carcinoma larynx
 - Carcinoma bladder
 - Carcinoma nasopharynx
 - Carcinoma maxilla
36. A 12-year-old girl has history of recurrent bulky stools and abdominal pain since 3 years of age. She has pallor and her weight and height are below the third percentile. Which of the following is the most appropriate investigation to make a specific diagnosis
- barium studies
 - D-xylose test
 - 24-hr fecal fat estimation
 - Endoscopy and small intestinal biopsy
37. A child underwent a tonsillectomy at 6 years of age with no complications. He underwent a preoperative screening for bleeding at the age of 12 years before an elective laparotomy, and was found to have a prolonged partial thromboplastin time but normal Prothrombin time. There was no family history of bleeding. The patient is likely to have:
- Acquired liver disease
 - Acquired vitamin K deficiency
 - Mild hemophilia A
 - Factor XII deficiency
38. Endoscopic biopsy from a case of *H. pylori* related early chronic gastritis is most likely to reveal:
- Multifocal atrophic gastritis
 - Erosive gastritis
 - Antral predominant gastritis
 - Gastric atrophy
39. All the following diseases cause massive splenomegaly **except**:
- Malaria
 - Kala azar
 - Idiopathic myelofibrosis
 - Infective endocarditis
40. A 10-day-old neonate is posted for pyloric stenosis surgery. The investigations report shows a serum calcium level of 6.0 mg /dL. What information would you like to know before you supplement calcium to this neonate:
- Blood glucose
 - Serum bilirubin
 - Serum albumin
 - Oxygen saturation
41. All the following drugs are used in the management of status epilepticus **except**:
- Phenytoin
 - Thiopentone sodium
 - Carbamazepine
 - Diazepam
42. Urgent reversal of warfarin therapy can be done by administration of:
- Platelet concentrates
 - Cryoprecipitate
 - Fresh frozen plasma
 - Packed red blood cell
43. Which of the following is generally not seen in idiopathic thrombocytopenic purpura (ITP):
- Petechiae, ecchymosis and bleeding
 - Palpable splenomegaly
 - More common in females
 - Increased megakaryocytes in the bone marrow
44. A young patient presenting with massive hematemesis was found to have splenomegaly. In this case the most likely source of bleeding is:
- Duodenal ulcer
 - Esophageal varices
 - Erosive gastritis
 - Gastric ulcer
45. Following resuscitation, a patient with bleeding esophageal varices should be treated initially with:
- Sclerotherapy
 - Sagstaken Blackmore tube
 - Propanolol
 - Surgery
46. In which of the states in India the maximum number of AIDS cases have been reported till now:
- Delhi
 - Kerala
 - Tamil Nadu
 - Bihar
47. Most sensitive test to detect hypothyroidism is:
- T₃
 - T₄
 - TSH
 - Radioactive iodine uptake
48. Following is not an anterior pituitary hormone:
- FSH
 - TSH
 - ACTH
 - ADH

49. HIV positive patients usually have:
- Low CD₄, low CD₈
 - High CD₄, low CD₈
 - Low CD₄, high CD₈
 - High CD₄, high CD₈
50. Which of the following mediators is released during late phase reaction of anaphylactic reaction:
- Eosinophil chemotactic factor (ECF-A)
 - Histamine
 - Leukotriene B₄
 - Angiotensinogen
51. IgE-mediated hypersensitivity is implicated in all of the following reactions *except*:
- Serum sickness
 - Latex anaphylaxis
 - Systemic reaction to insect sting
 - Asthma
52. Following are indications of surgery in endocarditis *except*:
- Congestive heart failure
 - Fungal endocarditis
 - History of injection drug use
 - Failure to respond to antibiotic therapy
53. Generalized lymphadenopathy is found in all *except*:
- Infectious mononucleosis
 - HIV-1
 - Malaria
 - Syphilis
54. Bacterial and viral meningitis are characterized by all *except*:
- Neck rigidity
 - Increased protein in CSF
 - Increased cells in CSF
 - Decreased CSF glucose
55. Which of the following hormones is given in cases of hypothyroidism:
- TSH
 - T₃
 - T₄
 - Thyroglobulin
56. All the following are features of diabetic ketoacidosis *except*:
- Glycosuria
 - Decreased peripheral utilization of glucose
 - Decreased lipolysis
 - Increased protein catabolism
57. A 37-year-old patient presented with generalized repetitive convulsions for 35 minutes. He is given 10 mg of diazepam intravenously. The next appropriate step would be to administer:
- Carbamazepine orally
 - Phenytoin intravenously
 - Phenobarbitone intravenously
 - Conazepam orally
58. Chronic type A gastritis is characterized by all the following clinical findings *except*:
- Pernicious anemia
 - Presence of parietal cell antibody
 - Antral involvement
 - Associated other autoimmune disorder
59. A person vaccinated against hepatitis B will have following serologic finding:
- Hepatitis B surface antigen (HBsAg)
 - Hepatitis B surface antibody (anti-HBs)
 - Both HBsAg and anti-HBs
 - Hepatitis Be antibody
60. All the following can precipitate hepatic encephalopathy in patients of cirrhosis:
- Gastrointestinal bleeding
 - SBP
 - Ampicillin therapy
 - Sedatives
61. A positive Chovstek's sign is found in:
- Hyperkalemia
 - Hypocalcemia
 - Acidosis
 - Hypoglycemia
62. In Pseudohypoparathyroidism the lab test reveal:
- Low calcium and high PTH (parathyroid hormone)
 - High calcium and low PTH
 - High calcium and high PTH
 - Low calcium and low PTH
63. All the following features may be in hyperthyroidism *except*:
- Atrial fibrillation
 - Pretibial myxedema
 - Constipation
 - Lid lag
64. Crackles are present in which of the following condition:
- Pleural effusion
 - Pneumothorax
 - Left ventricular failure
 - Emphysema
65. A 50-year-old asymptomatic patient has fasting blood glucose of 132 mg% on routine examination, he has:
- Normal report
 - Diabetes mellitus
 - Impaired fasting glucose
 - Impaired glucose tolerance
66. Which of the following tests is most appropriate to know glucose control in diabetic patients:
- Fasting glucose
 - Postprandial glucose
 - Urinary glucose
 - Glycated hemoglobin
67. A patient with lung malignancy has presented with nausea, vomiting, weakness and serum calcium value of 14.5 mg%. First line step would be to administer:
- Oral prednisolone
 - Mithramycin
 - Insulin
 - Intravenous saline and frusemide

68. The definitive diagnosis of pulmonary embolism is best made by:
 A. ABG analysis
 B. Chest X-ray
 C. Ventilation perfusion scan
 D. Pulmonary angiography
69. Clubbing may be seen in all the following conditions except:
 A. Fallot's tetralogy B. Eisenmenger syndrome
 C. Infective endocarditis D. Constrictive pericarditis
70. A female patient has presented with a history of menorrhagia and easy bruising. Lab studies show a normal PT, a prolonged APTT and prolonged bleeding time. The most likely diagnosis is:
 A. Hemophilia
 B. Factor IX deficiency
 C. von Willebrand's disease
 D. Severe hepatic dysfunction
71. A patient has serum calcium of 13.5 mg% on routine examination. Which of the following would be most appropriate test to diagnose primary hyperparathyroidism:
 A. Serum phosphate
 B. Ultrasound of the neck
 C. Serum parathyroid levels
 D. Serum ionized calcium
72. The side effects of thiazide diuretic therapy is:
 A. Hypoglycemia B. Hyperuricemia
 C. Cough D. Bronchospasm
73. All the following are causes of eosinophilia except:
 A. Ascaris infestation
 B. Nitrofurantoin therapy
 C. Enteric fever
 D. Allergic bronchopulmonary aspergillosis (ABPA)
74. Following are features of enteric fever except:
 A. Splenomegaly B. Rose spots
 C. Relative tachycardia D. Leukopenia
75. A patient has fasting blood sugar level of 98 mg/dL and has 182 mg% blood sugar level 2 hr after glucose load (GTT). A patient is having:
 A. Diabetes mellitus
 B. Impaired fasting glucose
 C. Impaired glucose tolerance
 D. Diabetic ketoacidosis
76. A patient has purpura with a platelet count of 2 lacs/mm³. Following may be the possibilities except:
 A. Henoch-Schönlein purpura
 B. Uremia
 C. ITP
 D. Vasculitis
77. Cryoprecipitate is rich in:
 A. All coagulation factors
 B. All coagulation factors and platelets
 C. Factor VIII, vWF and fibrinogen
 D. Platelets and leukocytes
78. Serum creatinine is generally normal in:
 A. Acute renal failure
 B. Nephrotic syndrome
 C. End-stage renal disease
 D. Nephritic syndrome
79. Following is true in Gilbert's syndrome:
 A. Raised alkaline phosphates
 B. Unconjugated hyperbilirubinemia
 C. High ALT and AST
 D. Tender hepatomegaly
80. Which of the following is not an oral manifestation of HIV:
 A. Hairy leukoplakia B. Papilloma virus (warts)
 C. Candidiasis D. P. cranii
81. Which of the following is not matched properly?
 A. Primary syphilis – chancre
 B. Tertiary syphilis – gumma
 C. Secondary syphilis – mulberry molars
 D. Congenital syphilis – Hutchinson's teeth
82. HIV can be confirmed and detected by:
 A. Polymerase chain reaction (PCR)
 B. Reverse transcriptase – PCR (RT-PCR)
 C. Real time PCR
 D. Mimic PCR
83. Patients infected with HIV may have:
 A. Elevated blastogenesis
 B. Depressed serum immunoglobulin levels
 C. High T4-T8 ratio
 D. Thrombocytopenia
84. AIDS is associated with depletion of:
 A. T4 lymphocytes B. T8 Lymphocytes
 C. B lymphocytes D. Monocytes
85. AIDS is frequently complicated by opportunistic infections, notably pneumonia due to:
 A. *Klebsiella pneumoniae*
 B. *Pneumocystis carinii*
 C. *Streptococcus pneumoniae*
 D. *Mycoplasma pneumoniae*
86. Diagnosis of typhoid fever in the first week is made by following test:
 A. Widal test B. Stool culture
 C. Urine culture D. Blood culture
87. Causes of chest pain include all except:
 A. Prinzmetal's angina B. Pericarditis
 C. Aortic stenosis D. Mitral stenosis

88. Which of the following is used in hypertensive crisis:
 A. Atenolol
 B. Thiazides
 C. Enalapril
 D. Nitroglycerine or nitroprusside
89. Multiple myeloma is commonly associated with *all except*:
 A. Anemia
 B. High serum calcium
 C. High serum alkaline phosphatase
 D. High ESR
90. Following type of insulin has longest duration of action:
 A. Regular
 B. Lispro
 C. Aspart
 D. Glargine
91. Hypoglycemia may occur in:
 A. Insulin overdose
 B. Insulinoma
 C. Severe hepatitis
 D. All the above
92. Polyurea is seen with:
 A. Diabetes mellitus
 B. Diabetes insipidus
 C. Hyperparathyroidism
 D. All the above
93. Earliest indication of diabetic nephropathy is:
 A. Hematuria
 B. Oliguria
 C. Proteinuria
 D. Raised serum creatinine
94. Hypovolemic shock occurs when intravenous volume is decreased by:
 A. 5%
 B. 15%
 C. 25%
 D. 40%
95. Which of the following drugs is bacteriostatic:
 A. Rifampicin
 B. Pyrazinamide
 C. Isoniazide
 D. Ethambutol
96. Dressler syndrome is a complication of:
 A. Hypertension
 B. Pericarditis
 C. Mitral stenosis
 D. Myocardial infarction
97. Most sensitive marker of MI is:
 A. Troponin T
 B. CPK
 C. SGOT
 D. LDH
98. All the following are true for hemophilia A *except*:
 A. Increased PT
 B. Increased APTT
 C. Soft tissue hematoma
 D. Decreased factor VIII
99. Which of the following is not used in the treatment of peptic ulcer:
 A. Omeprazole
 B. Itraconazole
 C. Rabepazole
 D. Pantaprazole
100. Which of the followings is used in CML:
 A. Cyclophosphamide
 B. Imatinib mesylate
 C. Prednisolone
 D. Chlorambucil
101. Normal daily excretion of protein in urine is:
 A. <150 mg
 B. <300 mg
 C. <500 mg
 D. <1000 mg
102. Carbamazepine is used to treat the following *except*:
 A. Partial seizures
 B. Trigeminal neuralgia
 C. Postherpetic neuralgia
 D. Tension headache
103. Déjà vu phenomenon is associated with:
 A. Absence seizures
 B. Grand mal seizures
 C. Simple partial seizures
 D. Complex partial seizures
104. Recently identified liver protein which regulates iron metabolism is:
 A. Hepadna
 B. Haptoglobin
 C. Hepcidin
 D. Transferrin receptor
105. Iron therapy is not indicated in:
 A. Chronic blood loss
 B. Pregnancy
 C. CRF
 D. Thalassemia minor
106. Anemia of chronic inflammation has following features *except*:
 A. Low ferritin
 B. Normal bone marrow iron
 C. Low serum iron
 D. Low TIBC
107. Infective endocarditis is least likely to occur in:
 A. VSD
 B. Fallot's tetralogy
 C. ASD
 D. Coarctation of aorta
108. Infective endocarditis is suggested by *all except*:
 A. Clubbing
 B. Splenomegaly
 C. Osler's node
 D. Berry's aneurysm
109. The cardiac valve affected in the infective endocarditis in IV drug abusers is:
 A. Mitral
 B. Tricuspid
 C. Aortic
 D. Pulmonary
110. Black water fever is a feature of:
 A. Kala azar
 B. Malaria
 C. Enteric fever
 D. Dengue fever
111. Which of the antitubercular drugs mostly interferes with antiretroviral therapy:
 A. INH
 B. Rifampicin
 C. PAS
 D. Streptomycin
112. Which of the following drugs can precipitate hemolysis in G6PD deficiency:
 A. Chloroquine
 B. Primaquine
 C. Mefloquine
 D. Quinine
113. Type of malaria generally not found in India:
 A. *P. ovale*
 B. *P. vivax*
 C. *P. malariae*
 D. *P. falciparum*

114. 'P' wave is absent in:
 A. Atrial flutter
 B. Atrial fibrillation
 C. Complete heart block
 D. Second degree heart block
115. Indicator of active replication of hepatitis B virus is:
 A. HBsAg B. HBcAg
 C. HBeAg D. Anti-HBe
116. Austin Flint murmur may be mistaken for:
 A. MR B. MS
 C. TS D. PR
117. Match the following:
 A. Graham Steel murmur 1. AR
 B. Mid-diastolic murmur 2. VSD
 C. Pansystolic murmur 3. PR
 D. Early diastolic murmur 4. MS
118. Collapsing pulse may be present in all the following except:
 A. AR B. PDA
 C. Severe anemia D. AS
119. Following drugs cause bronchodilatation except:
 A. Salbutamol B. Terbutaline
 C. Corticosteroids D. Ipratropium bromide
120. Clopidogrel is a following agent:
 A. Antiplatelet B. Thrombolytic
 C. Fibrinolytic D. Anticoagulant
121. Following are causes of hypocalcemia except:
 A. Hypomagnesemia B. Acute pancreatitis
 C. Milk alkali syndrome D. Hypoparathyroidism
122. Following is associated with high mortality if it occurs during pregnancy:
 A. Hepatitis A B. Hepatitis B
 C. Hepatitis C D. Hepatitis E
123. Drug of choice in trigeminal neuralgia is:
 A. Carbamazepine B. Phenobarbitone
 C. Beta blockers D. Verapamil
124. Pulmonary oligemia is characteristically seen in:
 A. ASD B. VSD
 C. PDA D. Fallot's tetralogy
125. All the following are associated with infranuclear facial palsy except:
 A. Bell's palsy B. Acoustic neuroma
 C. Cerebral infarction D. Middle ear disease
126. In adults, the spinal cord normally ends at:
 A. Lower border of L₁ B. Lower border of L₃
 C. Lower border of S₁ D. Lower border of L₅
127. Iron is present in all the following except:
 A. Myoglobin B. Cytochrome.
 C. Catalase. D. Pyruvate kinase
128. The National AIDS Control Program has the following components, except:
 A. Serosurveillance.
 B. Health education and information
 C. Screening of blood and blood products
 D. Banning of sexual contact with foreigners
129. The antibiotic cover is mandatory before extraction in the following condition of the heart:
 A. Ischemic heart disease (IHD).
 B. Hypertension
 C. Congestive cardiac failure
 D. Congenital heart disease
130. Blue sclera is characteristic of:
 A. Amelogenesis imperfecta
 B. Tetracycline hypoplasia
 C. Fluorosis
 D. Osteogenesis imperfecta
131. Bence Jones protein found in the urine may be suggestive of:
 A. Hyperparathyroidism B. Hodgkin's disease
 C. Multiple myeloma D. Christian's syndrome.
132. The syndrome which is associated with predisposition to the development of carcinoma of oral mucous membrane is:
 A. Gardner's syndrome
 B. Osler-Rendu-Weber syndrome
 C. Sturge-Weber syndrome
 D. Plummer-Vinson syndrome
133. The most preferred approach for pituitary surgery at present is:
 A. Transcranial B. Transethmoidal
 C. Transphenoidal D. Transcallosal
134. The normal range of serum osmolality (in mOsm/kg H₂O) is:
 A. 275 to 295 B. 310 to 330
 C. 350 to 375 D. 200 to 250
135. Afferent component of corneal reflex is mediated by:
 A. Vagus nerve
 B. Facial nerve
 C. Trigeminal nerve
 D. Glossopharyngeal nerve
136. Ground glass appearance in bone is seen in:
 A. Hyperparathyroidism B. Fibrous dysplasia
 C. Condensing osteitis D. Osteopetrosis
137. An affected male infant born to normal parents could be an example of all the following except:
 A. An autosomal dominant disorder
 B. An autosomal recessive disorder
 C. A polygenic disorder
 D. A vertically transmitted disorder

138. **Nevirapine is a:**
 A. Protease inhibitor
 B. Nucleoside reverse transcriptase inhibitor
 C. Non-nucleoside reverse transcriptase inhibitor
 D. Fusion inhibitor
139. **All the following are the known causes of osteoporosis except:**
 A. Fluorosis B. Hypogonadism
 C. Hyperthyroidism D. Hyperparathyroidism
140. **All the following are risk factors for atherosclerosis except:**
 A. Increased waist-hip ratio
 B. Hyperhomocysteinemia
 C. Decreased fibrinogen levels
 D. Decreased HDL levels
141. **All the following antibacterial agents act by inhibiting cell wall synthesis except?**
 A. Carbapenems B. Monobactams
 C. Cephamycins D. Nitrofurantoin
142. **A diabetic patient developed cellulitis due to *Staphylococcus aureus*, which was found to be methicillin resistant on the antibiotic sensitivity testing. All the following antibiotics will be appropriate except:**
 A. Vancomycin B. Imipenem
 C. Teichoplanin D. Linezolid
143. **Calcitonin is secreted by:**
 A. Thyroid gland B. Parathyroid gland
 C. Adrenal glands D. Ovaries
144. **Bisphosphonates act by:**
 A. Increasing the osteoid formation.
 B. Increasing the mineralization of osteoid.
 C. Decreasing the osteoclast-mediated resorption of bone
 D. Decreasing the parathyroid hormone secretion
145. **Brown tumors are seen in:**
 A. Hyperparathyroidism
 B. Pigmented villonodular synovitis
 C. Osteomalacia
 D. Neurofibromatosis
146. **Megaloblastic anemia due to folic acid deficiency is commonly due to:**
 A. Inadequate dietary intake
 B. Defective intestinal absorption
 C. Absence of folic acid binding protein in serum
 D. Absence of glutamic acid in the intestine
147. **Which one of the following drugs is an antipseudomonal penicillin?**
 A. Cephalexin B. Cloxacillin
 C. Piperacillin D. Dicloxacillin
148. **All of the following are therapeutic uses of penicillin G, except in:**
 A. Bacterial meningitis B. Rickettsial infection
 C. Syphilis D. Anthrax
149. **Which the following organs is not involved in calcium homeostasis?**
 A. Kidneys B. Skin
 C. Intestines D. Lungs
150. **Mycotic aneurysm is due to:**
 A. Bacterial infection B. Fungal infection
 C. Viral infection D. Mixed infection

Reference Laboratory Values

CHEMICAL CONSTITUENTS OF BLOOD

Albumin (s)	3.5-5.5 g/dL
Amino transferases (s)	
Aspartate (AST, SGOT)	0-35 U/L
Alanine (ALT, SGPT)	0-35 U/L
Amylase (s)	60-180 U/L
Arterial Blood Gases (p)	
-HCO ₃ ⁻	21-30 meq/L
-PCO ₂	35-45 mm Hg
-PO ₂	80-100 mm Hg
-pH	7.38 – 7.44
Bilirubin, Total (s)	0.3-1.0 mg/dL
-Direct	0.1-0.3 mg/dL
-Indirect	0.2-0.7 mg/dL
Calcium, Ionized (p)	4.5-5.6 mg/dL
Calcium total	9.0-10.5 mg/dL
Creatine Kinase (s)	
Females	20-170 U/L
Males	30-200 U/L
Creatine Kinase MB isoenzyme	< 6% of total CK
Creatinine (s)	<1.5 mg/dL
Ferritin (s)	
Female	10-200 ng/mL
Male	15-400 ng/mL
Folate (s)	3.1-17.5 mg/mL
Glucose(Fasting) (p)	
Normal	75-100 mg/dL
Impaired	100-125 mg/dL
Diabetes mellitus	> 126 mg/dL
Glucose, 2 hr Postprandial (p)	
Normal	<140 mg/dL
Impaired glucose tolerance	140-199 mg/dL
Diabetes mellitus	> 200 mg/dL

Hemoglobin A1c	upto 6% of total Hb
Iron (s)	50–150 mg/dL
Iron binding capacity (s)	250–370 mg/dL
Saturation	20-45 %
Lactate dehydrogenase (LDH) (s)	100-190 U/L
Lipase (s)	0-160 U/L
Osmolality (p)	285-295 mosmol/Kg serum water
Phosphatase, acid (s) (ACP)	0-5.5 U/L
Phosphatase, alkaline [ALP] (s)	30-120 U/L
Phosphorous, Inorganic (s)	3.0-4.5 mg/dL
Potassium (s)	3.5-5.0 mEq/L
Sodium (s)	136-145 mEq/L
Protein, total (s)	5.5-8.0 g/dL
Protein, fractions	
– Albumin	3.5-5.5 g/dL (50-60%)
– Globulin	2.0-3.5 g/dL (40-50%)
– α_1	0.2-0.4 g/dL
– α_2	0.5-0.9 g/dL
– β	0.6-1.1 g/dL
– γ	0.7-1.7 g/dL (13-23%)
Troponin I (s)	0-0.4 ng/mL
Troponin T (s)	0-0.1 ng/mL
Urea Nitrogen (s)	10-20 mg/dL
Urea (s)	20-40 mg/dL
Uric acid (s)	
Male	2.5-8.0 mg/dL
Female	1.5-6.0 mg/dL
Vitamin B ₁₂	250-900 ng/mL

HEMATOLOGICAL EVALUATION

Hemoglobin (p)	
Male	13-16 g/dL
Female	12-15 g/dL
Hematocrit (p)	
Male	42-52%
Female	37-48%
TLC	4000-11000/mm ³
DLC	
- Neutrophils	45-74%
- Lymphocytes	16-45%
- Monocytes	4-10%
- Eosinophils	0-7%
- Basophils	0-2%
Platelets	1.5-4.0 lacs/mm ³

ESR	
<i>Westergren <50 yrs</i>	
– Male	0-15 mm for 1 st hr
– Female	0-20 mm for 1 st hr
<i>Westergren >50 yrs</i>	
– Male	0-20 mm for 1 st hr
– Female	0-30 mm for 1 st hr
Bleeding Time	< 7 min
Prothombin Time (PT)	control ± 1 sec
Partial Thromboplastin Time	comparable to control (Activated PTT)
Thrombin Time	control ± 3 secs.
Mean Corpuscular Hb (MCH)	27-33 pg
Mean Corpuscular Hb Concentration (MCHC)	31-35 %
Mean Corpuscular Volume (MCV)	82-98 fL
Osmotic Fragility	
– Slight hemolysis	0.45-0.39%
– Complete hemolysis	0.33-0.3%
Reticulocyte	
– adults	0.5-1.5 %
– children	2.5-6.5 %

CEREBROSPINAL FLUID

CSF Pressure	50-180 mm H ₂ O
CSF Volume	~150 ml
Glucose	40-70 mg/dL
Protein	20-50 mg/dL
Cells total	< 5 per mm ³
DLC	
-Lymphocyte	60-70%
-Monocyte	30-50%
-Neutrophils	None

ATP 3 (ADULT TREATMENT PANEL) CLASSIFICATION OF LDL, TOTAL AND HDL CHOLESTEROL (mg/dL)

<i>LDL CHOLESTEROL</i>	
Optimal	< 100
Near normal	100-129
Borderline high	130-159
high	160-189
Very high	> 190
<i>TOTAL CHOLESTEROL</i>	
Desirable	<200
Borderline high	200-234
High	> 240

HDL CHOLESTEROL

Low	<40
High	>60

TRIGLYCERIDES

Normal	< 150 mg/dL
Borderline high	150-199 mg/dL
High	200-499 mg/dL
Very high	> 500 mg/dL

AVERAGE VALUES IN PULMONARY PHYSIOLOGY

<i>Parameters</i>	<i>Male</i>	<i>Female</i>
Forced vital capacity (FVC)	4.8 L	3.3 L
Forced expiratory volume in 1 second (FEV ₁)	3.8 L	2.8 L
FEV ₁ /FVC	76%	77%
Maximal expiratory flow rate	9.4 L/sec	6.1 L/sec
Total lung capacity (TLC)	6.4 L	4.9 L
Functional residual capacity (FRC)	2.2 L	2.6 L
Residual volume (RV)	1.5 L	1.2 L
Inspiratory capacity (IC)	4.8 L	3.7 L
Expiratory reserve volume	3.2 L	2.3 L
Vital capacity	1.7 L	1.4 L

Answers

CHAPTER 1: CLINICAL METHODS

Multiple Choice Questions

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. b | 2. b | 3. c | 4. c | 5. c |
| 6. b | 7. c | 8. e | 9. d | 10. b |
| 11. c | 12. b | 13. c | 14. c | 15. b |
| 16. c | 17. b | 18. b | 19. a | 20. c |
| 21. d | 22. d | 23. c | | |

Fill in the Blanks

1. Coarctation of aorta
2. Left ventricular failure
3. 4
4. Kussmaul's breathing
5. Tricuspid regurgitation
6. <60
7. >100
8. 12-16
9. Frequent ectopics
10. Hepatic failure
11. Carbon monoxide poisoning
12. Cheyne-Stokes respiration

CHAPTER 2: GASTROINTESTINAL AND HEPATOBILIARY SYSTEM

Multiple Choice Questions

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. C | 2. A | 3. B | 4. D | 5. C |
| 6. D | 7. D | 8. D | 9. B | 10. D |
| 11. D | 12. C | 13. C | 14. D | 15. C |
| 16. B | 17. D | 18. C | 19. D | 20. B |
| 21. A | 22. C | 23. D | 24. A | 25. B |
| 26. B | 27. D | 28. D | 29. C | 30. A |
| 31. B | 32. D | 33. D | 34. C | 35. D |
| 36. D | 37. C | 38. A | 39. B | 40. D |
| 41. C | 42. B | 43. C | 44. C | |

Fill in the Blanks

1. 10-12 mmHg
2. Caput medusae
3. 0.3-1 mg/dL
4. Liver
5. II, VII, IX, X
6. Alcoholic hepatitis
7. Hematemesis
8. Malabsorption (steatorrhea)
9. Liver
10. Fetor hepaticus
11. Ascites
12. Cirrhosis of liver
13. Cushing's ulcer
14. Curling's ulcer
15. Proximal small intestine (jejunum)

CHAPTER 3: HEMATOLOGICAL SYSTEM

Multiple Choice Questions

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. c | 2. c | 3. a | 4. d | 5. c |
| 6. d | 7. a | 8. b | 9. d | 10. a |
| 11. b | 12. b | 13. c | 14. b | 15. c |
| 16. b | 17. a | 18. a | 19. b | 20. b |
| 21. d | 22. b | 23. c | 24. d | 25. b |
| 26. d | 27. c | 28. b | | |

Fill in the Blanks

1. ALL
2. CML
3. PT
4. Heparin
5. Iron deficiency anemia
6. Low

7. CML
8. 10,000/ μ L
9. Hodgkin's disease
10. Hemolytic
11. vWD
12. Metastasis
13. Left supraclavicular LN due to metastasis
14. Vitamin B₁₂
15. Vitamin B₁₂
16. More than 8 cm from left costal margin
17. Cervical
18. Tuberculosis
19. Ferrous sulphate 200 mg tds
20. Raised
21. Iron deficiency anemia
22. 40-440/ μ L
23. 120 days
24. Megaloblastic
25. Malabsorption of vitamin B₁₂

CHAPTER 4: CARDIOVASCULAR SYSTEM

Multiple Choice Questions

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. b | 2. c | 3. b | 4. c | 5. b |
| 6. a | 7. a | 8. c | 9. c | 10. a |
| 11. d | 12. c | 13. d | 14. d | 15. b |
| 16. b | 17. b | 18. b | 19. b | 20. c |
| 21. a | | | | |

Fill in the Blanks

1. PDA
2. AR
3. Acute rheumatic fever
4. Eisenmenger syndrome
5. Pulmonary hypertension
6. VSD
7. PDA
8. Ostium secundum
9. Calcium channel blocker
10. Heart failure
11. 120-139 and 80-89 mmHg
12. No
13. SABE
14. Coarctation of aorta
15. SABE

16. AR
17. ASD
18. Loud
19. Left
20. Penicillin V, sulfadiazine, benzathine penicillin

CHAPTER 5: RESPIRATORY DISEASES

Multiple Choice Questions

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. d | 2. c | 3. d | 4. c | 5. a |
| 6. a | 7. d | 8. c | 9. a | 10. d |
| 11. c | 12. c | 13. c | 14. d | 15. c |
| 16. b | 17. c | 18. c | 19. d | 20. d |
| 21. c | 22. b | 23. d | 24. b | 25. b |

Fill in the Blanks

1. Right ventricle
2. Anaerobic
3. Pulmonary edema
4. S. pneumoniae
5. Gram-negative bacteria
6. Emphysema
7. Decreased
8. Crackles
9. Wheeze
10. Stridor
11. Ghon's lesion
12. Chyenne-Stokes breathing
13. COPD
14. Diethylcarbamazine
15. Emphysema
16. Pleuritis
17. Aegophony
18. PaO₂ 80-100 mmHg,
19. PaCO₂ 35-45 mmHg
20. pH 7.38-7.44

CHAPTER 6: RENAL DISEASES

Multiple Choice Questions

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. b | 2. b | 3. c | 4. d | 5. c |
| 6. b | 7. d | 8. c | 9. b | 10. b |
| 11. a | 12. d | 13. c | 14. c | 15. d |
| 16. a | 17. d | 18. a | | |

Fill in the Blanks

1. 3.5 g
2. 400 ml
3. Azotemia
4. 20-40 mg/dL
5. <1.5 mg/dL
6. 5.5-8 g/dL and 3.5-5.5 g/dL
7. 10-15

CHAPTER 7: NERVOUS SYSTEM

Multiple Choice Questions

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. a | 2. a | 3. a | 4. a | 5. c |
| 6. a | 7. d | 8. b | 9. c | 10. c |
| 11. d | 12. d | 13. d | 14. c | 15. c |
| 16. a | 17. c | 18. b | 19. a | 20. a |
| 21. d | 22. d | 23. c | 24. d | 25. c |
| 26. c | 27. c | | | |

Fill in the Blanks

1. Todd's palsy
2. Trigeminal neuralgia
3. Complication of injection therapy in trigeminal neuralgia
4. Common migraine
5. <5/mm³ lymphocytes
6. Tuberculous
7. Cryptococcal
8. Stapedius
9. Chorda tympani
10. Temporal and frontal lobes
11. Meningococcal meningitis
12. Meningococcal meningitis
13. Inhalation of 100% oxygen
14. Coronary artery disease and uncontrolled hypertension
15. Cribriform plate

CHAPTER 8: ENDOCRINE AND METABOLIC DISORDERS

Multiple Choice Questions

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. b | 2. d | 3. a | 4. c | 5. c |
| 6. b | 7. c | 8. c | 9. b | 10. c |
| 11. a | 12. c | 13. a | 14. b | 15. b |

- | | | | | |
|-------|-------|-------|-------|-------|
| 16. d | 17. d | 18. d | 19. b | 20. d |
| 21. d | 22. d | 23. d | 24. c | 25. c |

Fill in the Blanks

1. Cushing's syndrome
2. T₃
3. Hypocalcemia
4. 9-10.5 mg/dL
5. Hypocalcemic tetany
6. Graves' disease
7. Pseudohypoparathyroidism
8. Parathyroid adenoma
9. Chronic hypocalcemia due to hypoparathyroidism
10. >126 mg/dL
11. 100, 126 mg/dL
12. Kussmaul's breathing
13. 30-300 mg
14. ACE inhibitors
15. Regular insulin

CHAPTER 9: INFECTIONS

Multiple Choice Questions

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. b | 2. b | 3. b | 4. c | 5. c |
| 6. c | 7. a | 8. b | 9. c | 10. c |
| 11. b | 12. b | 13. b | 14. c | 15. b |
| 16. c | 17. c | 18. b | 19. b | 20. c |
| 21. c | 22. b | 23. b | 24. c | 25. d |
| 26. b | 27. b | 28. d | 29. c | 30. d |
| 31. c | 32. d | 33. c | 34. c | 35. b |
| 36. c | | | | |

Fill in the Blanks

1. Rubella
2. Congenital syphilis
3. Tertiary syphilis
4. Benzathine penicillin G
5. *Treponema pallidum*
6. Falciparum
7. Malaria
8. *Entamoeba histolytica*
9. Amoebic liver abscess
10. Diphtheria

CHAPTER 12: ANAPHYLAXIS AND DRUG ALLERGY

Multiple Choice Questions

- | | | | | |
|------|------|------|------|-------|
| 1. b | 2. b | 3. d | 4. b | 5. d |
| 6. a | 7. d | 8. a | 9. c | 10. b |

CHAPTER 13: NUTRITION

Multiple Choice Questions

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. b | 2. d | 3. d | 4. b | 5. a |
| 6. d | 7. d | 8. b | 9. c | 10. b |
| 11. c | 12. a | 13. a | 14. a | 15. a |
| 16. a | | | | |

Fill in the Blanks

- Niacin
- Thiamine
- Vitamin A
- Vitamin A
- Vitamin D
- Vitamin K
- Vitamin B₁₂
- Polished rice
- Riboflavin
- Pyridoxine
- Niacin

ANSWERS TO TEST PAPER

- | | | | | |
|-----------------------|-------|-------|-------|-------|
| 1. c | 2. d | 3. c | 4. c | 5. d |
| 6. a-4, b-3, c-2, d-1 | | | | |
| 7. a-2, b-1, c-4, d-3 | | | | |
| 8. a-2, b-1, c-4, d-3 | | | | |
| 9. d | 10. b | 11. c | 12. c | 13. b |
| 14. d | 15. c | 16. d | | |

- | | | | | |
|--|--------|--------|--------|--------|
| 17. a-3, b-1, c-2, d-4 | | | | |
| 18. b | 19. d | 20. d | | |
| 21. a-3, b-4, c-2, d-1 | | | | |
| 22. a-4, b-1, c-2, d-3 | | | | |
| 23. b | 24. d | 25. c | 26. c | 27. c |
| 28. a | 29. b | 30. b | 31. c | |
| 32. D (Flu like syndrome is usually seen in intermittent regime) | | | | |
| 33. a | 34. c | 35. c | | |
| 36. D (Biopsy can tell the specific cause of malabsorption) | | | | |
| 37. d | 38. c | 39. d | 40. c | 41. c |
| 42. c | 43. b | 44. b | 45. b | 46. c |
| 47. c | 48. d | 49. c | 50. c | 51. a |
| 52. c | 53. c | 54. d | 55. c | 56. c |
| 57. b | 58. c | 59. b | 60. c | 61. b |
| 62. a | 63. c | 64. c | 65. b | 66. d |
| 67. d | 68. d | 69. d | 70. c | 71. c |
| 72. b | 73. c | 74. c | 75. c | 76. c |
| 77. c | 78. b | 79. c | 80. d | 81. c |
| 82. b | 83. d | 84. a | 85. b | 86. d |
| 87. d | 88. d | 89. c | 90. d | 91. d |
| 92. d | 93. c | 94. d | 95. d | 96. d |
| 97. a | 98. a | 99. b | 100. b | 101. a |
| 102. d | 103. d | 104. c | 105. d | 106. d |
| 107. c | 108. d | 109. b | 110. b | 111. b |
| 112. b | 113. c | 114. b | 115. c | 116. b |
| 117. a-3, b-4, c-2, d-1 | | | | |
| 118. d | 119. c | 120. a | 121. c | 122. d |
| 123. a | 124. d | 125. c | 126. a | 127. d |
| 128. d | 129. d | 130. d | 131. c | 132. d |
| 133. c | 134. a | 135. c | 136. d | 137. a |
| 138. c | 139. a | 140. c | 141. d | 142. b |
| 143. a | 144. c | 145. a | 146. a | 147. c |
| 148. b | 149. d | 150. a | | |

Index

A

- Abdominal
 - bruit 81
 - distension 15, 16
 - examination 16
 - obesity 3
 - pain 15
- Abducens nerve 165
- Abnormal excretion of serum proteins 153
- Abnormalities of urinary sediments 153
- Accessory nerve 166
- Accommodation reflex 163
- Achondroplasia 3
- Acidic foods 11
- Acquired coagulation disorders 71
- Acromegaly 198
- Acute
 - AR 89
 - bacterial meningitis 174
 - coronary syndromes 99
 - diarrhea 25
 - erosive and hemorrhagic gastritis 22
 - gastritis due to infections 22
 - heart failure 104
 - hepatitis 35
 - leukemias 58
 - MR 87
 - myocardial infarction 100
 - pulmonary edema 104, 107, 243
 - renal failure 158
 - respiratory distress syndrome 252
 - rheumatic fever 83
 - severe asthma 133
 - viral hepatitis 35
- Addison's disease 198
- Adrenal crisis 245
- Advanced cardiac life support 255
- Agranulocytosis 72
- Airways obstruction 243
- Alkaline phosphatase 31
- Allergic bronchopulmonary aspergillosis 137
- Alteration in urine volume 153
- Altered bowel habits 15
- Aminotransferases 31
- Amoebiasis 226
- Amoebic liver abscess 227
- Anaphylaxis 245, 256
- Ancylostoma duodenale 52
- Anemia 50, 79
- Anemia of acute blood loss 57
- Angular
 - cheilitis 20
 - stomatitis 18, 52
- Anti-arrhythmic drugs 113
- Antimicrobial treatment 175
- Aortic
 - regurgitation 88
 - root repair 89
 - stenosis 87
 - valve replacement 89
- Aphthous ulcer 19
- Aplastic anemia 55
- Appendicitis 15
- Approach to case of headache 178
- Argyll Robertson pupil 163
- Arrhythmia 112
- Arterial
 - blood gas analysis 125, 135
 - pulse 80
- Ascites 9, 45
- Asthmatic attack 243
- Atrial
 - fibrillation 116
 - septal defect 109
 - tachycardia 116
- Atrioventricular
 - block 114
 - re-entrant tachycardia 116
- Atrophic thyroiditis 189
- Atypical facial pain 181
- Auscultation 81
- Autoimmune gastritis 22
- AV nodal re-entry tachycardia 116
- Avoidance of trigger factors 179

B

- Balanced diet 263
- Balloon tamponade 44
- Basilar migraine 179
- Bedrest 84
- Bell's palsy 183, 217
- Benign tertiary syphilis 225
- Berry aneurysms 110
- Biotin 266
- Bisferiens pulse 5
- Bleeding 15
- Blood
 - ammonia level 31
 - culture 90
 - pressure 5, 12, 80
 - tests 202
- Body mass index 3
- Bone marrow transplantation 55, 60
- Bradyarrhythmias 113
- Brain
 - death 169
 - stem 169
- Bronchial asthma 130
- Bronchiectasis 129
- Bronchogenic carcinoma 148
- Bronchopneumonia 126
- Brown tumors 196
- Brudzinski's sign 175
- Built of body 3
- Burkitt's lymphoma 64

C

- Calcium metabolism 193
- Candida albicans 20
- Candidal leukoplakia 20
- Carbimazole 192
- Carbohydrate 262
- Cardiac
 - arrest 253
 - catheterization 82
 - examination 81
 - syncope 111

Cardiogenic shock 250
 Cardiopulmonary resuscitation 253
 Cardiovascular
 collapse 253
 disorders 111
 evaluation 271
 syphilis 225
 system 78
 Carditis 83
 Carotenemia 7
 Carotidynia 179
 Casal's necklace 265
 Causes of
 cobalamin and folic acid deficiency 54
 CRF 159
 iron deficiency anemia 52
 seizures 171
 syncope 111
 Central nervous system 50, 94
 Cerebral malaria 230
 Cerebrospinal fluid examination 169
 Cerebrovascular
 accident 185, 245
 diseases 111, 185
 Cheilosis 18
 Chemoprophylaxis 144
 Chest
 pain 78, 122, 243
 X-ray 82
 Chickenpox 214
 Cholestatic jaundice 34
 Choline 266
 Chorea 83
 Chronic
 complications of malaria 229
 diarrhea 27
 gastritis 22
 heart failure 104
 hepatitis 39
 B 40
 C 40
 lymphocytic leukemia 61
 meningitis 177
 myeloid leukemia 60
 obstructive pulmonary disease 134
 renal failure 159
 viral hepatitis 40
 Chvostek's sign 193
 Cirrhosis of liver 41
 Classical migraine 178
 Clinical features of acute hepatitis 37
 Clot lysis 65
 Clubbing 7, 0
 Cluster headache 180
 Coagulation
 disorders 69

 factors 31
 inhibitors 66
 Coarctation of aorta 110
 Cobalamin 53
 deficiency 55
 Color vision 163
 Coma 3, 167
 Common migraine 178
 Community acquired pneumonia 126
 Complete urine analysis 154
 Complicated migraine 178
 Complications of
 diabetes 203
 pulmonary tuberculosis 144
 Confusional state 167
 Congenital
 heart disease 108
 syphilis 225
 Conjugated hyperbilirubinemia 34
 Conjunctiva 12
 Consciousness 167
 Control of heart failure 106
 Coronary
 arteriography 98
 artery
 bypass grafting 99
 disease 3, 96
 Correction of
 precipitating factors 108
 reticulocyte count 51
 Cough 120
 Crigler-Najjar syndromes 34
 CT scan and MRI 32
 Cushing's disease 199
 Cutaneous reactions 259
 Cyanosis 8, 79

D

Decline in glomerular filtration rate 154
 Delirium 3
 Dementia 3
 Dense proteins 11
 Dental
 procedures in pregnant women 246
 trauma 179
 Di George syndrome 195
 Diabetes
 insipidus 200
 mellitus 200
 Diabetic
 ketoacidosis 203
 nephropathy 204
 neuropathy 204
 Diarrhea 25
 Diet and nutrition 262
 Diphtheria 220

Directly observed treatment, short course 144
 Disorders of hemostasis 66
 Disseminated intravascular coagulation 71
 Doppler ultrasonography 138
 Drowsiness 167
 Drug
 allergy 256, 259
 resistant tuberculosis 143
 Dubin-Johnson and Rotor syndromes 34
 D-xylose test 30
 Dysentery and food poisoning 25
 Dysphagia 16, 20, 52
 Dyspnea 78, 120

E

Early syphilis 224
 Echocardiography 82, 90, 138
 Edema 9, 78-80
 Effects of hypertension on target organs 93
 Electrocardiogram 82
 Electroencephalography 169
 Embolectomy 139
 End stage renal disease 159
 Endocrinal evaluation 271
 Endocrine and metabolic disorders 188
 Endoscopic
 retrograde cholangiopancreatography 32
 therapy 22
 Energy 262
 Enteric fever 221
 Epilepsy 169
 Epstein-Barr virus 63
 Eradication of streptococci 84
 Erythema marginatum 83
 Esophageal
 disease 16
 dysphagia 21
 transection 44
 Essential hypertension 93
 Estimation of glomerular filtration rate 154
 Etiology of lung cancers 148
 Examination of
 comatose patient 167
 cranial nerves 162
 Excessive bleeding 246
 Extra-abdominal diseases 15

F

Facial
 migraine 179
 nerve 165
 palsy 181
 pain 180

Fat 262
 Fat soluble vitamins 264
 Fatigue 78, 79
 Fecal fat estimation 29
 Folate and vitamin B12 266
 Folic acid 54
 Food poisoning 26
 Forced
 expiratory volume 125
 vital capacity 125
 Functional platelet disorders 68
 Fungal meningitis 177

G

Gallstones 15
 Gastritis 22
 Gastrointestinal and hepatobiliary system 15
 German measles 213
 GI bleeding 16
 Gilbert's syndrome 34
 Gingival hyperplasia 20
 Glossitis 52
 Glassopharyngeal nerve 166
 Glossopharyngeal neuralgia 181
 Glucose lowering agents 205
 Glycated hemoglobin measurement 202
 Gonorrhea 223
 Graves' disease 191
 Gum hyperplasia 58
 Gynecological diseases 15

H

Halitophobia 11
 Halitosis 11
 Headache 178
 Health care associated pneumonia 126
 Heart
 burn 15
 failure 103
 Helicobacter pylori 63
 Hematemesis 16
 Hematological
 diseases 20
 evaluation 271
 system 50
 Hemoglobinopathy 56
 Hemolytic anemia 56
 Hemophilia 69
 Hemoptysis 79, 121
 Hepatic
 encephalopathy 44
 syndromes 260
 Hepatitis
 A 35, 39
 B 35, 39
 C 36

D 37
 E 37
 Hepatocellular
 carcinoma 45
 diseases 34
 Hepatorenal syndrome 45
 Herpes
 labialis 20
 simplex virus 20
 zoster 15, 214, 215
 Herpesvirus hominis 216
 Herpetic stomatitis 20
 HIV and tuberculosis 140
 Hodgkin's disease 61
 Horner's syndrome 163, 180
 Hoshimoto's thyroiditis 189
 Hospital acquired pneumonia 126
 Human immunodeficiency syndrome 231
 Hydrothorax 9
 Hyperbilirubinemia 7
 Hypercalcemia 194
 Hypereosinophilic syndrome 137
 Hyperglycemic hyperosmolar state 204
 Hyperkinetic pulse 5
 Hyperparathyroidism 196
 Hyperpyrexia 4
 Hypertension 92
 Hypertensive crisis 96, 242
 Hyperthyroidism 190
 Hypertrophic osteoarthropathy 8
 Hyperventilation 244
 Hypoadrenal shock 248
 Hypocalcemia 193
 Hypoglossal nerve 167
 Hypoglycemia 208, 245
 Hypoparathyroidism 195
 Hypopituitarism 3, 196
 Hypothyroidism 3, 188
 Hypovolemic shock 247

I

Idiopathic thrombocytopenic purpura 67
 Immunosuppressive therapy 55
 Impaired
 fasting glucose 202
 glucose tolerance 202
 hepatic metabolism 33
 Implications on dental practice 47, 118,
 160, 208, 238, 260, 269
 Infections 211
 Infectious mononucleosis 218
 Infective endocarditis 89
 Inferior vena caval filters 139
 Inherited coagulation disorders 69
 Insulin 207
 Intermittent fever 4

Interstitial
 lung disease 147
 pneumonia 127
 Intestinal obstruction 15
 Investigations in renal disorders 154
 Iron deficiency anemia 51, 52
 Ischemic heart disease 96

J

Jacksonian march 170
 Jarisch Herxheimer reaction 226
 Jaundice 7, 15, 16, 32
 Jugular venous
 pressure 6
 pulse 6, 81
 Junctional tachycardia 116

K

Kernig's sign 174
 Kidneys 17
 Kussmaul's sign 6
 Kwashiorkor 264

L

Lactate dehydrogenase 54
 Laryngeal paralysis 167
 Late latent syphilis 224
 Latent syphilis 224
 Left ventricle 108
 Leukemia 58
 Liver 30
 abscess 15
 biopsy 32
 disease 9
 function tests 30
 Lobar pneumonia 126
 Low output heart failure 105
 Lower
 half headache 179
 limbs 12
 Lung
 abscess 128
 cancers 148
 Lutembacher's syndrome 109
 Lymph nodes 10
 Lymphadenopathy 74
 Lymphomas 61

M

Malabsorption 28
 Malaria 228
 Management of
 cardiac arrest 253
 cardiogenic pulmonary edema 107

diabetes 205
 drug allergy 260
 hemorrhagic tendency in liver disease 44
 HIV infection 233
 Management steps in diabetic patients 207
 Marfan's syndrome 3
 Measles 211
 Medical emergencies in dental practice 241
 Medulla 169
 Megaloblastic anemia 53
 Melena 16
 Meningitis 174
 Mental and emotional status 3
 Methimazole 192
 Migraine 178
 equivalent 178
 Migrainous neuralgia 180
 Miliary
 pneumonia 127
 tuberculosis 140
 Milk products 11
 Minerals 263
 Mitral
 regurgitation 86
 stenosis 84
 valve
 repair 87
 replacement 85, 87
 valvotomy 85
 Mode of transmission 231
 Mother to child transmission of HIV infection 237
 Mouth wash 12
 Movements of abdominal wall 16
 Mucosal phase 28
 Multiple myeloma 64
 Mumps 212
 Myocardial infarction 15
 Myth about halitosis 12

N

Nausea and vomiting 15
 Neck rigidity 174
 Necrotizing ulcerative stomatitis 19
 Nephritic syndrome 156
 Nervous system 162
 Neurogenic shock 248
 Neurosyphilis 225
 Niacin 265
 Nodular lymphocyte predominant
 Hodgkin's disease 63
 Non-Hodgkin's lymphoma 63
 Nonsteroidal anti-inflammatory drugs 220
 Normal
 hematopoiesis 50

hemostasis 64
 Nutrition 262
 Nutritional
 deficiency 20
 evaluation 272

O

Obstructive
 jaundice 34
 shock 248
 Oculomotor nerve 164
 Odynophagia 21
 Olfactory nerve 162
 Ophthalmoplegic migraine 179
 Optic nerve 162
 Oral
 candidiasis 20
 cavity 12
 iron therapy 53
 manifestations in HIV disease 237
 ulcers 18
 Oropharyngeal dysphagia 21
 Orthopnea 78
 Orthostatic hypotension 242
 Osmotic diarrhea 27, 29
 Osteitis fibrosa cystica 196
 Osteum
 primum 109
 secundum 109
 Overproduction of bilirubin 33
 Owl's eye appearance 62

P

P. falciparum 4
 Palatal paralysis 166
 Pallor 7
 Palpitation 78
 Pancreatic exocrine functions 30
 Pancreatitis 15
 Pantothenic acid 266
 Paracentesis 45
 Parathyroid disorders 195
 Parenteral iron therapy 53
 Paroxysmal nocturnal dyspnea 78
 Partial seizures 170
 Patent ductus arteriosus 109
 Pathogenesis of coronary atherosclerosis 97
 Pathologic bad breath 11
 Pathological basis of headache 178
 Pathophysiology of
 coma 167
 halitosis 11
 unstable angina 99
 Patterson Kelly syndrome 52
 Peak expiratory flow rate 126

Pelvic
 inflammatory disease 223
 ulcer 15, 23
 Percutaneous transluminal coronary angioplasty 99
 Peripheral
 cyanosis 8
 signs of infective endocarditis 80
 Physiologic bad breath 11
 Pituitary gland 196
 Platelet
 disorders 67
 plug formation 65
 Pleural
 diseases 145
 effusion 146
 Pleurisy 145
 Pneumonia 15, 126
 Pneumothorax 146
 Portal
 hypertension 42
 venous pressure 43
 Post-herpetic neuralgia 181
 Post-primary pulmonary tuberculosis 140
 Postural
 hypotension 111, 242
 syncope 111
 Preoperative
 evaluation 271
 laboratory evaluation 272
 Prevention of
 HIV infection 237
 recurrent anaphylaxis 258
 variceal bleeding 44
 viral hepatitis 38
 Primary generalized seizures 262, 170
 hemostasis 65
 hyperparathyroidism 196
 pulmonary tuberculosis 139
 syphilis 224
 tuberculosis 139
 Prominent veins 17
 Prophylactic therapy 179
 Prophylaxis 91, 230
 Propranolol 192
 Propylthiouracil 192
 Protein 262
 energy malnutrition 263
 Proteinuria 153
 Pulmonary
 angiography 125, 138
 embolism 137
 eosinophilia 136
 evaluation 271
 function tests 134
 secretion examination 128
 stenosis 110

syndromes 260
tuberculosis 139
Pulse 4
rate 4
Pulsus alternans 5
Pupillary examination 163
Pyridoxine 266

R

Radio-femoral delay 5
Recommended dietary allowance 263
Red blood cells 50
transfusion 53
Reed-Sternberg cells 62
Regulation of calcium metabolism 193
Renal
diseases 68, 153
stones 15
syndromes 153, 260
Respiratory
diseases 120
failure 144
system examination 123
Rheumatic valvular heart disease 84
Rhythm 4
Riboflavin 265
Right atrium 108
Right ventricular hypertrophy 110
Route of infection 126
Rubella 213

S

Salicylates 84
Schamroth's window test 8
Scheme of general examination 12
Schilling test 29
Secondary
hemostasis 65
hyperparathyroidism 196
hypertension 93
syphilis 224
Secretory diarrhea 27, 29
Seizures 111, 244
Septic shock 249
Serological
studies 30
tests 128
Serum
albumin 31
bilirubin 30
biochemistry 154
enzymes 31
fructosamine measurement 202
Severe falciparum malaria 229

Shape of abdomen 16
Shock 247
Sick sinus syndrome 114
Sickle cell
anemia 56
disease 57
Sinus
bradycardia 113
venosus 109
Skin 17
Sleep 167
Spinal diseases 15
Spleen 17
Spontaneous bacterial peritonitis 46
Sputum 121
Stable angina 97
Staphylococcus aureus 90
Starling forces 9
Status epilepticus 173
Steroids 84
Stokes-Adams attacks 114
Stomatitis 18
Stress testing 98
Stroke 245
Stupor 3, 167
Subcutaneous nodules 83
Sudden cardiac death 253
Sugars 11
Supraventricular tachycardias 116
Symptoms and signs of
cardiovascular diseases 78
gastrointestinal diseases 15
Syncope 78, 79, 110, 111
Syphilis 224
Systolic or diastolic heart failure 105

T

Tachyarrhythmias 114
Takayasu's disease 110
Tension headache 180
Tertiary
hyperparathyroidism 196
syphilis 225
Tetralogy of Fallot 110
Thalassemias 56
Thiamine 264
pyrophosphate 264
Thyroid
crisis 192
disorders 188
function tests 169
Thyrotropin releasing hormone 188
Todd's paralysis 170
Tongue and bad breath 12

Tonic
clonic seizures 170
pupil 163
Transitory bad breath 11
Transmission of malaria 228
Treatment of
acute attack 179
bad breath 12
high grade lymphoma 64
low grade lymphoma 63
resistant malaria 230
urticaria and angioedema 258
variceal hemorrhage 43
Trench mouth 19
Tricuspid stenosis 6
Trigeminal
nerve 164
neuralgia 180
Trochlear nerve 164
Tropical eosinophilia 136
Trousseau's sign 193
Tuberculosis 139
Tuberculous meningitis 177
Types of
breath sound 123
heart failure 104
jaundice 33
pneumothorax 146
Typhoid fever 221

U

Ulcerative stomatitis 19
Ultrasonography 31
Umbilicus 16
Uncommon types of gastritis 23
Unconjugated hyperbilirubinemia 33
Unstable angina 99
Upper
gastrointestinal bleeding 43
GI endoscopy 30
Urine
bilirubin 30
test 202
Using
bad breath detective 11
halimeter 11

V

Vaccination 144
Vagal tone 116
Vagus nerve 166
Valsalva maneuver 116
Variants of migraine 179
Varicella 214

Vascular disorders 111
 Vasovagal syncope 111
 Ventilator associated pneumonia 126
 Ventricular
 fibrillation 117
 premature beats 117
 septal defect 108, 110
 tachyarrhythmias 117
 tachycardia 117
 Vessel wall abnormalities 67
 Vestibulocochlear nerve 165
 Vincent's infection 19
 Viral meningitis 177

Visual
 acuity 163
 field 163
 Vitamin 262, 264
 A 267
 B₁ 264
 B₂ 265
 B₃ 265
 B₆ 266
 C 266
 D 269
 E 267
 K 268

Volatile sulphur compounds 12
 von Willebrand's disease 69

W

Water soluble vitamins 264
 Waterhouse Friderichsen syndrome 174
 Weight loss 16
 Wheezing 122

X

Xylitol chewing gums 12